

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	2 of 623
	Status:			Effective Date:	15-May-2023
		Title:	Service Manual Erytra		



Diagnostic Grifols S.A.
 Passeig Fluvial 24
 08150 Parets del Vallés
 Barcelona, SPAIN

DIAGNOSTIC GRIFOLS, S.A.

Print date:	16-May-2023	<i>Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization</i>	
		Purpose:	Uncontrolled Copy

Changes from previous revision:

Revision	Change
September/2010	NA
July/2011	- New Erytra Diagnostic chapter. - New adjustment procedures, verifications procedures and tools. - Backup_Restore_DataBase application. - Information about SSD protection, date and time settings, installation of peripherals. - Installation, configuration and calibration of touchscreen ver.2 (3M) - Added procedures to replace Z chains and Y chain of PIP module. - Added probes offset verification. - Updated technical specifications. - Updated IQ procedure. - Updated OQ procedure. - Updated Preventive Maintenance. - Updated errors lists - Updated mechanical drawings references. - Updated electronic schematic references. - Updated software installation procedure. - Updated packing/unpacking procedures. - Updated aspiration pump verification. - Updated lubrication points. - Updated replacement procedures and verifications - Updated tachometer specification. - Minor changes in replacement and verifications procedures - Minor typographic errors.
July/2012	- IQ and OQ procedures can be found in Erytra IQ is REGD-0009727 document and Erytra OQ is REGD-0009728 document. - Updated OQ document. - Updated error lists. - Updated ErytraDiagnostic chapter. - Updated Technical Service Software chapter. - Updated spare part replacement PM -> Z CAT chain from 3 to 7 years replacement. - Updated probes parallelism verification procedure. - Updated parallelism adjustment in PIP chapter. - Updated Z probes chain replacement procedures. - Updated incubators height adjustment. - Updated fluidic procedures. - Updated how to report chapter. - Updated incubator adjustment/verification temperature to be done in PIP area. - Updated COM2 to COM3 in wiring list chapter General Module. - Updated touchscreen replacement procedure. - Updated adjustment distance of detection PIP / CAT. - Updated probe/wagon adjustment procedure. - Updated temperature probe position in incubator adjustment temperature. - Updated software installation procedures. - Updated functional characteristics. - Updated unpacking procedure. - Updated SSD structure section. - Added Erytra-LIS connection section. - Added new adjustment procedure probes offset. - Added new verification procedure probes offset. - Added troubleshooting to all modules. - Added new erytra.properties parameters. - Added new probe replacement procedure if impact detector is installed. - Added reference to Japanese and Norwegian language. - Added restore database backup procedure. - Added flatness verification card drawer in General Module chapter. - Added flatness verification samples drawer in General Module chapter. - Added flatness verification incubator in Incubator chapter. - Added flatness verification reagents drawer in General Module chapter. - Added PCA Level detector replacement (with and without impact detector). - Added previous scan of the rack in the verifications of the CAT Module. - Added specification for chronometer in CEN module. - Added new procedure backup/restore EEPROM adjustments. - Added lubrication point in user tray. - Removed reference to clog test in Fluidics Module. - RESPONSIBLE AUTHORITY changed into SUPERVISOR in first page of Service Manual and in ErytraDiagnostic chapter. - Minor cosmetic changes in PM template. - Minor typographic errors.

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	4 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

Revision	Change
September/2012	- Updated probe adjustment procedure in reagent drawers. - Updated probe adjustment procedure in incubators. - Updated remote connection section. - Updated incubator module replacements. - Updated CPD verifications. - Updated reagents and samples drawers keyboard replacement procedure. - Updated PIP robotic head removal procedure. - Updated level detection verification. - Updated mechanical drawings. - Updated files structure section. - Updated functional characteristics. - Added impact detector verification. - Added LIS communication errors. - Added username/password information of instrument computer. - Minor typographic errors.
February/2013	- Added dispensation pumps and expansion vessels verification. - Added Erytra On/Off switcher. - Added distribution blocks/probes washer racors replacement procedures. - Added Result -1 in samples tubes presence detector. - Added reference to Norwegian language. - Updated error codes tables. - Updated presence detector distance of detection adjustment. - Updated leak-tightness test. - Updated Technical Service software chapter. - Updated gripper in reader area verification. - Updated working conditions temperature. - Updated verification/adjustment when adjusting/removing reader module. - Updated level detection verification. - Updated touchscreen drivers installation procedures. - Updated technical specifications – dimensions. - Updated Windows registry structure section. - Updated fluidic chapter. - Updated fluidic procedures. - Updated dispensation pumps and proportional valves verification. - Updated aspiration pump verification. - Updated ErytraDiagnostic Warning and Fail results table. - Updated SSD image restore installation. - Updated incubator drawings. - Updated fluidic drawings. - Updated field length limitations in LIS section. - Updated LIS connection section.
March/2014	- Added touchscreen configuration. - Added PIP barcode configuration. - Added doors connectors in PCA GEN-BCKPL59 picture. - Added procedure to apply silicone in fittings. - Added Install OS procedure - Added procedure to add a delay to run automatically the software. - Added Erytra Database Manager section. - Added tables with encoders information. - Updated Preventive Maintenance. - Updated error codes. - Updated Technical Service Software chapter. - Updated adjustment procedures with new adjustment wizards. - Updated LIS communication protocol. - Updated How to enable and configure LIS connection section. - Updated database backup and restore procedures. - Updated archive section. - Updated samples drawers configuration. - Updated SERVICE_RACK_MODE configuration. - Updated Peltier pictures. - Updated probes offset verification. - Updated probes washer pictures. - Updated impact detector system verification. - Updated incubator temperature verification. - Updated dispensation pumps and expansion vessels verification. - Updated level detection PCA replacement. - Updated fluidic procedures. - Updated diluton cup verification. - Updated firmware update section. - Updated PIP, CAT, CUP troubleshooting. - Updated remote connection. - Updated language section.

Print date:	16-May-2023	Purpose:	Uncontrolled Copy
--------------------	-------------	-----------------	-------------------

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	5 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

March/2014	- Updated files structure section. - Updated dilution cup assembly. - Updated card drawers area, incubators area, card waste container area, centrifuges area verifications. - Updated erytra.properties parameters (DEFAULT_RACK_CONFIGURATION , DUAL_MODE, RESERVE_INCUBATOR_FOR_STAT). - Updated leak-tightness verification. - Updated touchscreen drivers configuration. - Updated Check home of the horns and Check home of the X-Y-Z1-Z2 axis sections. - Updated PCA INC-BL59 picture. - Updated ErytraDiagnostic chapter. - Updated Install All procedure - Updated Erytra ON/OFF switcher - Updated interconnection manifolds pictures. - Updated probes leadscrew replacement procedure. - Updated incubator mechanical and verification procedures. - Updated functional characteristics. - Updated card barcode reader mirror replacement. - Updated General module wiring table. - Updated mechanical drawings. - Updated fluidic diagram. - Updated electronic schematics. - Updated PCA GEN-BCKPL59 verification. - Minor typographic errors.
July/2015	- Updated troubleshooting sections. - Updated preventive maintenance. - Updated fluidic procedures. - Updated gripper parallelism adjustment. - Updated gripper parallelism verification. - Updated centrifuge verification. - Updated PIP adjustment on reagent drawers. - Updated Erytra Database Manager section. - Updated remote connection section. - Updated computer section. - Updated user tray and keyboards picture. - Updated computer verification. - Updated fbwfmgr section. - Updated touchscreen brightness value. - Updated SSD structure section. - Updated O.S. information. - Updated files structure section. - Updated prime test verification. - Updated LIS protocol. - Updated Drawers and Load cells test description of Erytra Diagnostic. - Updated how to report section. - Updated section temperature control system. - Updated SSD restore procedure. - Updated Operating System restoring procedure. - Updated Erytra software installation. - Updated Manual databases backup procedure. - Updated Database restore procedure. - Updated touchscreen drivers installation procedures. - Updated Pumps and electrovalves test in ErytraDiagnostic section. - Updated Centrifuges test in ErytraDiagnostic section. - Updated Warning and Fail results in Erytra Diagnostic software table. - Updated mechanical drawings. - Updated unpacking section. - Updated bottles configuration selector. - Updated drawers section. - Added sample barcode specifications. - Added fluidic drawer linear guide replacement procedure. - Added troubleshooting in LIS section. - Minor typographic errors.
March/2016	- Updated technical specifications. - Updated date and time setting section. - Updated computer section. - Updated fluidic procedures section. - Updated adjustment tools section. - Updated LIS connection section. - Updated SSD image restoring procedure. - Updated Operating System restoring procedure. - Updated description of log files section. - Updated mechanical drawings.

Print date:	16-May-2023	Purpose:	Uncontrolled Copy
--------------------	-------------	-----------------	-------------------

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization

Reference:	REGD-0006730	Version: 17.0; Most-Recent; Effective; CURRENT	Page 6 of 623	Effective Date: 15-May-2023
	Status: Effective			
Service Manual Erytra				
Title:				

March/2016	- Updated electronic schematics. - Minor typographic errors.
June/2017	- <i>Erytra</i>® logo updated to <i>erytra</i>®. - Added new software configuration parameters (erytra.properties). - Updated <i>Files structure</i> section. - Updated <i>Modular tabs</i> section with new Technical Service software functionalities. - Updated <i>Computer</i> section including PC EMBEDDED v4. - Updated SSD image restoring procedure. - Updated software installation procedure. - Updated Operating System restoring procedure. - Updated touchscreen driver installation procedure. - Updated <i>Peripheral installation</i> section. - Updated <i>Date and Time</i> setting section. - Updated <i>Gripper Adjustment in Card waste container area</i> procedure. - Updated adjustment procedure for the load cell of the card waste container. - Updated <i>Drawers and Load cells</i> test section in Erytra Diagnostic chapter. - Updated <i>Parts description</i> section in <i>Fluidic Module</i> chapter with external drain specifications. - Updated <i>Erytra Database Manager</i> section. - Updated <i>Erytra-LIS connection</i> section with new DianaServer and HostDemo features and new LIS configuration properties. - Updated <i>Erytra-LIS communication protocol</i> section. - Added <i>TeamViewer</i> section with guidelines on how to establish a remote connection. - Added <i>McAfee Application Control</i> section. - Removed references to the <i>C:</i> drive protection/unprotection from the corresponding procedures. - Updated definition of B-COM_RE connection port from COM3 to COM2. - Updated values for the accepted diameter reading values (security margin implemented). - Updated <i>Leak-tightness test</i> section. - Updated adjustment tool new codes and descriptions. - Updated images for Probes adjustment tool (Ver. 4). - Minor typographic errors.
March/2018	- Added new section <i>Completing the software installation</i>. - Updated <i>Software configurations</i> section with new Barcode reader configuration subsection. - Updated <i>Operation Qualification (OQ)</i> section. - Updated <i>Files structure</i> section - Updated <i>Technical service software overview</i> chapter with new PIP BCR configuration settings. - Updated probe adjustment procedure on incubators in <i>PIP Functional adjustments</i> section. - Updated <i>PIP Barcode readers configuration</i> with new BCR configuration customization procedure. - Updated <i>Fluidic procedures</i> section. - Updated adjustment procedure and verification for the load cell of the card waste container. - Updated syringe replacement procedure. - Updated <i>Leak-tightness test</i> section. - Updated <i>Diagnostic Application</i> chapter (formerly <i>Erytra Diagnostic</i>) with new application name, modified <i>Drawers and Load cells</i> test and modified pre-requisite text for <i>Pipetting precision and accuracy</i> test. - Updated <i>Preventive and corrective maintenance</i> section with new PM form, separated from the current Service Manual document under a new reference. - Added new section <i>Windows Security Updates installation</i>. - Updated <i>Description of log files</i> section. - Updated <i>TeamViewer</i> section. - Updated <i>Erytra-LIS connection</i> section with new DianaServer and HostDemo features and new LIS configuration properties. - Updated <i>Erytra-LIS communication protocol</i> section with new values for protocol fields, comments and removed <i>Field length limitations</i> subsection. - Updated <i>Erytra-LIS errors</i> section. - Updated values for the accepted diameter reading values. - Minor typographic errors.
May/2018	- Service Manual is unified for <i>CE</i> and <i>US</i> configurations. - PC Kontron is referred to as <i>PC EMBEDDED v2 / PC EMBEDDED – USA</i>. - Updated <i>Final setup</i> section. - Added printer specifications required by <i>Direct print</i> functionality. - Removed sections 9.3.3 <i>Dispensation pumps and proportional valves</i> and 9.4.4 <i>Proportional valves (EV11, EV12)</i> along with references to previous fluidic system. - Removed section 14.5 <i>Touchscreen version 1 (Fujitsu) driver installation</i> along with references to previous touchscreen component.
February/2020	- Updated section <i>1.4 TECHNICAL SPECIFICATIONS</i> indicating heat dispersion level in BTU/hour. - Updated section <i>1.6.2 SSD structure</i> and related troubleshooting in section 4.11, deleting the procedure to add a delay in the application launching process.

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization			Purpose:	Uncontrolled Copy
Print date:	16-May-2023			

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	8 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

November/2020	- Added new section <i>9.4.13 Syringe lock installation</i> describing the installation procedure for P/N 733168 SYRINGE LOCK. - Added new chapter <i>12. MECHANICAL DRAWINGS</i> describing new drawing models (3D) and the available viewing options. These drawings have replaced previous 2D models provided in .pdf format. - Updated section <i>15.2 SSD image restoring procedure</i> with modifications on the O.S. full recovery procedure, software installation procedure, removed programs installation section (no IHS-RUNTIME installer used), necessary tools to create a bootable USB drive, relevant data to be backed up and restored and final setup. Added note indicating compatibility restrictions for Windows 10 O.S. with existing PC models. - Added note in sections <i>15.2 SSD image restoring procedure</i> and <i>15.3 Operating System restoring procedure</i> requesting to scan the USB drive for viruses before connecting it to the instrument. - Updated section <i>15.3 Operating System restoring procedure</i> with new steps using Windows 10 O.S. Added note indicating compatibility restrictions for Windows 10 O.S. with existing PC models. - Updated section <i>15.4 Windows Security Updates installation</i> with the latest Windows updates included in the O.S. installation. - Removed previous section <i>Manual Databases Backup procedure</i> detailing the use of the removed Backup_Restore_DataBase bat file. - Updated section <i>15.5 Touchscreen driver installation and configuration</i> with information on the touchscreen drivers installation and touchscreen v2 model configuration. - Updated section <i>15.6 Database restore procedure</i> with new access procedure and added note with test counter considerations. - Updated section <i>15.7 Databases Backup with Boot menu procedure</i> with new Boot Application access and tool interface. - Updated section <i>15.8 Databases Restore with Boot menu procedure</i> with new Boot Application access, tool interface and test counter considerations. - Updated required computer specifications in section <i>15.9 Erytra Database Manager</i>. - Added new section <i>15.12 AppLocker application</i>, replacing previous <i>McAfee Application Control</i> section. - Added new section <i>15.17 Mapping a network drive</i> with procedure to access a shared folder mapping a network drive. - Updated centrifuge/incubator adjustment tool part numbers (733056 and 733068) and pictures. - References to Windows 7 Embedded O.S. replaced by Windows 10 Enterprise 2019 LTSC O.S. - References to <i>D:</i> drive replaced by <i>E:</i> drive. - Previous boot application accessing procedure ('T' pattern) has been replaced by new access through grid/diagnostic buttons in several sections. - Diana System registry path has been updated in several sections. - Whitelist update procedure using McAfee Application Control commands has been replaced by new procedure using AppLocker application commands in several sections. - Windows <i>Advanced sharing settings</i> are recommended to be reviewed during a peripheral installation (section 2.10) and while configuring access to shared folders (section 15.17). - Minor typographic errors.
July/2022	- Updated section <i>1.4 Technical Specifications</i>. New values heat and noise. - Updated in section <i>1.6.3 files estructure</i>. Described contents of E:\update folder and removed Tomcat version and folder E:\Log\winevt. - Added section <i>2.3 Instrument levelling</i>. - Updated section <i>2.7 SOFTWARE CONFIGURATIONS</i> with all the parameters in Erytra.properties file and providing further details on the system behaviour with respect to deleted or wrongly spelled parameters included in erytra.properties file. - Updated section <i>2.7.2 Language</i>. Added Catalan (ca_ES), Finnish (fi_FI), and Portuguese-Brazil (pt_BR) languages. - Updated section <i>2.7.4 Serial number</i>. The S/N can't be longer than 7 characters. - Updated section <i>2.10.2 remote connection</i>. Added note warning of possible incompatibilities with newest version of Internet Explorer and Flash Player. - Added new section <i>2.10.2.1 Alternative browser for remote connection</i>. - Updated section <i>4.1.2 Power Supply</i>. Added internal AC power cabling color coding standardization. - Updated section <i>4.1.6 Drawers</i>. Sample and Reagent racks modified sample tube maximum diameter allowed, adding a note stating that for tubes bigger than 16 mm a special samples holder need to be used. - Updated section <i>4.2.1.2 Touchscreen drawer lock</i>. Updated with new drawer. - Added section <i>4.4.6 RAM memory modules replacement</i>. - Added section <i>4.4.7 PC motherboard replacement</i>. - Added section <i>4.4.8 Card drawer electric lock replacement</i>. - Added section <i>4.4.9 Samples/Reagents drawer electric lock replacement</i>. - Added section <i>4.4.10 Fluidic drawer electric lock replacement</i>. Updated procedure in regards to the one described in INR-0059#0130 NEW ELECTRIC LOCK v2 adding emergency opening check when assembling electric lock. - Added section <i>4.4.11 User tray replacement</i>. - Updated section <i>4.9 Troubleshooting</i>. In table Computer /software issues it is not necessary to delete staterepository.xml file - Updated section <i>5.1 Description</i> of the Incubator module. Added information about hemmed edges.

Print date:	16-May-2023	Purpose:	Uncontrolled Copy
--------------------	-------------	-----------------	-------------------

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	9 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

July/2022	- Updated section 6.1 <i>Description</i> of the Centifuge module. Modified centrifuging time to 550 +/- 15s. - Updated section 6.3 <i>Verification</i> of the Centifuge module. Modified centrifuging time to 550 +/- 15s. - Added section 6.4.2 <i>Centrifuge Rotor replacement</i>. - Updated section 7.4.2 <i>Probes washer replacement</i> with indication about tubes routing. - Added section 7.4.6 <i>PIP X axis chain (CH1) replacement</i>. - Updated section 7.4.8 <i>PIP Z1 axis chain (CH3) / Z2 axis chain (CH4) replacement</i> with note considering using adhesive tape in shielded area. Added note about the tube/cable routing in the grounding clamp. Added checks after replacement. - Added section 7.4.10 <i>Horns chain replacement</i>. - Added section 7.4.11 <i>Toothed belt T2.5/245 (PIP X-axis) replacement</i>. - Added section 7.4.12 <i>Toothed belt T2.5/245 (PIP Y-axis) replacement</i>. - Added section 8.4.3 <i>CAT X chain replacement</i>. - Added section 8.4.4 <i>CAT Y chain replacement</i>. - Added section 8.4.5 <i>Gripper (Z) chain replacement</i>. - Added section 8.4.6 <i>Gripper replacement</i> with a consideration on applying sealant to the gripper screws. - Added section 8.4.7 <i>Toothed belt T2.5/245 (CAT X-axis) replacement</i> - Added section 8.4.8 <i>Toothed belt T2.5/245 (CAT Y-axis) replacement</i> - Updated section 9.1.1 <i>Parts description</i> of the Fluidic module. Added information about hemmed edges. - Updated section 9.2.2.3 <i>Pressure/Vacuum sensors</i>. Added caution note about sensors. Updated text about sensors assembly and added note about the text in TSS. - Updated section 9.3.1 <i>Pressure sensors</i>. Added caution note about sensors. Updated text about sensors assembly. - Updated section 9.4.1 <i>Pressure sensors (S3, S4, S5, S6) replacement</i>. Updated with new pressure sensors and added a caution note. Updated text about sensors assembly and added note about the text in TSS. Added checks after replacement. - Updated section 9.4.2 <i>CPD sensors (S1, S2) replacement</i>. Added picture CPD sensors assembly. Added checks after replacement. - Updated section 9.4.5 <i>Aspiration pump (PM2) replacement</i>. Updated order of the steps of the procedure. - Updated section 9.4.6 <i>Electrovalves replacement</i>. Updated order of the steps of the procedure. - Updated section 9.4.7 <i>Manifold replacement</i>. Added note to add a second layer of sealant product. - Updated section 9.4.8 <i>Manifold / distribution block (BK2, BK3) fitting replacement</i>. Added note to add a second layer of sealant product. - Updated section 9.4.9 <i>Load cell replacement</i>. Updated with new fluidic drawer. - Updated section 9.4.10 <i>Fitting of the tubes</i>. Added checks after replacement. - Updated section 9.4.11 <i>Syringe replacement</i>. Added removal of the syringe lock. Added checks after replacement. - Updated section 9.4.12 <i>Dilutor valve replacement</i>. Check no valve movement is seen. Added removal of the syringe lock. Added checks after replacement. - Updated section 9.4.13 <i>Syringe lock replacement</i>. Not in contact with valve. - Added section 9.4.15 <i>Coupling CPC bottles connector replacement</i>. - Added section 9.4.16 <i>Connection block (BK4, BK5) replacement</i>. - Added section 9.4.17 <i>Vacuum filter replacement</i>. - Added section 9.4.18 <i>Aspiration pump tubing replacement</i>. - Added section 9.4.19 <i>PCA FLU-CTRL59 board replacement</i>. - Updated section 9.9 <i>Troubleshooting</i>. Updated error code and description for: ERRORCODE FLU UNABLE TO READ DILUTOR PLUNGER POSITION. - Added section 10.4.1 <i>Dilution cup wells assembly replacement</i>. - Updated section 10.4.2 <i>Dilution cup motor replacement</i>. Updated with new dilution cup structure. Added checks after replacement. - Updated section 11.4.1 <i>Reader module removal</i>. Added checks after replacement. - Updated section 12 <i>Mechanical drawings</i>. Updated 3D and 2D drawings. - Updated section 12.1 <i>Requirements</i>. Added minimum version required of Creo view express (8.0) to view 3D drawings. - Updated section 14.2 <i>ITEMS AND CALENDAR</i>. Added table Equivalence between years in PM and tests under standard instrument use and added that both can be used for scheduling the PM.. - Updated section 14.3 <i>LUBRICATION POINTS</i>. Added indication not to lubricate in excess. Added indication on centering gripper spring assembly. - Updated section 15.3.1 <i>erytra software installation</i>. Added checkings for reader error and procedure to execute installer when updating from 4.3.1. - Updated the title and section 15.5 <i>Windows Security Updates</i> with latest security updates. - Added section 15.6 <i>Access to Windows 10 File Explorer</i>. - Updated section 15.11 <i>Erytra Database Manager application</i>. Added note warning of possible incompatibilities with newest version of Internet Explorer and Flash Player. - Added new section 15.11.1 <i>Alternative browser for EDM</i>. - Added section 15.14 <i>BIOS settings</i>. - Updated section 15.14.3 <i>Windows COM ports assignment</i>. Removed references to Windows 7.
-----------	--

Print date:	16-May-2023	Purpose:	Uncontrolled Copy
Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization			

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	10 of 623
	Status:				Effective Date: 15-May-2023
		Service Manual Erytra			
		Title:			
		Effective			

July/2022	- Updated section 15.15.1 <i>Updating the whitelist</i>. Added to check in the applocker log file successfully entered in audit mode. - Updated section 15.15.2 <i>Blocked executables</i>. Indicated that the blocked scripts are also registered in windows events log. - Removed section 15.16 <i>TeamViewer</i> and all references to TeamViewer in the manual. - Updated section 15.17 <i>Description of log files</i>. Added description of firewire.log and AppLocker.log. - Updated section 15.18 <i>Support pack</i>. It has been split in 2 parts, generating support pack from user software and generating support pack from OS shell, including the new support pack generator tool. Indicated LisTest.def and LisTest.bak files are not included in Support Pack. - Updated section 15.20.1 <i>DianaServer program</i>. Indicated new DianaServer configuration and added that software will set Diana Server as enabled when no previous configuration is found. - Updated section 15.20.3 <i>Files communication mode</i>. Indicated setting network as private if sharing an instrument folder. - Updated section 15.20.6 <i>How to enable and configure LIS connection</i>. New Diana Sever configuration. Indicated that Software installer will configure the system to start DianaServer automatically when user interface starts. - Updated section 15.21 <i>erytra - LIS communication protocol</i>. In sending of results: in Result status field added value "P" (preliminary results). Also indicated that comments included in remarks field of template will now be sent to LIS. Also indicated that the Serial Number information is sent in unidirectional mode. - Updated section 15.21.7 <i>erytra - LIS errors</i>. Added errors code 5000024 and 5000025.. - Added section 15.22 <i>Enabling Postgres logs (temporarily)</i>. - Updated all boot application screenshots (without TeamViewer). - Added password to access the BIOS. - Removed all the PCA electronic schematics, maintaining only the wiring diagram. - References to 'Windows 10 Enterprise 2019 LTSC O.S' replaced by 'Windows 10 Embedded (WES10) v2.1.0'. - References to 'Hard disk' replaced by 'SSD'. - Replaced <i>erytra</i>® for <i>Erytra</i>. - Removed references and images of parts/procedures without impact detection installed. - Minor text modifications.
April 2023	- Updated section 1.6.2 <i>SSD structure</i> with new SSD volume sizes and O.S. version checking procedure. - The folders ApplockerWorkspace from E:\, the folders eGalaxWorks, Loading and SistemaDianaRegistry from E:\Program files have been removed in section 1.6.3 <i>Files structure</i> according the new SW structure. The folder Support Pack has been added in E:\Program files. - In section 2.6 <i>Boot Application</i> added tool information to increase the timeout to enter Boot application to 1 hour on next reboot. The new Operating System recovery option from the boot application has also been included. - 2.7.2 <i>Language</i> section has been updated with the addition of Danish (Denmark), Slovak (Slovakia) and Slovenian (Slovenia) languages. - Added new parameters of the Erytra.properties file in sections 2.7.41 <i>AUTO_LIS_QUERY_ENABLED</i> 2.7.42 <i>AUTO_LIS_QUERY_MODE</i>, 2.7.43 and <i>AUTO_LIS_PERIOD_UPDATE</i> to allow a periodic automatic query to the LIS and its frequency. Also section 2.7.44 <i>NO_SAMPLE_ID_IF_NO_USER_LOGGED</i> to control the behaviour of the recognition of the reagent samples drawers when no user is logged in. - Section 2.7.41 <i>Barcode reader configuration</i> and its content have been moved to section 2.7.45 - Updated section 2.10.2 <i>Remote connection</i> and section 2.10.2.1 <i>Alternative browser for remote connection</i> removed according to SW version 4.4.0 requirements. - Section 4.4.8 <i>Card drawer electric lock replacement</i> has been updated with considerations to take into account when assembling the electric lock. - Section 4.4.9 <i>Samples/reagents drawer electric lock replacement</i> has been updated with considerations to take into account when assembling the electric lock. - Section 4.4.10 <i>Fluidic drawer electric lock</i> has been updated with considerations to take into account when assembling the electric lock. - Added section 6.4.3 <i>Remove centrifuge pot</i>. - Updated section 7.1 <i>DESCRIPTION</i> (PIP) as new information about the barcode reader is logged at the instrument driver (DG59Driver). - The replacement procedure in section 7.4.12 <i>Toothed belt T2.5/245</i> (PIP Y-axis), has been updated with the aim of not modifying the large belt tension. Any tension variation may affect the adjustments. - A note has been added in section 8.2.1.5 <i>Z-Axis/gripper belt tension adjustment</i>, to point out that incorrect belt tension can lead to an offset in the CAT-Z axis. - Section 8.2.1.6 and 8.3.1 <i>Gripper parallelism</i> updated. Threadlocking adhesive must be applied to the gripper screws. - The replacement procedure in section 8.4.8 <i>Toothed belt T2.5/245</i> (CAT Y-axis), has been updated with the aim of not modifying the large belt tension. Any tension variation may affect the adjustments. - Section 9.1.2.5 <i>Overpressure management</i> has been added. Describes the procedure when a dispensation pump has to be activated in a situation where pressure system is already high. - Former sections 9.4.11 <i>Syringe replacement</i> and 9.4.12 <i>Dilutor valve replacement</i> have been merged, now 9.4.11 <i>Syringe and valve replacement</i>. The procedure has been redefined to align it

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization		Purpose:	Uncontrolled Copy
Print date:	16-May-2023		

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	11 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

April 2023	with the manufacturer's practices and steps related to the syringe lock have been removed. MK set does not require this locking. - Section 9.4.13 <i>Syringe lock replacement</i> has been removed. MK valve and syringe assemblies do not require an additional locking element to prevent loosening in use. - The allowed range for "Pixels to mm Conversion Factor", in section 11.2.1 <i>Reader Adjustment</i>, has been updated, from Pix/mm*100: 1950 - 2250 to Pix/mm*100: 2050 - 2150. - Four new error codes to indicate that the container has not been primed have been added in Table 72. <i>List of errors related to Fluidics module</i>, section 9.9 TROUBLESHOOTING. - Added section 11.4.5 <i>Remove reader cover</i>. - Assembly drawings versions in section 12.2 <i>Drawings reference table</i> have been updated. - Gripper lubrication procedure in section 14.3.3 <i>CARD TRANSPORT ARM (CAT) MODULE</i> has been modified, adding instructions to disassemble only the lower part of the gripper. If only the lower part is disassembled, it is not necessary to readjust the gripper. - References to manual installation of the scripts for access to Windows explorer are removed from section 15.17.1 <i>Generating support pack</i> from OS Shell. SW version 4.4.0 installs them automatically. - Former sections 15.2 SSD image restoring procedure and 15.4 Operating System restoring procedure have been merged to the new section 15.2 Restoring procedures. - There has been added the new requirements of the OS 2.2.1, Windows license activation and the option to restore remotely the OS through OS Shell. - In section 15.3 <i>Erytra software installation</i>, information updated for SW v4.4.0. A workflow with the installation process has been added. - In section 15.3.2 <i>Final setup, installation steps</i> to access Windows File Explorer have been removed and levelling verification, camera reader adjustment and gripper adjustment in card reader area added. - Added section 15.3.3 <i>Software version verification</i>, to check which version of SW is installed. - Updated content and title of section 15.4 to <i>Windows security packages</i>, removing the installation procedure now included in SO 2.2.1. and updating the Table 93 Security Updates packages. - In section 15.5 <i>Access to Windows 10 File Explorer</i>, the former installation procedure has been removed and is now included in SW version 4.4.0 installation. Besides it is remarked that the shift key that performs the key combination to access Windows explorer is just the left shift key (and not the right one). - Updated section 15.7 <i>Backup</i>. All information regarding backups has been moved to this section. The subsection for performing backups from the boot application has been updated including new backup options. Now, Backup application allows selecting the backup destination, the type of backup (full or customized) and define a description. - In section 15.8 <i>Databases restore procedure</i>, warning messages with operation considerations have been added. - Updated section 15.9 <i>Databases restore with boot menu procedure</i>, including new restore options and adding warning messages with operation considerations.. - The content in section 15.10 <i>Erytra Database Manager application</i>, has been updated to the current version. Erytra Database Manager now has a single installer. - The former section 15.11.1 <i>Alternative browser for EDM</i>, has been removed. With SW version 4.4.0 it is not necessary to use this alternative browser. - Images referring to port assignment in section 15.13.3 <i>Windows COM ports assignment</i>, has been updated in order to correct a factual error. The images showed a COM port that was not in use and should never have been listed in the device manager. - Updated section 15.16 <i>Description of log files</i>. Adding the startup.log, where the access to Windows File Explorer will be logged and the erytraAir.log and. Files included with SW version 4.4.0. - Updated section 15.18 <i>Mapping a network drive</i>. Now for mapping drives an additional script has been created, which will read the mappedDrives files stored in "E:\Program Files\System Tools" and will try to create the mapped drives. - In section 15.19.1 <i>DianaServer program</i>. Added new LIS or BTM connection mode section. - Updated section 15.19.3 <i>Files communication mode</i>, removing the content referring to set the network as private. Now the networks are private by default. - Updated section 15.19.4 <i>TCP/IP communication mode</i>, indicating how to create an outbound firewall rule. - In section 15.19.5 <i>HostDemo program</i>, user interface has been updated and features improved. - In section 15.19.6 <i>How to enable and configure LIS connection</i>, addition of new erytra.properties configuration parameters: AUTO_LIS_QUERY_ENABLED, AUTO_LIS_QUERY_MODE and AUTO_LIS_PERIOD_UPDATE. - Section 15.20 <i>Share or access a shared folder</i> has been added. - The former section 15.21 <i>ERYTRA – LIS COMMUNICATION PROTOCOL</i> has been removed, except the subsection <i>Erytra – LIS errors</i>, moved to section 15.20.7. The document LIS INTERFACE SPECIFICATION has been created in order to collect all this information. - Minor text modifications.
------------	---

Print date:	16-May-2023	Purpose:	Uncontrolled Copy
Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization			

(this Service Manual starts on page 2)

This manual uses the following symbols to facilitate the location of specific sections.

	DANGER! Any kind of injury –critical or not– for people, that only can be diminished following the instructions shown.
	WARNING! Any kind of injury for people for whom protection systems are provided, but it is convenient the QUALIFIED TECHNICIAN knows it.
	CAUTION! Any kind of harmful situation for the equipment or other materials. This situation can be diminished using this indication or with other protections.
	NOTE: Used for clarification and complementary or emphasized information.

SUPERVISOR	Individual or group responsible for the use and maintenance of the equipment, and for ensuring that operators are adequately trained.
OPERATOR	Person operating the equipment for its intended purpose. The operator should have received training appropriate for this purpose.
QUALIFIED TECHNICIAN	Person responsible for the installation, reparation and special maintenance of the equipment who has received specific training for this purpose.
All restrictions referred to the SUPERVISOR are valid for the OPERATOR.	

This instrument is protected by international patents applicable to the entire machine or parts thereof.

The manufacturers reserve the right to take legal actions in order to protect their interests.

This manual should be used in conjunction with the Instructions for Use.

The information found in this document may be changed without prior notification.

Contents

1	INSTRUMENT GENERAL OVERVIEW	34
1.1	INTRODUCTION.....	34
1.2	SECURITY REMARKS	34
1.3	FUNCTIONAL CHARACTERISTICS	34
1.4	TECHNICAL SPECIFICATIONS.....	35
1.5	MODULAR DESCRIPTION.....	36
1.5.1	General overview	36
1.5.2	General module (GEN).....	36
1.5.3	Incubator module (INC).....	38
1.5.4	Centrifuge module (CEN).....	39
1.5.5	Pipetting robot arm module (PIP).....	39
1.5.6	Card transport robot arm module (CAT)	40
1.5.7	Fluidics module (FLU)	40
1.5.8	Dilution cup module (CUP).....	41
1.5.9	Reader module (LEC)	41
1.6	SOFTWARE STRUCTURE OVERVIEW	41
1.6.1	General overview	41
1.6.2	SSD structure	42
1.6.3	Files structure.....	43
1.6.4	Windows registry structure	47
2	SETTING UP AND STARTING THE INSTRUMENT	48
2.1	LOCATION REMARKS	48
2.2	UNPACKING.....	48
2.3	INSTRUMENT LEVELLING	53
2.4	INSTALLATION QUALIFICATION (IQ).....	55
2.5	COMPLETING THE SOFTWARE INSTALLATION	55
2.6	BOOT APPLICATION	56
2.7	SOFTWARE CONFIGURATIONS	58
2.7.1	SERVICE_RACK_MODE.....	59
2.7.2	LANGUAGE	59
2.7.3	INSTRUMENT_NAME	59
2.7.4	INSTRUMENT_SERIAL.....	59
2.7.5	WELL_NAME_FORMAT.....	60
2.7.6	INTERPRETATION_FORMAT.....	60
2.7.7	LIS_ACTIVATED.....	60
2.7.8	LIS_SHARED_MODE	60
2.7.9	LIS_AUTO_QUERY_BY_SAMPLE.....	60
2.7.10	RECEIVE_AND_SHOW_DEMOGRAPHIC_DATA.....	61
2.7.11	DIANA_SYSTEM_BATCH_CREATION.....	61
2.7.12	SAMPLE_RACKS_CONFIGURATION	61
2.7.13	RESULT_CONFIGURATION.....	62
2.7.14	DOUBLE_TUBE_ENABLED.....	63
2.7.15	QC_ENABLED	63
2.7.16	QC_HOURS_CADENCE	63
2.7.17	QC_NEW_LOT_WARNING_ENABLE	63
2.7.18	PASSWORD_REAUTHENTICATION.....	63
2.7.19	DISABLE_VALIDATION_BUTTON.....	64
2.7.20	DISABLE_EXPORT_BUTTON	64
2.7.21	DISABLE_CHANGE_RESULTS_CONFIRMATION.....	64
2.7.22	DUAL_MODE	64
2.7.23	PASSWORD_EXPIRATION_DAYS.....	64
2.7.24	DELTA_CHECK_INCONSISTENCIES_CONFIGURATION.....	64
2.7.25	DELTA_CHECK_HISTORICAL_FROM.....	65

2.7.26	ENABLE_AUTOVERIFICATION_QCPASS.....	65
2.7.27	SHOW_CONFIRMATION_WHEN_VALIDATE_MODIFIED_SAMPLEPROFILE	
S	65	
2.7.28	ARCHIVE_ENABLED.....	65
2.7.29	ARCHIVE_PATH.....	65
2.7.30	ARCHIVE_INTERVAL.....	66
2.7.31	RACK_BC_ENABLED.....	66
2.7.32	IMPACT_DETECTOR.....	66
2.7.33	RESERVE_INCUBATOR_FOR_STAT.....	66
2.7.34	DIRECT_PRINT.....	66
2.7.35	AUTOMATIC_RESULTS_VERIFICATION.....	67
2.7.36	FLUIDIC_SYSTEM_ALWAYS_READY.....	67
2.7.37	ISBT_ENABLED.....	67
2.7.38	DEFAULT_BC_IS_ISBT.....	67
2.7.39	ISBT_LIS_DIGITS.....	68
2.7.40	SEND_PENDING_TO_VALIDATE.....	68
2.7.41	AUTO_LIS_QUERY_ENABLED.....	68
2.7.42	AUTO_LIS_QUERY_MODE.....	68
2.7.43	AUTO_LIS_PERIOD_UPDATE.....	68
2.7.44	NO_SAMPLE_ID_IF_NO_USER_LOGGED.....	69
2.7.45	Barcode reader configuration.....	69
2.8	OPERATIONAL QUALIFICATION (OQ).....	69
2.9	DATE AND TIME SETTING.....	70
2.10	COMMUNICATIONS.....	70
2.10.1	LIS connection.....	70
2.10.2	Remote connection.....	71
2.11	PERIPHERAL INSTALLATION.....	80
2.12	DECONTAMINATION.....	80
2.13	UNINSTALLING AND SHIPPING.....	80
2.14	INSTRUMENT DISPOSAL.....	85
3	TECHNICAL SERVICE SOFTWARE OVERVIEW.....	86
3.1	INTRODUCTION.....	86
3.2	MODULAR TABS.....	86
3.2.1	Procedures tab.....	88
3.2.2	Incubators tab.....	90
3.2.3	Centrifuges tab.....	94
3.2.4	Dilution Cup tab.....	98
3.2.5	Gripper Arm tab.....	100
3.2.6	Drawers tab.....	108
3.2.7	Pip Arm tab.....	114
3.2.8	Fluidics tab.....	126
3.2.9	General tab.....	129
3.2.10	Firmware tab.....	131
3.2.11	Operations tab.....	135
4	GENERAL (GEN) MODULE.....	139
4.1	DESCRIPTION.....	139
4.1.1	PCA GEN-BCKPL59.....	139
4.1.2	Power supply.....	141
4.1.3	Computer.....	143
4.1.4	Touchscreen monitor.....	145
4.1.5	User tray.....	146
4.1.6	Drawers.....	146
4.1.7	Keyboards.....	149
4.1.8	Status LEDs.....	149
4.1.9	Doors.....	150

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	15 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

4.1.10	Temperature control system.....	151
4.1.11	Buzzer	152
4.1.12	Chassis.....	152
4.1.13	Cables and tubes rails.....	153
4.1.14	Covers	153
4.2	ADJUSTMENT PROCEDURES.....	156
4.2.1	Mechanical adjustments.....	156
4.2.2	Functional adjustments	162
4.3	VERIFICATIONS.....	167
4.3.1	Power supply (SN < 059-0001583)	167
4.3.2	PCA GEN-STDBY	167
4.3.3	PCA GEN-BCKPL59	167
4.3.4	Computer.....	167
4.3.5	Touchscreen monitor.....	168
4.3.6	Status LEDs	168
4.3.7	Ambient light.....	168
4.3.8	User tray.....	168
4.3.9	Temperature control system.....	168
4.3.10	Buzzer	168
4.3.11	Keyboards	169
4.3.12	Doors	169
4.3.13	Drawers	169
4.3.14	Reagent drawer flatness	170
4.3.15	Sample drawer flatness.....	170
4.3.16	Card drawer flatness	171
4.4	REPLACEMENT PROCEDURES.....	173
4.4.1	Power supply (SN < 059-0001583) replacement	173
4.4.2	Touchscreen monitor replacement.....	174
4.4.3	Reagents and samples drawers keyboard replacement.....	176
4.4.4	Card drawers keyboard replacement	177
4.4.5	PCA RM-ST59 replacement.....	178
4.4.6	RAM memory modules replacement.....	178
4.4.7	PC motherboard replacement	180
4.4.8	Card drawer electric lock replacement.....	183
4.4.9	Samples/reagents drawer electric lock replacement	185
4.4.10	Fluidic drawer electric lock replacement	186
4.4.11	User tray replacement.....	189
4.5	FUSES	191
4.6	LEDS	192
4.7	JUMPERS AND TEST POINTS	193
4.8	WIRING TABLE.....	193
4.9	TROUBLESHOOTING	202
5	INCUBATOR (INC) MODULE	210
5.1	DESCRIPTION.....	210
5.2	ADJUSTMENT PROCEDURES.....	212
5.2.1	Mechanical adjustments.....	212
5.2.2	Functional adjustments	213
5.3	VERIFICATIONS.....	216
5.3.1	Temperature	216
5.3.2	Movement.....	217
5.3.3	Incubator flatness	217
5.4	REPLACEMENT PROCEDURES.....	218
5.4.1	Linear guide.....	220
5.4.2	Peltier Module	222
5.4.3	PCA INC-BL59	223
5.5	FUSES	223
5.6	LEDS.....	224

Print date:	16-May-2023	Purpose:	Uncontrolled Copy

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization

5.7	JUMPERS AND TEST POINTS	224
5.8	WIRING TABLE.....	224
5.9	TROUBLESHOOTING	228
6	CENTRIFUGE (CEN) MODULE.....	232
6.1	DESCRIPTION.....	232
6.2	ADJUSTMENT PROCEDURES.....	234
6.2.1	Mechanical adjustments.....	234
6.2.2	Functional adjustments	235
6.3	VERIFICATIONS.....	236
6.4	REPLACEMENT PROCEDURES.....	237
6.4.1	Motor replacement	237
6.4.2	Centrifuge Rotor replacement	239
6.4.3	Remove centrifuge pot	241
6.5	FUSES	243
6.6	LEDS.....	244
6.7	JUMPERS AND TEST POINTS	244
6.8	WIRING TABLE.....	244
6.9	TROUBLESHOOTING	248
7	PIPETTING ROBOT ARM (PIP) MODULE.....	252
7.1	DESCRIPTION.....	252
7.2	ADJUSTMENT PROCEDURES.....	255
7.2.1	Mechanical adjustments.....	255
7.2.2	Functional adjustments	260
7.3	VERIFICATIONS.....	274
7.3.1	Movement.....	274
7.3.2	Check home of the horns	274
7.3.3	Check home of the X-Y-Z1-Z2 axis.....	274
7.3.4	Probes parallelism.....	275
7.3.5	Probes offset	276
7.3.6	Reagents drawer	276
7.3.7	Samples drawer	276
7.3.8	Incubators.....	277
7.3.9	Dilution cup.....	277
7.3.10	Barcode reader.....	278
7.3.11	Presence detector alignment	278
7.3.12	Level detection system.....	279
7.3.13	Impact detector system	280
7.4	REPLACEMENT PROCEDURES.....	281
7.4.1	Probes replacement	281
7.4.2	Probes washer replacement.....	282
7.4.3	Probes washer fitting replacement.....	283
7.4.4	Level detector PCA replacement	283
7.4.5	Leadscrews replacement	285
7.4.6	PIP X axis chain (CH1) replacement.....	292
7.4.7	PIP Y axis chain (CH2) replacement.....	295
7.4.8	PIP Z1 axis chain (CH3) / Z2 axis chain (CH4) replacement.....	299
7.4.9	Barcode reader mirror replacement	302
7.4.10	Horns chain replacement	303
7.4.11	Toothed belt T2.5/245 (PIP X-axis).....	304
7.4.12	Toothed belt T2.5/245 (PIP Y-axis).....	305
7.5	FUSES	307
7.6	LEDS.....	308
7.7	JUMPERS AND TEST POINTS	308
7.8	WIRING TABLE.....	309
7.9	TROUBLESHOOTING	315
8	CARD TRANSPORT (CAT) MODULE	323

8.1	DESCRIPTION.....	323
8.2	ADJUSTMENT PROCEDURES.....	325
8.2.1	Mechanical adjustments.....	325
8.2.2	Functional adjustments	329
8.3	VERIFICATIONS.....	340
8.3.1	Gripper parallelism	340
8.3.2	Card drawers area.....	341
8.3.3	Incubators area	342
8.3.4	Card waste container area	343
8.3.5	Centrifuges area.....	343
8.3.6	Card presence detector inclination.....	344
8.3.7	Card presence detector alignment – gripper parallelism	344
8.3.8	Card presence detector height detection	345
8.3.9	Card barcode reader inclination	346
8.3.10	Reader area	346
8.4	REPLACEMENT PROCEDURES.....	347
8.4.1	Gripper leadscrew replacement	347
8.4.2	Card barcode reader mirror replacement.....	349
8.4.3	CAT X chain replacement	350
8.4.4	CAT Y chain replacement	352
8.4.5	Gripper (Z) chain replacement	355
8.4.6	Gripper replacement.....	357
8.4.7	Toothed belt T2.5/245 (CAT X-axis)	359
8.4.8	Toothed belt T2.5/245 (CAT Y-axis)	359
8.5	FUSES	362
8.6	LEDS.....	362
8.7	JUMPERS AND TEST POINTS	362
8.8	WIRING TABLE.....	363
8.9	TROUBLESHOOTING	368
9	FLUIDICS (FLU) MODULE	375
9.1	DESCRIPTION.....	375
9.1.1	Parts description.....	375
9.1.2	Fluidic procedures	382
9.2	ADJUSTMENT PROCEDURES.....	392
9.2.1	Mechanical adjustments.....	392
9.2.2	Functional adjustments	393
9.3	VERIFICATIONS.....	397
9.3.1	Pressure sensors	397
9.3.2	CPD sensors	398
9.3.3	Dispensation pumps and expansion vessels	399
9.3.4	Aspiration pump	400
9.3.5	Electrovalves	400
9.3.6	Manifold.....	400
9.3.7	Load cell.....	400
9.4	REPLACEMENT PROCEDURES.....	400
9.4.1	Pressure sensors (S3, S4, S5, S6) replacement	401
9.4.2	CPD sensors (S1, S2) replacement	402
9.4.3	Dispensation pumps (PM1, PM3) replacement	403
9.4.4	Expansion vessels replacement.....	404
9.4.5	Aspiration pump (PM2) replacement.....	404
9.4.6	Electrovalves replacement	406
9.4.7	Manifold replacement.....	406
9.4.8	Manifold / distribution block (BK2, BK3) fitting replacement	407
9.4.9	Load cell replacement	408
9.4.10	Fitting of the tubes replacement.....	411
9.4.11	Syringe and valve replacement.....	412
9.4.12	Fluidic drawer linear guide replacement	417

9.4.13	Coupling CPC bottles connector O-ring replacement (F75, F78, F81 AND F84)	418
9.4.14	Connection block (BK4, BK5) replacement.....	419
9.4.15	Vacuum filter replacement.....	421
9.4.16	Aspiration pump tubing replacement.....	421
9.4.17	PCA FLU-CTRL59 board replacement	422
9.5	FUSES	423
9.6	LEDS.....	424
9.7	JUMPERS AND TEST POINTS	424
9.8	WIRING TABLE.....	425
9.9	TROUBLESHOOTING.....	430
10	DILUTION CUP (CUP) MODULE	449
10.1	DESCRIPTION.....	449
10.2	ADJUSTMENT PROCEDURES.....	450
10.2.1	Mechanical adjustments.....	450
10.2.2	Functional adjustments	452
10.3	VERIFICATIONS.....	452
10.4	REPLACEMENT PROCEDURES.....	453
10.4.1	Dilution cup wells assembly replacement	453
10.4.2	Dilution cup motor replacement	454
10.4.3	Fitting replacement.....	458
10.5	FUSES	458
10.6	LEDS.....	459
10.7	JUMPERS AND TEST POINTS	459
10.8	WIRING TABLE.....	459
10.9	TROUBLESHOOTING	463
11	CARD READER (LEC) MODULE.....	466
11.1	DESCRIPTION.....	466
11.2	ADJUSTMENT PROCEDURES.....	468
11.2.1	Functional adjustments	468
11.3	VERIFICATIONS.....	476
11.4	REPLACEMENT PROCEDURES.....	484
11.4.1	Reader module removal.....	484
11.4.2	Camera replacement.....	484
11.4.3	Reader illumination/PCA RD-IL421 replacement.....	485
11.4.4	Reader illumination pattern replacement	486
11.4.5	Remove reader cover.....	487
11.5	FUSES	488
11.6	LEDS.....	488
11.7	JUMPERS AND TEST POINTS	488
11.8	WIRING TABLE.....	488
11.9	TROUBLESHOOTING	490
12	MECHANICAL DRAWINGS	495
12.1	REQUIREMENTS	495
12.2	DRAWINGS REFERENCE TABLE	495
12.3	CREO VIEW EXPRESS INSTALLATION	496
12.3.1	Within Grifols company	496
12.3.2	External distributors.....	496
12.4	CREO VIEW CONFIGURATION.....	496
12.5	MODELS STRUCTURE AND NAVIGATION	498
12.5.1	Tree structure	498
12.5.2	Models navigation	499
12.6	ANIMATED ASSEMBLY PROCEDURES	500
12.7	SPARE PARTS SEARCHING	501
12.7.1	Finding a drawing part for a spare parts list's reference code	501

12.7.2	Finding the spare part list's reference code for a drawing part	502
13	DIAGNOSTIC APPLICATION.....	504
13.1	INTRODUCTION.....	504
13.2	OVERVIEW.....	505
13.3	TESTS DESCRIPTION	506
13.3.1	Initialization.....	507
13.3.2	Cards Management.....	507
13.3.3	Pipetting.....	510
13.3.4	Fluidics	514
13.3.5	Reader.....	517
13.3.6	Incubators.....	517
13.3.7	Centrifuges	518
13.3.8	Samples drawers.....	519
13.3.9	Reagents drawers	519
13.3.10	Cards drawers	519
13.3.11	Dilution cup.....	520
13.3.12	Miscellaneous	520
13.4	REPORT	521
14	PREVENTIVE AND CORRECTIVE MAINTENANCE.....	528
14.1	ADJUSTMENT TOOLS	528
14.2	ITEMS AND CALENDAR	528
14.3	LUBRICATION POINTS.....	529
14.3.1	GENERAL MODULE.....	529
14.3.2	PIPETTING ARM (PIP) MODULE.....	532
14.3.3	CARD TRANSPORT ARM (CAT) MODULE.....	533
14.3.4	FLUIDICS MODULE.....	536
14.4	REPORT OF CORRECTIVE MAINTENANCE SERVICES.....	537
14.5	HOW TO REPORT	538
15	ADDITIONAL INFORMATION	539
15.1	GUIDELINES ON ERROR CODES LOCATION	539
15.2	RESTORING PROCEDURE	539
15.2.1	From USB pen drive	540
15.2.2	From Boot application	542
15.2.3	Operating System recovery.....	543
15.2.4	Full recovery.....	545
15.2.5	Restoring configurations after a full recovery.....	547
15.2.6	Activate Windows License.....	550
15.2.7	OS and Software downgrade	556
15.3	ERYTRA SOFTWARE INSTALLATION	561
15.3.1	Installation steps.....	562
15.3.2	Final setup.....	565
15.3.3	Software version verification	566
15.4	WINDOWS SECURITY UPDATES	567
15.5	ACCESS TO WINDOWS 10 FILE EXPLORER.....	568
15.6	TOUCHSCREEN DRIVER INSTALLATION AND CONFIGURATION	571
15.6.1	Touchscreen driver installation	571
15.6.2	Touchscreen v2 driver configuration	571
15.7	BACKUP	573
15.7.1	with Boot menu procedure	573
15.7.2	with user interface or OS Shell backup option	575
15.7.3	manual backup	575
15.8	DATABASE RESTORE PROCEDURE	576
15.9	RESTORE WITH BOOT MENU PROCEDURE	578
15.10	ERYTRA DATABASE MANAGER APPLICATION.....	580
15.11	EEPROM BACKUP/RESTORE ADJUSTMENTS	581

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	20 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

15.12	FIRMWARE UPDATE	582
15.12.1	Update firmware in a valid PCA	583
15.12.2	Update firmware in a spare part PCA (CEN-SRM, INC-CUP, CAT-CM or PIP MotorControl)	583
15.12.3	Update firmware in a PCA with non valid firmware (CEN-SRM, INC-CUP, CAT-CM or PIP MotorControl)	584
15.13	BIOS SETTINGS	585
15.13.1	Check BIOS version	585
15.13.2	BIOS configuration	586
15.13.3	Windows COM ports assignment	587
15.14	APLOCKER APPLICATION	590
15.14.1	Updating the whitelist	590
15.14.2	Blocked executables	592
15.15	ERYTRA MACRO EXECUTOR	593
15.16	DESCRIPTION OF LOG FILES	594
15.17	SUPPORT PACK	595
15.17.1	Generating support pack from user software	596
15.17.2	Generating support pack from OS shell	597
15.18	MAPPING A NETWORK DRIVE	601
15.19	ERYTRA- LIS CONNECTION	602
15.19.1	DianaServer program	603
15.19.2	Serial communication mode	606
15.19.3	Files communication mode	606
15.19.4	TCP/IP communication mode	606
15.19.5	HostDemo program	609
15.19.6	How to enable and configure LIS connection	614
15.19.7	Erytra – LIS errors	617
15.20	SHARE OR ACCESS A SHARED FOLDER	621

Print date:	16-May-2023	Purpose:	Uncontrolled Copy
Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization			

Figures

FIGURE 1. DIAGRAM OF THE GENERAL STRUCTURE OF THE INSTRUMENT.....	36
FIGURE 2. INFORMATION OPTION	42
FIGURE 3. E DRIVE DIRECTORIES	43
FIGURE 4. E:\DATA SUBDIRECTORIES	43
FIGURE 5. E:\LOG SUBDIRECTORIES	44
FIGURE 6. E:\PROGRAM FILES SUBDIRECTORIES.....	45
FIGURE 7. E:\DATA\ERYTRA SUBDIRECTORIES.....	45
FIGURE 8. E:\DATA\SISTEMA DIANA SUBDIRECTORIES	46
FIGURE 9. CONFIG DIANA SYSTEM SOFTWARE	47
FIGURE 10. UPPER PART OF THE WOODEN PACKAGING	49
FIGURE 11. PACKING ASSEMBLY OF THE ERYTRA	49
FIGURE 12. HOOK-AND-LOOP FASTENERS.....	49
FIGURE 13. SMALL RAMP ATTACHED TO THE MAIN RAMP OF THE PACKAGING	49
FIGURE 14. ADHESIVE TAPE IN A FOOT OF THE ERYTRA	50
FIGURE 15. LOCATION OF THE NUT IN A FOOT OF ERYTRA	50
FIGURE 16. INSTRUCTIONS TO SCREW / UNSCREW THE FEET OF THE ERYTRA	50
FIGURE 17. FOAM AND WOODEN PROTECTION OF THE PACKAGING	51
FIGURE 18. DOOR HANDLES INSTALLED ON THE DOORS	51
FIGURE 19. LOCATION OF THE BOX PLUG	52
FIGURE 20. LOCATION OF THE RM COMB.....	52
FIGURE 21. LOCATION OF THE SUPPLEMENT COMB LATERAL	52
FIGURE 22. LOCATION OF THE SHIPPING CLAMP OF THE PIP ARM	52
FIGURE 23. LOCATION OF THE INTERNAL PACKAGING TRAY (LEFT) AND PLUG OF THE TRAY (RIGHT)	53
FIGURE 24. LOCATION OF THE SHIPPING CLAMP OF THE CAT ARM.....	53
FIGURE 25. LASER PIP ALIGNMENT	54
FIGURE 26. LASER CAT ALIGNMENT	54
FIGURE 27. UPPER NUT OF THE FOOT	54
FIGURE 28. LOWER NUT OF THE FOOT.....	55
FIGURE 29. BOOT APPLICATION INITIAL WINDOW	56
FIGURE 30. BOOT APPLICATION LOGIN WINDOW	57
FIGURE 31. BOOT APPLICATION PROTECTED/ADMINISTRATIVE OPTIONS.....	57
FIGURE 32. INFORMATION WINDOW.....	58
FIGURE 33. SAMPLES DRAWERS CONFIGURATION.....	62
FIGURE 34. DIRBASE PARAMETER CONFIGURATION.....	70
FIGURE 35 REMOTE CONNECTION INSTALLATION	71
FIGURE 36 REMOTE CONNECTION FINAL CHECKINGS	72
FIGURE 37 IP EDITION FOR REMOTE CONNECTION	72
FIGURE 38 STARTING REMOTE CONNECTION STEP 1	72
FIGURE 39 STARING REMOTE CONNECTION STEP 2.....	73
FIGURE 40 LOGIN INTO APPLICATION.....	73
FIGURE 41. FOAMS IN CARD DISPOSAL CONTAINER AND IN THE BOTTLES.....	81
FIGURE 42. DECONTAMINATION TOOL	81
FIGURE 43. LOCATION OF THE INTERNAL PACKAGING TRAY	82
FIGURE 44. LOCATION OF THE PLUG OF THE TRAY	82
FIGURE 45. DECONTAMINATION BOTTLE AND RACKS IN PACKAGING TRAY.....	82
FIGURE 46. POSITIONING OF THE SUPPLEMENT COMB LATERAL	82
FIGURE 47. POSITIONING OF THE RM COMB.....	83
FIGURE 48. POSITIONING OF THE SM COMB	83
FIGURE 49. POSITIONING OF THE BOX PLUG	83
FIGURE 50. MARKS FOR THE FEET LOCATED IN THE PALLET.....	84
FIGURE 51. RETRACTILE FILM IN THE PACKAGING	84
FIGURE 52. UPPER PART OF THE PACKAGING	85
FIGURE 53. TECHNICAL SERVICE ICON.....	86
FIGURE 54. BAR MENU OF THE TECHNICAL SERVICE SOFTWARE.....	86
FIGURE 55. EDIT SERIAL NUMBER WINDOW.....	86
FIGURE 56. PROCEDURES TAB IN TECHNICAL SERVICE SOFTWARE.....	88

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	22 of 623
	Status:		Title:		Effective Date: 15-May-2023
	Effective		Service Manual Erytra		

FIGURE 57. INCUBATOR X - INCUBATORS TAB IN TECHNICAL SERVICE SOFTWARE.....	90
FIGURE 58. HOME CYCLES - INCUBATORS TAB IN TECHNICAL SERVICE SOFTWARE	93
FIGURE 59. CENTRIFUGE X - CENTRIFUGES TAB IN TECHNICAL SERVICE SOFTWARE	94
FIGURE 60. HOME CYCLES - CENTRIFUGES TAB IN TECHNICAL SERVICE SOFTWARE	96
FIGURE 61. RANDOM POSITION - CENTRIFUGES TAB IN TECHNICAL SERVICE SOFTWARE	97
FIGURE 62. DILUTION CUP TAB IN TECHNICAL SERVICE SOFTWARE	98
FIGURE 63. MOVEMENT – GRIPPER ARM TAB IN TECHNICAL SERVICE SOFTWARE	100
FIGURE 64. PRESENCE DETECTOR – GRIPPER ARM TAB IN TECHNICAL SERVICE SOFTWARE	102
FIGURE 65. BCR – GRIPPER ARM TAB IN TECHNICAL SERVICE SOFTWARE	104
FIGURE 66. HOME CYCLES – GRIPPER ARM TAB IN TECHNICAL SERVICE SOFTWARE	106
FIGURE 67. RANDOM POSITION - GRIPPER ARM TAB IN TECHNICAL SERVICE SOFTWARE	107
FIGURE 68. CARD DRAWERS - DRAWERS TAB IN TECHNICAL SERVICE SOFTWARE.....	108
FIGURE 69. SAMPLE DRAWERS - DRAWERS TAB IN TECHNICAL SERVICE SOFTWARE	109
FIGURE 70. REAGENT DRAWERS – DRAWERS TAB IN TECHNICAL SERVICE SOFTWARE	110
FIGURE 71. FLUIDIC DRAWERS – DRAWERS TAB IN TECHNICAL SERVICE SOFTWARE	112
FIGURE 72. HOME CYCLES – DRAWERS TAB IN TECHNICAL SERVICE SOFTWARE	113
FIGURE 73. MOVEMENT - PIP ARM TAB IN TECHNICAL SERVICE SOFTWARE	114
FIGURE 74. REAGENT DRAWERS – PRESENCE DETECTOR – PIP ARM TAB IN TECHNICAL SERVICE SOFTWARE.....	116
FIGURE 75. SAMPLE DRAWERS – PRESENCE DETECTOR – PIP ARM TAB IN TECHNICAL SERVICE SOFTWARE.....	117
FIGURE 76. PRESENCE DETECTOR – PRESENCE DETECTOR – PIP ARM TAB IN TECHNICAL SERVICE SOFTWARE.....	118
FIGURE 77. LEVEL DETECTOR – PIP ARM TAB IN TECHNICAL SERVICE SOFTWARE	119
FIGURE 78. BCR – PIP ARM TAB IN TECHNICAL SERVICE SOFTWARE	121
FIGURE 79. CPD – PIP ARM TAB IN TECHNICAL SERVICE SOFTWARE	123
FIGURE 80. HOME CYCLES – PIP ARM TAB IN TECHNICAL SERVICE SOFTWARE	124
FIGURE 81. RANDOM POSITION – PIP ARM TAB IN TECHNICAL SERVICE SOFTWARE.....	125
FIGURE 82. FLUIDIC TAB IN TECHNICAL SERVICE SOFTWARE	126
FIGURE 83. GENERAL MODULE TAB IN TECHNICAL SERVICE SOFTWARE	129
FIGURE 84. PCA CONTROL TAB - FIRMWARE TAB IN TECHNICAL SERVICE SOFTWARE.....	131
FIGURE 85. EPROM CONTROL TAB - FIRMWARE TAB IN TECHNICAL SERVICE SOFTWARE	134
FIGURE 86. DILUTION CUP OPERATIONS TAB IN TECHNICAL SERVICE SOFTWARE.....	135
FIGURE 87. ASPIRE AND DISPENSE OPERATIONS TAB IN TECHNICAL SERVICE SOFTWARE	137
FIGURE 88. LOCATION OF THE DIFFERENT PCA MOTORCONTROL IN PCA GEN-BCKPL59	140
FIGURE 89. BLOCK DIAGRAM OF PCA GEN-BCKPL59	141
FIGURE 90. POWER SUPPLY SYSTEM ASSEMBLED IN AN INDEPENDENT PLATE.....	142
FIGURE 91. ERYTRA ON/OFF SWITCHER.....	143
FIGURE 92. PUSHBUTTON	143
FIGURE 93. COMPUTER CONNECTORS FOR PC EMBEDDED v3. FRONT VIEW	144
FIGURE 94. FIREWIRE PORTS IN PC EMBEDDED v4/v5. FRONT VIEW	144
FIGURE 95. COMPUTER CONNECTORS FOR PC EMBEDDED v2 / PC EMBEDDED - USA. FRONT VIEW (LEFT) AND LATERAL VIEW (RIGHT)	145
FIGURE 96. TOUCHSCREEN MONITOR	146
FIGURE 97. USER TRAY.....	146
FIGURE 98. TOP VIEW OF REAGENTS RACKS.....	147
FIGURE 99. SAMPLE RACK.....	148
FIGURE 100. SAMPLES/REAGENTS AND CARDS DRAWERS KEYBOARDS	149
FIGURE 101. SAMPLES DRAWERS KEYBOARD	149
FIGURE 102. GENERAL LED	150
FIGURE 103. UPPER AREA DOOR.....	150
FIGURE 104. INTERMEDIATE AREA DOOR.....	151
FIGURE 105. CELDUC SENSORS IN UPPER AREA DOOR (LEFT) AND IN INTERMEDIATE AREA DOOR (RIGHT)	151
FIGURE 106. TEMPERATURE SENSORS IN UPPER (LEFT) AND INTERMEDIATE (RIGHT) AREAS	151
FIGURE 107. FANS IN THE ELECTRONICS CABINET	152
FIGURE 108. LOCATION OF THE BUZZER IN PCA GEN-BCKPL59	152
FIGURE 109. CHASSIS OF ERYTRA.....	152
FIGURE 110. CABLES AND TUBES RAILS	153

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization			Purpose:	Uncontrolled Copy
Print date:	16-May-2023			

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	23 of 623
	Status:		Title:		Effective Date: 15-May-2023
	Effective		Service Manual Erytra		

FIGURE 111. FRONTAL COVERS.....	153
FIGURE 112. BACK COVERS	154
FIGURE 113. FLUIDIC CONNECTOR COVER	154
FIGURE 114. ELECTRONICS CONNECTORS COV	154
FIGURE 115. BLOCK DIAGRAM OF GENERAL MODULE.....	155
FIGURE 116. MAGNET OF THE DOORS.....	157
FIGURE 117. LOCK OF THE TOUCHSCREEN DRAWER	157
FIGURE 118. SLOTS TO INSERT THE FLAGS	157
FIGURE 119. LOCK OF THE USER TRAY.....	158
FIGURE 120. SLOTS TO INSERT THE PINS OF THE USER TRAY	158
FIGURE 121. SAFETY LATCH ADJUSTMENT WITH DRAWERS FRONT PANEL ALIGNMENT TOOL	159
FIGURE 122. LATCHES OF REAGENTS/SAMPLES DRAWERS (LEFT) AND CARDS DRAWERS (RIGHT)	159
FIGURE 123. COSMETIC FRONTAL PANELS SCREWS. SAMPLES/REAGENTS (LEFT) AND CARDS (RIGHT)	159
FIGURE 124. FRONT PANEL ADJUSTMENT WITH DRAWERS FRONT PANEL ALIGNMENT TOOL	160
FIGURE 125. SCREWS OF THE REAGENTS MOTOR.....	160
FIGURE 126. MAGNETIC BRAKE SCREWS.....	161
FIGURE 127. VERTICAL LINEAR GUIDE SCREWS - REAGENTS DRAWERS.....	161
FIGURE 128. VERTICAL LINEAR GUIDE SCREWS - SAMPLES DRAWERS	161
FIGURE 129. VERTICAL LINEAR GUIDE SCREWS – CARDS DRAWERS.....	162
FIGURE 130. CARD DRAWER FLATNESS ADJUSTMENT	162
FIGURE 131. MICROTOUCH MT 7 PROGRAM	163
FIGURE 132. TABLET PC SETTINGS.....	164
FIGURE 133. TABLET PC SETTINGS WINDOW	164
FIGURE 134. CALIBRATION SCREEN	165
FIGURE 135. SAVE CALIBRATION DATA.....	165
FIGURE 136. TOUCHSCREEN V2 BUTTONS	166
FIGURE 137. TOUCHSCREEN V3 BUTTONS	167
FIGURE 138. VERIFICATION DRAWERS WIZARD	169
FIGURE 139. FLATNESS VERIFICATION POINTS IN REAGENT DRAWER	170
FIGURE 140. LEFT VERIFICATION IN RACK 1 OF SAMPLE DRAWER.....	171
FIGURE 141. FLATNESS VERIFICATION POINTS IN SAMPLE DRAWER.....	171
FIGURE 142. REAR LEFT PART VERIFICATION IN RACK 1	172
FIGURE 143. REAR RIGHT PART VERIFICATION IN RACK 1	172
FIGURE 144. FRONTAL RIGHT PART (RIGHT PICTURE) AND FRONTAL LEFT PART (LEFT PICTURE) VERIFICATION IN RACK 4	172
FIGURE 145. INPUT CONNECTORS COVER OF THE PCA GEN-STDBY59.....	173
FIGURE 146. POSITION OF THE CABLES IN THE POWER SUPPLY AS INSTALLED ON THE INSTRUMENT .	174
FIGURE 147. SCREWS ATTACHING THE MONITOR TO THE TOUCHSCREEN DRAWER	174
FIGURE 148. HEXAGONAL SPACERS	174
FIGURE 149. USB PCA IN NEW TOUCHSCREEN MONITOR.....	175
FIGURE 150. USB PCA INSTALLATION PLACE	175
FIGURE 151. SCREWS ATTACHING THE MONITOR TO THE TOUCHSCREEN DRAWER	175
FIGURE 152. FROM LEFT TO RIGHT: POWER (ADAPTER), VGA, USB CABLES	176
FIGURE 153. TOUCHSCREEN DRAWER ASSEMBLY.....	176
FIGURE 154. REAR SET SCREWS IN TOUCHSCREEN DRAWER	177
FIGURE 155. POSITION OF THE PROTECTIVE COVER IN CARD DRAWERS KEYBOARD	177
FIGURE 156. POSITION OF THE NUTS OF CARD DRAWERS KEYBOARD.....	178
FIGURE 157. CONNECTORS, FUSES AND LEDs OF PCA GEN-BCKPL59 RELATED TO GENERAL MODULE.....	197
FIGURE 158. CONNECTORS, FUSES AND LEDs OF PCA GEN-STDBY (UNITS WITH SN < 059-0001583).....	198
FIGURE 159. LOCATION OF THE FUSE AND LEDs IN CEN-SRM / CAT-CM CONTROL	199
FIGURE 160. CONNECTORS OF PCA RM-ST59	199
FIGURE 161. CONNECTORS OF PCA GEN-FRNT59.....	200
FIGURE 162. CONNECTORS OF PCA GEN-TEMP59	200
FIGURE 163. CONNECTORS OF PCA GEN-LCK59	200
FIGURE 164. CONNECTORS OF PCA RM-CON59.....	200
FIGURE 165. CONNECTORS OF PCA GEN-LED59	201

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization		Purpose:	Uncontrolled Copy
Print date:	16-May-2023		

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	24 of 623
Status:	Effective	Title:	Service Manual Erytra		
				Effective Date:	15-May-2023

FIGURE 166. EDGE PROTECTORS ON INCUBATOR AREA	210
FIGURE 167. INCUBATOR MODULE BLOCK DIAGRAM	211
FIGURE 168. MAIN BLOCKS OF INCUBATOR MODULE	212
FIGURE 169. LOCATION OF THE SCREWS OF THE LATERALS OF THE INCUBATORS	213
FIGURE 170. JOYSTICK BUTTONS AND COUNTS SLIDING BAR	214
FIGURE 171. CORRECT ADJUSTMENT INCUBATOR POSITION	214
FIGURE 172. INCUBATOR ADJUSTMENT WIZARD (ADJUSTMENT MODE RT)	215
FIGURE 173. INCUBATOR ADJUSTMENT WIZARD (ADJUSTMENT MODE 37)	216
FIGURE 174. REGULATE ACTIONS SECTION.....	216
FIGURE 175. DISTANCE VERIFICATION BETWEEN GRIPPER AND INCUBATOR.....	217
FIGURE 176. LOCATION OF THE VERIFICATION POINTS OF THE INCUBATOR FLATNESS IN PIP AREA ...	218
FIGURE 177. INCUBATOR MODULE AND FRONTAL PROBES PLATFORM	218
FIGURE 178. INCUBATOR MODULE GROUND CABLE AND MOTOR FIXING SCREWS.....	219
FIGURE 179. INCUBATOR MOTOR-LINEAR GUIDE COUPLING	219
FIGURE 180. LOCATION OF THE SCREWS THAT ATTACH INCUBATOR MODULE TO ERYTRA CHASSIS ...	220
FIGURE 181. LOCATION OF THE FIXING CLAMPS	220
FIGURE 182. LOCATION OF THE FIXING CLAMPS	221
FIGURE 183. CHAIN, L-SHAPE HOME FLAG AND CABLE HOLDER	221
FIGURE 184. ASSEMBLY OF THE LINEAR GUIDE	221
FIGURE 185. HOME FLAG ALIGNMENT	221
FIGURE 186. PELTIER MODULE CABLE.....	222
FIGURE 187. UPPER SCREWS FIXING LEFT LATERAL INCUBATOR SUPPORT	222
FIGURE 188. UPPER SCREWS BUMPER AND GROUNDING CONNECTION	222
FIGURE 189. PELTIER MODULE ASSEMBLY	223
FIGURE 190. PCA INC-BL59 ASSEMBLY	223
FIGURE 191. LOCATION OF THE CONNECTOR, FUSES AND LEDs IN PCA GEN-BCKPL59 RELATED TO INCUBATOR MODULE.....	225
FIGURE 192. LOCATION OF THE FUSE AND LEDs IN PCA MotorControl	226
FIGURE 193. LOCATION OF THE CONNECTORS IN PCA INC-CON59	227
FIGURE 194. LOCATION OF THE CONNECTOR IN PCA INC-BL59	227
FIGURE 195. CENTRIFUGE MODULE BLOCK DIAGRAM	233
FIGURE 196. MAIN BLOCKS OF CENTRIFUGE MODULE	233
FIGURE 197. CENTRIFUGE REMOVAL	234
FIGURE 198. ROTOR HEIGHT ADJUSTMENT TOOLS	235
FIGURE 199. DISTANCE BETWEEN THE ROTOR AND THE BASIS OF THE COVER	235
FIGURE 200. CENTRIFUGE MOTOR SECTION IN TECHNICAL SERVICE SOFTWARE	236
FIGURE 201. REFLECTIVE ADHESIVE IN THE ROTOR OF THE CENTRIFUGE	236
FIGURE 202. MOVE ACTIONS SECTION IN TECHNICAL SERVICE SOFTWARE	236
FIGURE 203. SLOT OF THE CENTRIFUGE (LEFT) AND MEASUREMENT OF THE CENTRIFUGATION SPEED WITH A TACHOMETER (RIGHT).....	236
FIGURE 204. REMOVING THE MOTOR COVER OF THE CENTRIFUGE	237
FIGURE 205. MOTOR COVER-MOTOR ALIGNMENT	238
FIGURE 206. HEIGHT OF THE SETSCREW IN THE SILENTBLOCKS AND MOTOR COVER ALIGNMENT	238
FIGURE 207. LOCATION OF THE CONNECTORS, FUSES AND LEDs IN PCA GEN-BCKPL59 RELATED TO CENTRIFUGE MODULE	245
FIGURE 208. LOCATION OF THE FUSE AND LED IN PCA CEN-SRM CONTROL	246
FIGURE 209. LOCATION OF THE CONNECTORS IN PCA CEN-CON59	247
FIGURE 210. SAMPLE BARCODE SPECIFICATIONS	253
FIGURE 211. PIPETTING ROBOT ARM MODULE BLOCK DIAGRAM	254
FIGURE 212. MAIN BLOCKS OF PIP MODULE.....	255
FIGURE 213. ALLEN SCREWS OF THE PROBE WAGON - LEADSCREW	256
FIGURE 214. PARALLELISM CHECKING AT THE TOP OF THE ROBOTIC HEAD	256
FIGURE 215. EXTERNAL EDGES OF THE PROBES/WAGON ADJUSTMENT TOOL (VER. 2)	256
FIGURE 216. PARALLELISM CHECKING AT THE BOTTOM OF THE ROBOTIC HEAD.....	257
FIGURE 217. X-AXIS MOTOR SMALL BELT	257
FIGURE 218. X-AXIS MOTOR LARGE BELT	257
FIGURE 219. Y-AXIS MOTOR SMALL BELT	258
FIGURE 220. Y-AXIS MOTOR LARGE BELT	258
FIGURE 221. Z-AXIS MOTORS BELTS	259

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization	
Print date:	16-May-2023
Purpose:	Uncontrolled Copy

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	25 of 623
Status:	Effective	Title:	Service Manual Erytra		
				Effective Date:	15-May-2023

FIGURE 222. PRESENCE DETECTOR ASSEMBLY	259
FIGURE 223. PROBES OFFSET WIZARD IN TSS	260
FIGURE 224. ADJUSTMENT TOOL POSITION IN PROBES OFFSET ADJUSTMENT	260
FIGURE 225. REAGENT DRAWER SELECTION IN TSS.....	261
FIGURE 226. ADJUSTMENT OF PIP ARM ON REAGENTS DRAWERS –STEP	262
FIGURE 227. ADJUSTMENT OF PIP ARM ON REAGENTS DRAWERS – STEP 2	263
FIGURE 228. ADJUSTMENT OF PIP ARM OVER REAR PART OF SAMPLES DRAWER.....	264
FIGURE 229. ADJUSTMENT OF PIP ARM OVER FRONTAL PART OF SAMPLES DRAWER	265
FIGURE 230. ADJUSTMENT OF PIP ARM ON INCUBATORS	266
FIGURE 231. POSITION OF THE PROBE IN DILUTION CUP ADJUSTMENT.....	268
FIGURE 232. CORRECT POSITION OF BARCODE READER	268
FIGURE 233. POSITION OF THE HORNS IN FIRST STEP OF THE ADJUSTMENT.....	269
FIGURE 234. ADJUSTMENT IN Y AXIS OF LEFT BARCODE READER.....	269
FIGURE 235. ADJUSTMENT IN Y AXIS OF RIGHT BARCODE READER.....	270
FIGURE 236. ADJUSTMENT IN SAMPLES DRAWER OF THE BARCODE READERS	270
FIGURE 237. ADJUSTMENT TOOL POSITION ON REAGENT RACK	271
FIGURE 238. ACCESS TO PRESENCE DETECTOR POTENTIOMETER	271
FIGURE 239. LOCATION OF THE ADJUSTMENT AREA OF DETECTION OF PRESENCE DETECTOR	271
FIGURE 240. LOCATION OF THE ADJUSTMENT AREA OF NO DETECTION OF THE PRESENCE DETECTOR	272
FIGURE 241. BARCODE READERS CUSTOM CONFIGURATION.....	273
FIGURE 242. FIRST STEP OF THE PROBES PARALLELISM VERIFICATION	275
FIGURE 243. SECOND STEP OF THE PROBES PARALLELISM VERIFICATION	275
FIGURE 244. INFORMATION WINDOW IN PROBES OFFSET VERIFICATION.....	276
FIGURE 245. PROBES POSITION ON DILUTION CUP	277
FIGURE 246. PRESENCE DETECTOR ALIGNMENT VERIFICATION.....	278
FIGURE 247. MANUAL OPENING OF THE DRAWER.....	281
FIGURE 248. PROBE FITTING	281
FIGURE 249. PROBES WASHER SCREWS	282
FIGURE 250. PROBES WASHER TUBES C22 AND C24 ROUTING	282
FIGURE 251. SEALANT PRODUCT APPLICATION AREA	283
FIGURE 252. WAGON PROBE COVER FOR CHAIN CH4 (LEFT) AND CH3 (RIGHT)	284
FIGURE 253. TEFLON TUBE OF THE PROBE.....	284
FIGURE 254. LOCATION OF THE CONNECTOR IN THE LEVEL DETECTION PCA.....	284
FIGURE 255. LEVEL DETECTOR PCA REPLACEMENT STEPS	284
FIGURE 256. LOCATION OF THE CLAMPS IN THE PIP ROBOTIC HEAD	285
FIGURE 257. INTERCONNECTION MANIFOLD IN THE ROBOTIC HEAD	286
FIGURE 258. PROBES WASHER IN THE ROBOTIC HEAD	286
FIGURE 259. LATERAL COVER SCREWS OF THE Y ARM.....	286
FIGURE 260. LOCATION OF THE SCREWS THAT ATTACH Y CHAIN TO Z TOWER/ROBOTIC HEAD	286
FIGURE 261. LOCATION OF THE RIBBON CABLES CONNECTORS AT PCA PIP-TOW59	287
FIGURE 262. Y ARM GUIDE FRONT END STOPPER SCREWS	287
FIGURE 263. REAR GEAR Y BELT (LEFT) AND FIXING SCREWS OF THE TENSOR ASSEMBLY (RIGHT)....	287
FIGURE 264. REMOVING THE PIP ROBOT HEAD	288
FIGURE 265. BELT AND MOTOR PINION OF THE HORNS	288
FIGURE 266. FRONTAL ROBOTIC HEAD STOPPER LOCATION	288
FIGURE 267. SET SCREW OF THE LEADSCREW HOLDER	289
FIGURE 268. POSITION OF THE HOME FLAG RING AND BARCODE READER STRUCTURE PLATE IN THE LEADSCREW.....	289
FIGURE 269. LEADSCREW HOLDER	289
FIGURE 270. LEADSCREW INSTALL	289
FIGURE 271. PULLEY OF THE MOTOR OF THE HORNS	290
FIGURE 272. PROBES BELTS	290
FIGURE 273. LOCATION OF THE SCREWS THAT ATTACH Y LINEAR GUIDE WAGON TO THE PIP ROBOTIC HEAD	290
FIGURE 274. LOCATION OF THE SET SCREW TO FIX THE LINEAR GUIDE WAGON.....	291
FIGURE 275. ROBOTIC HEAD AND LEADSCREW HOLDERS	291
FIGURE 276. LOCATION OF THE SET SCREW TO FIX THE LINEAR GUIDE WAGON.....	292
FIGURE 277. TUBING	292

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization	
Print date:	16-May-2023
Purpose:	Uncontrolled Copy

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	26 of 623
	Status:		Title:		Effective Date: 15-May-2023
	Effective		Service Manual Erytra		

FIGURE 278. LAST CHAIN SLOT	292
FIGURE 279. FLAT CABLES	293
FIGURE 280. LATERAL REMOVAL	293
FIGURE 281. TUBING	293
FIGURE 282. FLAT CABLES	293
FIGURE 283. CHAIN SCREWS.....	294
FIGURE 284. TEFLON TUBES	294
FIGURE 285. TEFLON TUBES UNDER THE BLOCKS	294
FIGURE 286. LOCATION OF THE CLAMPS IN THE PIP ROBOTIC HEAD	295
FIGURE 287. INTERCONNECTION MANIFOLD IN THE ROBOTIC HEAD	296
FIGURE 288. LATERAL COVER SCREWS OF Y ARM	296
FIGURE 289. LOCATION OF THE SCREWS THAT ATTACH Y CHAIN TO Z TOWER/ROBOTIC HEAD	296
FIGURE 290. LOCATION OF THE RIBBON CABLES CONNECTORS AT PCA PIP-TOW59	296
FIGURE 291. INTERCONNECTION MANIFOLDS AT THE REAR PART OF PIP ARM	297
FIGURE 292. TIE-WRAPS IN THE LATERAL COVER OF THE Y ARM.....	297
FIGURE 293. LOCATION OF THE SCREWS THAT ATTACH THE Y CHAIN TO THE LATERAL COVER OF THE PIP ARM.....	297
FIGURE 294. LOCATION OF THE SCREWS OF THE REAR COVER OF THE PIP ARM.....	297
FIGURE 295. TIE-WRAP UNDER REAR COVER OF PIP ARM (LEFT) AND CONNECTORS IN PCA PIP-CONY59 (RIGHT)	298
FIGURE 296. PLATE ATTACHING Y CHAIN TO Z TOWER/ROBOTIC HEAD	298
FIGURE 297. INTERCONNECTION MANIFOLD IN THE ROBOTIC HEAD	299
FIGURE 298. LATERAL COVER OF PIP ROBOTIC HEAD	299
FIGURE 299. REAR COVER OF PIP ROBOTIC HEAD.....	299
FIGURE 300. LATERAL COVER SCREWS OF Y ARM	300
FIGURE 301. CONNECTOR LOCATION IN PCA PIP-TOW59 FOR CHAIN CH4 (LEFT) AND CH3 (RIGHT)	300
FIGURE 302. WAGON PROBE COVER FOR CHAIN CH4 (LEFT) AND CH3 (RIGHT)	300
FIGURE 303. TEFLON TUBE OF THE PROBE.....	301
FIGURE 304. CONNECTOR LOCATION IN LEVEL DETECTION PCA.....	301
FIGURE 305. GROUND CONNECTION.....	301
FIGURE 306. MIRROR FIRST FACE (LEFT) AND MIRROR SECOND FACE (RIGHT)	302
FIGURE 307. LEFT COVER SCREWS.....	303
FIGURE 308. CABLE, TIE AND CHAIN SREW	303
FIGURE 309. CHAIN SCREW.....	303
FIGURE 310. HORNS CABLING	304
FIGURE 311. FINAL ASSEMBLY.....	304
FIGURE 312. SMALL BELT.....	304
FIGURE 313. PIP LARGE BELT	305
FIGURE 314. LARGE BELT (PINION)	305
FIGURE 315. LOCATION OF THE SCREWS OF THE REAR COVER OF THE PIP ARM.....	305
FIGURE 316. SUPPORT SCREWS.....	306
FIGURE 317. CABLING AND CABLE CLAMP	306
FIGURE 318. MOTOR SCREW.....	306
FIGURE 319 TENSION SCREW.....	307
FIGURE 320. CONNECTORS, FUSES AND LEDs OF PCA GEN-BCKPL59 RELATED TO PIP MODULE	311
FIGURE 321. LOCATION OF THE FUSE AND LEDs IN PCA PIP CONTROL.....	312
FIGURE 322. LOCATION OF THE CONNECTORS IN PCA GEN-CONX59	312
FIGURE 323. LOCATION OF THE CONNECTORS IN PCA PIP-CONY59 RELATED TO PIP MODULE	313
FIGURE 324. LOCATION OF THE CONNECTORS IN PCA PIP-TOW59	313
FIGURE 325. LOCATION OF THE CONNECTOR IN PCA PIP-P1OSC59 (IMPACT DETECTOR)	314
FIGURE 326. LOCATION OF THE CONNECTOR IN PCA PIP-P2OSC59 (IMPACT DETECTOR)	314
FIGURE 327. LOCATION OF THE CONNECTORS IN PCA PIP-BCR59	314
FIGURE 328. CAT MODULE BLOCK DIAGRAM	324
FIGURE 329. MAIN BLOCKS OF THE CAT MODULE.....	325
FIGURE 330. X-AXIS MOTOR SMALL BELT	325
FIGURE 331. X-AXIS MOTOR LARGE BELT	326
FIGURE 332. Y-AXIS MOTOR SMALL BELT	326
FIGURE 333. Y-AXIS MOTOR LARGE BELT	327

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization		Purpose:	Uncontrolled Copy
Print date:	16-May-2023		

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	27 of 623
	Status:		Title:		Effective Date: 15-May-2023
	Effective		Service Manual Erytra		

FIGURE 334. Z-AXIS MOTOR BELT	327
FIGURE 335. GRIPPER PARALLELISM ADJUSTMENT	328
FIGURE 336. CARD PRESENCE DETECTOR INCLINATION	329
FIGURE 337. CARD DRAWER SELECTION IN TSS	330
FIGURE 338. REAR POSITION OF CARD DRAWER ADJUSTMENT TOOL IN THE DRAWER	330
FIGURE 339. FRONT POSITION OF THE CARDS DRAWER ADJUSTMENT TOOL IN THE DRAWER	331
FIGURE 340. INCUBATOR ADJUSTMENT SELECTION IN TSS	332
FIGURE 341. INCUBATOR ADJUSTMENT IN TSS.....	332
FIGURE 342. GRIPPER ADJUSTMENT IN CARD WASTE CONTAINER (REMOVED COVER)	334
FIGURE 343. CENTRIFUGE/INCUBATOR ADJUSTMENT TOOL (LEFT PICTURE) AND CENTRIFUGE ADJUSTMENT TAB IN TS (RIGHT PICTURE)	335
FIGURE 344. GRIPPER WIZARD ADJUSTMENT IN CENTRIFUGE IN TSS.....	335
FIGURE 345. CARD PRESENCE DETECTOR PARALLELISM ADJUSTMENT	336
FIGURE 346. CARD PRESENCE DETECTOR HEIGHT ADJUSTMENT AREAS.....	337
FIGURE 347. PRESENCE DETECTOR OVER ADJUSTMENT AREA 2.....	337
FIGURE 348. BARCODE READER INCLINATION ADJUSTMENT	338
FIGURE 349. ACQUISITION (CENTRAL) POSITION WIZARD – SCREEN 1	339
FIGURE 350. GRIPPER ADJUSTMENT IN CARD READER AREA	339
FIGURE 351. ACQUISITION (CENTRAL) POSITION WIZARD – SCREEN 2	339
FIGURE 352. GRIPPER PARALLELISM VERIFICATION	340
FIGURE 353. GRIPPER ADJUSTMENT IN CARD DRAWERS VERIFICATION	341
FIGURE 354. GRIPPER ADJUSTMENT IN INCUBATOR VERIFICATION	342
FIGURE 355. GRIPPER ADJUSTMENT IN CENTRIFUGE VERIFICATION	344
FIGURE 356. CARD PRESENCE DETECTOR PARALLELISM CHECKING.....	344
FIGURE 357. CARD PRESENCE DETECTOR HEIGHT ADJUSTMENT	345
FIGURE 358. PRESENCE DETECTED OVER ADJUSTMENT AREA 2	345
FIGURE 359. GRIPPER ADJUSTMENT IN CARD READER AREA	346
FIGURE 360. READ/CHECK CARD TEST CONFIGURATION WINDOW AND WELLS PICTURE	347
FIGURE 361. LOCATION OF THE SCREWS THAT ATTACH Y CHAIN TO Z TOWER/ROBOTIC HEAD	347
FIGURE 362. LOCATION OF Y CHAIN RIBBON CABLES AT PCA CAT-TOW59	348
FIGURE 363. Y ARM GUIDE	348
FIGURE 364. GRIPPER GUIDE ASSEMBLY AND LEADSCREW NUT FLANGE.....	348
FIGURE 365. LEADSCREW REMOVING PROCESS	349
FIGURE 366. LEADSCREW.....	349
FIGURE 367. ADJUSTMENT OF THE LEADSCREW IN THE LEADSCREW HOLDER.....	349
FIGURE 368. BARCODE READER MIRROR	350
FIGURE 369. FIRST FACE OF THE MIRROR (LEFT) AND SECOND FACE OF THE MIRROR (RIGHT)	350
FIGURE 370. INTERCONNECTION COVER	350
FIGURE 371. FLAT CABLES AND TIES.....	351
FIGURE 372. X CHAIN SLOT	351
FIGURE 373. FLAT CABLES AND CABLE CLAMP	351
FIGURE 374. X CHAIN SCREWS.....	352
FIGURE 375. RIGHT COVER OF THE CAT MODULE	352
FIGURE 376. FLAT CABLES.....	353
FIGURE 377. Y CHAIN SLOT	353
FIGURE 378. Y CHAIN SCREWS AND CABLE CLAMP.....	353
FIGURE 379. INTERCONNECTION COVER SCREWS.....	354
FIGURE 380. FLAT CABLES AND CABLE CLAMP	354
FIGURE 381. RIGHT COVER OF THE CAT MODULE	355
FIGURE 382. LEFT COVER OF THE CAT MODULE	355
FIGURE 383. BARCODE READER CABLE.....	355
FIGURE 384. METALLIC PLATE SCREW	356
FIGURE 385. GRIPPER CABLING (1)	356
FIGURE 386. GRIPPER CABLING (2)	356
FIGURE 387. RIGHT COVER OF THE CAT MODULE	357
FIGURE 388. BARCODE READER CABLE.....	357
FIGURE 389. METALLIC PLATE SCREW	357
FIGURE 390. GRIPPER CABLING	358
FIGURE 391. SMALL BELT.....	359

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization	
Print date:	16-May-2023
Purpose:	Uncontrolled Copy

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	28 of 623
Status:	Effective	Title:	Service Manual Erytra		
				Effective Date:	15-May-2023

FIGURE 392. CAT LARGE BELT.....	359
FIGURE 393. LARGE BELT (PINION)	359
FIGURE 394. INTERCONNECTION COVER SCREWS.....	360
FIGURE 395. SUPPORT SCREWS.....	360
FIGURE 396. CABLING AND CABLE CLAMP	360
FIGURE 397. MOTOR SCREW.....	361
FIGURE 398. PULLEY TIGHTENING SCREW.....	361
FIGURE 399. CONNECTORS, FUSES AND LEDs OF PCA GEN-BCKPL59 RELATED TO CAT MODULE	364
FIGURE 400. LOCATION OF THE FUSE AND LEDs IN PCA CAT-CM CONTROL	365
FIGURE 401. LOCATION OF THE CONNECTORS IN PCA GEN-CONX59	365
FIGURE 402. LOCATION OF THE CONNECTORS IN PCA CAT-CONY59.....	366
FIGURE 403. LOCATION OF THE CONNECTORS IN PCA CAT-TOW59.....	366
FIGURE 404. LOCATION OF THE CONNECTORS IN PCA PINZA57	367
FIGURE 405. DISPENSATION PUMPS	375
FIGURE 406. EXPANSION VESSELS	375
FIGURE 407. BOTTLES CONFIGURATION SELECTOR	376
FIGURE 408. BOTTLES CONFIGURATION SELECTOR POSITIONS – EXTERNAL DRAIN NOT ACTIVATED (LEFT PICTURE), EXTERNAL DRAIN ACTIVATED (RIGHT PICTURE)	376
FIGURE 409. ASPIRATION PUMP	377
FIGURE 410. FLUIDIC DRAWER EDGE PROTECTORS	378
FIGURE 411. CPDs IN THE INTERCONNECTION MANIFOLD	378
FIGURE 412. LOCATION OF THE INTERCONNECTION MANIFOLDS, BK2 (LEFT) AND BK3 (RIGHT).....	379
FIGURE 413. LOCATION OF THE INTERCONNECTION MANIFOLD (C).....	379
FIGURE 414. FLUIDIC MODULE SUPPORT WITH ALL THE COMPONENTS.....	380
FIGURE 415. FLUIDIC BLOCK DIAGRAM.....	381
FIGURE 416. MAIN BLOCKS OF FLUIDICS MODULE.....	382
FIGURE 417. LOCATION OF THE LATCH IN THE BOTTLES / CARDS DISPOSAL CONTAINER DRAWERS....	393
FIGURE 418. LOCATION OF THE SCREWS THAT ATTACH THE FRONT PANEL TO THE DRAWER.....	393
FIGURE 419. SETUP OF THE ADJUSTMENT OF THE PRESSURE SENSORS.....	396
FIGURE 420. LEAK-TIGHTNESS WIZARD.....	398
FIGURE 421. BLOCKING THE END OF A TUBE.....	398
FIGURE 422. CPD SENSORS	402
FIGURE 423. CPD SENSORS POSITION INSIDE A BLOCK.....	403
FIGURE 424. DISPENSATION PUMP SCREWS.....	403
FIGURE 425. O-RINGS IN DISPENSATION PUMP SUPPORT	404
FIGURE 426. EXPANSION VESSELS SCREWS.....	404
FIGURE 427. ASPIRATION PUMP CABLE CONNECTION.....	405
FIGURE 428. TUBES AND GROUND CABLE.....	405
FIGURE 429. PLATE ASSEMBLY.....	405
FIGURE 430. ELECTROVALVE SCREWS.....	406
FIGURE 431. SEALANT PRODUCT APPLICATION AREA	407
FIGURE 432. SCREWS HOLDING THE LOWER REAR COVER	408
FIGURE 433. LOAD CELLS CONNECTIONS TO PCA FLU-CTRL59	408
FIGURE 434. NUTS ATTACHING THE MANIFOLD BASIS TO THE INSTRUMENT.....	408
FIGURE 435. CLAMPS HOLDING LOAD CELL CABLE.	409
FIGURE 436. CONNECTOR AND CLAMP BELOW THE MANIFOLD BASIS.....	409
FIGURE 437. LOCATION OF SCREWS ATTACHING THE DRAWER FRAME TO THE BASIS OF THE INSTRUMENT.	409
FIGURE 438. SCREWS HOLDING THE LOAD CELL	410
FIGURE 439. LOAD CELL REMOVAL	410
FIGURE 440. POSITIONING OF THE NEW LOAD CELL	410
FIGURE 441. LOCK RING (LEFT) AND FERRULE (RIGHT) TUBE ASSEMBLY.....	411
FIGURE 442. LOCK RING AND FERRULE ASSEMBLY.....	411
FIGURE 443. SCREWS FIXING FLUIDIC DRAWER TO LOWER BASIS.....	417
FIGURE 444. LINEAR GUIDE SCREWS ON FLUIDIC DRAWER LATERAL.....	418
FIGURE 445. LINEAR GUIDE SCREWS ON DRAWER MAIN LATERAL PART	418
FIGURE 446. CPC BOTTLES CONNECTOR LEVER	418
FIGURE 447. CPC BOTTLES CONNECTOR O-RING	419

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization		Purpose:	Uncontrolled Copy
Print date:	16-May-2023		

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	29 of 623
Status:	Effective	Title:	Service Manual Erytra		
				Effective Date:	15-May-2023

FIGURE 448. TUBING	419
FIGURE 449. SCREWS ATTACHING THE 2 BLOCKS TOGETHER	419
FIGURE 450. SCREWS ATTACHING THE 2 BLOCKS TO THE DISTRIBUTOR	420
FIGURE 451. CPD SENSORS	420
FIGURE 452. CPD SENSORS POSITION INSIDE A BLOCK.....	420
FIGURE 453. ASPIRATION PUMP TUBING.....	421
FIGURE 454. MANIFOLD SCREWS.....	422
FIGURE 455. ASPIRATION PUMP TUBING IN THE MANIFOLD.....	422
FIGURE 456. LOCATION OF THE CONNECTORS, FUSES AND LEDs IN PCA GEN-BCKPL59 RELATED TO FLUIDICS MODULE	427
FIGURE 457. LOCATION OF THE CONNECTORS, FUSE AND LEDs IN PCA FLU-CTRL59	428
FIGURE 458. LOCATION OF THE CONNECTORS IN PCA PIP-CONY59 RELATED TO FLUIDICS MODULE	429
FIGURE 459. BLOCK DIAGRAM OF DILUTION CUP MODULE	449
FIGURE 460. MAIN BLOCKS OF DILUTION CUP MODULE	450
FIGURE 461. SETSCREW INSERTED IN THE SILENTBLOCK	451
FIGURE 462. DILUTION CUP ALIGNMENT TOOL FIXED INTO DILUTION CUP STRUCTURE	451
FIGURE 463. PARALLELISM ADJUSTMENT OF THE DILUTION CUP MODULE	451
FIGURE 464. CHECKING OF THE CORRECT ADJUSTMENT OF THE DILUTION CUP	452
FIGURE 465. LOCATION OF THE BASIS OF THE MOTOR (LEFT) AND RPM MEASUREMENT WITH A TACHOMETER DURING STIRRING (RIGHT).....	452
FIGURE 466. DILUTION CUP SCEWS	453
FIGURE 467. REAR PART OF DILUTION CUP WELLS.....	458
FIGURE 468. LOCATION OF THE CONNECTOR, FUSES AND LEDs IN PCA GEN-BCKPL59 RELATED TO DILUTION CUP MODULE	460
FIGURE 469. LOCATION OF THE FUSE AND LEDs IN PCA MOTORCONTROL	461
FIGURE 470. LOCATION OF THE CONNECTORS IN PCA CUP-ST59	462
FIGURE 471. READER RAW IMAGE (LEFT) AND MICROTUBE IMAGE (RIGHT)	466
FIGURE 472. LEC MODULE BLOCK DIAGRAM	467
FIGURE 473. SUMMARY OF ADJUSTMENTS OF LEC MODULE	469
FIGURE 474. FIXING SCREW OF THE DIAPHRAGM	469
FIGURE 475. DIAPHRAGM ADJUSTMENT SCREEN ON TSS	470
FIGURE 476. READER ADJUSTMENT TOOL-PATTERN V2 (PN: 731398) ATTACHED TO READER ADJUSTMENT TOOL-SUPPORT V2 (PN: 234212)	470
FIGURE 477. FIXING SCREW OF THE FOCUS.....	471
FIGURE 478. FOCUS ADJUSTMENT SCREEN ON TSS.....	471
FIGURE 479. ILLUMINATION POSITION ADJUSTMENT SCREEN ON TSS	472
FIGURE 480. ILLUMINATION DIMENSIONS ADJUSTMENT SCREEN ON TSS.....	472
FIGURE 481. ILLUMINATION ROTATION VERIFICATION SCREEN ON TSS	473
FIGURE 482. ILLUMINATION PERSPECTIVE ADJUSTMENT SCREEN ON TSS.....	474
FIGURE 483. ILLUMINATION INTENSITY ADJUSTMENT SCREEN ON TSS	474
FIGURE 484. ILLUMINATION COLOR ADJUSTMENT SCREEN ON TSS.....	475
FIGURE 485. CONVERSION FACTOR ADJUSTMENT SCREEN ON TSS.....	475
FIGURE 486. DIAPHRAGM APERTURE CHECK INITIAL SCREEN ON TSS	477
FIGURE 487. DIAPHRAGM APERTURE CHECK SCREEN ON TSS	478
FIGURE 488. OPTICAL FOCUS CHECK SCREEN ON TSS	478
FIGURE 489. ILLUMINATION POSITION CHECK SCREEN ON TSS.....	479
FIGURE 490. ILLUMINATION DIMENSIONS CHECK SCREEN ON TSS	479
FIGURE 491. ILLUMINATION ROTATION CHECK SCREEN ON TSS.....	480
FIGURE 492. ILLUMINATION PERSPECTIVE CHECK SCREEN ON TSS	481
FIGURE 493. ILLUMINATION INTENSITY CHECK SCREEN ON TSS	481
FIGURE 494. ILLUMINATION COLOR CHECK SCREEN ON TSS	482
FIGURE 495. ILLUMINATION UNIFORMITY CHECK SCREEN ON TSS	482
FIGURE 496. COLOR UNIFORMITY CHECK SCREEN ON TSS	483
FIGURE 497. ILLUMINATION CLEANLINESS CHECK SCREEN ON TSS.....	483
FIGURE 498. USB AND FIREWIRE CABLES. READER AND GROUNDING FIXING SCREWS.....	484
FIGURE 499. FIREWIRE/IEEE1394 CAMERA CABLE	484
FIGURE 500. CAMERA ASSEMBLY	485
FIGURE 501. CAMERA AND OPTICS SET SCREWS ASSEMBLY	485

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization	
Print date:	16-May-2023
Purpose:	Uncontrolled Copy

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	30 of 623
Status:	Effective	Title:	Service Manual Erytra		
				Effective Date:	15-May-2023

FIGURE 502. PCA RD-IL421 v3 ASSEMBLY.....	486
FIGURE 503. READER ILLUMINATION PATTERN ASSEMBLY	486
FIGURE 504. LOCATION OF THE CONNECTORS IN PCA RD-IL421 v3.....	489
FIGURE 505. MAIN STRUCTURE CODE.....	496
FIGURE 506. CREO VIEW OPTIONS.....	497
FIGURE 507. HIGHLIGHTED PART DESCRIPTION AFTER SELECTING A DRAWING PART.....	498
FIGURE 508. MAIN STRUCTURE CODE.....	498
FIGURE 509. TREE STRUCTURE COMPONENT SELECTION.....	499
FIGURE 510. TRANSFORM OPTIONS IN CREO VIEW EXPRESS TOOLBAR	500
FIGURE 511. FIGURES AREA.....	500
FIGURE 512. PLAYBACK CONTROL BUTTONS FOR AN ANIMATED ASSEMBLY PROCEDURE	501
FIGURE 513. FINDING A DRAWING PART FROM A KNOWN CODE	502
FIGURE 514. FIND A CODE BY SELECTING A DRAWING PART.....	502
FIGURE 515. DIAGNOSTIC ICON	504
FIGURE 516. DIAGNOSTIC APPLICATION	505
FIGURE 517. START BUTTON IN DIAGNOSTIC APPLICATION.....	506
FIGURE 518. INITIALIZATION OPTION IN DIAGNOSTIC APPLICATION	507
FIGURE 519. CARDS MANAGEMENT OPTIONS IN DIAGNOSTIC APPLICATION.....	507
FIGURE 520. PIPETTING OPTIONS IN DIAGNOSTIC APPLICATION.....	510
FIGURE 521. FLUIDICS OPTIONS IN DIAGNOSTIC APPLICATION	514
FIGURE 522. READER OPTION IN DIAGNOSTIC APPLICATION	517
FIGURE 523. INCUBATORS OPTIONS IN DIAGNOSTIC APPLICATION.....	517
FIGURE 524. CENTRIFUGES OPTIONS IN DIAGNOSTIC APPLICATION	518
FIGURE 525. SAMPLES DRAWERS OPTION IN DIAGNOSTIC APPLICATION.....	519
FIGURE 526. REAGENTS DRAWERS OPTION IN DIAGNOSTIC APPLICATION.....	519
FIGURE 527. CARDS DRAWERS OPTION IN DIAGNOSTIC APPLICATION	520
FIGURE 528. DILUTION CUP OPTION IN DIAGNOSTIC APPLICATION.....	520
FIGURE 529. MISCELANEOUS OPTION IN DIAGNOSTIC APPLICATION	520
FIGURE 530. PRINTER BUTTON IN DIAGNOSTIC APPLICATION.....	521
FIGURE 531. REPORT EXAMPLE IN DIAGNOSTIC APPLICATION.....	522
FIGURE 532. LUBRICATION POINT IN TOUCHSCREEN DRAWER	529
FIGURE 533. LUBRICATION POINTS IN TOUCHSCREEN DRAWER.....	529
FIGURE 534. LUBRICATION POINTS IN USER TRAY	530
FIGURE 535. LUBRICATION POINTS IN UPPER DOOR	530
FIGURE 536. LUBRICATION POINTS IN INTERMEDIATE DOOR.....	530
FIGURE 537. LUBRICATION POINT IN REAGENT DRAWER (LATCH)	531
FIGURE 538. LUBRICATION POINT IN REAGENT DRAWER (LINEAR GUIDES)	531
FIGURE 539. LUBRICATION POINT IN REAGENT DRAWER (STIRRED REAGENTS PLATFORM).....	531
FIGURE 540. LUBRICATION POINT IN SAMPLE DRAWER (LATCH)	531
FIGURE 541. LUBRICATION POINT IN SAMPLE DRAWER (LINEAR GUIDE)	531
FIGURE 542. LUBRICATION POINT IN CARD DRAWER (LATCH).....	532
FIGURE 543. LUBRICATION POINT IN CARD DRAWER (LINEAR GUIDES)	532
FIGURE 544. LUBRICATION POINT IN PIP MODULE (HORNS).....	532
FIGURE 545. LUBRICATION POINT IN PIP MODULE (X-AXIS LINEAR GUIDE).....	532
FIGURE 546. LUBRICATION POINT IN PIP MODULE (Y-AXIS LINEAR GUIDE).....	532
FIGURE 547. LUBRICATION POINTS IN PIP MODULE (LEADSCREWS)	533
FIGURE 548. LUBRICATION POINT IN CAT MODULE (X-AXIS LINEAR GUIDE)	533
FIGURE 549. LUBRICATION POINT IN CAT MODULE (Y-AXIS LINEAR GUIDE)	533
FIGURE 550. LUBRICATION POINT IN CAT MODULE (LEADSCREW).....	534
FIGURE 551. LUBRICATION POINT IN CAT MODULE (GRIPPER LINEAR GUIDE).....	534
FIGURE 552. LUBRICATION POINT IN CAT MODULE (GRIPPER PINS)	535
FIGURE 553. LUBRICATION POINT IN CAT MODULE (GRIPPER CENTERING SYSTEM)	535
FIGURE 554. LUBRICATION POINT IN FLUIDICS DRAWER (LOAD CELL)	536
FIGURE 555. LUBRICATION POINT IN FLUIDICS DRAWER (TILTING BASE JOINTS).....	536
FIGURE 556. LUBRICATION POINT IN FLUIDICS DRAWER (LATCH)	536
FIGURE 557. LUBRICATION POINT IN FLUIDICS DRAWER (LINEAR GUIDE)	536
FIGURE 558. EXAMPLE OF USER ERROR MESSAGE	539
FIGURE 559. ERYTRA RECOVERY USB SOFTWARE.....	541
FIGURE 560. BOOT APPLICATION INITIAL WINDOW	544

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization		
Print date:	16-May-2023	Purpose: Uncontrolled Copy

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	31 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

FIGURE 561. BOOT APPLICATION LOGIN WINDOW	544
FIGURE 562 BOOT APPLICATION LOGIN WINDOW	562
FIGURE 563 VIEW OF TECHNICAL SERVICE SOFTWARE WITH ACCESS TO WINDOWS 10 FILE EXPLORER	568
FIGURE 564 REQUEST FOR ADMIN USER CREDENTIALS IN DIAGNOSTIC APPLICATION MODE.....	569
FIGURE 565 VIEW OF DIAGNOSTIC APPLICATION WITH ACCESS TO WINDOWS 10 FILE EXPLORER	569
FIGURE 566 REQUEST FOR ADMIN USER CREDENTIALS IN USER APPLICATION MODE.....	570
FIGURE 567 VIEW OF USER APPLICATION WITH ACCESS TO WINDOWS 10 FILE EXPLORER	571
FIGURE 568. MICROTOUCH MT 7 PROGRAM	572
FIGURE 569. MT7 CONTROL PANEL.....	572
FIGURE 570. BACKUP ICON	573
FIGURE 571. BACKUP DESCRIPTION AND LOG FILES CHOICE	575
FIGURE 572. RESTORE _DATABase SOFTWARE WINDOW	576
FIGURE 573. RESTORE ICON	578
FIGURE 574. RESTORE BACKUP WINDOW.....	578
FIGURE 575. EPROM CONTROL WINDOW IN TSS	582
FIGURE 576. FIRMWARE UPDATE WINDOW IN TECHNICAL SERVICE SOFTWARE.....	583
FIGURE 577 BOOT LOGO PAGE	585
FIGURE 578 UEFI VERSION.....	585
FIGURE 579 NEW BOOT LOGO PAGE	586
FIGURE 580 SUPER IO CONFIGURATION SELECTION	586
FIGURE 581 COM 4 PORT DISABLING	586
FIGURE 582 COM 6 PORT DISABLING	587
FIGURE 583 LPT1 PORT DISABLING	587
FIGURE 584 BIOS EXIT TAB	587
FIGURE 585 RUNNING “DEVMGMT.MSC” AS ADMINISTRATOR.....	587
FIGURE 586 PORTS ASSIGNMENT	588
FIGURE 587 UNINSTALLING NUVOTON COMMUNICATIONS PORT (COM4)	588
FIGURE 588 UNINSTALLING NUVOTON COMMUNICATIONS PORT (COM6)	588
FIGURE 589 UNINSTALLING ECP PRINTER PORT (LPT1).....	589
FIGURE 590 SILICON LABS CP210X USB TO UART BRIDGE SELECTION.....	589
FIGURE 591 PORT SETTINGS TAB	589
FIGURE 592 COM 4 SELECTION IN COM PORT NUMBER COMBO	590
FIGURE 593 EXPECTED PORT ASSIGNMENT IN WINDOWS 10	590
FIGURE 594. APPLOCKER APPLICATION ENTERING UPDATE MODE	591
FIGURE 595. APPLOCKER APPLICATION EXITING UPDATE MODE.....	592
FIGURE 596. EXAMPLE OF BLOCKED APPLICATION (I).....	592
FIGURE 597. EXAMPLE OF BLOCKED APPLICATION (II).....	592
FIGURE 598. ERYTRA MACRO EXECUTOR MAIN SCREEN.....	593
FIGURE 599.COMBO BOX IN ERYTRA MACRO EXECUTOR MAIN SCREEN.....	593
FIGURE 600.SUPPORT PACK BUTTON IN TECHNICAL SERVICE ACTION TAB (USER SOFTWARE)	596
FIGURE 601.SUPPORT PACK MENU.....	596
FIGURE 602 SUPPORT PACK FOLDER.....	598
FIGURE 603 SUPPORT PACK START DATE INTRODUCTION.....	598
FIGURE 604 SUPPORT PACK END DATE INTRODUCTION	598
FIGURE 605 SUPPORT PACK "INCLUDE RAW IMAGE FILES" QUESTION	599
FIGURE 606 SUPPORT PACK "INCLUDE TAN FILES" QUESTION	599
FIGURE 607 SUPPORT PACK GENERATION	600
FIGURE 608 SUPPORT PACK GENERATED SUCCESSFULLY	600
FIGURE 609. MAPPING A NETWORK UNIT	601
FIGURE 610.DIANASERVER CONFIGURATION OPTIONS IN DIANA O.	603
FIGURE 611.DIANASERVER ICON.....	603
FIGURE 612.DIANASERVER CONFIGURATION WINDOW	604
FIGURE 613.HOSTDEMO PROGRAM	610

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization	
Print date:	16-May-2023
Purpose:	Uncontrolled Copy

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	32 of 623
	Status:		Effective		Effective Date: 15-May-2023
		Service Manual Erytra			
		Title:			

Tables

TABLE 1. TECHNICAL SPECIFICATIONS.....	35
TABLE 2. SSD STRUCTURE	42
TABLE 3. LIST OF OPTIONS IN PROCEDURES TAB	89
TABLE 4. LIST OF OPTIONS IN INCUBATOR X - INCUBATORS TAB	92
TABLE 5. LIST OF OPTIONS IN HOME CYCLES - INCUBATORS TAB	93
TABLE 6. LIST OF OPTIONS IN CENTRIFUGE X - CENTRIFUGES TAB	95
TABLE 7. LIST OF OPTIONS IN HOME CYCLES - CENTRIFUGES TAB	96
TABLE 8. LIST OF OPTIONS IN RANDOM POSITION - CENTRIFUGES TAB.....	97
TABLE 9. LIST OF OPTIONS IN DILUTION CUP TAB	99
TABLE 10. LIST OF OPTIONS IN MOVEMENT - GRIPPER ARM TAB.....	102
TABLE 11. LIST OF OPTIONS IN PRESENCE DETECTOR – GRIPPER ARM TAB.....	103
TABLE 12. LIST OF OPTIONS IN BCR – GRIPPER ARM TAB.....	105
TABLE 13. LIST OF OPTIONS IN HOME CYCLES - GRIPPER ARM TAB	106
TABLE 14. LIST OF OPTIONS IN RANDOM POSITION – GRIPPER ARM TAB.....	107
TABLE 15. LIST OF OPTIONS IN CARD DRAWERS - DRAWERS TAB	108
TABLE 16. LIST OF OPTIONS IN SAMPLE DRAWERS - DRAWERS TAB.....	109
TABLE 17. LIST OF OPTIONS IN REAGENT DRAWERS – DRAWERS TAB.....	111
TABLE 18. LIST OF OPTIONS IN FLUIDICS DRAWERS – DRAWERS TAB.....	112
TABLE 19. LIST OF OPTIONS IN HOME CYCLES – DRAWERS TAB	113
TABLE 20. LIST OF OPTIONS IN MOVEMENT - PIP ARM TAB.....	115
TABLE 21. LIST OF OPTIONS IN REAGENT DRAWERS – PRESENCE DETECTOR – PIP ARM TAB	116
TABLE 22. LIST OF OPTIONS IN SAMPLE DRAWERS – PRESENCE DETECTOR – PIP ARM TAB.....	117
TABLE 23. LIST OF OPTIONS IN PRESENCE DETECTOR – PRESENCE DETECTOR – PIP ARM TAB.....	118
TABLE 24. LIST OF OPTIONS IN LEVEL DETECTOR – PIP ARM TAB	120
TABLE 25. LIST OF OPTIONS IN BCR – PIP ARM TAB	122
TABLE 26. LIST OF OPTIONS IN CPD – PIP ARM TAB	123
TABLE 27. LIST OF OPTIONS IN HOME CYCLES – PIP ARM TAB.....	124
TABLE 28. LIST OF OPTIONS IN RANDOM POSITION – PIP ARM TAB	125
TABLE 29. LIST OF OPTIONS IN FLUIDIC TAB.....	128
TABLE 30. LIST OF OPTIONS IN GENERAL MODULE TAB	130
TABLE 31. LIST OF OPTIONS IN PCA CONTROL - FIRMWARE TAB.....	133
TABLE 32. LIST OF OPTIONS IN EPROM CONTROL - FIRMWARE TAB	134
TABLE 33. LIST OF OPTIONS IN DILUTION CUP OPERATIONS TAB	136
TABLE 34. LIST OF OPTIONS IN ASPIRE AND DISPENSE OPERATIONS TAB.....	138
TABLE 35. LIST OF CABLES OF THE POWER SUPPLY	173
TABLE 36. LIST OF FUSES OF GENERAL MODULE	191
TABLE 37. LIST OF LEDs OF GENERAL MODULE	192
TABLE 38. LIST OF JUMPERS AND TEST POINTS OF GENERAL MODULE	193
TABLE 39. LIST OF WIRING OF GENERAL MODULE	196
TABLE 40. LIST OF ERRORS RELATED TO GENERAL MODULE.	205
TABLE 41. GENERAL MODULE TROUBLESHOOTING.....	209
TABLE 42. INC AXIS ENCODER RESOLUTION.....	211
TABLE 43. LIST OF FUSES OF INCUBATOR MODULE	223
TABLE 44. LIST OF LEDs OF INCUBATOR MODULE	224
TABLE 45. LIST OF WIRING OF INCUBATOR MODULE	224
TABLE 46. LIST OF ERRORS RELATED TO INCUBATOR MODULE.....	229
TABLE 47. INCUBATOR MODULE TROUBLESHOOTING	231
TABLE 48. CEN ENCODER RESOLUTION.....	232
TABLE 49. LIST OF FUSES OF CENTRIFUGE MODULE	243
TABLE 50. LIST OF LEDs OF CENTRIFUGE MODULE	244
TABLE 51. LIST OF WIRING OF CENTRIFUGE MODULE	244
TABLE 52. LIST OF ERRORS RELATED TO CENTRIFUGE MODULE	250
TABLE 53. CENTRIFUGE MODULE TROUBLESHOOTING.....	251
TABLE 54. PIP AXES ENCODER RESOLUTION	252
TABLE 55. LEVEL DETECTION SYSTEM VALUES IN DRY SITUATION.....	279
TABLE 56. LEVEL DETECTION SYSTEM VALUES IN PRIME SITUATION.....	280

Print date:	16-May-2023	Purpose:	Uncontrolled Copy
	Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization		

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	33 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

TABLE 57. LIST OF FUSES OF PIP MODULE	307
TABLE 58. LIST OF LEDs OF PIP MODULE	308
TABLE 59. LIST OF WIRING OF PIP MODULE	310
TABLE 60. LIST OF ERRORS RELATED TO PIP MODULE	319
TABLE 61. PIP MODULE TROUBLESHOOTING	322
TABLE 62. CAT AXES ENCODER RESOLUTION.....	323
TABLE 63. LIST OF FUSES OF CAT MODULE.....	362
TABLE 64. LIST OF LEDs OF CAT MODULE	362
TABLE 65. LIST OF WIRING OF CAT MODULE	363
TABLE 66. LIST OF ERRORS RELATED TO CAT MODULE.....	371
TABLE 67. CAT MODULE TROUBLESHOOTING	374
TABLE 68. CONFIGURATION OF THE DILUTORS	397
TABLE 69. LIST OF FUSES OF FLUIDICS MODULE	423
TABLE 70. LIST OF LEDs OF FLUIDICS MODULE	424
TABLE 71. LIST OF WIRING OF FLUIDICS MODULE	426
TABLE 72. LIST OF ERRORS RELATED TO FLUIDICS MODULE	441
TABLE 73. LIST OF ERRORS (3 LAST DIGITS) RELATED TO FLUIDICS MODULE	444
TABLE 74. FLUIDICS MODULE TROUBLESHOOTING.....	448
TABLE 75. LIST OF FUSES OF DILUTION CUP MODULE.....	458
TABLE 76. LIST OF LEDs OF DILUTION CUP MODULE	459
TABLE 77. LIST OF WIRING OF DILUTION CUP MODULE.....	459
TABLE 78. LIST OF ERRORS RELATED TO DILUTION CUP MODULE	464
TABLE 79. DILUTION CUP MODULE TROUBLESHOOTING	465
TABLE 80. SUMMARY OF VERIFICATIONS	477
TABLE 81. LIST OF FUSES OF READER MODULE	488
TABLE 82. LIST OF LEDs OF READER MODULE	488
TABLE 83. LIST OF WIRING OF READER MODULE	488
TABLE 84. LIST OF ERRORS RELATED TO READER MODULE	493
TABLE 85. READER MODULE TROUBLESHOOTING.....	494
TABLE 86. 3D DRAWINGS REFERENCE TABLE	495
TABLE 87. 2D DRAWINGS REFERENCE TABLE	496
TABLE 88. AVAILABLE BUTTONS IN DIAGNOSTIC APPLICATION.....	506
TABLE 89. WARNING AND FAIL RESULTS IN DIAGNOSTIC APPLICATION	527
TABLE 90. ERYTRA ADJUSTMENT TOOLS	528
TABLE 91. EQUIVALENCE BETWEEN YEARS IN PM AND TESTS UNDER STANDARD INSTRUMENT USE ..	528
TABLE 92. MAINTENANCE SERVICES LIST	537
TABLE 93. SECURITY UPDATES PACKAGES.....	567
TABLE 94. DIANA SYSTEM ERRORS RELATED TO LIS COMMUNICATION	617
TABLE 95. LIST OF ERRORS RELATED TO ERYTRA – LIS COMMUNICATION MODULE.....	619
TABLE 96. LIS TROUBLESHOOTING	620

1 INSTRUMENT GENERAL OVERVIEW

1.1 INTRODUCTION

Configurable as a standalone instrument or interfaced to the customer's Laboratory Information System (LIS), *Erytra* automates many of the test processing functions and data management requirements using gel cards and digital image processing.

Samples contained in standard immunohaematology laboratory tubes are introduced into the instrument and are processed using cards and reagents also introduced, according to the instructions for use. Results are automatically read, interpreted and registered. Results can be reviewed, printed or exported upon user request.

This manual is oriented to the **QUALIFIED TECHNICIAN** and contains the necessary information for a proper and secure manipulation of the instrument.

Read carefully all the information of this manual before starting the manipulation of the instrument. It is important to read the information concerning maintenance, correct use and liquids management.



NOTE:
All the operations described in this manual must be carried out only for a **QUALIFIED TECHNICIAN**.

1.2 SECURITY REMARKS

All the information concerning security, marking and regulatory of the instrument is described in Instructions for Use manual.

1.3 FUNCTIONAL CHARACTERISTICS

Functional characteristics of *Erytra* can be found in Instructions for Use manual.

1.4 TECHNICAL SPECIFICATIONS

The following table summarizes the main technical specifications of *Erytra*.

Model	<i>Erytra</i>	
Supply	Voltage	100-240V AC
	Frequency	50-60 Hz
	Maximum input power	600W
	Standby power	30VA
	Fuses	2 x T6.3A L 250V,5x20mm
Protection against electric shock	Class I	
Installation category	Overvoltage category II	
Dimensions	700 mm (depth) x 1060 mm (width) x 1785 mm (height)	
	725 mm (depth, with door handles installed) x 1060 mm (width) x 1785 mm (height)	
	Packaging: 800 mm (depth) x 1200 mm (width) x 1800 mm (height)	
Weight	Approximately 350 Kg (with washing and waste containers full)	
Working conditions	Indoors use	
	Temperature	15 °C to 30 °C
	Maximum relative humidity	85 % (non condensing)
	Maximum altitude	3000 m (9842 ft)
	Maximum fluctuations of the power supply	±10 % of the nominal power supply
	Pollution degree range	2
Storage and transport conditions	Temperature	-29 °C to 60 °C
	Maximum relative humidity	85 % (non condensing)
Sound level	58,9 dBA	
Heat dispersion (per hour)*	Stand By	0,005 kWh / 18 kJ / 4,3 kcal / 17,1 BTU
	Start Up / Operation	0,171 kWh / 615,6 kJ / 147,1 kcal / 583,5 BTU
	Ready for Workload	0,138 kWh / 496,8 kJ / 118,7 kcal / 470,9 BTU
Heat dispersion (per hour)**	Stand By	0,006 kWh / 21,6 kJ / 5,2 kcal / 20,5 BTU
	Start Up / Operation	0,126 kWh / 453,6 kJ / 108,4 kcal / 429,9 BTU
	Ready for Workload	0,121 kWh / 435,6 kJ / 104,1 kcal / 412,9 BTU

Table 1. Technical specifications

* with old electronics cabinet

** with new electronics cabinet (S/N ≥ 059-0001583)

- Status LEDs: allows the operator to know the instrument status without looking at the software screen. The instrument contains status LEDs in the sample drawers, in the pushbutton and on the top of the instrument (General LED).
- Temperature control system: system to check that the internal temperature does not exceed defined values.
- Buzzer: it alerts acoustically of any incidence to the operator.
- Structure of the instrument: chassis, cables, tubes rails and external covers. In the base of the instrument there are 4 rotating supporting wheels, one in each corner, to allow moving the instrument.

1.5.2.1 Embedded computer and touchscreen monitor

An embedded computer and a touchscreen monitor are integrated in the instrument, thus there is no need to use an external computer to control the instrument.

There are some ports available to allow the user to connect external peripherals and to connect the instrument to a Laboratory Information System (LIS). These ports are:

- 2 USB ports located at the front part of the instrument.
- 4 USB ports located at the rear part of the instrument.
- 1 RS232 port located at the rear part of the instrument.
- 1 ETHERNET/LAN connection located at the rear part of the instrument.

The screen is installed in a drawer located under samples and reagents area. This drawer allows hiding completely the screen inside the instrument when is not in use. Also the drawer allows the operator to adjust its position/inclination.

1.5.2.2 Supply and Communication bus

There is a system to provide a standby power supply to the computer in order to, when it is pressed the pushbutton to switch it on, activate the ATX control of the computer and allow the system to turn on. Once the instrument is switched on, main supply voltages (24V and 5V) in Erytra are generated by the power supply system.

Regarding the communication system, the instrument has 3 different types of communication buses: RS232, RS485 and I2C. The different parts of the instrument that use each communication bus are described in chapters 4 to 11.

1.5.2.3 Samples management module

This module is located in the upper area of the instrument. The instrument has 4 sample drawers that are labelled as 3, 4, 5 and 6 in front of the instrument. Each drawer has a safety latch (controlled electronically) to monitor open/close status and assure closed/working position. A button is located just below each drawer to activate the latch, opening the drawer.

Each sample drawer allocates 2 samples racks. The correct centering of different diameter tubes in their lodgings is assured by centering clips. The racks have an opening window to allow reading the barcode of the samples tubes.

1.5.2.4 Reagents management module

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	38 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

This module is also located in the upper area of the instrument. The instrument has 2 reagents drawers, labelled as 1 and 2 in front of the instrument. Each drawer has a safety latch (controlled electronically) to monitor open/close status and assure closed/working position. As well as the samples drawers, a button is located just below each drawer to activate the latch, opening the drawer.

Each reagents drawer allocates 2 racks (18 vial rack and 6 vials and 3 diluents rack). The reagents racks have a window to allow reading the barcode of the vials/bottles. All vial lodgings have a spring clip to fix its position against the reading window. The spring clips avoid any movement of the vial in its lodging. All the vials lodgings place the vial/bottle with a certain inclination to minimise dead volumes.

Both drawers include a rotating mechanism in order to stir the vials of the 18 vials rack all together.

1.5.2.5 Cards management module

This module is located in the middle area of the instrument and includes 4 drawers, labelled as 1, 2, 3 and 4 in front of the instrument. Each drawer is able to allocate 4 card holders, orientated in landscape mode.

Also 4 safety latches (controlled electronically) are included, one per each drawer, that assure the drawer is closed in the working position and allows close/open status monitoring.

Drawers are opened by means of a keyboard including buttons for each drawer. This keyboard is located under the drawers.

1.5.2.6 Doors

An upper area door gives to the technician / operator the access to the samples and reagents management modules. Once opened, it keeps the position and does not close down by its own weight. The door does not have a latch, it has a sensor that monitors open/close status.

An intermediate area door gives to the technician the access to the cards management modules. The door does not have a latch, it has a sensor that monitors open/close status. In order to prevent the access to the user to the cards management area, the door is closed by means of a screw located in the sensor area.

1.5.3 Incubator module (INC)

There are 4 independent incubator blocks, each block having capacity for 12 cards which are positioned equidistantly in the block. A Peltier module unit (formed by Peltier cell, fan and heatsink) attached to incubator block is used as reversible heat/cool pump to warm up or cool the block to the desired temperature.

The other function of the incubator module is to transport the cards between the upper and intermediate level moving vertically the incubator block. The incubator movement is performed by a servocontrolled motor, with a home position in the lower position of the incubator detected by means of a light barrier. The higher position of the incubator is determined by an adjustment.

Print date:	16-May-2023	Purpose:	Uncontrolled Copy
Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization			

1.5.4 Centrifuge module (CEN)

Centrifuge module (CEN) is integrated by 2 identical centrifuges. Each centrifuge has a capacity of 12 cards. Main parts of each centrifuge are the rotor with the card holders, the motor and the case.

The motors are DC brushless motors (one per each centrifuge) with a home position detected by means of a Hall effect sensor. The motors are isolated with silentblocks to reduce a possible transmission of vibration to the rest of the instrument.

The rotor is the rotating part of the centrifuge and is attached to the motor axis. It holds the card holders that contain the cards to be centrifuged.

Centrifuge design ensures that any even number of cards can be introduced in a way that the centrifuge is not unbalanced. To balance the centrifuge, when an odd number of cards are centrifuged, an empty card is used. There is no specific card to balance the centrifuge, it will be used a card waiting to be used in card drawers. There is not unbalance detector.

1.5.5 Pipetting robot arm module (PIP)

The pipetting robot arm module (PIP) is allocated in the upper area of the instrument and is responsible for pipetting samples and reagents into cards (with the appropriate controls and level detection) and identification and presence detection of samples and reagents.

The PIP module can be decomposed in the following submodules:

- X-axis: it allows the movement along X-axis (left to right in the instrument).
- Y-axis: it allows the movement along Y-axis (back to front in the instrument).
- Robot head: it contains the following items:
 - Probes including probes washer, liquid level detector system as well as related mechanics and electronics.
 - Barcode readers and related mechanics and electronics.
 - Presence detector and related mechanics and electronics.
- Correct pipetting detectors (CPD), located at the rear part of the robot arm, and digital pumps (dilutors) are explained in the fluidic module (FLU).

This module can read barcodes of each sample tube, vial or bottle. Samples and reagents identification can be done automatically or manually. Having 2 barcode readers allows reading the barcodes from each side of the rack and they are located in such a way that ensures that when one reader is centered with a sample tube the other one is not focused to any other tube.

The identification includes 2 main tasks:

- Presence detection: detects the presence of tubes and vials/bottles in the drawers.
- Barcode reading: reading of the barcode of each tube/vial/bottle.

In order to perform the pipetting actions, the robot arm includes 2 probes. Using the level detection system (capacitive sensing, based on frequency variations detection), the instrument calculates the available amount of reagent in each vial in order to give the operator feedback of the available and required reagent volume. This estimation is performed before each aspiration and in each reagent drawer identification. For this purpose, a double detection is performed. The system monitors the position of both detections and raises an error if they are different. If impact detector system is installed, it is possible to detect and protect probes from undesired impacts which are frequently caused by improper user operation.

Probe internal wash is required to remove any trace of aspirated liquid (extra volume of liquid aspirated and rests of this liquid in the tubing) in order to minimise the risk of carry over. Internal

probe wash is performed with Solution A. Probe external wash is required to remove any trace of aspirated liquid in the probe external surface in order to minimise the risk of carry over. External wash is performed with Solution B.

1.5.6 Card transport robot arm module (CAT)

Card transport robot arm module (CAT) is allocated in the intermediate floor of the instrument and is responsible for card transportation between cards drawers, incubators, centrifuges and reader, as well as release cards to waste container and card inventory (mapping and identification).

Card transport is required as a link between functionalities:

- From card load/unload to card identification.
- From card identification to card revision.
- From card revision to card pipetting and incubation (includes card lift).
- From card incubation to card centrifugation (includes card lift).
- From card centrifugation to card reading.
- From card reading to card return.

The CAT module can be decomposed in the following submodules:

- X-axis: it allows the movement along X-axis (left to right).
- Y-axis: it allows the movement along Y-axis (back to front).
- Robot head: it contains the following items:
 - Gripper (and related mechanics and electronics).
 - Barcode reader (and related mechanics and electronics).
 - Presence detector (and related mechanics and electronics).

Cards identification includes 2 main tasks:

- Presence detection: detection of cards in any location.
- BC reading: reading of the barcode of each card. The reader is in a fixed position and the card is placed in front of it by the Card Transport system.

After being processed, cards are normally disposed to a waste container except if the result obtained requires a review from the operator or there are some wells not used. In these cases the cards are left in a special area (service rack) and they can be recovered by the operator.

1.5.7 Fluidics module (FLU)

The fluidics module can be decomposed in the following submodules:

1. Manifold: acrylic block (PMMA) where fluidic components are integrated. These components are electrovalves, dispensation pumps, expansion vessels, sensors and bottles configuration selector.
- Aspiration Pump: responsible for removing waste solution from probes washer and dilution cup.
- Electronics: provide interface between fluidic components (electrovalves, pumps, sensors) and the instrument control. Additionally, it provides interface with lockers and with bottle weighting system.
- Tubing: provide connection between fluidic components.
- Module support: integrates module components and gives integration with the rest of the instrument.

- Interconnection manifolds: small manifolds used to interconnect tubing in order to allow mechanical module union. There are 3 units, one to join tubing that comes from fluidics module to the tubing of X-axis module, one to join tubing that comes from X-axis module to the tubing of Y-axis module and one one to join tubing that comes from Y-axis module to the tubing of Z-axis module.
- Bottles and cards disposal container drawers: there is one drawer per each bottle and card disposal container, located in the lower part of the instrument.
- Dilutors: responsible for aspirating and dispensing samples, reagents, diluents and dilutions at the required accuracy.
- Correct Pipetting Detectors (CPD): sensors to check possible obstructions in the system.

Dilution cup (covered in dilution cup module) and probes washer (covered in PIP module) are fluidic components but are considered as independent module or included in other modules.

1.5.8 Dilution cup module (CUP)

The dilution cup has 2 independent and identical wells, so 2 independent dilutions can be prepared simultaneously or a single dilution can be processed in any of the 2 wells. To homogenize the dilution, a stirring system is provided.

Each well has 2 fittings for the tubes required to wash and drain. For each well, one fitting (located at the upper part of the well) allows filling the well with Solution A or B, while the other one (located at the bottom part of the well) allows emptying to the waste bottle.

1.5.9 Reader module (LEC)

Reading module for gel cards tests, including image capturing and processing in order to obtain the level of agglutination of every single microtube, as well as applying the necessary safeguards to assure proper results.

1.6 SOFTWARE STRUCTURE OVERVIEW

In this chapter is detailed an overview of the software structure of the *Erytra*: general overview, SSD structure of the computer, a description of the files structure related to the software and a register structure description.

1.6.1 General overview

Erytra is provided with an embedded computer that contains all the software applications installed. Installation processes are required in update operations or during some corrective maintenance procedures.

1.6.2 SSD structure

The instrument contains 1 SSD of 256 GB (SSD) divided in 7 volumes. The following table summarizes the main characteristics of each drive.

Drive	Name	Size	Contents
C:\	SYSTEM	69,4 GB	Windows 10
D:\	APPS	1 GB	Internal applications data
E:\	DATA	159,9 GB	Software of the instrument
R:\	DGRECOVERY	15 GB	OS Image Windows 10 v2.2.1 (only allowed in maintenance mode)
--	RECOVERY	499 MB	Internal Recovery feature
--	EFI PARTITION	100 MB	Boot loader feature
V:\	DGFILESMTGMT	7,1 GB	Default unit to save reports (Virtual unit)

Table 2. SSD structure

Operating system (O.S.) of the instrument is Windows 10 Enterprise 2019 LTSC version 2.1.1. To check Windows 10 version installed in the instrument, press the grid (■) button during system startup, log in as *adminUser* (case sensitive, *admin* password) and the Boot Application options will be displayed. Once in there, press the arrow in the right side of the screen to switch to page 2/2 of the menu, then select the *Information* icon among the Boot Application options.



Figure 2. Information option

After switching on the instrument, O.S. is loaded and login is automatically performed. Default username and password for *Erytra* computer are (small letters):

Username: application

Password: root

Administrator username and password:

Username: admin
Password: dgroot

When the instrument is switched on, *Erytra* application is automatically launched.

1.6.3 Files structure

Main application files (program files, data and logs) are found in *E:* drive.

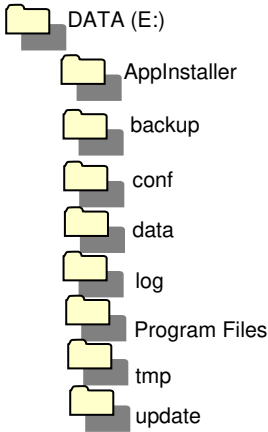


Figure 3. E drive directories

- **Folder *E:\backup*:** default folder where the backups are stored.



NOTE:

The backup destination folder can be changed by indicating the desired path in *Backup.conf* file in *E:\Program Files\System Tools*.

- **Folder *E:\Conf*:** contains *erytra.properties* file and logo picture (*logo.jpg*, up to 200x45 pixels), if any. It also contains *erytra.properties.backup* file, used by the system when the *erytra.properties* file doesn't exist or can't be read.
- **Folder *E:\data*:** contains:

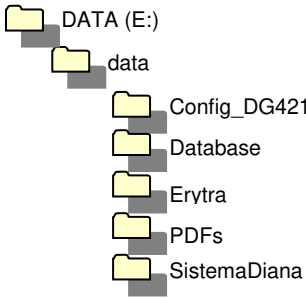


Figure 4. E:\data subdirectories

- E:\data\Config_DG421*: contains Reader module configuration files.
- E:\data\Database*: contains *Erytra* databases files and data (samples,...).
- E:\data\Erytra*: contains *Erytra* data (configuration files, pictures, logs,...) and *staterepository.xml* file.

-E:\data\PDFs: contains the PDF report files requested through the user application. The name of the files is *PAC00XY.pdf*, where *X* is the report type (1 for Traceability report, 2 for Reading report and 3 for Results List report) and *Y* is a 13-digit timestamp in milliseconds. PDF report files older than 3 hours will be deleted when a new report is requested.

-E:\data\SistemaDiana: contains DianaSystem files (Working directory, DianaServer).

- Folder E:\Log: contains,

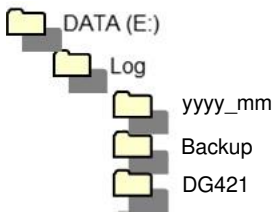


Figure 5. E:\Log subdirectories

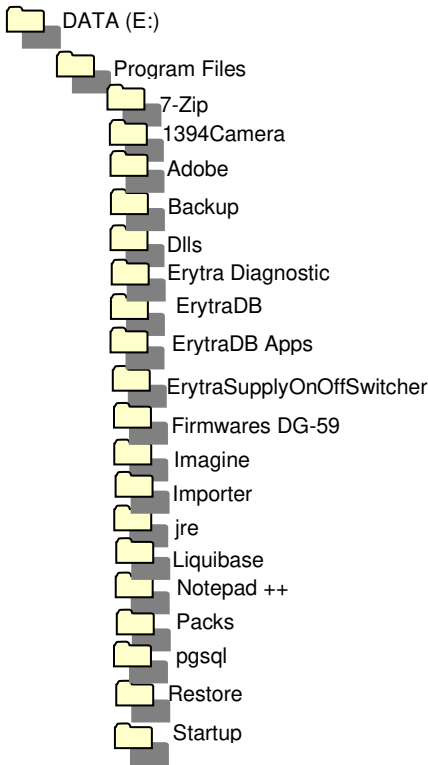
-E:\Log: contains log files of the instrument.

-E:\Log\yyyy_mm (where yyyy corresponds to the year and mm to the month when the folder was created): contains old web application server (Tomcat) console log files (stdout.log files).

-E:\Log\Backup: contains backup of log files.

-E:\Log\DG421: contains reader log files.

- Folder E:\Program Files: contains,



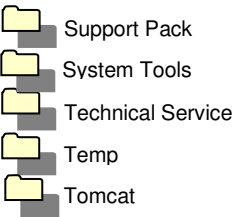


Figure 6. E:\Program Files subdirectories

- EnvironmentVariables.bat: add environment variables to the O.S.
 - E:\Program Files\Dlls: contains Erytra application libraries.
 - E:\Program Files\Erytra Diagnostic: contains Diagnostic application.
 - E:\Program Files\ErytraDB: contains Erytra (factory) databases backup files and a batch file that deletes current databases and restores these backups.
 - E:\Program Files\ErytraDB Apps: includes the Restore_DataBase tool.
 - E:\Program Files\Firmwares DG-59: contains all Erytra modules firmwares and related kernels.
 - E:\Program Files\Importer: contains Erytra database importer (new tests,...).
 - E:\Program Files\jre: contains java runtime environment files.
 - E:\Program Files\Liquibase: includes Liquibase application.
 - E:\Program Files\Packs: includes languagepack and reagentpack files.
 - E:\Program Files\pgsql: contains PostgreSQL files.
 - E:\Program Files\SistemaDianaRegistry: includes the necessary files to set Windows registry key configuration for Sistema Diana.
 - E:\Program Files\Startup: contains Erytra application start up.
 - E:\Program Files\System Tools: contains batch files to perform automatic backups.
 - E:\Program Files\Technical Service: contains Technical Service application files.
 - E:\Program Files\Temp: temporary folder.
 - E:\Program Files\tomcat: contains Web server (Tomcat), related application files and erytra elfexis® web applications.
- **Folder E:\update:** Any software installer copied to this folder will be automatically executed on reboot.
 - **Folder E:\data\Erytra:** contains,

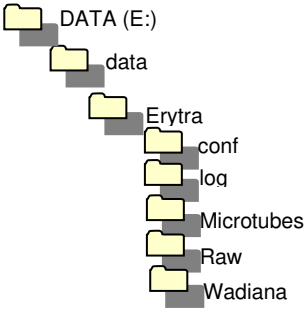


Figure 7. E:\data\Erytra subdirectories

- staterepository.xml* file: file where the disabled modules are stored.
- E:\data\Erytra\conf*: contains Instrument configuration files: geometry, adjustments (*PIPCalibrations.txt* and *CATCalibrations.txt* files),etc. and *testcounter.txt* file registering the number of executed (completed or cancelled) tests in the current database.
- E:\data\Erytra\log*: not in use (software versions ≥ 2.00).
- E:\data\Erytra\Microtubes*: contains (individual) microtubes jpg and bmp pictures.
Microtubes jpg files are organized into different folders as follows:
 - 1st folder level: card type, card barcode digits 1 to 4.
 - 2nd folder level: card lot, card barcode digits 9 to 11.
 - 3rd folder level: card lot, card barcode digits 12 to 13.
 - 4th folder level: last 2 digits of card serial number, card barcode digits 17 to 18.
- E:\data\Erytra\Raw*: contains reader raw images acquired last 14 days (2 wells per picture, pixels information as they were acquired by the sensor -8bits).
- E:\data\Erytra\Wadiana*: contains Diana System Connectivity definition and messages files.

Folder E:\data\SistemaDiana: contains,

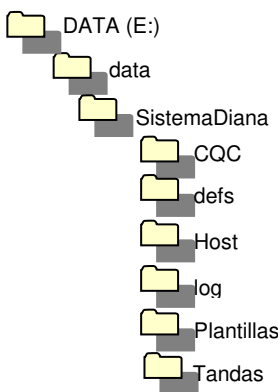


Figure 8. E:\data\SistemaDiana subdirectories

- E:\data\SistemaDiana\CQC*: contains Diana System quality control files.
- E:\data\SistemaDiana\defs*: contains Diana System definition files (cards, wells,...).
- E:\data\SistemaDiana\Host*: contains Diana System LIS database.
- E:\data\SistemaDiana\log*: contains Diana System programs log and logs backups.
- E:\data\SistemaDiana\Plantillas*: contains Diana System templates.
- E:\data\SistemaDiana\Tandas*: contains Diana System batches files (**.tan* and **.bmp*).

1.6.4 Windows registry structure

In case it is necessary to modify the Diana System working directory and/or the language of Diana System, run *E:\data\SistemaDiana\Config.exe* file as administrator (before running it, the rest of applications must be closed). In the main tab it is possible to define both parameters and the selection is stored in Windows registry:

- Working directory: *HKEY_LOCAL_MACHINE* → *SOFTWARE* → *WOW6432Node* → *Grifols* → *SistemaDiana* → *General* → *DirBase* parameter.
- Diana System language: *HKEY_LOCAL_MACHINE* → *SOFTWARE* → *WOW6432Node* → *Grifols* → *SistemaDiana* → *General* → *NumIdioma* parameter

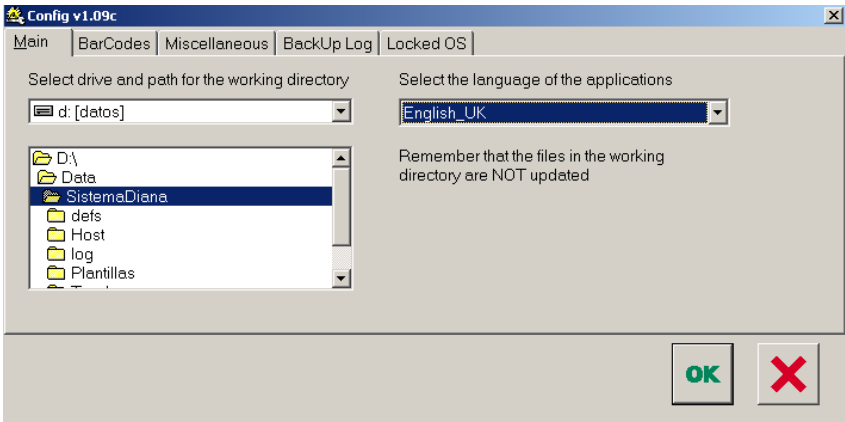


Figure 9. Config Diana System software

2 SETTING UP AND STARTING THE INSTRUMENT

2.1 LOCATION REMARKS

Before placing the instrument in the installation site it is necessary to pay attention to the following requirements:

- The installation process must be carried out by a **QUALIFIED TECHNICIAN**.
- Space requirements (with the transportation pallet):
 - Size: 1210mm wide x 2010mm height x 800mm deep
 - Weight: 500 Kg. (approximately).
- Place the instrument so that it is over a non inclined surface.
- Do not place the instrument out of doors.
- Do not place the instrument over a flammable material surface.
- Do not place the instrument so that it is difficult to access for maintenance or servicing.
- Do not block air openings nor place the instrument on a soft surface which might block them.
- Do not place the instrument where it could get wet.
- Do not position the instrument so that it is difficult to operate the mains switch.
- Do not let the instrument or its flexible cord comes into contact with surfaces which are too hot.
- Do not place anything on the top of the instrument.
- Do not locate the instrument in an area of direct sunlight.
- Locate the instrument near a fluid drain access.
- Do not place the instrument over any electric or electronic equipment since failures in the fluidic circuit may damage them.
- Appliance coupler or plug shall be readily and easy accessible.



CAUTION!

Disregard may lead to malfunction of and cause damage to the *Erytra*.

2.2 UNPACKING

The instrument is dispatched in a wooden box on a transportation pallet. Before unpacking the instrument, the pallet should be placed on the floor, as close as possible to the installation site.

To unpack the instrument it is necessary minimum 2 people (3 recommended), 1 of them must be a **QUALIFIED TECHNICIAN**. To unpack the instrument, remove the packing by following the instructions below:

Tools:

- Open end spanner number 12 (for arms shipping clamps) and number 19 (for feet).
- Electric screwdriver with Phillips screw head.

Procedure:

1. Place the packaging as close as possible to the installation site.

2. Remove upper part of wooden packaging.

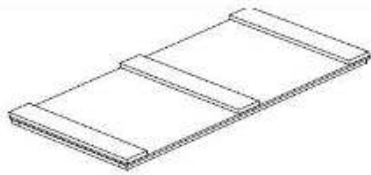


Figure 10. Upper part of the wooden packaging

3. Remove the planar lateral (narrow lateral without wooden strips) of the packaging and place it attached to the pallet in order to act as a ramp. Place upper part of the wooden packaging under the ramp.

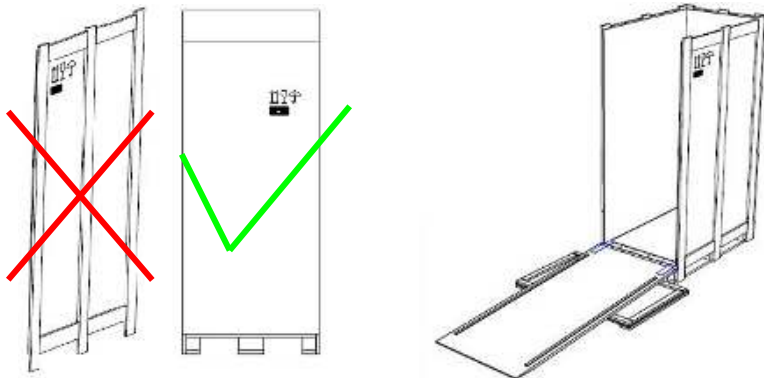


Figure 11. Packing assembly of the Erytra

4. Fix the ramp with the two hook-and-loop fasteners.



Figure 12. Hook-and-loop fasteners

5. Dismantle the rest of the laterals of the wooden box.
6. Remove the wooden small ramp included inside the packaging (above the instrument) and place it at the end of the main ramp. Stick it to the main ramp with adhesive tape.



Figure 13. Small ramp attached to the main ramp of the packaging

- 7. Remove door handles that are also located above the instrument.
- 8. Remove aluminium anti-humidity cover that is wrapped around the instrument.
- 9. Remove the retractile film and protective foams located at each corner of the *Erytra*.
- 10. Remove the adhesive tape of the feet.



Figure 14. Adhesive tape in a foot of the Erytra

- 11. Move up the instrument turning in counterclockwise direction the nut of each foot with an open end spanner number 19. Move down the feet in a crossed way.



Figure 15. Location of the nut in a foot of Erytra



Figure 16. Instructions to screw / unscrew the feet of the Erytra

- 12. Once the instrument is lifted over its feet, remove the foam and the wooden board located underneath the *Erytra*.



Figure 17. Foam and wooden protection of the packaging

- 13. Move down the *Erytra* by means of the nut of the feet (turning them in clockwise direction, use the open end spanner number 19) until wheels rest on the pallet.
- 14. Move carefully by the ramp the instrument to the floor, with minimum two people avoiding that the instrument moves by itself for the ramp.



DANGER!

Weight of the instrument is quite considerable and it is recommended to strictly follow instructions depicted in this chapter.

- 15. Place the instrument in the installation site.
- 16. Remove frontal foams located in the user tray and in the touchscreen monitor.
- 17. Remove protective adhesive tape of the frontal USB connectors.
- 18. Install the door handles. The screws for install them are located in a bag attached to the door handles themselves. To install the door handle of the intermediate door (remove its fixing screw), the support of the celduc must be removed. Once the door handle is installed, place again the support of the celduc attached to the door and the door fixing screw as well.



Figure 18. Door handles installed on the doors

- 19. Open all the fluidic drawers (push the lever located in the front of the drawer, at the bottom area) and remove the foams located on both sides of the card disposal container and in the handles of the bottles. Remove the contents of the card disposal container: antihumidity bags, power cords, decontamination tool, disposable bags for card waste and documentation. Remove the protective transparent adhesive of identification labels in fluidic drawers surface and solution A/B and waste bottles (lids and bottles). Close the drawers.
- 20. Open upper door and remove the box plug between the RM comb and SM comb and the X-axis support arm.



Figure 19. Location of the box plug

21. Remove the SM comb between the RM comb and the supplement comb lateral.

22. Remove external barcode reader box and then the RM comb.



Figure 20. Location of the RM comb

23. Remove supplement comb lateral.



Figure 21. Location of the supplement comb lateral

24. Remove the shipping clamp of PIP arm (use an open end spanner number 12) which prevents the arm to move.



Figure 22. Location of the shipping clamp of the PIP arm

25. Remove adhesive tape of the dilution cup wells and remove its fixing foams.

26. Remove the decontamination bottle, 2 samples racks, 2 diluents racks and 2 reagents racks of the internal packaging tray. Remove protective transparent adhesive of decontamination bottle identification labels (lid and bottle).
27. Remove the internal packaging tray and the plug of the tray.

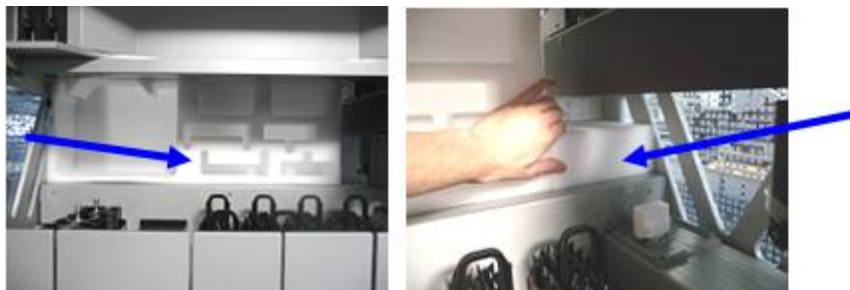


Figure 23. Location of the internal packaging tray (left) and plug of the tray (right)

28. Open user tray and remove the screw of the intermediate door. Open the intermediate door and remove the shipping clamp of the CAT arm (use an open end spanner number 12) which prevents the arm to move. Check also inside of the instrument for any foreign objects. Close the intermediate door and put its fixing screw.



Figure 24. Location of the shipping clamp of the CAT arm

29. Remove protective transparent adhesive from surface of reagents-samples keyboard and card keyboard.
30. Move down the frontal feet of the instrument until they rest on the ground.

2.3 INSTRUMENT LEVELLING

Instrument levelling is recommended to be performed once the instrument has been set in its corresponding position in the laboratory: during its initial installation or when an instrument in use has been moved from its physical location due to a technical intervention.

Instrument levelling can be done by means of the front feet of the instrument:

1. Switch on the instrument and access the Bootapplication menu.
2. Make sure the rear feet of the instrument are resting on the ground.
3. Open Technical Service software and perform a home of the PIP and CAT arms.
4. Go to *Procedures\Checks\PIP* section and perform 7.3.11 *Presence detector alignment* check routine using P/N 234213 Probes adjustment tool (Ver. 4).
5. Activate the PIP presence detector (*PIP Arm* tab → *Presence Detector* subtab, then select *Set on*).
6. Check that the laser is over the hole of the tool.



Figure 25. Laser PIP alignment

- 7. Perform 8.3.7 Card Presence detector alignment – gripper parallelism check routine using P/N 233878 Cards drawer adjustment tool (Ver.2).
- 8. Activate the CAT presence detector (Gripper Arm tab → Presence Detector subtab, then select Set on).
- 9. Check that the laser is over the central slot of the tool.



Figure 26. Laser CAT alignment

Should the check for the laser position for the PIP or CAT modules FAIL, then it is needed to level the instrument:

- 10. With both lasers on over the corresponding adjustment tools, access the upper nut of the left or right front instrument feet.
- 11. Loosen the upper nut by turning it upwards (counter clockwise).



Figure 27. Upper nut of the foot

- 12. Access the lower nut of the foot, turning it clockwise or counter clockwise (depending on the correction needed for each instrument) until the laser beams fit in the corresponding adjustment tools for each module.

- Turning clockwise left front foot: Lasers move to the left.
- Turning clockwise right front foot: Lasers move to the right.



Figure 28. Lower nut of the foot

- 13. Switch off the PIP and CAT presence detector (Set off button).
- 14. Remove the adjustment tools from the instrument.
- 15. Tighten back the upper nut.

2.4 INSTALLATION QUALIFICATION (IQ)

During the Installation Qualification (IQ) phase, all of the activities associated with properly installing the instrument at the users' site are documented. The IQ procedure should be executed during the installation of the instrument at customer site. For further details, consult *Erytra* IQ REGD-0009727 current version document.

Once IQ procedure has successfully finished, Operational Qualification (OQ) procedure should be performed.

2.5 COMPLETING THE SOFTWARE INSTALLATION

Software versions $\geq 4.1.0$ are pre-installed in *Erytra* instruments from factory. Once the instrument has been unpacked and placed in the corresponding position in the lab, the software installation should be finished. In order to do it, the software installer file will be automatically launched the first time that the instrument is switched on (*erytraW10-INSTALLER-x.x.x.exe*, which should be installed following the installation procedure described in section 15.3).

2.6 BOOT APPLICATION

When switching on the instrument, the boot application is launched, showing the Erytra logo and bottom corner buttons on screen.



Figure 29. Boot Application initial window

From there, the protected/administrative options can be accessed by pressing the bottom-left () 'grid' button and entering the corresponding credentials. Additionally, the Diagnostic menu can be accessed by pressing the bottom-right () 'diagnostic' button. The Diagnostic Menu includes options which are mainly focused on the end user.



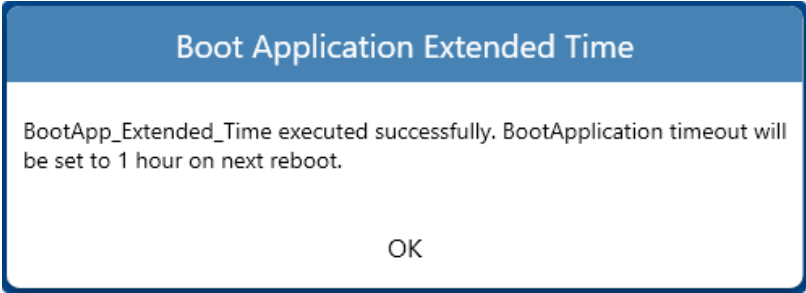
NOTE:

There is a limited time of 5 seconds to select the desired menu before the application is launched. At this point, the grid and diagnostic buttons will not be available and the user application login screen will be shown.



NOTE:

A file named `BootApp_Extended_time.vsb` (E:\Program Files\System Tools) can be executed to increase the timeout to enter Boot application to 1 hour only on next reboot.



By preseing the grid () button, the Boot Application login window will be displayed.



Figure 30. Boot Application login window

Login should be done with the following credentials (case sensitive):

- **User:** *adminUser*
- **Password:** *admin*



NOTE:

Only Grifols authorized service personnel (FSE or TAS users) is intended to enter the protected/administrative options within the Boot Application. Thus, the above login credentials cannot be changed and should not be provided to users other than Grifols authorized service personnel.

After logging in, the Boot Application options will be displayed in 2 screens.

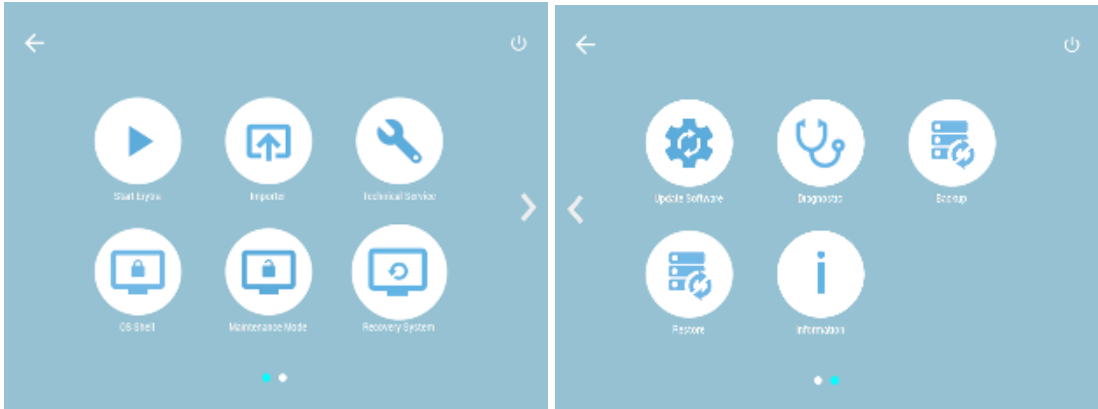


Figure 31. Boot Application protected/administrative options

The available options are listed below:

- **Start Erytra:** this button will start running the Erytra Software program.
- **Importer:** this button will open the Importer application.
- **Technical Service:** this button will open the Technical Service application. Further details on the application can be seen in section 3.
- **OS Shell:** this button will open OS Shell mode.
- **Maintenance Mode:** this button will open Maintenance mode.
- **Update Software:** this button will allow updating the SW version in the instrument. The system will automatically check if there is a removable device connected with a folder named *installers* inside. If so, the system will execute the software installer within that folder. Further details on the procedure can be seen in section 15.3.
- **Recovery system:** this button allows to perform Operating System Full Recovery or OS Recovery from instrument SSD and Boot Application; without the need of connecting a USB pen drive. This new feature allows to update or recover the OS remotely. Further details on the procedure can be seen in section

- **Diagnostic:** this button will open Diagnostic application. Further details on the application can be seen in section 13.
- **Backup:** this option will start the Backup application. Further details on the application can be seen in section 15.7.
- **Restore:** this option will start the Restore application. Further details on the application can be seen in section 15.8.
- **Information:** this functionality will show the information of the current OS and SW version.

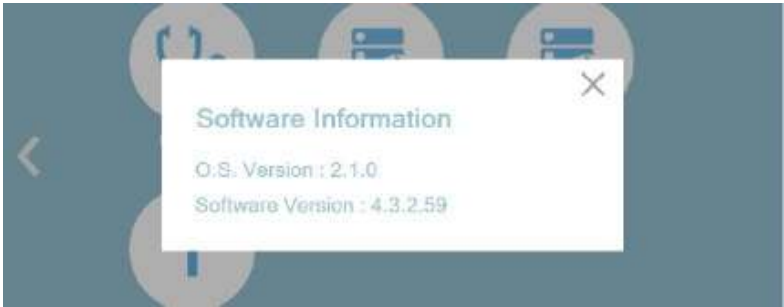


Figure 32. Information window

- **Backwards:** arrow located on the top left corner of the screen. When pressed, it goes back to the Boot Application initial window.
- **Shutdown:** when pressing the button located in the top right corner of the screen, a confirmation message appears informing that the system will be shut down. Press Yes to confirm the action.

2.7 SOFTWARE CONFIGURATIONS

Before performing OQ procedure, it is recommended to define some parameters of the software like language, *Erytra* serial number, samples drawers configuration, LIS connection configuration, etc. These configurations are defined in *erytra.properties* file located in *E:\conf*.

NOTE:

When modifying the default configuration, check that spelling of the written parameters is correct. Otherwise the default value for the wrongly spelt parameter will be used without being changed in *erytra.properties* file.

NOTE:

If a properties parameter is deleted while modifying the default configuration, the default value for that parameter will be used and written in the file.

NOTE:

Before modifying any parameter, check that software is not running and the MSDOS command windows are closed.

2.7.1 SERVICE_RACK_MODE

Defines the number of racks used as service rack. It is possible to work with the following configurations:

- **NORMAL:** there are 2 service racks in position rack 4 of the card drawer 3 and rack 4 of the card drawer 4. This is the default option.
- **HALF:** there are 8 service racks placed in position racks 1,2,3 and 4 of card drawers 3 and 4.
- **NONE:** there is no service rack is defined, all cards will be discarded directly in the waste cards container.

2.7.2 LANGUAGE

Parameter to define User Software language.

LANGUAGE=xx_XX

where **xx_XX** is the defined language. Available languages are:

ca_ES: catalan language
cs_CZ: czech language
da_DK: danish language
de_DE: german language
el_GR: greek language
en_GB: english (UK) language (default in CE configuration)
en_US: English (US) language (default in US configuration)
es_ES: spanish language
fi_FI: finnish language
fr_FR: french language
it_IT: italian language
ja_JP: japanese language
nb_NO: norwegian language
pl_PL: polish language
pt_BR: portuguese (Brasil) language
pt_PT: portuguese (Portugal) language
ru_RU: russian language
sk_SK: slovak language
sl_SI: slovenian language
sv_SE: swedish language
zh_SG: chinese language

This parameter can be defined also by means of Importer application.

2.7.3 INSTRUMENT_NAME

Parameter to define instrument name (maximum 10 characters). This parameter can be defined also by means of Importer application.

2.7.4 INSTRUMENT_SERIAL

This parameter defines the serial number of the instrument (maximum 7 characters). By default it is set in a *000xxxx* format.

This parameter can also be defined by means of Technical Service Software (*Run* menu → *Edit serial number* option).

2.7.5 WELL_NAME_FORMAT

Defines the standard used to designate well names.

- **TRADITIONAL**: it uses the traditional nomenclature. This is the default option.
- **ISBT**: it uses the ISBT nomenclature

2.7.6 INTERPRETATION_FORMAT

Defines the standard used to designate interpretation results.

- **TRADITIONAL**: it uses the traditional nomenclature. This is the default option.
- **ISBT**: it uses the ISBT nomenclature
- **TRADITIONAL_EXTENDED**: it uses an extended traditional nomenclature

2.7.7 LIS_ACTIVATED

Defines whether the system connects to LIS or not.

- **TRUE**: the LIS connection is activated. If the system is connected to LIS, the samples shall be exported to LIS.
- **FALSE**: the LIS connection is not activated. This is the default option.

2.7.8 LIS_SHARED_MODE

Defines the behaviour of the instrument when receiving a request for a sample that is not present.

- **FALSE**: sample requests with sample missing will be hold waiting for sample (default value).
- **TRUE**: sample request with missing sample will be sent back to LIS as cancelled. Recommended option when there are several instruments connected together to the LIS.

When several instruments share the same LIS connection, *LIS_SHARED_MODE* parameter must be set to *TRUE*. Otherwise, all the workload would be sent to only one Erytra and not received by the others.

In case there is only one Erytra connected to LIS, it is recommended to set *LIS_SHARED_MODE* parameter to *FALSE*.

2.7.9 LIS_AUTO_QUERY_BY_SAMPLE

Defines if a workload query to LIS is executed after samples drawer recognition. Available options are:

- **true**: Once a sample drawer is identified or a sample tube is manually identified, automatically request to LIS all pending work for every identified sample inside the instrument.
- **false**: No automatic request to LIS will be performed after a sample drawer is

- identified or a sample tube is manually identified (default value).

2.7.10 RECEIVE_AND_SHOW_DEMOGRAPHIC_DATA

Allows defining if sample demographic data sent from LIS is to be stored in the system and shown to the operator. The available options are:

- **TRUE:** to receive and show the demographic data.
- **FALSE:** to not show the demographic data (default option).

This parameter is only available in CE configuration. In US configuration, parameter is not shown in the properties file and it is considered as false.

2.7.11 DIANA_SYSTEM_BATCH_CREATION

Defines whether the system creates Diana System batches or not. It can be configured as:

- **TRUE:** the tan (batch) file creation is activated
- **FALSE:** the tan (batch) file creation is not activated. This is the default option.

2.7.12 SAMPLE_RACKS_CONFIGURATION

In order to define the tubes contained in a sample rack, different configurations are used depending on barcode labels attached to the sample racks.

It is possible to configure 7 different barcode labels. Configuration of each barcode label has 4 parameters:

BCv=w, x, y, z where,

- **v:** number from 1 to 7
- **w:** identifies the sample tube type, related to level detection sensitivity. Available options are:
 - o NORMAL
 - o PLUNGER
 - o PEDRIATIC
- **x:** sample holder barcode.
- **y:** sample tube external diameter (limited from 9 to 16 mm). It may be used to ignore the tube diameter readings given by the presence detector and use a configured value instead. Available options are:
 - o If set to 0, the system considers tube diameters given by the presence detector.
 - o Otherwise, all of the tube diameters for the holder will be set to the configured value (in mm).
- **z:** sample tube bottom height (limited from -2 to 100 mm). It indicates the height offset in mm (number without decimals) added to the position set by the probe adjustment.



NOTE:

A negative height offset means that the probe will be moved down further than the position set by the adjustment. A positive height offset means that the probe movement will be stopped before reaching the position set by the adjustment.

Also there is another parameter to define the default rack configuration (grey label):

DEFAULT_RACK_CONFIGURATION= w, x, y, z where,

- **w:** identifies the sample tube type, related to level detection sensitivity. Available options are:
 - o NORMAL
 - o PLUNGER
 - o PEDRIATIC
- **x:** sample holder barcode. Default value is GY.
- **y:** sample tube external diameter (limited from 9 to 16 mm). It may be used to ignore the tube diameter readings given by the presence detector and use a configured value instead. Available options are:
 - o If set to 0, the system considers tube diameters given by the presence detector.
 - o Otherwise, all of the tube diameters for the holder will be set to the configured value (in mm).
- **z:** sample tube bottom height (limited from -2 to 100 mm). It indicates the height offset in mm (number without decimals) added to the position set by the probe adjustment.

**NOTE:**

A negative height offset means that the probe will be moved down further than the position set by the adjustment. A positive height offset means that the probe movement will be stopped before reaching the position set by the adjustment.

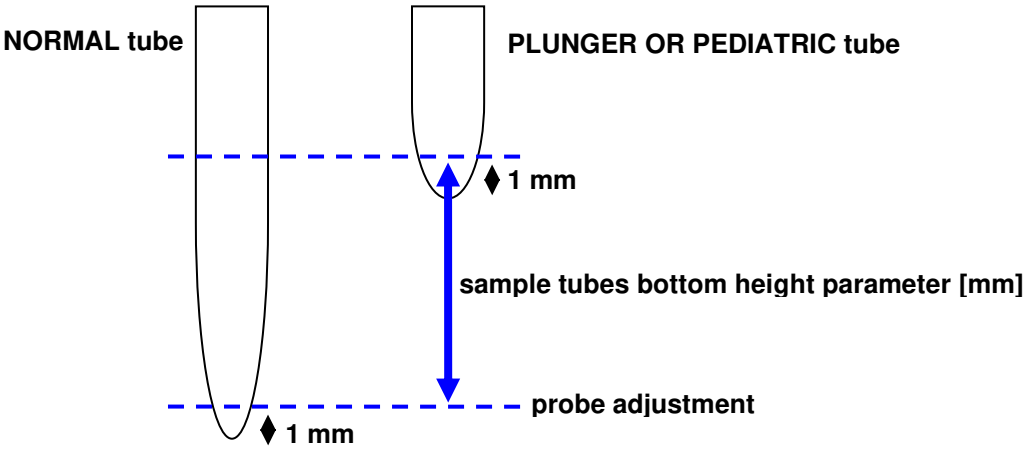


Figure 33. Samples drawers configuration

DEFAULT_RACK_CONFIGURATION will be applied to those holders with the barcode configured in this parameter (by default GY, matching grey label provided with the instrument). If rack barcodes configuration has been set to *not required*, (see *Barcodes on sample holders* section below) DEFAULT_RACK_CONFIGURATION settings will be applied also to those holders without barcode or with a barcode not successfully decoded.

2.7.13 RESULT_CONFIGURATION

Parameter to define the microtube's results that shall be checked by the operator before validation. That is, the cards will be left at the service rack so the operator can visualt check the results. Multiple options shall be selected by entering them separated by commas. Available options are:

- **SRW:** Weak well results (+/-). Only for CE.
- **SRDL:** Doubtful results (?).
- **SRDP:** Double population (DP) results (mix field).
- **SROD:** Operator defined special results. This special results are defined through the importer application.
- **RI:** Results with reading incidences.
- **PI:** Results with process incidences.
- **CSI:** Results with sample or card incidences.
- **SRI:** Interpretation special results (NI or DISCREPANT or AUTOCONTROL NOT NEG or DAT NOT NEG results).
- **All:** All processed cards.

By default, all the partial cards are loaded on the Service Rack. The default option is:

- CE: SRW, SRDP and SRDL.
- US: SRDP and SRDL.

2.7.14 DOUBLE_TUBE_ENABLED

Parameter to define if double sample tubes (1 with serum and 1 with RBC's) are used.

- Options: **true/false**.
- Default option: **false**.

2.7.15 QC_ENABLED

Parameter to define if Quality Controls (QCs) are used on the system.

- Options: **true/false**.
- Default option: **true**.

2.7.16 QC_HOURS_CADENCE

Parameter to define the periodicity in hours for the QC.

- Options: any integer value.
- Default option: 8.

2.7.17 QC_NEW_LOT_WARNING_ENABLE

Parameter to define if the system shows a warning when a reagent or card newly detected lot has not passed any QC protocol.

- Options: **true/false**.
- Default option: **false**.

2.7.18 PASSWORD_REAUTHENTICATION

Parameter to allow enabling or disabling the authentication request (user/password) for some of the actions performed by the operator.

- Options: **true/false**.
- Default option: **true**.

2.7.19 DISABLE_VALIDATION_BUTTON

Parameter to define if the Validation button in the Validation list is disabled.

- Options: **true/false**.
- Default option: **false**.

2.7.20 DISABLE_EXPORT_BUTTON

Parameter to define if the Export button in the Validation list is disabled.

- Options: **true/false**.
- Default option: **false**.

2.7.21 DISABLE_CHANGE_RESULTS_CONFIRMATION

Parameter to define if the confirmation message when the user modifies results is disabled.

- Options: **true/false**.
- Default option: **false**.

2.7.22 DUAL_MODE

Parameter to configure the usage of external drain. When activated, waste 1 bottle position is used as solution A bottle and waste 2 bottle position is used as solution B bottle. An external drain must be connected to the instrument when activating this parameter and also a configuration must be applied to the manifold (check chapter 9).

where available options are:

- **true**: external drain is activated.
- **false**: external drain is deactivated (default value).

2.7.23 PASSWORD_EXPIRATION_DAYS

Parameter to define after how many days the user passwords will expire. Default is 90 days.

2.7.24 DELTA_CHECK_INCONSISTENCIES_CONFIGURATION

Parameter to define the inconsistencies on results to be checked. Available options are:

- **None**: no Inconsistencies are checked.
- **Sample**: Inconsistencies are checked on Sample tube BC basis.
- **Patient**: Inconsistencies are checked on Patient ID basis.

2.7.25 DELTA_CHECK_HISTORICAL_FROM

Parameter to define the time interval where Inconsistencies are checked within exported SampleProfiles.

Available options are:

- **ALL**: Inconsistencies are checked within all data stored in database.
- **yyyy/mm/dd**: Inconsistencies are checked within all SampleProfiles processed from defined date onwards.
- **any number**: Inconsistencies are checked within all SampleProfiles processed within the defined number of days prior the current date. Default option 0.

2.7.26 ENABLE_AUTOVERIFICATION_QCPASS

Parameter to define if a QC Pass protocol is automatically validated and exported.

- Options: **NONE, ONLY VALIDATE, VALIDATE AND EXPORT.**
- Default option: **ONLY VALIDATE.**

2.7.27 SHOW_CONFIRMATION_WHEN_VALIDATE_MODIFIED_SAM PLEPROFILES

Parameter to define if the system shows a confirmation window (not requiring the username and password of the user) whenever the user validates any modified result.

- Options: **true/false.**
- Default option: **true.**

2.7.28 ARCHIVE_ENABLED

Parameter to activate the archive option. This option moves the contents of *E:\data\Erytra\Microtubes* and *E:\Log* folders to the defined folder in *ARCHIVE_PATH* parameter.

- Options: **true/false.**
- Default option: **false.**

2.7.29 ARCHIVE_PATH

Parameter to define the path where the above mentioned files will be moved.

- Options: Any valid path. Any '\' symbol must be written as '\\' (for instance, E:\\ or C:\\Program Files\\...). There is no limitation regarding the number of introduced characters. Note that '\' symbol would be added automatically in the defined path by the software before the : symbol when the software is launched.
- Default option: **NONE.**



NOTE:

Recommended path destination format is NTFS.

2.7.30 ARCHIVE_INTERVAL

Parameter to define the number of days that files to be archived remain in the corresponding folder of the instrument (recommended value ≤ 14). After the number of days defined (checked by the software every hour), the files are moved to the folder defined in *archive_path* parameter. If the parameter is defined as 0, the files are continuously moved to the corresponding folder. Negative values are not allowed.

- Options: Any positive integer number or **NONE**.
- Default option: **NONE**.

2.7.31 RACK_BC_ENABLED

Parameter to configure sample holders barcode requirement. Available options are:

- **true**: All sample holders must have a barcode meeting one of those defined in *erytra.properties* in order to be able to use the samples in that holder (default value, recommended).
- **false**: Barcode is not required for any holder. If a holder barcode is found, the corresponding tube settings will be applied. If no holder barcode is found, DEFAULT_RACK_CONFIGURATION will be applied.

2.7.32 IMPACT_DETECTOR

Parameter to define if impact detector is installed in the system.

- Options: **true/false**.
- Default option: **true**.

2.7.33 RESERVE_INCUBATOR_FOR_STAT

Parameter to configure whether there is one incubator reserved for STAT or not.

- Options: **true/false**.
- Default option: **true**.

2.7.34 DIRECT_PRINT



NOTE:

Direct print functionality requires the following specifications to be met by the printer device being used:

- At least 128MB memory
- Compatibility with PCL5, PCL6 or PostScript languages

Available options are:

- **true**: any report requested by the operator will be sent directly to the printer without being displayed on screen.

- **false:** any report requested by the operator will be displayed on screen in PDF format (default value).

2.7.35 AUTOMATIC_RESULTS_VERIFICATION

Parameter to define if SampleProfiles without incidences or special results, and with QC status exported or QC disabled, are automatically validated and exported.

- Options: **true/false**.
- Default option: **false**.

2.7.36 FLUIDIC_SYSTEM_ALWAYS_READY

This parameter is used to select the fluidic procedure to be performed after 1 hour of fluidic inactivity. At that moment, when the parameter is set to *true*, the system will perform a quick internal and external wash of the probes and dilution cup, repeating these actions every hour for 24 consecutive hours as long as the fluidic inactivity condition is met. Otherwise (when set to *false*) a full rinsing procedure will be performed and no further fluidic procedures will take place afterwards until required by the operator. This rinsing procedure will also be executed when the parameter is set to true and the system has performed 24 consecutive inactivity washes.

Available options are:

- **true:** a quick washing procedure will be performed after each hour of fluidic inactivity for 24 consecutive hours (default value).
- **false:** a rinsing procedure will be performed after 1 hour of fluidic inactivity.

When this parameter is set to *true* and the last performed fluidic action has been an inactivity wash, the system will empty and then prime both probe circuits before executing an action required by the operator which involves any fluidic procedure.

2.7.37 ISBT_ENABLED

Parameter to define if ISBT 128 DIN barcodes are detected and interpreted by the system.

- Options: **true/false**.
- Default option: **false**.

2.7.38 DEFAULT_BC_IS_ISBT

Available options are:

- **true:** any manually introduced barcode is considered as ISBT type (16 characters expected).
- **false:** any manually introduced barcode is considered as non-ISBT type (default value).

When manually identifying a sample tube barcode, the manual identification window prompted by the software includes a checkbox allowing to indicate whether the barcode is ISBT type or not (ticked by default when the DEFAULT_BC_IS_ISBT parameter is enabled and vice versa).

2.7.39 ISBT_LIS_DIGITS

Parameter to define how ISBT BCs are shown in Worksheet, IC-Samples, Validation list, Validation detail, Database list and Database detail screens and in Reports, and, if LIS is enabled, received from LIS or sent to LIS. Options are:

- **DIN:** system communication to/from LIS consists of 13-character ISBT codes.
- **FULL:** system communication to/from LIS consists of 16-character ISBT codes (default value, includes all the available characters).

2.7.40 SEND_PENDING_TO_VALIDATE

Parameter to inform LIS that the execution of the requested test has been completed. The status P (Preliminary results) in Results Status field will be transmitted to LIS when this feature is enabled.

This parameter will be automatically added and set to disabled in erytra.properties when any application starts (user application, Technical Service,...), if it is not found (no manual action is required to add it).

- Options: **true/false**.
- Default option: **false**.

2.7.41 AUTO_LIS_QUERY_ENABLED

It allows enabling and disabling the periodic automatic query to LIS.

Values allowed:

- true: Auto lis query feature is enabled.
- false: Auto lis query feature is disabled. Default value.

2.7.42 AUTO_LIS_QUERY_MODE

Parameter to set the query mode to Query All or to Query by Sample (when AUTO_LIS_QUERY_ENABLED=true).

Values allowed:

- QALL: Query All will be sent to LIS (when AUTO_LIS_QUERY_ENABLED=true).
- QBSAMPLE: Query by Sample will be sent to LIS. (when AUTO_LIS_QUERY_ENABLED=true). Default value

2.7.43 AUTO_LIS_PERIOD_UPDATE

Parameter to define the amount of time in minutes that will elapse between each periodic automatic query (when AUTO_LIS_QUERY_ENABLED=true).

Values allowed: integers from 1 to 60.

Default value: 5.

Default value will be applied if a value out of the allowed range is found.

2.7.44 NO_SAMPLE_ID_IF_NO_USER_LOGGED

The values allowed are:

- **true:** Sample drawers will not be opened when there is no user logged in.
If a sample drawer that was already opened is closed when there is no user logged in, the samples in that drawer will not be identified.
- **false:** Sample drawers can be opened when there is no user logged in. The samples in the sample drawers closed when there is no user logged in will be identified and available in the system. Default value.

2.7.45 Barcode reader configuration

PIP barcode readers should be configured/checked to match the sample tube barcode characteristics used in each site (barcode type, checksum, etc.). While most of the barcode types currently used to label sample tubes are enabled by default, the default configuration can be modified by following the instructions detailed in section 7.2.2.9.

2.8 OPERATIONAL QUALIFICATION (OQ)

Operational Qualification (OQ) testing is performed to verify that the instrument or instrument modules operate as intended. OQ procedure should be performed in the following cases:

- After IQ procedure.
- After PM procedure.
- After a corrective maintenance, executing the part related to the module involved in the corrective action and the functional checking.

For further details, consult *Erytra* OQ REGD-0009728 current version document.

First of all, ensure that the mains switch (located at the back of the instrument) is turned off. Then plug the mains cable, supplied with the instrument, to *Erytra* first and to the mains later.



DANGER!

Electromagnetic compatibility and electrical safety tests have been performed using the cord supplied by the manufacturer. Using cables different to the one supplied may degrade performances of the instrument concerning electromagnetic compatibility and safety.



NOTE:

If the instrument has been stored under different environmental conditions than operation conditions, wait for at least 1 hour under operation conditions before turning the instrument on.

Before turning on the instrument, check that at least one empty waste bottle is connected and both wash solution bottles are filled with expected solutions and connected. Check also that card waste container is correctly placed.

Then, turn the mains switch on and push the button located in the front of the instrument. The LED of this button changes to green colour and after that, O.S. is loaded. Finally run Technical Service software.

2.9 DATE AND TIME SETTING

Follow the next steps in order to set the date and time shown by the O.S:

- 1. Switch on the instrument and press the grid () button during system startup (login credentials will be required), then select the *Maintenance Mode* icon among the Boot Application options.
- 2. Once PC reboots and desktop is available, go to *Start - Windows System - Control Panel*.
- 3. Set the appropriate time zone in *Clock and Region - Date and Time - Change Time zone...* option. If available for the selected time zone, the *Automatically adjust clock for daylight saving changes* option is enabled by default.
- 4. If displayed and enabled, disable the *Notify me when the clock changes* option in *Date and Time* window.
- 5. Set the appropriate date and time in *Clock and Region - Date and Time - Change Date and Time...* option (password for admin user will be required).
- 6. Switch off the instrument.

2.10 COMMUNICATIONS

1 RS232 port, 1 ETHERNET/LAN connection and 6 USB connections are available.



NOTE:

Use only shielded cables to ensure the proper performance of the equipment.

2.10.1 LIS connection

Communication of *Erytra* with LIS is performed using *DianaServer* software. Please refer to section 15.19 in this Service Manual for further details.
Diana System working directory must be checked to confirm that it is properly configured (default value is *E:\data\SistemaDiana*, run *E:\data\SistemaDiana\Config.exe*, main tab).

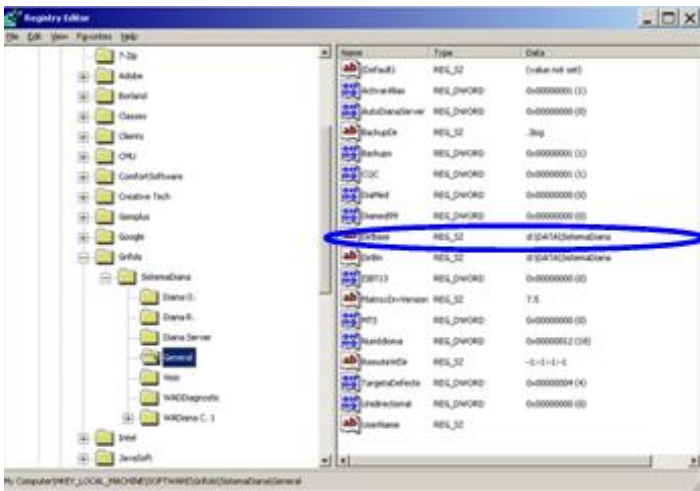


Figure 34. DirBase parameter configuration

Default Diana System template used to transmit results to LIS is *erytra.pln* (*E:\data\SistemaDiana\Plantillas*). This template is provided empty and must be configured. To edit the template, *Diana O.* software (*Listas.exe*) located in *E:\data\SistemaDiana* can be used.

Template name can be configured by means of Windows registry parameter *PlantillaDefecte* found in *HKEY_LOCAL_MACHINE → SOFTWARE → WOW6432Node → Grifols → SistemaDiana → Diana O.*

Importer application (*Profiles* tab → *Edit Profile .pln* name option) also allows the association of a Diana System template to each Technique (to be used in LIS communications in uni-directional transmissions and when LIS has not requested any template).

2.10.2 Remote connection

Remote connection allows enabled users to access the instruments from any computer in the network. This functionality is used on external computers and therefore, its configuration may be different from that of the instrument. From January 2021 onwards, depending on the computer setup, Flash Player or Internet Explorer browser could be upgraded to a version not compatible with remote connection, which would result in an availability problem. This potential unavailability will cause that once the user launches the application, the login window will not be available and consequently, it will not be possible to connect the instrument remotely.

Remoteconnection.zip is a tool that can be installed in those computers in which this functionality is not available due to the browser's version or configuration. It contains controlled Chrome browser and Flash Player plugin with its updates disabled to guarantee the availability of remote connection. In addition, a whitelisting procedure has been applied, which will not allow the navigation to other URLs different than the URL of user application.

INSTALLATION PROCEDURE

In order to install the alternative browser for remote connection, please follow the steps listed below:

- 1. Copy RemoteConnection.zip in any folder of the computer where the remote connection is required to be installed.
- 2. Unzip RemoteConnection.zip.
- 3. Run *..\RemoteConnection\Erytra\Installer_RemoteConnection.bat* with admin privileges.
- 4. Once the installation has finished, press Enter to end the installation process.

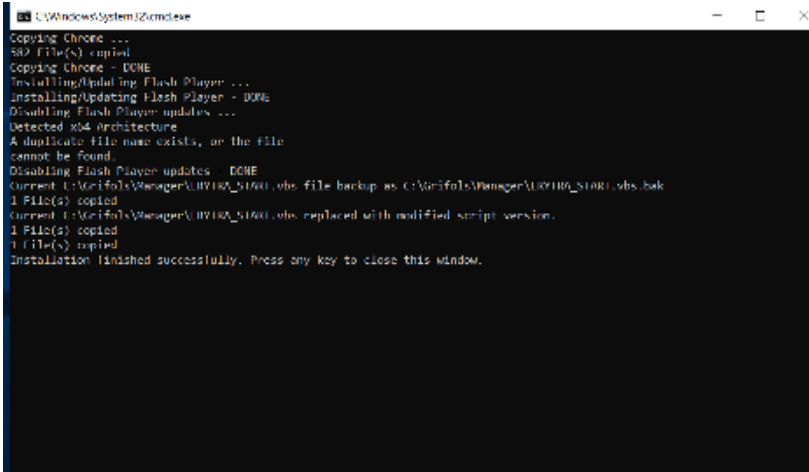


Figure 35 Remote connection installation

5. Check that ERYTRA_REMOTE.bat has been copied into C:\Grifols\Manager path.
6. Check that erytra_remote.txt has been copied into C:\Grifols\Manager path.

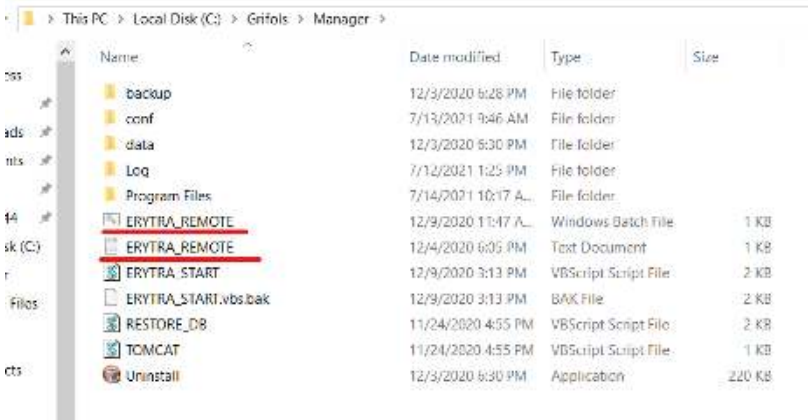


Figure 36 Remote connection final checkings

7. Open C:\Grifols\Manager\erytra_remote.txt and edit the IP address of the instrument (same IP address as the one that was previously used when connecting using the old browser).

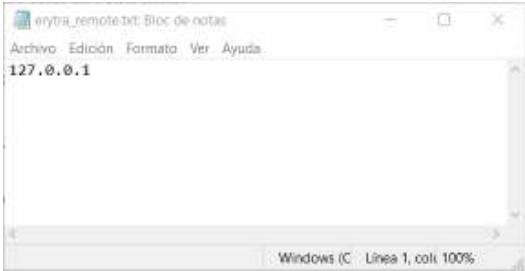


Figure 37 IP edition for remote connection

The remote connection functionality using the alternative browser can be started by running C:\Grifols\Manager\ERYTRA_REMOTE.bat and logging in the application with the corresponding credentials.

For the first time that the remote connection is started, the following steps should be performed:

1. Click on “Advanced settings” and click on “Proceed to URL (unsafe)”.

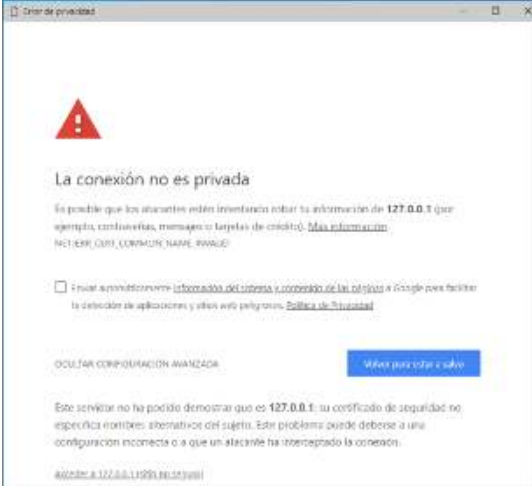


Figure 38 Starting remote connection step 1

2. Click on “Get Flash” and then click on “Allow” to allow Flash to run on this site.



Figure 39 Staring remote connection step 2

3. Login to the application as usual.

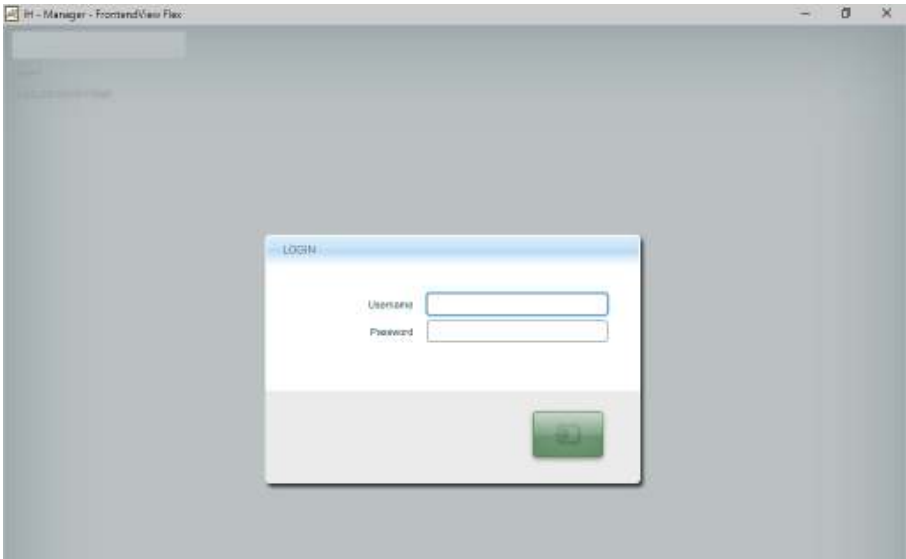


Figure 40 Login into application

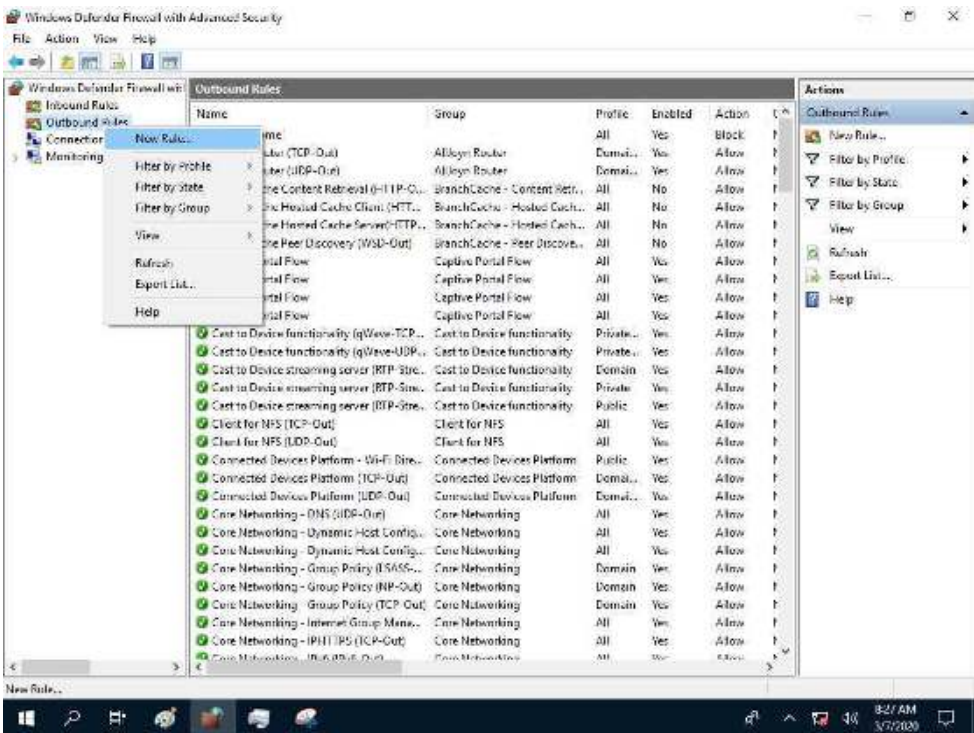
WINDOWS FIREWALL CONFIGURATION

Once the software has been installed, it is recommended to block outbound connections of the alternative browser by means of Windows firewall configuration in the remote computer. With this, access from the alternative browser to IP addresses different than the one configured for the analyzer will not be allowed, increasing the security of the system by blocking unwanted connections. To do so, the following steps should be executed:

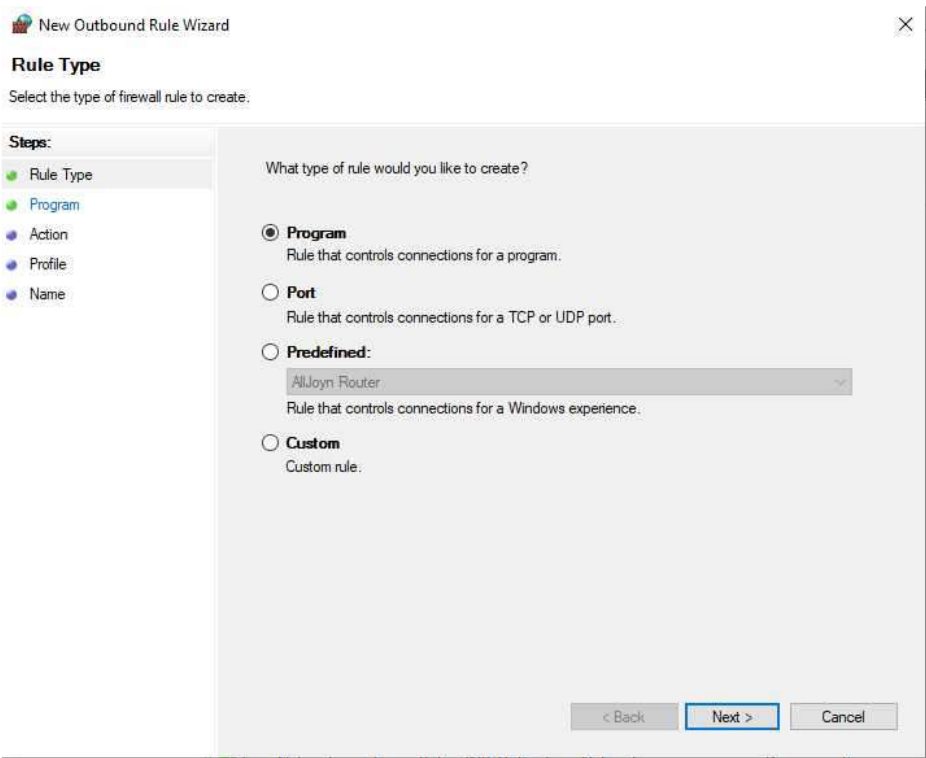
- 1. Go to Windows Menu / Windows Administrative tools / Windows Defender Firewall with Advanced Security. Right click and select Run as administrator option (credentials may be requested).



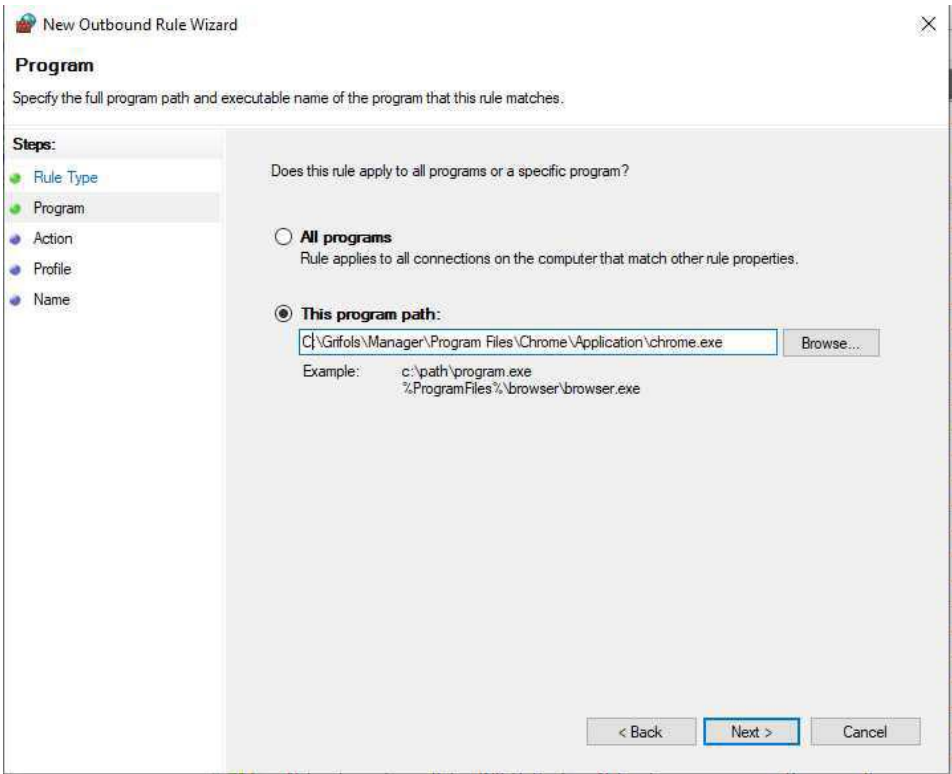
- 1) Right click on *Outbound Rules* and select *New Rule*.



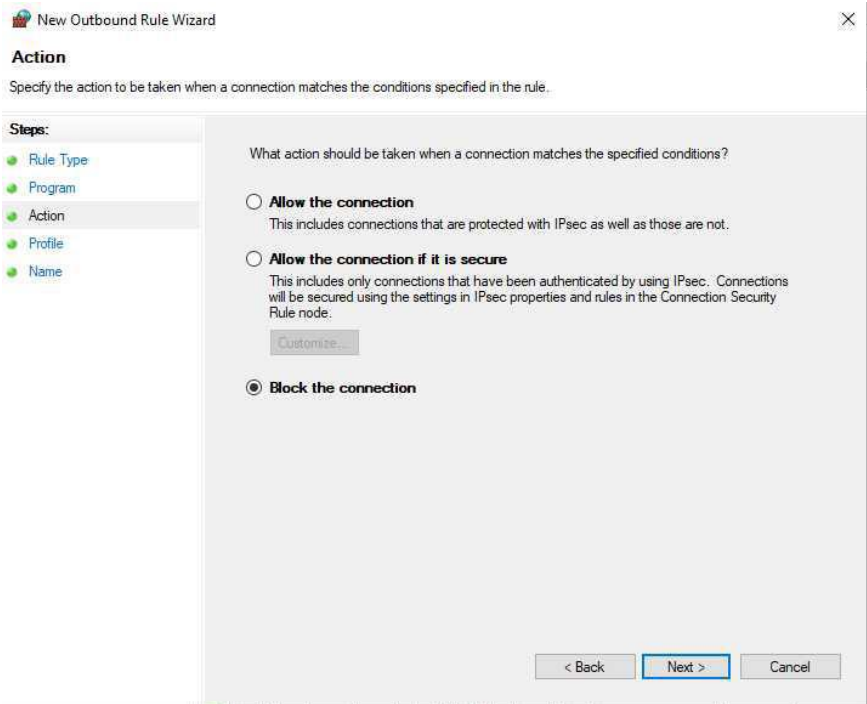
2) Select *Program* (selected by default) and click on *Next*.



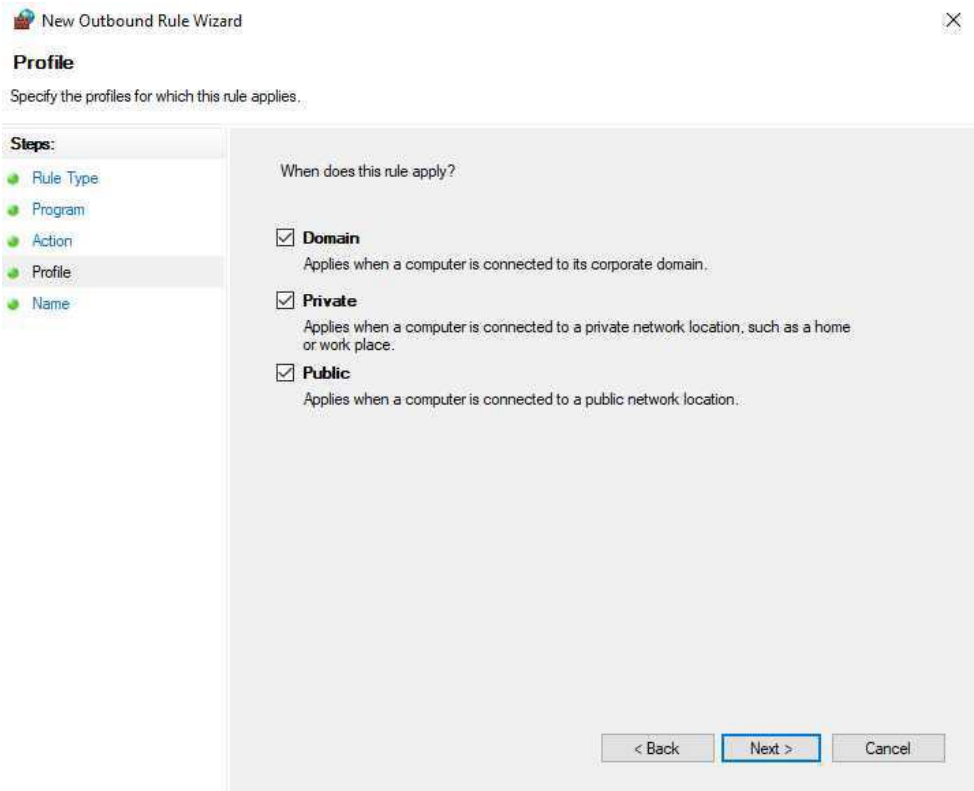
3) Select *This program path:* option and enter the path of the Chrome application:
C:\Grifols\Manager\Program Files\Chrome\Application\chrome.exe.



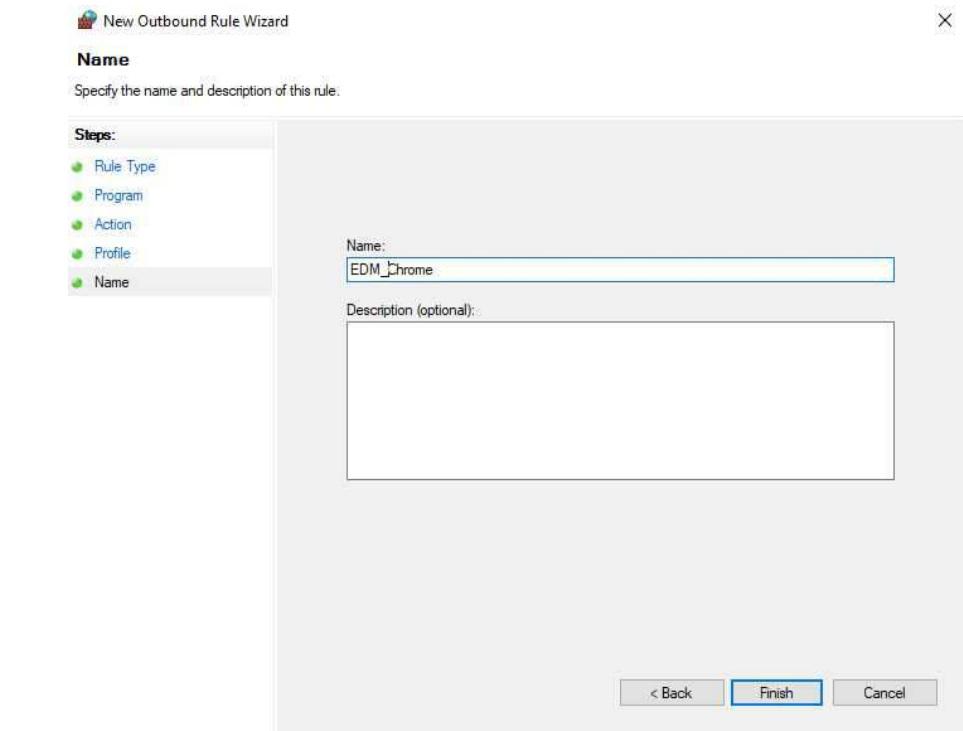
- 4) Click on Next.
- 5) Select *Block the connection* option and click on *Next*.



- 6) Do not change the default settings and click on Next.

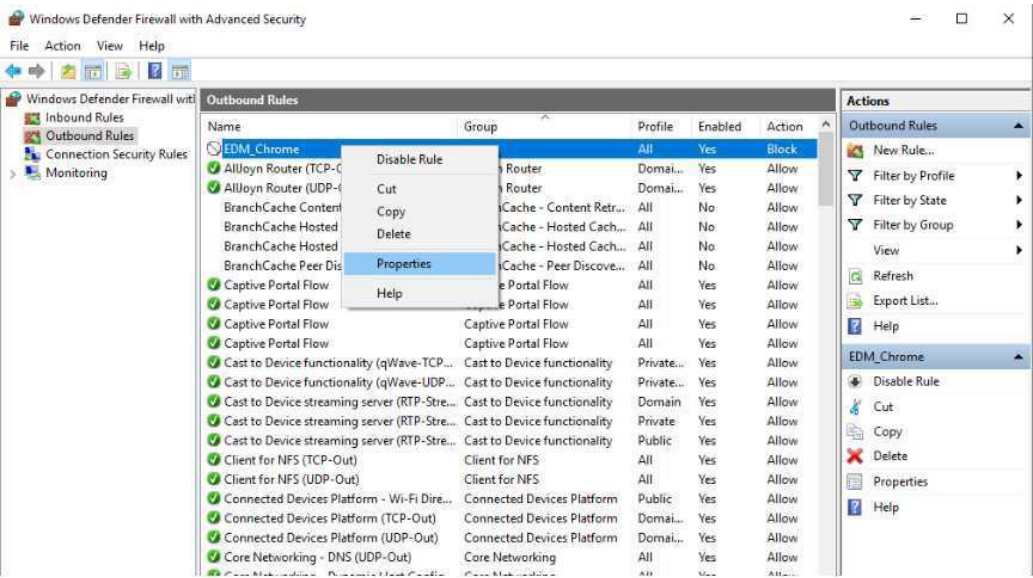


7) Give a name to the created rule. For instance, *EDM_Chrome*.

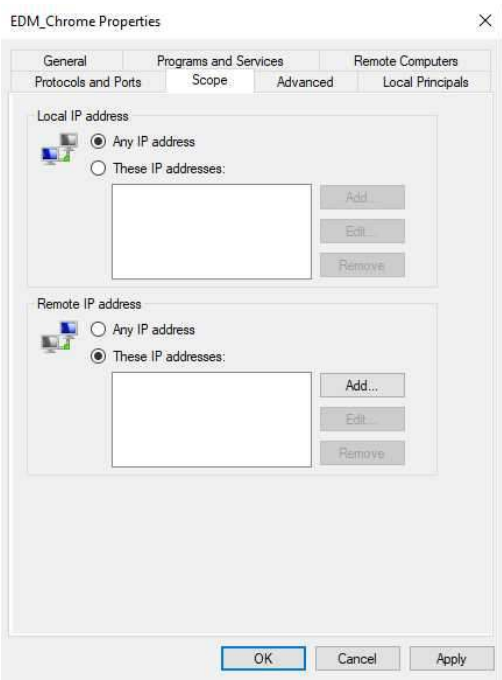


8) Click on *Finish*. The created rule will then be shown among the available Outbound rules.

9) Right click on the created rule and select *Properties*.

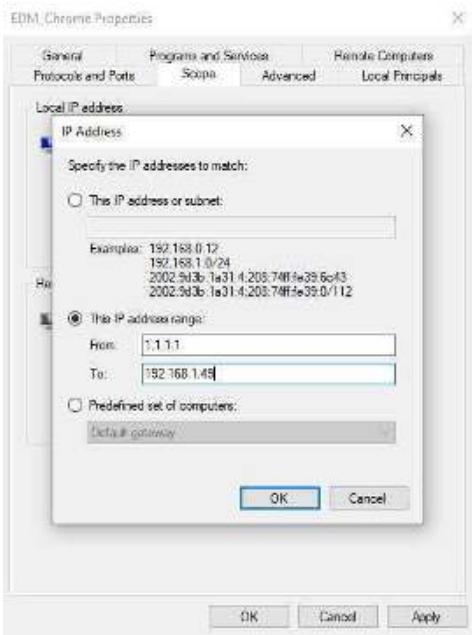


10) Go to *Scope* tab and select *These IP addresses* on *Remote IP address* section. Click on *Add...*



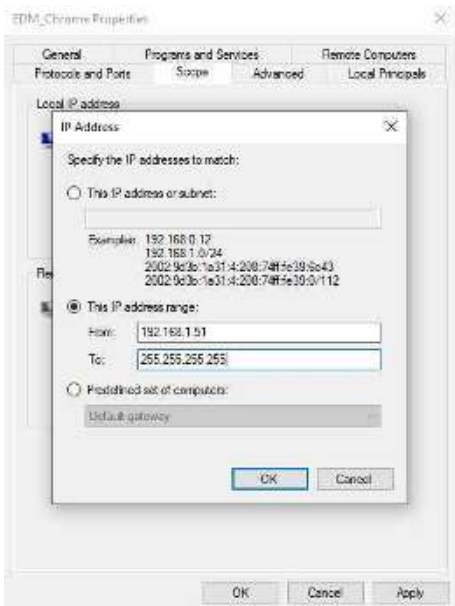
11) Select *This IP address range* option and add the following IP addresses:

From: 1.1.1.1
To: IP configured in (C:\Grifols\Manager\erytra_remote.txt) – 1. For instance, if configured IP is 192.168.1.50, type 192.168.1.49.

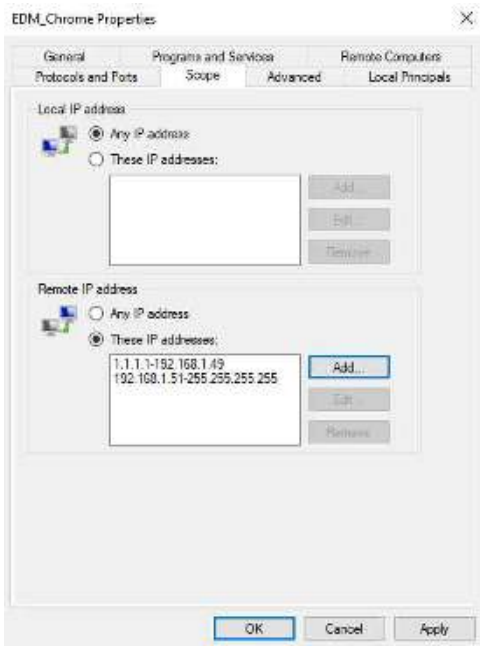


12) Click OK.
13) Click *Add...* in *Remote IP addresses* section once again.
14) Select *This IP address range* option and add the following IP addresses:

From: IP configured in (C:\Grifols\Manager\erytra_remote.txt) + 1. For instance, if configured IP is 192.168.1.50, type 192.168.1.51.
To: 255.255.255.255.



Click *OK*, then *Apply* and *OK* again in the EDM Chrome Properties window.



To avoid the warnings related to SSL certificate when using the remote connection functionality, the following steps must be taken:

- 1) Open C:\Grifols\Manager\ERYTRA_REMOTE.bat file in edition mode by means of Notepad++ or similar software.
- 2) Edit line 15 by adding **--ignore-certificate-errors** after “%RUNTIME_BASE%\Program Files\Chrome\Application\chrome.exe”. Keep the rest of the line 15 as it is:

“%RUNTIME_BASE%\Program Files\Chrome\Application\chrome.exe” **--ignore-certificate-errors** --user-data-dir="%RUNTIME_BASE%\Program Files\Chrome\Application\userData" --app=https://%ip_erytra%:8443/IHManager

- 3) Save ERYTRA_REMOTE.bat file and close the edition software.

2.11 PERIPHERAL INSTALLATION

To install a peripheral device, system volumes shall be unprotected. If connected, it is also recommended to disconnect the instrument from the network during the installation procedure.

Unprotect:

- Switch on the instrument and press the grid (■) button during system startup (login credentials will be required), then select the *Maintenance Mode* icon among the Boot Application options.
- PC will be rebooted and Maintenance Mode will be accessed in the subsequent startup.



CAUTION!
Once the computer has been restarted, drives are not protected.

- Once desktop is available, connect a mouse and a keyboard.
- Press *Windows Start* button and type *Command Prompt*. Right click at the top result and select *Run as administrator*, then type administrator password *dgroot*.
- Enter the command: *applocker bu* (Begin Update). The AppLocker Manager graphical interface will be opened in a separate window.
- Wait for 10 seconds so that the change to update mode is completed.
- Install the drivers of the peripheral device.
- At this point, consider reviewing the *Advanced sharing settings* in *Control Panel – Network and Internet – Network and Sharing Center*. Note that file and printer sharing option is turned off by default.

Protect

- Type *eu* (End Update) in the AppLocker Manager graphical interface window.
- Wait for 10 seconds so that the change to protected mode is completed.
- Shut down the system.
- The volumes will be protected and the system will shut down.

2.12 DECONTAMINATION

Before manipulating the instrument for any maintenance purpose or before preparing it for storing and packaging, it is necessary to decontaminate the instrument. All procedures for such purpose are described in Instructions for Use manual.

2.13 UNINSTALLING AND SHIPPING

If the instrument needs to be stored for a long time, it is recommended to decontaminate the fluidics circuit and package it as for transport. Space requirements are described in previous sections.



WARNING!
The instrument must be decontaminated before transport or storage.



WARNING!
The liquids need to be removed from the fluidics circuit when uninstalling the instrument.



CAUTION!
Use only original packaging to transport the instrument.

When software is closed, the instrument automatically performs a wash of the fluidic system and empties the liquid of the circuit. Nevertheless if software is not closed properly or in the fluidic circuit remains some liquid, it is possible to empty the circuit with Technical Service software. To do so, enter in Technical Service software, start the *Driver* and press *fluidic empty* button in *Fluidics* tab.

Once the circuit is empty, remove the liquids from solution A and B bottles and from waste bottles. Empty card disposal tray if necessary.
Two people (three recommended) are required to pack the instrument and at least one of them must be a **QUALIFIED TECHNICIAN**. To pack the instrument, follow instructions below:

- 1. Open all the fluidic drawers and place the foams in the card disposal container and in the handles of the bottles.



Figure 41. Foams in card disposal container and in the bottles

- 2. Place inside the card disposal container 2 antihumidity bags, power cords, decontamination tool, disposable bags for card waste and documentation related to the instrument (manuals, etc.). Close fluidic drawers.



Figure 42. Decontamination tool

- 3. Move PIP arm to the right of the instrument. Place the internal packaging tray in the upper floor, over PIP platform and in left position.



Figure 43. Location of the internal packaging tray

- Place the plug of the tray between internal packaging tray and right lateral.



Figure 44. Location of the plug of the tray

- Place the decontamination bottle, 2 samples racks, 2 diluents racks and 2 reagents racks in the packaging tray.



Figure 45. Decontamination bottle and racks in packaging tray

- Cover with some adhesive tape the dilution cup wells and place the fixing foams of the dilution cup.
- Move the PIP robotic head to the front of the instrument and remove the 2 samples racks of the drawer labelled as 6 of the upper area to place the supplement comb lateral as shown in the following picture.



Figure 46. Positioning of the supplement comb lateral

8. Place again the samples racks in the drawer labelled as 6 of the upper area and place the RM comb so it touches the drawers frontal and the lateral of PIP platform.



Figure 47. Positioning of the RM comb

9. Place the external barcode reader box in its position in the RM comb. Place 1 antihumidity bag at each side of the box.
10. Place the SM comb between the RM comb and the supplement comb lateral.



Figure 48. Positioning of the SM comb

11. Place the box plug between the RM comb and SM comb and the X-axis support arm.



Figure 49. Positioning of the box plug

12. Place the frontal foams located in the user tray and in the touchscreen monitor.
13. Place the 2 shipping clamps (use an open end spanner number 12) which prevent the arms to move. Check that the shipping clamps are blocking the arm and the robot head at the same time (there is only one position of the robotic head that matches with a hole in the arm to block both at the same time).
14. Remove the door handles and place the screws in a bag. Attach the bag to the door handles.
15. Move up manually the feet in order to avoid that touch the surface.
16. Place the pallet of the packaging as close as possible where the instrument is located.
17. Place the planar lateral (narrow lateral without wooden strips) of the packaging attached to the pallet in order to act as a ramp. Place upper part of the wooden packaging under the ramp.
18. Fix the ramp with the two hook-and-loop fasteners.

- 19. Place the wooden small ramp included inside the packaging at the end of the main ramp. Stick it with adhesive tape.
- 20. Push the *Erytra* through the ramp until it is over the pallet.
- 21. Put the instrument in the position where the feet of the instrument are over the marks that are located in the pallet.

**NOTE:**

It is very important that the instrument will be as much centered as possible over the pallet.

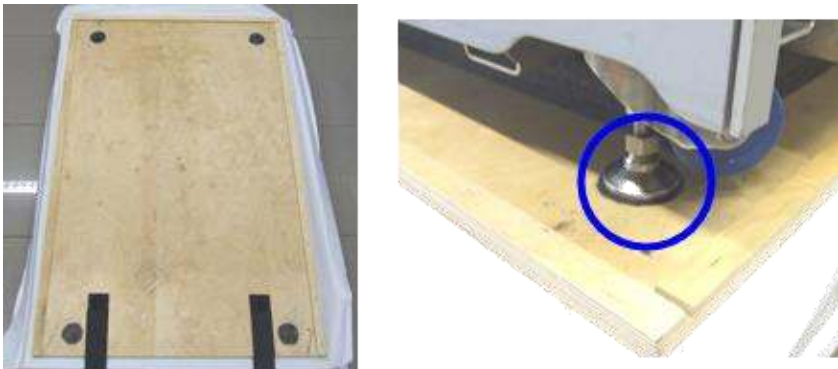


Figure 50. Marks for the feet located in the pallet

- 22. Move up the instrument turning in counterclockwise direction the nut of each feet with an open end spanner number 19. Move down the foot in a crossed way.
- 23. Place the foam and the wooden board included in the packaging underneath the instrument.
- 24. Move down the *Erytra* by means of the nut of the feet (turning them in clockwise direction, use the open end spanner number 19) until the instrument rests on the wooden board over the foam.
- 25. Put adhesive tape to the feet to avoid that could fall down during the shipment.
- 26. Place the protective foams at each corner of the *Erytra*. Fix them with retractile film.



Figure 51. Retractable film in the packaging

- 27. Place the aluminium anti-humidity cover.
- 28. Place the door handles and the small ramp inside their protective packaging and place them over the instrument.
- 29. Assemble the wooden box. There are 2 considerations in this step:

- a. The laterals of the packaging must be levelled with the basis of the pallet in the rear part of the packaging (the rear part is the opposite where is the main ramp).
- b. The flat side of the upper part of the packaging must be faced with the frontal part of the packaging (the one where is the main ramp).

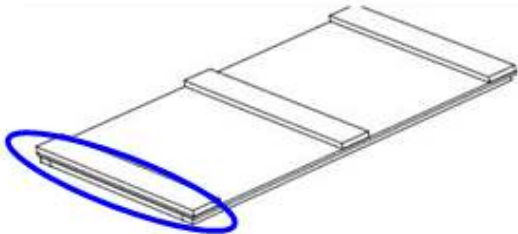


Figure 52. Upper part of the packaging

2.14 INSTRUMENT DISPOSAL

If the instrument is to be disposed of, all possible residues of reagents and samples must be removed and it must be cleaned and totally decontaminated. Then, it must be sent to an authorized centre for such purpose.



CAUTION:
When the equipment is at the end of its lifetime, the User must take into account all local regulations to dispose it.



NOTE:
When disposing of the instrument, *Erytra* should be considered an electronic residue.

3 TECHNICAL SERVICE SOFTWARE OVERVIEW

3.1 INTRODUCTION

This chapter contains an overview of the Technical Service software that allows the **QUALIFIED TECHNICIAN** to adjust and verify all the modules of the instrument.

In order to access Technical Service software, the boot menu must be accessed. For that, switch on the PC, wait for the Erytra logo to appear on screen and then press the grid (■) button. Log in as adminUser (case sensitive, admin password) and the Boot Application options will be displayed. Once in there, select the Technical Service icon among the Boot Application options.

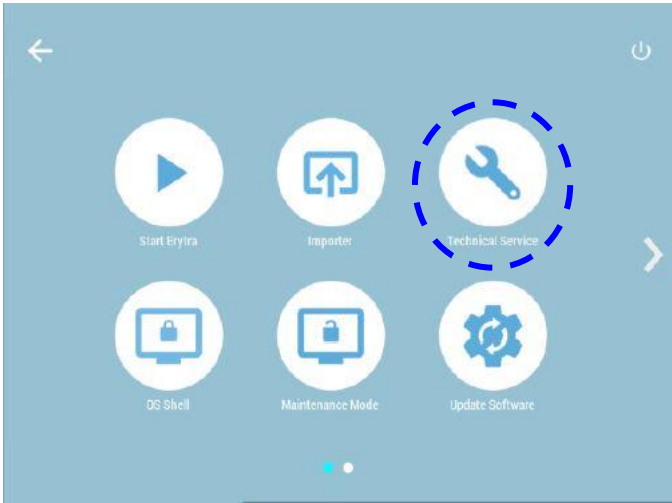


Figure 53. Technical Service icon

Once Technical Service software is opened, start the *Driver* (select *Run* → *Start* as detailed in the next section).

3.2 MODULAR TABS

There is a bar menu at the top of the application, which manages the configuration of the application.



Figure 54. Bar menu of the Technical Service software

- *Run*:
 - *Start*: it starts *Driver* application, enables serial port communication.
 - *Stop*: it stops *Driver* application, disables serial port communication.
 - *Edit serial number*: option to enter serial number of the instrument (maximum 7 characters). It is stored in *erytra.properties* file (*E:\conf*).



Figure 55. Edit serial number window

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	87 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

- *Erytra MacroExecutor*: it opens graphical interface designed to run macros (see chapter 15).
 - *Erytra On/Off switcher*: it runs application to control by software the outputs of the power supply (see chapter 4).
 - *Start Windows Keyboard*: it opens Windows keyboard.
 - *Exit*: it ends program execution.
- *Mode: Driver*: it enables serial port communication.
- *Help: Software Version*: it shows software version.

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization			Purpose:	Uncontrolled Copy
Print date:	16-May-2023			

3.2.1 Procedures tab

This tab includes verifications and adjustments of the different modules of the *Erytra*. Adjustment of each module is done through a wizard to facilitate the **QUALIFIED TECHNICIAN** to monitor the different steps of the adjustment process.

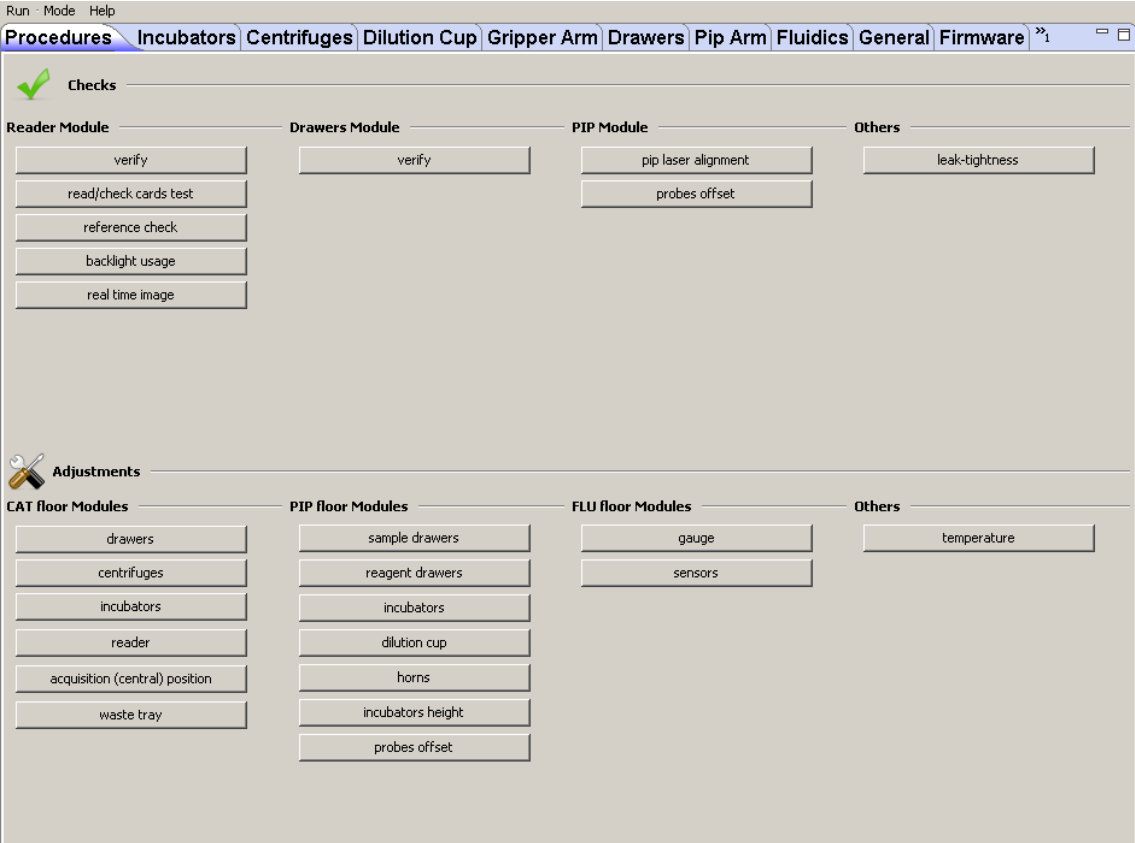


Figure 56. Procedures tab in Technical Service software

Option	Description
Checks	
Reader Module	
verify	Verification of the camera/reader adjustment.
read/check cards test	Perform card initial check and results reading position (microtubes color can be selected).
reference check	Check adjustment of the gripper inside reader module.
backlight usage	Check time [hours] that reader backlight has been switched on.
real time image	It shows real time image of reader module.
Drawers Module	
verify	Verification of the drawers status (samples, reagents and cards).
PIP Module	
pip laser alignment	Check alignment of presence detector laser of PIP arm.
probes offset	Check the adjustment of the probes offset.
Others	
leak-tightness	Check fluidic circuit between dilutors and probes.

Option	Description
Adjustments	
Modules CAT floor	
drawers	Adjustment wizard of card drawers (gripper adjustment in drawers).
centrifuges	Adjustment wizard of centrifuges (gripper adjustment in centrifuges).
incubators	Adjustment wizard of incubators (gripper adjustment in incubators).
Reader	Adjustment wizard of the camera/reader module.
acquisition (central) position	Adjustment wizard of the gripper inside reader module.
waste tray	Adjustment wizard of card waste tray (gripper adjustment in card waste tray).
Modules PIP floor	
sample drawers	Adjustment wizard of samples drawers (probes adjustment in samples drawers).
reagent drawers	Adjustment wizard of reagents drawers (probes adjustment in reagents drawers).
incubators	Adjustment wizard of incubators (probes adjustment in incubators).
dilution cup	Adjustment wizard of dilution cup (probes adjustment in dilution cup).
horns	Adjustment wizard of the horns (horns adjustment in reagents/samples drawers).
incubators height	Adjustment wizard of the maximum height of the incubators.
probes offset	Adjustment wizard to set both probes at same level in Z-axis.
Modules FLU floor	
gauge	Adjustment wizard of the load cells (bottles of solution A and B, cards waste container and bottles of waste).
sensors	Adjustment wizard of the pressure/vaccum sensors located in the manifold (S3, S4, S5 and S6).
Others	
temperature	Adjustment wizard of the incubators temperature.

Table 3. List of options in Procedures tab

3.2.2 Incubators tab

Incubators tab includes all the parameters and functions related to the incubators module. It contains 5 subtabs, one per each incubator and a Home Cycles tab.

Incubator 1,2,3,4 tab

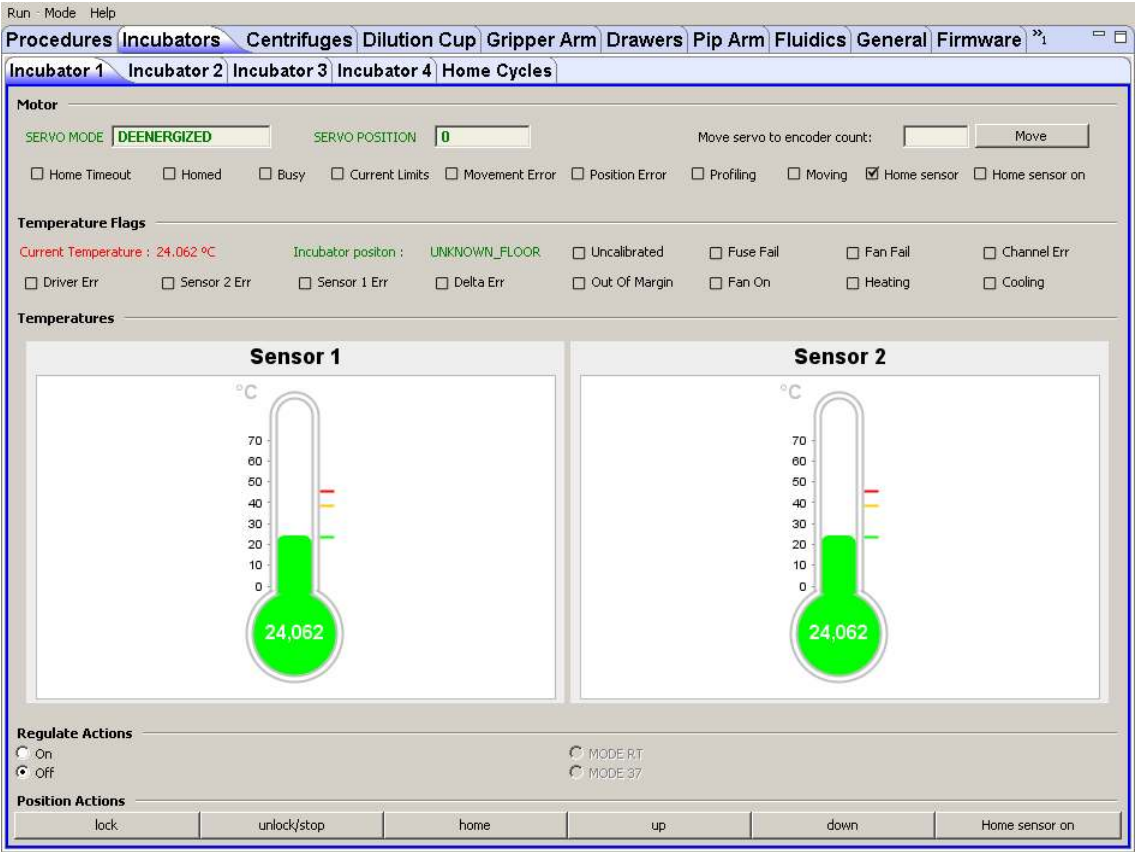


Figure 57. Incubator x - Incubators tab in Technical Service software

Option	Description
Motor	
SERVO MODE	Describes motor movement. There are 4 possible status: - Position Control: motor energized, maintaining the actual position. - Deenergized: motor deenergized. - Integral: performing a movement maintaining the speed but without a profile (initial position, acceleration, maximum speed and final position). It is like a home movement. - Trapezoidal: performing a movement with a profile (initial position, acceleration, maximum speed and final position).
SERVO POSITION	Indicates motor encoder counts.
Move	Incubator moves to the number of counts written in the <i>Move servo to encoder count</i> text box.

Option	Description
Motor	
Home Timeout	Home process has been aborted due to exceeding an encoder count timeout with no change detected at the home sensor.
Homed	Home position has been detected.
Busy	Servo is executing a process.
Current Limits	Current limit has been detected on Servo motor. The hardware automatically unconfigures servo-driver.
Movement Error	Profiled movement has been aborted due to detection of non-correct movement of the motor. Hardware automatically de-energizes servo.
Position Error	Servo is in Position Control Mode and the position error to the actual target exceeds the servo EEPROM settings.
Profiling	Servo is executing trapezoidal movement or integral velocity movement.
Moving	Movement is detected on the servo encoder (encoder position has changed from last time position was checked).
Home sensor	Indicates, during a home process, that the home position has been detected.
Home sensor on	During a home process, it is activated when the home sensor is on.
Temperature Flags	
Present Temperature	Average temperature of the 2 temperature sensors.
Incubator position	Information about where is located the incubator. There are 3 possibilities: - Probes: incubator placed in the upper area - Gripper: incubator placed in the intermediate area - UNKNOWN_FLOOR: if a home process has not been performed.
Uncalibrated	Indicates that the incubator is not calibrated
Fuse Fail	Indicates that temperature regulator fuse is blown.
Fan Fail	Indicates that an error happened when communicating with the Peltier cell fan.
Channel Err	Indicates that an error happened when communicating with incubator multiplexer peripheral.
Driver Err	Indicates an error happened when communicating with the temperature regulator auxiliary peripheral.
Sensor 2 Err	Indicates that an error happened when communicating with the temperature regulator sensor 2.
Sensor 1 Err	Indicates that an error happened when communicating with the temperature regulator sensor 1.
Delta Err	Indicates that temperature difference between sensors 1 and 2 exceeds 1°C.
Out Of Margin	Indicates that the Incubator temperature is out of regulation margin (only for regulation modes): 3°C for RT and 1°C for 37°C.

Option	Description
Temperature Flags	
Fan On	Indicates that Peltier cell fan is on
Heating	Indicates that Peltier cell is in heating mode (increasing regulator temperature).
Cooling	Indicates that Peltier cell is in cooling mode (decreasing regulator temperature).
Temperatures	
Sensor 1	Indicates the temperature measured by the sensor 1 in real time.
Sensor 2	Indicates the temperature measured by the sensor 2 in real time.
Regulate Actions	
On	Activates the temperature regulation system.
Off	Deactivates the temperature regulation system.
Mode RT	When the temperature regulation system is on, the temperature is regulated to 24°C.
Mode 37	When the temperature regulation system is on, the temperature is regulated to 37°C.
Position Actions	
lock	Lock the servo motor.
unlock/stop	Unlock the servo motor or to stop the current movement.
home	Perform a home movement.
up	Move the incubator to the upper position (samples area).
down	Move the incubator to the lower position (cards area).
Home sensor on	Activate home optical barrier.

Table 4. List of options in Incubator x - Incubators tab

Home Cycles tab

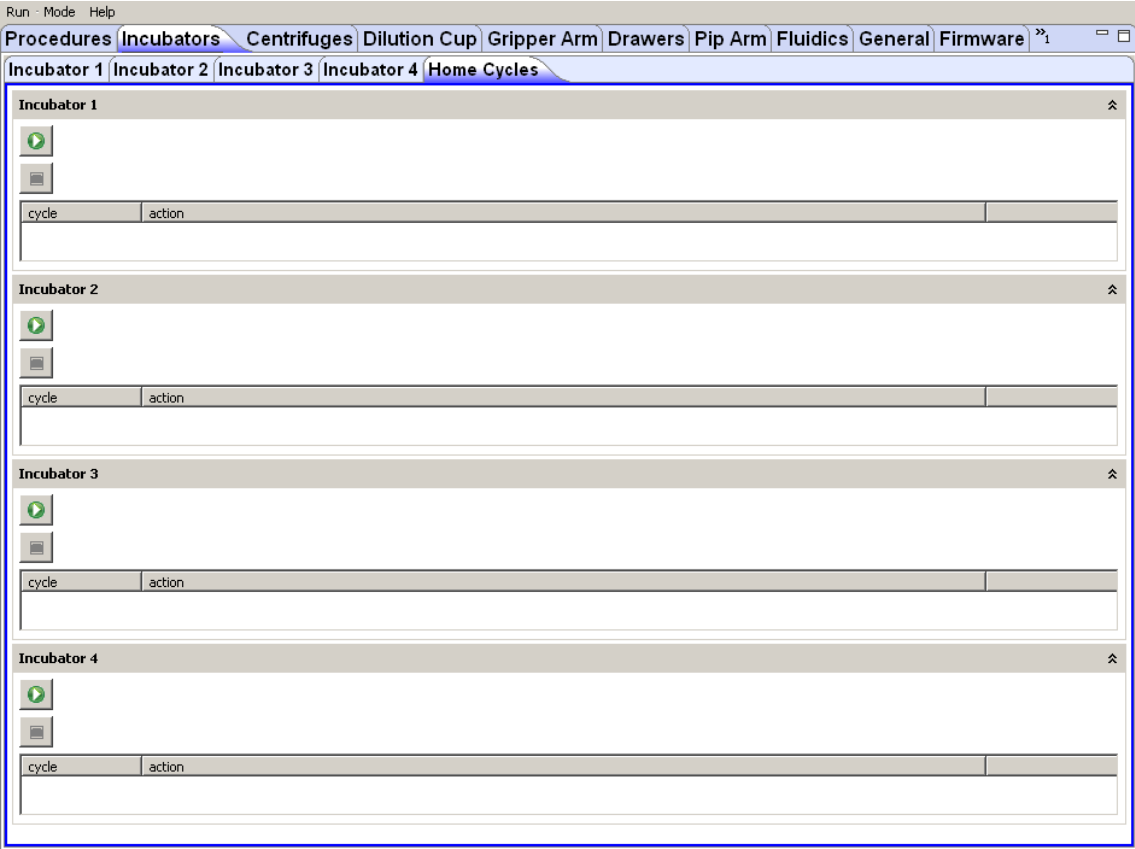


Figure 58. Home Cycles - Incubators tab in Technical Service software

Option	Description
Incubator 1, 2, 3, 4	
	Perform continuous home movements of the selected incubator.
	Stop the continuous home movements of the selected incubator.

Table 5. List of options in Home Cycles - Incubators tab

3.2.3 Centrifuges tab

Centrifuges tab includes all the parameters and functions related to the centrifuges module. It contains 4 subtabs, one per each centrifuge, Home Cycles and Random Position.

Centrifuge 1,2 tab

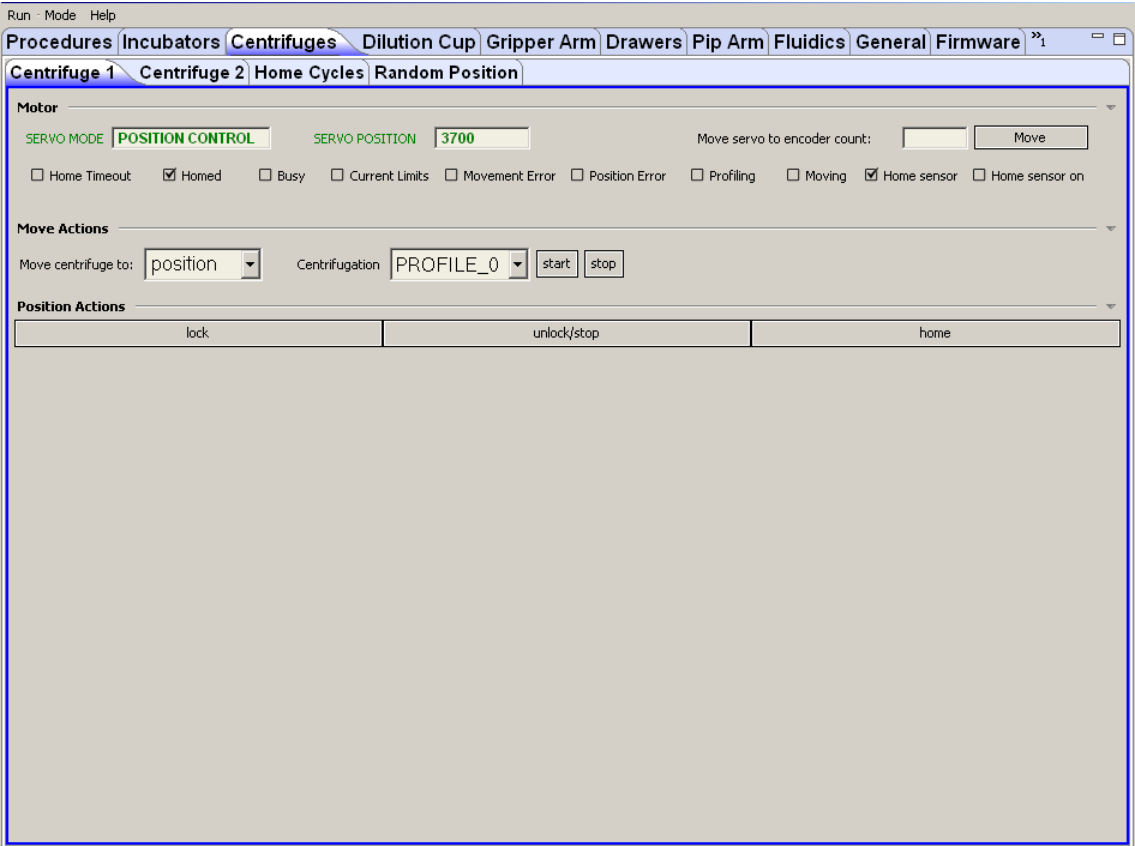


Figure 59. Centrifuge x - Centrifuges tab in Technical Service software

Option	Description
Motor	
SERVO MODE	Describes motor movement. There are 4 possible status: - Position Control: motor energized, maintaining the actual position. - Deenergized: motor deenergized. - Integral: performing a movement maintaining the speed but without a profile (initial position, acceleration, maximum speed and final position). It is like a home movement. - Trapezoidal: performing a movement with a profile (initial position, acceleration, maximum speed and final position).
SERVO POSITION	Indicates motor encoder counts.
Move	Centrifuge moves to the number of counts written in the <i>Move servo to encoder count</i> text box.

Option	Description
Motor	
Home Timeout	Home process has been aborted due to exceeding an encoder count timeout with no change detected at the home sensor.
Homed	Home position has been detected.
Busy	Servo is executing a process.
Current Limits	Current limit has been detected on Servo motor. The hardware automatically unconfigures servo-driver.
Movement Error	Profiled movement has been aborted due to detection of non-correct movement of the motor. Hardware automatically de-energizes servo.
Position Error	Servo is in Position Control Mode and the position error to the actual target exceeds the servo EEPROM settings.
Profiling	Servo is executing trapezoidal movement or integral velocity movement.
Moving	Movement is detected on the servo encoder (encoder position has changed from last time position was checked).
Home sensor	Indicates, during a home process, that the home position has been detected.
Home sensor on	During a home process, it is activated when the home sensor is on.
Move Actions	
Move centrifuge to	Move centrifuge to any of the selected slots (1 to 12) to the entry/exit card position.
Centrifugation	Selection list to choose programmed parameters depending on the Profile selected: • Profile_0: 1206 rpm, 540s • Profile_1: 1206 rpm, 120s • Profile_2: Not used • Profile_3: Not used
start	Initiate the centrifugation with the selected profile.
stop	Cancel/stop current centrifugation.
Position Actions	
lock	Lock the servo motor.
unlock/stop	Unlock the servo motor or to stop the current movement.
home	Perform a home movement.

Table 6. List of options in Centrifuge x - Centrifuges tab

Home Cycles tab

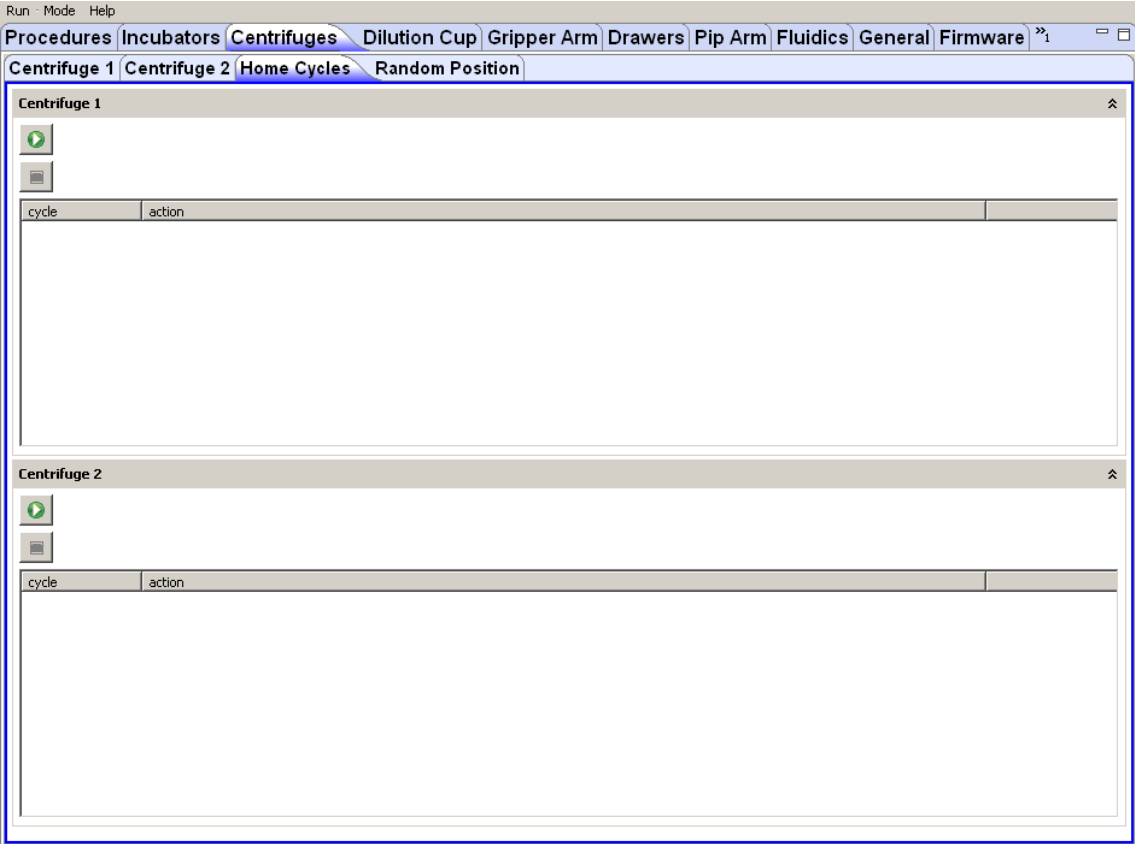


Figure 60. Home Cycles - Centrifuges tab in Technical Service software

Option	Description
Centrifuge 1, 2	
	Perform continuous home movements of the selected centrifuge.
	Stop the continuous home movements of the selected centrifuge.

Table 7. List of options in Home Cycles - Centrifuges tab

Random Position tab

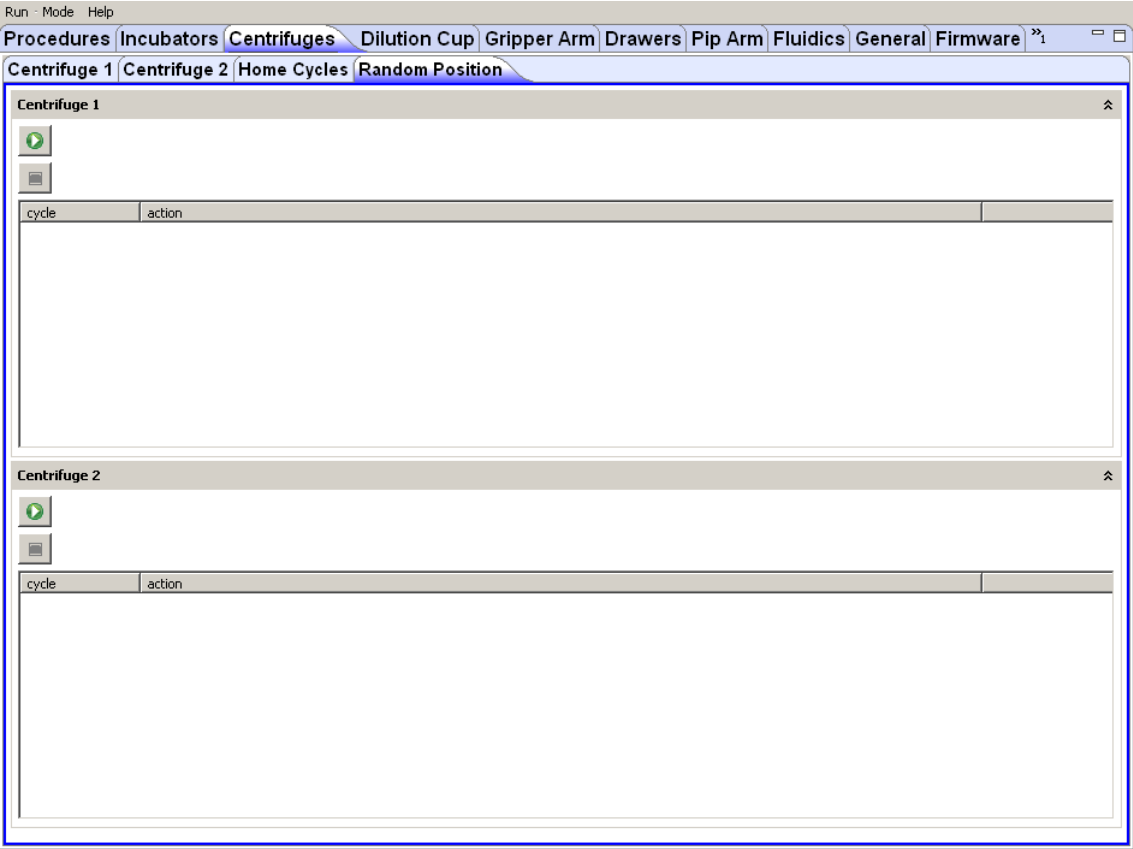


Figure 61. Random Position - Centrifuges tab in Technical Service software

Option	Description
Centrifuge 1, 2	
	Perform random movements of the selected centrifuge.
	Stop the random movements of the selected centrifuge.

Table 8. List of options in Random Position - Centrifuges tab

3.2.4 Dilution Cup tab

Dilution cup tab includes all the parameters and functions related to the dilution cup system.



Figure 62. Dilution Cup tab in Technical Service software

Option	Description
Motor	
SERVO MODE	Describes motor movement. There are 4 possible status: - Position Control: motor energized, maintaining the actual position. - Deenergized: motor deenergized. - Integral: performing a movement maintaining the speed but without a profile (initial position, acceleration, maximum speed and final position). It is like a home movement. - Trapezoidal: performing a movement with a profile (initial position, acceleration, maximum speed and final position).
SERVO POSITION	Indicates motor encoder counts.
Busy	Servo is executing a process.
Current Limits	Current limit has been detected on Servo motor. The hardware automatically unconfigures servo-driver.

Option	Description
Motor	
Movement Error	Profiled movement has been aborted due to detection of non-correct movement of the motor. Hardware automatically de-energizes servo.
Position Error	Servo is in Position Control Mode and the position error to the actual target exceeds the servo EEPROM settings.
Profiling	Servo is executing trapezoidal movement or integral velocity movement.
Moving	Movement is detected on the servo encoder (encoder position has changed from last time position was checked).
Home sensor	Indicates, during a home process, that the home position has been detected.
Home sensor on	During a home process, it is activated when the home sensor is on.
Stirrer Actions	
It is possible to select 4 different stirring profiles: • Profile_0: 800 rpm, 9s • Profile_1: Not used • Profile_2: Not used • Profile_3: Not used	
Spin	Initiate the stirring process of the dilution cup.
stop	Stop the stirring process of the dilution cup.

Table 9. List of options in Dilution cup tab

3.2.5 Gripper Arm tab

Gripper Arm tab includes all the parameters and functions related to the CAT arm. It includes 5 subtabs: Movement, Presence Detector, BCR, Home Cycles and Random Position.

Movement tab

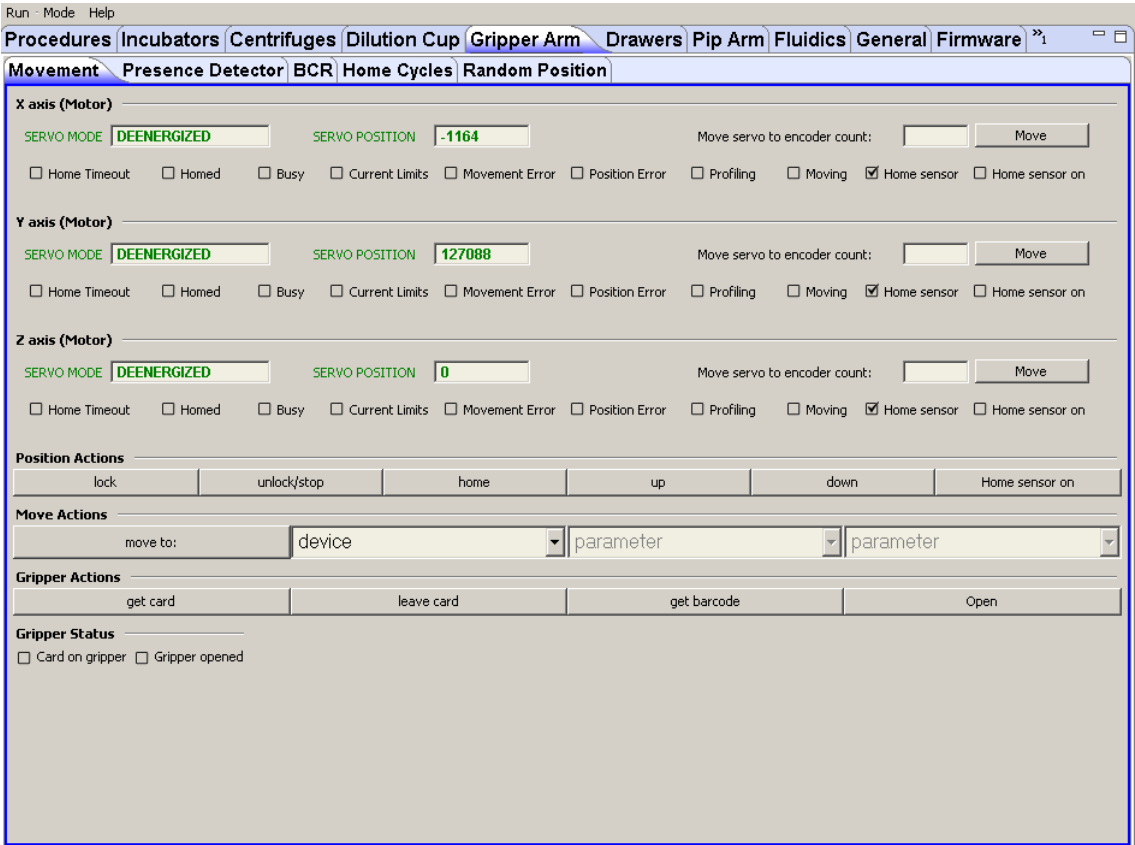


Figure 63. Movement – Gripper Arm tab in Technical Service software

Option	Description
X axis , Y axis or Z axis (Motor)	
SERVO MODE	Describes motor movement. There are 4 possible status: - Position Control: motor energized, maintaining the actual position. - Deenergized: motor deenergized. - Integral: performing a movement maintaining the speed but without a profile (initial position, acceleration, maximum speed and final position). It is like a home movement. - Trapezoidal: performing a movement with a profile (initial position, acceleration, maximum speed and final position).
SERVO POSITION	Indicates motor encoder counts.

Option	Description																		
X axis , Y axis or Z axis (Motor)																			
Move	Selected axis moves to the number of counts introduced in the <i>Move servo to encoder count</i> text box.																		
Home Timeout	Home process has been aborted due to exceeding an encoder count timeout with no change detected at the home sensor.																		
Homed	Home position has been detected.																		
Busy	Servo is executing a process.																		
Current Limits	Current limit has been detected on Servo motor. The hardware automatically unconfigures servo-driver.																		
Movement Error	Profiled movement has been aborted due to detection of non-correct movement of the motor. Hardware automatically de-energizes servo.																		
Position Error	Servo is in Position Control Mode and the position error to the actual target exceeds the servo EEPROM settings.																		
Profiling	Servo is executing trapezoidal movement or integral velocity movement.																		
Moving	Movement is detected on the servo encoder (encoder position has changed from last time position was checked).																		
Home sensor	Indicates, during a home process, that the home position has been detected.																		
Home sensor on	During a home process, it is activated when the home sensor is on.																		
Position Actions																			
lock	Lock a servo motor (X, Y, Z or all of them).																		
unlock/stop	Unlock a servo motor (X, Y, Z or all of them) or to stop the current movement.																		
home	Perform a home movement (X, Y, Z or all of them).																		
up	Move up the gripper.																		
down	Move down the gripper when CAT is over reader area.																		
Home sensor on	Activate home optical barriers.																		
Move Actions																			
move to:	Button to move the gripper arm to any of the following allowed positions:																		
	<table><tr><th>First combo box</th><th>Second combo box</th><th>Third combo box</th></tr><tr><td>INCUBATOR_1 INCUBATOR_2 INCUBATOR_3 INCUBATOR_4</td><td>SLOT_1 ... SLOT_12</td><td>INACTIVE</td></tr><tr><td>CENTRIFUGE_1 CENTRIFUGE_2</td><td>SLOT_1 ... SLOT_12</td><td>INACTIVE</td></tr><tr><td>CARD_DRAWER_1 CARD_DRAWER_2 CARD_DRAWER_3 CARD_DRAWER_4</td><td>RACK_1 ... RACK_4</td><td>CARD_1 ... CARD_25</td></tr><tr><td>CAMERA</td><td>INTEGRITY READ</td><td>WELLS_1_2 WELLS_3_4 WELLS_5_6 WELLS_7_8</td></tr><tr><td>WASTE_TRAY</td><td>INACTIVE</td><td>INACTIVE</td></tr></table>	First combo box	Second combo box	Third combo box	INCUBATOR_1 INCUBATOR_2 INCUBATOR_3 INCUBATOR_4	SLOT_1 ... SLOT_12	INACTIVE	CENTRIFUGE_1 CENTRIFUGE_2	SLOT_1 ... SLOT_12	INACTIVE	CARD_DRAWER_1 CARD_DRAWER_2 CARD_DRAWER_3 CARD_DRAWER_4	RACK_1 ... RACK_4	CARD_1 ... CARD_25	CAMERA	INTEGRITY READ	WELLS_1_2 WELLS_3_4 WELLS_5_6 WELLS_7_8	WASTE_TRAY	INACTIVE	INACTIVE
	First combo box	Second combo box	Third combo box																
	INCUBATOR_1 INCUBATOR_2 INCUBATOR_3 INCUBATOR_4	SLOT_1 ... SLOT_12	INACTIVE																
	CENTRIFUGE_1 CENTRIFUGE_2	SLOT_1 ... SLOT_12	INACTIVE																
	CARD_DRAWER_1 CARD_DRAWER_2 CARD_DRAWER_3 CARD_DRAWER_4	RACK_1 ... RACK_4	CARD_1 ... CARD_25																
	CAMERA	INTEGRITY READ	WELLS_1_2 WELLS_3_4 WELLS_5_6 WELLS_7_8																
WASTE_TRAY	INACTIVE	INACTIVE																	

Option	Description
Gripper Actions	
get card	Move down the gripper and take a card in the position where it is placed the arm.
leave card	Move down the gripper and leave a card in the position where it is placed the arm.
get barcode	Activate the barcode reader, returning the read of the barcode. Previously a card must be taken with the get card button.
Open	Open gripper fingers.
Gripper Status	
Card on gripper	Indicates if it is detected a card on the gripper.
Gripper opened	Indicates if gripper fingers are opened.

Table 10. List of options in Movement - Gripper Arm tab

Presence detector tab

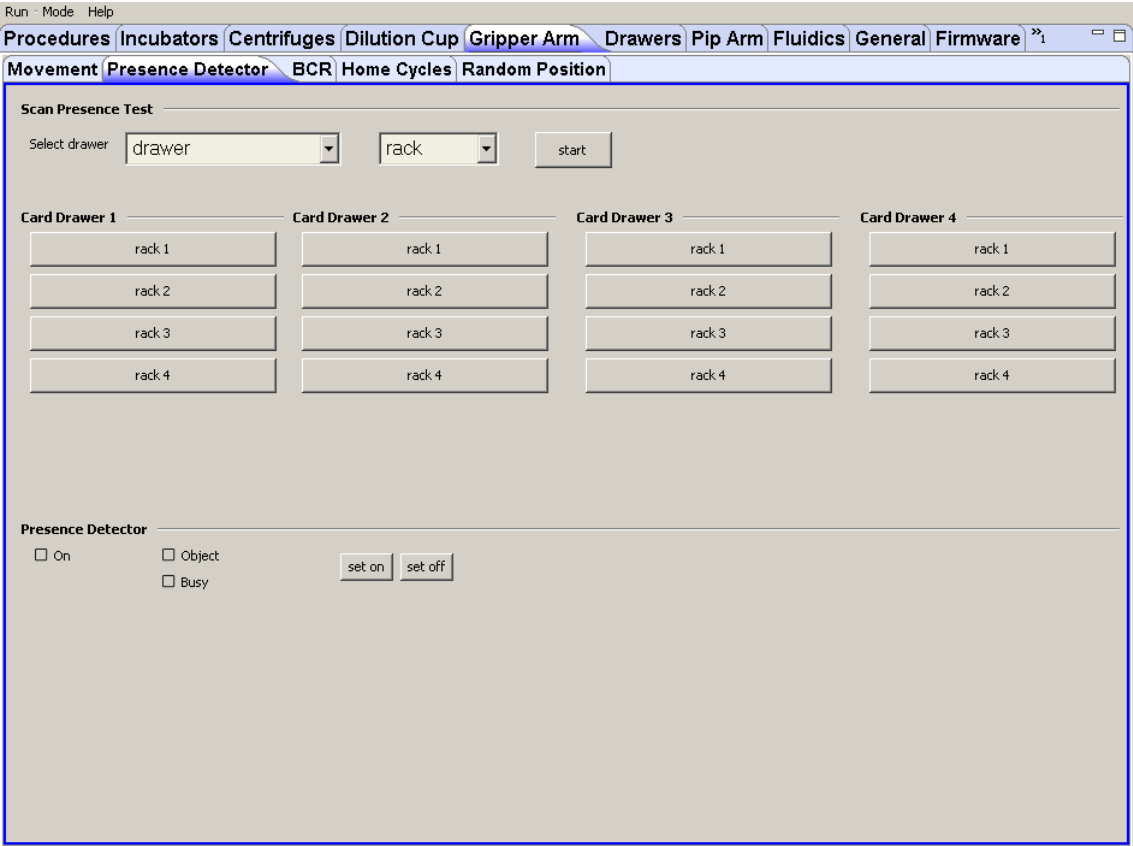


Figure 64. Presence Detector – Gripper Arm tab in Technical Service software

Option	Description				
Scan Presence Test					
Select drawer	List to select a position in order to perform a detection card test. The allowed positions are:				
	<table><tr><th>First combo box</th><th>Second combo box</th></tr><tr><td>CARD_DRAWER_1 CARD_DRAWER_2 CARD_DRAWER_3 CARD_DRAWER_4</td><td>RACK_1 ... RACK_4</td></tr></table>	First combo box	Second combo box	CARD_DRAWER_1 CARD_DRAWER_2 CARD_DRAWER_3 CARD_DRAWER_4	RACK_1 ... RACK_4
	First combo box	Second combo box			
CARD_DRAWER_1 CARD_DRAWER_2 CARD_DRAWER_3 CARD_DRAWER_4	RACK_1 ... RACK_4				
start	Move CAT arm to the selected position and it performs a presence detection card test.				
Card Drawer (1,2,3,4)					
rack 1	Show status of the card rack 1 located in the selected card drawer (1, 2, 3 or 4). It indicates the number of detected and undetected cards and the position.				
rack 2	Show status of the card rack 2 located in the selected card drawer (1, 2, 3 or 4). It indicates the number of detected and undetected cards and the position.				
rack 3	Show status of the card rack 3 located in the selected card drawer (1, 2, 3 or 4). It indicates the number of detected and undetected cards and the position.				
rack 4	Show status of the card rack 4 located in the selected card drawer (1, 2, 3 or 4). It indicates the number of detected and undetected cards and the position.				
Presence Detector					
On	Indicate if the presence detector laser beam is on.				
Object	Indicates if an object is detected in front of the laser beam.				
Busy	Indicates if presence detector is executing a scanning.				
set on	Activate the laser beam.				
set off	Deactivate the laser beam.				

Table 11. List of options in Presence Detector – Gripper Arm tab

BCR tab

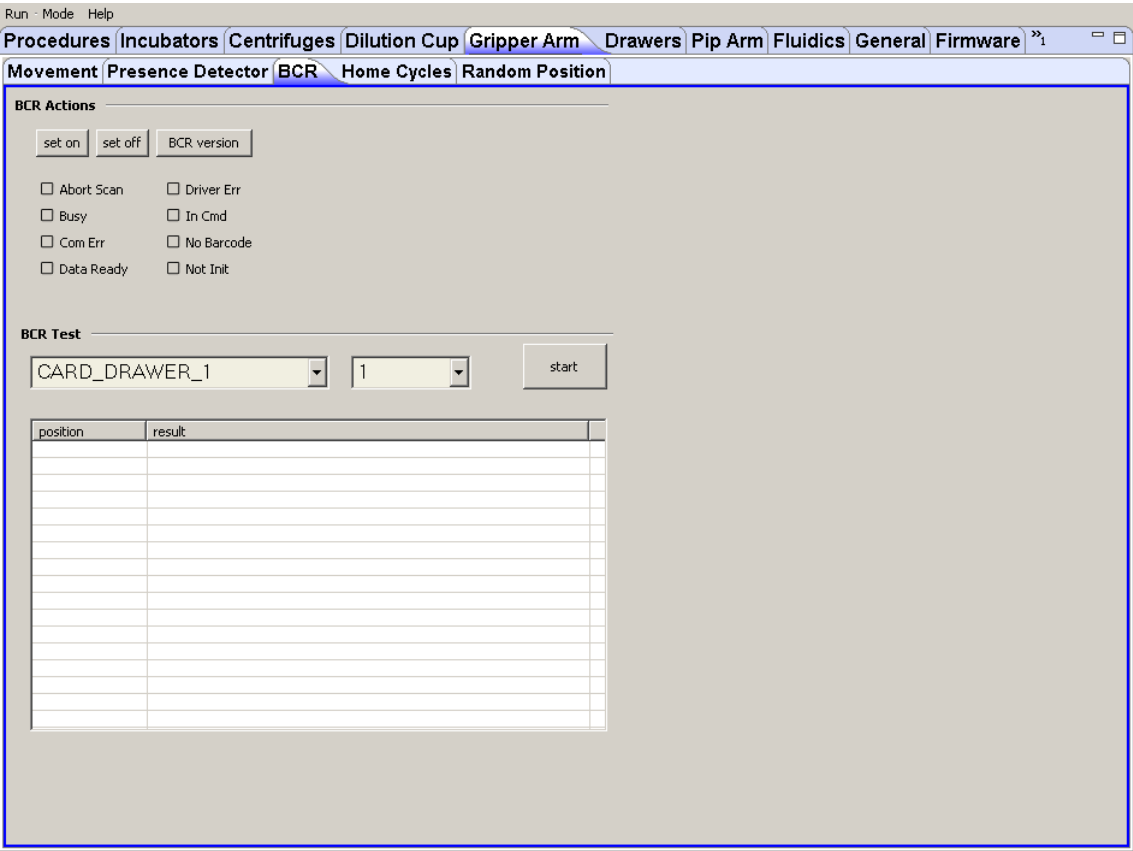


Figure 65. BCR – Gripper Arm tab in Technical Service software

Option	Description
BCR Actions	
set on	Activate barcode reader.
set off	Deactivate barcode reader.
BCR version	Returns barcode reader version.
Abort scan	Indicates that a scanning operation has been aborted.
Busy	Barcode reader is on.
Com Err	Indicates a barcode reader communication error.
Data Ready	Indicates that information obtained by the barcode reader is ready for being recovered by software.
Driver Err	Indicates a barcode reader communication error.
In Cmd	Indicates that there is communication between microcontroller and barcode reader.
No Barcode	No barcode is read.
Not Init	Barcode reader is not initialized.

Option	Description										
BCR Test											
start	Button to initiate the process of barcode reading. Previously, a detection test is performed to check where cards are placed. Allowed positions are: <table><tr><th>First combo box</th><th>Second combo box</th></tr><tr><td>CARD_DRAWER_1</td><td>RACK_1</td></tr><tr><td>CARD_DRAWER_2</td><td>RACK_2</td></tr><tr><td>CARD_DRAWER_3</td><td>RACK_3</td></tr><tr><td>CARD_DRAWER_4</td><td>RACK_4</td></tr></table>	First combo box	Second combo box	CARD_DRAWER_1	RACK_1	CARD_DRAWER_2	RACK_2	CARD_DRAWER_3	RACK_3	CARD_DRAWER_4	RACK_4
First combo box	Second combo box										
CARD_DRAWER_1	RACK_1										
CARD_DRAWER_2	RACK_2										
CARD_DRAWER_3	RACK_3										
CARD_DRAWER_4	RACK_4										
text box	Results of barcode reading process are displayed. In the first column (<i>position</i>) is shown the position of the card detected and in the second column (<i>result</i>) is shown the barcode.										

Table 12. List of options in BCR – Gripper Arm tab

Home Cycles tab

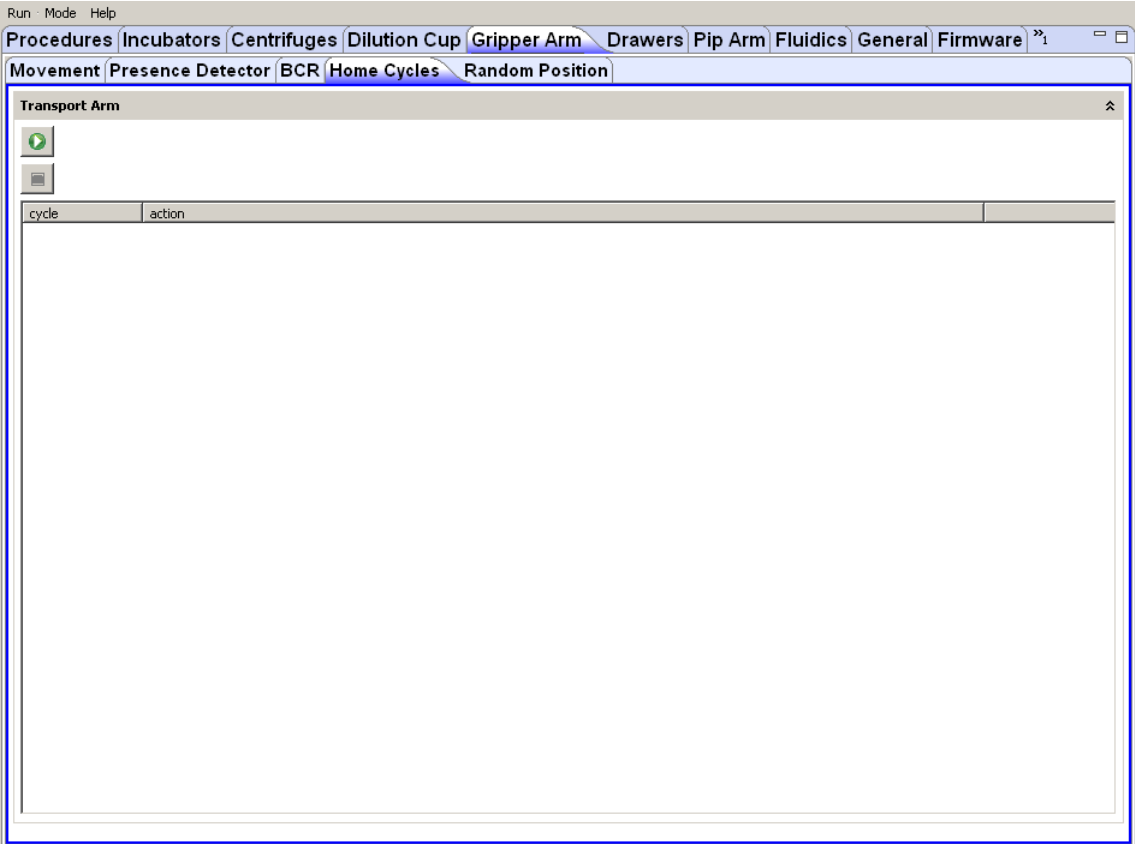


Figure 66. Home Cycles – Gripper Arm tab in Technical Service software

Option	Description
Transport Arm	
	Perform continuous home movements of the CAT arm.
	Stop continuous home movements of the CAT arm.

Table 13. List of options in Home Cycles - Gripper Arm tab

Random Position tab

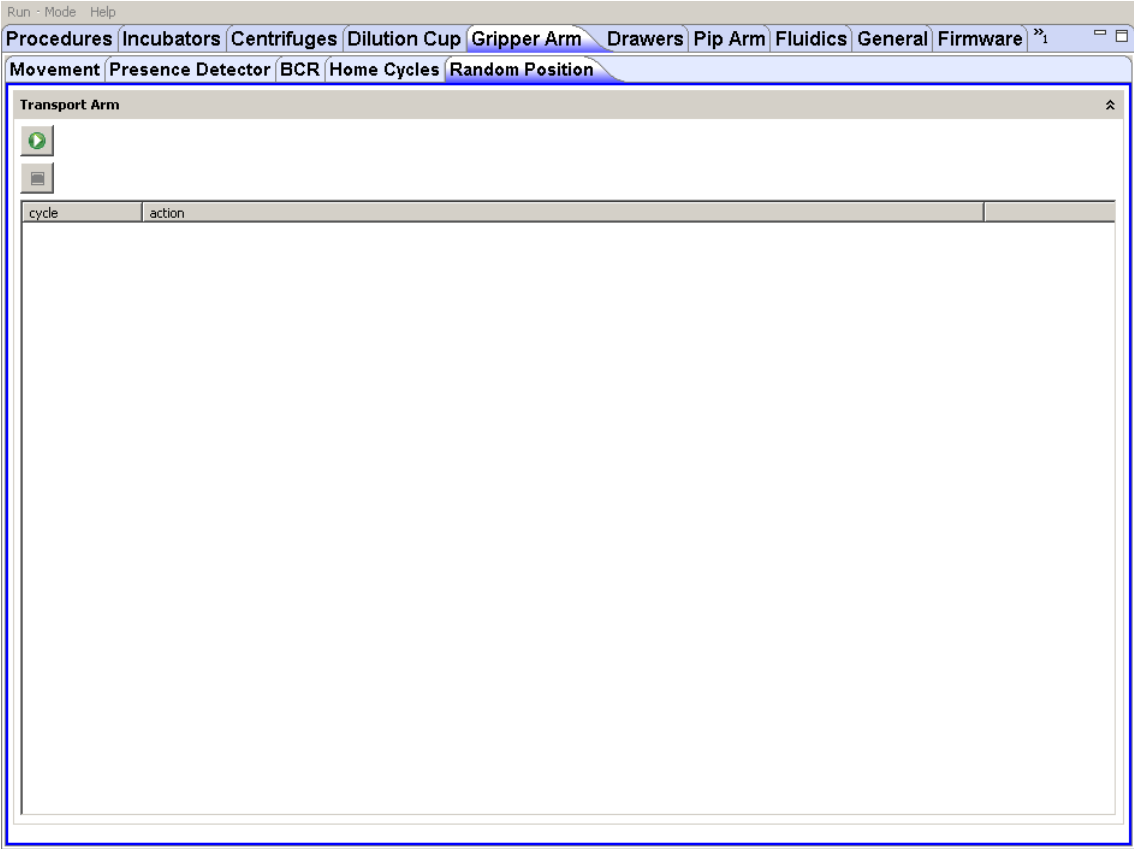


Figure 67. Random Position - Gripper Arm tab in Technical Service software

Option		Description
Transport Arm		
		Perform random movements of the CATarm.
		Stop random movements of the CAT arm.

Table 14. List of options in Random Position – Gripper Arm tab

3.2.6 Drawers tab

Drawers tab includes all the parameters and functions related to the drawers. It contains 5 subtabs: Card Drawers, Sample Drawers, Reagent Drawers, Fluidic Drawers and Home Cycles.

Card Drawers tab

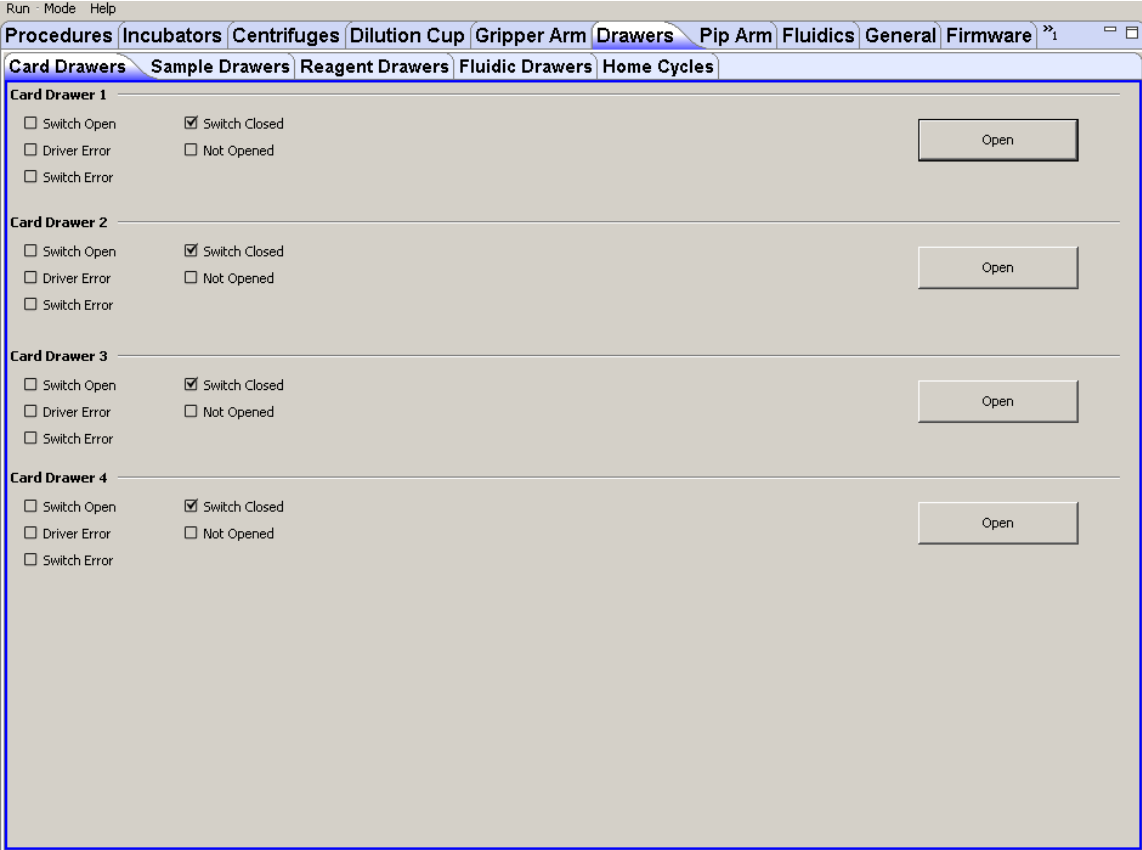


Figure 68. Card Drawers - Drawers tab in Technical Service software

Option	Description
Card Drawer (1,2,3,4)	
Switch Open	Indicates that the drawer is open.
Switch Closed	Indicates that the drawer is closed.
Switch Error	Indicates that an error condition (is open and closed at the same time or not opened nor closed at the same time).
Driver Error	Indicates that an error happened when communicating with the lock auxiliary peripheral, through I2C bus.
Not opened	Indicates that the lock has not opened within 1 second after an open command has been executed.
Open	Open the selected drawer.

Table 15. List of options in Card Drawers - Drawers tab

Sample Drawers tab

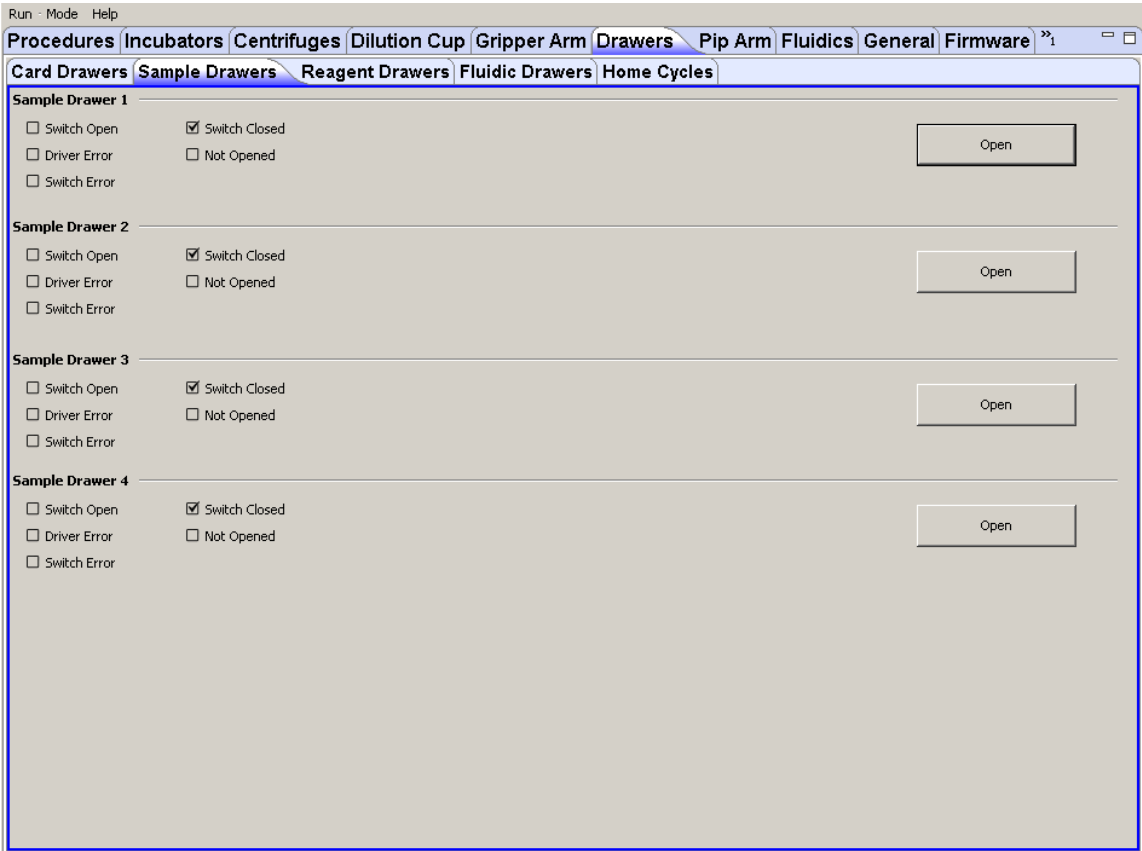


Figure 69. Sample Drawers - Drawers tab in Technical Service software

Option	Description
Sample Drawer (1,2,3,4)	
Switch Open	Indicates that the drawer is open.
Switch Closed	Indicates that the drawer is closed.
Switch Error	Indicates that an error condition (is open and closed at the same time or not opened nor closed at the same time).
Driver Error	Indicates that an error happened when communicating with the lock auxiliary peripheral, through I2C bus.
Not opened	Indicates that the lock has not opened within 1 second after an open command has been executed.
Open	Open the selected drawer.

Table 16. List of options in Sample Drawers - Drawers tab

Reagent Drawers tab

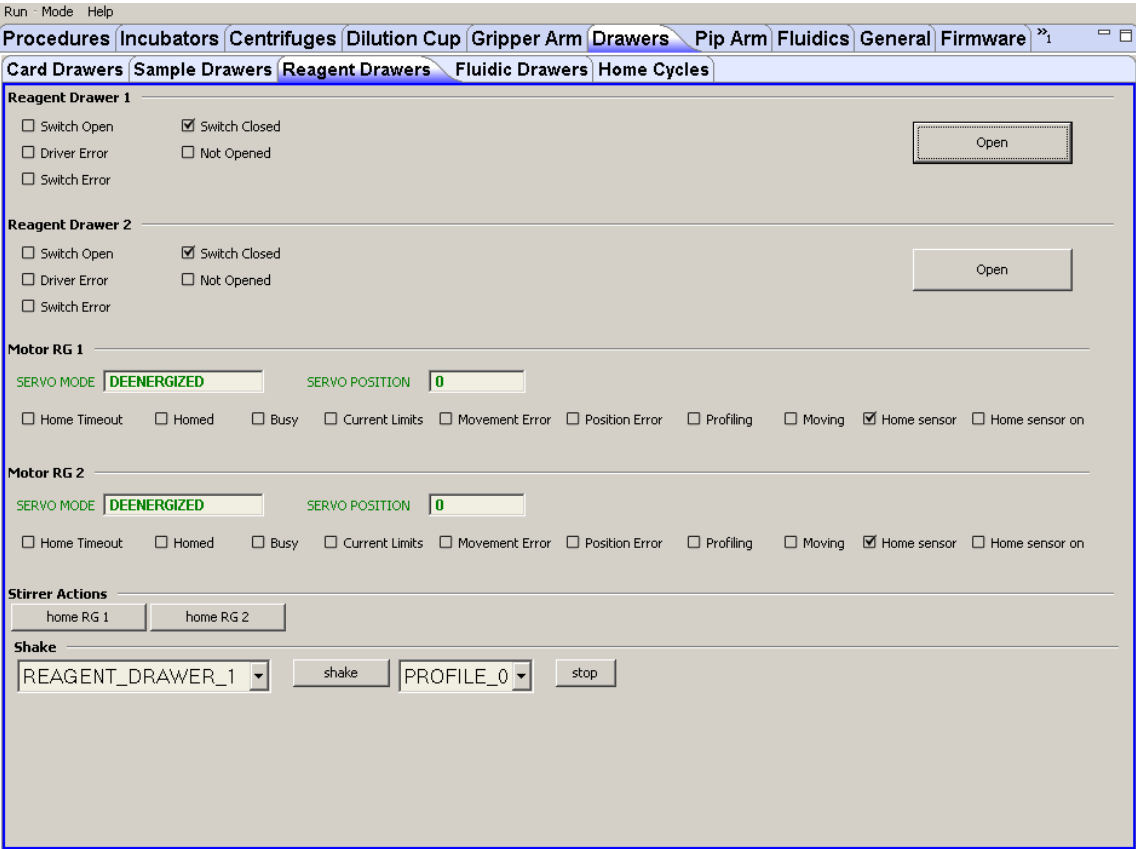


Figure 70. Reagent Drawers – Drawers tab in Technical Service software

Option	Description
Reagent Drawer (1, 2)	
Switch Open	Indicates that the drawer is open.
Switch Closed	Indicates that the drawer is closed.
Switch Error	Indicates that an error condition (is open and closed at the same time or not opened nor closed at the same time).
Driver Error	Indicates that an error happened when communicating with the lock auxiliary peripheral, through I2C bus.
Not opened	Indicates that the lock has not opened within 1 second after an open command has been executed.
Open	Button to open the selected drawer.
Motor RG (1, 2)	
SERVO MODE	Describes motor movement. Position Control: motor energized, maintaining the actual position. - Deenergized: motor deenergized. - Integral: performing a movement maintaining the speed but without a profile (initial position, acceleration, maximum speed and final position). It is like a home movement. - Trapezoidal: performing a movement with a profile (initial position, acceleration, maximum speed and final position).

Option	Description						
Motor RG (1, 2)							
SERVO POSITION	Indicates the encoder counts of the motor.						
Home Timeout	Home process has been aborted due to exceeding an encoder count timeout with no change detected at the home sensor.						
Homed	Home position has been detected.						
Busy	Servo is executing a process.						
Current Limits	Current limit has been detected on Servo motor. The hardware automatically unconfigures servo-driver.						
Movement Error	Profiled movement has been aborted due to detection of non-correct movement of the motor. Hardware automatically de-energizes servo.						
Position Error	Servo is in Position Control Mode and the position error to the actual target exceeds the servo EEPROM settings.						
Profiling	Servo is executing trapezoidal movement or integral velocity movement.						
Moving	Movement is detected on the servo encoder (encoder position has changed from last time position was checked).						
Home sensor	Indicates, during a home process, that the home position has been detected.						
Home sensor on	During a home process, it is activated when the home sensor is on.						
Stirrer Actions							
home RG1	Perform a home of the rack 2 of reagent drawer 1.						
home RG2	Perform a home of the rack 2 of reagent drawer 2.						
Shake							
shake	Initiate the stirring process with the selected profile: PROFILE_0: Increases speed during 5 seconds reaching 600 rpm, then decelerates during 5 more seconds until it stops. PROFILE_1: Not used PROFILE_2: Not used PROFILE_3: Not used <table border="1"> <tr> <th>First combo box</th><th>Second combo box</th></tr> <tr> <td>REAGENT_DRAWER_1</td><td>PROFILE_0 ...</td></tr> <tr> <td>REAGENT_DRAWER_2</td><td>PROFILE_4</td></tr> </table>	First combo box	Second combo box	REAGENT_DRAWER_1	PROFILE_0 ...	REAGENT_DRAWER_2	PROFILE_4
First combo box	Second combo box						
REAGENT_DRAWER_1	PROFILE_0 ...						
REAGENT_DRAWER_2	PROFILE_4						
stop	Cancel/stop the stirring process.						

Table 17. List of options in Reagent Drawers – Drawers tab

Fluidic Drawers tab

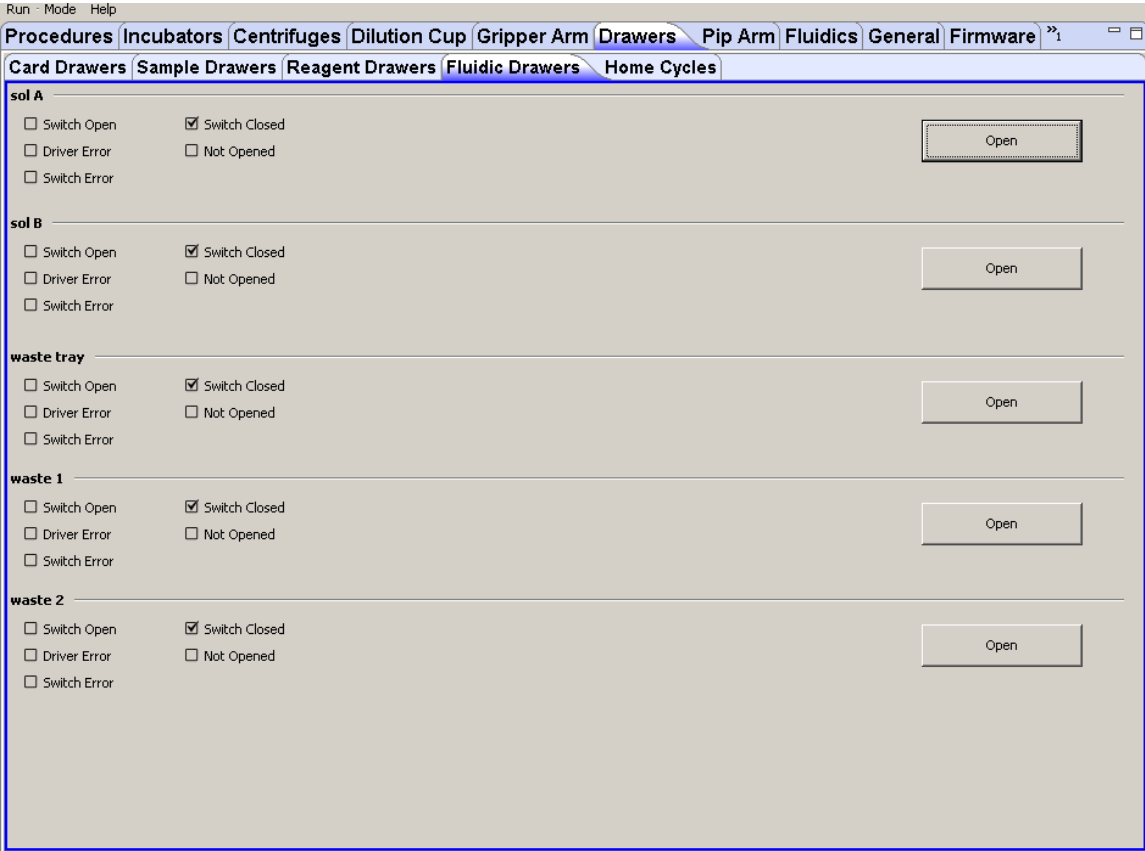


Figure 71. Fluidic Drawers – Drawers tab in Technical Service software

Option	Description
sol A, sol B, waste tray, waste 1 and waste 2	
Switch Open	Indicates that the drawer is open.
Switch Closed	Indicates that the drawer is closed.
Switch Error	Indicates that an error condition (is opened and closed at the same time or not opened nor closed at the same time).
Driver Error	Indicates that an error happened when communicating with the lock auxiliary peripheral, through I2C bus.
Not opened	Indicates that the lock has not opened within 1 second after an open command has been executed.
Open	Open the selected drawer.

Table 18. List of options in Fluidics Drawers – Drawers tab

Home Cycles

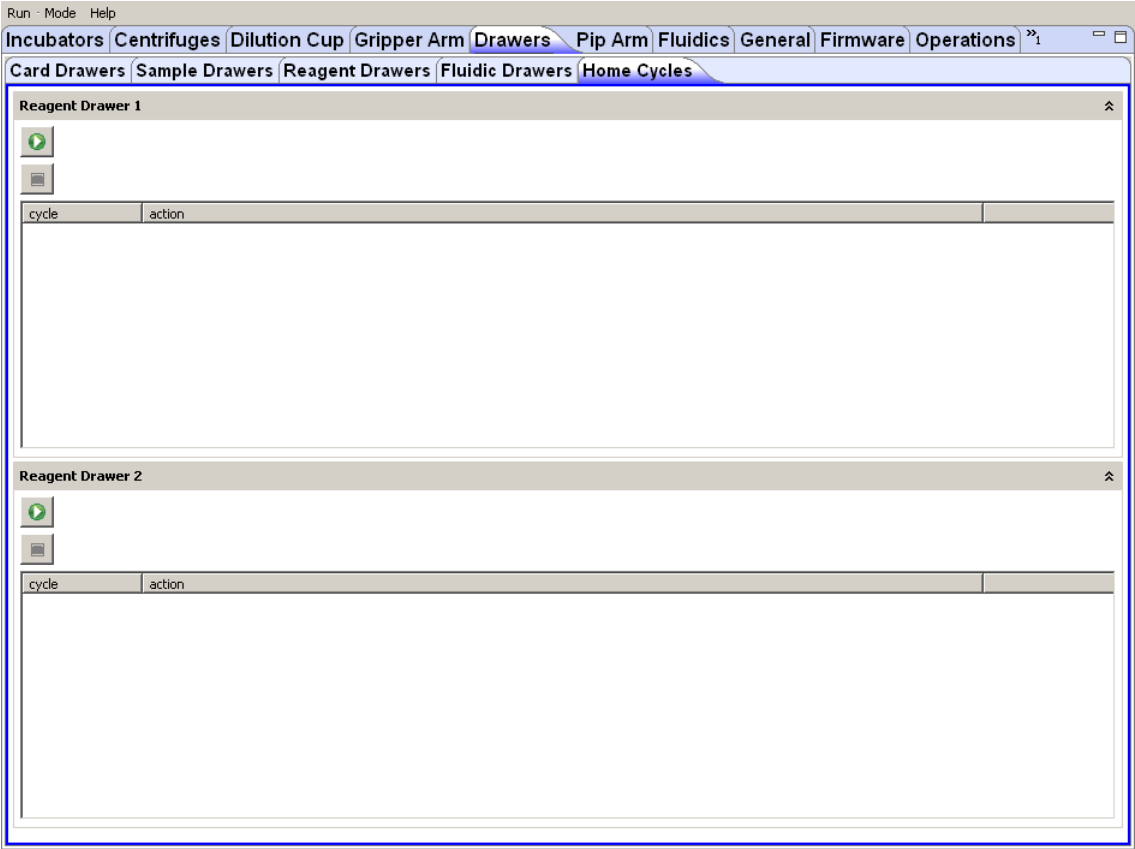


Figure 72. Home Cycles – Drawers tab in Technical Service software

Option	Description
Reagent Drawer 1, 2	
	Perform home cycles of the reagent drawer selected (rack 2).
	Stop the home cycles of the reagent drawer selected (rack 2).

Table 19. List of options in Home Cycles – Drawers tab

3.2.7 Pip Arm tab

Pip Arm tab includes all the parameters and functions related to the pipetting arm. It contains 7 tabs: Movement, Presence Detector, Level Detector, BCR, CPD, Home Cycles and Random Position.

Movement tab

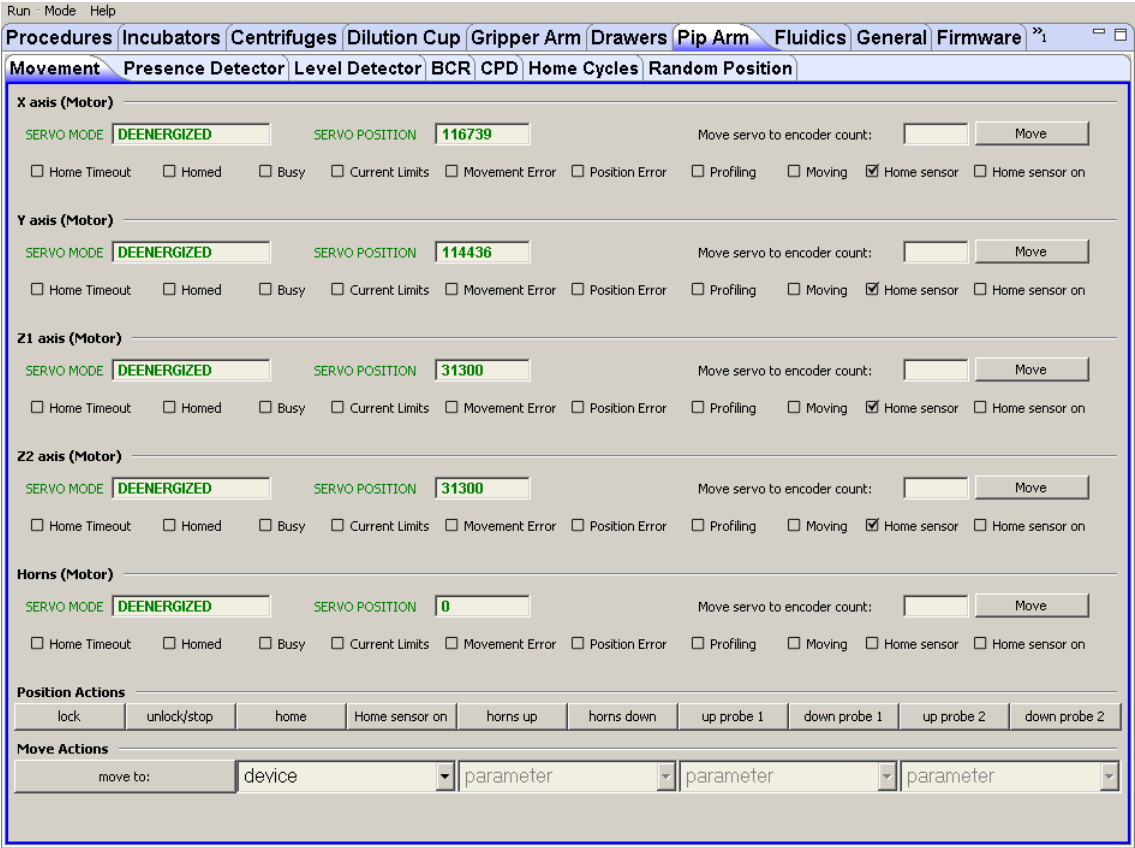


Figure 73. Movement - PIP Arm tab in Technical Service software

Option	Description
X axis, Y axis, Z1 axis, Z2 axis and Horns (Motor)	
SERVO MODE	Describes motor movement. There are 4 possible status: - Position Control: motor energized, maintaining the actual position. - Deenergized: motor deenergized. - Integral: performing a movement maintaining the speed but without a profile (initial position, acceleration, maximum speed and final position). It is like a home movement. - Trapezoidal: performing a movement with a profile (initial position, acceleration, maximum speed and final position).
SERVO POSITION	Indicates motor encoder counts.
Move	The selected axis moves to the number of counts written in the <i>Move servo to encoder count</i> text box.
Home Timeout	Home process has been aborted due to exceeding an encoder count timeout with no change detected at the home sensor.
Homed	Home position has been detected.

Option	Description			
X axis, Y axis, Z1 axis, Z2 axis and Horns (Motor)				
Busy	Servo is executing a process.			
Current Limits	Current limit has been detected on Servo motor. The hardware automatically unconfigures servo-driver.			
Movement Error	Profiled movement has been aborted due to detection of non-correct movement of the motor. Hardware automatically de-energizes servo.			
Position Error	Servo is in Position Control Mode and the position error to the actual target exceeds the servo EEPROM settings.			
Profiling	Servo executing trapezoidal movement / integral velocity movement.			
Moving	Movement is detected on the servo encoder (encoder position has changed from last time position was checked).			
Home sensor	Indicates, during a home process, that the home position has been detected.			
Home sensor on	During a home process, it is activated when the home sensor is on.			
Position Actions				
lock	Lock a servo motor (X, Y, Z1, Z2, Horns or all of them).			
unlock/stop	Unlock a servo motor (X, Y, Z1, Z2, Horns or all of them) or to stop the current movement.			
home	Perform a home movement. It is possible to select X, Y, Z1, Z2, horns or all of them.			
Home sensor on	Activate home optical barriers.			
horns up	Move the horns to the upper position.			
horns down	Move the horns to the lower position.			
up probe 1	Move up probe 1.			
down probe 1	Move down probe 1 until adjustment position (depending on where the PIP module is).			
up probe 2	Move up probe 2.			
down probe 2	Move down probe 2 until adjustment position (depending on where the PIP module is).			
Move Actions				
move to	Button to move the PIP arm to any of the following allowed positions:			
	First combo box	Second combo box	Third combo box	Fourth combo box
	SAMPLE_DRAWER_1 SAMPLE_DRAWER_2 SAMPLE_DRAWER_3 SAMPLE_DRAWER_4	RACK_1, RACK_2	POS_1 ... POS_12	PROBE_1, PROBE_2
	REAGENT_DRAWER_1 REAGENT_DRAWER_2		POS_1 ... POS_9	
	INCUBATOR_1 INCUBATOR_2 INCUBATOR_3 INCUBATOR_4	SLOT_1... SLOT_12	WELL_1... WELL_8	
	DILUTION_CUPS	PROBE_1, PROBE_2	ASPIRE DISPENSE	INACTIVE

Table 20. List of options in Movement - PIP Arm tab

Presence Detector tab

There are 3 subtabs inside Presence Detector tab: Reagent drawers, Sample drawers and Presence Detector tabs.

Reagent drawers tab

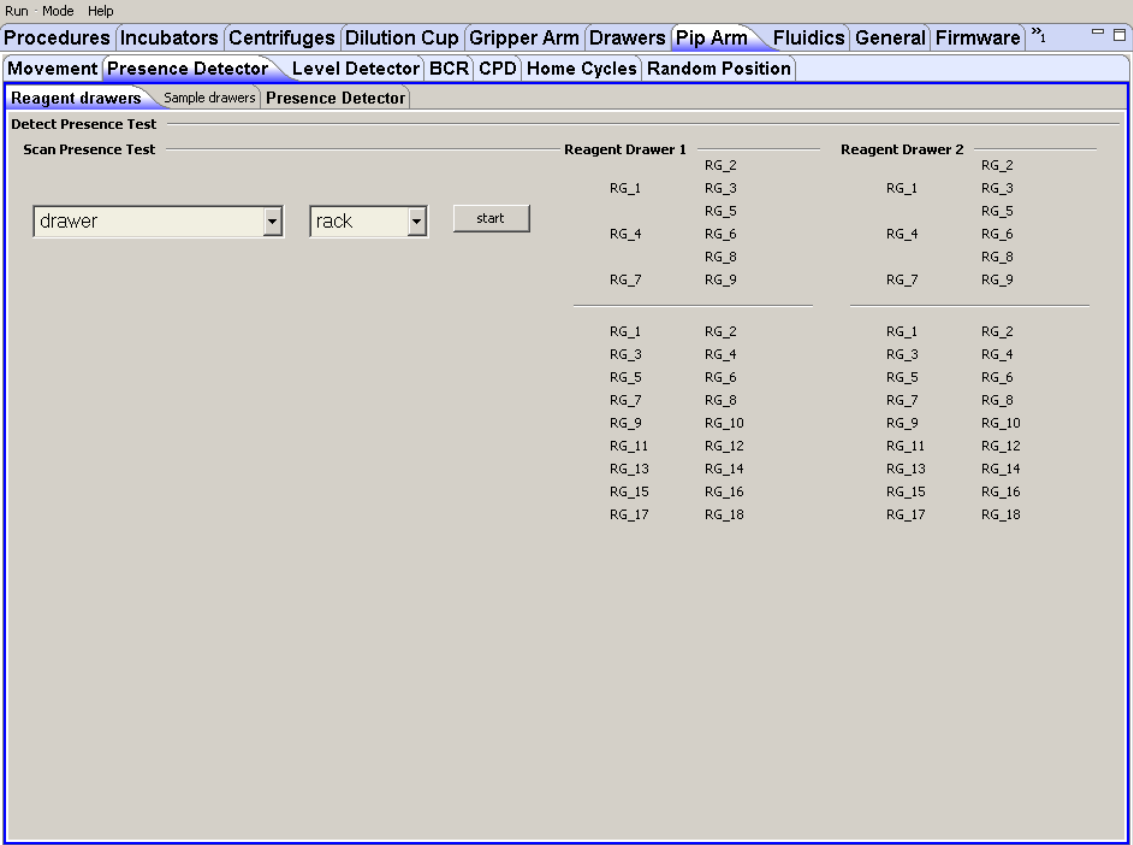


Figure 74. Reagent drawers – Presence Detector – Pip Arm tab in Technical Service software

Option	Description						
Detect Presence Test							
Scan Presence Test							
start	Initiate the detection process in the reagent rack selected. Allowed positions are: <table><tr><th>First combo box</th><th>Second combo box</th></tr><tr><td>REAGENT_DRAWER_1</td><td>RACK_1</td></tr><tr><td>REAGENT_DRAWER_2</td><td>RACK_2</td></tr></table>	First combo box	Second combo box	REAGENT_DRAWER_1	RACK_1	REAGENT_DRAWER_2	RACK_2
First combo box	Second combo box						
REAGENT_DRAWER_1	RACK_1						
REAGENT_DRAWER_2	RACK_2						
Reagent Drawer 1, 2	Indicates the presence of a bottle/vial in a position of each drawer of the reagents positions.						

Table 21. List of options in Reagent drawers – Presence Detector – Pip Arm tab

Sample drawers tab

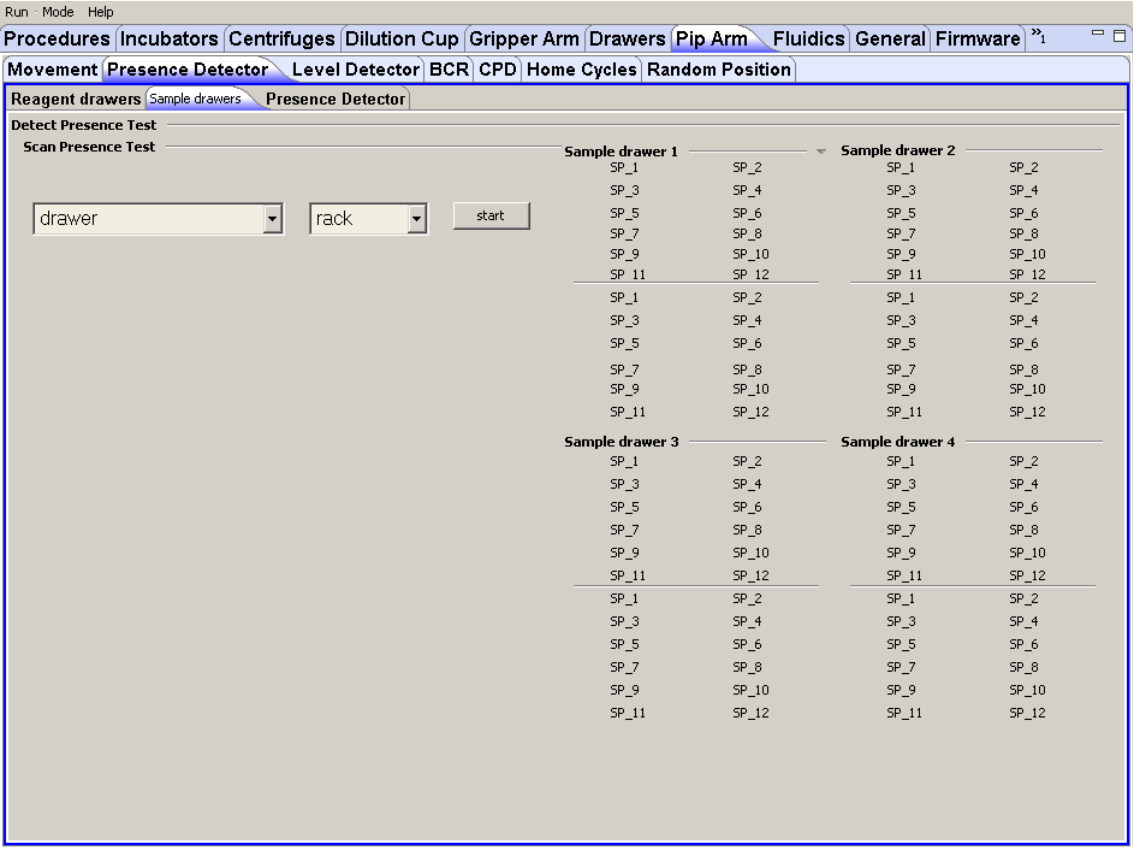


Figure 75. Sample drawers – Presence Detector – Pip Arm tab in Technical Service software

Option	Description				
Detect Presence Test					
Scan Presence Test					
start	Initiate the detection process in the sample rack selected. Allowed positions are:<table><tr><th>First combo box</th><th>Second combo box</th></tr><tr><td>SAMPLE_DRAWER_1 SAMPLE_DRAWER_2 SAMPLE_DRAWER_3 SAMPLE_DRAWER_4</td><td>RACK_1 ... RACK_2</td></tr></table>	First combo box	Second combo box	SAMPLE_DRAWER_1 SAMPLE_DRAWER_2 SAMPLE_DRAWER_3 SAMPLE_DRAWER_4	RACK_1 ... RACK_2
First combo box	Second combo box				
SAMPLE_DRAWER_1 SAMPLE_DRAWER_2 SAMPLE_DRAWER_3 SAMPLE_DRAWER_4	RACK_1 ... RACK_2				
Sample drawer 1, 2, 3, 4	Indicates the presence of a tube in a position of each drawer of the samples positions. If detected, it returns the diameter of the tube (-1 if diameter can not be calculated).				

Table 22. List of options in Sample drawers – Presence Detector – Pip Arm tab

Presence Detector tab

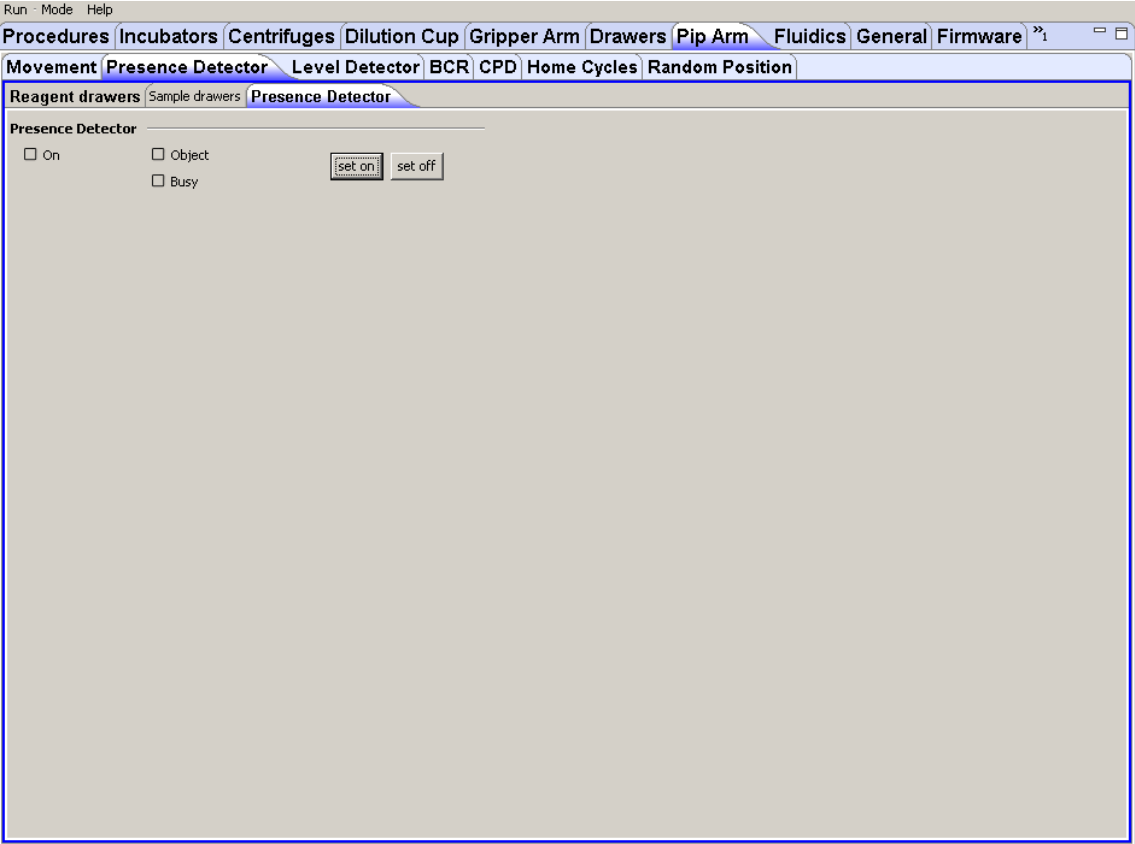


Figure 76. Presence Detector – Presence Detector – Pip Arm tab in Technical Service software

Option	Description
On	Indicates that the presence detector laser beam is on.
Object	Indicates that an object is detected in front of the laser beam.
Busy	Indicates that presence detector is executing a scanning.
set on	Activate the laser beam.
set off	Deactivate the laser beam.

Table 23. List of options in Presence Detector – Presence Detector – Pip Arm tab

Level Detector tab

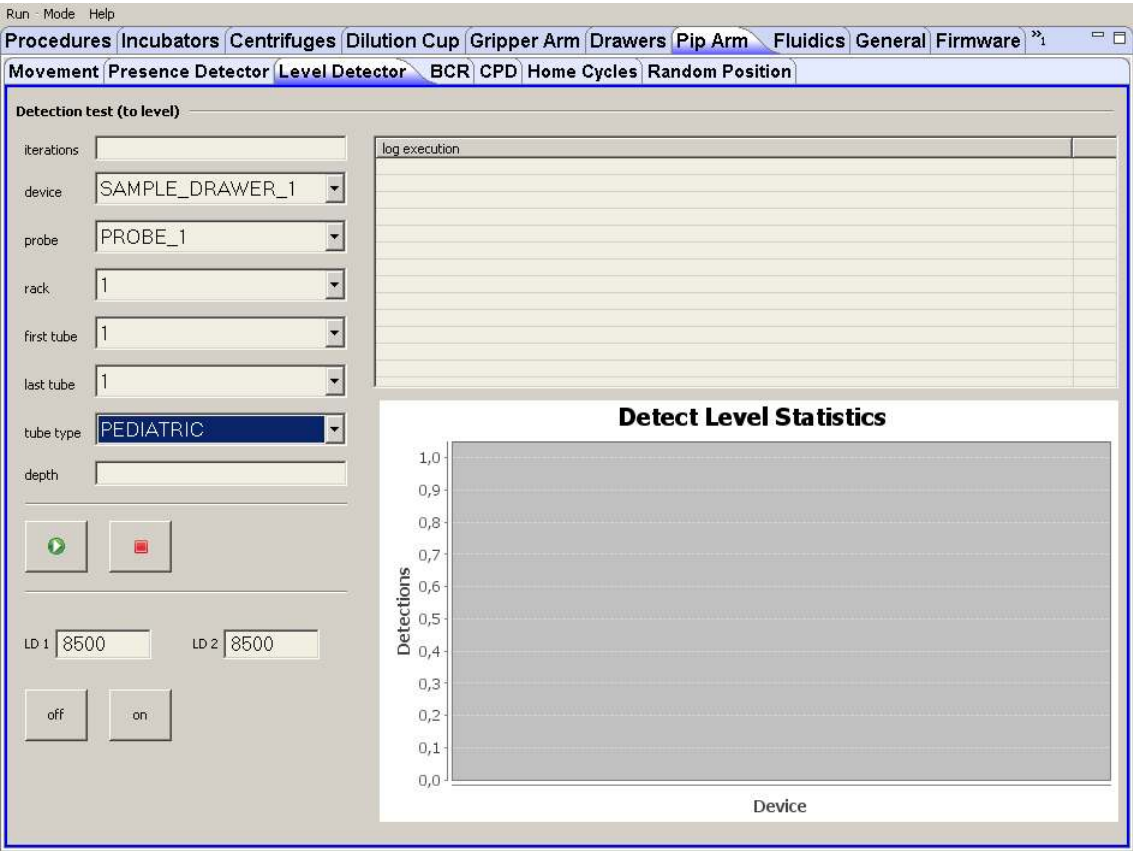


Figure 77. Level Detector – Pip Arm tab in Technical Service software

Option	Description																																						
Detection test (to level)																																							
iterations	Text box to enter the number of times to execute the level detection test.																																						
device, probe, rack/position, first tube, last tube	Combo box to select the position where to perform the level detection test. Allowed positions are: <table><tr><th>device</th><th>probe</th><th>rack/position</th><th>first tube</th><th>last tube</th></tr><tr><td>REAGENT_DRAWER_1</td><td rowspan="8">PROBE_1 PROBE_2</td><td>RACK_1</td><td>1..9</td><td>1..9</td></tr><tr><td></td><td>RACK_2</td><td>1..18</td><td>1..18</td></tr><tr><td>REAGENT_DRAWER_2</td><td>RACK_1</td><td>1..9</td><td>1..9</td></tr><tr><td></td><td>RACK_2</td><td>1..18</td><td>1..18</td></tr><tr><td>SAMPLE_DRAWER_1</td><td rowspan="3">RACK_1 RACK_2</td><td rowspan="3">1..12</td><td rowspan="3">1..12</td></tr><tr><td>SAMPLE_DRAWER_2</td></tr><tr><td>SAMPLE_DRAWER_3</td></tr><tr><td>SAMPLE_DRAWER_4</td><td></td><td></td><td></td><td></td></tr><tr><td>DILUTION_CUPS</td><td></td><td>1,2</td><td colspan="2">INACTIVE</td></tr></table>	device	probe	rack/position	first tube	last tube	REAGENT_DRAWER_1	PROBE_1 PROBE_2	RACK_1	1..9	1..9		RACK_2	1..18	1..18	REAGENT_DRAWER_2	RACK_1	1..9	1..9		RACK_2	1..18	1..18	SAMPLE_DRAWER_1	RACK_1 RACK_2	1..12	1..12	SAMPLE_DRAWER_2	SAMPLE_DRAWER_3	SAMPLE_DRAWER_4					DILUTION_CUPS		1,2	INACTIVE	
device	probe	rack/position	first tube	last tube																																			
REAGENT_DRAWER_1	PROBE_1 PROBE_2	RACK_1	1..9	1..9																																			
		RACK_2	1..18	1..18																																			
REAGENT_DRAWER_2		RACK_1	1..9	1..9																																			
		RACK_2	1..18	1..18																																			
SAMPLE_DRAWER_1		RACK_1 RACK_2	1..12	1..12																																			
SAMPLE_DRAWER_2																																							
SAMPLE_DRAWER_3																																							
SAMPLE_DRAWER_4																																							
DILUTION_CUPS		1,2	INACTIVE																																				
tube type	Select level detection sensitivity: NORMAL or PEDIATRIC																																						
depth	This box only appears if PEDIATRIC tube type is selected. Indicates the maximum depth that the probe can move down when entering into the tube (the value is the distance in mm from the standard adjustment Z position).																																						
	Initiate xx times (indicated in iterations field) a test of level detection.																																						
	Stop detection level test.																																						

Option	Description
Detection test (to level)	
LD 1	Box where the level detection counts of the probe 1 are displayed.
LD 2	Box where the level detection counts of the probe 2 are displayed.
off	Deactivate the level detection system.
on	Activate the level detection system.
Option	Description
Detection test (to level)	
log execution	It details the steps of the level detection test.
Detect Level Statistics	Graph where the statistics of the level detections are shown.
	Possible results are:
	Success: It has been detected level.
	Fail: It has not been detected level.
	Error: Error while detecting level.
	Impact: It has been detected an impact.

Table 24. List of options in Level Detector – Pip Arm tab

BCR tab

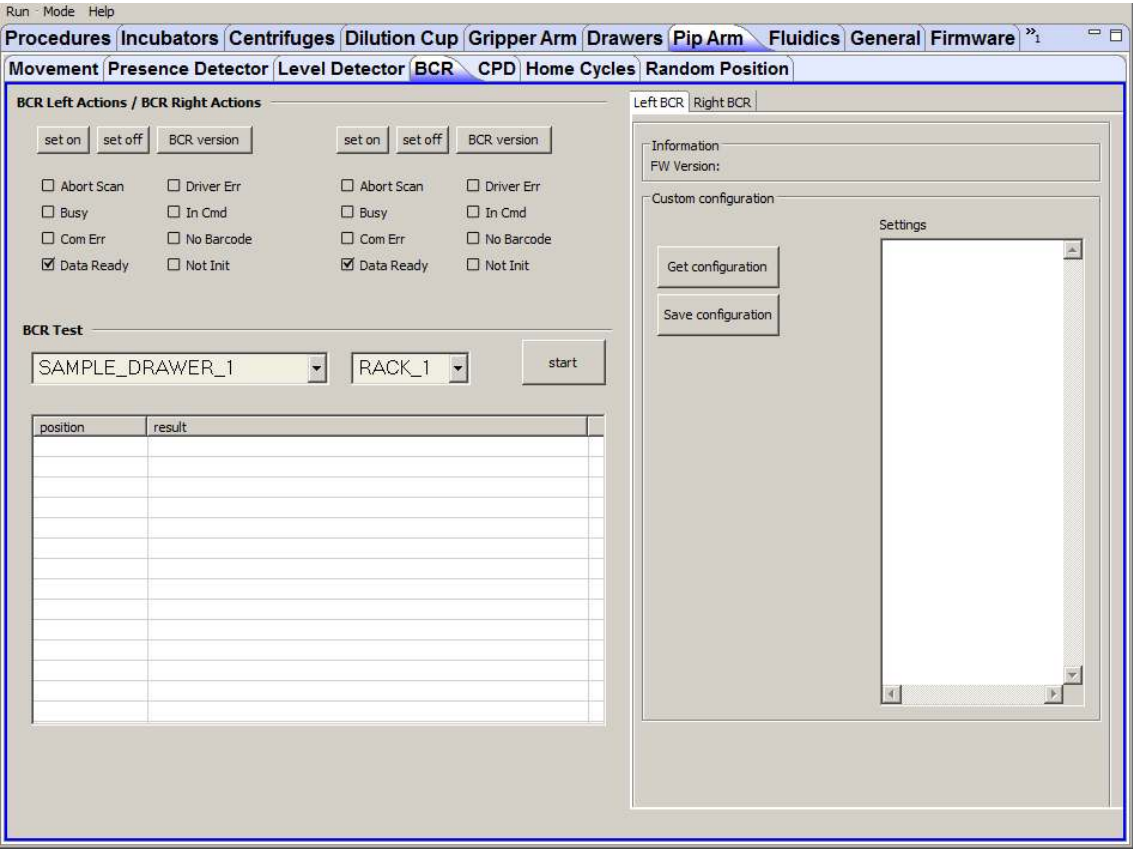


Figure 78. BCR – Pip Arm tab in Technical Service software

Option	Description
BCR Left Actions / BCR Right Actions	
set on	Activate barcode reader.
set off	Deactivate the barcode reader.
BCR version	Returns the barcode reader version.
Abort Scan	Indicates that a scanning operation has been aborted.
Busy	Barcode reader is on.
Com Err	Indicates a barcode reader communication error.
Data Ready	Indicates that information obtained by the barcode reader is ready for being recovered by software.
Driver Err	Indicates a barcode reader communication error.
In Cmd	Indicates that there is communication between microcontroller and barcode reader.
No Barcode	No barcode is read.
Not Init	Barcode reader is not initialized.

Option	Description				
BCR Test					
start	Initiate the process of barcode reading. Previously, a detection test is performed to check where tubes, vials or bottles are placed. Allowed positions are: <table> <tr> <th>First combo box</th><th>Second combo box</th></tr> <tr> <td> SAMPLE_DRAWER_1 SAMPLE_DRAWER_2 SAMPLE_DRAWER_3 SAMPLE_DRAWER_4 REAGENT_DRAWER_1 REAGENT_DRAWER_2 </td><td> RACK_1 RACK_2 </td></tr> </table>	First combo box	Second combo box	SAMPLE_DRAWER_1 SAMPLE_DRAWER_2 SAMPLE_DRAWER_3 SAMPLE_DRAWER_4 REAGENT_DRAWER_1 REAGENT_DRAWER_2	RACK_1 RACK_2
First combo box	Second combo box				
SAMPLE_DRAWER_1 SAMPLE_DRAWER_2 SAMPLE_DRAWER_3 SAMPLE_DRAWER_4 REAGENT_DRAWER_1 REAGENT_DRAWER_2	RACK_1 RACK_2				
text box	Results of barcode reading process are displayed. In the first column (<i>position</i>) is shown the position of the vial/bottle/tube detected and in the second column (<i>result</i>) is shown the barcode.				
Settings for BCR Left / BCR Right - Information					
FW version	Firmware version of each barcode reader device.				
Settings for BCR Left / BCR Right – Custom configuration					
Settings	Textbox where the customized left/right BCR configuration is shown. Modifications related to this configuration should be directly written on the textbox. The resulting customized configuration will be used to modify the default BCR configuration.				
Get configuration	Reads and shows the saved parameters related to the customized configuration of the selected BCR device.				
Save configuration	Writes and saves the parameters detailed in the <i>Settings textbox</i> as the customized configuration for the selected BCR device.				

Table 25. List of options in BCR – Pip Arm tab

CPD tab

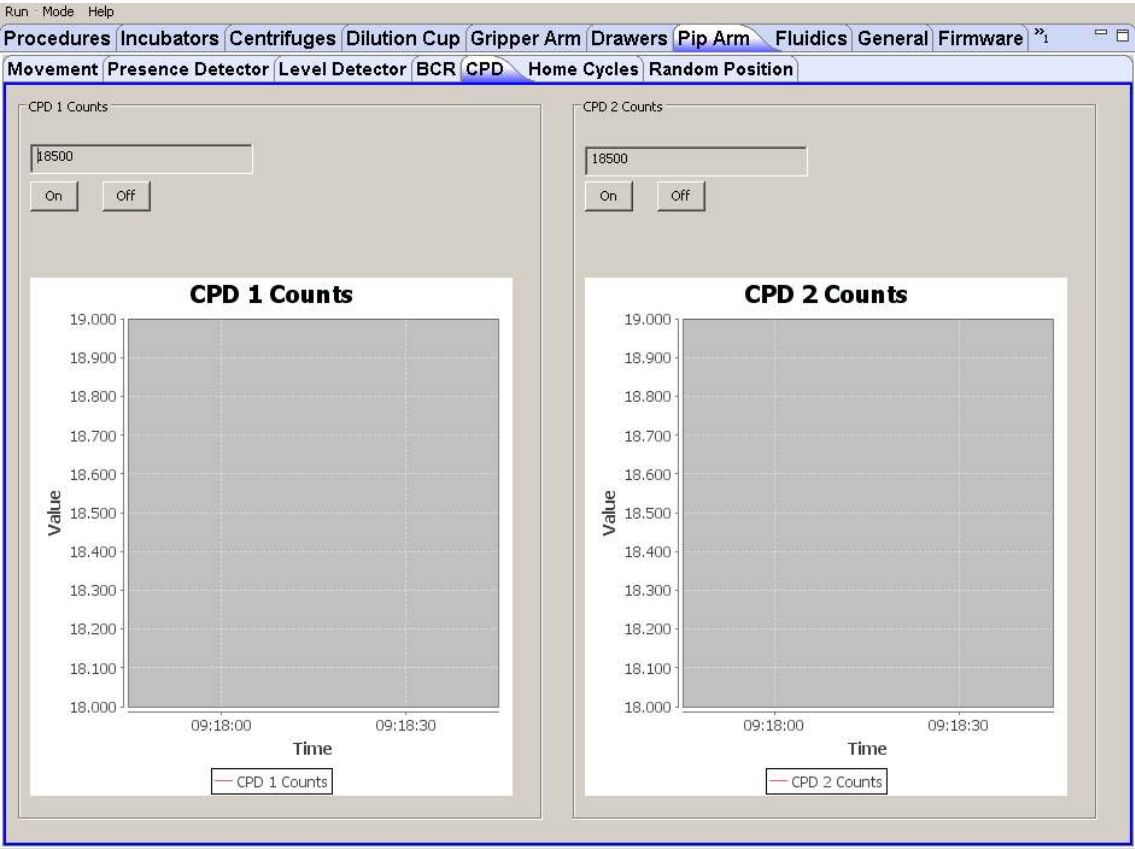


Figure 79. CPD – Pip Arm tab in Technical Service software

Option	Description
CPD 1 / CPD 2 Counts	
On	Activates CPD sensor, showing counts in the text box and in the graphs at the bottom of the screen.
Off	Deactivates reading counts process of CPD sensor.

Table 26. List of options in CPD – Pip Arm tab

Home Cycles tab

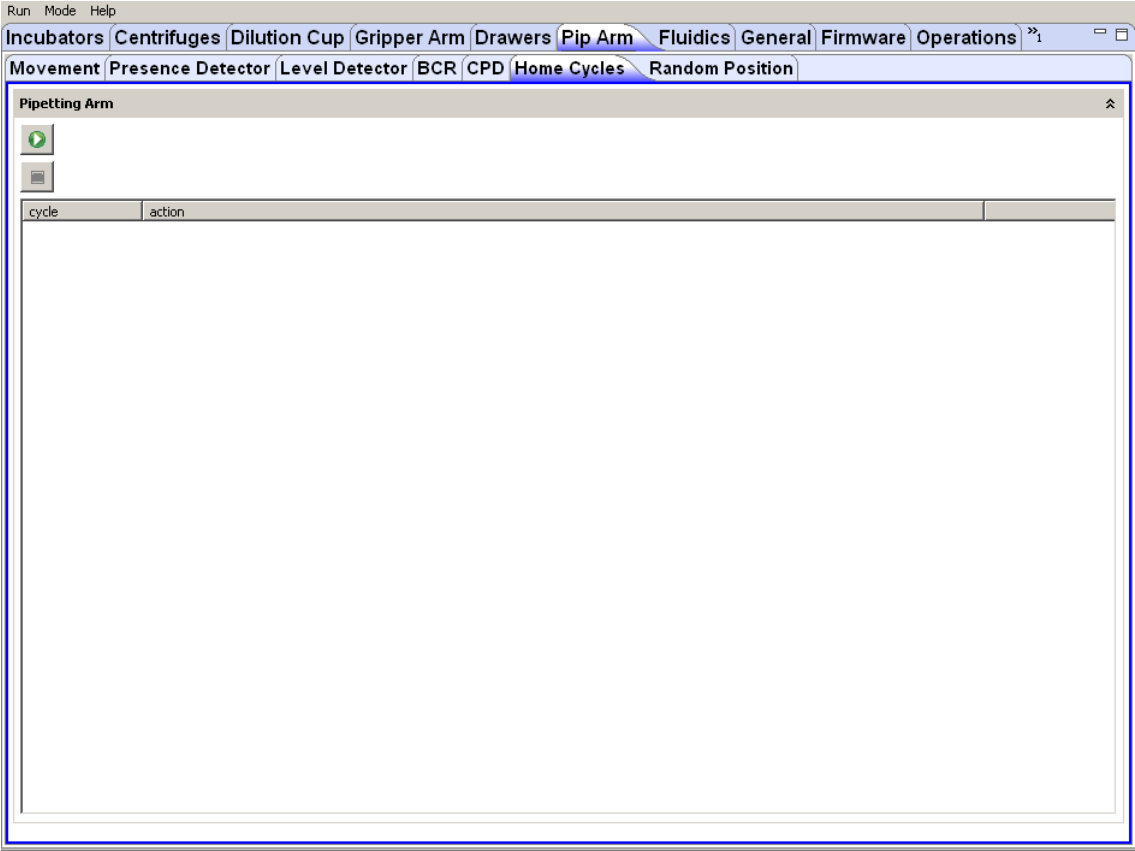


Figure 80. Home Cycles – Pip Arm tab in Technical Service software

Option	Description
Pipetting Arm	
	Perform home cycles of the pipetting robot.
	Stop the home cycles of the pipetting robot.

Table 27. List of options in Home Cycles – Pip Arm tab

Random Position tab



Figure 81. Random Position – Pip Arm tab in Technical Service software

Option	Description
Pipetting Arm	
	Perform random position movements of the pipetting robot.
	Stop random position movements of the pipetting robot.

Table 28. List of options in Random Position – Pip Arm tab

3.2.8 Fluidics tab

Fluidic tab includes all the parameters and functions related to the fluidics.

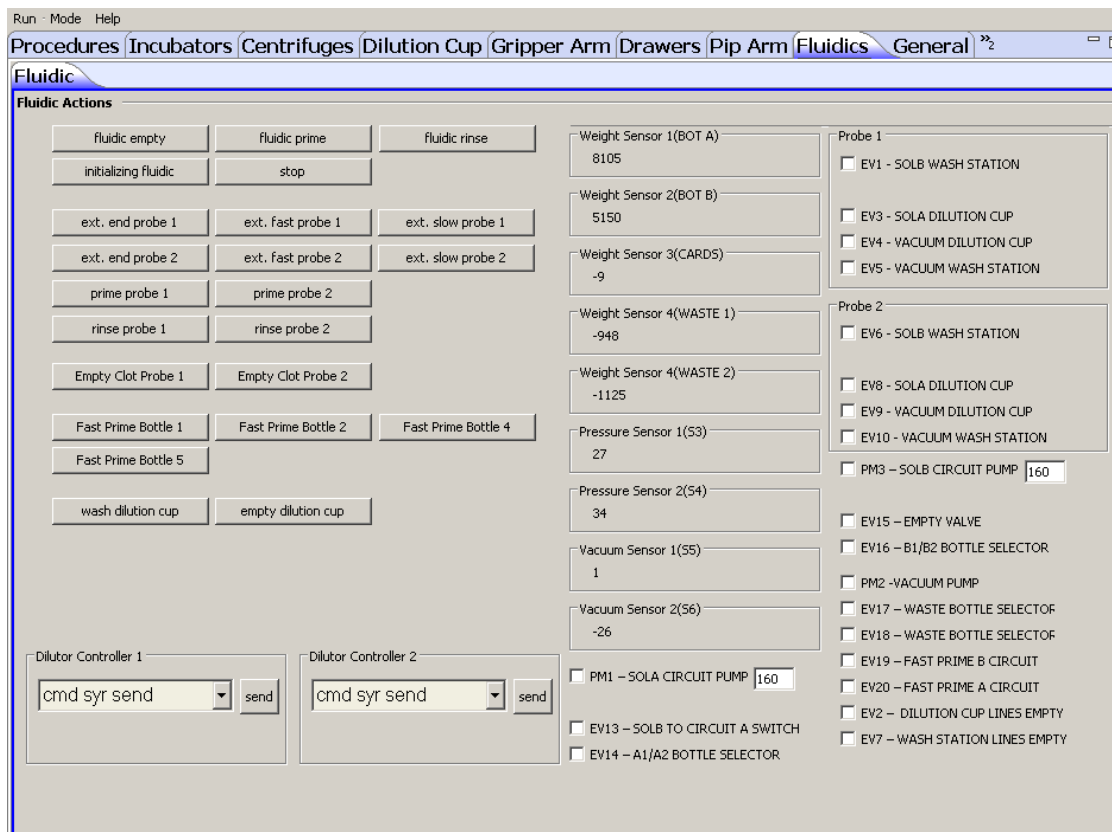


Figure 82. Fluidic tab in Technical Service software

Option	Description
Fluidic Actions	
fluidic empty	Empty all the fluidic circuit.
fluidic prime	Prime fluidic circuit between manifold, probes washer and dilution cup.
fluidic rinse	Wash with solution B fluidic circuit between manifold, probes washer and dilution cup.
initializing fluidic	Initialize the fluidics (home of the PIP arm and dilutors initialization).
stop	Stop the process that is being performed.
ext. end probe 1	External wash of the end of probe 1 (fluidic circuit must be primed).
ext. fast probe 1	External fast wash of probe 1 (fluidic circuit must be primed).
ext. slow probe 1	External slow wash of probe 1 (fluidic circuit must be primed).
ext. end probe 2	External wash of the end of probe 2 (fluidic circuit must be primed).
ext. fast probe 2	External fast wash of probe 2 (fluidic circuit must be primed).
ext. slow probe 2	External slow wash of probe 2 (fluidic circuit must be primed).
prime probe 1	Prime with solution A the fluidic circuit of probe 1 (fluidic circuit must be primed).
prime probe 2	Prime with solution A the fluidic circuit of probe 2 (fluidic circuit must be primed).
rinse probe 1	Rinse with solution B the fluidic circuit of probe 1 (fluidic circuit must be primed).

Option	Description
Fluidic Actions	
rinse probe 2	Rinse with solution B the fluidic circuit of probe 2 (fluidic circuit must be primed).
wake up probe 1	Empty and prime probe 1 circuit (same procedure as the one performed during routine after an inactivity wash)
wake up probe 2	Empty and prime probe 2 circuit (same procedure as the one performed during routine after an inactivity wash)
Empty Clot Probe 1	Routine to try to remove a clot from probe 1
Empty Clot Probe 2	Routine to try to remove a clot from probe 2
Fast Prime Bottle 1	Fill with liquid fluidic path between bottle 1 and waste bottle/external drain
Fast Prime Bottle 2	Fill with liquid fluidic path between bottle 2 and waste bottle/external drain
Fast Prime Bottle 3	Fill with liquid fluidic path between bottle 3 and external drain (present if dual mode is activated)
Fast Prime Bottle 4	Fill with liquid fluidic path between bottle 4 and external drain (present if dual mode is activated)
wash dilution cup	Wash with solution A the dilution cup (fluidic circuit must be primed).
empty dilution cup	Empty the dilution cup (fluidic circuit should be primed previously).
send - Dilutor Controller 1 / 2	Send the command selected in the text box to the dilutor. INIT: initialize the dilutor PRIME_PROBE_HALF: 5 cycles of aspiration and dispensation of 1 ml from from the bottle A to the probe. VENT_PROBE_HALF: 5 cycles of aspiration and dispensation of 1 ml from the probe to the bottle A. ASPIRATE_1ml_FROM_PROBE: perform an aspiration of 1ml from the probe, rotating properly the valve. ASPIRATE_1ml_FROM_SOL_A: perform an aspiration of 1ml from the solution A bottle, rotating properly the valve. DISPENSE_TO_PROBE: perform a dispensation of the liquid that there is in the syringe to the probe, rotating properly the valve. ASPIRATE_GAP: perform an aspiration of a gap ABORT: abort the movement of the dilutor
Weight Sensor 1 (BOT A)	Return the value (in ml) of the weight of the load cell placed in the solution A drawer.
Weight Sensor 2 (BOT B)	Return the value (in ml) of the weight of the load cell placed in the solution B drawer.
Weight Sensor 3 (CARDS):	Return the value (cards number) of the weight of the load cell placed in the cards disposal tray drawer.
Weight Sensor 4 (WASTE 1):	Return the value (in ml) of the weight of the load cell placed in the waste 1 drawer.
Weight Sensor 5 (WASTE 2):	Return the value (in ml) of the weight of the load cell placed in the waste 2 drawer.
Pressure Sensor 1 (S3):	Return the value (in mbar) of pressure sensor S3.
Pressure Sensor 2 (S4):	Return the value (in mbar) of pressure sensor S4.
Vacuum Sensor 1 (S5):	Return the value (in mbar) of vacuum sensor S5.
Vacuum Sensor 2 (S6):	Return the value (in mbar) of vacuum sensor S6.
PM1 – SOLA CIRCUIT PUMP	Activates PM1 dispensation pump (solution A). There is a textbox where the working pressure can be modified. Allowed values: 160 mbar (default) 180 mbar

Option	Description
Fluidic Actions	
EV13 – SOLB TO CIRCUIT A SWITCH	Activates EV13.
EV14 – A1/A2 BOTTLE SELECTOR	Activates EV14.
EV1 – SOLB WASH STATION	Activates EV1.
EV3 – SOLA DILUTION CUP	Activates EV3.
EV4 – VACUUM DILUTION CUP	Activates EV4.
EV5 – VACUUM WASH STATION	Activates EV5.
EV6 – SOLB WASH STATION	Activates EV6.
EV8 – SOLA DILUTION CUP	Activates EV8.
EV9 – VACUUM DILUTION CUP	Activates EV9.
EV10 – VACUUM WASH STATION	Activates EV10.
PM3 – SOLB CIRCUIT PUMP	Activates PM3 dispensation pump (solution B). There is a textbox where the working pressure can be modified. Allowed values: 160 mbar (default) 180 mbar
EV15 – EMPTY VALVE	Activates EV15.
EV16 – B1/B2 BOTTLE SELECTOR	Activates EV16.
PM2 – VACUUM PUMP	Activates PM2 aspiration pump.
EV17 – WASTE BOTTLE SELECTOR	Activates EV17.
EV18 – WASTE BOTTLE SELECTOR	Activates EV18.
EV19 – FAST PRIME B CIRCUIT	Activates EV19.
EV20 – FAST PRIME A CIRCUIT	Activates EV20.
EV2 – DILUTION CUP LINES EMPTY	Activates EV2.
EV7 – WASH STATION LINES EMPTY	Activates EV7.

Table 29. List of options in Fluidic tab

3.2.9 General tab

General Module tab includes all the parameters and functions related to the general module as the general LED, drawer lights, temperatures, etc.

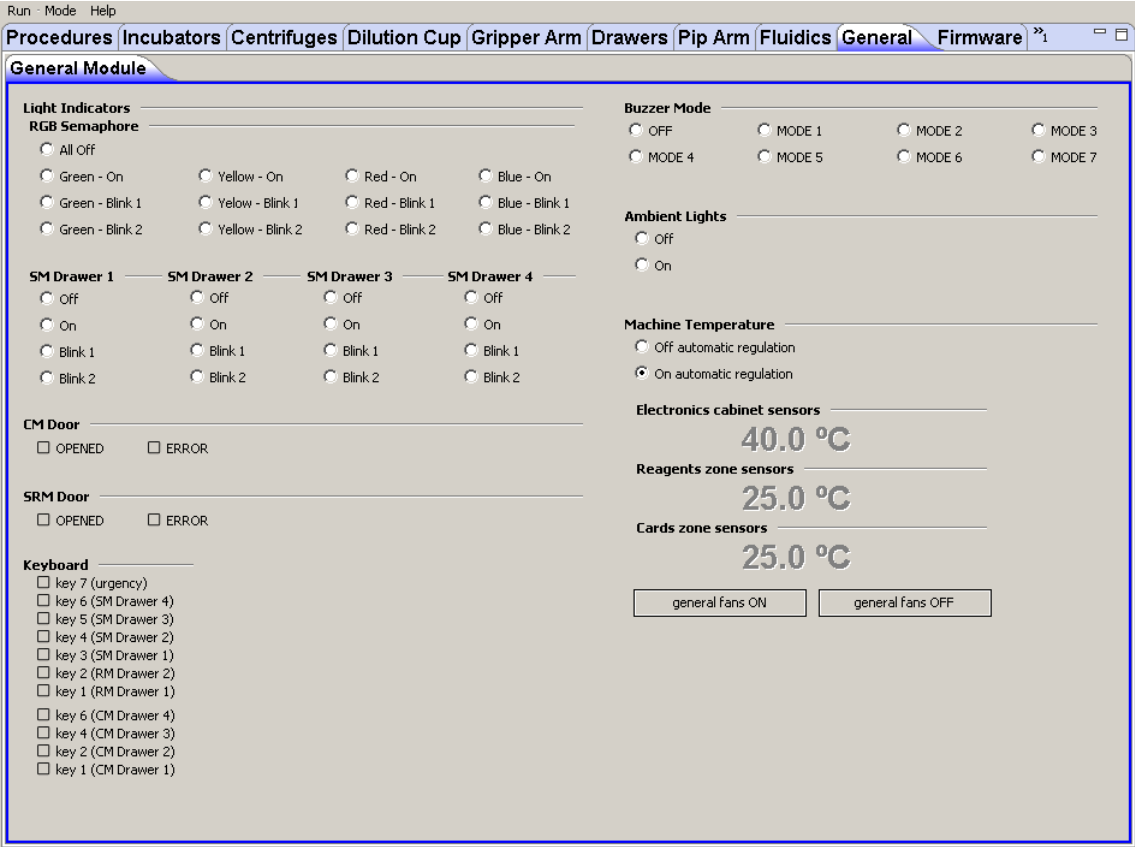


Figure 83. General Module tab in Technical Service software

Option	Description
Light Indicators	
RBG Semaphore	
All Off	Deactivates any light in the general LED.
Green – On	Activates the green light in the general LED.
Green – Blink 1	Activates the blink frequency 1 with the green light in the general LED.
Green – Blink 2	Activates the blink frequency 2 with the green light in the general LED.
Yellow – On	Activates the yellow light in the status LED.
Yellow – Blink 1	Activates the blink frequency 1 with the yellow light in the general LED.
Yellow – Blink 2	Activates the blink frequency 2 with the yellow light in the general LED.
Red – On	Activates the red light in the general LED.
Red – Blink 1	Activates the blink frequency 1 with the red light in the general LED.
Red – Blink 2	Activates the blink frequency 2 with the red light in the general LED.
Blue – On	Activates the blue light in the general LED.
Blue – Blink 1	Activates the blink frequency 1 with the blue light in the general LED.
Blue – Blink 2	Activates the blink frequency 2 with the blue light in the general LED.
SM Drawer (1, 2, 3, 4)	
Off	Switch off the light in the selected drawer indicator.
On	Switch on the light in the selected drawer indicator.
Blink 1	Activates the blink frequency 1 light in the selected drawer indicator.
Blink 2	Activates the blink frequency 2 light in the selected drawer indicator.

Option	Description
CM Door	
OPENED	Inform that intermediate door is opened.
ERROR	Inform that an error is detected in the intermediate door detector.
SRM Door	
OPENED	Inform that upper door is opened.
ERROR	Inform that an error is detected in the upper door detector.
Keyboard	
key 7 (urgency)	Inform that urgency button is pressed.
key 6 (SM Drawer 4)	Inform that sample drawer 4 button is pressed.
key 5 (SM Drawer 3)	Inform that sample drawer 3 button is pressed.
key 4 (SM Drawer 2)	Inform that sample drawer 2 button is pressed.
key 3 (SM Drawer 1)	Inform that sample drawer 1 button is pressed.
key 2 (RM Drawer 2)	Inform that reagent drawer 2 button is pressed.
key 1 (RM Drawer 1)	Inform that reagent drawer 1 button is pressed.
key 6 (CM Drawer 4)	Inform that card drawer 4 button is pressed.
key 4 (CM Drawer 3)	Inform that card drawer 3 button is pressed.
key 2 (CM Drawer 2)	Inform that card drawer 2 button is pressed.
key 1 (CM Drawer 1)	Inform that card drawer 1 button is pressed.
Buzzer Mode	
OFF	Deactivate the buzzer.
MODE 1	Activate buzzer with mode 1 sound.
MODE 2	Activate buzzer with mode 2 sound.
MODE 3	Activate buzzer with mode 3 sound.
MODE 4	Activate buzzer with mode 4 sound.
MODE 5	Activate buzzer with mode 5 sound.
MODE 6	Activate buzzer with mode 6 sound.
MODE 7	Activate buzzer with mode 7 sound.
Ambient Lights	
Off	Switch off ambient lights (PIP and CAT areas).
On	Switch on ambient lights.
Machine Temperature	
Off automatic regulation	Deactivate temperature regulation that switches on automatically the fans of the electronic cabinet when reaching 30°C.
On automatic regulation	Activate temperature regulation that switches on automatically the fans of the electronic cabinet when reaching 30°C.
Electronics cabinet sensors	Information about temperature in real time of the upper part of the rear part, where it is located the power supply, computer, PCA GEN-BCKPL59 and PCAs Motorcontrol.
Reagents zone sensors	Information about temperature in real time of the upper area of the instrument.
Cards zone sensors	Information about temperature in real time of the intermediate area of the instrument.
general fans ON	Activate the fans located in the backpanel.
general fans OFF	Deactivate the fans located in the backpanel.

Table 30. List of options in General Module tab

3.2.10 Firmware tab

Firmware tab includes option related to the firmwares of the PCAs. There are 2 tabs inside the Firmware tab: PCA Control and Eprom control.

PCA control tab

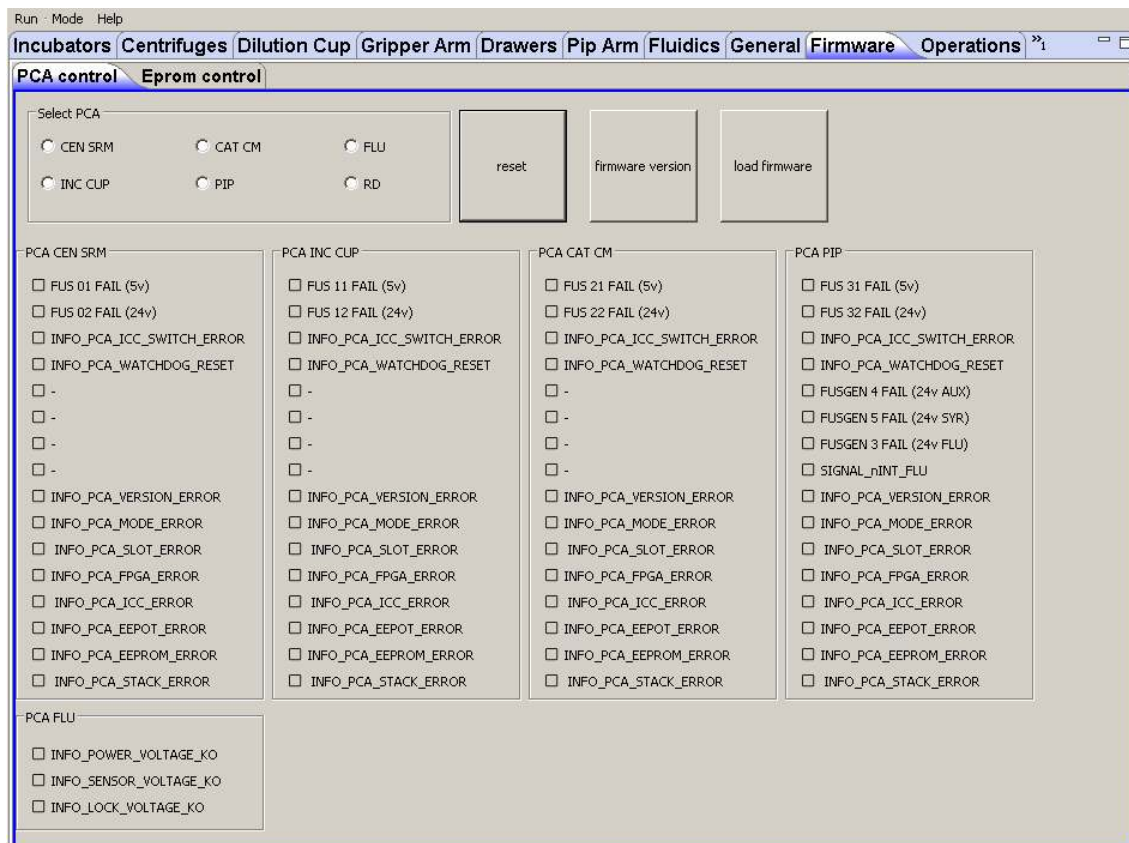


Figure 84. PCA control tab - Firmware tab in Technical Service software

Option	Description
Select PCA	
CEN SRM	Select PCA CEN-SRM Control
CAT CM	Select PCA CAT-CM Control
FLU	Select PCA FLU-CTRL59
INC CUP	Select PCA INC-CUP Control
PIP	Select PCA PIP Control
RD	Select PCA RD-IL421
reset	Perform a reset of the selected PCA
firmware version	Return firmware version of the selected PCA
load firmware	Load a firmware. It opens a new window.
PCA CEN SRM	
FUS 01 FAIL (5v)	Fus 01 (5V) of the PCA GEN-BCKPL59 fails
FUS 02 FAIL (24v)	Fus 02 (24V) of the PCA GEN-BCKPL59 fails
INFO_PCA_ICC_SWITCH_ERROR	PCA CEN-SRM Control is not communicating or not working properly with the external IIC switch.
INFO_PCA_WATCHDOG_RESET	Last PCA CEN-SRM Control reset was a watchdog reset.

Option	Description
PCA CEN SRM	
INFO_PCA_VERSION_ERROR	PCA HW version is erroneous and not compatible with the FW present on it.
INFO_PCA_MODE_ERROR	PCA operating mode is erroneous.
INFO_PCA_SLOT_ERROR	PCA CEN-SRM Control is inserted in an erroneous PCA GEN-BCKPL59 slot.
INFO_PCA_FPGA_ERROR	PCA FPGA programming procedure did not complete successfully.
INFO_PCA_ICC_ERROR	ICC error detected on the Control PCA ICC devices.
INFO_PCA_EEPOT_ERROR	PCA ICC non-volatile potentiometer is not working properly.
INFO_PCA_EEPROM_ERROR	Error during a PCA EEPROM read/write operation.
INFO_PCA_STACK_ERROR	PCA stack has grown over the security margin.
PCA INC CUP	
FUS 11 FAIL (5v)	Fus 11 (5V) of the PCA GEN-BCKPL59 fails
FUS 12 FAIL (24v)	Fus 12 (24V) of the PCA GEN-BCKPL59 fails
INFO_PCA_ICC_SWITCH_ERROR	PCA INC-CUP Control is not communicating or working properly with the external IIC switch.
INFO_PCA_WATCHDOG_RESET	Last PCA INC-CUP Control reset was a Watchdog reset.
INFO_PCA_VERSION_ERROR	PCA HW version is erroneous and not compatible with the FW present on it.
INFO_PCA_MODE_ERROR	PCA operating mode is erroneous.
INFO_PCA_SLOT_ERROR	PCA INC-CUP Control is inserted in an erroneous PCA GEN-BCKPL59 slot.
INFO_PCA_FPGA_ERROR	PCA FPGA programming procedure did not complete successfully.
INFO_PCA_ICC_ERROR	ICC error detected on the Control PCA ICC devices.
INFO_PCA_EEPOT_ERROR	PCA ICC non-volatile potentiometer is not working properly.
INFO_PCA_EEPROM_ERROR	Error during a PCA EEPROM read/write operation.
INFO_PCA_STACK_ERROR	PCA stack has grown over the security margin.
PCA CAT CM	
FUS 21 FAIL (5v)	Fus 21 (5V) of the PCA GEN-BCKPL59 fails.
FUS 22 FAIL (24v)	Fus 22 (24V) of the PCA GEN-BCKPL59 fails.
INFO_PCA_ICC_SWITCH_ERROR	PCA CAT-CM Control is not communicating or working properly with the external IIC switch.
INFO_PCA_WATCHDOG_RESET	Last PCA CAT-CM Control reset was a Watchdog reset.
INFO_PCA_VERSION_ERROR	PCA HW version is erroneous and not compatible with the FW present on it.
INFO_PCA_MODE_ERROR	PCA operating mode is erroneous.
INFO_PCA_SLOT_ERROR	PCA CAT-CM Control is inserted in an erroneous PCA GEN-BCKPL59 slot.
INFO_PCA_FPGA_ERROR	PCA FPGA programming procedure did not complete successfully.
INFO_PCA_ICC_ERROR	ICC error detected on the Control PCA ICC devices.
INFO_PCA_EEPOT_ERROR	PCA ICC non-volatile potentiometer is not working properly.
INFO_PCA_EEPROM_ERROR	Error during a PCA EEPROM read/write operation.
INFO_PCA_STACK_ERROR	PCA stack has grown over the security margin.

Option	Description
PCA PIP	
FUS 31 FAIL (5v)	Fus 01 (5V) of the PCA GEN-BCKPL59 fails.
FUS 32 FAIL (24v)	Fus 02 (24V) of the PCA GEN-BCKPL59 fails.
INFO_PCA_ICC_SWITCH_ERROR	Control_PCA is not communicating or working properly with the external IIC switch.
INFO_PCA_WATCHDOG_RESET	Last Control_PCA reset was a Watchdog reset.
FUSGEN 4 FAIL (24v AUX)	FUSGEN4 (24V) of the PCA GEN-BCKPL59 fails.
FUSGEN 5 FAIL (24v SYR)	FUSGEN5 (24V) of the PCA GEN-BCKPL59 fails.
FUSGEN 3 FAIL (24v FLU)	FUSGEN3 (24V) of the PCA GEN-BCKPL59 fails.
SIGNAL_nINT_FLU	Signal nINT_FLU (auxiliary signal coming from fluidics module) is active.
INFO_PCA_VERSION_ERROR	PCA HW version is erroneous and not compatible with the FW present on it.
INFO_PCA_MODE_ERROR	PCA operating mode is erroneous.
INFO_PCA_SLOT_ERROR	PCA PIP Control is inserted in an erroneous PCA GEN-BCKPL59 slot.
INFO_PCA_FPGA_ERROR	PCA FPGA programming procedure did not complete successfully.
INFO_PCA_ICC_ERROR	ICC error detected on the Control PCA ICC devices.
INFO_PCA_EEPOT_ERROR	PCA ICC non-volatile potentiometer is not working properly.
INFO_PCA_EEPROM_ERROR	Error during a PCA EEPROM read/write operation.
INFO_PCA_STACK_ERROR	PCA stack has grown over the security margin.
PCA FLU	
INFO_POWER_VOLTAGE_KO	No power supply at PCA FLU-CTRL59.
INFO_SENSOR_VOLTAGE_KO	Fluidic sensors voltage error.
INFO_LOCK_VOLTAGE_KO	Fluidic lockers voltage error.

Table 31. List of options in PCA control - Firmware tab

Eprom control

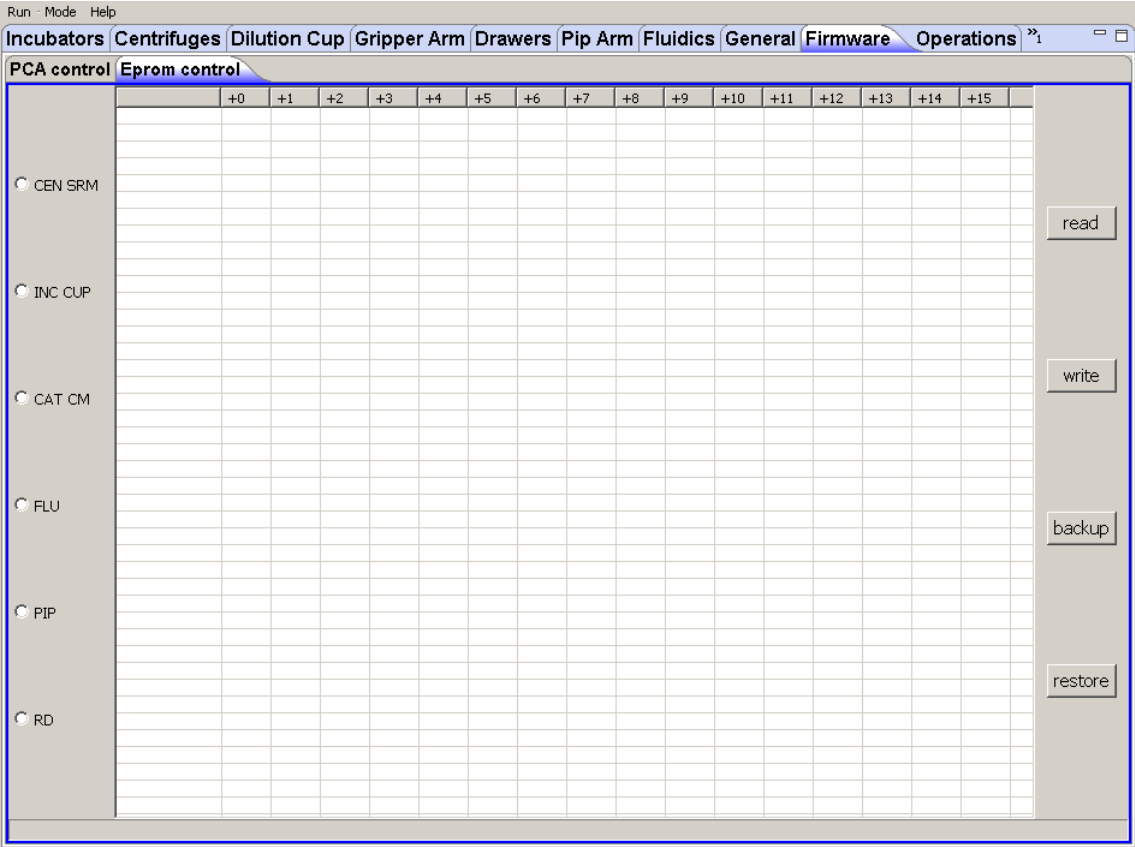


Figure 85. Eprom control tab - Firmware tab in Technical Service software

Option	Description
CEN SRM	Select CEN SRM MotorControl PCA
INC CUP	Select INC CUP MotorControl PCA
CAT CM	Select CAT-CM MotorControl PCA
FLU	Select FLU-CTRL59 PCA
PIP	Select PIP MotorControl PCA
RD	Select RD-IL421 PCA
read	Read EEPROM values of the selected PCA.
write	Write the values shown on the screen to the EEPROM of the selected PCA.
backup	Save into a file the contents of the EEPROM of the selected PCA.
restore	Restore from a file the contents of the EEPROM of the selected PCA.

Table 32. List of options in Eprom control - Firmware tab

3.2.11 Operations tab

Dilution Cup Operations tab

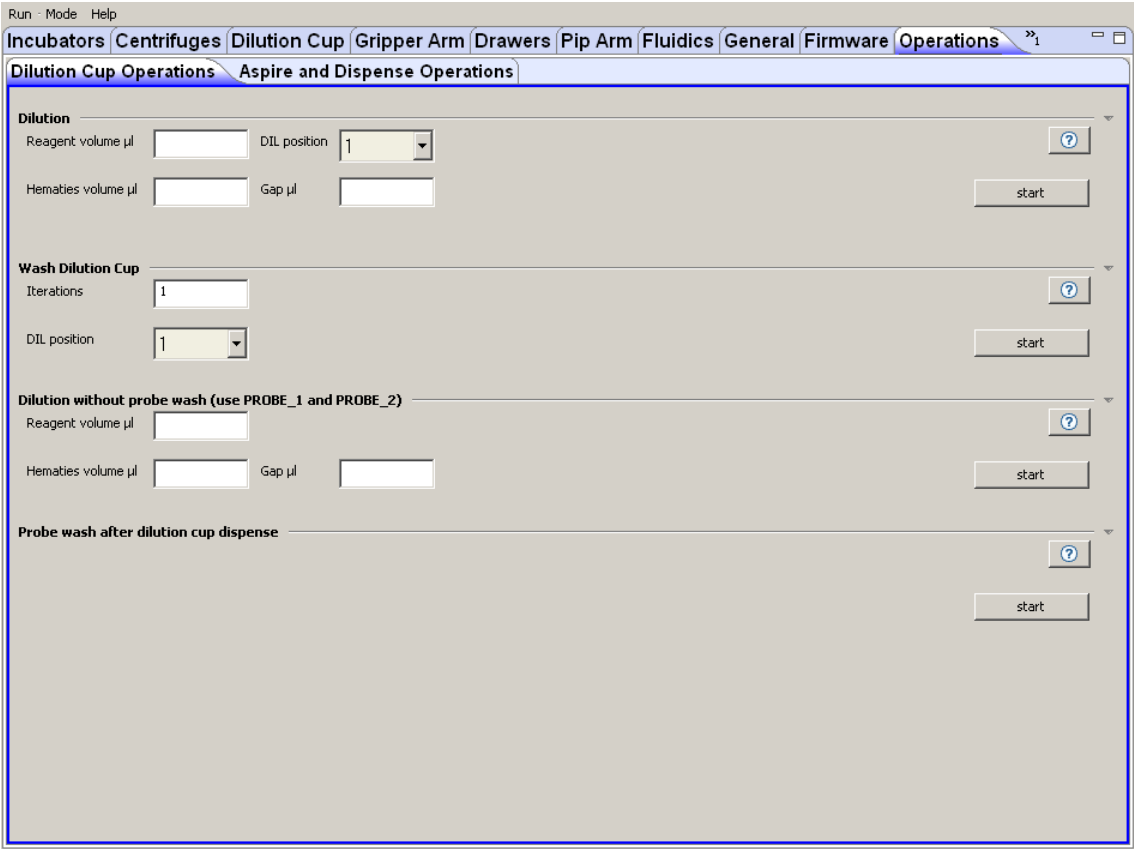


Figure 86. Dilution Cup Operations tab in Technical Service software

Option	Description
Dilution	
Reagent volume µl	Text box to enter the volume [µl] of the reagent to perform the dilution. Suggested value = 500.
DIL position	Dilution cup well (1 or 2) where to perform the dilution.
Hematies volume µl	Text box to enter the volume [µl] of the hematies to perform the dilution. Suggested value = 5.
Gap µl	Text box to enter the air gap [µl] to perform the dilution. Suggested value = 5.
?	Information about the prerequisites of the process: - sample tube in sample drawer 1, rack 1, position 1. - diluent bottle in reagent drawer 1, rack 1, position 1.
Start	Initiates the dilution process.
Wash Dilution Cup	
Iterations	Text box to enter the number of times to repeat the wash.
DIL position	Dilution cup well (1 or 2) where to perform the wash.
?	Information about the process.
Start	Initiates the washing process.
Dilution without probe wash (use PROBE_1 and PROBE_2)	
Reagent volumen µl	Text box to enter the volume [µl] of the reagent to perform the dilution. Suggested value = 500.

Option	Description
Dilution without probe wash (use PROBE_1 and PROBE_2)	
Hematies volume µl	Text box to enter the volume [µl] of the hematies to perform the dilution. Suggested value = 5.
Gap µl	Text box to enter the air gap [µl] to perform the dilution. Suggested value = 5.
?	Information about the prerequisites of the process: - sample tube in sample drawer 1, rack 1, position 1. - sample tube in sample drawer 1, rack 1, position 2. - diluent bottle in reagent drawer 1, rack 1, position 1.
start	Initiates the dilution process.
Probe wash after dilution cup dispense	
?	Information about the process. It is performed only in probe 1.
start	Initiates the washing process.

Table 33. List of options in Dilution Cup Operations tab

Aspire and Dispense Operations tab

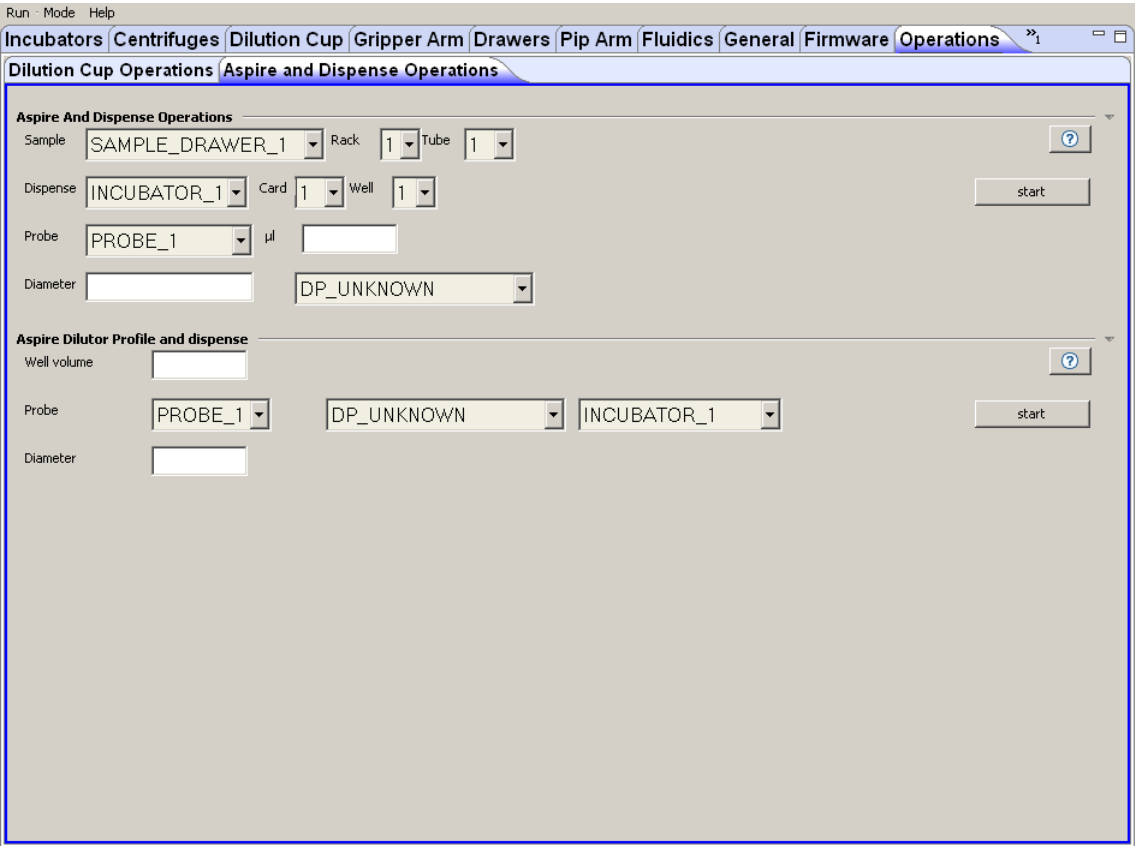


Figure 87. Aspire and Dispense Operations tab in Technical Service software

Option	Description
Aspire And Dispense Operations	
Sample	Select sample drawer: 1 to 4.
Rack	Select rack of the drawer: 1 or 2.
Tube	Select position of the tube: 1 to 12.
Dispense	Select incubator where to dispense: 1 to 4.
Card	Select incubator slot where to dispense: 1 to 12.
Well	Select well position in the card where to dispense: 1 to 8.
Probe	Select probe used: 1 or 2.
µl	Text box to enter the serum aspirated [µl] to perform the operation. Suggested value = 25.
Diameter	Text box to enter the diameter of the sample tube.
Selection box	Select dilutor profile.
?	Information about the process. It is needed: - sample tube in the selected position. - incubator selected with a card in the selected position in the upper floor.
start	Initiates the process.

Option	Description
Aspire Dilutor Profile and dispense	
Well volume	Text box to enter the volume [µl] of the serum to dispense into the card well. Suggested value = 25.
Probe	Probe (1 or 2) used in the process.
First selection box	Select the dilutor profile.
Second selection box	Select the incubator to dispense.
Diameter	Text box to enter the diameter of the sample tube.
?	Information about the process. It is needed: - sample tube in sample drawer 1, rack 1, position 1. - incubator selected filled with cards in the upper floor.
Start	Initiates the process.

Table 34. List of options in Aspire and Dispense Operations tab

4 GENERAL (GEN) MODULE

In this chapter, General module is described. It includes an overall description of the module, adjustments, replacement procedures, fuses, LEDs, jumpers and test points description, wiring list, electronic schematics and a troubleshooting section.

4.1 DESCRIPTION

An overall description of the parts included in the general module is explained in this chapter.

4.1.1 PCA GEN-BCKPL59

Main PCA of the instrument, located at the upper part of the backpanel. This PCA has the following functionalities:

- Convert between RS-232 (computer) and RS-485 (PCAs internal bus) protocol.
- Digital pumps RS-485 interface (convert 3 digital I/O signals of PCA PIP control into RS485 two-wire bus).
- Support all PCA MotorControl. They control specific modules of the instrument and some of these PCAs can control 2 or more modules.
- Provide a connection for PCA Fluidics Control.
- Incorporate all necessary electronic interfaces between PCAs MotorControl and the rest of PCAs, including connection.
- Distribute power supply (with the appropriate protections) to PCAs MotorControl, and the rest of PCAs interfaces and PCAs.

RS-232 (PC-side) to 4-wire RS485 (machine-side) converter

PCA GEN-BCKPL59 has a RS-232 to 4-wire RS485 converter. This converter, optically isolated, only converts TxD and RxD signals of the RS-232 port, with a maximum baudrate of 115200 bauds. PC-side is supplied from +5VPC input supply, whereas machine-side is supplied from +5V supply rail.

PC-side connection (TxD, RxD and GND) is routed to a standard DB-9 shielded female connector, in such a way that a pin to pin DB-9 cable is required to connect PCA GEN-BCKPL59 to a PC. Configuration of the pins is as follows,

- Transmission to PC : Pin 2
- Reception from PC : Pin 3
- GND : Pin 5
- Chassis : Connector shield

Machine-side 4-wire RS485 signals are routed to the RS485 pins of all PCA MotorControl slots and also to the RS485 pins of the PCA Fluidics Control connector.

PCA MotorControl support

PCA GEN-BCKPL59 provides 4 DIN connectors (slots) to connect 4 PCA MotorControl units. There are 2 protections against connection of a PCA MotorControl in a wrong slot,

- 2 PCA MotorControl digital inputs of each slot are reserved for slot-addressing purposes.
- These two slot-addressing PCA MotorControl digital inputs are the same at all slots. Addressing is as follows:

- PCA CEN-SRM Control: Address 0 ("00") – SLOT 0.
- PCA INC-CUP Control: Address 1 ("01") – SLOT 1.
- PCA CAT-CM Control: Address 2 ("10") – SLOT 2.
- PCA PIP Control: Address 3 ("11") – SLOT 3.

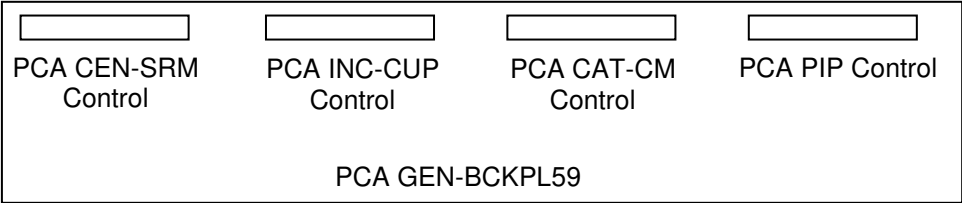


Figure 88. Location of the different PCA MotorControl in PCA GEN-BCKPL59

I2C Bus

Communication between PCA GEN-BCKPL59 and some of the different modules of the instrument is performed through I2C buses. This bus only requires two bus lines and each device connected to the bus is software-addressable by a unique address.

The different I2C buses of the instrument are the following ones:

- SM safety latches + RM safety latches + GEN SM&RM floor Temperature Sensor.
- CM safety latches + GEN CM floor Temperature Sensor.
- PIP (Probe1 Level Detector activation + Probe2 Level Detector activation + Probe1 Correct Pipetting Detector (activation and signal) + Probe2 Correct Pipetting Detector (activation and signal) + Barcode Readers activation + Barcode readers axis home sensor activation and current limit sensing + Presence Detector activation).
- Incubator 1 (Peltier Cell + Peltier Fan + 2 x Temperature Sensors).
- Incubator 2 (Peltier Cell + Peltier Fan + 2 x Temperature Sensors).
- Incubator 3 (Peltier Cell + Peltier Fan + 2 x Temperature Sensors).
- Incubator 4 (Peltier Cell + Peltier Fan + 2 x Temperature Sensors).
- FLU safety latches.

Power Supply input connection

PCA GEN-BCKPL59 has two supply-input connectors:

- 1 connector to input +24V and +5V power supplies, coming directly from the Power Supply system. Both supplies share ground connection (common ground plane).
- 1 connector to input +5VPC, that comes from PCA GEN-STDBY511 +5VPC output.

Block diagram of the PCA GEN-BCKPL59 is shown below:

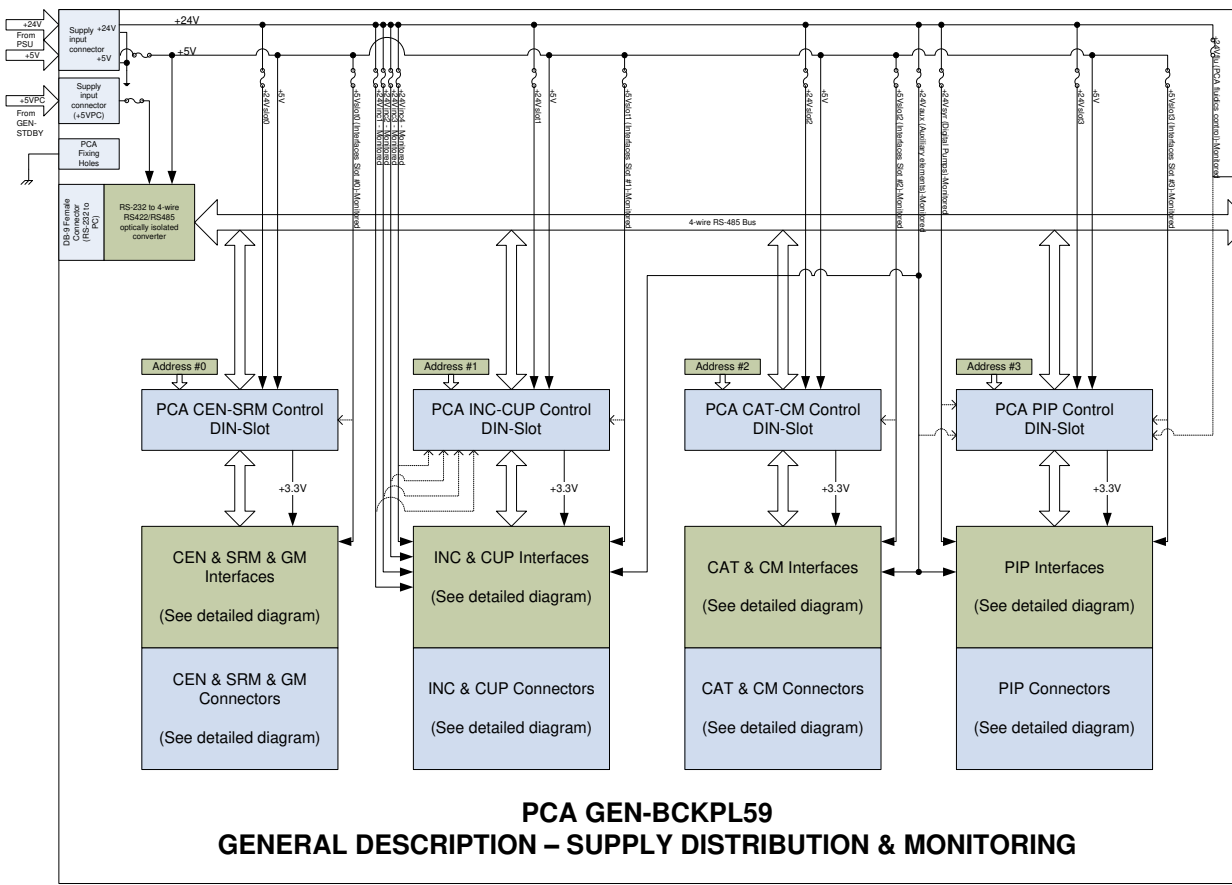


Figure 89. Block diagram of PCA GEN-BCKPL59

4.1.2 Power supply

Located at the upper part of the instrument, it connects *Erytra* to the mains AC voltage.

It is a switched power supply system which automatically adapts to the mains supply voltage, thus does not require any type of manual switching. It has a standby output voltage in order to supply standby-mode electronics that is kept on as long as mains voltage is applied at the power supply input.

Internal AC power cabling is composed of 3 cables (Ground, Line and Neutral). Each of these 3 cables can be identified by its color and has a specific position when connected to the mains filter (position identified/marked in the mains filter label).

1. Brown: Line (P)
2. Light blue: Neutral (N)
3. Yellow-green: Ground (PE)



The power supply system consists of two individual power supply units: one providing 24VDC and another one providing 12VDC. Additionally, a voltage converter PCA provides a 5VDC output signal using a 24VDC input.

The whole power supply system is assembled in an independent plate, which is attached to the top of the electronics box (above the PC) by means of 2 hand screws.

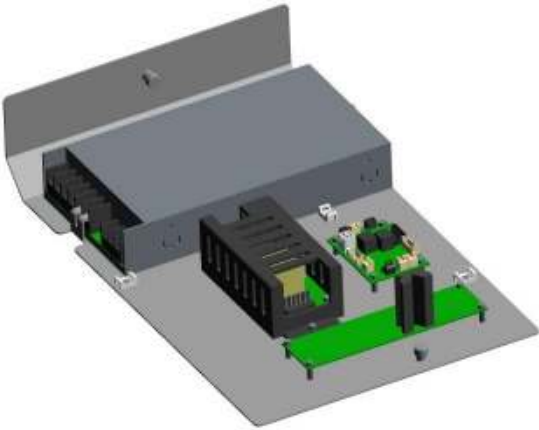


Figure 90. Power supply system assembled in an independent plate

- Specifications of the power supply:
- Input range: 85 – 264 VAC, 47-64 Hz.
 - Power Factor Correction.
 - Output voltages:
 - +24VDC $\pm 2\%$ (23,52V to 24,48V): Motors, syringes, solenoids, electrovalves, pumps and Peltier cells.
 - +5VDC $\pm 4\%$ (4,8V to 5,2V): Control electronics logic.
 - +12V $\pm 2\%$ (11,76V to 12,24V): PC supply.
 - -12V $\pm 2\%$ (-11,76V to -12,24V): PC supply.
 - +5V $\pm 4\%$ (4,8V to 5,2V): PC supply.
 - +3.3V $\pm 4\%$ (3,17V to 3,43V): PC supply.
 - +5V $\pm 4\%$ (4,8V to 5,2V): Standby power.

Erytra On/Off Switcher

24V and 5V power supply outputs can be activated/deactivated by Technical Service Software.



NOTE:

For units with sn < 059-0001583, jumper JP3 of PCA GEN-STDBY59 must be shortcircuited to use this function.

1. In Technical Service Software, go to *Run* menu → *Erytra On/Off Switcher* option. A new window will appear on screen.



Figure 91. Erytra On/Off Switcher

2. To switch off the 5V and 24V power supply outputs, press *All voltages OFF* button. This action will not switch off the computer. Then in order to switch on the 5V and 24V power supply outputs, press *All voltages ON* button.
3. For units with sn < 059-0001583, also it is possible to switch off only the 5V or 24V power supply output pressing *5V OFF* or *24V OFF* button, respectively. Then in order to switch them on press *5V ON* or *24V ON* button, respectively.

4.1.3 Computer

The computer contains the O.S. (Windows 10 Embedded (WES10) v2.1.0) and *Erytra* software. It is located in the upper part of the instrument, under the power supply system.

Computer is switched on by means of the pushbutton that allows switching on the instrument and setting it to standby mode. It is located between the upper floor and the intermediate one. It includes a green LED that is used as a status LED:

- OFF: instrument switched off.
- ON: instrument PC-system (PC, screen, ports) is switched on.



Figure 92. Pushbutton

PC EMBEDDED v3, v4 and v5 (PC ASRock)

Connectors of the computer are the following ones:

- 8 USB ports:
 - USB 3.0: USB 1, USB 3, USB 6 and USB 7 (related to the USB ports available at the rear part of the instrument).
 - USB 2.0: USB 0, USB 2, USB 4 and USB 5 (related to internal communications and USB ports available at the front part of the instrument).
- ATX POWER
- COM 1 and COM 3
- VGA
- FW (FireWire/IEEE1394)
- LAN1
- PC ATX POWER
- PC FEATURE

For a detailed wiring list, check section 4.8 Wiring table in this chapter.



Figure 93. Computer connectors for PC EMBEDDED v3. Front view

Connections for PC EMBEDDED v4 and PC EMBEDDED v5 should be done in the same way as indicated in the picture above for PC EMBEDDED v3. The only difference is that two ports of the built-in firewire PCIe are available in PC EMBEDDED v4 and PC EMBEDDED v5 (one remains unused), instead of the three ports available in PC EMBEDDED v3 (two of them remain unused).



Figure 94. Firewire ports in PC EMBEDDED v4/v5. Front view

PC EMBEDDED v2 / PC EMBEDDED - USA (PC Kontron)

Connectors of the computer are the following ones:

- 8 USB 2.0 ports:
 - USB 1, USB 3, USB 6 and USB 7 (related to the USB ports available at the rear part of the instrument).
 - USB 0, USB 2, USB 4 and USB 5 (related to internal communications and USB ports available at the front part of the instrument).
- ATX POWER
- COM 1 and COM 3

- VGA
- FW (FireWire/IEEE1394)
- LAN1
- PC ATX POWER
- PC FEATURE

For a detailed wiring list, check section 4.8 Wiring table in this chapter.

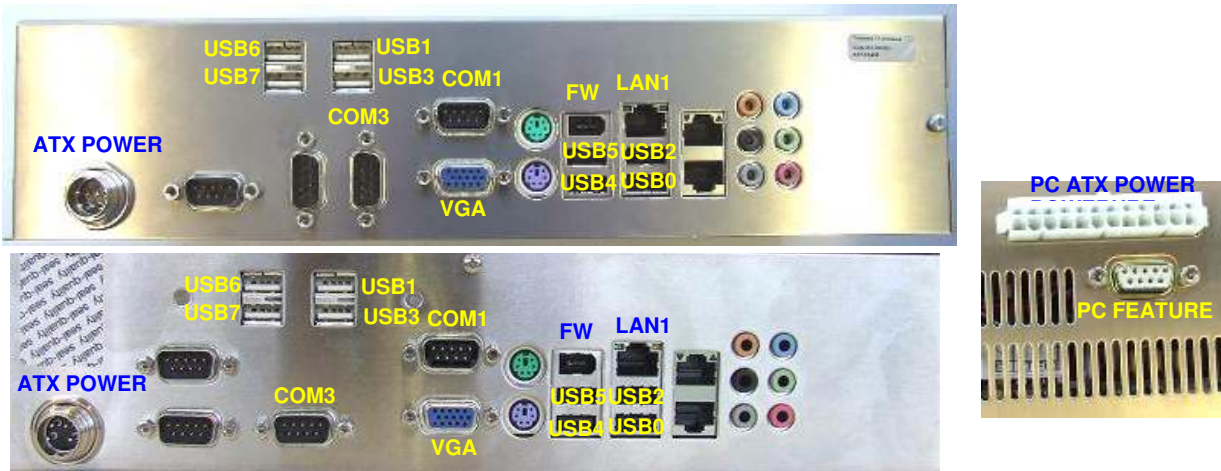


Figure 95. Computer connectors for PC EMBEDDED v2 / PC EMBEDDED - USA. Front view (left) and lateral view (right)

Both PC models have a direct connection with the reader module by means of a FireWire protocol for the camera control and with an USB protocol for the reader backlight and for the communication emulating a serial port.



NOTE:

If the PC battery needs to be replaced for any of the above described models, the new battery shall be a 3V CR2032-type, with a maximum operating temperature $\geq 80^{\circ}\text{C}$.

4.1.4 Touchscreen monitor

Located in a drawer under the loading area of samples, it is the main operator interface. The mechanism of the drawer allows the operator to tilt the screen and to hide it completely inside the instrument when it is not in use. In order to extract the touchscreen monitor, the frontal panel must be pushed once and then pulled from the frontal panel. When the touchscreen monitor is outside completely of the instrument, it can be tilted.

The monitor is TFT LCD 15" with a resolution of 1024 x 768 pixels that incorporates a touchscreen. The monitor is connected to the computer using a VGA protocol and the touchscreen is controlled by an internal USB port that is not accessible externally.



Figure 96. Touchscreen monitor

4.1.5 User tray

Located under reagents loading area, it helps the operator to handle samples and reagents. It has a usable surface of 340mm width x 200mm depth. The mechanism of the drawer allows the operator to hide it completely inside the instrument when it is not in use.

In order to extract the user tray, the frontal panel must be pushed once and then pulled from the frontal panel.



Figure 97. User tray

4.1.6 Drawers

There are 4 types of drawers in the *Erytra*: reagents, samples, cards and bottles drawers. In this chapter, reagents, samples and cards drawers are explained. Bottles drawers are explained on Fluidic chapter 9.

Samples and reagents drawers are controlled with an I2C bus. In this bus is also attached the temperature sensors of the upper floor. The same communication bus applies to the cards drawers, attached as well to the temperature sensors of the intermediate floor.

The system assigns one address to each reagent/samples drawer. PCA GEN-BCKPL59 provides a '000' binary address at SRM-LCK interface connector, as the address of the first PCA GEN-LCK59 connected to the safety latch bus. The output of each PCA GEN-LCK59 is input address+1 and this is the address provided to the next PCA GEN-LCK59 and so on. The same applies to the cards drawers.

REAGENT DRAWERS

The structure of the module is formed by the following elements:

- Moving part: it supports the reagent racks (located between centerers) and it allows the operator to load/unload reagent racks. It includes a cosmetic front panel and the mechanics and electronics to perform the required reagent stirring.
- Fixed part: it is responsible for allowing the movement of the moving part outside the instrument by means of a linear guide and to attach the module to it. It also includes the cable that connects the moving part to the electronics.
- Safety latch: it controls open/close status, it allows opening the drawer electronically and it assures closed position repeatability.

There are 2 reagents drawers located over the user tray, on the left of the instrument and labelled as 1 and 2. Each drawer allocates 2 reagents racks, oriented in a way that the reagents are located in two rows parallel to drawer movement. The racks are manufactured from plastic and are washable in Laboratory Glassware Washer.



Figure 98. Top view of reagents racks

In each drawer, the frontal rack allows to an orbital stirring of the reagents. Each rack for stirred reagents allocates 18 vials and each rack for non-stirred reagents allocates 6 vials and 3 diluent bottles. There is a Hall effect sensor in order to determine the home position of the stirring platform.

Racks centerers are placed on the drawers in order to do not place incorrectly the reagents racks.

Reagents drawers incorporate a magnetic braking system in order to avoid the possibility during its movement of a splashing process of the reagents.

Each reagent/diluent position in the rack has a window to allow reading (vertical) barcode of the vials/bottles.

There are clips in order to avoid any movement of the vials/bottles on its lodgings. All the vials lodgings place the vial/bottle with an inclination to minimise dead volumes.

A safety latch (controlled electronically), located at the rear part of the drawer, assures drawer is closed in the working position and allows open/close status monitoring. The latch is attached to the support, while the bolt is attached to the moving part. The bolt position is adjustable in order to adjust the cosmetic front panel.

SAMPLE DRAWERS

The structure of the module is formed by the following elements:

- Moving part: it supports the sample racks and allows the operator to load/unload samples racks. It includes a cosmetic front panel.
- Fixed part: it is responsible for allowing the movement of the moving part outside the instrument by means of a linear guide and to attach the module to it.
- Safety latch: it controls open/close status, it allows opening the drawer electronically and it assures closed position repeatability.

There are 4 samples drawers located over the touchscreen monitor, on the right of the instrument and labelled as 3, 4, 5 and 6. Each drawer allocates 2 samples racks oriented in a way that samples are located in two rows parallel to drawer movement. The racks are manufactured from injected plastic and are washable in Laboratory Glassware Washer.



Figure 99. Sample rack

To ensure the correct position of the racks in the drawer, there are positioning devices that hold the rack by their 4 corners. Due to the rack symmetry, a 180° rotation in the drawer is allowed.

Each rack allows placing 12 samples tubes. The distance between the center of the tubes in the rack is 25mm in X and Y axis. The correct centering of tubes with different diameter in their lodging is assured by centering clips. Allowed dimensions of the tubes (maximum tube height: 100mm) are:

- Glass tubes (external diameter 9 to slightly smaller than 17mm)
- Plastic tubes (external diameter 9 to slightly smaller than 17mm)



NOTE:

To load sample tubes with an external diameter larger than 16mm and slightly smaller than 17mm, the special samples holder for tubes > 16mm need to be used.

Samples drawers incorporate a magnetic braking system in order to avoid the possibility, during its movement, of a splashing process, aerosolization, bubble formation, breakage or remixing of a centrifuged specimen.

A safety latch (controlled electronically), located at the rear part of the drawer, assures drawer is closed in the working position and allows open/close status monitoring. The latch is attached to the support, while the bolt is attached to the moving part. The bolt position is adjustable in order to adjust the cosmetic front panel.

CARD DRAWERS

The structure of the module is formed by the following elements:

- Moving part: it supports the card holders and allows the operator to load/unload cards holders.
- Fixed part: they are responsible for allowing the movement of the moving part outside the instrument and to attach the module to it.
- Safety latch: it controls open/close status, it allows opening the drawer electronically and it assures closed position repeatability.

There are 4 card drawers located on the intermediate floor and labelled as 1, 2, 3 and 4 in the keyboard below. Each drawer can allocate 4 25-card holders, which can be properly placed on the drawer by means of centerers attached to its surface. Additionally, card holders are retained by means of a plastic bar attached to the vertical linear guide on the left side of each drawer and a plastic clip on the opposite side (one for each rack). Should a holder be positioned with its card barcodes facing the right side of the instrument, the software will warn the user.

A safety latch (controlled electronically) located at the rear part of the drawer, assures the drawer position when closed and allows the open/close status monitoring. The latch is located in the rear support, while the bolt is attached to the moving part of the drawer. The bolt position can be adjusted in order to modify the depth of the drawer inside the instrument

4.1.7 Keyboards

Keyboards allow the operator to open the drawers of the samples, reagents and cards by means of a button located under each drawer. It is possible also to know samples drawers status by means of a LED located at the right side of each opening button. There is a button on the right side of the keyboard to request the opening of a samples drawer in order to introduce a STAT sample.

Samples/Reagents keyboard is connected directly to the PCA GEN-BCKPL59 with a cable that transports each button signal/line in parallel. Communication between PCA GEN-BCKPL59 and the PCA CEN-SRM Control is performed with an I2C bus. Identical communication system is used for the Cards keyboard (between PCA GEN-BCKPL59 and PCA CAT-CM Control).



Figure 100. Samples/Reagents and Cards drawers keyboards

4.1.8 Status LEDs

Status LEDs allow the operator to know the instrument status without looking at the software screen. The instrument contains the following status LEDs:

- Samples drawers: the availability of each drawer is highlighted in blue color.



Figure 101. Samples drawers keyboard

Samples drawers status LEDs are connected directly to the PCA GEN-BCKPL59 with the same cable that transports information of the Samples/Reagents keyboard.

Communication between PCA GEN-BCKPL59 and the PCA CEN-SRM Control is performed with an I2C bus.

- General LED: located at the top right of the instrument, it shows by means of 3 different colors (green, yellow and red) if there is any alarm in the system allowing to check the instrument status without the need to open screen drawer.



Figure 102. General LED

General LED is connected directly with a cable to the PCA GEN-BCKPL59. Communication between PCA GEN-BCKPL59 and the PCA CEN-SRM Control is performed with an I2C bus.

4.1.9 Doors

There are 2 doors that give technician access to the instrument for servicing purposes:

- Upper area door: located where pipetting is performed. It opens manually upwards and keeps its position until it is closed down by the operator/technician. The door does not have a latch.



Figure 103. Upper area door

- Intermediate area door: located where card transport is performed. It opens downwards manually by its own weight with a cushioned movement. The door does not have a latch, it is closed by means of a screw located in the sensor area. To open the door, the screw must be removed and the user tray must be opened.



Figure 104. Intermediate area door

Status of the doors is monitored by means of one sensor (Celduc sensor) for each door. This Hall effect sensor is split in two parts: one attached to the door and the other one attached to the instrument, so when the door is closed the part attached to the door is in contact with the fixed part.

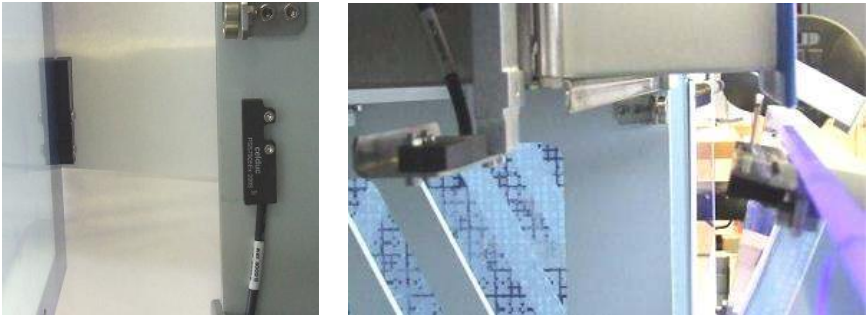


Figure 105. Celduc sensors in upper area door (left) and in intermediate area door (right)

4.1.10 Temperature control system

There are 3 areas in the *Erytra* where the temperature is monitored. 2 of them are located in the lower part of each floor of the instrument, where there are located a PCA with 2 temperatures sensors (PCA GEN_TEMP59). The other area is located in the electronics cabinet, where there is a temperature sensor on the PCA GEN-BKCPL59 (U21), close to the RM_2 connector. Sensors are LM92 with an accuracy of $\pm 0.33^{\circ}\text{C}$.

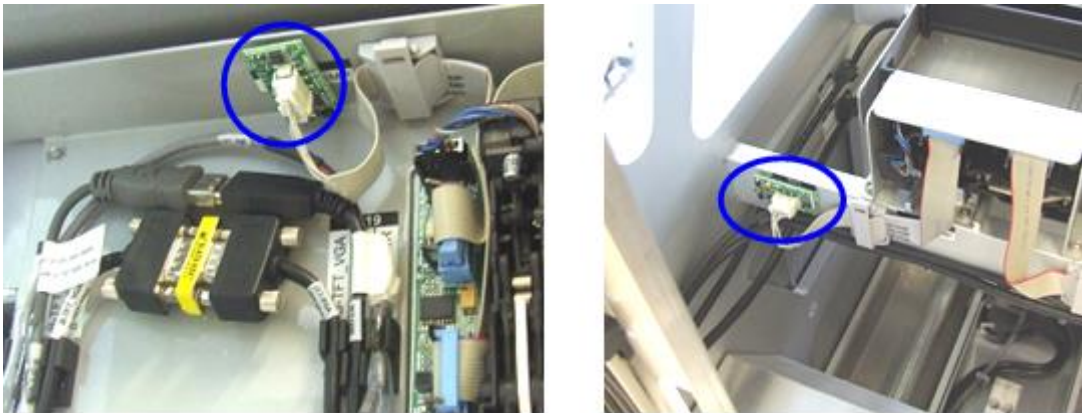


Figure 106. Temperature sensors in upper (left) and intermediate (right) areas

There is a forced air extraction system in the electronics area activated by electronics cabinet temperature (30°C) while instrument is working. This extraction system consists of two fans that extract the heat outside the instrument, one located in the right right side of the electronics cabinet

(looking from the rear side of the instrument) and another one in the central part, assembled on a plate providing rigidity of the whole cabinet.

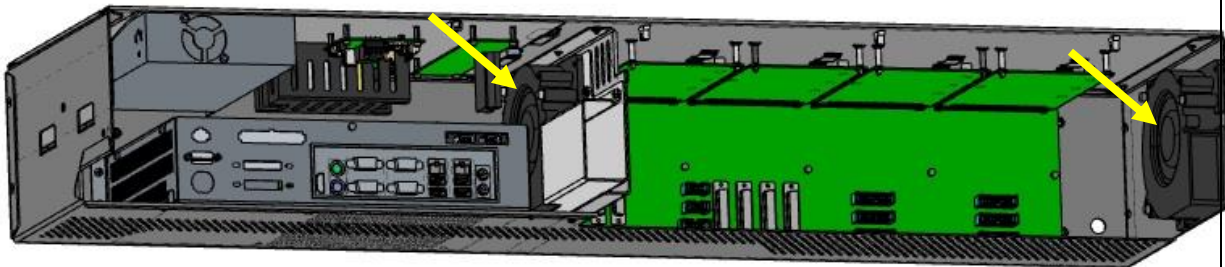


Figure 107. Fans in the electronics cabinet

Communication buses of the temperature control system of the upper and intermediate area are through I2C buses, sharing same bus with the reagents/samples drawers (upper area) and cards drawers (intermediate area).

4.1.11 Buzzer

The instrument incorporates a buzzer to alert of any incidence to the operator. It is placed at the PCA GEN-BCKPL59, under the slot number 2.



Figure 108. Location of the buzzer in PCA GEN-BCKPL59

4.1.12 Chassis

It is formed by 2 laterals (made of aluminium) held together with crossbars (made of aluminium) and the X axis of the upper and intermediate floors. It is mounted on the instrument base.

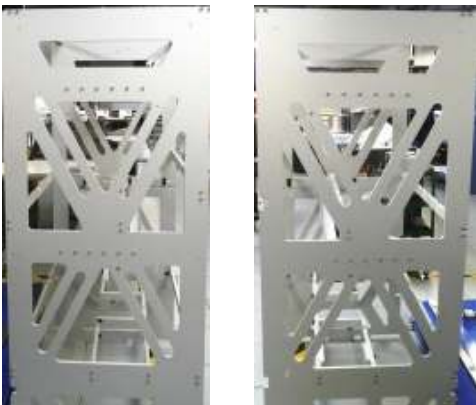


Figure 109. Chassis of Erytra

4.1.13 Cables and tubes rails

The rails route the tubes/cables in order to distribute them throughout the instrument. They are located at the rear part of the instrument, attached to the chassis crossbeams. There are 3 rails, one at each side of the *Erytra* and another one centered.

In each tie wrap of the rails there are labels indicating the codes of the cables/tubes in order to facilitate the identification.

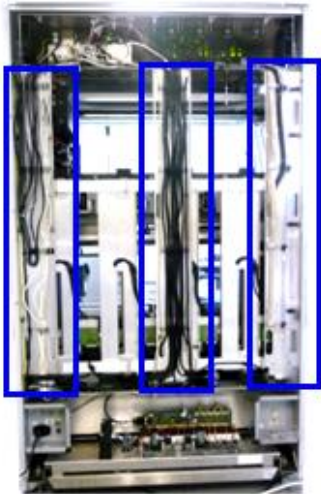


Figure 110. Cables and tubes rails

4.1.14 Covers

Aesthetical covers of the instrument. There are the following covers:

- Upper and intermediate area laterals: covered by a transparent lateral attached to the chassis.
- Lower area laterals: covered by an opaque lateral attached to the chassis.
- Top cover: it protects the instrument from spills.
- Frontal covers: include all frontal areas that are not covered by the frontal of the drawers and doors. One of the frontal covers includes the pushbutton and 2 USB ports.

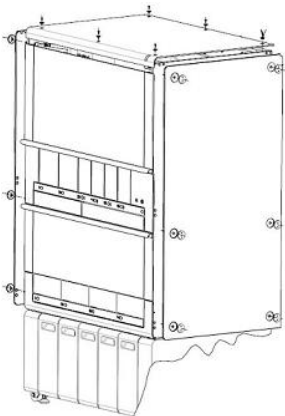


Figure 111. Frontal covers

- Back covers: There are 3 back covers, one per each area (upper, intermediate and lower). Upper and lower covers have a matrix of holes in order to allow heat extraction.

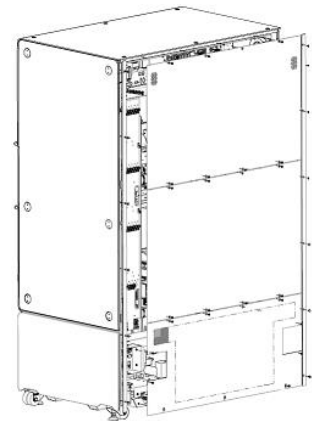


Figure 112. Back covers

- Connectors covers: They are attached to instrument laterals and are independent from lower back covers. Connector covers allow room for external connected cables, therefore when those are connected they do not jut out the instrument footprint. In the lower area, there are 2 specific covers:

- Fluidics: includes a fitting to connect the fluidic system to the laboratory drain.



Figure 113. Fluidic connector cover

- Electronics: to allow mains connection and external available ports (4 USB ports, 1 Ethernet connector and 1 serial port).



Figure 114. Electronics connectors cov

General block diagram of General module is shown in the following figure:

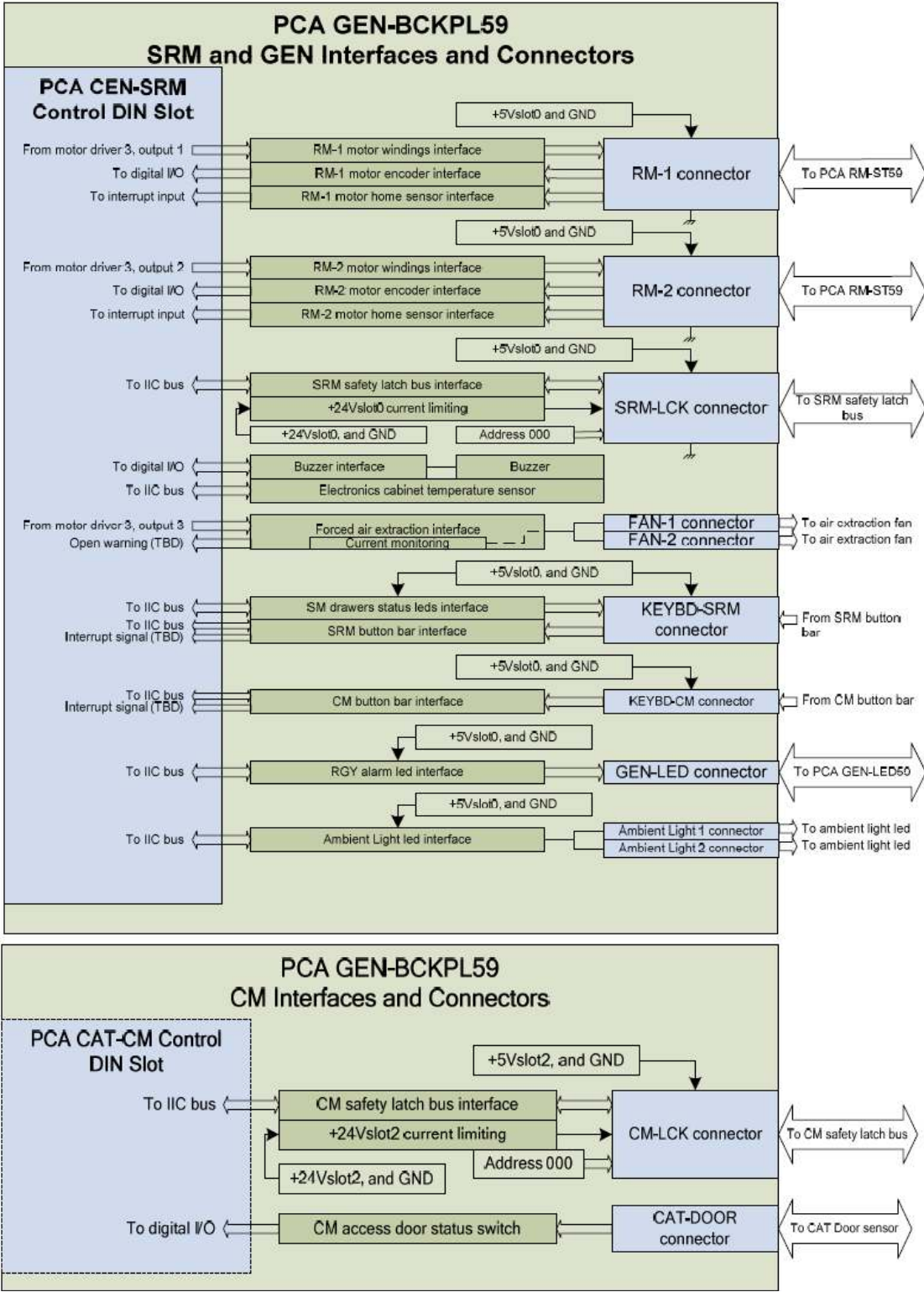


Figure 115. Block diagram of General module

PCAs related to General module are:

- **PCA GEN-BCKPL59:** detailed description on section 4.1.1.
- **PCA CEN-SRM Control:** located at the backpanel (slot 0), controls all the signals used in the centrifuge module and samples/reagents management module. It contains the firmware of both modules.
- **PCA CAT-CM Control:** located at the backpanel (slot 2), controls all the signals used in the card transport module and card management module. It contains the firmware of both modules.
- **PCA GEN-STDBY511:** it contains the standby power supply system and interfaces the Power Supply system to the computer (PC) and PCA GEN-BCKPL59. It is able to activate/deactivate the power supply of the PC and electronics control unit (power and control).
- **PCA GEN-LED59:** it contains a high-power LED and by means of current fixing resistors, it is possible to obtain 3 different colors (red, green and blue). One additional color, yellow, is obtained with a combination of red and green. There are 3 PCAs on the instrument:
 - 1 PCA located at the top right of the instrument (Status LED)
 - 1 PCA located under the electronics drawer (Ambient light)
 - 1 PCA located under the intermediate floor (Ambient light)
- **PCA GEN-FRNT59:** it supports and interfaces the standby switch button and 2 USB Type A ports located at the front of the instrument. This PCA is located under the cover of the dilution cup, at the front of the instrument.
- **PCA GEN-TEMP59:** it contains the temperature sensors to monitor temperature at both instrument floors. There are 2 PCA located at:
 - Between the upper and the intermediate floor, at the rear part of the instrument, under the dilution cup area.
 - At the rear part of the card drawer number 4, in the intermediate floor.
- **PCA GEN-LCK59:** it supports latch electronic control. It contains the safety latch interface (activate / deactivate) and it monitors its status (opened/closed). This PCA is located at the rear part of the reagents/samples/cards drawers.
- **PCA RM-ST59:** it interfaces the stirrer motor and the home detector of the stirrer. It is located under the reagents drawers.
- **PCA RM-CON59:** it interconnects the PCA RM-ST59 with the PCA GEN-BCKPL59.

4.2 ADJUSTMENT PROCEDURES

In this section, adjustments related to General module are explained. There are 2 types of adjustments: mechanicals and functionals.

4.2.1 Mechanical adjustments

Mechanicals adjustments related to General module are detailed.

4.2.1.1 Upper / Intermediate area door

This adjustment is required to set the reagents-samples and cards door front panel depth into the instrument in order to align it with the rest of the instrument frontal.

Prerequisite: NA

Adjustment tool: NA

Procedure:

1. In order to adjust the parallelism of the door, loose the screws that attach the magnets to the *Erytra* chassis.



Figure 116. Magnet of the doors

2. Adjust the position of the magnets by means of the slotted holes in order to place the reagents and samples door parallel to the chassis once it is closed.

4.2.1.2 Touchscreen drawer lock

This adjustment is required to set the touchscreen monitor housing front panel depth into the instrument in order to align it with the rest of the instrument frontal.

Prerequisite: NA

Adjustment tool: NA

Procedure:

1. To adjust the position of the touchscreen monitor inside its drawer, loose the 2 screws of the lock and adjust the position by means of its slotted holes. Access to the lock screws is reached through CAT area.

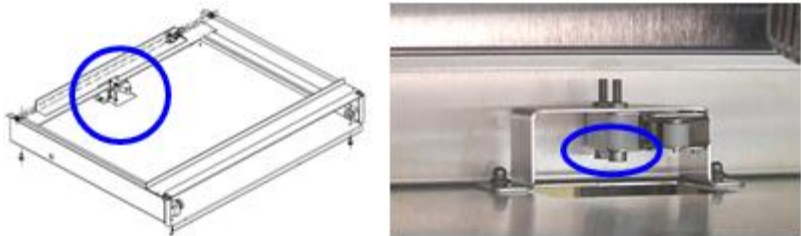


Figure 117. Lock of the touchscreen drawer

1. Insert the screen support plate into the instrument. Check that the two flags at the end of the screen support plate are inserted into the two slots.



Figure 118. Slots to insert the flags

2. Check that touchscreen monitor is aligned with the frontal of the instrument in closed position.

4.2.1.3 User tray drawer lock

This adjustment is required to set the user tray front panel depth into the instrument in order to align it with the rest of the instrument frontal.

Prerequisite: NA

Adjustment tool: NA

Procedure:

2. Extract the user tray from the instrument. Remove the screws located under the reagents/samples keyboard and extract the user tray.
3. Loose the 2 screws of the lock located at the rear part of the user tray and adjust the position by means of its slotted holes.



Figure 119. Lock of the user tray

4. Insert the user tray into the instrument. Check that the two pins at the end of the user tray are inserted into the two slots.



Figure 120. Slots to insert the pins of the user tray

5. Tighten the screws that attach the user tray to the instrument.
6. Check that the user tray is aligned with the frontal of the instrument in closed position.

4.2.1.4 Safety latch reagents, samples and cards drawers

This adjustment is required to set the depth of the drawer front panel into the instrument in order to align it with the rest of the drawers and the instrument frontal.

Prerequisite: NA

Adjustment tool: Drawer front panel alignment tool (PN: 233824)

Procedure:

1. Open the drawer to be adjusted. The safety latch is located in the rear part of the drawer. Open the upper door (reagent/samples drawer) or the intermediate door (cards drawer) to have better access.
2. Loose the nut attached to the safety latch, and loose or screw the safety latch in order to align the drawer. To check if the drawer is aligned, place the alignment tool and close the

drawer to check if the frontal panel is aligned with the front of the instrument. Once the drawer is aligned, tighten the nut.



Figure 121. Safety latch adjustment with drawers front panel alignment tool

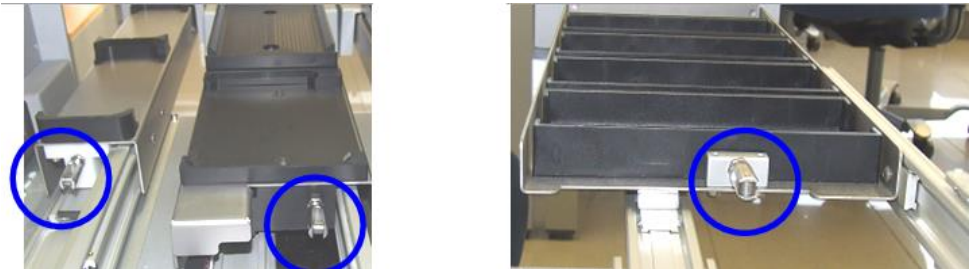


Figure 122. Latches of reagents/samples drawers (left) and cards drawers (right)

- 3. Close the drawer and check its alignment.
- 4. Perform probes adjustment in reagents/samples drawers or gripper adjustment in cards drawers depending on which safety latch has been adjusted.

4.2.1.5 Front panel of reagents, samples and cards drawers

This adjustment is required to set the front panel of the drawers in order to align it with the rest of the drawers and the instrument frontal.

Prerequisite: NA
Adjustment tool: Drawer front panel alignment tool (PN: 233824)
Procedure:

- 1. Open the drawer to be adjusted.
- 2. Remove the front panel cover by means of the screws attached to each side.
- 3. Loose the 4 screws that are attaching the front panel to the drawer.

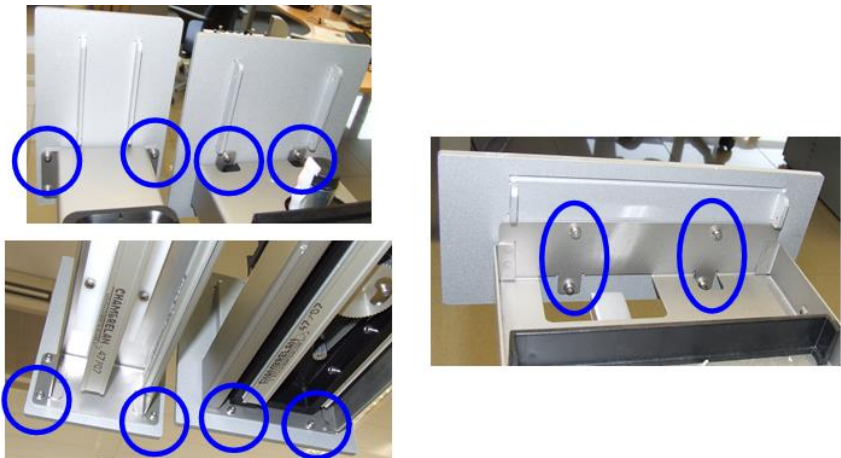


Figure 123. Cosmetic frontal panels screws. Samples/reagents (left) and cards (right)

4. Place the alignment tool and move in Z and X axis the front panel in order to align it. Once the frontal panel is aligned, tighten the 4 screws.



Figure 124. Front panel adjustment with drawers front panel alignment tool

5. Place the frontal panel cover and tighten the screws.
6. Close the drawer and check its alignment.

4.2.1.6 Tension of the belt of the reagents drawers

The purpose of this adjustment is to adjust the tension of the belt of the motors located under the reagents drawers.

Prerequisite: NA

Adjustment tool: NA

Procedure:

1. Open the reagents drawer to be adjusted.
2. Loose the screws of the motor
3. Move the motor to adjust the belt tension and tighten the screw of the slotted hole. Then tighten the other screw.
4. Verify reagents stirring process (check section 4.3.13)

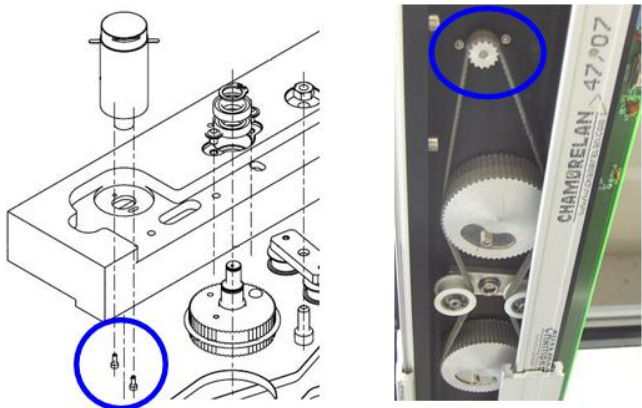


Figure 125. Screws of the reagents motor

4.2.1.7 Reagents drawers flatness

The purpose of this adjustment is to adjust the flatness of the samples drawers.

Prerequisite: NA

Adjustment tool: NA

Procedure:

1. Open the reagent drawer.

2. Remove the plate of the magnetic brake of the reagent drawer.



Figure 126. Magnetic brake screws

3. Loose the 4 screws of the vertical linear guide and then modify manually the position of the drawer (there are slotted holes) to adjust the flatness and finally tighten the screws. Difference in counts between the maximum and the minimum encoder counts in 4 different points should be ≤ 1047 (refer to section 4.3.14).



Figure 127. Vertical linear guide screws - reagents drawers

4. Install the plate of the magnetic brake.
5. Perform probes adjustment in the reagent drawer.

4.2.1.8 Samples drawers flatness

The purpose of this adjustment is to adjust the flatness of the samples drawers.

Prerequisite: NA

Adjustment tool: NA

Procedure:

1. Open the sample drawer to be verified and remove the front panel cover by means of its 4 screws to get access to the screws of the vertical linear guide.
2. Loose the 4 screws of the vertical linear guide, modify manually the position of the drawer (there are slotted holes) to adjust the flatness and finally tighten the screws. Difference in encoder counts between left and right part of the previous positions should be ≤ 1047 (refer to section 4.3.15).



Figure 128. Vertical linear guide screws - samples drawers

3. Install the front panel cover.
4. Perform probes adjustment in the samples drawer.

4.2.1.9 Cards drawers flatness

The purpose of this adjustment is to adjust the flatness of the cards drawers.

Prerequisite: NA

Adjustment tool: NA

Procedure:

1. Open the card drawer.
2. Remove the front panel cover by means of its 2 screws.
3. Loose the 4 screws that attach the vertical linear guide to the drawer.



Figure 129. Vertical linear guide screws – cards drawers

4. Modify manually the position of the drawer due to the slotted holes of the cover and tighten the screws of the linear guide to adjust the flatness. Difference in encoder counts between left and right part of the same rack (1 and 4) is ≤ 600 (refer to section 4.3.16).



Figure 130. Card drawer flatness adjustment

5. Install the front panel cover.
6. Perform gripper adjustment in the drawer.

4.2.2 Functional adjustments

Functional adjustments related to General module are detailed.

4.2.2.1 Touchscreen

This procedure is needed to calibrate the touchscreen, in order to make sure that the point touched on the screen matches with the point where the cursor of the O.S. is placed.



NOTE:

The touchscreen calibration process must be done in Maintenance mode of the software. Once the calibration is finished, shut the system down so that the next time it can be started in the desired mode.

Prerequisite: NA
Adjustment tool: NA
Procedure:

Touchscreen v2 (SN < 059-0001515)

1. Switch on the instrument and press the grid () button during system startup (login credentials will be required), then select the *Maintenance Mode* icon among the Boot Application options.
2. Once PC reboots and desktop is available, press Windows *Start* button and select *MicroTouch MT 7* program, then click on *Control Panel* among the drop down options.

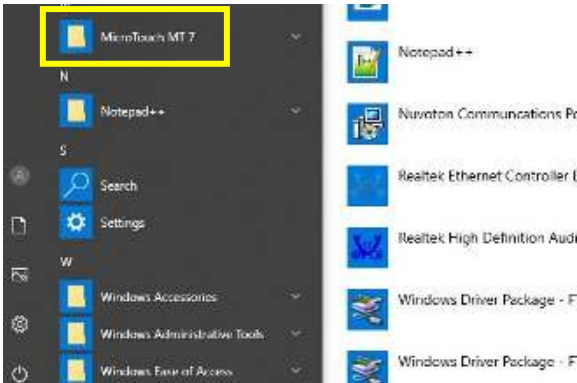


Figure 131. MicroTouch MT 7 program

3. Select *Main* tab and click on *Calibrate* button. A target will be displayed: 
4. Touch the target centre during several seconds and then lift your finger off (the target will change to red colour). Repeat this step for the next 2 calibration points.
5. Once you have calibrated the touchscreen, a dialog box will appear giving the option to test, accept or cancel the performed calibration. Finish the calibration process by pressing *Accept* button (do not test).
6. Verify the calibration accuracy. If it is not accurate enough then repeat the calibration and verify again.
7. If suitable accuracy cannot be achieved, perform a 'Linearization':
 - a. In Micro Touch MT 7 Control Panel, select the *Tools* tab and click on *Advanced Options*.
 - b. Check the *Enable manual linearization* option an click on *OK*.
 - c. Select the *Controller* tab and click on *Linearize* button.
 - d. 25 calibration targets  (one at a time) will appear. Press with a pointer over the centre during several seconds until changes to red and remove the pointer.
 - e. Finish the process by pressing 'Accept' (do not test).
 - f. Verify calibration accuracy and, if considered necessary, perform touchscreen calibration again.
8. Switch off the instrument.

Touchscreen v3 (SN ≥ 059-0001515)

1. Switch on the instrument and press the grid () button during system startup (login credentials will be required), then select the *Maintenance Mode* icon among the Boot Application options.
2. Once PC reboots and desktop is available, go to *Start - Windows System - Control Panel*.
3. Select Hardware and Sound - *Tablet PC Settings - Calibrate the screen for pen or touch input* option.

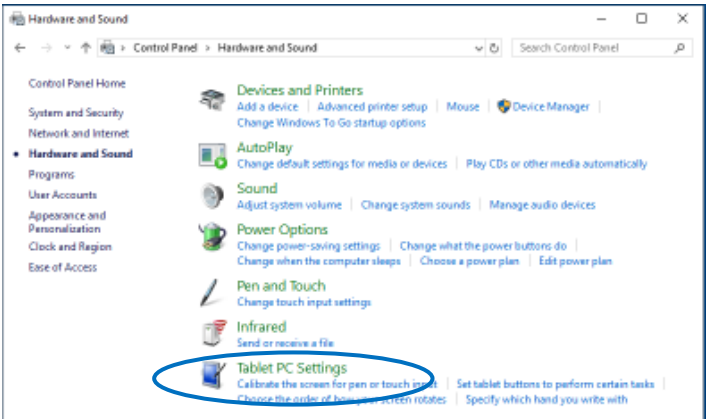


Figure 132. Tablet PC Settings

4. *Tablet PC Settings* window will be displayed. Select *Calibrate* option in *Display* tab and enter the admin user password in *User Account control* window (if displayed).

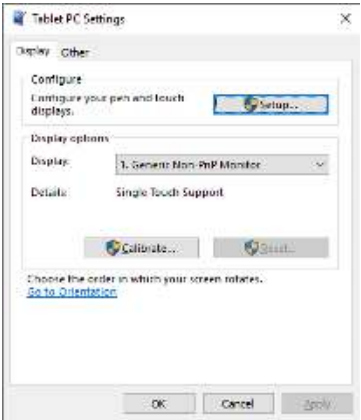


Figure 133. Tablet PC Settings window

5. Calibration process will start, requesting to tap the crosshair each time that it appears on screen.



Figure 134. Calibration screen

It will display 16 crosshairs (4 in each corner) the first time the touchscreen is calibrated (or when *Tablet PC Settings – Reset* option has been selected) and just 4 crosshairs the next time (only inner lines are displayed).

- 6. Once the calibration process finishes, a new *Digitizer Calibration Tool* message window asking whether data should be saved will be displayed. Select *Yes*.

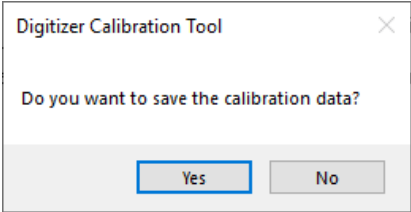


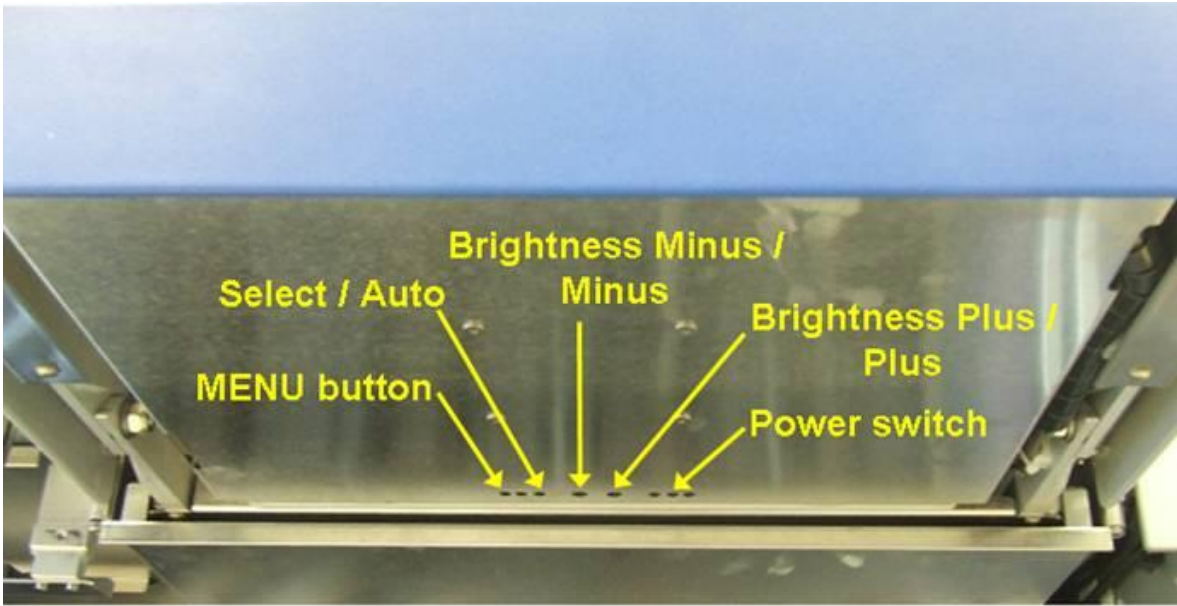
Figure 135. Save calibration data

- 7. Click on *Tablet PC Settings – OK* button to close this window.
- 8. Close *Hardware and Sound* window.
- 9. Switch off the instrument.

4.2.2.2 Touchscreen monitor menu

The holes in the rear part of the touchscreen drawer allow accessing the monitor menu buttons, so that settings such as brightness, contrast, etc. can be customized.. Default brightness values are set to 100 for the monitor with touchscreen v2 and 50 for the one with touchscreen v3.

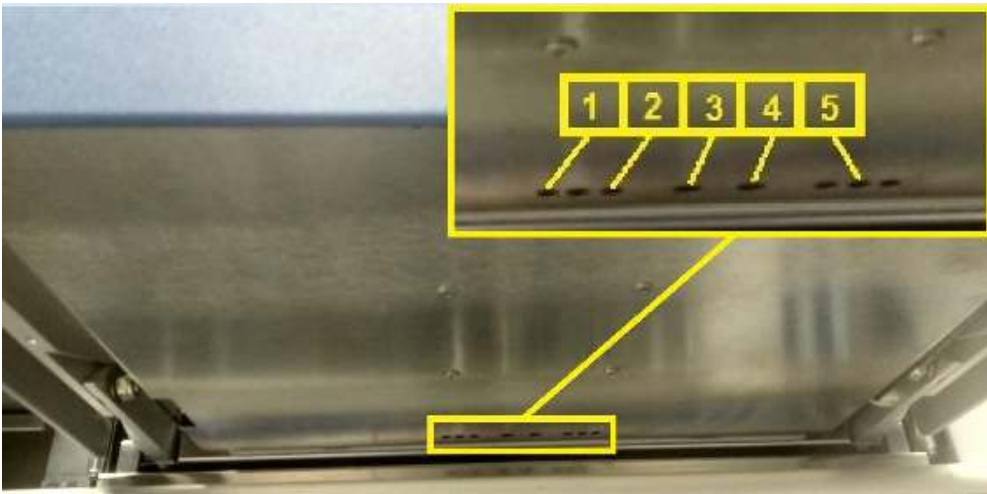
Touchscreen v2 (SN < 059-0001515)



No./ Icon	Control	Function
❶ 	Menu button	Displays the OSD menus.
❷ 	Select / Auto	Select- To select the adjustment items from OSD menus. Auto- To activate the "Auto Adjustment" function to obtain an optimum image.
❸ 	Brightness Minus / Minus	1. Decreases the brightness of the display image. 2. Decreases value of the adjustment items.
❹ 	Brightness Plus / Plus	1. Increases the brightness of the display image. 2. Increases value of the adjustment items.
❺ 	Power Switch	Switches on/off the power of the LCD display.

Figure 136. Touchscreen v2 buttons

Touchscreen v3 (SN ≥ 059-0001515)



Button #	Function	Description
1	MENU	Activate menu / Select menu
2	SELECT	Menu Exit / Auto adjust
3	DOWN	Cursor Control down / Display monitor information
4	UP	Cursor Control up / Source Change (VGA, DVI, Auto select)
5	POWER	Power On / Off

Figure 137. Touchscreen v3 buttons

4.3 VERIFICATIONS

In this section, verifications of the parts of General module are detailed.

4.3.1 Power supply (SN < 059-0001583)

1. Check fuses of the mains connector.
2. Check that all the cables are correctly connected.
3. Connect *Erytra* to the mains and switch on the button from mains connector.
4. Switch on the instrument and check voltages at the output of power supply with a calibrated multimeter:
 - a. +24VDC $\pm$ 2% (23,52V to 24,48V): Motors, syringes, solenoids, electrovalves, pumps and Peltier cells.
 - b. +5VDC $\pm$ 4% (4,8V to 5,2V): Control electronics logic.
 - c. +12V $\pm$ 2% (11,76V to 12,24V): PC supply.
 - d. -12V $\pm$ 2% (-11,76V to -12,24V): PC supply.
 - e. +5V $\pm$ 4% (4,8V to 5,2V): PC supply.
 - f. +3.3V $\pm$ 4% (3,17V to 3,43V): PC supply.
 - g. +5V $\pm$ 4% (4,8V to 5,2V): Standby power.

4.3.2 PCA GEN-STDBY

1. Check that all the cables are correctly connected.
2. Connect the instrument to the mains and switch on the instrument.
3. Check that the LED is ON.

4.3.3 PCA GEN-BCKPL59

1. Check that all the cables are correctly connected.
2. Check that all fuses are present.
3. Connect *Erytra* to the mains and switch on the instrument.
4. Check status of fuses by means of their LEDs.

4.3.4 Computer

1. Check that the computer is switched on when it is pressed the push button. Once is pressed, green LED of the push button is lighted.
2. Check that O.S. is loaded correctly and that there are no problems in the hardware (*Start* menu $\rightarrow$ *Control Panel* $\rightarrow$ *System and Security* $\rightarrow$ *Device Manager*).

4.3.5 Touchscreen monitor

1. Check that the mechanism of the touchscreen monitor opens smoothly.
2. Check that it is possible to tilt the touchscreen monitor when it is extracted completely from its drawer.
3. Check that when the touchscreen monitor is tilted is not possible to enter it in its drawer.
4. Check that the lock system works properly.
5. Check that the point where you put the pointer on the screen is the same that the point where the cursor is placed.

4.3.6 Status LEDs

1. Open Technical Service software, go to *General* tab and switch on the different colours of the upper general LED. Then switch OFF the light.
2. In same tab, check in the same way the LEDs of the keyboard of the samples drawers.

4.3.7 Ambient light

1. In Technical Service software, go to *General* tab and switch on the ambient lights. Check visually that they are ON.
2. Switch OFF the lights, checking visually that they are OFF.

4.3.8 User tray

1. Check that the user tray opens smoothly.
2. Check that the lock system works properly.

4.3.9 Temperature control system

1. Open Technical Service software, go to *General* tab and press *general fans ON* button. Check that the fans are switched ON.
2. Press *general fans OFF* button and check that they are switched OFF.
3. Check that the temperature in the electronics cabinet, reagents and samples floor and cards floor returned by the software is updated and correct.

4.3.10 Buzzer

1. Open Technical Service software, go to *General* tab and activate the buzzer (*Buzzer Mode section*) with MODE1..MODE7 options checking its sound.

4.3.11 Keyboards

1. Open Technical Service software, go to *General* tab and press the buttons of the keyboard and check that its associated key (under *Keyboard* section) is activated. Check that the drawer is opened (before pressing different buttons wait 2 seconds).

4.3.12 Doors

1. Open Technical Service software, go to *General* tab and check that the status of the upper area door is correctly detected (*SRM Door* section).
2. Open Technical Service software, go to *General* tab and check that the status of the intermediate area door is correctly detected (*CM Door* section).

4.3.13 Drawers

OPEN/CLOSE

1. In Technical Service software, perform a home of the PIP arm (*Pip Arm* tab → *Movement* tab → *home* button and then select all axes) .
2. In *Procedures* tab, press *verify* button of *Drawers Module* located in *Checks* section.
3. It will start a wizard in order to check if the reagents/samples/cards drawers are detected in their opened/closed positions.
4. Close all the reagents/samples/cards drawers and press *Next>* button.
5. Open all the reagents/samples/cards drawers and press *Next>* button.
6. Close all drawers and press *Next>* button.
7. A screen appears showing the cards detection status. In case status detection has been performed correctly, pass result is shown.

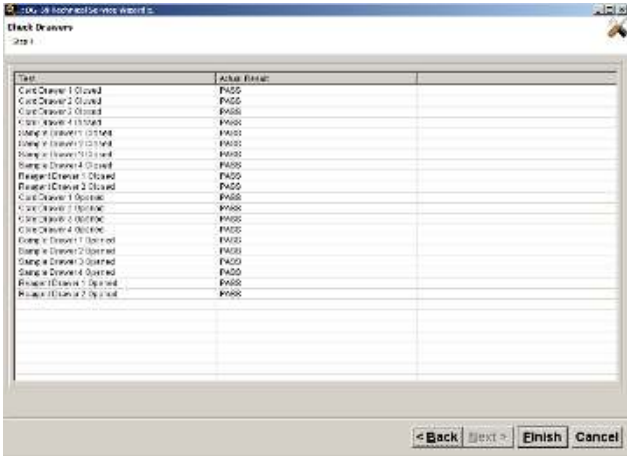


Figure 138. Verification drawers wizard

8. Press *Finish* and *OK* in the information window.
9. Press *OK* button in the confirmation window that indicates that home movements will be performed.
10. Press *OK* button to finish the verification procedure.

REAGENTS STIRRING

1. Open Technical Service software, go to *Home Cycles* subtab in *Drawers* tab.
2. Press *start* button for *Reagent Drawer 1* and/or *Reagent Drawer 2* in order to start cycles of home detection of the reagent racks. Check that no errors appear.
3. In Technical Service software, go to *Reagents Drawers* subtab in *Drawers* tab.
4. Select *REAGENT_DRAWER_1* and/or *REAGENT_DRAWER_2* with *PROFILE_0* and press *Shake* in order to start a stirring process. Check that no errors appear.

4.3.14 Reagent drawer flatness

1. Remove the reagents racks of the drawer to be verified.
2. Remove the reagents racks platforms from the cover of the drawer.
3. Perform a home of probe 1 using *home* button in TSS, *Pip Arm* tab → *Movement* subtab → *Position Actions* section and then selecting *z1* axis.
4. De-energize PIP robotic head using *unlock/stop* button in TSS, *Pip Arm* tab → *Movement* subtab → *Position Actions* and then selecting *z1* axis.
5. Move manually robotic head over the drawer to be checked.
6. Lower probe 1 manually (by means of its belt) until the probe touches gently the following positions, checking in all of them Z1 encoder counts.
 - a. Position 4 of rack 1
 - b. Between positions 5 and 6 of rack 1
 - c. Position 9 of rack 2
 - d. Position 10 of rack 2



Figure 139. Flatness verification points in reagent drawer

7. Check that the difference between the maximum and the minimum encoder counts of the previous 4 points is ≤ 1047 . Otherwise the reagent drawer flatness should be adjusted.

4.3.15 Sample drawer flatness

1. Remove the samples racks of the drawer to be verified.
2. Perform a home of probe 1 using *home* button in TSS, *Pip Arm* tab → *Movement* subtab → *Position Actions* section and then selecting *z1* axis.
3. De-energize PIP robotic head using *unlock/stop* button in TSS, *Pip Arm* tab → *Movement* subtab → *Position Actions* and then selecting *z1* axis.
4. Move manually robotic head over the drawer to be checked.
5. Lower probe 1 manually (by means of its belt) until the probe touches gently the position corresponding to sample tube 1 in rack 1.



Figure 140. Left verification in rack 1 of sample drawer

- 6. Check Z1 encoder counts in TSS.
- 7. Repeat same procedure for position of sample tube 2 in rack 1. Check that the difference in counts between the two positions is ≤ 1047 encoder counts.
- 8. Finally compare if Z1 encoder counts between positions of samples tubes 11 and 12 in rack 2 is also ≤ 1047 encoder counts by means of lowering probe 1 manually (by means of its belt) until the probe touches gently these positions.



Figure 141. Flatness verification points in sample drawer

- 9. In case the difference of any of the two comparisons between left and right side is > 1047 encoder counts, the flatness of the drawer should be adjusted.

4.3.16 Card drawer flatness

- 1. Remove the card racks of the drawer to be verified if any.
- 2. Perform a home of gripper using *home* button in TSS, *Gripper Arm* tab $\rightarrow$ *Movement* subtab $\rightarrow$ *Position Actions* section and then selecting Z axis.
- 3. De-energize CAT robotic head using *unlock/stop* button in TSS, *Gripper Arm* tab $\rightarrow$ *Movement* subtab $\rightarrow$ *Position Actions* section.
- 4. Move manually robotic head over the drawer to be checked.
- 5. Lower gripper manually (by means of its belt) until gripper rear main surface touches gently the left part of the rear centerer of rack 1 of the selected drawer.

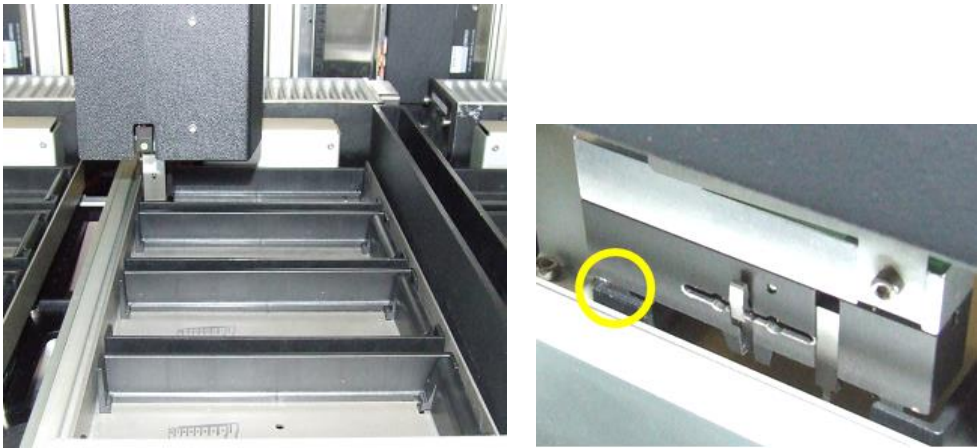


Figure 142. Rear left part verification in rack 1

6. Check Z encoder counts in TSS.
7. Perform a home of the gripper, unlock the axes by means of TSS and then lower the gripper manually (by means of its belt) until gripper rear main surface touches gently the right part of the rear center of rack 1 of the selected drawer.



Figure 143. Rear right part verification in rack 1

8. Check Z encoder counts in TSS.
9. Repeat previous steps (from 4 to 7) in in the frontal center of rack 4 of selected drawer.

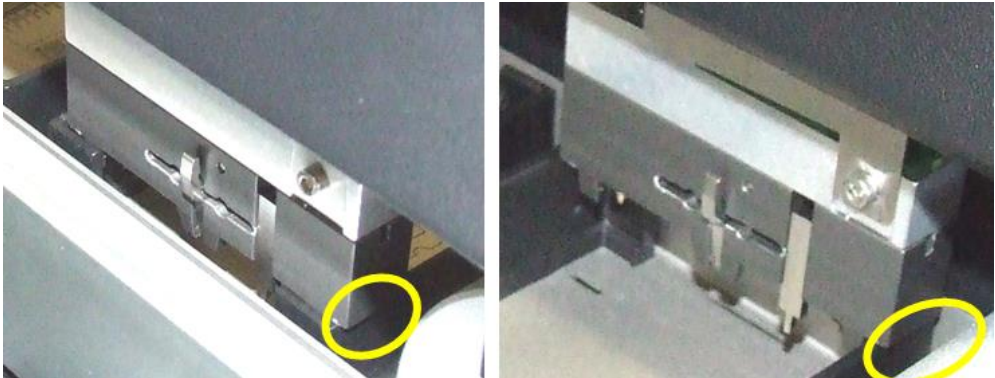


Figure 144. Frontal right part (right picture) and frontal left part (left picture) verification in rack 4

10. Check if the difference in counts between left and right part of the same rack is ≤ 600 . Otherwise adjust the flatness of the drawer.

4.4 REPLACEMENT PROCEDURES

In this section, are detailed replacement procedures for different parts of the general module

4.4.1 Power supply (SN < 059-0001583) replacement

The procedure to replace the power supply is the following:

1. Switch off the instrument and disconnect it from the mains.
2. Take out the rear upper cover to get access to the electronic drawer and disconnect the output cables from the power supply.
3. Remove the input connectors cover of the PCA GEN-STDBY59 by means of its 2 screws.

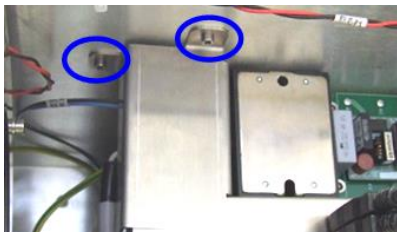


Figure 145. Input connectors cover of the PCA GEN-STDBY59

4. Disconnect the ELE-AC cable from the PCA GEN-STDBY59.
5. Unscrew the PCA GEN-STDBY59 that is attached to the power supply.
6. Remove top cover of the instrument by means of its 4 screws.
7. Remove the 4 screws that hold the power supply to the upper part of the instrument. Be careful when the 4 screws are removed because the power supply can fall over the computer.
8. Remove from the instrument the power supply. Before to remove completely, disconnect the ground cable and the ELE-AC cable from the input of the power supply
9. Place the new power supply and connect all the cables.
10. Repeat steps 1 to 7 in reverse mode.

CABLE	COLOR	OUTPUT	VOLTAGE
ELE-V1	Orange	V1 +	+5V
	Blue	V1 -	
ELE-V1_EARTH	Yellow / Green	V1 -	-
ELE-V2-	Black	V2 -	+24V
	Black	V2 -	
	Black	V2 -	
ELE-V2+	Red	V2 +	+24V
	Red	V2 +	
	Red	V2 +	
ELE-V2_EARTH	Yellow / Green	V2 -	-

CABLE	COLOR	OUTPUT	VOLTAGE
ELE-V3+	Red	V3 +	+5V
	Red	V3 +	
ELE-V3-	Black	V3 -	+5V
	Black	V3 -	
ELE-V4	Orange	V4 +	+3V3
	Black	V4 -	
ELE-V5	Yellow	V5 +	+12V
	Black	V5 -	
ELE-V6	Black	V6 +	-12V
	Blue	V6 -	

Table 35. List of cables of the power supply

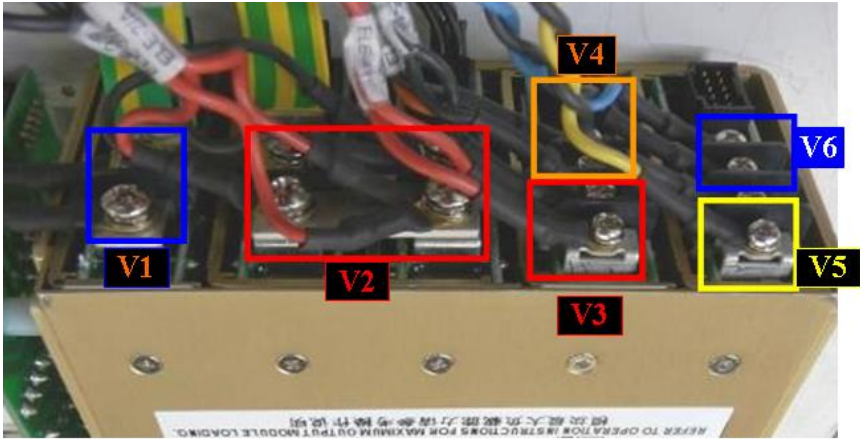


Figure 146. Position of the cables in the power supply as installed on the instrument

4.4.2 Touchscreen monitor replacement

Procedure to replace the touchscreen monitor is listed below for each touchscreen model:

Touchscreen v2 (SN < 059-0001515)

- 1. Make sure that the instrument is switched off and disconnect it from the mains.
- 2. Extract the touchscreen drawer from the instrument by pushing it and then pulling it outwards.
- 3. Remove the 6 screws attaching the monitor cover to the touchscreen drawer (3 at each side, underneath the drawer).
- 4. Remove the 4 screws attaching the monitor to the touchscreen drawer.

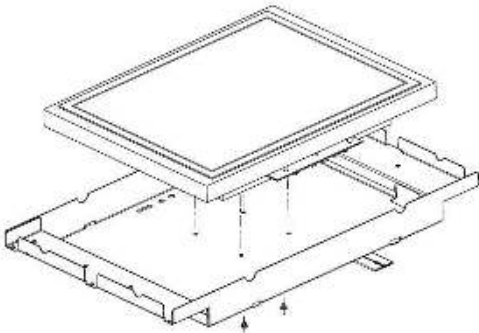


Figure 147. Screws attaching the monitor to the touchscreen drawer

- 5. Carefully lift up the monitor and disconnect the VGA, USB and power cables.
- 6. Remove the USB PCA located in the replaced touchscreen monitor and take the 2 hexagonal spacers. Note that one of the spacers is larger than the other one.

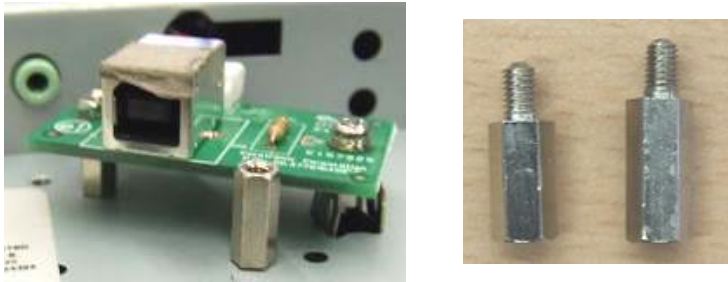


Figure 148. Hexagonal spacers

7. Remove the USB PCA from the new touchscreen monitor and the plate attached.



Figure 149. USB PCA in new touchscreen monitor

8. Install the USB PCA in the new touchscreen monitor in the same place than the replaced touchscreen monitor using the hexagonal spacers. The larger spacer is the one acting as a stopper.

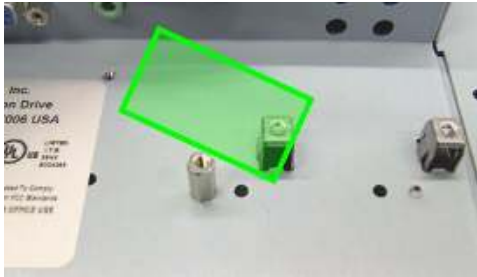


Figure 150. USB PCA installation place

9. Place the new touchscreen monitor, connect the cables and repeat steps 1 to 4 in reverse order.
10. Perform driver installation (refer to chapter 15), if necessary; execute adjustment procedure described in section 4.2.2.1 of this chapter.

Touchscreen v3 (SN ≥ 059-0001515)

1. Make sure that the instrument is switched off and disconnect it from the mains.
2. Extract the touchscreen drawer from the instrument by pushing it and then pulling it outwards.
3. Remove the 6 screws attaching the monitor cover to the touchscreen drawer (3 at each side, underneath the drawer)..
4. Remove the 4 screws attaching the monitor to the touchscreen drawer.

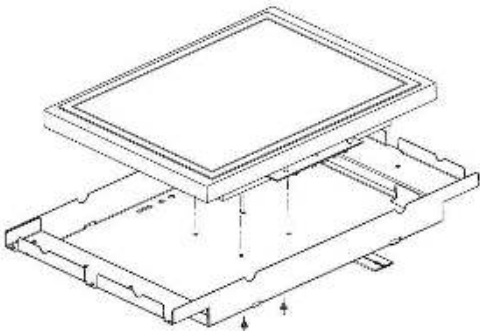


Figure 151. Screws attaching the monitor to the touchscreen drawer

- 5. Carefully lift up the monitor and disconnect the VGA, USB and power cables.
- 6. Remove the monitor, placing the new one in its place. Make sure that the PCA including the monitor configuration buttons is on the inner part of the drawer.
- 7. Carefully lift up the monitor and connect the VGA, USB and power cables, using the adapter cable for the last one. Consider using a cable tie to hold the 3 cables together in the space underneath the lower side of the monitor.



Figure 152. From left to right: Power (adapter), VGA, USB cables

- 8. Attach the monitor to the touchscreen drawer using the 4 previously removed screws.
- 9. Place the monitor cover back in its place, using the previously removed 6 screws.
- 10. Perform driver installation (refer to chapter 15), if necessary; execute adjustment procedure described in section 4.2.2.1 of this chapter.

4.4.3 Reagents and samples drawers keyboard replacement

Procedure to replace the keyboard of the reagents/samples drawers is the following:

- 1. Extract the user tray from the instrument removing the 2 screws that attach it to the handrail.
- 2. Disconnect the VGA, power and USB cables from the computer.
- 3. Remove the 2 frontal screws of the touchscreen drawer and the 2 frontal set screws located rear them.

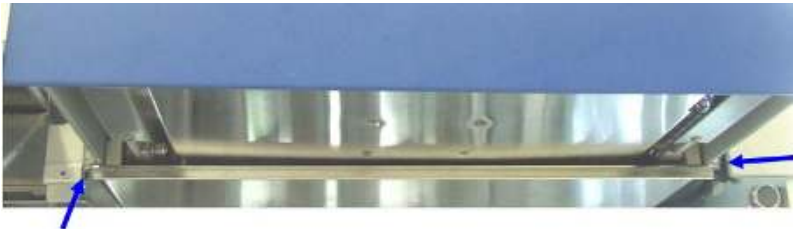


Figure 153. Touchscreen drawer assembly

- 4. Remove the 2 set screws located at the rear part of the drawer. To get a better access, it is recommended to remove the blue right lateral cover of the instrument and the PCA GEN-LED59 (ambient light) of the card drawers area.

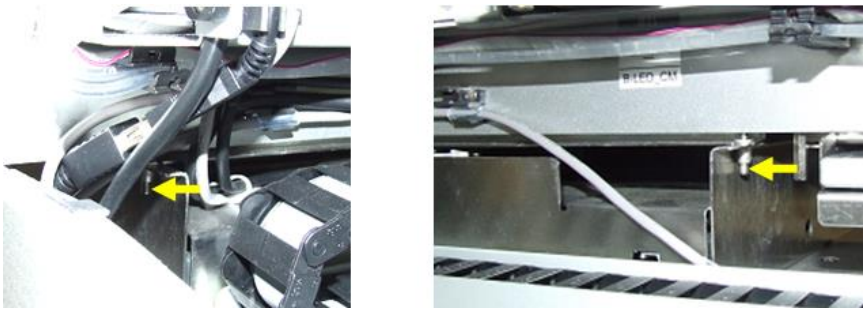


Figure 154. Rear set screws in touchscreen drawer

- 5. Extract as much as possible the touchscreen drawer from the instrument.
- 6. Remove the keyboard cable protective cover by means of loosening the 4 screws and moving the cover to the right and pulling it.
- 7. Remove the 4 nuts that attach the screws of the keyboard to the handrail.
- 8. Before removing completely the keyboard from the instrument, disconnect the keyboard cable.
- 9. Place the new keyboard and repeat steps 1 to 4 in reverse mode.

4.4.4 Card drawers keyboard replacement

Procedure to replace the keyboard of the card drawers is the following:

- 1. Open completely all the card drawers in order to get a better access to the screws.
- 2. Remove the keyboard cable protective cover located under card drawer 4 by means of loosening the 4 screws and moving the cover to the right and pulling it.



Figure 155. Position of the protective cover in card drawers keyboard

- 3. Remove the 4 nuts that attach the screws of the keyboard to the handrail. In order to remove the nut close to the drawer 3, it is recommended to take out the pot of the centrifuge 2.

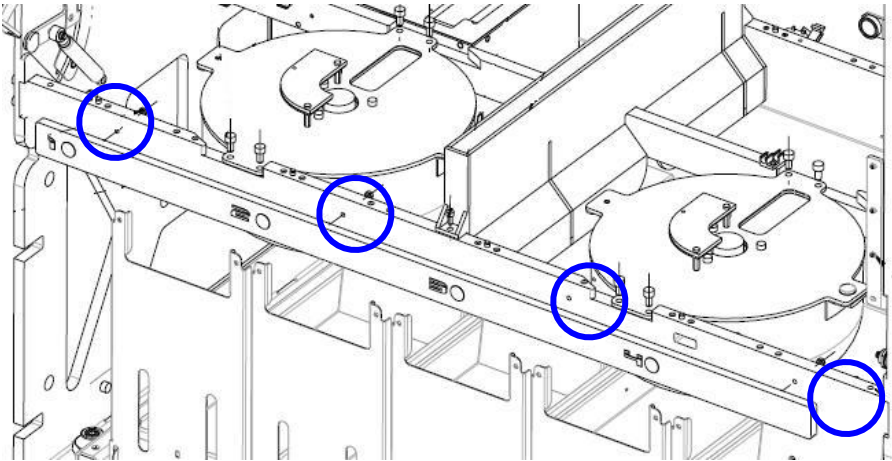


Figure 156. Position of the nuts of card drawers keyboard

- Before removing completely the keyboard from the instrument, disconnect the keyboard cable.
- Place the new keyboard and repeat steps 1 to 4 in reverse mode.

4.4.5 PCA RM-ST59 replacement

The procedure to replace the PCA RM-ST59, located under the reagent drawer, is the following:

- Open the reagents drawer that contains the PCA RM-ST59 to be replaced.
- Remove the reagents racks.
- Remove both reagents platforms.
- Take out the cosmetic cover of the motor.
- Remove carefully the basis of the drawer by means of its screws.
- Disconnect the cable from the motor (power and encoder) and the cable located at the end of the drawer.
- Replace the PCA removing its 4 screws and repeat the steps 1 to 6 in reverse mode.
- Verify the performance of the reagents stirring function.

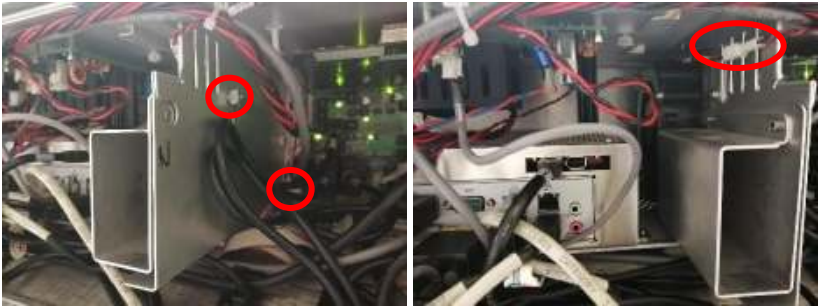
4.4.6 RAM memory modules replacement

To replace the RAM memory modules follow the next steps:

- Switch off the instrument and disconnect it from the mains.
- Remove the screws that attach the instrument upper rear cover, then remove it.
- For instruments with SN < 059-0001583 (refer to INR-0059#0111: NEW ELECTRONICS CABINET) remove 2 screws of the metallic plate.



4. For instruments with SN $\geq$ 059-0001583, to access the screws on the right side of the computer cabinet, remove the fan and its support by removing 2 screws. Disconnect the fan cable.



5. Remove 4 screws that attach ASROCK computer cabinet (rear side of the instrument).



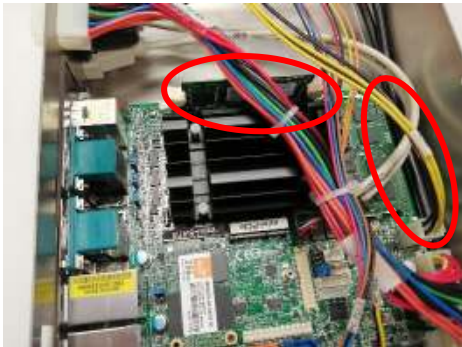
6. Disconnect all ASROCK computer cabling and remove the computer cabinet.



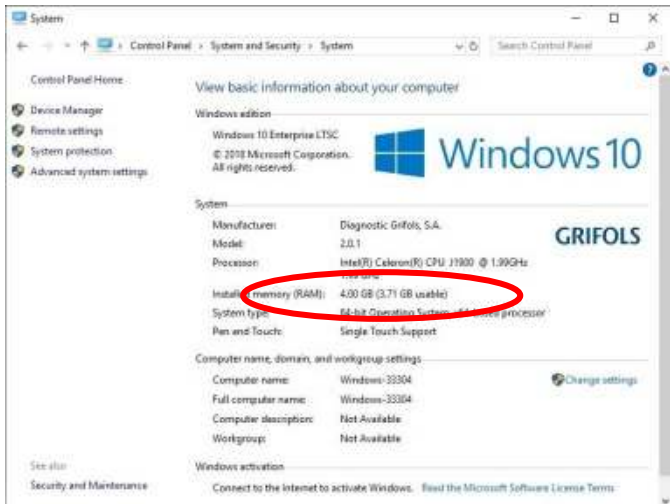
7. Remove 6 screws of the cover of the ASROCK computer cabinet to have access to the 2 RAM memories, pull apart the cover carefully as there is one cable connected.



8. Remove the RAM memories from their slots in the ASROCK computer.



9. Install the new RAM memories and reassemble all the parts again.
10. Finally, switch on the instrument, enter the boot application menu and choose Maintenance Mode. Then check in *Control Panel -> System and Security -> System* that 4GB of RAM are detected.



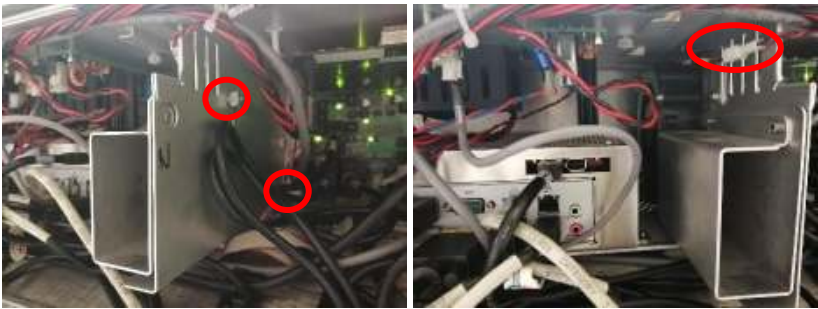
4.4.7 PC motherboard replacement

To replace the PC motherboard, follow next steps:

1. Switch off the instrument and disconnect it from the mains.
2. Remove the screws that attach the instrument upper rear cover, then remove it.
3. For instruments with SN < 059-0001583 (refer to INR-0059#0111: NEW ELECTRONICS CABINET) remove 2 screws of the metallic plate.



4. For instruments with SN $\geq$ 059-0001583, to access the screws on the right side of the computer cabinet, remove the fan and its support by removing 2 screws. Disconnect the fan cable.



5. Remove 4 screws that attach ASROCK computer cabinet (rear side of the instrument).



6. Disconnect all ASROCK computer cabling and remove the computer cabinet.



7. Remove 6 screws of the cover of the ASROCK computer cabinet to have access to the PC motherboard, pull apart the cover carefully as there is one cable connected.



8. Disconnect all the cables connected to the ASROCK computer and remove the board by unscrewing the 4 screws attaching it to its base plate in the computer cabinet.



9. Install the new PC motherboard and reassemble all the parts again.
10. Connect an external keyboard, switch on the instrument and press *Del/Supr* key during the initialization of the computer to enter in the BIOS/UEFI menu.

**NOTE:**
A password to access the BIOS has been set.
When requested enter password: TSDGIH.

11. Once the BIOS software is opened, go to the *Main tab* and check in *UEFI version* parameter that the BIOS firmware version is *IMB-150 P2.20L*.
12. If a new SSD has been installed in the new motherboard, proceed with a *Full Recovery* procedure as described in the Service Manual.
13. If the SSD from the replaced motherboard will be used with the new one, proceed with an *Operating System Recovery* procedure as described in the Service Manual.

4.4.8 Card drawer electric lock replacement

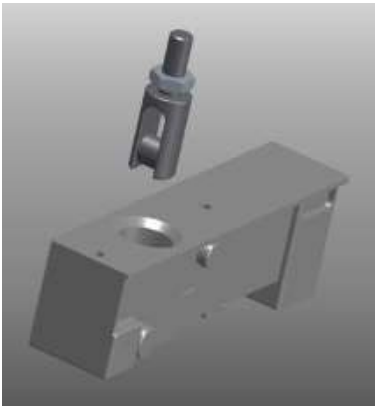
Previous considerations

When assembling P/N 734238 ELECTRIC LOCK v2, the following considerations should be considered in order to ensure a correct assembly of the system:

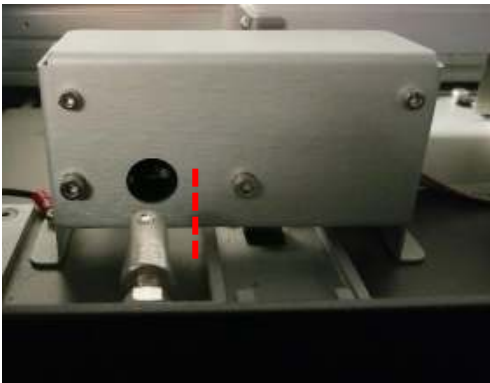
- Ensure a parallel positioning of the electric lock when assembled.



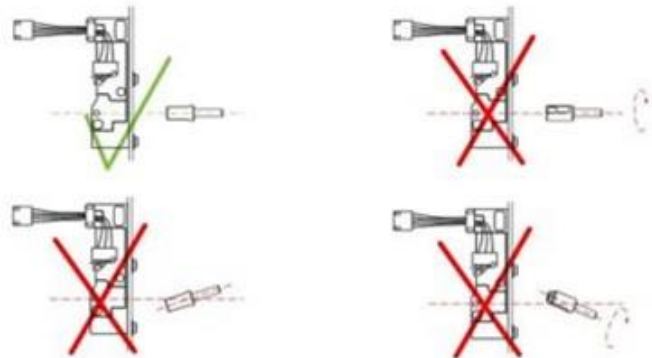
- Initially, the screws of the electric lock should be lightly tightened by hand, so the latch will help to define electric lock final position once inserted.



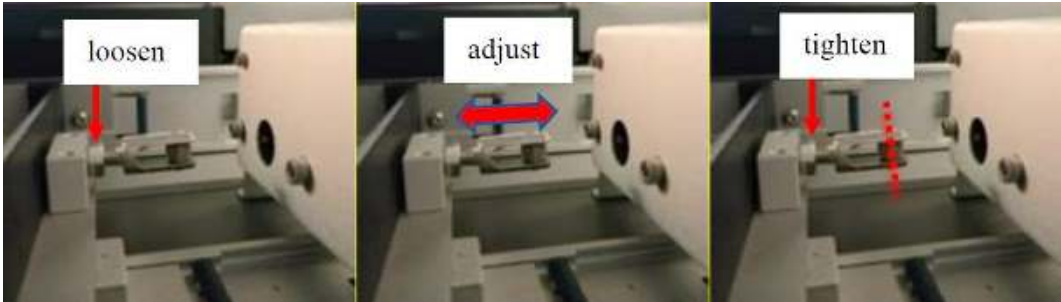
- Electric lock opening and latch must be centered in relation to each other.



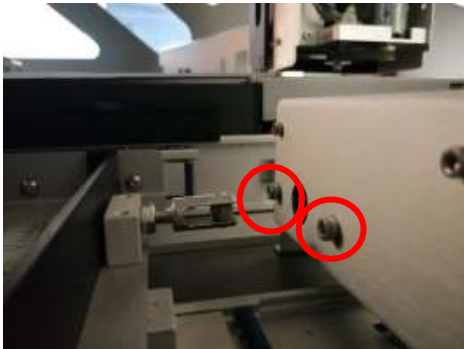
- Screw the latch nut securing the latch against rotations. Parallelism between electric lock opening and the latch must be ensured. Make sure the latch is not rotated and its bolard is in vertical position.



- Before tightening completely the screws of the electric lock, gently close the door/drawer for which the lock is being replaced. The closing force is directly related to the correct alignment and parallelism of the components. If necessary, loosen the nut and adjust the thread length of the latch. Then lock its final position with the nut, taking into account the above-indicated instructions.

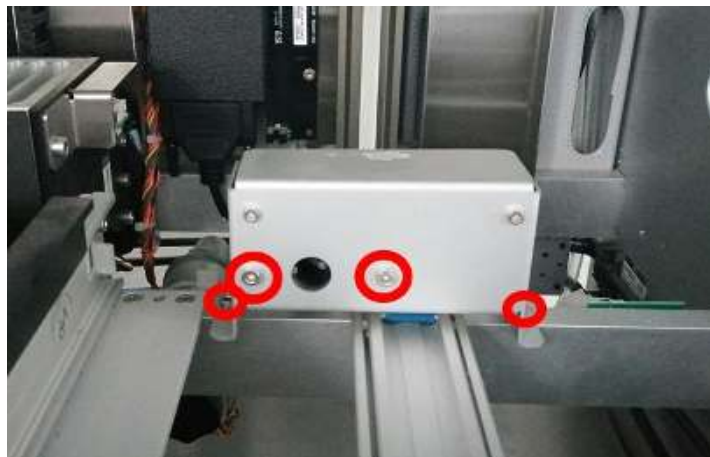


- Open the door/ drawer and tighten completely the screws of the electric lock.



To replace the card drawer electric lock follow the next steps:

1. Using Technical Service software, open the drawer corresponding to the electric lock to be replaced and move the incubator located behind the opened drawer to the upper position.
2. Switch off the instrument and disconnect it from the mains.
3. Pull out the user tray and the screen, then remove the screw of the intermediate door and open intermediate door.
4. Remove the 2 screws that assemble the electric lock cover and the 2 screws that assemble the electric lock.



5. Pull apart the electric lock cover carefully. Disconnect electric lock cable and remove it.



6. Assemble the new electric lock (do it without completely tightening the screws, then close the drawer placing the locker in a position where the drawer latch can be properly inserted) then tighten its screws and reassemble the rest of the parts.
7. Once all the parts are assembled make sure that the cabling in the electric lock area does not interfere with the incubator movement.
8. Check that the drawer closes/opens correctly.
9. Adjust the CAT arm (gripper) on the drawer corresponding to the electric lock that has been replaced.

4.4.9 Samples/reagents drawer electric lock replacement

See *previous considerations* in section 4.4.8 before replacing the electric lock.
To replace the samples/reagents drawer electric lock follow the next steps:

1. Switch off the instrument and disconnect it from the mains.

2. Open manually the drawer by pushing the lever upwards.



3. Remove the 2 screws that assemble electric lock.



4. Disconnect the electric lock cable and remove it.



5. Assemble the new electric lock and reassemble the rest of the parts. When doing so, make sure that the electric lock cable does not interfere with the path of the incubator located behind.
6. Once all the parts are assembled check that the drawer closes/opens correctly.
7. Adjust the PIP arm (probe) on the drawer corresponding to the electric lock that has been replaced.

4.4.10 Fluidic drawer electric lock replacement

See *previous considerations* in section 4.4.8 before replacing the electric lock.
To replace the fluidic drawer electric lock follow the next steps:

1. With the instrument switched off, open manually the drawer corresponding to the electric lock to be replaced by pushing the lever located under the bottle drawer.

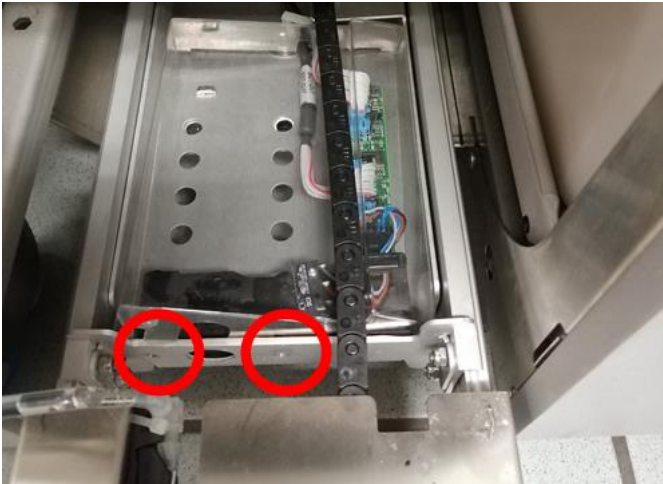
2. Disconnect and remove the bottle from the drawer, then remove the front cover of the drawer by removing its 3 screws.



3. Pull apart the container tube and remove the container basis.



4. Loosen the 2 screws of the electric lock.



5. Disengage 1 slot of the chain so that the plastic cover can be easily removed afterwards.



6. Carefully remove the plastic cover.



- 7. Disconnect the electric lock cable and remove the electric lock.
- 8. Assemble the new electric lock.



NOTE:
When assembling the new lock, consider all the tips listed in the *Previous considerations* section of this document.



CAUTION!

Make sure that the lever of the drawer metallic plate is inserted in the hole for the manual emergency opening of the electric lock.



9. Check that the bottle drawer closes/opens correctly manually through the emergency opening (an external latch can be used to test it).



10. Remove the external latch.
11. Reassemble the rest of the parts.
12. Once all the parts are assembled check that the bottle drawer closes/opens correctly.
13. Check that the load cell adjustment of the bottle drawer is correct.

4.4.11 User tray replacement

To replace the User tray follow the next steps:

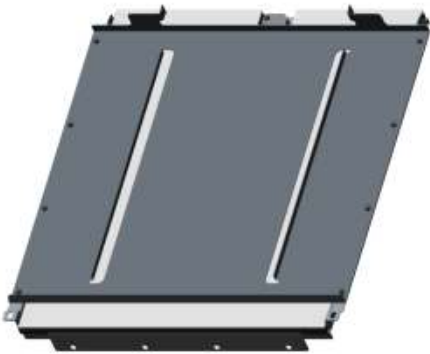
1. Switch off the instrument and disconnect it from the mains.
2. Open Intermediate area door.
3. Pull the user tray out, and remove the cosmetic front panel, unscrewing the 4 screws on its back side.



4. Remove the 2 allen screws attaching the user tray to the instrument.



5. Pull the whole module out of the instrument.



6. Insert the new user tray making sure that the rear flanges enter the corresponding slots in the instrument.



7. Screw the user tray to the instrument, using the 2 previously removed screws.
8. Attach the cosmetic front panel, using the 4 previously removed screws.

4.5 FUSES

Fuses related to General module are detailed in following table,

PCA	FUSE	DESCRIPTION
PCA GEN-BCKPL59	FUSGEN1 FUSGEN2 FUSGEN4 FUS01 FUS02 FUS21 FUS22	MST250 0034.6620 3.15A/250V MST250 0034.6615 1A/250V MST250 0034.6615 1A/250V MST250 0034.6615 1A/250V MST250 0034.6620 3.15A/250V MST250 0034.6615 1A/250V MST250 0034.6620 3.15A/250V
PCA CEN-SRM Control	FUS1	MSF250 0034.6013 0.8A/250V
PCA CAT-CM Control	FUS1	MSF250 0034.6013 0.8A/250V
PCA GEN-STDBY	FUS1	MSF250 0034.6018 F2.5A/250V
PCA GEN-LED59	NA	NA
PCA GEN-FRNT59	NA	NA
PCA GEN-TEMP59	NA	NA
PCA GEN-LCK59	NA	NA
PCA RM-ST59	NA	NA
MAINS CONNECTOR	NA	2xT 6,3A L 250 V, 5x20 mm

Table 36. List of fuses of General module

4.6 LEDS

LEDs related to General module are detailed in following table,

PCA	LED	DESCRIPTION
PCA GEN-BCKPL59	LED1 LED2 LED4 LED5 LED6 LED13 LED14	Green 5V (FUSGEN1) Green 5V (FUSGEN2) Green 24V (FUSGEN4) Green 5V (FUS01) Green 24V (FUS02) Green 5V (FUS21) Green 24V (FUS22)
PCA CEN-SRM Control	LED1	Yellow When the instrument is switched on, LED is ON and a few seconds later is OFF. This means that the FPGA has been loaded correctly with the appropriate data.
	LED2	Green VMotor Ok. It indicates that the voltage supplied to the motor is $\geq 12V$
PCA CAT-CM Control	LED1	Yellow When the instrument is switched on, LED is ON and a few seconds later is OFF. This means that the FPGA has been loaded correctly with the appropriate data.
	LED2	Green VMotor Ok. It indicates that the voltage supplied to the motor is $\geq 12V$
PCA GEN-STDBY	LED1	Green It indicates that there is AC voltage at the input.
PCA GEN-LED59	NA	NA
PCA GEN-FRNT59	NA	NA
PCA GEN-TEMP59	NA	NA
PCA GEN-LCK59	NA	NA
PCA RM-ST59	NA	NA

Table 37. List of LEDs of General module

4.7 JUMPERS AND TEST POINTS

Jumpers and test points related to General module are detailed in the following table,

PCA	JUMPER	TEST POINT
PCA GEN-STDBY (SN < 059-0001583)	JP1	Not present: Standby button controls the switch on of the power supply.
		Present: Power supply is always receiving the mains voltage. Instrument is switched on by means of the switch located in the mains connector. Computer is not switched on.
	JP3	Not present: Electronic power is activated when the push button is pressed.
		Present: Electronic power is controlled by software.

Table 38. List of jumpers and test points of General module

4.8 WIRING TABLE

In the following table, there is detailed all the cables related to General module. A wiring diagram can be found attached to Service Manual.

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
ELE-F1	H	FAN ELEC	FAN1	PCA GEN-BCKPL59
ELE-F2 + ELE-EXT	H	FAN PC	FAN2	PCA GEN-BCKPL59
ELE_+5V	H14	PCA GEN-STDBY	H2	PCA GEN-BCKPL59
ELE-V1 , ELE-V2-, ELE-V2+	V1+, V2-, V2+	POWER SUPPLY	H1	PCA GEN-BCKPL59
ELE-AC	FASE+NEUTRAL	POWER SUPPLY	H2	PCA GEN-STDBY59
ELE-GND	EARTH	POWER SUPPLY	-	CHASSIS
B-AC PWR	OUTPUT	POWER LINE FILTER	H1-PH+N	PCA GEN-STDBY59
B-FIL_AC	PH+N	MAINS	INPUT	POWER LINE FILTER
B-FIL_EA	EARTH	MAINS	INPUT	POWER LINE FILTER
B-EARTH	EARTH	MAINS	H1-EARTH	PCA GEN-STDBY59
ELE-V4,ELE-V3+,ELE-V3-	V3+,V3-,V4+	POWER SUPPLY	H3	PCA GEN-STDBY
ELE-V5,ELE-V6	V5+,V5-,V6+,V6-	POWER SUPPLY	H4	PCA GEN-STDBY

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
ELE-ATX	ATX POWER	COMPUTER	H5	PCA GEN-STDBY
ELE-V1s	V1s	POWER SUPPLY	H9	PCA GEN-STDBY
ELE-V2s	V2s	POWER SUPPLY	H10	PCA GEN-STDBY
ELE-V3s/V4s	V3/4s	POWER SUPPLY	H11	PCA GEN-STDBY
ELE-FEAT	Pc Feature	COMPUTER	H12	PCA GEN-STDBY
ELE-COM	COM1	COMPUTER	H3	PCA GEN-BCKPL59
B52_BUTTON	-	PUSHBUTTON	H4	PCA GEN-FRNT59
B-PWR_FR	ATX	COMPUTER	H5	PCA GEN-FRNT59
B-USB1_FR	USB4	COMPUTER	H3	PCA GEN-FRNT59
B-USB2_FR	USB5	COMPUTER	H2	PCA GEN-FRNT59
B-TFT_USB	USB0	COMPUTER	USB	TFT SCREEN
B-TFT_VGA	VGA	COMPUTER	ADAPTOR DB15	NA
TFT-VGA	VGA	TFT SCREEN	ADAPTOR DB15	NA
B-TFT_PWR	H13	PCA GEN-STDBY59	DC In +12V	TFT SCREEN
B-LAN_RE	LAN1	COMPUTER	LAN	ELECTRONIC COVER
B-COM_RE	COM2	COMPUTER	COM	ELECTRONIC COVER
B-USB2_RE	USB1	COMPUTER	USB1	ELECTRONIC COVER
B-USB1_RE	USB3	COMPUTER	USB3	ELECTRONIC COVER
B-USB4_RE	USB6	COMPUTER	USB6	ELECTRONIC COVER
B-USB3_RE	USB7	COMPUTER	USB7	ELECTRONIC COVER
B-LED_SRM	H1	PCA GEN-LED59 SRM	GEN_LED SR M	PCA GEN-BCKPL59
B-LED_CM	H1	PCA GEN-LED59 CM	GEN_LED CM	PCA GEN-BCKPL59

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
B-LED	H1	PCA GEN-LED59	GEN_LED	PCA GEN-BCKPL59
B-KCM	-	KEYBOARD CM	KEYBD_CM	PCA GEN-BCKPL59
B-KSRM	-	KEYBOARD SRM	KEYBD_SRM	PCA GEN-BCKPL59
B-CM_LOCK	H1 FROM MASTER	PCA GEN-LCK59 CARD DRAWER 1	CM_LCK	PCA GEN-BCKPL59
B-CM1/2	H2 TO SLAVE	PCA GEN-LCK59 CARD DRAWER 1	H1 FROM MASTER	PCA GEN-LCK59 CARD DRAWER 2
NA	H3	PCA GEN-LCK59 CARD DRAWER 1	H	SAFETY LOCK
B-CM2/3	H2 TO SLAVE	PCA GEN-LCK59 CARD DRAWER 2	H1 FROM MASTER	PCA GEN-LCK59 CARD DRAWER 3
NA	H3	PCA GEN-LCK59 CARD DRAWER 2	H	SAFETY LOCK
B-CM3/4	H2	PCA GEN-LCK59 CARD DRAWER 3	H1	PCA GEN-LCK59 CARD DRAWER 4
NA	H3	PCA GEN-LCK59 CARD DRAWER 3	H	SAFETY LOCK
B-CM4/TEMP	H2	PCA GEN-LCK59 CARD DRAWER 4	H1	PCA GEN-TEMP59
NA	H3	PCA GEN-LCK59 CARD DRAWER 4	H	SAFETY LOCK
B_SRM_LOCK	H1 FROM MASTER	PCA GEN-LCK59 REAGENT DRAWER 1	SRM_LCK	PCA GEN-BCKPL59
B-RM1/2	H2 TO SLAVE	PCA GEN-LCK59 REAGENT DRAWER 1	H1 FROM MASTER	PCA GEN-LCK59 REAGENT DRAWER 2
B-STIR1	RM_1	PCA GEN-BCKPL59	H1	PCA RM-CON59 REAGENT DRAWER 1
NA	H2	PCA RM-CON59 REAGENT DRAWER 1	H1	PCA RM-ST59 REAGENT DRAWER 1
NA	H2	PCA RM-ST59 REAGENT DRAWER 1	ENCODER	REAGENT MOTOR
NA	H3	PCA RM-ST59 REAGENT DRAWER 1	POWER	REAGENT MOTOR
B-STIR2	RM_2	PCA GEN-BCKPL59	H1	PCA RM-CON59 REAGENT DRAWER 2
NA	H2	PCA RM-CON59 REAGENT DRAWER 2	H1	PCA RM-ST59 REAGENT DRAWER 2
NA	H2	PCA RM-ST59 REAGENT DRAWER 2	ENCODER	REAGENT MOTOR
NA	H3	PCA RM-ST59 REAGENT DRAWER 2	POWER	REAGENT MOTOR

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
B-SRM_LOCK	SRM_LCK	PCA GEN-BCKPL59	H1	PCA GEN-LCK59 REAGENT DRAWER 1
B-RM1/2	H2	PCA GEN-LCK59 REAGENT DRAWER 1	H1	PCA GEN-LCK59 REAGENT DRAWER 2
NA	H3	PCA GEN-LCK59 REAGENT DRAWER 1	H	SAFETY LOCK
B-RM2/SM1	H2	PCA GEN-LCK59 REAGENT DRAWER 2	H1	PCA GEN-LCK59 SAMPLE DRAWER 1
NA	H3	PCA GEN-LCK59 REAGENT DRAWER 2	H	SAFETY LOCK
B-SM1/2	H2	PCA GEN-LCK59 SAMPLE DRAWER 1	H1	PCA GEN-LCK59 SAMPLE DRAWER 2
NA	H3	PCA GEN-LCK59 SAMPLE DRAWER 1	H	SAFETY LOCK
B-SM2/3	H2	PCA GEN-LCK59 SAMPLE DRAWER 2	H1	PCA GEN-LCK59 SAMPLE DRAWER 3
NA	H3	PCA GEN-LCK59 SAMPLE DRAWER 2	H	SAFETY LOCK
B-SM3/4	H2	PCA GEN-LCK59 SAMPLE DRAWER 3	H1	PCA GEN-LCK59 SAMPLE DRAWER 4
NA	H3	PCA GEN-LCK59 SAMPLE DRAWER 3	H	SAFETY LOCK
B-SM4/TEMP	H2	PCA GEN-LCK59 SAMPLE DRAWER 4	H1	PCA GEN-TEMP59
NA	H3	PCA GEN-LCK59 SAMPLE DRAWER 4	H	SAFETY LOCK
B-DOOR_SRM	NA	SRM DOOR SENSOR	PIP_DOOR	PCA GEN-BCKPL59
B-DOOR_CM	NA	CM DOOR SENSOR	CAT_DOOR	PCA GEN-BCKPL59

Table 39. List of wiring of General module

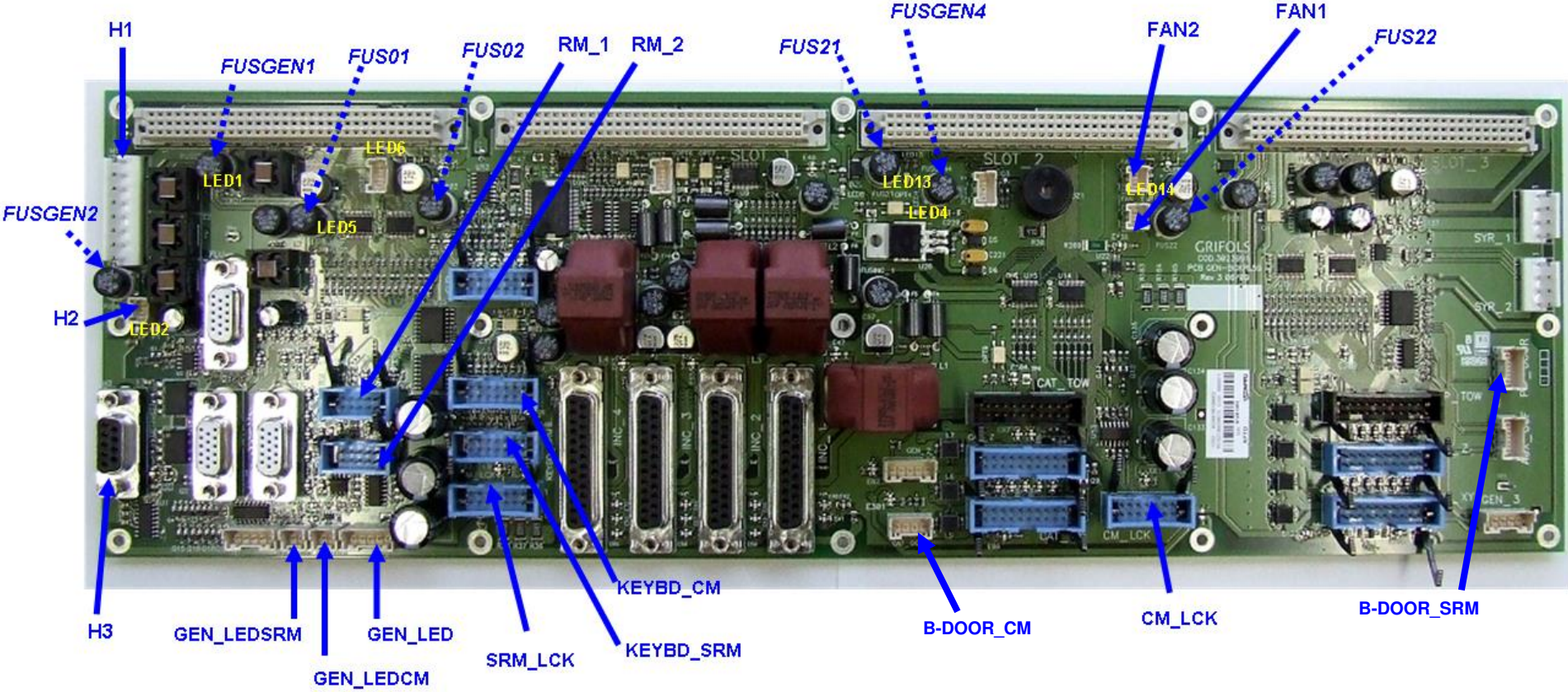


Figure 157. Connectors, fuses and LEDs of PCA GEN-BCKPL59 related to General module

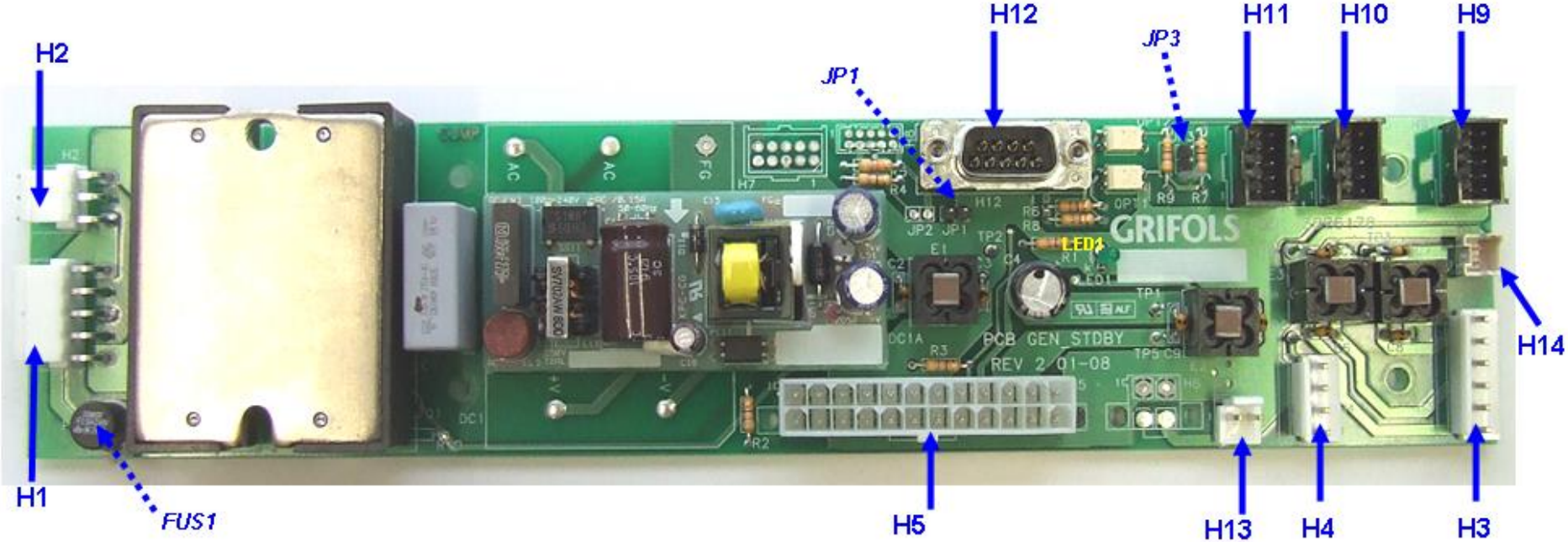


Figure 158. Connectors, fuses and LEDs of PCA GEN-STDBY (units with sn < 059-0001583)



Figure 160. Connectors of PCA RM-ST59

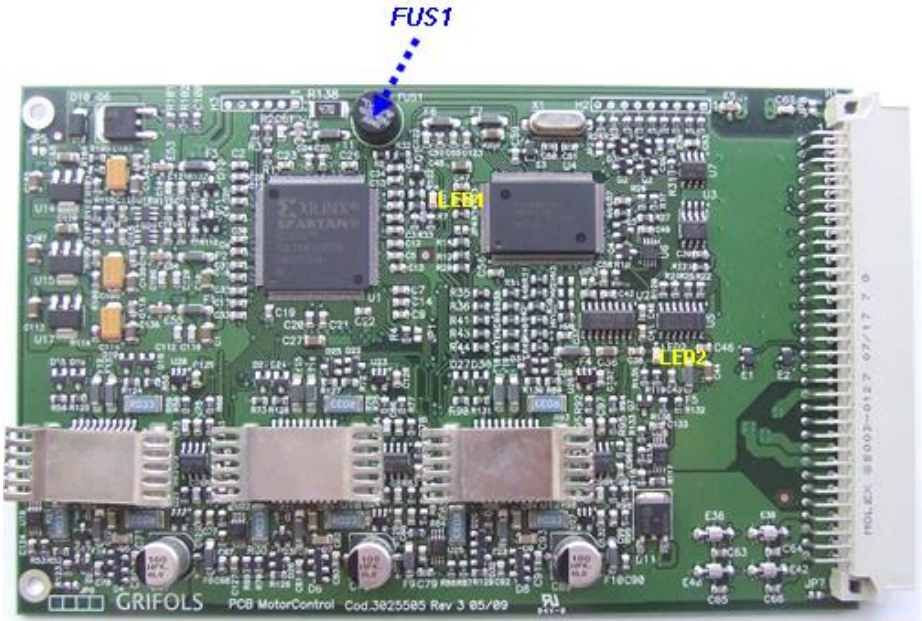


Figure 159. Location of the fuse and LEDs in CEN-SRM / CAT-CM Control

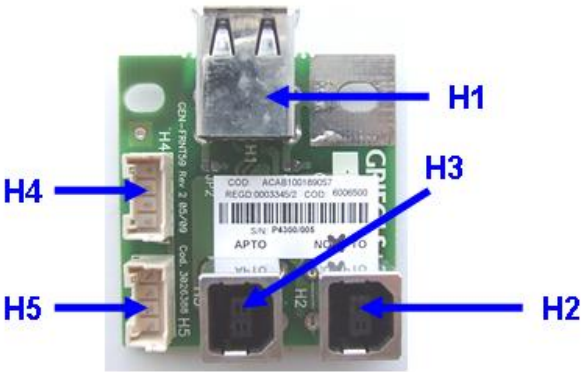


Figure 161. Connectors of PCA GEN-FRNT59



Figure 162. Connectors of PCA GEN-TEMP59

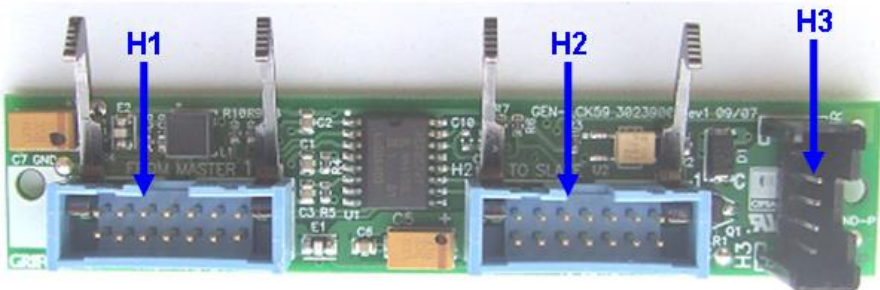


Figure 163. Connectors of PCA GEN-LCK59

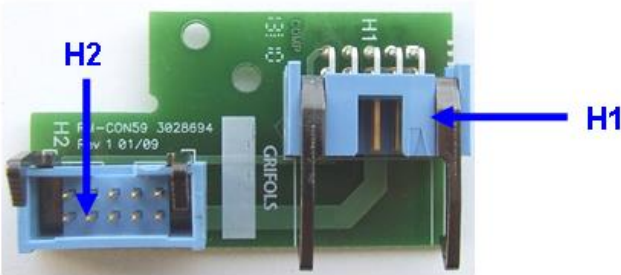


Figure 164. Connectors of PCA RM-CON59

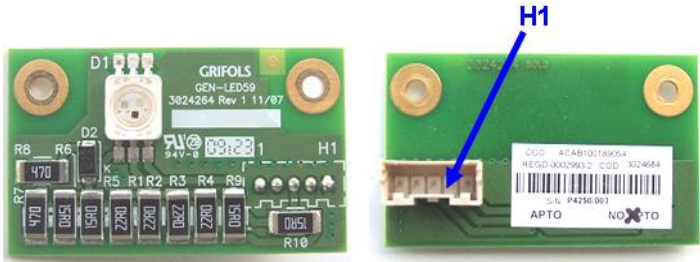


Figure 165. Connectors of PCA GEN-LED59

4.9 TROUBLESHOOTING

Module	Device	Error code	Error	Cause	Text	Options
SRM (Samples And Reagents Management Module) (09)	Samples and reagents lockers (01) Drawer 1 (02) Drawer 2 (03) Drawer 3 (04) Drawer 4 (05) Drawer 5 (06) Drawer 6	003000	ERRORCODE_LEN	FW has received a command with incorrect length	Unrecoverable error. Please press OK to RESTART.	OK
		005000	ERRORCODE_ARG	Command has wrong arguments		
		039000	SWITCH_ERR	Locker error detected		
		040000	ERRORCODE_DRAWER_UNKNOWN_ERROR	Drawer unknown error	NA	NA
		006000	ERRORCODE_BUSY	Command is addressed to a busy device		
		038000	ERRORCODE_DRAWER_OPEN_SYNC_TIMEOUT	Drawer has been busy for more than its timeout margin		
		088000	ERRORCODE_PRECONDITION_FAILED	A precondition is not reach to perform the action		
	Reagents stirrer motor (16) Drawer 1 (17) Drawer 2	003000	ERRORCODE_LEN	FW has received a command with incorrect length	Error during reagents stirring process or homogenization in dilution cup. Please choose one of the following options:	RETRY CANCEL Works (it leads to: <i>Reagent drawer has been disabled. Please reboot the system as soon as possible. If problem persists, contact Technical Service.</i>)
		005000	ERRORCODE_ARG	Command has wrong arguments		
		006000	ERRORCODE_BUSY	Command is addressed to a busy device		
		007000	ERRORCODE_NOTHOMED	Move command is addressed to a not homed device		
		012000	ERRORCODE_SPINBADPROFILE	Command includes a spin profile that is invalid for the addressed device		
		013000	ERRORCODE_NOTSPINNING	The addressed device is not performing the requested spin		
		032000	CURRENT_LIMIT	Current limit at servo device		
		033000	POSITION_ERROR	A position error has been produced at a servo device		
		034000	MOVEMENT_ERROR	A movement error has been produced at a servo device		
		035000	HOME_TIMEOUT	A home timeout has been produced at a servo device		

Module	Device	Error code	Error	Cause	Text	Options
General module (06)	Keyboard (01)	info	KEYBD1_DRIVER_ERR	Keyboard driver error	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		info	KEYBD2_DRIVER_ERR	Keyboard driver error		
	Lights (03)	info	AMBIENT_DRIVER_ERR	Ambient lights driver error		
		info	DRIVER_ERR	Driver error		
	Backpanel Temp sensors & fans (04)	info	SENSOR_ERR	Error detected in the value measured by the sensor – electronics cabinet		
		info	DRIVER_ERR	Electronics cabinet fan is in Circuit open, Driver error		
		info	CURRENT_LIMIT	Electronics cabinet fan is in Circuit open, Current limit error		
	Instrument temp sensors (04)	info	SENSOR_ERROR	Error detected in the value measured by the sensor		
CM (Cards Management Module) (01)	Cards lockers (00) Drawer 1 (01) Drawer 2 (02) Drawer 3 (03) Drawer 4	003000	ERRORCODE_LEN	FW has received a command with incorrect length	Unrecoverable error. Please press OK to RESTART.	OK
		005000	ERRORCODE_ARG	Command has wrong arguments		
		039000	SWITCH_ERR	Locker error detected		
		040000	ERRORCODE_DRAWER_UNKNOWN_ERROR	Drawer unknown error	NA	NA
		006000	ERRORCODE_BUSY	Command is addressed to a busy device		
		038000	ERROR_CODE_DRAWER_OPEN_SYNC_TIMEOUT	Drawer has been busy for more than its timeout margin		
		088000	ERRORCODE_PRECONDITION_FAILED	A precondition is not reach to perform the action		

Module	Device	Error code	Error	Cause	Text	Options
Any (xx)	Any (xx)	002000	ERRORCODE_CMD	Unknown command code received	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		003000	ERRORCODE_LEN	FW has received a command with incorrect length		
		004000	ERRORCODE_DEV	Command is addressed to an incorrect device		
		005000	ERRORCODE_ARG	Command has wrong arguments		
		014000	ERRORCODE_HW_DEV	The addressed device has a problem with its related HW		
		015000	ERRORCODE_HW_HWFAIL	There is a problem with PCA HW		
		016000	ERRORCODE_HW_HWNOTSET	PCA setup error		
Non firmware errors (xx)	Non firmware errors (xx)	000000	ERRORCODE_UNKNOWN	Unknown error		
		008000	ERRORCODE_FW_NOTSAFE	Movement not safe		
		017000	ERRORCODE_FW_UNKNOWN_ERROR	Unknown error		
		061000	ERRORCODE_INVALID_ARGUMENTS	Invalid arguments		
		062000	ERRORCODE_FWCOMM_TIMEOUT	Communication timeout		
		063000	ERRORCODE_FWCOMM_CONNECTION_LOST	Communication lost		

Module	Device	Error code	Error	Cause	Text	Options
Non firmware errors (xx) 0	Non firmware errors (xx) 240	064000	ERRORCODE_FWCOMM_COMM_ERROR	Communication error	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		088000	ERRORCODE_PRECONDITION_FAILED	Previous condition for the action is not met		
		089000	ERRORCODE_STARTUP_ERROR	Driver startup error		
		091000	ERRORCODE_ALREADY_STARTED	Another instance of Driver is already opened		
		090000	ERRORCODE_UNEXPECTED_ERROR	A Driver unexpected error has been produced		

Table 40. List of errors related to General module.

POWER SUPPLY ISSUES

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Instrument does not switch on	AC cable disconnected	Connect AC cable to mains connector
	Switch of mains connector is in OFF position	Switch on mains connector
	Fuses of mains connector are blown	Replace fuses
	Cables of mains connector disconnected / damaged	Check / replace cables
	Pushbutton is damaged	Check / replace pushbutton
	Fuse of PCA GEN-STDBY59 blown (sn < 059-0001583)	Check / replace FUS1 of PCA GEN-STDBY59
	PCA GEN-FRNT59 damaged	Check / replace PCA GEN-FRNT59
	PCA GEN-STDBY59 damaged (sn < 059-0001583)	Check / replace PCA GEN-STDBY59
	Power supply damaged	Replace power supply
Output voltages do not match with expected ones	Power supply damaged	Replace power supply

COMPUTER / SOFTWARE ISSUES

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Not possible to run the computer	Power supply failure	Check / replace power supply
		Check / replace PCA GEN-STDBY59 (sn < 059-0001583)
	Hardware error	Replace computer
		Replace SSD
Computer is switched off automatically	Fan of the computer is blocked	Check internal cables of the computer
	Electronic cabinet fans are not working	Check/replace electronic cabinet fans
O.S. errors	Software error	Install default image of SSD
	Wrong configurations	Check applied configurations
USB connector does not work	Hardware error	Check/replace wiring
	Software error	Install default image of SSD
	Computer USB plug does not work	Replace computer
Low SSD space	No space in SSD	Backup and delete files
Adobe Air installer reports that installer is corrupted	Software error	Check configuration of O.S. date
		Check that ErytraLogin is not running in the background when installing Adobe Air
Java error when starting the Driver in TSS or login into application	Adjustment files are not found	Check that <i>PIPCalibrations.txt</i> and <i>CATCalibrations.txt</i> are present in <i>E:\data\Erytra\conf</i> and they are <i>not read only</i> .
java.lang.NullPointerException when starting Erytra software		Delete <i>E:\data\Erytra\startup.check</i> file
Server not available in Erytra Login	Incorrect <i>erytra.properties</i> configuration	Check <i>erytra.properties</i> configuration
	Erytra database is not found	Install Erytra database
	Erytra database folder is corrupted	Rename <i>E:\data\Database</i> folder and run <i>E:\Program Files\Liquibase\createdb.bat</i> to create new database structure. Then restore a database backup.
	Erytra software files corrupted	Reinstall O.S and Erytra software

TOUCHSCREEN ISSUES

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Flickering	VGA cable is not connected properly	Check VGA connection
	VGA cable is damaged	Replace VGA cable
	Touchscreen is damaged	Replace touchscreen monitor
Image has a red/green color	VGA cable is not connected properly	Check VGA connection
	VGA cable is damaged	Replace VGA cable
	Touchscreen is damaged	Replace touchscreen monitor
Cursor on the screen does not match where it is pointed on the screen	Screws of the touchscreen cover are too tight	Loose a little bit touchscreen cover screws
	Incorrect drivers installed	Check/Install proper drivers
	Touchscreen is not calibrated	Calibrate touchscreen
	Touchscreen is damaged	Replace touchscreen monitor
Hard to insert / extract touchscreen monitor into/from the instrument	Drawer lock is not correctly placed	Check/adjust drawer lock position
	Touchscreen cover is not correctly placed	Check cover position and tighten/loose its screws
	Cables of the touchscreen monitor are not correctly placed	Check cables position
Impossible to insert touchscreen monitor into the instrument	Touchscreen monitor can not be placed flat	Check that touchscreen monitor can be placed completely flat before to insert it into the instrument
		Lubricate tilting latch lock

UPPER/INTERMEDIATE AREA DOOR ISSUES

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Door status not detected	Both parts of the celduc sensor are not in contact when door is closed	Check assembly of celduc sensor
	Celduc sensor is damaged	Replace celduc sensor
Noise when opening/closing the door	Lack of lubrication	Lubricate the door
Door is not aligned	Incorrect assembly of the magnets of the doors	Adjust the position of the door

USER TRAY ISSUES

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Hard to insert / extract user tray into/from the instrument	User tray is not lubricated	Lubricate user tray
	Drawer lock is not correctly placed	Check/adjust drawer lock position
User tray exists completely from the drawer	Stopper broken	Replace user tray

STATUS LED ISSUES

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Status LED does not switch on	PCA GEN-LED59 damaged	Replace PCA GEN-LED59
	Cable damaged	Replace cable (B-LED)
A color is not activated	PCA GEN-LED59 damaged	Replace PCA GEN-LED59

AMBIENT LIGHTS ISSUES

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Upper/Intermediate area ambient light does not switch on	PCA GEN-LED59 damaged	Replace PCA GEN-LED59
	Cable damaged	Replace cable (B-LED_SRM or B-LED_CM)

SAMPLE DRAWERS ISSUES

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Drawer not opened SD3: 0903006000 SD3: 0903038000 SD3: 0903039000 SD4: 0904006000 SD4: 0904038000 SD4: 0904039000 SD5: 0905006000 SD5: 0905038000 SD5: 0905039000 SD6: 0906006000 SD6: 0906038000 SD6: 0906039000	Keyboard damaged	Check/replace keyboard
	Keyboard cable damaged	Check/replace B-KSRM cable
	Cable between locks damaged	Check / replace cable between locks
	Lid electric lock damaged	Check/replace lid electric lock
	PCA GEN-LCK59 damaged	Check/replace PCA GEN-LCK59
Incorrect drawer opened	Keyboard damaged	Check/replace keyboard
	Keyboard cable damaged	Check/replace keyboard cable
	Cable between locks damaged	Check / replace cable between locks
	PCA GEN-LCK59 damaged	Check/replace PCA GEN-LCK59
	PCA GEN-BCKPL59 damaged	Check/replace PCA GEN-BCKPL59
Sample rack is too loose / too tight over the drawer	Centerers of the drawer are not correctly placed	Adjust centerers position and then readjust probe in the drawer
Cosmetic front panel is not aligned	Cosmetic front panel screws are loose	Align cosmetic front panel
Drawer is not aligned	Safety latch depth position is not correct	Adjust safety latch position and then readjust probe in the drawer
Hard to open / close the drawer	Lack of lubrication	Lubricate
	Linear guides damaged	Replace linear guides
Sample tube tilted	Sample holder of rack damaged	Replace sample rack

CARD DRAWERS ISSUES

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Drawer not opened CD1: 0100006000 CD1: 0100038000 CD1: 0100039000 CD2: 0101006000 CD2: 0101038000 CD2: 0101039000 CD3: 0102006000 CD3: 0102038000 CD3: 0102039000 CD4: 0103006000 CD4: 0103038000 CD4: 0103039000	Keyboard damaged	Check/replace keyboard
	Keyboard cable damaged	Check/replace B-KCM cable
	Cable between locks damaged	Check / replace cable between locks
	Lid electric lock damaged	Check/replace lid electric lock
	PCA GEN-LCK59 damaged	Check/replace PCA GEN-LCK59
Incorrect drawer opened	Keyboard damaged	Check/replace keyboard
	Keyboard cable damaged	Check/replace B-KCM cable
	Cable between locks damaged	Check / replace cable between locks
	PCA GEN-LCK59 damaged	Check/replace PCA GEN-LCK59
	PCA GEN-BCKPL59 damaged	Check/replace PCA GEN-BCKPL59
Card rack is not correctly placed over the drawer	Centerers of the drawer are not correctly placed	Check centerers position
Cosmetic front panel is not aligned	Cosmetic front panel screws are loose	Align cosmetic front panel

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Drawer is not aligned	Safety latch depth position is not correct	Adjust safety latch position and then readjust probe in the drawer
Hard to open / close the drawer	Lack of lubrication	Lubricate
	Linear guides damaged	Replace linear guides

REAGENT DRAWERS ISSUES

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Movement error / Current limit error RD1: 0916032000 RD1: 0916034000 RD2: 0917032000 RD2: 0917034000	Fuse blown	Check / replace FUS01 or FUS02
	Dirt under reagent rack platform	Clean under reagent rack platform
	Feet of the rack platform are not lubricated properly	Lubricate with grease underneath the feet of rack platform
	Motor belt is too tight	Check/adjust motor belt tension
	Motor belt is damaged	Replace motor belt
	Motor/encoder damaged	Check/replace motor
	Ball bearing damaged	Check/replace ball bearing
	PCA RM-ST59 failure	Check/replace PCA RM-ST59
	Cable between PCA RM-ST59 and PCA RM-CON59 damaged	Check/replace cable
Home position is not detected RD1: 0916035000 RD2: 0917035000	PCA CEN-SRM Control failure	Check/replace PCA CEN-SRM Control
	Hall effect sensor tilted	Check position of Hall effect sensor in PCA RM-ST59
	Cable between PCA RM-ST59 and PCA RM-CON59 damaged	Check/replace cable
Drawer not opened RD1: 0901006000 RD1: 0901038000 RD1: 0901039000 RD2: 0902006000 RD2: 0902038000 RD2: 0902039000	PCA RM-ST59 failure	Check/replace PCA RM-ST59
	Keyboard damaged	Check/replace keyboard
	Keyboard cable damaged	Check/replace B-KSRM cable
	Cable between locks damaged	Check / replace cable between locks
	Lid electric lock damaged	Check/replace lid electric lock
Incorrect drawer opened	PCA GEN-LCK59 damaged	Check/replace PCA GEN-LCK59
	Keyboard damaged	Check/replace keyboard
	Keyboard cable damaged	Check/replace keyboard cable
	Cable between locks damaged	Check / replace cable between locks
	PCA GEN-LCK59 damaged	Check/replace PCA GEN-LCK59
Cosmetic front panel is not aligned	PCA GEN-BCKPL59 damaged	Check/replace PCA GEN-BCKPL59
	Cosmetic front panel screws are loose	Align cosmetic front panel
Drawer is not aligned	Safety latch depth position is not correct	Adjust safety latch position and then readjust probe in the drawer
Hard to open / close the drawer	Lack of lubrication	Lubricate
	Linear guides damaged	Replace linear guides

Table 41. General module troubleshooting

5 INCUBATOR (INC) MODULE

5.1 DESCRIPTION

There are 2 main functions of this module:

- Incubate the cards to the defined temperature.
- Move cards from intermediate level to upper level and vice-versa (lift) retaining the incubation block/card position during the operations performed in each level.

There are 4 independent incubator blocks located in the rear part of the instrument. Each block has a capacity for 12 cards positioned equidistantly.

The edges of the metallic plates where the incubators move upwards and downwards along the linear guides have been hemmed in order to avoid injuries to the field engineer when accessing this area. However there are available edge protectors to be installed in instruments without hemmed edges to mitigate this risk.

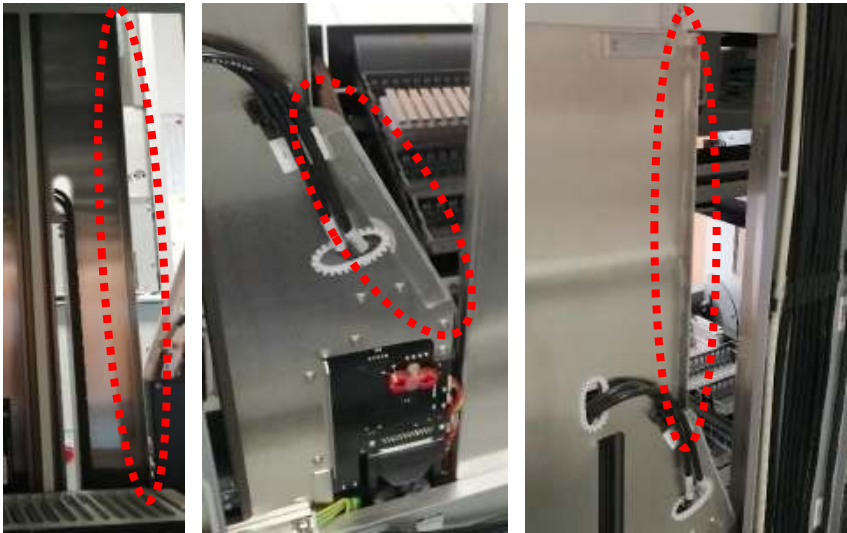


Figure 166. Edge protectors on incubator area

Incubation procedure has 2 parameters: temperature and time. Temperature is controlled by firmware, while the time is controlled by software.

Two temperatures are allowed: RT (room temperature), that is $24\text{ }^{\circ}\text{C} \pm 3^{\circ}\text{C}$, and $37\text{ }^{\circ}\text{C} \pm 1^{\circ}\text{C}$. Temperature is modified by a Peltier module unit (formed by Peltier cell, fan and heat sink; 24VDC, 3.3A) attached to incubator block (bottom). This module unit is used as reversible heat pump to warm up or cool down the block to the desired temperature. Temperature is monitored by two temperature sensors in order to protect against single fail condition. These sensors are assembled in a PCA INC-BL59 attached to incubator right side and protected by a bumper from unexpected impacts with CAT arm. Incubator built-in isolation material all around and the heat transmission U shape.

Incubator block target temperature shall be reached within 20 minutes at any of the allowed environmental conditions and taking under consideration that internal instrument temperature may be different from external temperature.

Servocontrolled system is used in order to achieve required movement speed and positioning precision. This system assures that incubator position is kept without powering the motor up to 70N. The motor transmits the movement to guide by means of a coupling.

Encoder resolution for incubator axis is as follows:

INC Axis	Encoder counts / mm
Z	851.851 counts/mm

Table 42. INC axis encoder resolution

There is an optical home sensor assembled at the right-bottom end of PCA INC-CON59 that determines lower position of the incubator (card floor) and that is used as home reference. A small L-shape plate assembled in the guide wagon is used as home flag. Upper level/pipetting floor position of the incubator is determined with the incubator height adjustment.

Cards are retained in incubator slot by means frontal probes platform (PIP floor) including a window per each incubator with a width lower than card width so it retains for both ends. This platform also has a hole per each incubator that is used by PIP arm presence detector to find out if incubator is at pipetting floor during user software initialization.

Peltier module and temperature sensor are controlled/signals between PCA INC-CON59 and PCA GEN-BCKPL59 are transmitted though I2C bus.

General block diagram of the module is shown in the following figure,

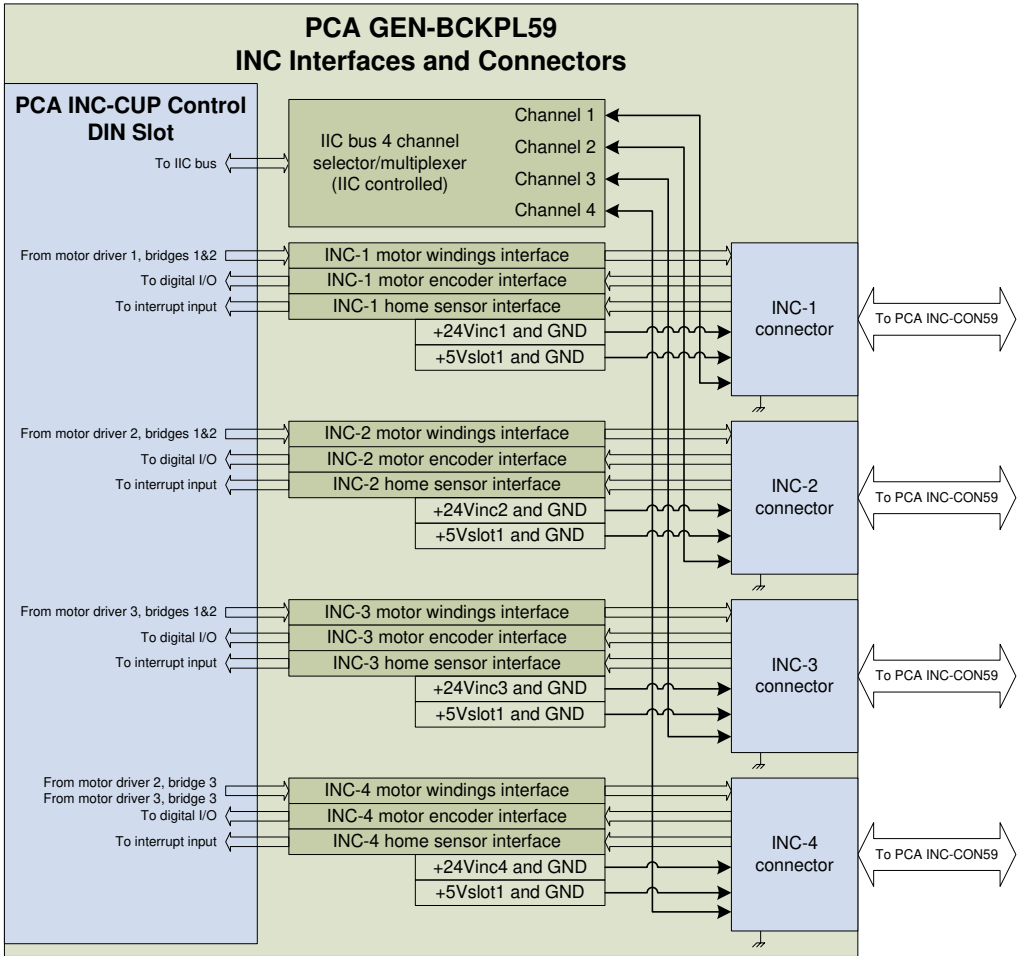


Figure 167. Incubator module block diagram

PCAs related to Incubator module are:

- **PCA GEN-BCKPL59**: main PCA of the instrument, located in the backpanel.
- **PCA INC-CUP Control**: located in the backpanel (slot 1), attached to PCA GEN-BCKPL59 (slot 1). It controls signals used in Incubator module and Dilution cup module. It contains the firmware of both modules.
- **PCA INC-CON59**: 1 PCA for each incubator, located in incubator panel and protected by a cover. It contains Peltier module driver, Peltier module connections, lift home sensor, motor connections and INC-BL59 connections.
- **PCA INC-BL59**: There is 1 PCA for each incubator, located attached to the incubator block. It contains 2 temperature sensors (LM92 sensors).

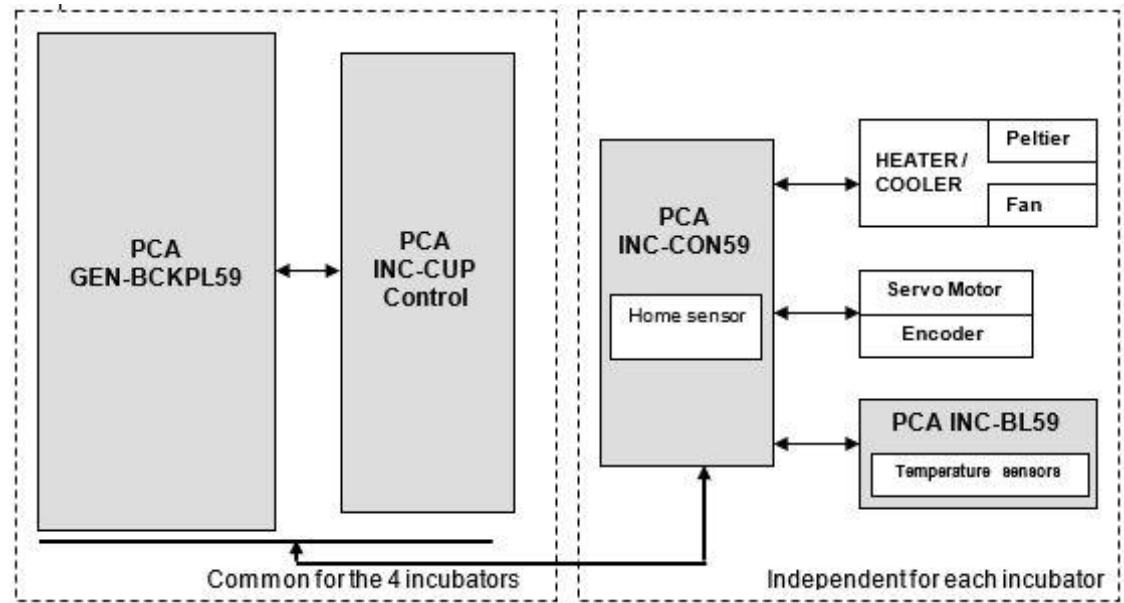


Figure 168. Main blocks of Incubator module.

5.2 ADJUSTMENT PROCEDURES

In this section, adjustments related to the incubator are explained.

5.2.1 Mechanical adjustments

Mechanicals adjustments related to Incubator module are detailed.

5.2.1.1 Incubator flatness

Purpose of this adjustment is to set the flatness of the incubators.

Prerequisite: NA

Adjustment tool: NA

Procedure:

1. Loose the screws of the incubator laterals and modify manually the position of the incubator to fulfil that distance between incubator surface and gripper is $\leq 0.4\text{mm}$ over slots 1 and 12 of the incubator to be adjusted (refer to section 5.3.3). Then tighten the screws.

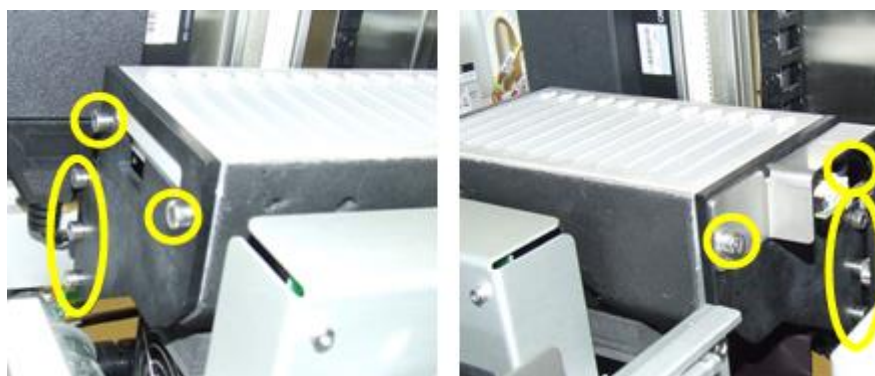


Figure 169. Location of the screws of the laterals of the incubators

2. Loose the screws of the incubator laterals and modify manually the position of the incubator to fulfil that difference in Z gripper encoder counts between slots 1 and 12 of the incubator to be adjusted is ≤ 600 (refer to section 5.3.3). Then tighten the screws.
3. Loose the screws of the incubator laterals and modify manually the position of the incubator to fulfil that the difference between the maximum and the minimum encoder counts of probe 1 over the defined points in the incubator to be adjusted is ≤ 1047 (refer to section 5.3.3). Then tighten the screws.
4. In case flatness of the incubator is adjusted, check/perform incubator height adjustment, probes adjustment in incubator and gripper adjustment in incubator.

5.2.2 Functional adjustments

Incubator height and temperature adjustment are detailed in this section.

5.2.2.1 Incubator height

This procedure is needed to adjust incubator position at upper/pipetting floor (maximum height of each incubator). Adjustment is then performed with the incubator in the upper module.



CAUTION!

Check CAT arm position to avoid it to interfere with incubator movement and lead to a crash.

Prerequisite:

- Incubators homed.

Adjustment tool: NA

Procedure:

1. Open intermediate door (remove the screw to open it) and place manually 3 cards in the incubator to be adjusted. Place a card in slots 1, 6 and 12.
2. Go to *Procedures* tab in Technical Service software → *PIP floor Modules* section → *incubators height* button.
3. An adjustment wizard is open:
 - a. Select the incubator to be adjusted and press *Next>* button.
 - b. Move CAT arm so it does not interfere with incubator movement. Move up the incubator by means of the joystick buttons the amount of counts selected and indicated by the software. It is possible to modify the amount of counts sliding the bar.

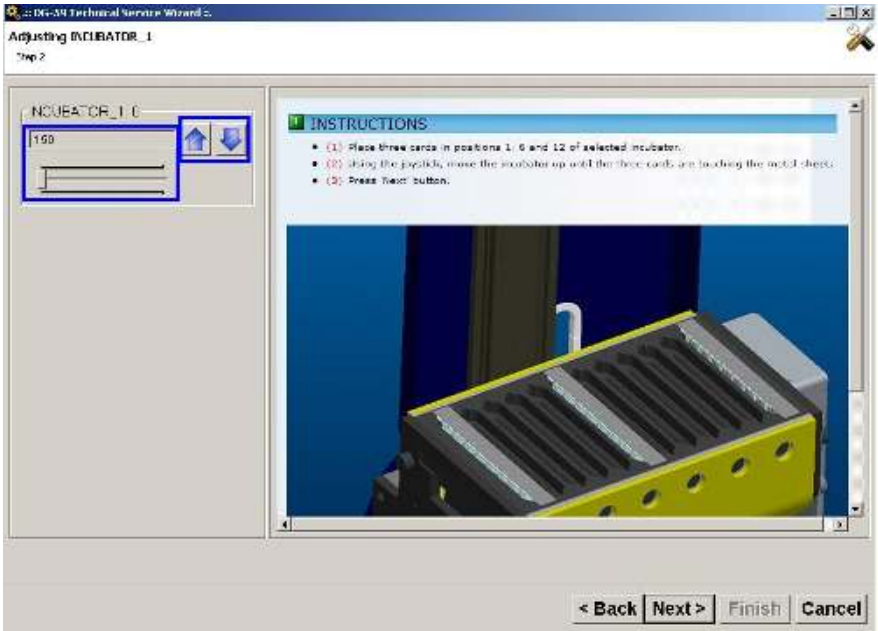


Figure 170. Joystick buttons and counts sliding bar

- c. Correct adjustment position of the incubator is achieved when the cards placed in the incubator are just touching the surface of the platform.



CAUTION!
Check that incubator block does not crash with the platform, avoiding deforming the platform or forcing incubator motor.

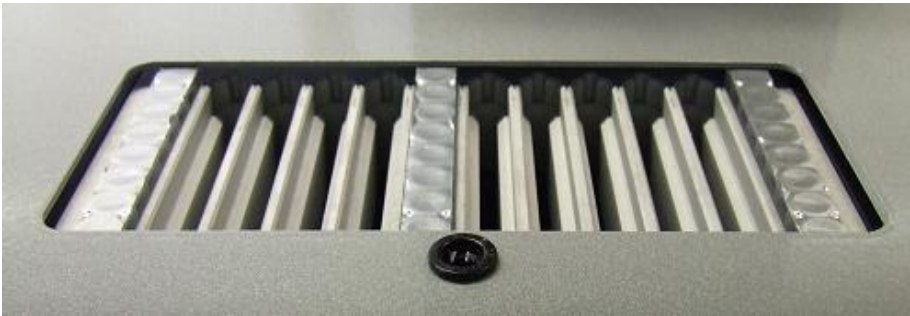


Figure 171. Correct adjustment incubator position

- If cards are not touching at the same time, check the assembly of the platform flatness or incubator flatness.
- d. Once the incubator is in the correct position, press *Next>* button.

- e. A screen appears showing the parameters to be stored. Press *Finish* button and press *OK* on the confirmation window.
- f. Once the adjustment is stored, press *OK* on the information window.
- g. Press *OK* in the confirmation window that indicates that home movements will be performed.
- h. Press *OK* to finish the adjustment procedure.

**NOTE:**
Adjustment stored in *E:\data\Erytra\conf\PIPCalibrations.txt*

5.2.2.2 Temperature adjustment

This procedure is to adjust the temperatures (24°C and 37°C) for each incubator module. The incubators can be adjusted at any environmental temperature between 15°C to 30°C.

Prerequisites:

- Upper and intermediate area doors closed
- All the covers placed
- Incubators homed

Adjustment tool: Digital thermometer HIBOK-15 MiniDIN 4v (PN: 230903) and Temperature probe PT-100 for cards – MiniDIN 4v (PN: 230905).

Procedure:

- 1. Place the card probe in slot 6 of the incubator that is going to be adjusted.
- 2. In Technical Service software go to *Procedures* tab → *Adjustments* → *Others* → *temperature* in order to start the adjustment wizard.
- 3. Select the incubator to adjust and press *Next>* button.
- 4. The instrument activates automatically the temperature regulation in RT mode. Then follow the instructions of the wizard:
 - a. Wait until the temperature is stabilized.
 - b. Write (maximum 2 decimals separated by dot) in *Set minimum* textbox of the software the measured temperature in the calibrated thermometer and press *Next>* button.

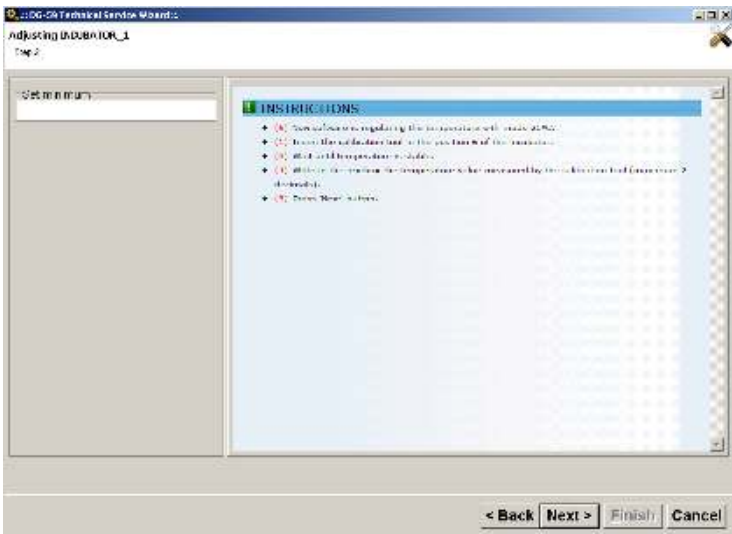


Figure 172. Incubator adjustment wizard (adjustment mode RT)

- 5. The instrument activates automatically the temperature regulation in 37°C mode. Then follow the instructions of the wizard:

- a. Wait until the temperature is stabilized.
- b. Write in *Set maximum* textbox of the software the measured temperature in the calibrated thermometer and press *Next>* button.

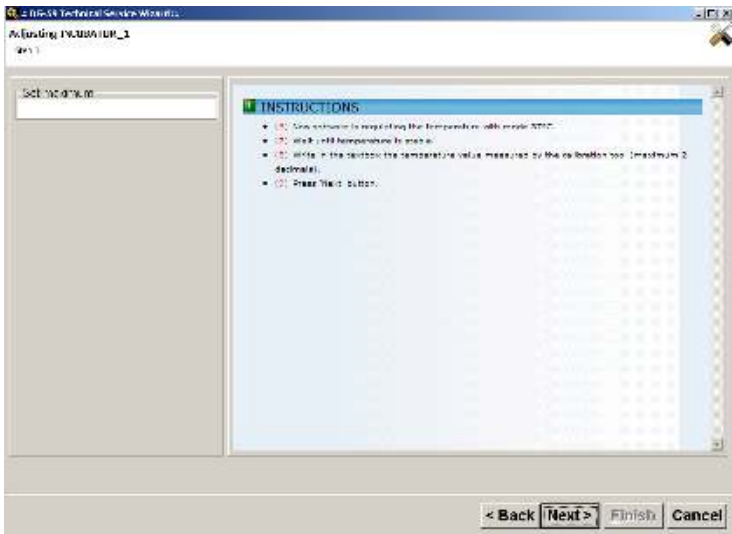


Figure 173. Incubator adjustment wizard (adjustment mode 37)

- 6. A screen appears showing the parameters to be stored. Press *Finish* button and press *OK* button on the information window.
- 7. Press *OK* button in the confirmation window that indicates that home movements will be performed. Incubator will perform a home movement.
- 8. Press *OK* button to finish the adjustment procedure.



NOTE:
Adjustment stored in PCA INC-CUP Control. In case of replacing this PCA, the adjustment must be restored (if a previous backup was performed). Otherwise the adjustment must be performed.

5.3 VERIFICATIONS

In this section are detailed the verifications of incubator functions: temperature and movement.

5.3.1 Temperature

- 1. Place a temperature card probe in incubator slot 6 of the incubator to verify.
- 2. Move to PIP area the incubator to verify, checking that movement does not interfere with CAT arm.
- 3. Check that upper and intermediate area doors are closed and all the covers are placed.
- 4. In Technical Service software, go to Incubators tab and select the incubator tab to verify. Then in *Regulate Actions*, activate *On* option and select 24 °C temperature (*MODE RT*).

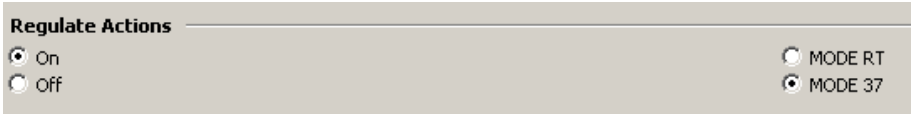


Figure 174. Regulate Actions section

5. Wait until the temperature is stabilized (maximum 20 minutes checking that *Out of Margin* flag is deactivated) and check that measured temperature with the calibrated thermometer and temperature on the software are $24^{\circ}\text{C} \pm 3^{\circ}\text{C}$.
6. Change temperature regulation to 37°C (*MODE 37*) in Technical Service software.
7. Wait until the temperature is stabilized (maximum 20 minutes checking that *Out of Margin* flag is deactivated) and check that measured temperature with the calibrated thermometer and temperature on the software are $37^{\circ}\text{C} \pm 1^{\circ}\text{C}$.
8. Change temperature regulation to 24°C (*MODE RT*) in Technical Service software.
9. Wait until the temperature is stabilized (maximum 20 minutes checking that *Out of Margin* flag is deactivated) and check that the temperature is $24^{\circ}\text{C} \pm 3^{\circ}\text{C}$.
10. Deactivate temperature regulation (Incubator tab → *Regulate Actions*, select *Off* option).

5.3.2 Movement

1. Make a home of CAT arm axes (Click *Home* button in *Gripper Arm* tab → *Movement* subtab → *Position Actions* area).
2. In Technical Service software, go to *Incubators* tab and select the subtab of the incubator to verify. Then press *home* button and check that *Homed* indicator is activated when incubator reaches home position.



CAUTION!

Check CAT arm position to avoid it interfering with incubator movement and lead to a crash.

3. In Technical Service software, go to *Incubators* tab → *Home Cycles* tab. Press *start* button of the incubator to verify. Perform home cycles for 5 minutes checking that no errors appear during movement.

5.3.3 Incubator flatness

1. Perform a home of CAT robotic head and a home of the incubator to verify by means of TSS, checking that movements between them do not interfere.
2. Move CAT robotic head to slot 1 of the incubator to verify by means of TSS, *Gripper Arm* tab → *Movement* subtab → *Move Actions* section.
3. De-energize Z axis using *unlock/stop* button in TSS, *Gripper Arm* tab → *Movement* subtab → *Position Actions* section.
4. Lower gripper manually (by means of its belt) until gripper touches gently the surface of the incubator.
5. Check that distance between gripper and incubator surface is $\leq 0.4\text{mm}$.



Figure 175. Distance verification between gripper and incubator

6. Perform a home of Z-axis and repeat previous procedure in slot 12, checking that distance between gripper and incubator surface is $\leq 0.4\text{mm}$.

- 7. Perform a home of Z-axis and move CAT robotic head to slot 1 of the incubator to be verified by means of TSS, *Gripper Arm* tab → *Movement* subtab → *Move Actions* section.
- 8. Unlock Z-axis using *unlock/stop* button in TSS, *Gripper Arm* tab → *Movement* subtab → *Position Actions* section.
- 9. Lower gripper manually (by means of its belt) until gripper touches gently the surface of the incubator.
- 10. Check Z encoder counts in TSS.
- 11. Perform a home of Z-axis and repeat points 8 to 11 but in slot 12 of the incubator.
- 12. Check that difference in Z encoder counts between slot 1 and slot 12 is ≤ 600 .
- 13. After that, move the incubator to PIP area (checking where the gripper arm is located in order to avoid that incubator crashes with gripper arm) by means of TSS, *Gripper Arm* tab → *Movement* subtab → *Gripper Actions* section → *up* button.
- 14. Perform a home of PIP robotic head.
- 15. Unlock PIP robotic head using *unlock/stop* button in TSS, *Pip Arm* tab → *Movement* subtab → *Position Actions* section.
- 16. Move manually PIP robotic head and lower probe 1 in the following points, checking Z1 encoder counts.



Figure 176. Location of the verification points of the incubator flatness in PIP area

- 17. Check that difference between Z1 encoder counts in previous 2 points is ≤ 1047 .

5.4 REPLACEMENT PROCEDURES

For some replacement procedures, it is recommended to disassemble the whole incubator module from the instrument for a convenient access to all the parts.

In order to disassemble the desired incubator module from the instrument, follow next steps:

- 1. Switch off the instrument and disconnect it from the mains.
- 2. Remove upper and middle rear covers.
- 3. Remove the screw that attaches incubator module to frontal probes platform, located in PIP floor.



Figure 177. Incubator module and frontal probes platform

- 4. Unplug the DB25 connector from PCA INC-CON59 (cable from PCA INC-CON59 to PCA GEN-BCKPL59).
- 5. Disconnect the ground cable that it is located between motor and incubator chassis from motor side.



Figure 178. Incubator module ground cable and motor fixing screws

- 6. Loose the 4 screws that attach the motor with the motor flange and move the motor to the right, so incubator is free to move by the linear guide.



CAUTION!
Hold the incubator in order to do not allow it to move free by the linear guide.

- 7. Move up manually the incubator block until is placed in the middle of the linear guide (to gain access to the two bottom screws fixing the module to the instrument chassis).
- 8. Move motor to the left to couple the system again and fix motor position tightening the four screws that attach it to motor flange.
- 9. Remove the two bottom screws that attach the incubator module to the chassis of the Erytra.
- 10. Loose again the 4 screws that attach the motor with the motor flange and move the motor to the right, so the incubator is free to move by the linear guide.



CAUTION!
Hold the incubator in order to do not allow it to move free by the linear guide.

- 11. Remove motor-linear guide coupling (it is not fixed and can easily fall from its location when motor and linear guide are not coupled).



Figure 179. Incubator motor-linear guide coupling

- 12. Remove the 4 screws that attach the motor with motor flange and remove the flange.

13. Remove the 2 upper screws that attach incubator module to the chassis of the *Erytra*.



Figure 180. Location of the screws that attach incubator module to *Erytra* chassis

- 14. Unplug the motor (motor power and encoder) from PCA INC-CON59 and remove the motor.
- 15. Remove the whole module by the CAT floor. It is advisable to first make the module to lay horizontally on cards drawers and then extract it handling the module from instrument front side (intermediate floor door).
- 16. In order to assemble the module, repeat steps from 1 to 15 in reverse mode. Couple motor-linear guide so resulting incubator position assures that home plate is located higher than home optical barrier.
- 17. Check that all incubator module cables and any other cable that may have been affected by removal/assembly are properly routed and fixed. Pay special attention to the SRM and CM locks ribbon cables and the cables routed through instrument laterals.
- 18. Check/perform incubator flatness, incubator height adjustment, probe and gripper adjustment in the incubator.

5.4.1 Linear guide

Once the incubator module is removed, follow next steps:

- 1. Remove the screw and the clamp fixing PCA INC-BL59 cable.
- 2. Unplug cable from PCA INC-BL59.
- 3. Unplug cable from Peltier module connector and remove the connector from incubator block.



Figure 181. Location of the fixing clamps

- 4. Remove the six screws (3 per side) fixing incubator block to linear guide wagon and remove the incubator block.

5. Remove the 4 fixing clamps that are attaching the linear guide to the incubator plate. Note that each set of clamps is screwed to a rectangular part at the back of incubator plate that will get free when clamps are removed. Note also that the ground cable, the nut and washers fixing it will get free as well.

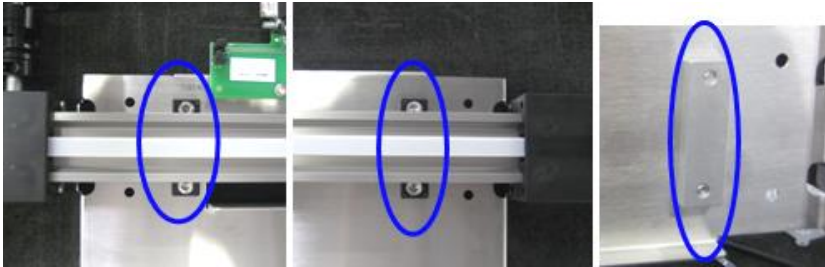


Figure 182. Location of the fixing clamps

6. Remove the guide.
7. Unscrew the chain and the L-shape home flag from the wagon. Unscrew cable holder from the wagon.



Figure 183. Chain, L-shape home flag and cable holder

8. Assemble cable holder to new guide wagon.
9. Assemble the chain and the L-shape home flag to new guide wagon.
10. Assemble new guide fixing it with the four clamps (place the back part and screw the clamps; remember to assembled ground cable to the bottom clamp set). The correct assembly is placing the linear guide over the top pins of the incubator plate.



Figure 184. Assembly of the linear guide

11. Check L-shape home flag is centered with home optical barrier. Should this not be the case, loose the two screw fixing L-shape home flag to align it.



Figure 185. Home flag alignment

12. Insert Peltier module cable in its incubator block position and plug the cable.
13. Plug cable from PCA INC-BL59.

- 14. Attach incubator block to linear guide wagon with the six screws (three per side).
- 15. Check that the incubator is placed perpendicular with the linear guide.
- 16. Assemble the clamp fixing PCA INC-BL59 cable to incubator block.

5.4.2 Peltier Module

In order to replace a Peltier module, it is not necessary to remove the whole incubator module, just the incubator block. Follow next steps:

- 1. Open Technical Service software, start the *Driver* and perform a home of the incubator which Peltier module is going to be replaced.
- 2. Move incubator module 150000 encoder counts.
- 3. Switch off the instrument and disconnect it from the mains.
- 4. Disconnect Peltier cable.

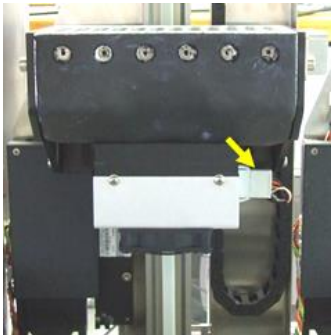


Figure 186. Peltier module cable

- 5. Remove the 2 upper screws fixing left lateral incubator support to incubator module.



Figure 187. Upper screws fixing left lateral incubator support

- 6. Remove grounding connection and bumper (two upper screws fixing right lateral incubator) from the right side incubator module.



Figure 188. Upper screws bumper and grounding connection

- 7. Disconnect cable from PCA INC-BL59.
- 8. Remove the 3 screws fixing right lateral incubator support to linear guide platform.
- 9. Extract carefully incubator module from its position.

10. Remove the 4 screws fixing Peltier Module to incubator and remove the module.



Figure 189. Peltier module assembly

- 11. Replace thermal silicon from incubator U-shape Peltier module area.
- 12. Assemble new Peltier module following previous steps in reverse order.
- 13. Check/perform incubator flatness, incubator height adjustment, probe and gripper adjustment in the incubator and incubator temperature adjustment.

5.4.3 PCA INC-BL59

Once the incubator block is removed from the instrument (check steps 1-8 in section 5.4.2), follow next steps:

- 1. Remove the 2 screws that attach PCA INC-BL59 to incubator block and remove PCA.



Figure 190. PCA INC-BL59 assembly

- 2. Assemble new PCA INC-BL59.
- 3. Install incubator block in the instrument and check/perform incubator flatness, incubator height adjustment, probe and gripper adjustment in the incubator and incubator temperature adjustment.

5.5 FUSES

Fuses related to Incubator module are detailed in the following table,

PCA	FUSE NAME	DESCRIPTION
PCA GEN-BCKPL59	FUS11	MST250 0034.6615 1A/250V
	FUS12	MST250 0034.6620 3.15A/250V
	FUSINC-1	MST250 0034.6620 3.15A/250V
	FUSINC-2	MST250 0034.6620 3.15A/250V
	FUSINC-3	MST250 0034.6620 3.15A/250V
	FUSINC-4	MST250 0034.6620 3.15A/250V
PCA INC-CUP Control	FUS1	MSF250 0034.6013 0.8A/250V
PCA INC-CON59	NA	NA
PCA INC-BL59	NA	NA

Table 43. List of fuses of Incubator module

5.6 LEDS

LEDs related to Incubator module are detailed in the following table,

PCA	LED	DESCRIPTION
PCA GEN-BCKPL59	LED7 LED8 LED9 LED10 LED11 LED12	Green 5V (FUS11) Green 24V (FUS12) Green 24V (FUSINC-1) Green 24V (FUSINC-2) Green 24V (FUSINC-3) Green 24V (FUSINC-4)
PCA INC-CUP Control	LED1	Yellow When the instrument is switched on, LED is ON and a few seconds later is OFF. This means that the FPGA has been loaded correctly with the appropriate data.
	LED2	Green VMotor Ok. It indicates that the voltage supplied to the motor is $\geq 12V$
PCA INC-CON59	NA	NA
PCA INC-BL59	NA	NA

Table 44. List of LEDs of Incubator module

5.7 JUMPERS AND TEST POINTS

Neither jumpers nor test points are related to Incubator module.

5.8 WIRING TABLE

In the following table, there is detailed all the cables related to Incubator module. A wiring diagram can be found attached to Service Manual.

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
B-INC1	H1	PCA INC-CON59	INC_1	PCA GEN-BCKPL59
B-INC2	H1	PCA INC-CON59	INC_2	PCA GEN-BCKPL59
B-INC3	H1	PCA INC-CON59	INC_3	PCA GEN-BCKPL59
B-INC4	H1	PCA INC-CON59	INC_4	PCA GEN-BCKPL59
INC-ENC	H2	PCA INC-CON59	ENCODER	INCUBATOR MOTOR
INC-MOT	H3	PCA INC-CON59	POWER	INCUBATOR MOTOR
INC-PELT	H4	PCA INC-CON59	PELTIER + FAN	PELTIER MODULE
INC-TEMP	H5	PCA INC-CON59	H1	PCA INC-BL59

Table 45. List of wiring of Incubator module

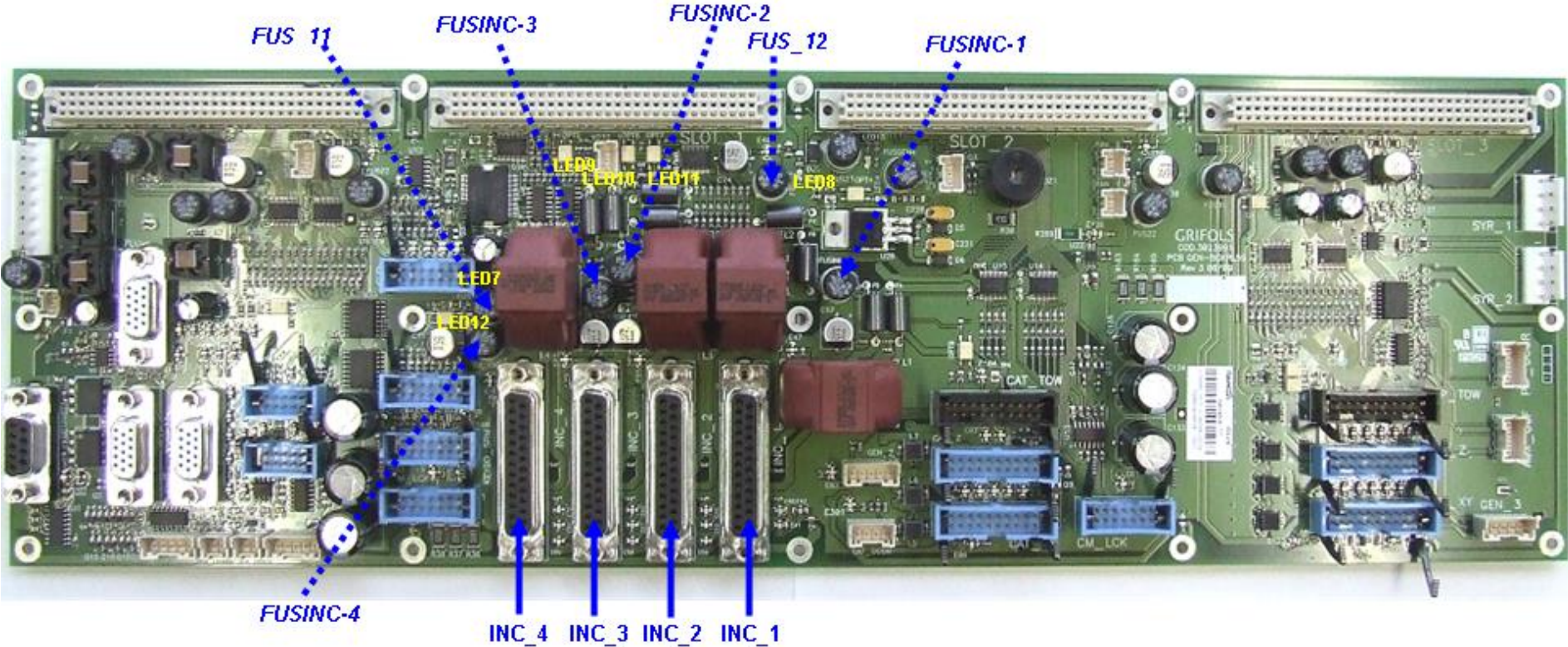


Figure 191. Location of the connector, fuses and LEDs in PCA GEN-BCKPL59 related to Incubator module

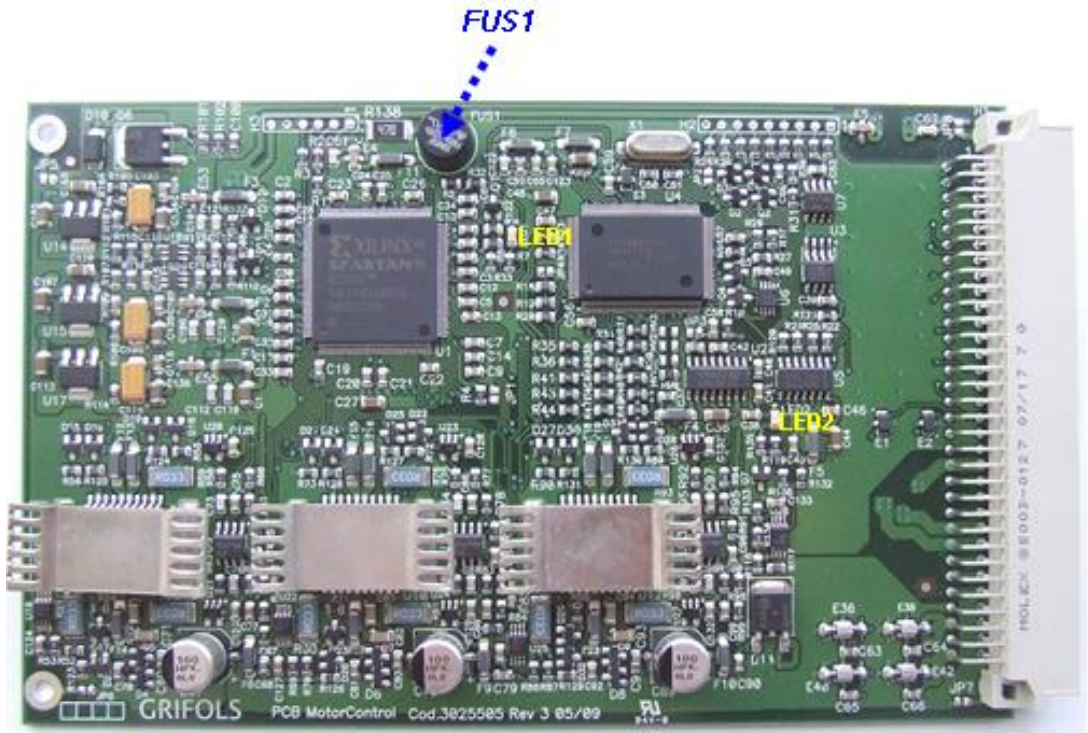


Figure 192. Location of the fuse and LEDs in PCA MotorControl

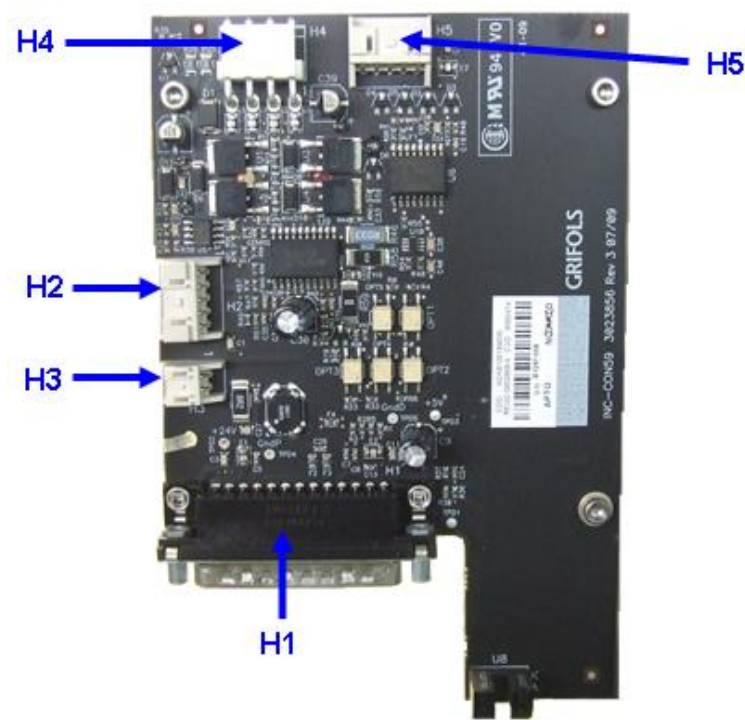


Figure 193. Location of the connectors in PCA INC-CON59

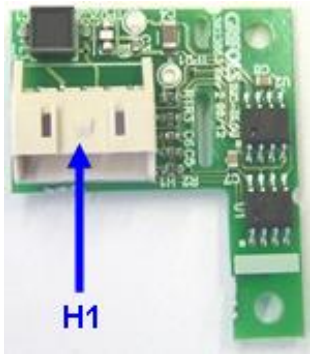


Figure 194. Location of the connector in PCA INC-BL59

5.9 TROUBLESHOOTING

Module	Device	Error code	Error	Cause	Text	Options
INC (07)	Incubator motor (20) Incub 1 (36) Incub 2 (52) Incub 3 (68) Incub 4	003000	ERRORCODE_LEN	FW has received a command with incorrect length	Error when trying to move an incubator up/down. Please choose one of the following options:	RETRY CANCEL Works (it leads to: <i>Incubator {1, 2, 3 or 4} has been disabled</i>)
		005000	ERRORCODE_ARG	Command has wrong arguments		
		006000	ERRORCODE_BUSY	Command is addressed to a busy device		
		007000	ERRORCODE_NOTHOME D	Move command is addressed to a not homed device		
		032000	CURRENT_LIMIT	Current limit at servo device		
		033000	POSITION_ERROR	A position error has been produced at a servo device		
		034000	MOVEMENT_ERROR	A movement error has been produced at a servo device		
		035000	HOME_TIMEOUT	A home timeout has been produced at a servo device		
		036000	ERRORCODE_SRV_SYN C_TIMEOUT	A synchronization error has been produced at a servo device		
		037000	ERRORCODE_SRV_UNK NOWN_ERROR	An unknown error has been produced at a servo device		
	Temperature regulator (16) Incub 1 (32) Incub 2 (48) Incub 3 (64) Incub 4	info	FUSE_FAIL	Incubator temperature regulator is at Fuse fail, Fan fail, Channel error, Driver Error, Sensor 1 error or Sensor 2 error	Incubator {1, 2, 3 or 4} has been disabled	OK
		info	FAN_FAIL			
		info	DRIVER_ERR			
		info	PWM_ERR			
		info	SENSOR_2_ERR			
		info	SENSOR_1_ERR			
		info	DELTA_ERR			
		info	CHANNEL_ERR			
		info	OUT_OF_MARGIN (if Ramping=false)	Incubator temperature is out of margins		

Module	Device	Error code	Error	Cause	Text	Options
Any (xx)	Any (xx)	002000	ERRORCODE_CMD	Unknown command code received	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		003000	ERRORCODE_LEN	FW has received incorrect command length		
		004000	ERRORCODE_DEV	Command addressed to an incorrect device		
		005000	ERRORCODE_ARG	Command has wrong arguments		
		014000	ERRORCODE_HW_DEV	Addressed device has a problem with its related HW		
		015000	ERRORCODE_HW_HWF AIL	There is a problem with PCA HW		
		016000	ERRORCODE_HW_HWN OTSET	PCA setup error		
0	Non firmware errors (xx) 240	000000	ERRORCODE_UNKNOW N	Unknown error	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		008000	ERRORCODE_FW_NOTS AFE	Movement not safe		
		017000	ERRORCODE_FW_UNKN OWN_ERROR	Unknown error		
		061000	ERRORCODE_INVALID_ ARGUMENTS	Invalid arguments		
		062000	ERRORCODE_FWCOMM _TIMEOUT	Communication timeout		
		063000	ERRORCODE_FWCOMM _CONNECTION_LOST	Communication lost		
		064000	ERRORCODE_FWCOMM _COMM_ERROR	Communication error		
		088000	ERRORCODE_PRECOND ITION_FAILED	Previous condition for the action is not met		
		089000	ERRORCODE_STARTUP _ERROR	Driver startup error		
		091000	ERRORCODE_ALREADY _STARTED	Another instance of Driver is already opened		
		090000	ERRORCODE_UNEXPEC TED_ERROR	A Driver unexpected error has been produced		

Table 46. List of errors related to Incubator module.

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Incubator disabled when running tests	No heating / cooling	Check no heating / cooling section in this troubleshooting
	Temperature sensors error	Check temperature sensors error section in this troubleshooting
	Incorrect temperature regulation	Check Incorrect temperature regulation section in this troubleshooting
Incubator never used when running tests	Reserved for STAT samples	Check <i>RESERVE_INCUBATOR_FOR_STAT</i> parameter in <i>E:\conflerytra.properties</i> file
	No heating / cooling	Check no heating / cooling section in this troubleshooting
	Temperature sensors error	Check temperature sensors error section in this troubleshooting
	Incorrect temperature regulation	Check Incorrect temperature regulation section in this troubleshooting
No heating / cooling	Fuse blown	Check / replace FUSINC-x
	Peltier cell failure	Check / replace Peltier module
	Peltier fan failure	
	Cable damaged	Check / replace incubator chain
	PCA INC-CON59 failure	Check / replace PCA INC-CON59
	Cable damaged	Check / replace B-INCx cable
	PCA INC-CUP Control failure	Check / replace PCA INC-CUP Control
	PCA GEN-BCKPL59 failure	Replace PCA GEN-BCKPL59
	Temperature sensor/s damaged	Check / replace PCA INC-BL59
	Cable damaged	Check / replace incubator chain
Temperature sensors error	PCA INC-CON59 failure	Check / replace PCA INC-CON59
	Cable damaged	Check / replace B-INCx cable
	PCA INC-CUP Control failure	Check / replace PCA INC-CUP Control
	PCA GEN-BCKPL59 failure	Check / replace PCA GEN-BCKPL59
Incorrect temperature regulation	Temperature of the room out of specifications	Check room conditions
	Wrong adjustment of incubator temperature	Check / readjust incubator temperature
	Temperature sensor/s damaged	Check / replace PCA INC-BL59
	Cable damaged	Check / replace incubator chain
	PCA INC-CON59 failure	Check / replace PCA INC-CON59
	Cable damaged	Check / replace B-INCx cable
	PCA INC-CUP Control failure	Check / replace PCA INC-CUP Control
	PCA GEN-BCKPL59 failure	Check / replace PCA GEN-BCKPL59

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Movement error / Current limit error INC1: 0720032000 INC2: 0736032000 INC3: 0752032000 INC4: 0768032000 INC1: 0720034000 INC2: 0736034000 INC3: 0752034000 INC4: 0768034000	Temperature of the room out of specifications	Check room conditions
	Incubator movement is interfered	Check if something is interfering in incubator movement
	Incubator crashes with the probes platform	Adjust incubator height
	Fuse blown	Check / replace FUS11 or FUS12
	Motor and linear guide are not correctly coupled	Check coupling between motor and linear guide
	Motor/encoder damaged	Check / replace motor
	PCA INC-CON59 failure	Check / replace PCA INC-CON59
	Cable damaged	Check / replace B-INCx cable
	PCA INC-CUP Control failure	Check / replace PCA INC-CUP Control
	PCA GEN-BCKPL59 failure	Check / replace PCA GEN-BCKPL59
Home position is not detected INC1: 0720007000 INC2: 0736007000 INC3: 0752007000 INC4: 0768007000 INC1: 0720035000 INC2: 0736035000 INC3: 0752035000 INC4: 0768035000	Home flag is not aligned with optical barrier	Check position of the home flag
	Incubator position is lower than optical barrier	Move manually incubator over optical barrier position
	Motor does not turn	Check / replace motor
	Failure in optical barrier	Check / replace PCA INC-CON59
	Cable damaged	Check / replace B-INCx cable
	PCA INC-CUP Control failure	Check / replace PCA INC-CUP Control
	PCA GEN-BCKPL59 failure	Check / replace PCA GEN-BCKPL59

Table 47. Incubator module troubleshooting

6 CENTRIFUGE (CEN) MODULE

6.1 DESCRIPTION

Main function of this module is the centrifugation of the cards. There are 2 centrifuges located in the lower part of the instrument, under card drawers. Each centrifuge has a capacity of 12 cards.

Structure of the centrifuge is formed by following elements:

- Top cover: it holds the whole centrifuge module to the instrument and supports the centrifuge motor structure and the casing. It has an opening to allow the gripper of the CAT arm to load/unload cards.
- Motor structure: formed by a motor, a holder and a set of silentblocks.
- Rotor and card holders: the rotor is the rotating part of the centrifuge and it is attached to the motor shaft. Holders are located in the rotor and they are responsible for allocating the cards to be centrifuged.
- Casing: attached to the base, it covers the complete structure in order to improve operator safety and power consumption.
- Electronics: provide interface with the instrument control.

Centrifuge motors are servo controlled and have a reference position, which is detected by Hall effect sensors. Home process allows the motors to move to the desired position once the known reference is found. Spinning direction is clockwise. Motor is isolated dynamically with silentblocks to reduce a possible transmission of vibration to the rest of the instrument.

Encoder resolution for centrifuge motor is as follows:

CEN	Encoder counts / turn
Rotation	4000 counts/turn

Table 48. CEN encoder resolution

There is no unbalance detector, thus a non pippeted card is used to balance the centrifuge when an odd number of cards are centrifuged. There is no specific card to balance the centrifuge, it is used a card located in any of the card drawers.

Centrifugation process has 4 parameters: acceleration, speed, brake cycle and time and all of them are controlled by firmware. Values of the centrifugation are the following ones:

- Centrifuging speed: 1206 rpm $\pm$ 36 rpm.
- Centrifugation duration: 550s $\pm$ 15s (including braking time).

Time monitoring of the centrifuge is controlled by computer clock. Once firmware sends the command to start a centrifugation, software stores initial time. When firmware sends the command to stop the centrifugation, software compares final time with initial one and it checks if there is any deviation.

All card holders are suspended and unbalanced to assure correct position for centrifuging the cards when centrifuge is rotating, and assure that CAT arm can grab the card when the centrifuge is stopped. When a centrifugation is running, centrifugal force is applied aligned to microtube axis due to holders rotation. Positioning is forced by rotation itself, no external device is used to position the centrifuge holder.

General block diagram of the module is shown in the following figure,

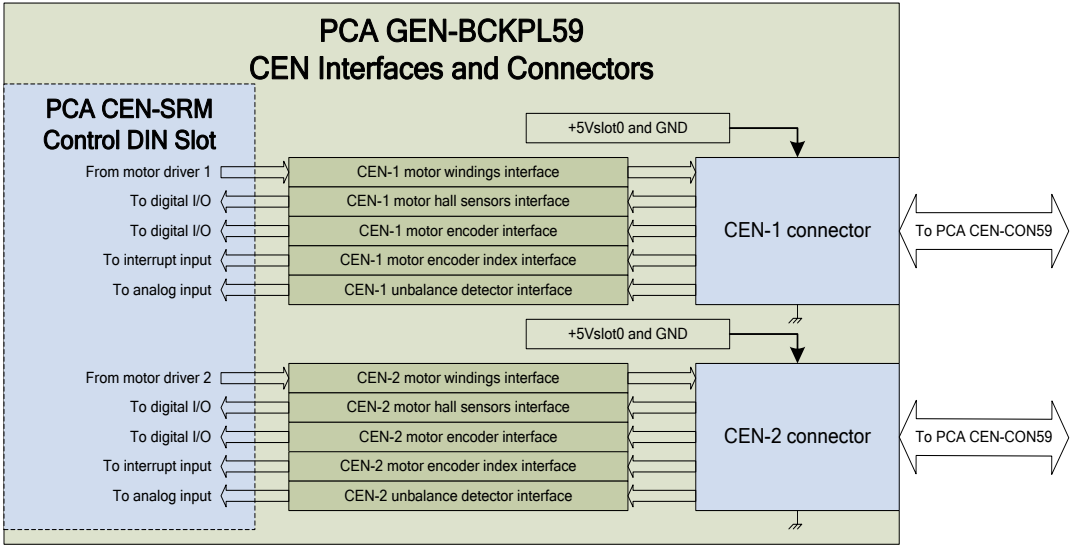


Figure 195. Centrifuge module block diagram

PCAs related to the centrifuge module are:

- **PCA GEN-BCKPL59:** main PCA of the instrument, located in the backpanel.
- **PCA CEN-SRM Control:** located in the backpanel (slot 0), attached to PCA GEN-BCKPL59. It controls signals used in Centrifuge module and samples/reagents drawers management module. It contains the firmware of both modules.
- **PCA CEN-CON59:** located over the centrifuge, this PCA contains the centrifuge motor and encoder interface.

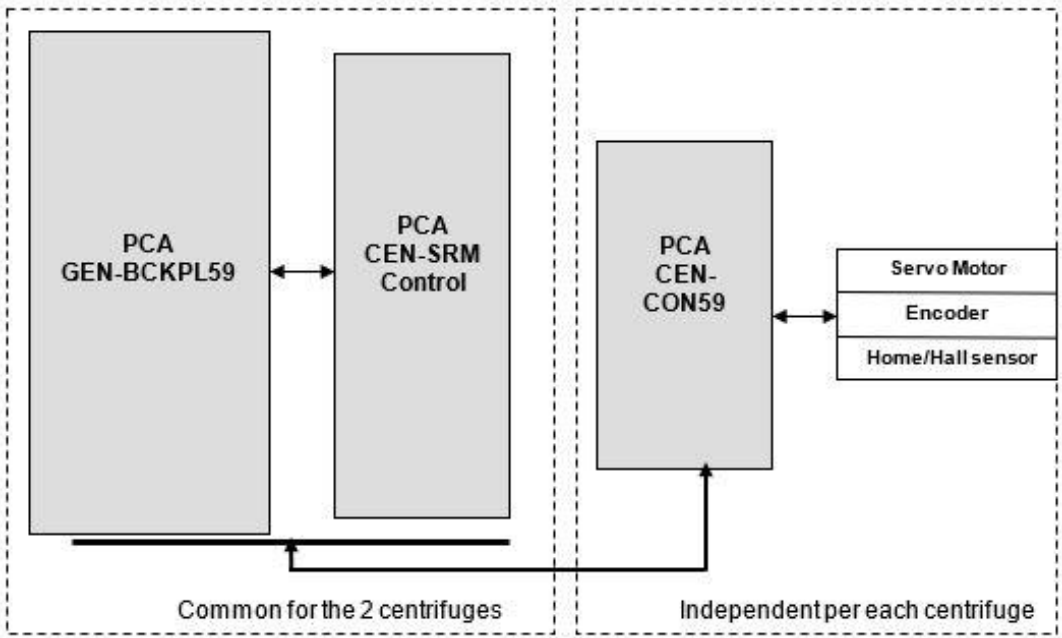


Figure 196. Main blocks of Centrifuge module

6.2 ADJUSTMENT PROCEDURES

In this section, adjustments related to Centrifuge module are explained.

6.2.1 Mechanical adjustments

Mechanical adjustments related to Centrifuge module are detailed in this section.

6.2.1.1 Rotor height adjustment

Rotor height mechanical adjustment is required to set the rotor vertical position in the motor shaft in order to allow the card transport robot arm to grab cards from the centrifuge.

Prerequisite: Extract centrifuge from the instrument, following the next steps:

1. Open completely solution A and solution B bottle drawers (for centrifuge 1) or Waste 1 and Waste 2 bottle drawers (for centrifuge 2) using the software or manually.
2. Remove the bottles and close bottle drawers.
3. Open intermediate door, opening first user tray and then removing the screw of the intermediate door.
4. Open completely card drawers 1 and 2 (for centrifuge 1) or card drawers 3 and 4 (for centrifuge 2).



WARNING!

It is recommended to turn off the instrument at this point.

5. Switch off the instrument. Disconnect cable B-CEN1 (for centrifuge 1) or B-CEN2 (for centrifuge 2), located at the rear part of the centrifuge module.
6. Remove the 4 screws that attach the centrifuge to the chassis.



Figure 197. Centrifuge removal

7. Close the card drawers.
8. Open bottles drawers in order to have access to the centrifuge module and rotate the centrifuge in clockwise direction until it is free from its position. Remove it from the instrument by the bottles drawers area.
9. Remove the centrifuge cover by means of the lateral flap located at one side of the cover and the screw located at the other side.
10. Place the centrifuge cover over the pot with the motor axis outside the pot.
11. Remove all the centrifuge card holders.
12. Loose the nut and extract the rotor.

Adjustment tool: Centrifuge rotor adjustment tool (PN: 233821)

Procedure:

1. Place centrifuge rotor height adjustment tools over the centrifuge cover at each side and place the rotor over them.

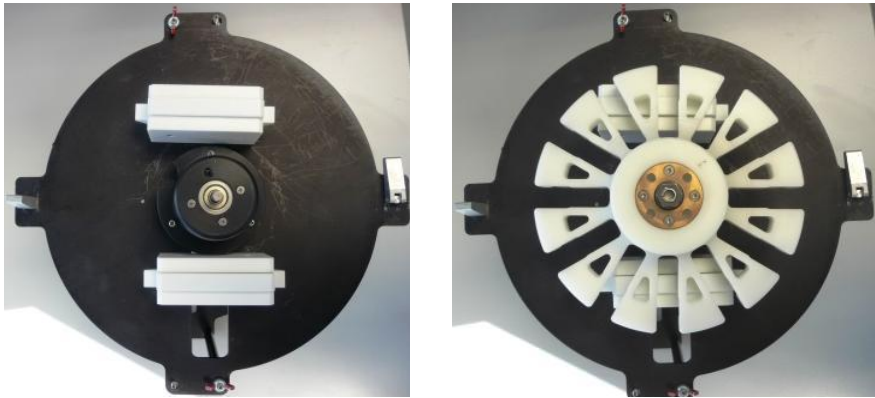


Figure 198. Rotor height adjustment tools

2. Screw the nut to the centrifuge motor axis. Remove the centrifuge rotor adjustment tools (PN: 233821) and check that distance between rotor and basis of the cover is $81.3\text{mm} \pm 0.5\text{mm}$.

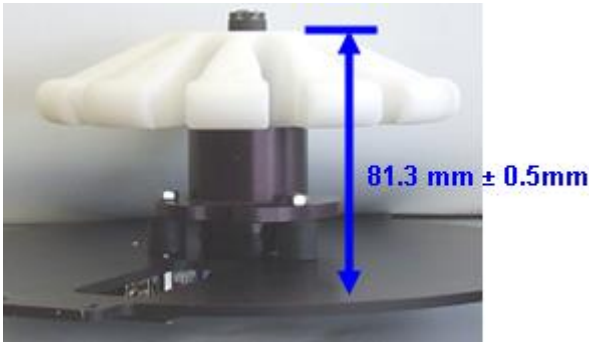


Figure 199. Distance between the rotor and the basis of the cover

3. Remove the cover from the pot and place the centrifuge card holders in the rotor.
4. Place correctly the centrifuge cover in the pot (with the rotor inside the pot) and lock them with the lateral flap and the screw.
5. Install the centrifuge module in the instrument, placing the 4 screws that attach the centrifuge to the structure of the instrument.
6. Connect cable B-CEN1 (for centrifuge 1) or B-CEN2 (for centrifuge 2).
7. Perform gripper adjustment in the centrifuge.

6.2.2 Functional adjustments

There are not functional adjustments in Centrifuge module. Functional adjustments related to gripper inside centrifuge are detailed in chapter 8.

6.3 VERIFICATIONS

- 1. In Technical Service software, go to *Centrifuges* tab. Then select *Centrifuge 1* tab or *Centrifuge 2* tab and press *home* button. Wait until rotor stops its movement and check that *Homed* flag is activated.



Figure 200. Centrifuge motor section in Technical Service software

- 2. Place a reflective adhesive in the rotor between two consecutive holders.



Figure 201. Reflective adhesive in the rotor of the centrifuge

Then, in *Centrifuge 1* tab or *Centrifuge 2* tab, select *PROFILE_0* in *Move centrifuge to* section, and press *start* button.

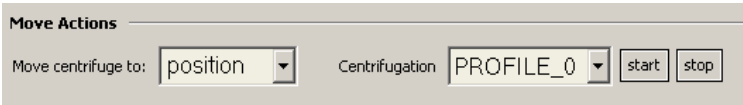


Figure 202. Move Actions section in Technical Service software

Place a tachometer vertically over the slot of the centrifuge in order to measure the rpm. Check that the centrifugation speed is 1206 rpm ± 36 rpm with duration of 550s ± 15s (including braking time, so measure the time between centrifuge encoder counts start to change until they stop). Check that there are no errors in the software.

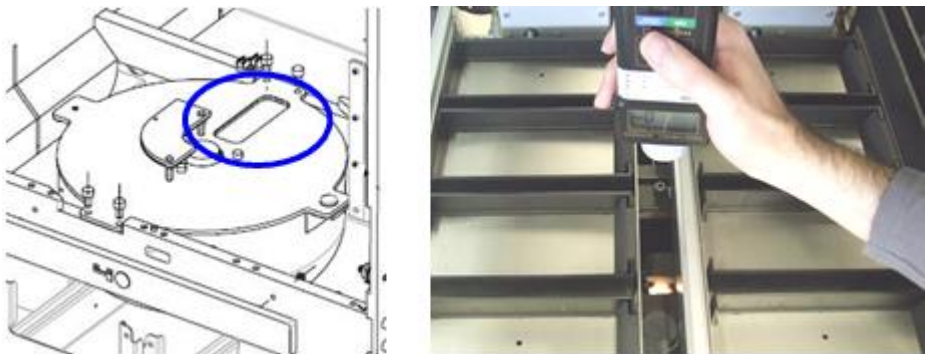


Figure 203. Slot of the centrifuge (left) and measurement of the centrifugation speed with a tachometer (right)

- 3. Remove the reflective adhesive placed in the rotor.



NOTE:
The use of any tachometer is possible as long as it complies with following specifications:

Type: Optical
Scale division (rpm): ≤ 1 rpm
Range (rpm): 1 – 99999 rpm
Maximum uncertainty permitted: ± 3 rpm
Minimum detection distance: 5 cm with visible white light



NOTE:
The use of any chronometer is possible as long as it complies with having at least a precision about 1 second in 10 minutes.

6.4 REPLACEMENT PROCEDURES

In this section, the motor replacement is detailed.

6.4.1 Motor replacement

1. Remove centrifuge module from the instrument and centrifuge rotor following steps detailed in section 6.2.1.1.
2. Remove cover of PCA CEN-CON59.
3. Disconnect cables CEN-1, CEN-2 and CEN-3 from PCA CEN-CON59.
4. Remove the 3 nuts that attach motor cover to the silentblocks (blocking the set screw with an Allen key and then loosening the nut).



Figure 204. Removing the motor cover of the centrifuge

5. Remove the motor with the cover.
6. Remove the 4 screws that attach the cover to the motor.
7. Replace the motor.

8. Attach the cover to the new motor by means of the 4 screws aligning motor cables with cover flat section.



Figure 205. Motor cover-motor alignment

9. Place the motor with the cover in the silentblocks and screw the nuts to the setscrews (block the set screw with an Allen key and screw the nut). Check that the setscrew is 10 mm above the basis of the silentblock. Pay attention that the cover is not symmetrical; the rounded slot must be positioned in front of the cover slot (see below picture).

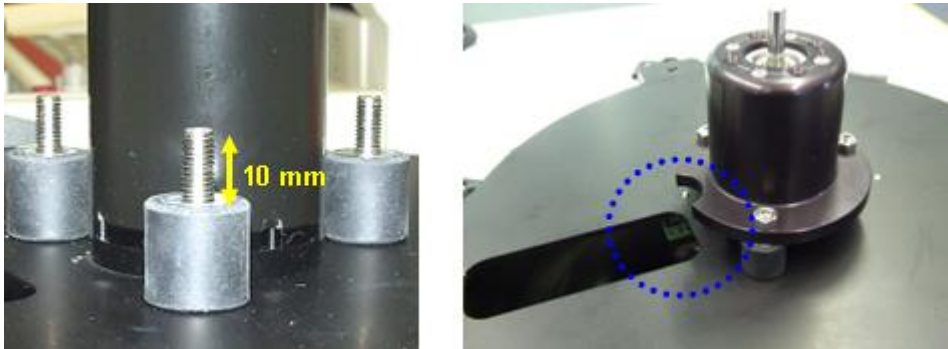


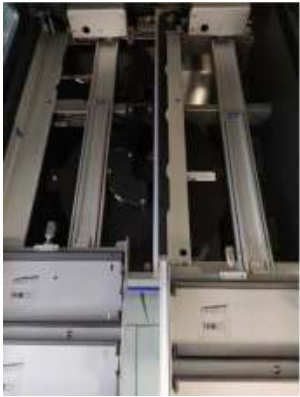
Figure 206. Height of the setscrew in the silentblocks and motor cover alignment

10. Connect cables CEN-1, CEN-2 and CEN-3 to PCA CEN-CON59.
11. Place the cover of the PCA CEN-CON59 and install the whole module in the instrument.
12. Perform gripper adjustment in centrifuge.

6.4.2 Centrifuge Rotor replacement

To replace the centrifuge rotor, follow next steps:

1. Open completely solution A and solution B bottle drawers (for centrifuge 1) or Waste 1 and Waste 2 bottle drawers (for centrifuge 2).
2. Remove the bottles and close bottle drawers.
3. Open intermediate door, opening first user tray and then removing the screw of the intermediate door.
4. Using Technical Service software, open completely card drawers 1 and 2 (for centrifuge 1) or card drawers 3 and 4 (for centrifuge 2).



5. Switch off the instrument. Disconnect cable B-CEN1 (for centrifuge 1) or B-CEN2 (for centrifuge 2), located at the rear part of the centrifuge module.



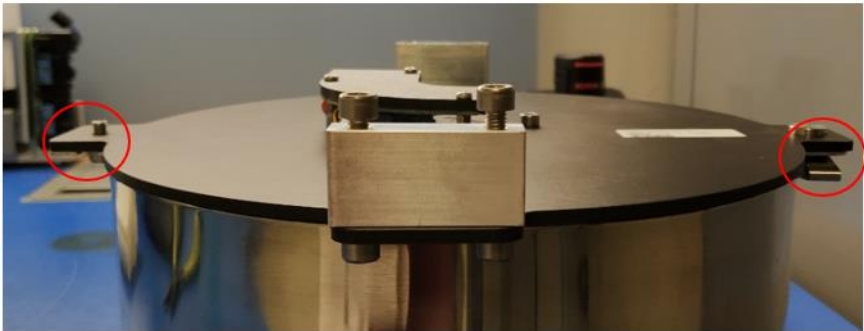
6. Remove the 4 screws that attach the centrifuge to the chassis.



- 7. Close the card drawers.
- 8. Open bottle drawers in order to have access to the centrifuge module and rotate the centrifuge in clockwise direction until it is free from its position. Carefully remove it from the instrument by the bottles drawers area.



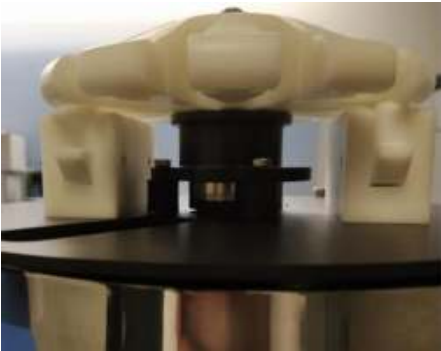
- 9. Remove the centrifuge cover by means of the lateral flap located at one side of the cover and the screw located on the other side.



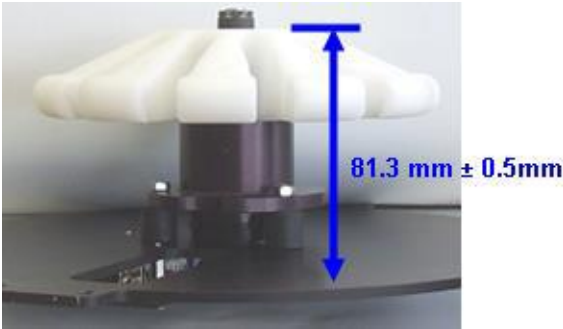
- 10. Place the centrifuge cover over the pot with the motor axis outside the pot.
- 11. Remove all the centrifuge card holders.
- 12. Loose the nut and extract the rotor.



- Place centrifuge rotor height adjustment tools (P/N 233821) over the centrifuge cover at each side and place the new centrifuge rotor over them.



- Screw the nut to the centrifuge motor axis. Remove the centrifuge rotor adjustment tools (P/N 233821) and check that distance between rotor and basis of the cover is 81.3mm $\pm$ 0.5mm.



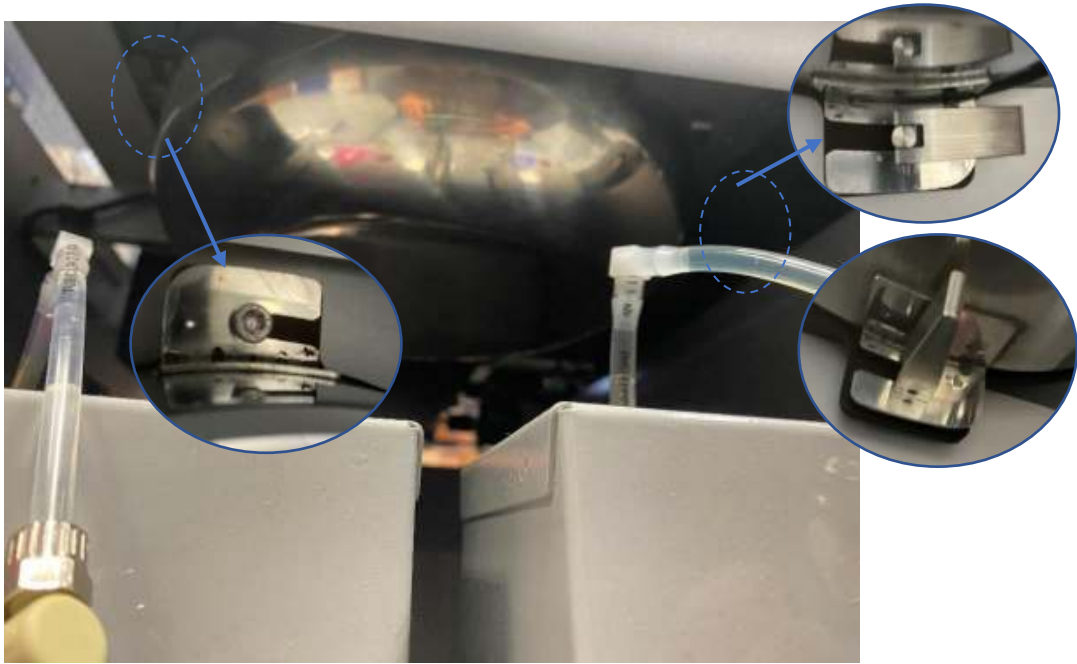
- Remove the cover from the pot and place the centrifuge card holders in the rotor.
- At this point, it is recommended to manually spin the new rotor to ensure that holders can freely move up and down once placed in the rotor.
- Place the centrifuge cover in the pot (with the rotor inside the pot) and lock them with the lateral flap and the screw.
- Install the centrifuge module in the instrument, placing the 4 screws that attach the centrifuge to the structure of the instrument.
- Connect cable B-CEN1 (for centrifuge 1) or B-CEN2 (for centrifuge 2).
- Perform gripper adjustment in the centrifuge which rotor has been replaced.

6.4.3 Remove centrifugue pot

In order to remove the pot of centrifuge 1 follow the next steps:

- Open fluidic drawer A and B and remove the two solutions.

2. Loosen the screw (at the left side), and pull down the lever (on the right side)



3. Turn the centrifuge pot clockwise while pressing slightly upwards until it is released.



4. Once the pot has been released, it can be removed from the instrument through the opening above the fluidic drawers.



To remove the pot of centrifuge 2 the same procedure must be carried out by opening the other 3 fluidic drawers.

6.5 FUSES

Fuses related to Centrifuge module are detailed in the following table,

PCA	FUSE NAME	DESCRIPTION
PCA GEN-BCKPL59	FUS01 FUS02	MST250 0034.6615 1A/250V MST250 0034.6620 3.15A/250V
PCA CEN-SRM Control	FUS1	MSF250 0034.6013 0.8A/250V
PCA CEN-CON59	NA	NA

Table 49.List of fuses of Centrifuge module

6.6 LEDS

LEDs related to Centrifuge module are detailed in the following table,

PCA	LED	DESCRIPTION
PCA GEN-BCKPL59	LED5	Green 5V (FUS01)
	LED6	Green 24V (FUS02)
PCA CEN-SRM Control	LED1	Yellow When the instrument is switched on, LED is ON and a few seconds later is OFF. This means that the FPGA has been loaded correctly with the appropriate data.
	LED2	Green VMotor Ok. It indicates that the voltage supplied to the motor is $\geq 12V$
PCA CEN-CON59	NA	NA

Table 50. List of LEDs of Centrifuge module

6.7 JUMPERS AND TEST POINTS

Neither jumpers nor test points are related to Centrifuge module.

6.8 WIRING TABLE

In the following table, there is detailed all the cables related to Centrifuge module. A wiring diagram can be found attached to Service Manual.

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
B-CEN1	H3	PCA CEN-CON59	CEN1	PCA GEN-BCKPL59
B-CEN2	H3	PCA CEN-CON59	CEN2	PCA GEN-BCKPL59
CEN-MOT	H1	PCA CEN-CON59	NA	MOTOR POWER
CEN-HALL	H2	PCA CEN-CON59	NA	MOTOR HALL SENSORS
CEN-ENC	H4	PCA CEN-CON59	NA	MOTOR ENCODER

Table 51. List of wiring of Centrifuge module

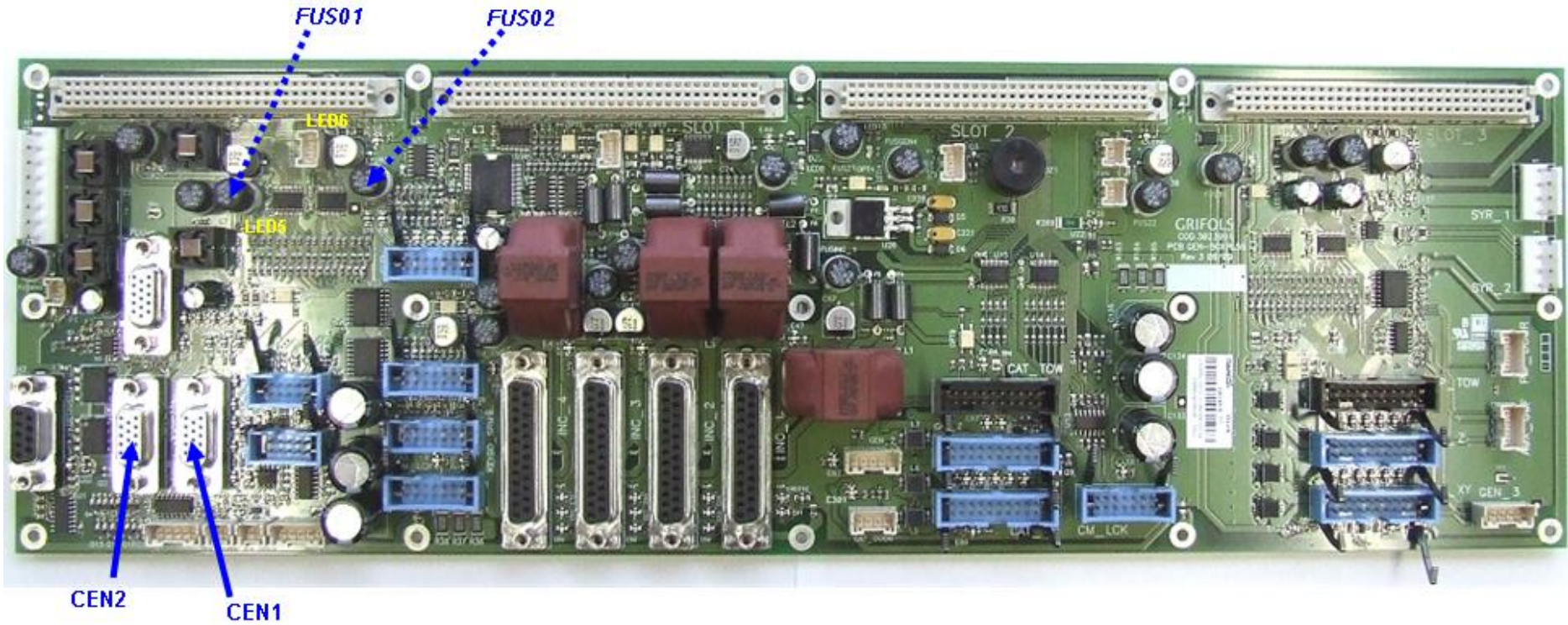


Figure 207. Location of the connectors, fuses and LEDs in PCA GEN-BCKPL59 related to Centrifuge module

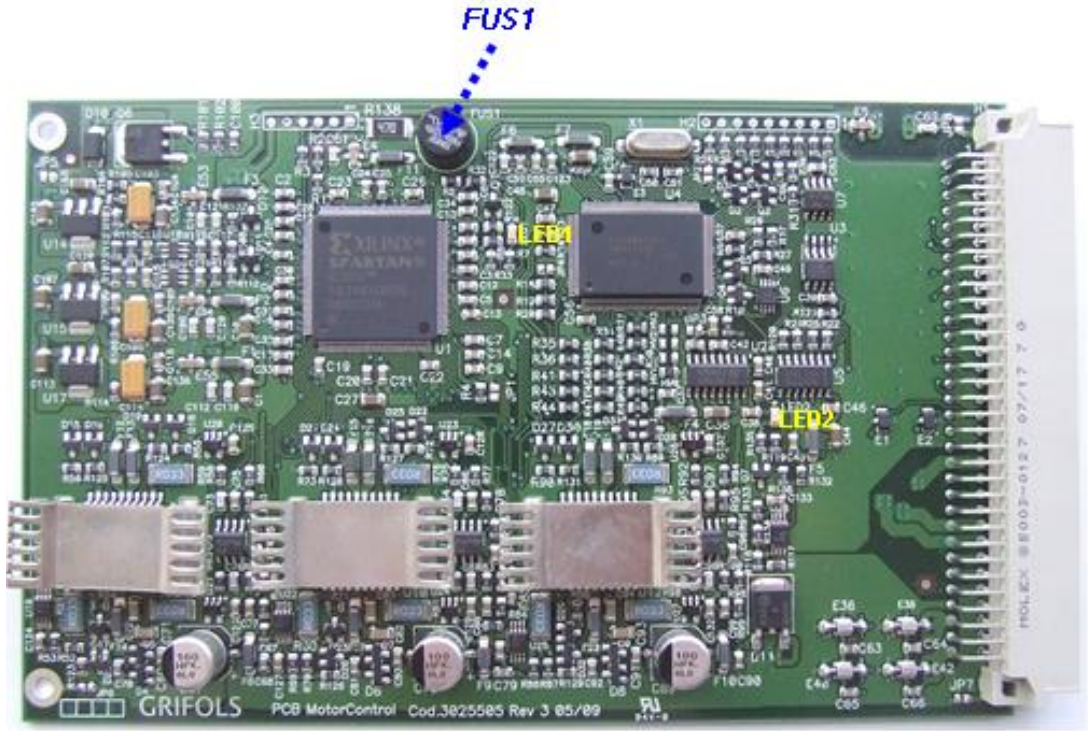


Figure 208. Location of the fuse and LED in PCA CEN-SRM Control

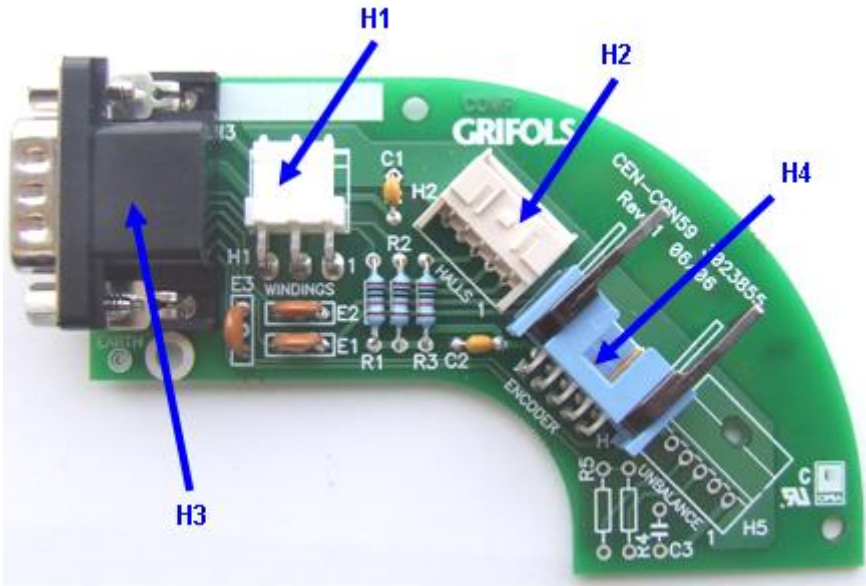


Figure 209. Location of the connectors in PCA CEN-CON59

6.9 TROUBLESHOOTING

Module	Device	Error code	Error	Cause	Text	Options
Centrifuge (03)	Centrifuge motor (00) Centr 1 (01) Centr 2	003000	ERRORCODE_LEN	FW has received a command with incorrect length	Centrifuge {1 or 2} has been disabled	OK
		005000	ERRORCODE_ARG	Command has wrong arguments		
		006000	ERRORCODE_BUSY	Command is addressed to a busy device		
		007000	ERRORCODE_NOTHOMED	Move command is addressed to a not homed device		
		012000	ERRORCODE_SPINBADPROFILE	Command includes a spin profile that is invalid for the addressed device		
		013000	ERRORCODE_NOTSPINNING	The addressed device is not performing the requested spin		
		033000	POSITION_ERROR	A position error has been produced at a servo device		
		035000	HOME_TIMEOUT	A home timeout has been produced at a servo device		
		036000	ERRORCODE_SRV_SYNC_TIMEOUT	A synchronization error has been produced at a servo device		
		037000	ERRORCODE_SRV_UNKNOWN_ERROR	An unknown error has been produced at a servo device		
		032000	CURRENT_LIMIT	Current limit at servo device	<i>When loading/unloading a card:</i> Error during centrifuge positioning. <i>During centrifugation:</i> Centrifuge {1 or 2} has been disabled	RETRY CANCEL Works (it leads to: <i>Centrifuge {1 or 2} has been disabled</i>) OK

Module	Device	Error code	Error	Cause	Text	Options
Centrifuge (03)	Centrifuge motor (00) Centr 1 (01) Centr 2	034000	MOVEMENT_ERROR	A movement error has been produced at a servo device	When loading/unloading a card: Error during centrifuge positioning.	RETRY CANCEL Works (it leads to: <i>Centrifuge {1 or 2} has been disabled</i>)
					During centrifugation: Centrifuge {1 or 2} has been disabled	OK
Any (xx)	Any (xx)	002000	ERRORCODE_CMD	Unknown command code received	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		003000	ERRORCODE_LEN	FW has received a command with incorrect length		
		004000	ERRORCODE_DEV	Command is addressed to an incorrect device		
		005000	ERRORCODE_ARG	Command has wrong arguments		
		014000	ERRORCODE_HW_DEV	The addressed device has a problem with its related HW		
		015000	ERRORCODE_HW_HWFALL	There is a problem with PCA HW		
		016000	ERRORCODE_HW_HWNOT SET	PCA setup error		

Module	Device	Error code	Error	Cause	Text	Options
0	240	000000	ERRORCODE_UNKNOW N	Unknown error	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		008000	ERRORCODE_FW_NOTS AFE	Movement not safe		
		017000	ERRORCODE_FW_UNKN OWN_ERROR	Unknown error		
		061000	ERRORCODE_INVALID_ ARGUMENTS	Invalid arguments		
		062000	ERRORCODE_FWCOMM_ TIMEOUT	Communication timeout		
		063000	ERRORCODE_FWCOMM_ CONNECTION_LOST	Communication lost		
		064000	ERRORCODE_FWCOMM_ COMM_ERROR	Communication error		
		088000	ERRORCODE_PRECOND ITION_FAILED	Previous condition for the action is not met		
		089000	ERRORCODE_STARTUP_ ERROR	Driver startup error		
		091000	ERRORCODE_ALREADY_ STARTED	Another instance of Driver is already opened		
		090000	ERRORCODE_UNEXPEC TED_ERROR	A Driver unexpected error has been produced		

Table 52. List of errors related to Centrifuge module

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Centrifuge disabled	Centrifuge disabled was already disabled in software	Check centrifuge status in Erytra software
	Movement error / Current limit error	Check Movement error / Current limit error section in this troubleshooting
	Home position is not detected	Check Home position is not detected section in this troubleshooting
Movement error / Current limit error CEN1: 0300032000 CEN2: 0301032000 CEN1: 0300034000 CEN2: 0301034000	Fuse blown	Check / replace FUS01 or FUS02
	Rotor is not correctly assembled	Check rotor assembly
	Motor/encoder damaged	Check / replace motor
	PCA CEN-CON59 failure	Check / replace PCA CEN-CON59
	Check wiring	Check / replace wiring
	PCA CEN-SRM Control failure	Check / replace PCA CEN-SRM Control
Home position is not detected CEN1: 0300007000 CEN2: 0301007000 CEN1: 0300035000 CEN2: 0301035000	PCA GEN-BCKPL59 failure	Check / replace PCA GEN-BCKPL59
	Fuse blown	Check / replace FUS01 or FUS02
	Failure in home detector	Check / replace motor
	PCA CEN-CON59 failure	Check / replace PCA CEN-CON59
	Check wiring	Check / replace wiring
	PCA CEN-SRM Control failure	Check / replace PCA CEN-SRM Control
Centrifuging speed / time is not the correct one	PCA GEN-BCKPL59 failure	Check / replace PCA GEN-BCKPL59
	Motor/encoder damaged	Check / replace motor
	Check wiring	Check / replace wiring
	PCA CEN-SRM Control failure	Check / replace PCA CEN-SRM Control
	PCA GEN-BCKPL59 failure	Check / replace PCA GEN-BCKPL59

Table 53. Centrifuge module troubleshooting

7 PIPETTING ROBOT ARM (PIP) MODULE

7.1 DESCRIPTION

The functions of this module are:

- Presence detection and identification of samples and reagents.
- Pipetting samples and reagents into cards, including appropriate controls and level detection.

To do so, this module has a five-axis robotic head: X, Y, Z1 (Probe 1), Z2 (Probe 2) and horns (involved in barcode reading).

Robotic head X-axis structure is formed by a mechanized aluminium plate screwed to the instrument laterals. A linear guide screwed to the plate allows movement of X-axis. X-axis is moved by a servo-controlled motor assembled in the left area. This motor turns and provokes that a small belt moves. This belt transmits the movement by means of a gear to a large belt attached to X wagon (assembled on the linear guide). Large belt tension is adjusted by modifying X end pulley position. Home position is defined at the left and using an L-shaped part assembled in X wagon that will get in an optical barrier. PIP module and CAT module built in same X-axis.

Y-axis is moved by a servo-controlled motor assembled in back axis area. This motor turns and provokes that small belt moves. This belt transmits the movement by means of a gear to a large belt attached to Y wagon (assembled on a linear guide). Small belt tension can be adjusted modifying motor position (slotted hole). Large belt tension is adjusted by modifying Y end pulley position. Home position is defined at the back and using and L-shaped part assembled in Y wagon that will get in an optical barrier.

Z axes (composed of probe 1 axis, probe 2 axis and horns axis) are moved by servo-controlled motors assembled in robotic head bottom. Motor turns and provokes that a small belt moves. This belt provokes that a pulley fixed to a leadscrew with a set screw turns and this one that the probes or horns move. Small belt tension can be adjusted modifying motor position (slotted hole).

Encoders resolution for the PIP axes is as follows:

PIP Axis	Encoder counts / mm
X	285.714 counts/mm
Y	285.714 counts/mm
Z1	2095.2 counts/mm
Z2	2095.2 counts/mm
Horns	2095.2 counts/mm

Table 54. PIP axes encoder resolution

Horns home position is defined at Z axis top and using an L-shaped part assembled in leadscrew nut that will get in an optical barrier placed in PCA PIP-TOW59. Once horns activate the optical barrier, they move up. Probes home positions are defined at the top Z axis. Using L-shaped parts assembled in the probe wagons, they get in optical barriers placed in PCA PIP-TOW59. Once probes activate the optical barrier, they move to the position adjusted in the probes offset adjustment.

Level detection system is based on a capacitive system (RC oscillator) where the probe is part of the capacitor. When probe touches the surface of liquid, its capacitance is modified and a decrease in the oscillation frequency is obtained. There are 2 different PCA, one per each probe, sharing PCB and all components except those related to generate the oscillation frequency, in order to have a different frequency at each probe and to avoid any coupling. The level detector system is used also to detect if card foil is present when first pipetting into a well. If impact detector

system is installed, it is possible to detect and protect probes from undesired impacts which are frequently caused by improper user operation.

Presence detection of samples and reagents is performed by means of a laser detection system. The laser is placed in a fixed position in the robot head, no movement is required. The robot arm passes over the racks with samples and reagents and with the detection of the reflected light of the laser, the system is able to detect whether is located a tube, vial or bottle in the scanned rack. Also the reflected light of the laser gives information of the sample tubes diameter. To scan the reagents racks, X-movements of the PIP arm are performed whereas to scan the samples racks, Y-movements of the PIP arm are performed.

Identification of samples and reagents is performed by means of two barcode readers assembled at each side of the horns located in the robotic head. Barcode reader reference is Symbol SE-955 (s/n < 059-0001335) or Zebra SE-965 (s/n ≥ 059-0001335) and the communication is through RS232. The robot arm passes over the rack and the horns move down, activating the barcode reader to read the barcode labels. The tubes, vials and bottles must be faced with the barcode label outside the rack in order to let the system read it. Allowed types of barcodes are the following ones:

- Code 39 (check digit enabled)
- Interleaved 2 of 5 (check digit enabled)
- Code 128
- ISBT 128
- GS1 – 128 (formerly UCC/EAN-128)
- Codabar
- Others upon barcode readers configuration.

Sample barcode specifications (* based on code type 128 barcode) are the following ones:

- Minimum resolution*: 0.15 mm.
- Position*: 3mm from top/bottom
- Quiet zone*: 2 mm.
- Maximum length*: 45 mm.
- Minimum width*: 8mm.
- Slope*: 20°
- Maximum number of characters: 24.

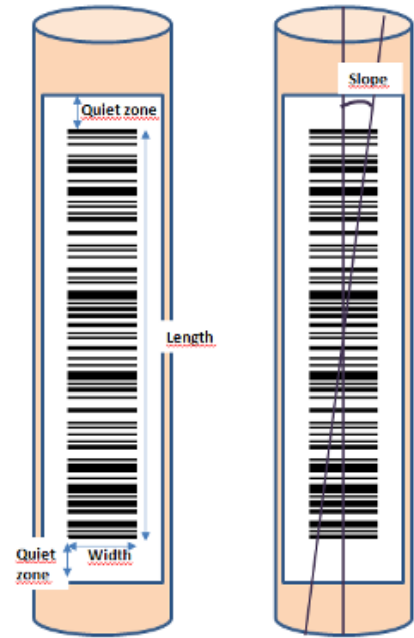


Figure 210. Sample barcode specifications

The number of PIP barcodes reading retries for sample tubes and reagents and the total number of samples and reagents barcodes read in the current session are logged at instrument Driver (DG59Driver) log when the shutdown process is performed.

Block diagram of the module is shown in following figure:

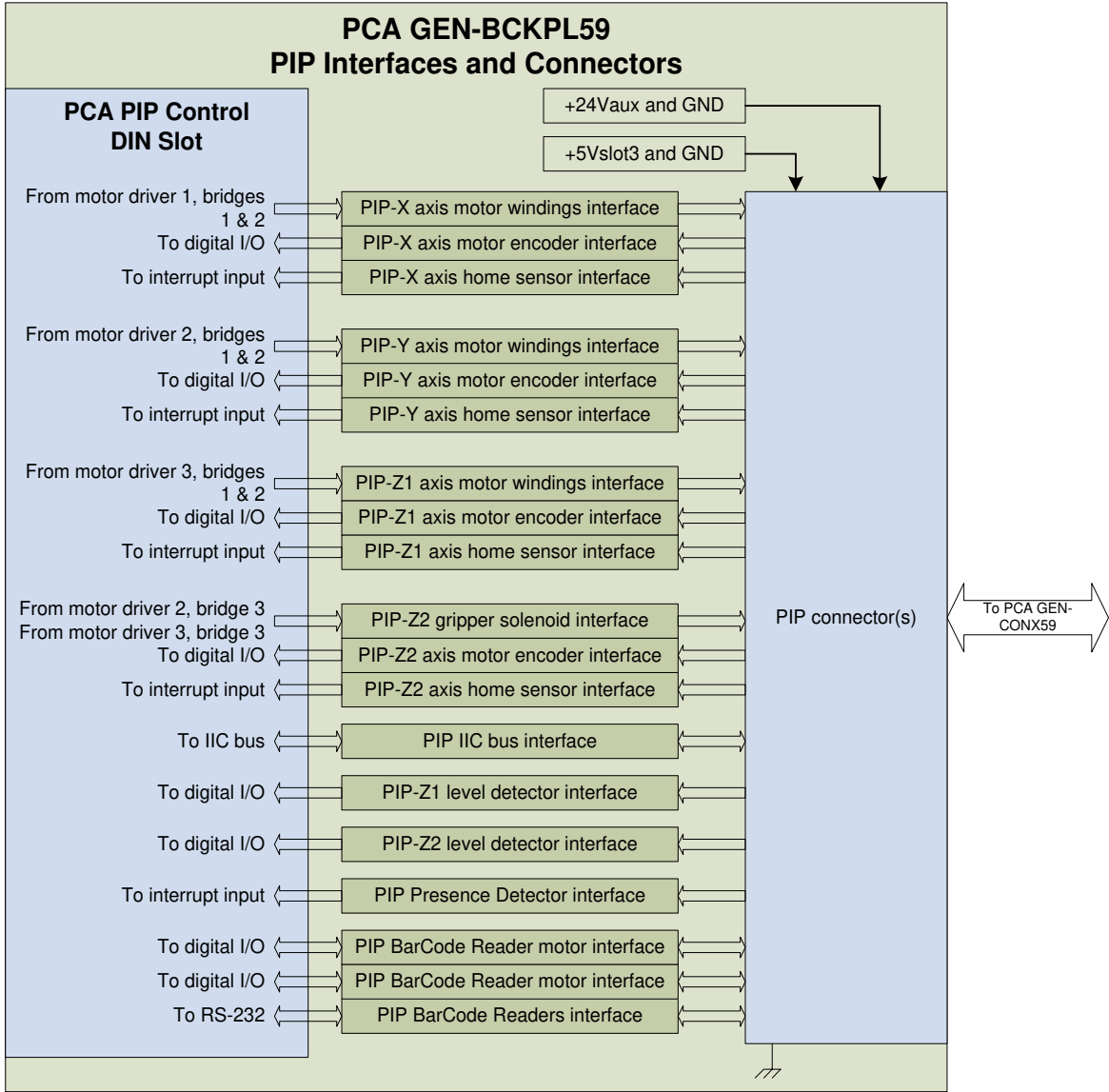


Figure 211. Pipetting robot arm module block diagram

PCAs related to PIP module are:

- **PCA GEN-BCKPL59:** main PCA of the instrument, located at backpanel.
- **PCA PIP Control:** located at backpanel (slot 3), attached to PCA GEN-BCKPL59. It controls signals used in pipetting module. It contains the firmware of this module.
- **PCA GEN-CONX59:** This PCA is located at X axis left side (looking from the front of the instrument). It contains X axis home sensor and X axis motor powering and encoder.
- **PCA PIP-CONY59:** Located at Y axis back (looking from the front of the instrument). It contains Y axis home optical barrier (power and signal), Y axis motor powering and encoder. Other parts are detailed on the fluidics module.
- **PCA PIP-TOW59:** Located inside robotic head. It contains probes and horns home optical barriers (power and signal), probes axis motor powering and encoder, horns axis motor powering and encoder and barcode reader signals.
- **PCA PIP-P1OSC59:** Located inside robotic head, over Z1 probe. It contains level detector oscillator and sends oscillator signal to PCA PIP-TOW59. Oscillation frequency for Z1 probe is different than for Z2 probe.

- PCA PIP-P2OSC59:** Located inside robotic head, over Z2 probe. It contains level detector oscillator and sends oscillator signal to PCA PIP-TOW59.
- PCA PIP-BCR59:** Located inside robotic head, it connects barcode readers interfaces with PCA PIP-TOW59.

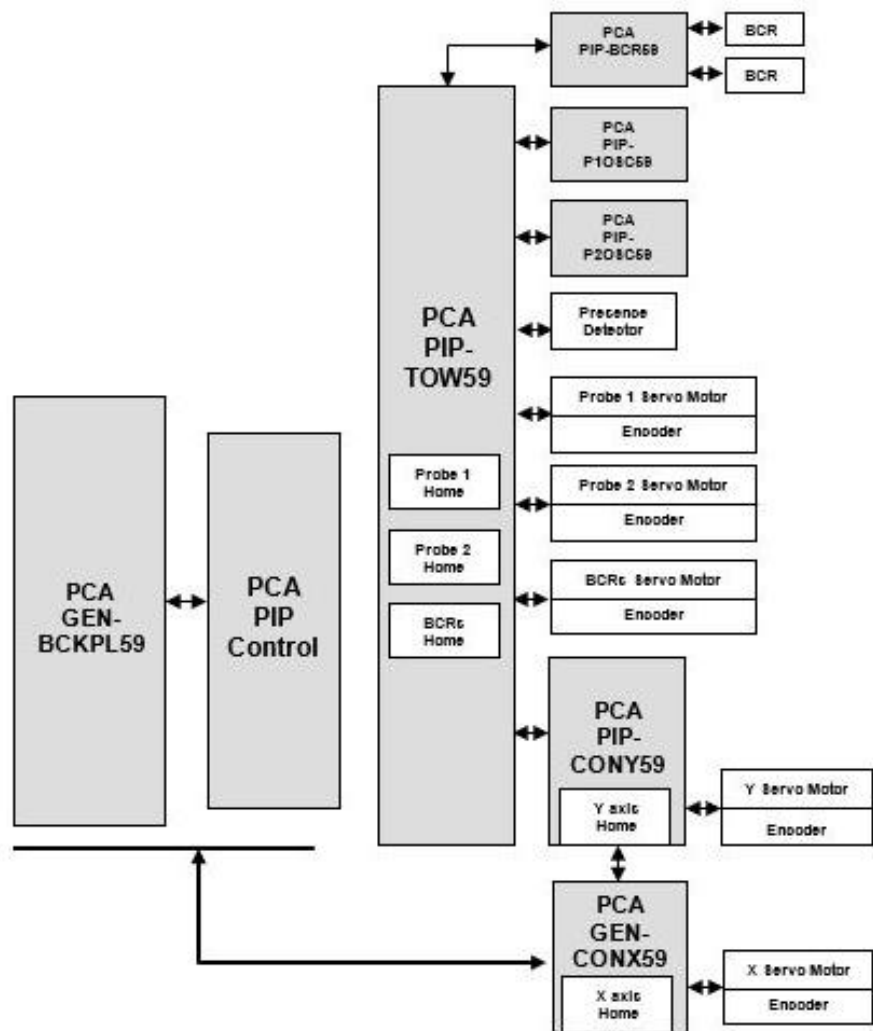


Figure 212. Main blocks of PIP module

This module uses a bus I2C for Probe1 Level Detector activation + Probe2 Level Detector activation + Probe1 Correct Pipetting Detector (activation and signal) + Probe2 Correct Pipetting Detector (activation and signal) + Barcode Readers activation + Barcode readers axis home sensor activation and current limit sensing + Presence Detector activation.

Communication with both barcode readers are through a RS232 bus.

7.2 ADJUSTMENT PROCEDURES

In this section, adjustments related to PIP module are explained. There are 2 types of adjustments: mechanicals and functionals.

7.2.1 Mechanical adjustments

Mechanical adjustments related to PIP module are detailed.

7.2.1.1 Probes parallelism

This adjustment is performed to ensure the parallelism between both probes. It must be performed if:

- Probe leadscrew is replaced.
- Probe wagon is replaced.

Prerequisite: NA
Adjustment tool: Probes/wagon adjustment tool (Ver. 2) (P/N: 233901)
Procedure:

1. Loose the 2 Allen screws of the probes washer and the 3 Allen screws that attach each probe wagon to the leadscrew.



Figure 213. Allen screws of the probe wagon - leadscrew

2. Place the tool between the upper part of the two leadscrews, checking also that both probe wagons are located at the top of the robotic head.

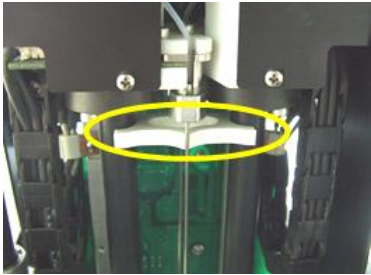


Figure 214. Parallelism checking at the top of the robotic head

3. Adjust the position of both probe wagons until both probes touch the external edges of the tool. Then tighten the screws of the probe wagon – leadscrew and remove the tool.



Figure 215. External edges of the probes/wagon adjustment tool (Ver. 2)

4. Move down manually (by means of their belts) both probes and adjust the position of the probes washer, checking that both probes are parallel in X-Y axes. Tighten the screws of the probes washer. In order to verify the parallelism, check section 7.3.4 in this chapter.



Figure 216. Parallelism checking at the bottom of the robotic head

5. Check alignment between home flag and home sensor.

7.2.1.2 X-Axis small belt tension adjustment

No manual adjustment of tension of X-axis small belt is required. Assembly and silentblocks assure proper belt tension.

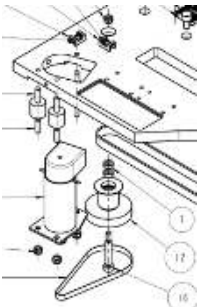


Figure 217. X-axis motor small belt

7.2.1.3 X-Axis large belt tension adjustment

Adjustment performed to set large belt tension of PIP arm X-axis.

Prerequisite: NA
Adjustment tool: NA
Procedure:

- 1. Screw pulley tightening screw to allow pulley movement.
- 2. Loose both pulley fixing screws.
- 3. Tighten the belt unscrewing the pulley tightening screw and fix its position tightening both fixing screws.

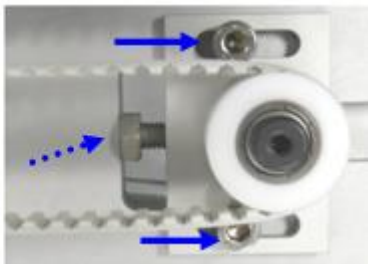
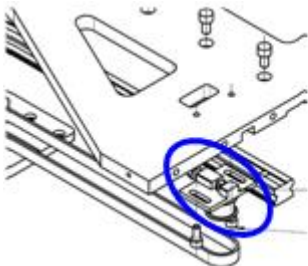


Figure 218. X-axis motor large belt

7.2.1.4 Y-Axis small belt tension adjustment

This adjustment is performed to set small belt tension of PIP arm Y-axis.

Prerequisite: NA
Adjustment tool: NA
Procedure:

1. Remove Y arm motor cover (remove 4 fixing screws).
2. Unplug Y arm ribbon cables from PIP-CONY59 board.
3. Remove the screws fixing Y motor-belt assembly to Y arm and remove the assembly.
4. Loose Y motor slotted hole screw.

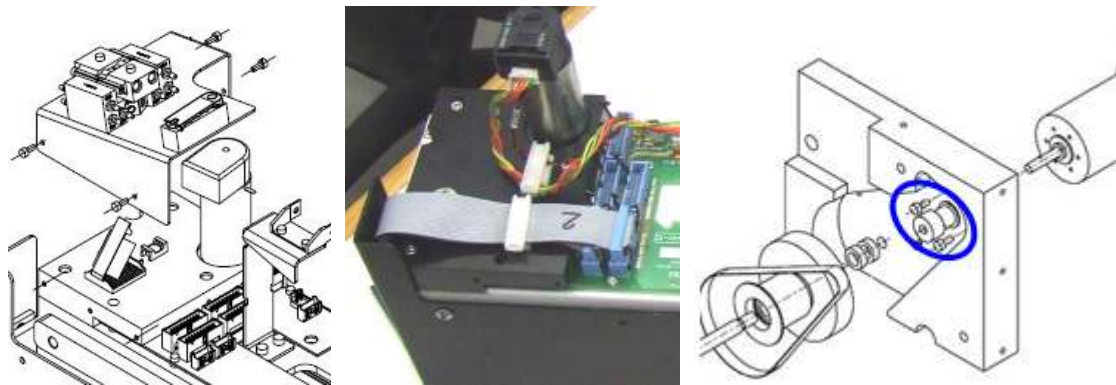


Figure 219. Y-axis motor small belt

5. Move Y motor to adjust belt tension.
6. Fix motor position tightening slotted hole screw
7. Assemble Y motor-belt assembly to Y arm tightening the three screws.
8. Connect Y-arm ribbon cables to PCA PIP-CONY59.
9. Assemble Y motor cover.

7.2.1.5 Y-Axis large belt tension adjustment

This adjustment is performed to adjust tension of the large belt of Y-axis of PIP arm.

Prerequisite: NA
Adjustment tool: NA
Procedure:

1. Screw pulley tightening screw to allow pulley movement.
2. Loose both pulley fixing screws.
3. Tighten the belt unscrewing the pulley tightening screw and fix its position tightening both fixing screws.

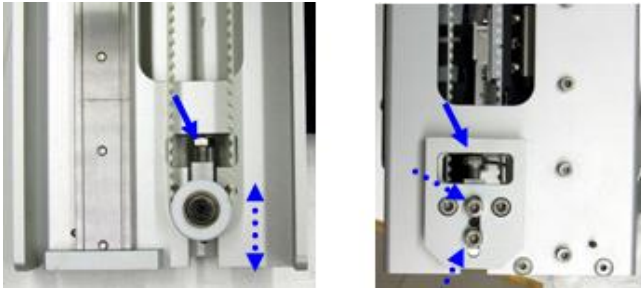


Figure 220. Y-axis motor large belt

7.2.1.6 Z-Axis belt tension adjustment

The adjustment method applies for the three Z-axis motors (Z1 probe, Z2 probe and Z horns). It is performed to adjust tension of the belt of any of the Z-axis of the PIP arm.

Prerequisite: NA
Adjustment tool: NA
Procedure:

1. Loose the 2 screws that are attaching the motor to PIP robot bottom base.
2. Push motor pinion in order to adjust the tension of the motor belt and tighten the screw of the slotted hole. Then tighten the other screw.



Figure 221. Z-axis motors belts

7.2.1.7 Presence detector inclination

Adjustment of the inclination of the presence detector system. It must be performed if:

- Presence detector is replaced.
- Wrong detections are observed.

Prerequisite: NA
Adjustment tool: NA
Procedure:

1. Loose the nut of the screw that attaches presence detector to PIP robot head.
2. Move presence detector so pin of presence detector is inserted in its location.

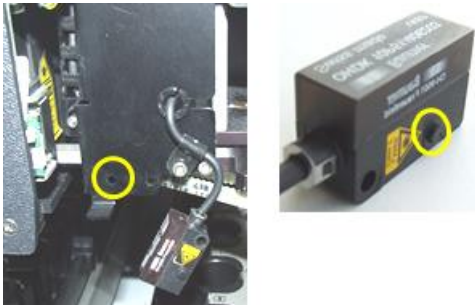


Figure 222. Presence detector assembly

3. Tighten the nut of the screw that attaches the presence detector to presence detector holder plate.
4. Verify position of presence detector (check section 7.3.11)

7.2.2 Functional adjustments

Functional adjustments related to PIP module are detailed.

7.2.2.1 Probes offset

This adjustment is required to define exact position of probes washer respect of the probes and to compensate possible difference in Z position of both probes. It must be performed/restored if:

- The probe wagon is replaced.
- The leadscrew of a probe is replaced.
- PCA PIP-TOW59 is replaced.
- PCA PIP Control is replaced.

Prerequisites:

- Fluidic circuit empty.
- PIP homed.

Adjustment tool: Probes adjustment tool (Ver. 4) (PN: 234213)

Procedure:

1. In Technical Service software, go to *Procedures* tab. Then select *probes offset* in *PIP floor Modules* section. A precondition message is shown on the screen, advising that the fluidic circuit must be empty (*Fluidics* tab → *fluidic empty*).
2. After accepting the message, an adjustment wizard window is opened.



Figure 223. Probes offset wizard in TSS

3. Place the adjustment tool (holding it with your hand) as indicated on following picture, checking that adjustment tool holes match with probes washer exit holes and that the adjustment tool is completely flat when inserted in this position.



Figure 224. Adjustment tool position in probes offset adjustment

4. Press *Next>* button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press *OK* button. Probes will move down automatically until they detect tool surface.
5. Once the process is finished, a screen appears showing the parameters to be stored. Remove the tool from the probes washer, press *Finish* button and press *OK* on the information window.

6. Press *OK* in the confirmation window that indicates that home movements will be performed.
7. Press *OK* to finish the adjustment procedure.



NOTE:

Probes offset adjustment is stored in PCA PIP Control. In case of replacing this PCA, the adjustment can be restored (if a previous backup was performed). Otherwise the adjustment must be performed.

7.2.2.2 Reagents drawers

Adjustment of the probes position on reagent drawers. It must be performed if:

- It is removed the reagent drawer from the instrument.
- It is removed the PIP robotic head.
- It is replaced the probe wagon or the leadscrew.
- It is modified the position of the latch of the reagent drawer.

Prerequisite:

- Offset between probes adjustment performed.
- Reagent drawer flatness verification.
- PIP homed.

Adjustment tool: Probes adjustment tool (Ver. 4) (PN: 234213)

Procedure:

1. In Technical Service software go to *Procedures* tab → *PIP floor Modules* section → *reagent drawers* button. An adjustment wizard is opened.
2. Select the reagent drawer to adjust and press *Next>* button.

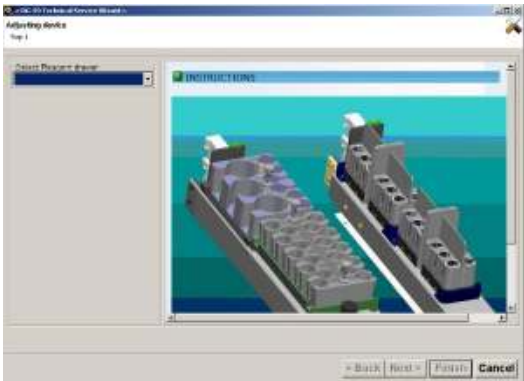


Figure 225. Reagent drawer selection in TSS

3. A warning message informing that the instrument will perform movements/settings appears on the screen. Press *OK* button. Rack 2 of selected drawer will perform a home movement. Insert the adjustment tool over positions 1 and 5 in rack 2 of selected drawer. Place manually PIP arm over adjustment tool and with the help of the screen joysticks (or manually), move PIP arm until Z1 probe is above the adjustment area and close to the upper left corner (not touching tool surface and not further than 6 mm from tool surface or adjustment edges), as it is shown in following pictures.



NOTE:

Z1 probe must not touch the tool surface.

**NOTE:**
Check that adjustment area of the tool is over holder 1 in reagent rack.

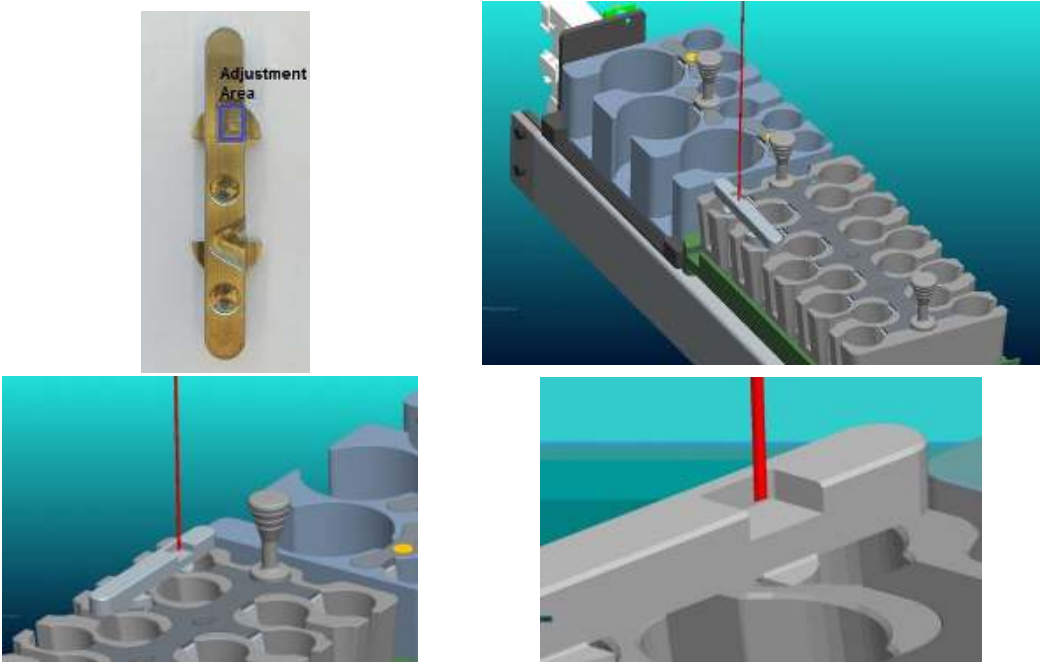


Figure 226. Adjustment of PIP arm on reagents drawers –step

- 4. Press *Next>* button. Z1 probe is moved automatically to find the adjustment position (upper left corner on the adjustment area). First, probe 1 is moved automatically in Z axis and, by means of level detection, it is detected when probe 1 touches the tool surface. Then probe 1 is moved automatically in X axis to detect, by means of level detection, left wall of the adjustment tool. Finally, probe 1 is moved automatically in Y axis to detect the upper wall of the adjustment tool. When procedure finishes, probe 1 is homed.
- 5. Then place adjustment tool over positions 4 and 6 in rack 1 of the selected drawer, pushing the tool to the right in its location. Place manually PIP arm over adjustment tool and with the help of the screen joysticks (or manually), move PIP arm until Z1 probe is above the adjustment area (not touching tool surface) avoiding the hole, as it is shown in following picture.

**NOTE:**
Only Z coordinate will be adjusted in this step. Adjustment tool is free in X and Y axes when placed into adjustment position. Check that adjustment tool remains flat over adjustment position.

**NOTE:**
Z1 probe must not touch surface of the tool.

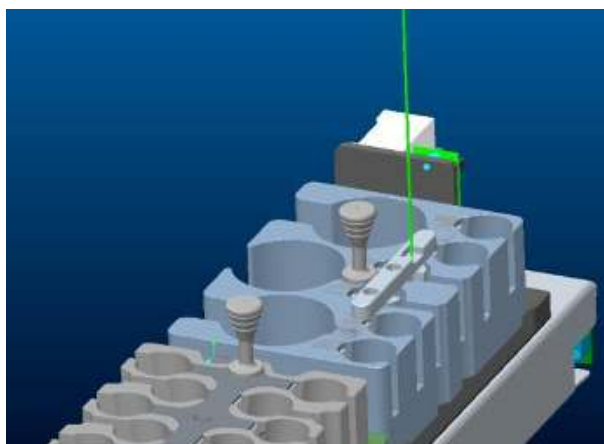


Figure 227. Adjustment of PIP arm on reagents drawers – step 2

6. Press *Next>* button. Z1 probe is moved automatically to find the adjustment position (surface on the adjustment area). Probe 1 is moved automatically in Z axis and, by means of level detection, it is detected when probe 1 touches the adjustment area tool surface. When procedure finishes, probe 1 is homed.
7. Once the process is finished, a screen appears showing the parameters to be stored. Press *Finish* button and press *OK* on the confirmation window. When the adjustments are stored, press *OK* on the information window.
8. Press *OK* in the confirmation window that indicates that home movements will be performed.
9. Press *OK* to finish the adjustment procedure.



NOTE:

Adjustment stored in *E:\data\Erytra\conf\PIPCalibrations.txt*.

10. Repeat same procedure for the other reagent drawer if necessary.



CAUTION!

Remove the adjustment tool from the instrument when adjustment procedure finishes.

7.2.2.3 Samples drawers

Adjustment of the position of the probes in samples drawers. It must be performed if:

- It is removed the sample drawer from the instrument.
- It is removed the PIP robotic head.
- It is replaced the probe wagon or the leadscrew.
- It is modified the position of the latch of the samples drawer.
- It is modified the position of the centerers of the samples drawer.

Prerequisite:

- Offset between probes adjustment performed.
- Sample drawer flatness verification.
- Home of PIP arm: X, Y, Z1, Z2 and horns.

Adjustment tool: Probes adjustment tool (Ver. 4) (PN: 234213)

Procedure:

1. In Technical Service software, go to *Procedures* tab → *PIP floor Modules* section → *sample drawers* button. An adjustment wizard is opened.
2. Select the sample drawer to adjust and press *Next>* button.
3. Insert adjustment tool over positions 2 and 4 in rack 1 of selected drawer. Place manually PIP arm over the adjustment tool and with the help of the screen joysticks (or manually), move PIP arm until Z1 probe is above the adjustment area and close to the upper left corner (not touching tool surface and not further than 6 mm from tool surface or adjustment edges), as it is shown in the following pictures.

**NOTE:**
Z1 probe must not touch the tool surface.

**NOTE:**
Check that adjustment area of the tool is over holder 2 in the sample rack.

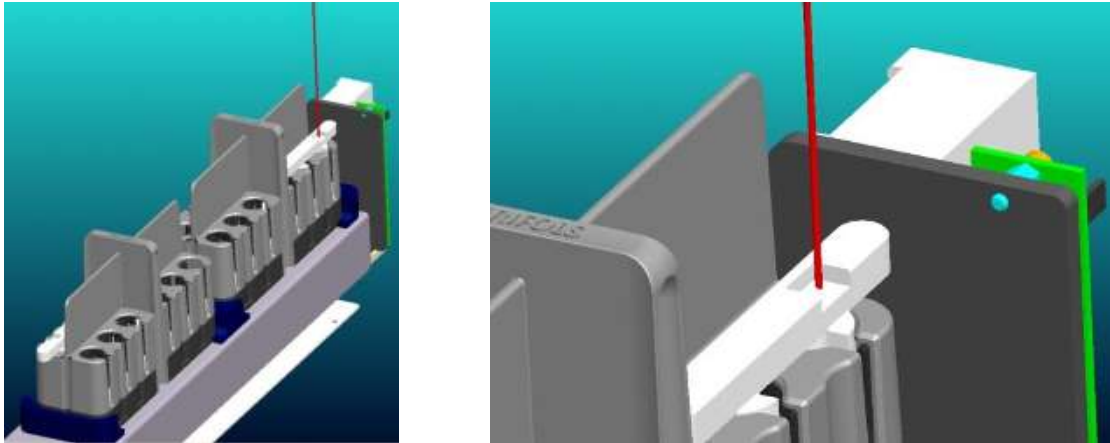


Figure 228. Adjustment of PIP arm over rear part of samples drawer

4. Press *Next>* button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press *OK* button. Z1 probe is moved automatically to find the adjustment position (upper left corner on the adjustment area) by means of level detection. First, probe 1 is moved automatically in Z axis until it is detected that probe 1 touches the tool surface. Then probe 1 is moved automatically in X axis to detect the left wall of the adjustment tool. Finally, probe 1 is moved automatically in Y axis to detect upper wall of the adjustment tool. When procedure finishes, probe 1 is homed.
5. Then insert the adjustment tool over positions 9 and 11 in the rack 2 of the selected drawer. Place manually the PIP arm over the adjustment tool and with the help of the screen joysticks (or manually), move the PIP arm until Z1 probe is above the adjustment area and close to the lower right corner, as it is shown in the following pictures.

**NOTE:**
Z1 probe must not touch surface of the tool.

**NOTE:**
Check that adjustment area of the tool is over holder 11 in sample rack.

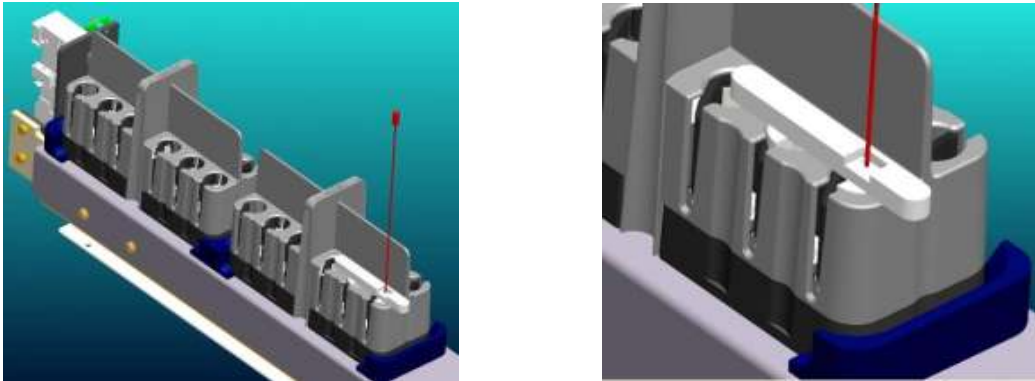


Figure 229. Adjustment of PIP arm over frontal part of samples drawer

6. Press *Next>* button. Z1 probe is moved automatically to find the adjustment position (lower right corner on the adjustment area) by means of level detection. First, the probe is moved automatically in Z axis until it is detected that probe touches the surface of the tool. Then the probe is moved automatically in X axis to detect the right wall of the adjustment tool. Finally, the probe is moved automatically in Y axis to detect the lower wall of the adjustment tool.
7. Once the process is finished, a screen appears showing the parameters to be stored. Press *Finish* button and press *OK* on the confirmation window. When the adjustments are stored, press *OK* on the information window.
8. Press *OK* in the confirmation window that indicates that home movements will be performed.
9. Press *OK* to finish the adjustment procedure.

**NOTE:**
Adjustment stored in *E:\data\Erytra\conf\PIPCalibrations.txt*.

10. Repeat same procedure for other samples drawers if necessary.

**CAUTION!**
Remove the adjustment tool from the instrument when adjustment procedure finishes.

7.2.2.4 Incubators

Adjustment of the position of the probes on each incubator. It must be performed if:

- It is removed the incubator from the instrument.
- It is removed the PIP robotic head.
- It is replaced the probe wagon or the leadscrew.

Prerequisite:

- Offset between probes adjustment performed.
- Incubator flatness verification.
- Adjustment of the height of the incubator performed.
- Incubator in home position.

- PIP homed.

Adjustment tool: Probes adjustment tool (Ver. 4) (PN: 234213)

Procedure:

1. In Technical Service software, go to *Procedures* tab → *PIP floor Modules* section → *incubators* button. A precondition message on screen appears, indicating the requirement of the incubator height adjustment. If *OK* button is pressed, a wizard adjustment is opened.
2. Select the incubator to adjust. A message appears on screen informing that it is necessary to place the adjustment tool in the sixth slot of the selected incubator and that when *Next* button is pressed, the selected incubator will move up. Place the tool in the sixth slot (once in this position, it is recommended to push it as far as possible to the rear part of the slot) and then press *OK* button and then press *Next>* button. The incubator will move to the intermediate area.



CAUTION!

Check that CAT arm is not over the incubator to adjust (interfering with its movement).

3. Place manually PIP arm over the adjustment tool and with the help of the screen joysticks (or manually), move the PIP arm until the Z1 probe is above the adjustment area and close to the upper right corner (not touching tool surface and not further than 6 mm from tool surface or adjustment edges), as it is shown in the following pictures.

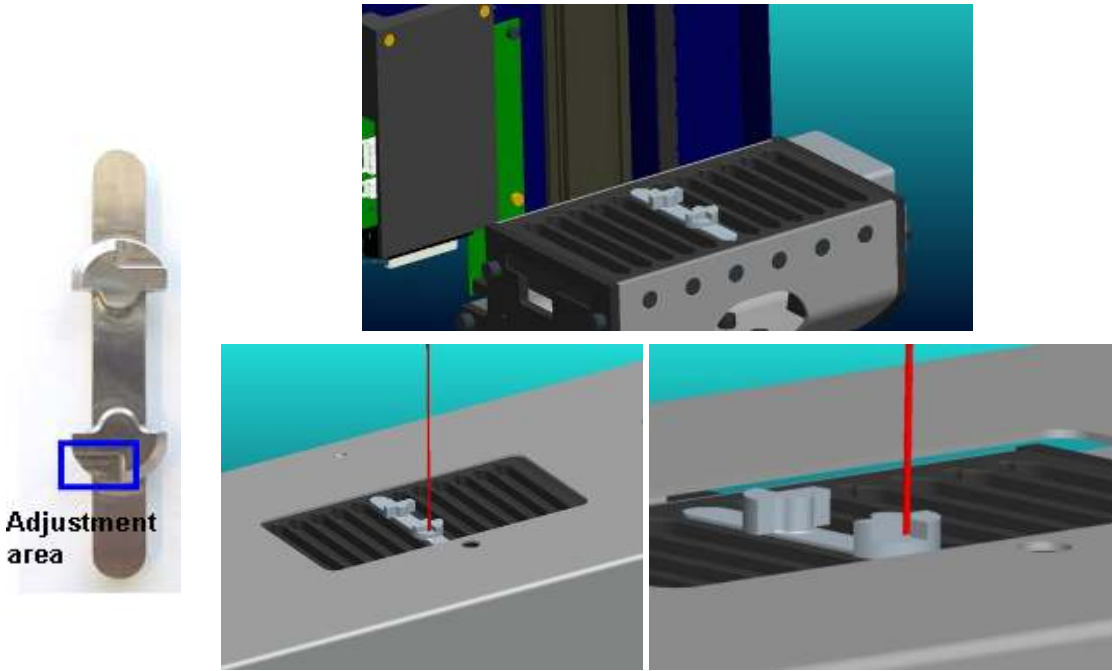


Figure 230. Adjustment of PIP arm on incubators

4. Press *Next>* button. Z1 probe is moved automatically to find the adjustment position (upper right corner on the adjustment area) by means of level detection. First, the probe is moved automatically in Z axis until it is detected that probe touches the surface of the tool. Then the probe is moved automatically in X axis to detect the right wall of the adjustment tool. Finally, the probe is moved automatically in Y axis to detect the upper wall of the adjustment tool.
5. After locating the adjustment area, Z1 probe is placed automatically over well 1 position of slot 6 and then the probe is moved automatically in Z axis until it is detected that probe touches the surface of the tool. Then Z1 probe is placed automatically over well 8 position

- of slot 6 and then the probe is moved automatically in Z axis until it is detected that probe touches the surface of the tool.
- Once the process is finished, a screen appears showing the parameters to be stored. Press *Finish* button and press *OK* on the confirmation window. When the adjustments are stored, press *OK* on the information window.
 - Press *OK* in the confirmation window that indicates that home movements will be performed. The incubator will automatically move to the home position (CAT area).
 - Press *OK* to finish the adjustment procedure.



NOTE:

Adjustment stored in *E:\data\Erytra\conf\PIPCalibrations.txt*.

- Repeat same procedure for other incubators if necessary.



CAUTION!

Remove the adjustment tool from the instrument when adjustment procedure finishes.

7.2.2.5 Dilution cup

Adjustment of the position of the probes in the dilution cup. It must be performed if:

- It is removed the dilution cup module.
- It is removed the PIP robotic head.
- It is replaced the probe wagon or the leadscrew.

Prerequisite:

- Offset between probes adjustment performed.
- Dilution cup parallelism adjustment performed.
- PIP homed.

Adjustment tool: NA

Procedure:

- In Technical Service software, go to *Procedures* tab → *PIP floor Modules* section → *dilution cup* button. It will open a wizard adjustment. Press *Next>* button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press *OK* button.
- Place manually PIP arm over the dilution cup and with the help of the screen joysticks (or manually), lower Z1 probe until is inserted in the center notch located on the surface of the dilution cup, touching the bottom of the hole but without moving the cup.

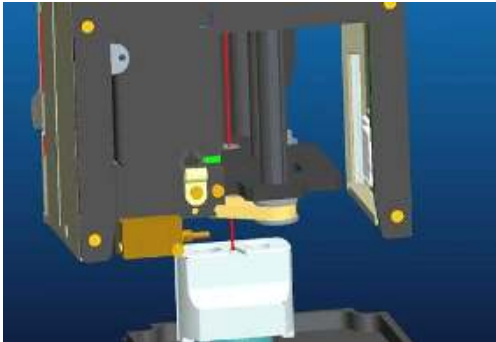


Figure 231. Position of the probe in dilution cup adjustment

3. Press *Next>* button. A screen appears showing the parameters to be stored. Press *Finish* button and press *OK* on the confirmation window. When the adjustments are stored, press *OK* on the information window.
4. Press *OK* in the confirmation window that indicates that home movements will be performed.
5. Press *OK* to finish the adjustment procedure.



NOTE:

Adjustment stored in *E:\data\Erytra\conf\PIPCalibrations.txt*.

7.2.2.6 Barcode reader inclination position

Adjustment of the laser beam position of the barcode reader. It must be performed if a barcode reader is removed.

Prerequisite: NA

Adjustment tool: NA

Procedure:

1. Loose the 2 screws that attach the barcode reader to the horns.
2. Activate the barcode reader. To do so, open Technical Service software and go to *Pip Arm* tab → *BCR* tab → *set on* button, activating the desired barcode reader.
3. Move manually the barcode reader in order to center the laser light in the mirror, and then tighten both screws.



Figure 232. Correct position of barcode reader

4. Deactivate the barcode reader by means of *set off* button of the desired barcode reader (Technical Service software, *Pip Arm* tab → *BCR* tab).

7.2.2.7 Horns position

Adjustment of the position of barcode reading in the drawers. It must be performed if:

- It is removed the PIP robotic head.
- It is replaced the horns wagon or the horns leadscrew.
- It is modified the position of the latch of the reagent drawer.

Prerequisite:

- Adjustment of the barcode reader inclination position.
- Adjustment of the reagents and samples drawers.

- PIP homed.

Adjustment tool: NA
Procedure:

1. In Technical Service software, go to *Procedures* tab → *PIP floor Modules* section → *horns* button. It will open a wizard adjustment.
2. Check that the barcode readers are ON and then press *Next>* button.
3. Adjustment of the horns X and Z axis: Place a diluent bottle in diluent position 4 of the reagent rack 1 of drawer 1, with barcode in the middle of reading window. Move by means of joystick buttons in the software or manually the PIP arm in X axis until horns are centered in reagents rack 1 of drawer 1. Move down the horns until laser beam of barcode reader illuminates whole barcode label. Then press *Next>* button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press *OK* button.

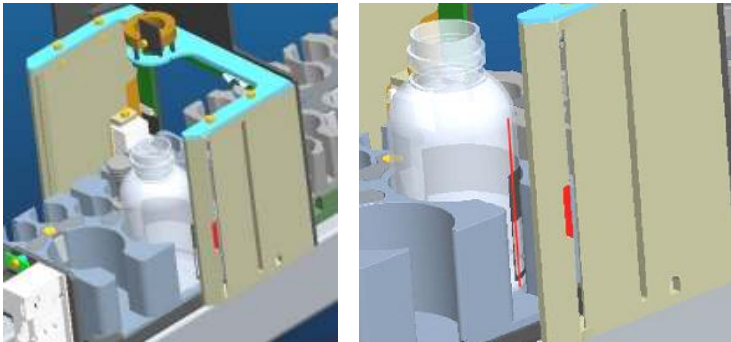


Figure 233. Position of the horns in first step of the adjustment

4. Adjustment in Y axis of left barcode reader: move PIP arm in Y axis until laser beam of left barcode reader illuminates the centre of the frontal handle of reagent drawer 1 - rack 2. Then press *Next>* button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press *OK* button.

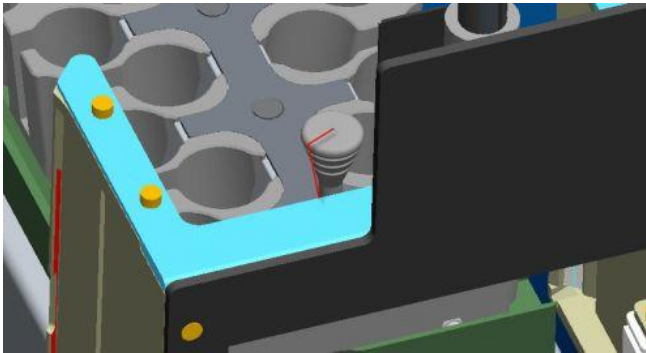


Figure 234. Adjustment in Y axis of left barcode reader

5. Adjustment in Y axis of the right barcode reader: move PIP arm in Y axis until laser beam of right barcode reader illuminates the centre of the frontal handle of reagent drawer 1 - rack 2. Then press *Next>* button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press *OK* button.

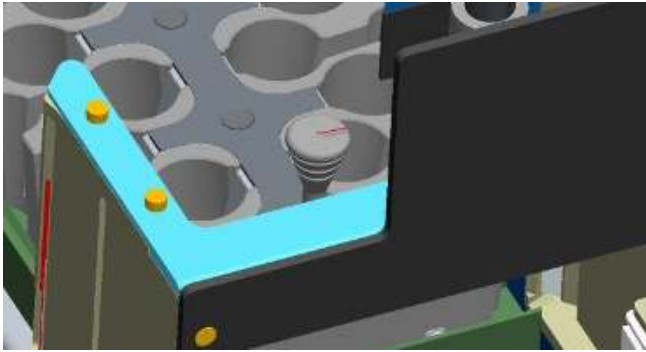


Figure 235. Adjustment in Y axis of right barcode reader

6. Adjustment in X axis of samples drawer 1: Move by means of the joystick buttons in the software or manually the PIP arm in X axis until horns are centered in samples drawer 1 – rack 1. Then press *Next>* button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press *OK* button.

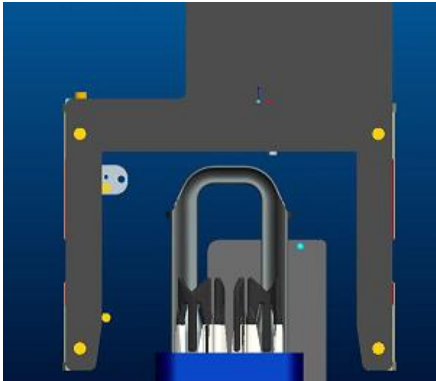


Figure 236. Adjustment in samples drawer of the barcode readers

7. Once the process is finished, a screen appears showing the parameters to be stored. Press *Finish* button and press *OK* on the confirmation window. When the adjustments are stored, press *OK* on the information window.
8. Press *OK* in the confirmation window that indicates that home movements will be performed.
9. Press *OK* to finish the adjustment procedure.

**NOTE:**
Adjustment stored in *E:\data\Erytra\conf\PIPCalibrations.txt*.

7.2.2.8 Presence detector distance of detection

Adjustment of distance detection of presence detector system. It must be performed if presence detector is replaced.

**NOTE:**
The PIP presence detector device comprises two LEDs:
- Green LED should always be ON during the scanning routines, indicating that presence detector device is activated.
- Orange LED will only light (ON mode) when presence is detected.

Prerequisite: Adjustment of presence detector inclination.
Adjustment tool: Probes adjustment tool (Ver. 4) (PN: 234213)
Procedure:

1. Deenergize PIP arm (In Technical Service software, go to *Pip Arm* tab → *Movement* tab → *Unlock/stop* button).
2. Place the adjustment tool over a reagent rack, as it is shown in following picture.

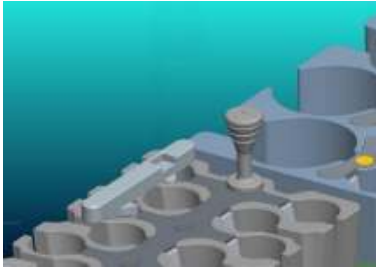


Figure 237. Adjustment tool position on reagent rack

3. Activate presence detector (in Technical Service software, *Pip Arm* tab → *Presence Detector* tab → *Presence Detector* subtab → *set on* button).
4. Turn slowly the presence detector screw counter-clockwise (remove the sealant if any) until a 'click' is heard.

The potentiometer is located in one side of the presence detector. In order to get a better access, it is recommended to move up the horns.

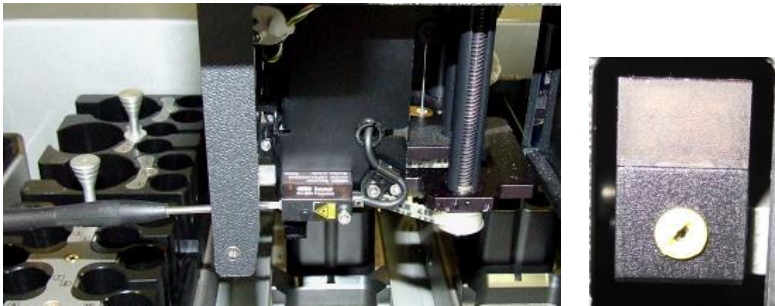


Figure 238. Access to presence detector potentiometer

5. Move manually the PIP arm until the laser beam of the presence detector illuminates the higher adjustment surface of the adjustment area of the tool (avoiding the hole).



Figure 239. Location of the adjustment area of detection of presence detector

6. Turn slowly the presence detector screw clockwise until the point where presence is detected (orange LED is ON and not blinking).
7. Move manually the PIP arm until the laser beam of the presence detector illuminates the black surface of the reagent rack.

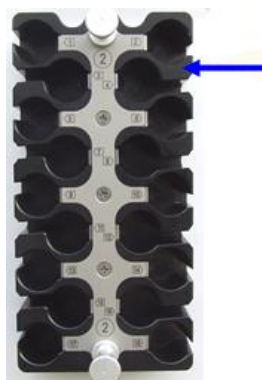


Figure 240. Location of the adjustment area of no detection of the presence detector

8. Check that presence is not detected. This is indicated by means of the LEDs of the presence detector:
 - Green LED: ON. The presence detector system is ON.
 - Orange LED: OFF. The system has not detected presence.
9. Move manually the PIP arm alternatively between the higher adjustment surface of the tool and the black surface of the reagent rack, checking that the orange LED is continuously ON and OFF, respectively. Should this not be the case, repeat the previous procedure.
10. Deactivate the laser of the presence detector (in Technical Service software, *Pip Arm* tab → *Presence Detector* tab → *Presence Detector* subtab → *set off* button).
11. Consider applying a sealant to the presence detector screw.

7.2.2.9 Barcode readers configuration

Barcode reader devices installed in the PIP module use a default configuration which is loaded every time that the instrument is restarted. The enabled barcode types by default are:

- Code 39 (check digit enabled)
- Interleaved 2 of 5 (check digit enabled)
- Code 128
- ISBT 128
- GS1 – 128 (formerly UCC/EAN-128)
- Codabar

In order to adapt the system to a specific lab configuration where different barcode types are used, the default configuration can be modified by means of Technical Service software. All the introduced changes will be applied for each barcode reader device after the default configuration has been loaded, modifying it accordingly.



NOTE:

The same barcode reader configuration is applied to read sample and reagent barcodes.

In order to program a customized barcode reader configuration, follow the below listed steps:

1. In Technical Service software, go to *Pip Arm* tab → *BCR* tab.
2. Select the device where the custom configuration is going to be applied (*Left BCR* or *Right BCR*).

3. Press *Get configuration* so that the current customized configuration (if any) is displayed in the *Settings* box.
4. Use the *Settings* box to write/delete the desired modifications to be applied with respect to the default configuration.



NOTE:
New settings should be introduced by writing the *Parameter number* corresponding to the desired barcode type/feature, followed by a code representing the action to be implemented (enable or disable). Parameter number for each barcode type/feature can be checked in the *Symbol/SE955/SE965 Integration Guide* barcode reader manual.

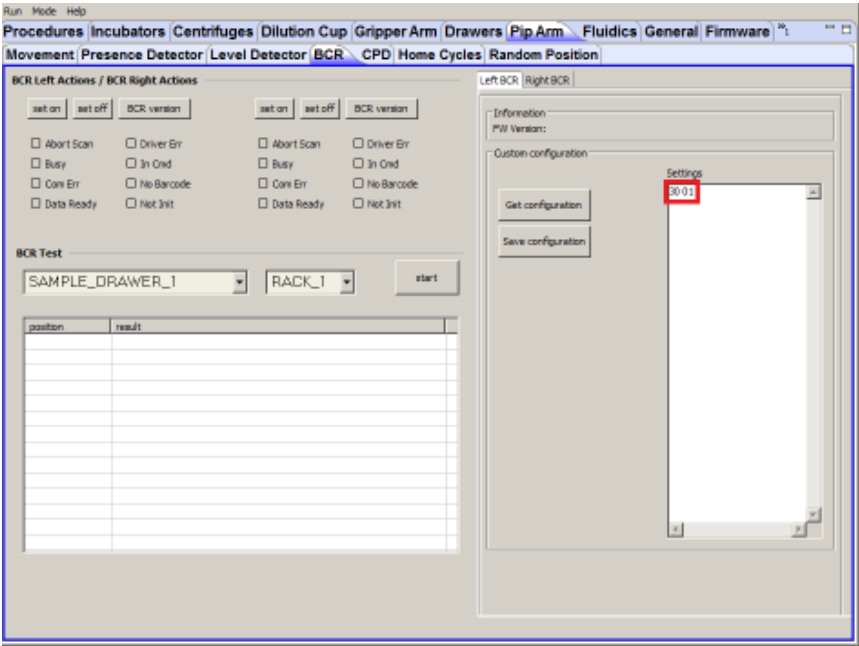


Figure 241. Barcode readers custom configuration

5. Click on *Save configuration* to save the introduced changes.



NOTE:
Changes applied with respect to the default configuration are saved in *E:\data\Erytra\conf* using independent files for each barcode reader device. These files are named *LeftBCCustom.cfg* and *RightBCCustom.cfg* and apply for the left and right barcode reader device (respectively). These files are included in the backup functionality available in the user software.

6. Press *Get configuration* and check that the new barcode reader configuration is properly read and displayed in the *Settings* box.
7. Repeat previous steps 3 to 6 for the other barcode reader device.



NOTE:
The same configuration should be applied for both left and right barcode reader devices.

8. Perform sample barcode readings (*Pip Arm* tab → *BCR* tab) to verify the applied configuration.

7.3 VERIFICATIONS

In this section are detailed all the verifications of the different parts of PIP arm.

7.3.1 Movement

1. In Technical Service software, go to *Pip Arm* tab → *Movement* subtab → *Position Actions* section.
2. Press *home* button and select all axes.
3. Press *start* button in *Pip Arm* tab → *Random Position* and perform random movements during 5 minutes, checking that no errors appear.
4. Press *home* button (*Pip Arm* tab → *Movement* subtab → *Position Actions* section) and select all axes.
5. Press *unlock/stop* button selecting all axes and move manually PIP head over a reagent drawer, assuring that horns are centered over the drawer.
6. Press *horns down* and *horns up* button 10 times checking that no errors appear.

7.3.2 Check home of the horns

1. In Technical Service software, go to *Pip Arm* tab → *Movement* subtab.
2. Press *Home sensor on* button to activate the home sensor.
3. Move up and down manually the horns by means of its belt checking that when the home flag cuts the home sensor, its *Home sensor* flag is activated.
4. Press *home* button selecting *horns* option and check that home movement and detection are performed correctly, without errors.
5. Repeat 10 times the step 4.

7.3.3 Check home of the X-Y-Z1-Z2 axis

1. In Technical Service software, go to *Pip Arm* tab → *Movement* subtab.
2. Press *Home sensor on* button to activate the home sensors.
3. Move manually X and Y PIP axes checking that when the home flag cuts the home sensor, its related *Home sensor* flag is activated.
4. In Technical Service software, go to *Pip Arm* tab → *Home Cycles* subtab.
5. Press *start* button and perform homes during 5 minutes. Check that home movement and detection for all axes are performed correctly, without errors.

7.3.4 Probes parallelism

1. Place Probes adjustment tool (Ver. 4) (PN: 234213) in a sample drawer, rack 2, between positions 9 to 11 checking that the hole of the tool is over position 11.



Figure 242. First step of the probes parallelism verification

2. With Technical Service Software (TSS), move Pip Arm over the sample drawer where the tool is placed, rack 2, position 11, probe 1 (Pip Arm tab → Movement subtab → Move Actions section).
3. Unlock the probe 1 motor (unlock/stop button → z1 axis in TSS) and lower manually, by means of its belt, probe 1 checking that it fits with the hole.
4. Perform a home of probe 1 by means of TSS.
5. Place the Probes adjustment tool (Ver. 4) (PN: 234213) in the same sample drawer, rack 2, between positions 10 to 12 checking that the hole of the tool is over position 10.

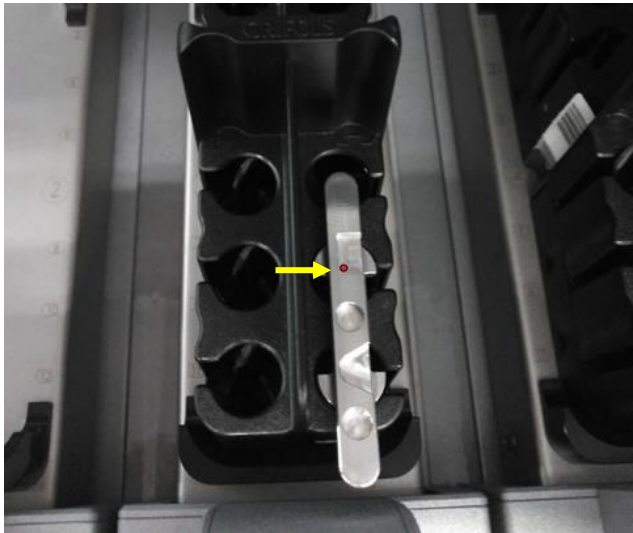


Figure 243. Second step of the probes parallelism verification

6. With Technical Service Software (TSS), move the Pip Arm over the sample drawer where the tool is placed, rack 2, position 9, probe 1 (Pip Arm tab → Movement subtab → Move Actions section).
7. Unlock the probe 2 motor (unlock/stop button → Z2 axis in TSS) and lower manually, by means of its belt, probe 2 checking that it fits with the hole.
8. Perform a home of probe 2 by means of TSS.

7.3.5 Probes offset

1. Perform a home of PIP axes: X, Y, Z1, Z2 and horns.
2. Unlock PIP axes (*Pip Arm* tab → *Movement* tab → *unlock/stop* button and select all axes).
3. Move manually PIP robotic head to the front of the instrument.
4. Press *probes offset* button in TSS → *Procedures* tab → *Checks* section.
5. A precondition message informing that probes will move down 31300 encoder counts appears on screen. If *OK* button is pressed, probes move down to that position.
6. After the automatic movement, following message on the screen is shown:

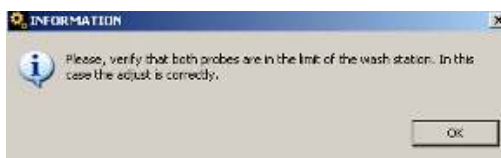


Figure 244. Information window in probes offset verification

7. Check visually that both probes are levelled with the probes washer exit holes and then press *OK* button. The probes are homed automatically.

7.3.6 Reagents drawer

1. Perform a home of PIP axes: X, Y, Z1, Z2 and horns.
2. Place vials in position 2 of rack 1 and position 17 of rack 2 of the reagent drawer to verify.
3. In Technical Service software, go to *Pip Arm* tab → *Movement* tab → and select the reagent drawer to verify, position rack1, pos_2, probe_1 and press *move to* button.
4. Once the robot is placed over the indicated position, write *150000* encoder counts in *Move servo to encoder count* box in Z1 axis Motor and press *Move* button.
5. Check that Z1 probe is centered in Y axis with the vial place.
6. Perform a home of probe Z1 (*Pip Arm* tab → *Movement* tab → *home* button and select Z1 axis).
7. Repeat steps from 1 to 5 for rack2, pos_17.
8. Perform a home of PIP axes: X, Y, Z1, Z2 and horns.

7.3.7 Samples drawer

1. Perform a home of all axes of PIP arm: X, Y, Z1, Z2 and horns.
2. Place samples tubes in position 1 of rack 1 and position 11 of rack 2 of the sample drawer to verify.
3. In Technical Service software, go to *Pip Arm* tab → *Movement* tab → and select the sample drawer to verify, position rack1, pos_1, probe_1 and press *move to* button.
4. Once the robot is placed over the indicated position, deenergize Z1 axis (*unlock/stop* button and select Z1 axis) and move down manually the probe by means of its belt until enters into the tube.
5. Check that Z1 probe is centered in Y axis with the sample tube placed and 0.5mm from the center of the tube to the right in X axis.
6. Perform a home of probe Z1 (*Pip Arm* tab → *Movement* tab → *home* button and select Z1 axis).
7. Repeat steps from 1 to 5 for the rack2, pos_11.
8. Perform a home of all axes of PIP arm: X, Y, Z1, Z2 and horns.

7.3.8 Incubators

1. Perform a home of the incubator.

	CAUTION! Check that CAT arm is not interfering with incubator movement.
---	--

2. Place 2 cards in slots 1 and 3 of the incubator and move the incubator to the PIP floor.
3. In Technical Service software, go to *Pip Arm* tab → *Movement* tab and select the incubator to verify, position *SLOT_1*, *WELL_1*, *PROBE_1* and press *move to* button.
4. Press *down probe 1* and *down probe 2* in order to move both probes until dispensation position inside the cards. Both probes should perforate the cards film.
5. Perform a home of both probes (*Pip Arm* tab → *home* button and select Z1 and Z2 axis).
6. In Technical Service software, go to *Pip Arm* tab → *Movement* tab and select the incubator to verify, position *SLOT_1*, *WELL_8*, *PROBE_1* and press *move to* button.
7. Press *down probe 1* and *down probe 2* in order to move both probes until dispensation position inside the cards. Both probes should perforate the cards film.
8. Perform a home of both probes (*Pip Arm* tab → *home* button and select Z1 and Z2 axes).
9. Check that holes in the film card are centered in X axis and shifted to the rear part of the instrument in Y axis 1 mm as maximum.
10. Perform a home of the incubator. Be careful with position of CAT arm in order to do not crash with the incubator.
11. Remove the cards from the incubator.

7.3.9 Dilution cup

1. In Technical Service software, perform a home of the PIP arm (*Pip Arm* tab → *Movement* tab → *home* button and then select all axes) .
2. Move PIP over dilution cup: *Pip Arm* tab → *Movement* tab and select *DILUTION_CUPS*, *PROBE_1*, *ASPIRE* and press *move to* button.
3. Press *down probe 1* button and *down probe 2* button.
4. In Technical Service software, go to *Dilution Cup* tab and press *Spin* button (check that *PROFILE_0* is selected). Check that walls of the wells are not touching the probes.
5. Once stirring process finishes, go to *Pip Arm* tab and perform a home of the probes.
6. Then in *Movement* subtab, select *DILUTION_CUPS*, *probe_1*, *DISPENSE* and press *move to* button.
7. Press *down probe 1* button and *down probe 2* button.
8. Check that position of both probes is close to the rear part of the wells of the dilution cup in Y axis and centered in X axis. Check that probes do not touch the walls of the wells.



Figure 245. Probes position on dilution cup

- In Technical Service software, go to *Dilution Cup* tab and press *Spin* button (check that *PROFILE_0* is selected). Check that probes do not touch the walls of the wells.
- Perform a home of both probes (*Pip Arm* tab → *Movement* tab → *home* button and select Z1 and Z2 axis).

7.3.10 Barcode reader

Reagents drawer:

- Place some vials in both racks of a reagents drawer, with barcodes faced to reading windows.
- In Technical Service software, go to *Pip Arm* tab → *BCR* tab and select the reagent drawer to verify and *rack1*. Then press *start* button.
- Once reading process is finished, check that all the barcodes are correctly read.
- In Technical Service software, go to *Pip Arm* tab → *BCR* tab and select the reagents drawer to verify and *rack2*. Then press *start* button.
- Once the reading process is finished, check that all the barcodes are correctly read.

Samples drawer:

- Place some allowed tubes in both samples racks of a samples drawer, with barcodes faced outside to reading windows.
- In Technical Service software, go to *Pip Arm* tab → *BCR* tab and select the samples drawer to verify and *rack1*. Then press *start* button.
- Once reading process is finished, check that all barcodes are correctly read.
- In Technical Service software, go to *Pip Arm* tab → *BCR* tab and select the samples drawer to verify and *rack2*. Then press *start* button.
- Once the reading process is finished, check that all barcodes are correctly read.

7.3.11 Presence detector alignment

- Perform a home of PIP axes.
- Place Probes adjustment tool (Ver. 4) (PN: 234213) in position 9 and 11 of the rack 2 of samples drawer 1.
- In Technical Service software, go to *Procedures* tab → *Checks* section and press *pip laser alignment* button. Press *Ok* button.
- PIP arm is moved over samples drawer 1 – rack 2 and presence detector laser is switched on. Check that laser fits inside the hole of the adjustment tool.



Figure 246. Presence detector alignment verification

Reagents drawer:

1. Check that no vials/bottles are present in the racks of a reagent drawer.
2. In Technical Service software, go to *Pip Arm* tab → *Presence Detector* tab → *Reagent drawers* tab and select the reagents drawer to verify and *rack1*. Then press *start* button.
3. Once reading process is finished, select *rack2* and press *start* button.
4. Once reading process is finished, check that no vials are detected.
5. Place some vials in both racks of a reagent drawer, with distilled water.
6. In Technical Service software, go to *Pip Arm* tab → *Presence Detector* tab → *Reagent Drawers* tab and select the reagents drawer to verify and *rack1*. Then press *start* button.
7. Once reading process is finished, select *rack2* and press *start* button.
8. Once reading process is finished, check that vials are correctly detected.



NOTE:

Check that surface of reagent rack is dry in order to avoid false detections by the presence of liquid.

Samples drawer:

1. Check that no tubes are present in the racks of a sample drawer.
2. In Technical Service software, go to *Pip Arm* tab → *Presence Detector* tab → *Sample Drawers* tab and select the samples drawer to verify and *rack1*. Then press *start* button.
3. Once reading process is finished, select *rack2* and press *start* button.
4. Once reading process is finished, check that no tubes are detected.
5. Place some allowed tubes in both racks of samples drawer to verify, with distilled water.
6. In Technical Service software, go to *Pip Arm* tab → *Presence Detector* tab → *Sample Drawers* tab and select the samples drawer to verify and *rack1*. Then press *start* button.
7. Once reading process is finished, select *rack2* and press *start* button.
8. Once reading process is finished, check that tubes are correctly detected (0, -2mm):
 - Diameter value shown on screen is 1 mm smaller than the detected one. This value cannot be equal or bigger than the real one.
 - If the diameter value shown on screen is smaller than the real one, it can be accepted until up to 2mm smaller than the real one.

7.3.12 Level detection system

1. Empty the fluidic circuit from the dilutor to the probe (syringe, valve, tubing and probe). Use the *fluidic empty* button in *Fluidics* tab in Technical Service software.
2. Place both probes in home position (*Pip Arm* tab → *Movement* subtab → *home* button and select Z1 and Z2) and go to *Pip Arm* tab → *Level Detector* subtab.
3. Select *PROBE_1* in the *probe* selection box and press *on* button.
4. Select *PROBE_2* in the *probe* selection box and press *on* button.
5. Check that level detection system values are the following ones:

	Probe 1	Probe 2
Probes not installed	6300 ± 300	8400 ± 500
Probes installed	6700 ± 300	9000 ± 500

Table 55. Level detection system values in dry situation

6. If the probes are not installed, move up-down 3 times the probes holder assembly to simulate impacts, then check that level detection values match with the ones of previous table (*probes not installed* case).
7. Prime the fluidic circuit (in TSS → *Fluidics* tab → *initializing* fluidic option and then *fluidic prime* option)

- Prime probe 1 and probe 2 fluidic circuits. (*Fluidics* tab → *prime probe 1* button and *prime probe 2* button).
- Check that level detection system values are the following ones:

Probe 1	Probe 2
7600 ± 300	10400 ± 500

Table 56. Level detection system values in prime situation

- Perform a home of reagent racks 2 of both reagent drawers.
- Place a vial with saline solution in position 1 of rack 2 of both reagents drawers.
- In Technical Service software, go to *Pip Arm* tab → *Level Detector* subtab. Introduce 5 in iterations field, select *REAGENT_DRAWER_1*, *PROBE_1*, rack 2, first tube 1, last tube 1. Press *start* button and check that all the detections are ok.
- Repeat previous procedure for reagent drawer 2 and probe 2 (select *REAGENT_DRAWER_2*, *PROBE_2*, rack 2, first tube 1, last tube 1).
- Place a sample tube with saline solution in position 1 of rack 2 in the 4 samples drawers.
- In Technical Service software, go to *Pip Arm* tab → *Level Detector* subtab. Introduce 5 in iterations field, select *SAMPLE_DRAWER_1*, *PROBE_1*, rack 2, first tube 1, last tube 1 and tube type NORMAL (in case of using a pediatric tube, select *PEDIATRIC* and then define the difference between the maximum depth of the pediatric tube and a normal one). Press *start* button and check that all the detections are ok.
- Repeat the previous procedure for sample drawer 2 and probe 2 (select *SAMPLE_DRAWER_2*, *PROBE_2*, rack 2, first tube 1, last tube 1 and tube type).
- Repeat the previous procedure for sample drawer 3 and probe 1 (select *SAMPLE_DRAWER_3*, *PROBE_1*, rack 2, first tube 1, last tube 1 and tube type).
- Repeat the previous procedure for sample drawer 4 and probe 2 (select *SAMPLE_DRAWER_4*, *PROBE_2*, rack 2, first tube 1, last tube 1 and tube type).

7.3.13 Impact detector system

In case impact detector system is installed on the instrument (installed from factory in instruments with S/N ≥ 059-0000611), verification procedure is as follows:

- Remove probe 1 and check value of level detection system in *Pip Arm* tab → *Level Detector* subtab. Move up-down 3 times the probes holder assembly, checking that the probes holder rests over the pad of the PCA and that the final value of level detection system is the same as the initial one ± 30 counts. Install again probe 1.
- Remove probe 2 and check value of level detection system in *Pip Arm* tab → *Level Detector* subtab. Move up-down 3 times the probes holder assembly, checking that the probes holder rests over the pad of the PCA and that the final value of level detection system is the same as the initial one ± 30 counts. Install again probe 2.
- Check level detection system as stated on section 7.3.12.
- Place a diluent with cap in reagent drawer 1 – rack 1 – pos 4.
- Perform a home of the PIP arm (in TSS, *Pip Arm* tab → *Movement* subtab → *home* button and select all axes).
- Go to *Pip Arm* tab → *Level Detector* subtab. Write 3 in iterations field, select *REAGENT_DRAWER_1*, *PROBE_1*, rack 1, first tube 4 and last tube 4. Press *start* button and check that impact is detected 3 times (in the graph of the window or in *log execution* section).
- Place a diluent with cap in reagent drawer 2 – rack 1 – pos 4.
- Perform a home of the PIP arm (in TSS, *Pip Arm* tab → *Movement* subtab → *home* button and select all axes).
- Go to *Pip Arm* tab → *Level Detector* subtab. Write 3 in iterations field, select *REAGENT_DRAWER_2*, *PROBE_2*, rack 1, first tube 4 and last tube 4. Press *start* button and check that impact is detected 3 times (in the graph of the window or in *log execution* section).

7.4 REPLACEMENT PROCEDURES

7.4.1 Probes replacement

Before replacing a probe, follow next steps:

1. Switch off the instrument and disconnect it from the mains.
2. Open one of the samples drawer:
 - With Technical Service software, go to *Drawers* tab → *Sample Drawers* subtab and press *Open* button in the selected drawer.
 - Or manually, with the emergency open located in the lock at the rear part of the drawer.



Figure 247. Manual opening of the drawer

3. Remove the racks of the opened drawer and then move manually the PIP arm over the drawer at the frontal part of the instrument to get a better access.
4. Move down the horns:
 - With Technical Service software, go to *Pip Arm* tab → *Movement* tab and press *horns down* button
 - Or manually, by means of rotating the horns belt located under the PIP arm.

Z1 probe

5. Move down Z2 probe to get a better access of the Z1 probe:
 - With Technical Service software, move down Z2 probe to 250000 encoder counts. Write this counts number in *Move servo to encoder count* box from *Z2 axis (Motor)* and then press *Move* button.
 - Manually, rotating the Z2 probe belt located under the PIP arm.
6. Move down Z1 probe to 220000 encoder counts
 - In Technical Service software, write this encoder counts number in *Move servo to encoder count* box from *Z1 axis (Motor)* and then press *Move* button.
 - Manually, rotating the Z1 probe belt located under the PIP arm.
7. Unscrew the fitting - it is possible to use P/N 233898 OMNIFIT TOOL (for tube connection Ver 2) for unscrewing it - and remove the probe from its position.



Figure 248. Probe fitting

8. Insert new probe first in the probe holder and then in its hole of the probes washer. The upper part of the probe has a flat area that must be faced properly inside the probe holder. Once the probe is correctly placed, tighten probe fitting to the maximum. It is possible to use P/N 233898 OMNIFIT TOOL (for tube connection Ver 2) for tightening it.
9. Perform a home of all the axes of PIP arm.

Z2 probe

10. Move down Z2 probe to 220000 encoder counts
 - In Technical Service software, write this encoder counts number in Move servo to encoder count box from Z2 axis (Motor) and then press *Move* button.
 - Manually, rotating the Z2 probe belt located under the PIP arm.
11. Unscrew the fitting and remove the probe from its position.
12. Insert new probe first in the probe holder and then in its hole of the probes washer. The upper part of the probe has a flat area that must be faced properly inside the probe holder. Once the probe is correctly placed, tighten probe fitting to the maximum. It is possible to use P/N 233898 OMNIFIT TOOL (for tube connection Ver 2) for tightening it.
13. Perform a home of all the axes of PIP arm.

7.4.2 Probes washer replacement

1. Remove the 2 screws that attach the probes washer to the PIP robotic head. To get a better access is recommended to move up the horns.



Figure 249. Probes washer screws

2. Disconnect all tubes from the probes washer.
3. Replace the probes washer. Check status of the end of the tubes and connect the tubes to the fittings of the washer. Make sure the tubes routing of C22 and C24 is as below image.



Figure 250. Probes washer tubes C22 and C24 routing

4. Slightly tighten the 2 screws of the probes washer.
5. Move down manually both probes (by means of their belts) and check that both probes are aligned.

6. Tighten the 2 screws of the probes washer to the maximum and then ¼ of turn more.

**WARNING!**

Avoid the possibility of pricking yourself with the probes when tightening the screws of the probes washer.

7. Move up manually both probes (by means of their belts).

7.4.3 Probes washer fitting replacement

Probes washer fittings are sealed with acetoxysilicone, recommended for use on plastic pipe threaded fittings (such as Loctite 5331). To replace a fitting, follow the next steps:

1. Check that the fluidic system is empty.
2. Disconnect the tube from the fitting to be replaced.
3. Unscrew the fitting from the probes washer and clean its allocation.

**CAUTION!**

Check that no seal product gets into the probes washer during cleaning process.

4. Place the new fitting in its allocation only tightening it a little bit, so most of the thread is out of its allocation.
5. Put sealant product around the fitting thread. Sealant product should be spread only on the fitting thread as uniformly as possible from the middle of fitting thread toward the union point with nut. To ensure a better and ease spreading use a pointed end object.

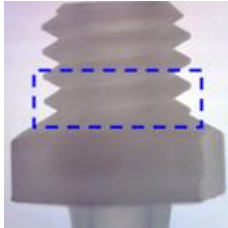


Figure 251. Sealant product application area

6. Tighten the fitting manually to the maximum and then ¼ turn more in its allocation.

**CAUTION!**

Check that the seal product does not block the connection between the fitting and the probes washer.

7.4.4 Level detector PCA replacement

1. Switch off the instrument and disconnect it from the mains.
2. Remove the 3 screws of the cover of the wagon probe and carefully apart it from its position as much as possible.

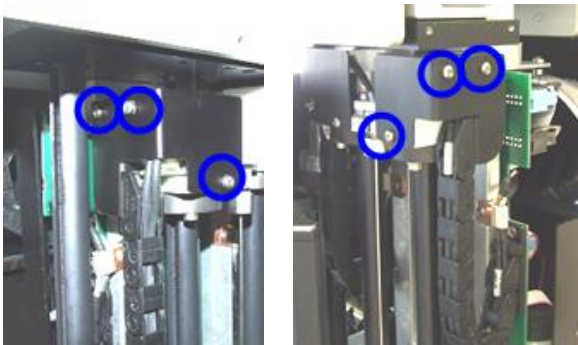


Figure 252. Wagon probe cover for chain CH4 (left) and CH3 (right)

3. In order to remove the teflon tube from its position unscrew the fitting from the PCA.



Figure 253. Teflon tube of the probe

4. Disconnect carefully the cable (level detection signals) of the chain to be replaced connected to the level detection PCA (PCA PIP-P1OSC59 or PCA PIP-P2OSC59).



Figure 254. Location of the connector in the level detection PCA

5. Remove the PCA from its position:
- Unscrew the nut of the probe holder and push the probe holder to the top. Then remove the 2 screws of the PCA.

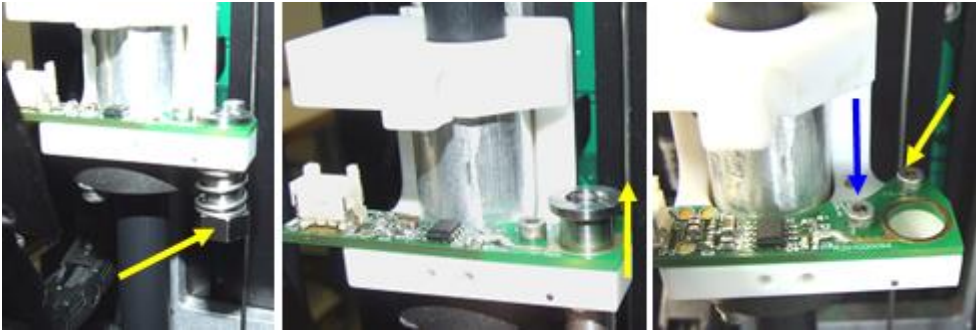


Figure 255. Level detector PCA replacement steps

- Place the new PCA over the wagon, checking that the hole of the PCA is aligned with the hole of the wagon.
- Insert the probes holder through the PCA hole and then place the spring around the probes holder. Tighten manually the nut of the probes holder assembly.
- Move up-down 3 times the probes holder assembly, checking that the probes holder rests over the pad of the PCA.
- Connect level detection cable, install the probe and screw the teflon tube fitting. Finally place the wagon probe cover.

6. Verify level detection system (refer to section 7.3.12)

7.4.5 Leadscrews replacement

Set up:

To replace the leadscrews it is necessary to remove the PIP robotic head from the instrument.

1. Switch off the instrument, disconnect it from the mains, remove upper rear cover of the instrument and open upper area door.
2. Remove both probes.
3. Free from its clamps the fluidic tubing from the probes washer to the Y chain.

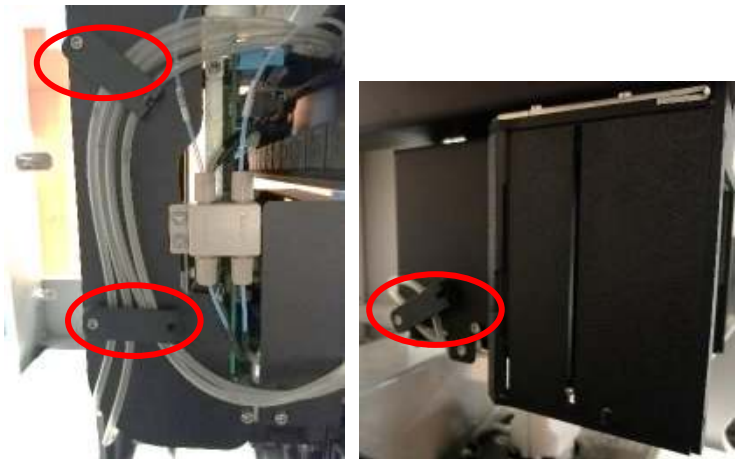


Figure 256. Location of the clamps in the PIP robotic head



NOTE:
Make sure that the position of the white tubing clamp is not modified while removing or inserting the fluidic tubes (see image above).

4. Disconnect, from the interconnection manifold located at the rear part of the PIP robotic head, the tubes of the probes that come from Y-chain.



Figure 257. Interconnection manifold in the robotic head

5. Remove the 2 screws of the probes washer.



Figure 258. Probes washer in the robotic head

6. Remove the 2 screws from the Y arm lateral cover.



Figure 259. Lateral cover screws of the Y arm

7. From the rear part of the instrument remove the 2 screws that attach Y-chain to PIP robotic head.



Figure 260. Location of the screws that attach Y chain to Z tower/robotic head

8. Disconnect Y chain ribbon cables from PCA PIP-TOW59.

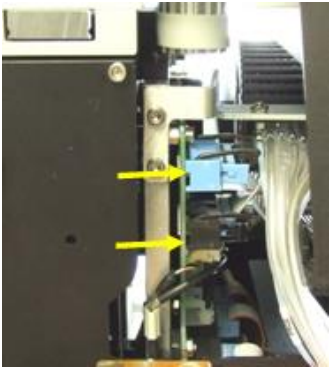


Figure 261. Location of the ribbon cables connectors at PCA PIP-TOW59

9. From the front of the instrument remove Y-arm linear guide front end stopper (remove 2 screws).

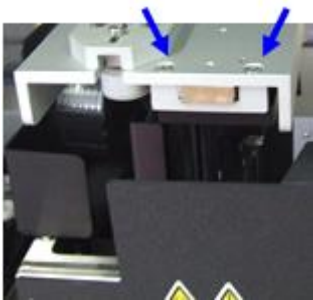


Figure 262. Y arm guide front end stopper screws

10. Take out Y-belt from its position. First remove the 2 fixing screws of the tensor assembly and then free the belt from the rear gear.



Figure 263. Rear gear Y belt (left) and fixing screws of the tensor assembly (right)

CAUTION!

Be careful to do not damage Y axis home barrier when placing the belt.

11. Remove carefully PIP robotic head by the front of the instrument, opening a drawer if necessary.



Figure 264. Removing the PIP robot head

Horns leadscrew

In order to remove the horns leadscrew from PIP robotic head follow the next steps:

- 1. Remove the belt and loose the set screw from the horns leadscrew pulley and remove it. Horns leadscrew pulley and belt are the ones close to the presence detector.

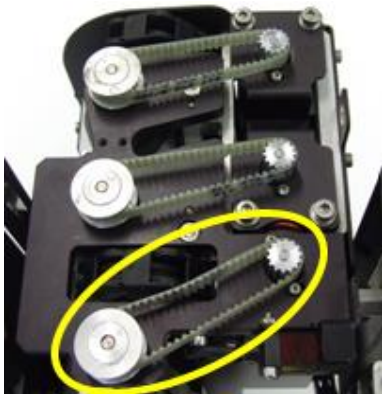


Figure 265. Belt and motor pinion of the horns

- 2. Remove Y axis frontal robotic head stopper in order to have access to the leadscrew holder. Screws are located under the robotic head stopper.

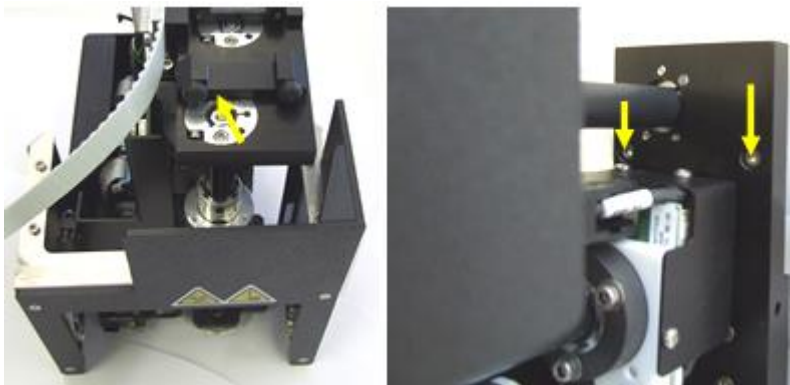


Figure 266. Frontal robotic head stopper location

- 3. Remove the screws that attach the leadscrew holder to the robotic head and the 3 screws that attach the leadscrew to the home flag ring and the horns structure.
- 4. Remove the leadscrew upwards until the leadscrew holder is free from its position. If it is hard to remove, hit smoothly from the bottom upwards with a rubber hammer.
- 5. Remove the leadscrew holder from the leadscrew by means of loosening the set screw.

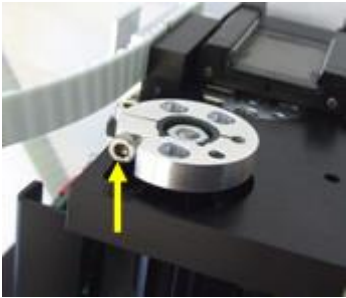


Figure 267. Set screw of the leadscrew holder

6. Finally, take out the leadscrew from the horns removing the home flag ring.

Install new horns leadscrew into the PIP robotic head following previous steps 1-6 in reverse mode. Some considerations must be taken into account:

1. Place the leadscrew between the home flag ring and barcode readers structure plate. Use the Horns adjustment tool (BCR) (PN 233819) in order to align it.



Figure 268. Position of the home flag ring and barcode reader structure plate in the leadscrew

2. When the leadscrew holder is placed in the leadscrew, it must be placed at the same level of the leadscrew.

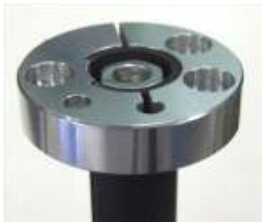


Figure 269. Leadscrew holder

3. If it is hard to place the leadscrew holder in its position on the PIP robotic head, hit smoothly from the top downwards with a rubber hammer facing correctly the other side of the leadscrew.

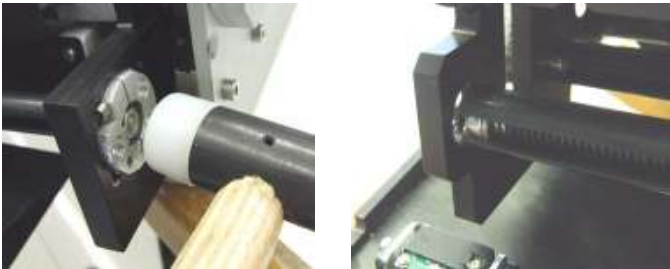


Figure 270. Leadscrew install

4. Place the horns leadscrew pulley at the same level than the leadscrew axis. Then tighten the set screw of the pulley in the flat side of the leadscrew axis.



Figure 271. Pulley of the motor of the horns

5. Place the belt and adjust its tension as indicated in section 7.2.1.6.
6. Check the probes offset.
7. Check probes parallelism.
8. Perform probes adjustment in all positions: reagents drawers, samples drawers, incubators and dilution cup.
9. Perform adjustments verifications related to the PIP module, refer to section 7.3 Verifications.

Probes leadscrew

1. Remove the belt of the probe and loose the setscrew from the probe leadscrew pulley motor and remove it.

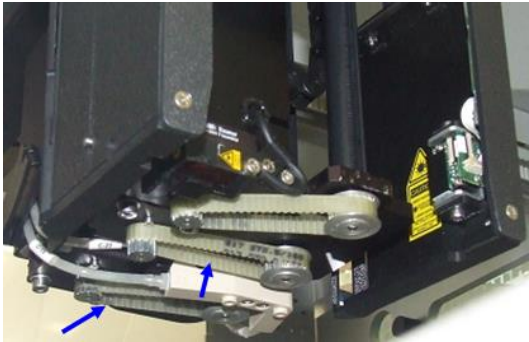


Figure 272. Probes belts

2. Move down the two probes wagons manually and remove Y axis robotic head rear stopper in order to have full access to the leadscrew holder in case replacing probe 1 leadscrew.



WARNING!

Be careful to do not prick yourself with the probes, if assembled.

3. Remove the 4 screws that attach Y axis linear guide wagon to the PIP robotic head.



Figure 273. Location of the screws that attach Y linear guide wagon to the PIP robotic head

4. Loose (if present) the set screw that fixes the position of the ball bearing wagon. To get access to the set screw, remove the plate that holds the Y-axis belt by means of its 2 screws. Then loose the set screw and remove the ball bearing wagon. Before removing the ball bearing wagon, mark with a pen its orientation in order to have a reference when it will be installed again.

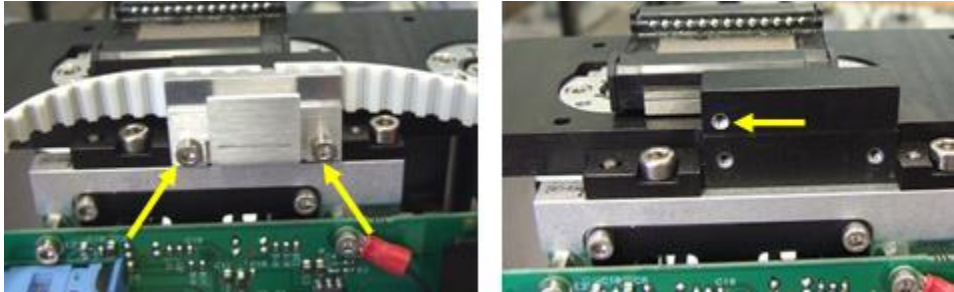


Figure 274. Location of the set screw to fix the linear guide wagon

5. Remove the 3 screws (depending on the probe to be replaced) that attach the leadscrew holder to the robotic head and the 3 screws that attach the leadscrew to the probe wagon.

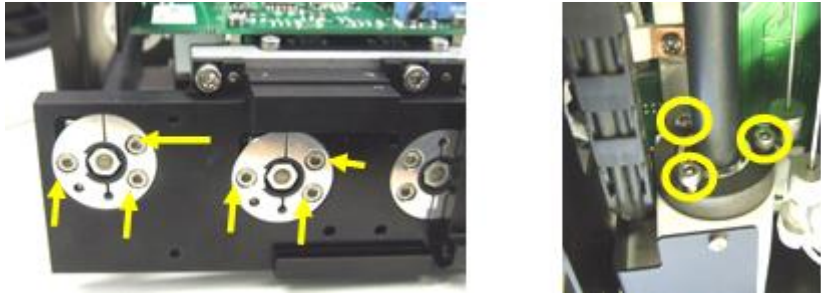


Figure 275. Robotic head and leadscrew holders

6. Remove the leadscrew upwards until the leadscrew holder is free from its position. If it is hard to remove, hit smoothly from the bottom upwards with a rubber hammer.
7. Remove the leadscrew holder from the leadscrew loosening its set screw.
8. Finally, take out the leadscrew downwards

Install the probe leadscrew into the PIP robotic head previous steps in reverse mode. Some considerations must be taken into account:

- When the leadscrew holder is placed in the leadscrew, it must be placed at the same level of the leadscrew.
- If it is hard to place the leadscrew holder in its position on the PIP robotic head, hit smoothly from the top downwards with a rubber hammer facing correctly the other side of the leadscrew.
- When placing the linear guide wagon, and before placing the screws that attach it to the PIP robotic head, fix the position of the linear guide wagon with the set screw included in the leadscrew spare part. To get access to the set screw allocation, remove the plate that holds the Y-axis belt by means of its 2 screws and then tighten the set screw against the linear guide wagon. Place the screws that attach Y linear guide wagon to the PIP robotic head, tighten them and finally remove the set screw. Install the Y-axis belt.

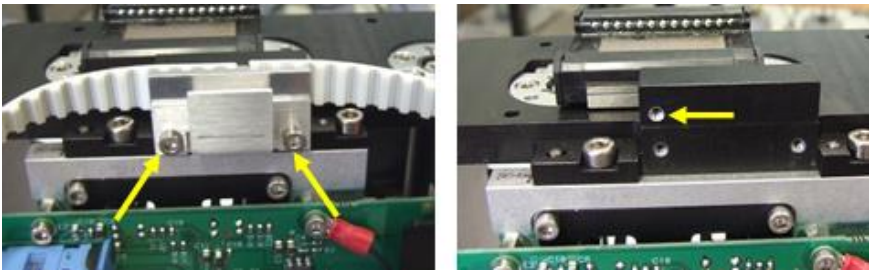


Figure 276. Location of the set screw to fix the linear guide wagon

- Place the motor pulley at the same level than the leadscrew axis. Then tighten the set screw of the motor pulley to block it.
- Place the belt and adjust its tension as indicated in section 7.2.1.6 Z-Axis belt tension adjustment.
- Check the probes offset.
- Check probes parallelism.
- Perform probes adjustment in all positions: reagents drawers, samples drawers, incubators and dilution cup.
- Perform adjustments verifications related to the PIP module, refer to section 7.3 Verifications.

7.4.6 PIP X axis chain (CH1) replacement

1. Switch off the instrument and disconnect it from the mains.
2. Open upper door.
3. Disconnect tubing from the connection block and open the clamp.

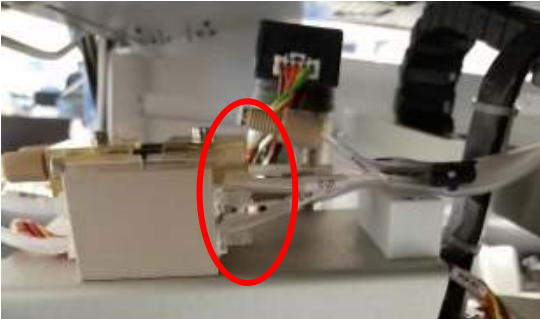


Figure 277. Tubing

4. Disengage the last slot of the chain.



Figure 278. Last chain slot

5. Disconnect the 3 flat cables from PCA PIP-CONY59 and remove the ties.



Figure 279. Flat cables

6. Remove instrument left lateral methacrylate cover.



Figure 280. Lateral removal

7. Disconnect 4 tubes from connection block.



Figure 281. Tubing

8. Disconnect the 3 flat cables from PCA GEN-CONX59.



Figure 282. Flat cables

9. Remove 2 screws that attach the chain.



Figure 283. Chain screws

10. Disconnect Teflon tubes from valves.



Figure 284. Teflon tubes

11. Remove the screws of the blocks over the dilutors (side of the teflon tube) to be able to pull apart the teflon tubes.

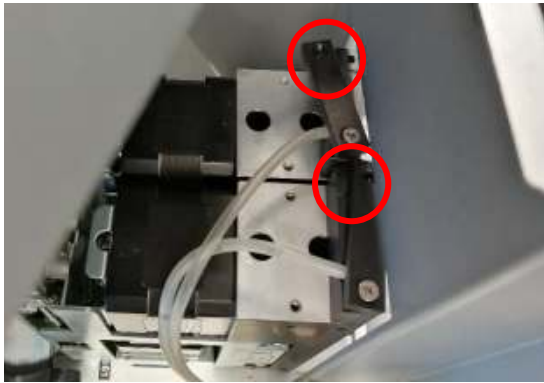


Figure 285. Teflon tubes under the blocks

12. Remove the PIP X chain.
13. Place new PIP X chain and ssemble back all the parts.

CAUTION!
Do not swap ribbon cables in PCA GEN-CONX59 or PCA PIP-CONY59 when connecting them (it could damage different parts of the instrument), specially cables labelled as XB-PIP_XY and XB-PIP_Z that have both a blue connector. Cable labelled as XB-PIP_TOW has a black connector. Cables should match the connector colour when connected to the PCA.

- 14. Perform a Prime to check the fluidic system.
- 15. Perform Leak-tightness test.

7.4.7 PIP Y axis chain (CH2) replacement

- 1. Switch off the instrument and disconnect it from the mains.
- 2. Free from its clamps the fluidic tubing from the probes washer to the Y axis chain.

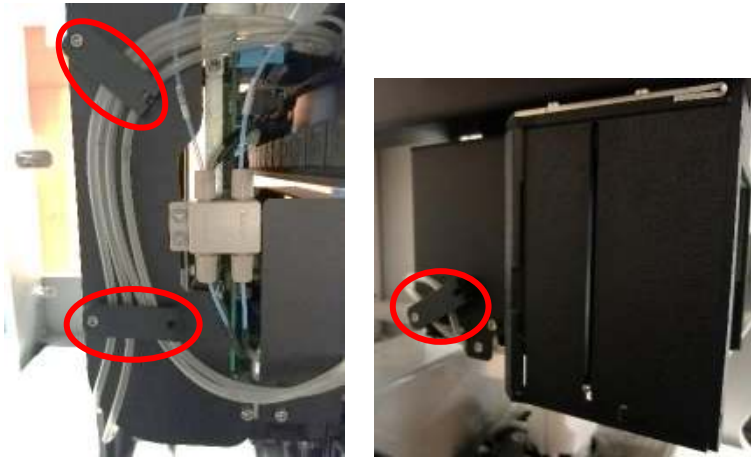


Figure 286. Location of the clamps in the PIP robotic head

NOTE:
Make sure that the position of the tubing clamp is not modified while removing or inserting the fluidic tubes (see image above).

- 3. Remove the 2 screws of the probes washer, pull it apart it from its position and disconnect all the tubes from their fittings.
- 4. Disconnect, from the interconnection manifold located in the robotic head, the tubes of the probes that come from the Y axis chain.



Figure 287. Interconnection manifold in the robotic head

5. Remove the 2 screws from the lateral cover of Y arm.



Figure 288. Lateral cover screws of Y arm

6. Remove the screws that attach Y chain to Z tower/robotic head.

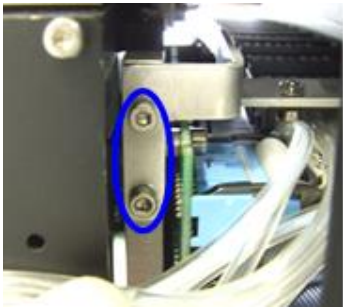


Figure 289. Location of the screws that attach Y chain to Z tower/robotic head

7. Disconnect Y chain ribbon cables from PCA PIP-TOW59.

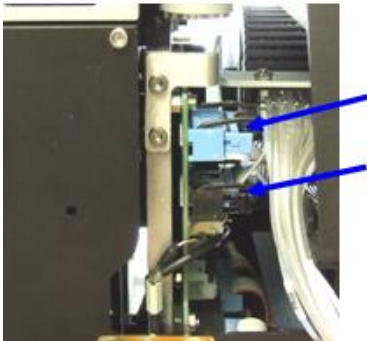


Figure 290. Location of the ribbon cables connectors at PCA PIP-TOW59

8. Disconnect the tubes that come from the PIP Y axis chain to the interconnection manifolds located at the rear part of the PIP arm.

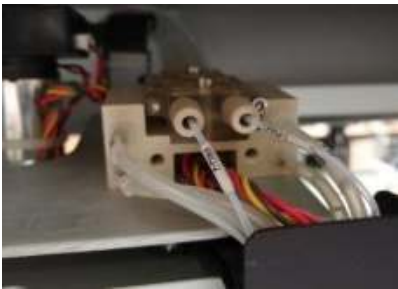


Figure 291. Interconnection manifolds at the rear part of PIP arm

9. Open the tie-wraps attached to the lateral cover of the Y arm in order to release the ribbon cables and tubing from the Y axis chain.

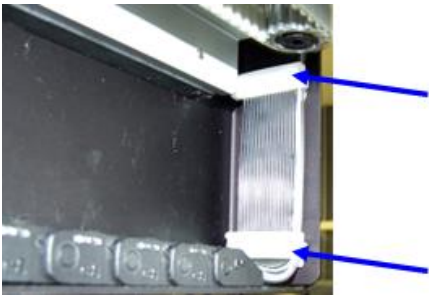


Figure 292. Tie-wraps in the lateral cover of the Y arm

10. Remove the screws that attach the PIP Y axis chain to the lateral cover of the PIP arm.



Figure 293. Location of the screws that attach the Y chain to the lateral cover of the PIP arm

11. Remove the screws of the rear cover of PIP arm and apart the cover from its position.



Figure 294. Location of the screws of the rear cover of the PIP arm

12. Open the tie-wrap in order to release the ribbon cables that come from the Y axis chain to the PCA PIP-CONY59, and disconnect the cables from the PCA.

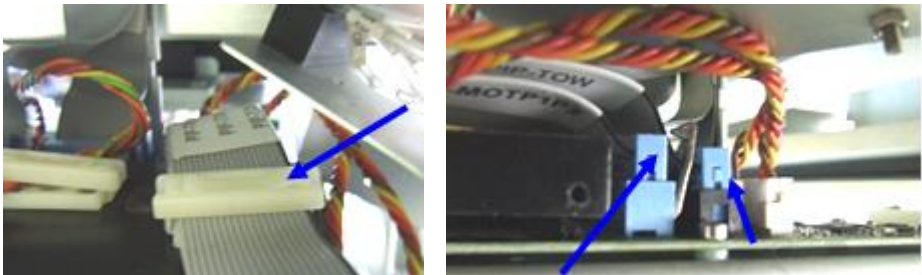


Figure 295. Tie-wrap under rear cover of PIP arm (left) and connectors in PCA PIP-CONY59 (right)

13. Remove completely the chain from the instrument. Then remove the plate that attaches Y chain to Z tower/robotic head from the chain in order to place it in the new Y chain.



Figure 296. Plate attaching Y chain to Z tower/robotic head

14. Repeat previous steps in reverse mode in order to install the new chain.

CAUTION!

Do not swap ribbon cables in PCA PIP-CONY59 or PCA PIP-TOW59 when connecting them (it could damage different parts of the instrument).

Cables should match the connector colour when connected to the PCA: cable labelled as PIP-TOW black and PIP-Z blue.

15. Perform a Prime to check the fluidic system.
16. Perform Leak-tightness test.

7.4.8 PIP Z1 axis chain (CH3) / Z2 axis chain (CH4) replacement

- 1. Switch off the instrument and disconnect it from the mains.
- 2. Disconnect, from the interconnection manifold located in the robotic head, the tube of the probe that comes from the chain to be replaced.

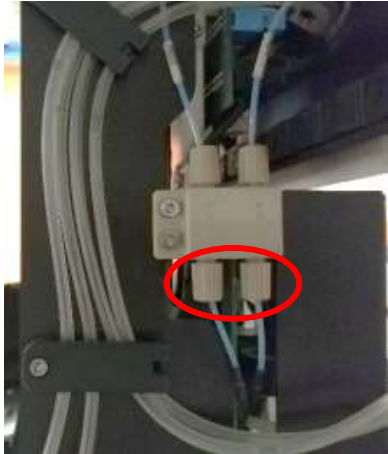


Figure 297. Interconnection manifold in the robotic head

- 3. Move down manually (by means of its belt) the horns and remove the lateral cover of the PIP robotic head.

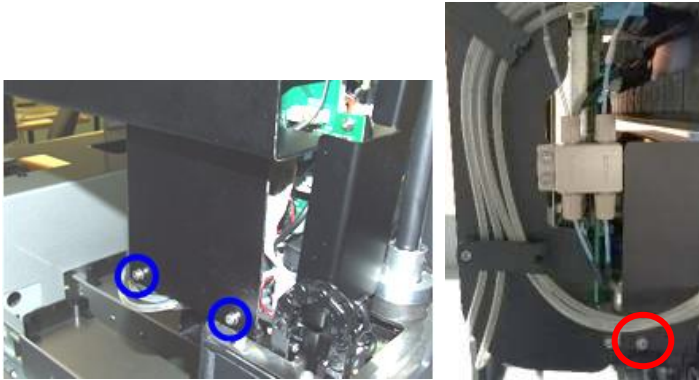


Figure 298. Lateral cover of PIP robotic head

- 4. In case that the chain to be replaced is the PIP Z1 axis chain (CH3), remove also the rear cover of the PIP robotic head.



Figure 299. Rear cover of PIP robotic head

5. Remove the 2 screws from the lateral cover of the Y arm.

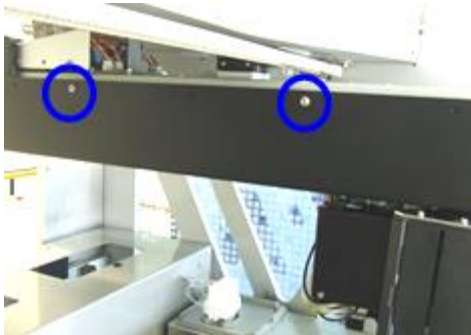


Figure 300. Lateral cover screws of Y arm

6. Disconnect level detection cable of the chain to be replaced from PCA PIP-TOW59.

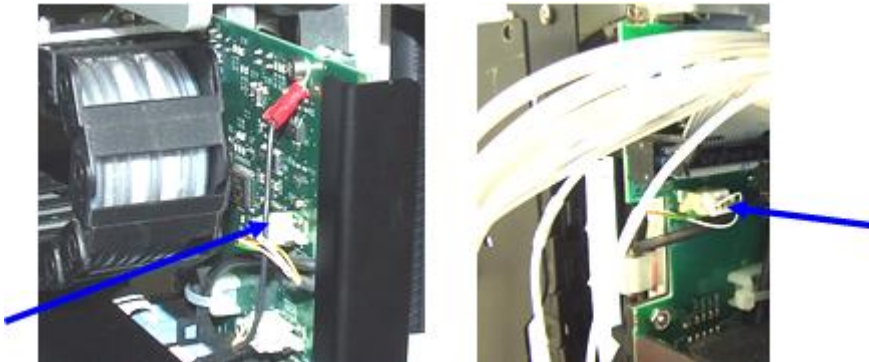


Figure 301. Connector location in PCA PIP-TOW59 for chain CH4 (left) and CH3 (right)

7. Cut the two tie-wraps located at PCA PIP-TOW59 fixing the position of the tubes.
8. Remove the probe related with the chain to be replaced.
9. Remove the 3 screws of the cover of the wagon probe and carefully apart it from its position as much as possible.



Figure 302. Wagon probe cover for chain CH4 (left) and CH3 (right)

10. Unscrew the fitting from the PCA in order to remove the teflon tube from its position.



Figure 303. Teflon tube of the probe

11. Disconnect carefully the cable (level detection signals) of the chain to be replaced connected to the level detection PCA (PCA PIP-P1OSC59 or PCA PIP-P2OSC59).



Figure 304. Connector location in level detection PCA

12. Unscrew the ground connection of the chain to be replaced.



Figure 305. Ground connection

- 13. Remove the chain to be replaced from the instrument.
- 14. Remove the cover of the probe wagon from the chain and install it in the new chain.
- 15. Place the new chain and assemble back all the parts.



NOTE:

When installing the ground connection clamp, make sure that the level detection cable and the Teflon tube are tightened without squeezing them (place the teflon tube at the back of the clamp) but also making sure that no movement is allowed by pulling them slightly up and down.



NOTE:

Once the ground connection is done, consider placing adhesive tape around the remaining visible shielded area of the Teflon tube and the level detection cable to avoid undesired contact of this shielded area with other parts of the instrument..

- 16. Perform a fluidic prime to check that the Teflon tube is not squeezed.
- 17. Perform Leak-tightness test.
- 18. Check level detection.

7.4.9 Barcode reader mirror replacement

- 1. Remove the mirror through its slot located at the horns rear corner.
- 2. Place the new mirror facing the first face to the barcode reader. In order to know which is the first face of the mirror, touch the mirror with an object. If the tip of the object it seems that is touching the reflected image, it is indicating that this side of the mirror is the first face.



Figure 306. Mirror first face (left) and mirror second face (right)

7.4.10 Horns chain replacement

- 1. Switch off the instrument and disconnect it from the mains.
- 2. Open upper door.
- 3. Remove PIP module left cover.

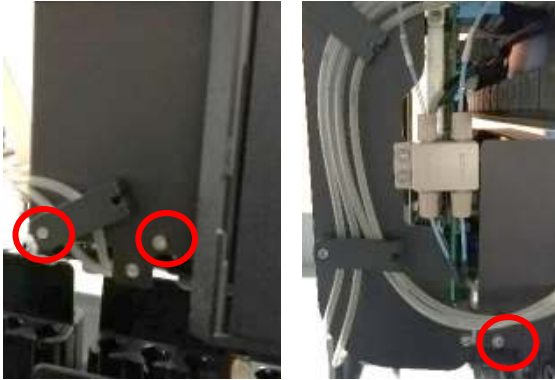


Figure 307. Left cover screws

- 4. Disconnect BCR cable, then remove the tie and the screw attaching the chain.

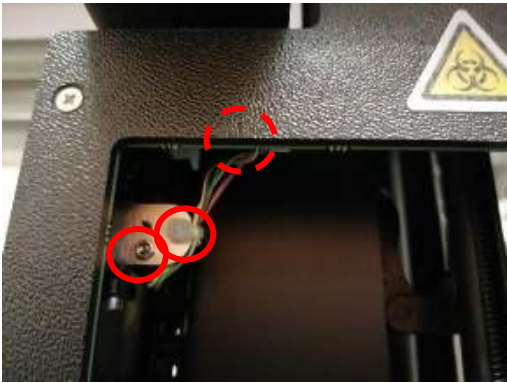


Figure 308. Cable, tie and chain srew

- 5. Remove the other screw attaching the chain.

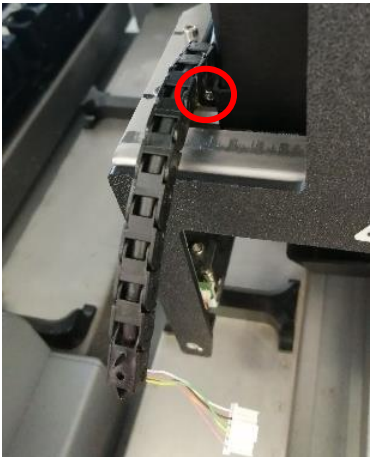


Figure 309. Chain screw

6. Disconnect BCR cable and ground cable.



Figure 310. Horns cabling

7. Remove horns chain.
8. Place the new chain.
9. Assemble back all the parts
10. Place the tie assuring that the cable does not interfere with any part of the instrument when moving upwards/downwards.



Figure 311. Final assembly

11. Verify card barcode reading performance.

7.4.11 Toothed belt T2.5/245 (PIP X-axis)

1. Switch off the instrument and disconnect it from the mains.
2. Open upper door.
3. Remove small belt.



Figure 312. Small belt

- 4. Place the new small belt.

7.4.12 Toothed belt T2.5/245 (PIP Y-axis)

- 1. Switch off the instrument and disconnect it from the mains.
- 2. Open upper door.

From the front of the instrument, take out the Y-belt from its position by removing the fixing screw of the pulley.



Figure 313. PIP large belt

- 3. Remove the large belt from the pinion. The belt will be hang on the CAT robotic head.



Figure 314. Large belt (pinion)

- 4. Remove the upper rear cover.
- 5. Remove the screws of the rear cover of PIP arm.



Figure 315. Location of the screws of the rear cover of the PIP arm

- 6. Disconnect tubing from the Y chain and and pull apart the cover.
- 7. Remove motor support screws.



Figure 316. Support screws

- 8. Disconnect 2 flat cables, open cable clamp and pull apart the cables.



Figure 317. Cabling and cable clamp

- 9. Remove the motor screw (slotted hole).



Figure 318. Motor screw

- 10. Move the motor to be able to remove the small belt.
- 11. Place the new small and large belts and assemble back all the parts.



NOTE:
If the pulley tightening screw is in the same position before disassembly, the belt tension will remain unchanged, else belt tension must be adjusted by modifying the end pulley position.



Figure 319 Tension screw

NOTE:
Any variation of the belt tension may modify probe adjustments in all positions.

- 12. Adjust tension of the small belt.
- 13. Perform probe positionings check.

7.5 FUSES

Fuses related to PIP module are detailed in the following table,

PCA	FUSE NAME	DESCRIPTION
PCA GEN-BCKPL59	FUS31 FUS32	MST250 0034.6615 1A/250V MST250 0034.6620 3.15A/250V3
PCA PIP Control	FUS1	MSF250 0034.6013 0.8A/250V
PCA PIP-P1OSC59	NA	NA
PCA PIP-P2OSC59	NA	NA
PCA PIP-TOW59	NA	NA
PCA PIP-BCR59	NA	NA
PCA PIP-CONY59	NA	NA
PCA GEN-CONX59	NA	NA

Table 57. List of fuses of PIP module

7.6 LEDS

LEDs related to PIP module are detailed in the following table,

PCA	LED	DESCRIPTION
PCA GEN-BCKPL59	LED15 LED16	Green 5V (FUS31) Green 24V (FUS32)
PCA PIP Control	LED1	Yellow When the instrument is switched on, LED is ON and a few seconds later is OFF. This means that the FPGA has been loaded correctly with the appropriate data.
	LED2	Green VMotor Ok. It indicates that the voltage supplied to the motor is ≥ 12V
PCA PIP-P1OSC59	NA	NA
PCA PIP-P2OSC59	NA	NA
PCA PIP-TOW59	NA	NA
PCA PIP-BCR59	NA	NA
PCA PIP-CONY59	NA	NA
PCA GEN-CONX59	NA	NA

Table 58. List of LEDs of PIP module

7.7 JUMPERS AND TEST POINTS

Neither jumpers nor test points are related to PIP module.

7.8 WIRING TABLE

In the following table, there is detailed all the cables related to PIP module. A wiring diagram can be found attached to Service Manual.

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
BRX-EXT + BRX-MotX	H1 MotX	PCA GEN-CONX59	Power motor	X AXIS MOTOR
BRX-EncX	H2 EncX	PCA GEN-CONX59	Encoder motor	X AXIS MOTOR
B-PIP_XY	H3 In CAT/PIP-XY	PCA GEN-CONX59	PIP_XY	PCA GEN-BCKPL59
B-PIP_Z	H4 In CAT/PIP-Z	PCA GEN-CONX59	PIP_Z	PCA GEN-BCKPL59
B-PIP_TOW	H5 In CAT/PIP-TOW	PCA GEN-CONX59	PIP_TOW	PCA GEN-BCKPL59
XB-PIP_TOW	H8 Out CAT/PIP-TOW	PCA GEN-CONX59	H4	PCA PIP-CONY59
XB-PIP_Z	H7 Out CAT/PIP-Z	PCA GEN-CONX59	H3	PCA PIP-CONY59
XB-PIP_XY	H6 Out CAT/PIP-XY	PCA GEN-CONX59	H5	PCA PIP-CONY59
PIP-EncY	H8 MotY	PCA PIP-CONY59	Encoder motor	Y AXIS MOTOR
PIP-MotY	H9 EncY	PCA PIP-CONY59	Power motor	Y AXIS MOTOR
PIP-Z	H1 Out PIP-Z	PCA PIP-CONY59	H2 PIP-Z	PCA PIP-TOW59
PIP-TOW	H2 Out PIP-TOW	PCA PIP-CONY59	H1 PIP-TOW	PCA PIP-TOW59
PIP-Z1	H4 PIP-Z1	PCA PIP-TOW59	H1	PCA PIP-P1OSC59
PIP-Z2	H6 PIP-Z2	PCA PIP-TOW59	H1	PCA PIP-P2OSC59
PIP-BCR	H7 BCR	PCA PIP-TOW59	H2	PCA PIP-BCR59
PIP-PresDet	H8 PRES_DET	PCA PIP-TOW59	NA	LASER DETECT
PIP-EncZ1	H9 ENC_Z1	PCA PIP-TOW59	Encoder motor	P1 AXIS MOTOR
PIP-MotZ1	H10 MOT_Z1	PCA PIP-TOW59	Power motor	P1 AXIS MOTOR

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
PIP-EncZ2	H11 ENC-Z2	PCA PIP-TOW59	Encoder motor	P2 AXIS MOTOR
PIP-MotZ2	H12 MOT-Z2	PCA PIP-TOW59	Power motor	P2 AXIS MOTOR
PIP-EncBCR	H13 ENC_BCR	PCA PIP-TOW59	Encoder motor	BCR AXIS MOTOR
PIP-MotBCR	H14 MOT_BCR	PCA PIP-TOW59	Power motor	BCR AXIS MOTOR
NA	H1	PCA PIP-BCR59	NA	BARCODE READER
NA	H3	PCA PIP-BCR59	NA	BARCODE READER

Table 59. List of wiring of PIP module



Figure 320. Connectors, fuses and LEDs of PCA GEN-BCKPL59 related to PIP module

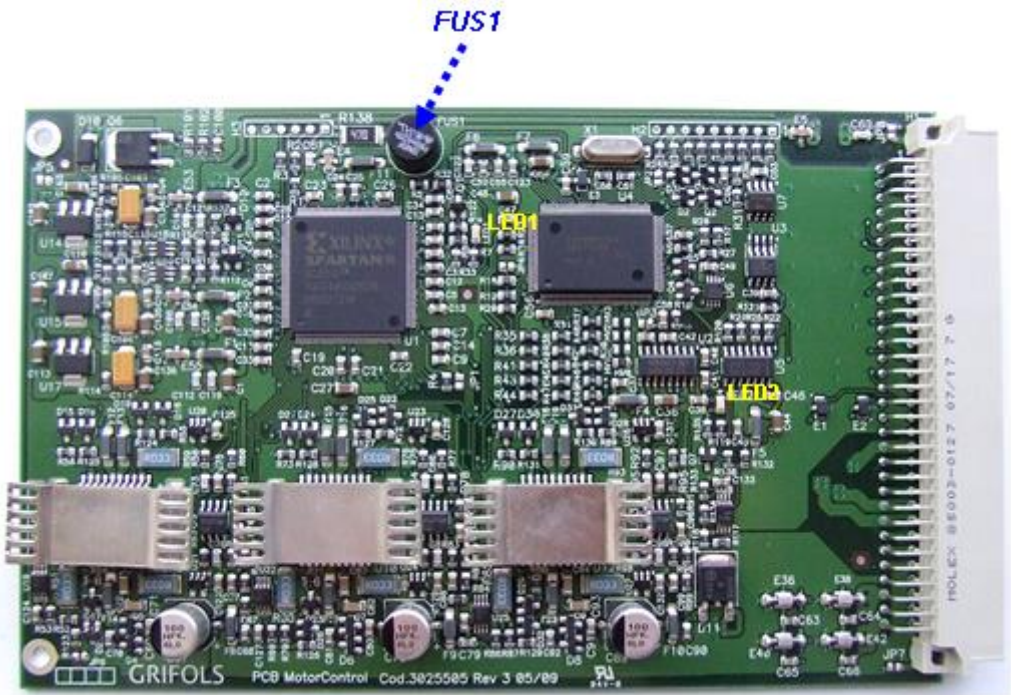


Figure 321. Location of the fuse and LEDs in PCA PIP Control

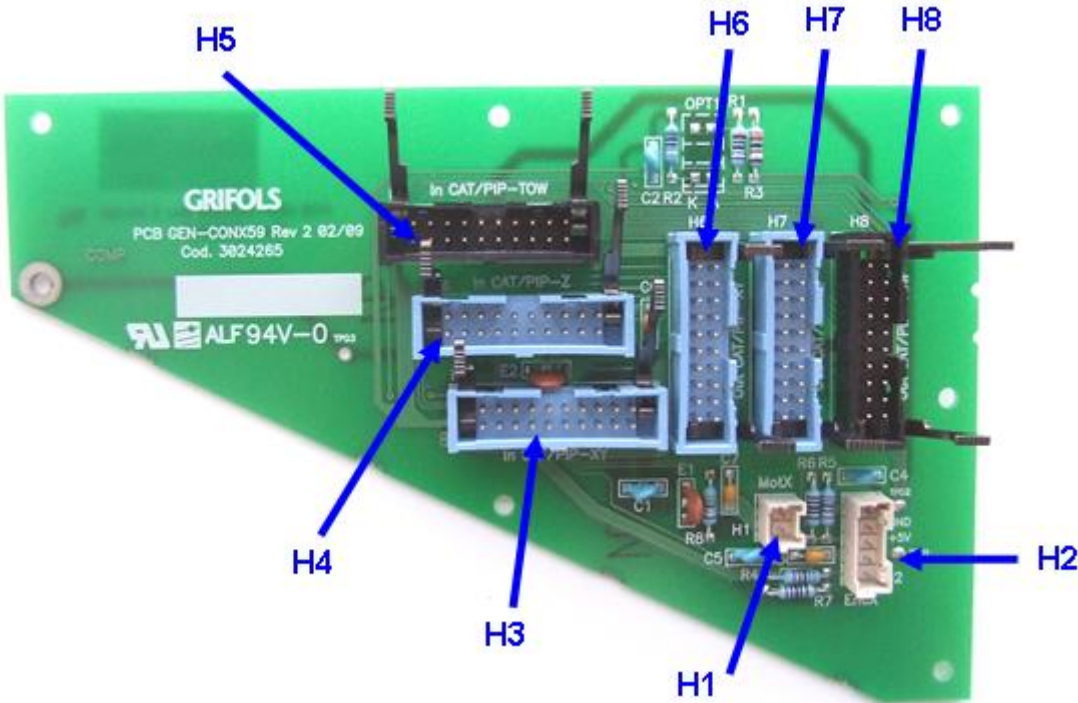


Figure 322. Location of the connectors in PCA GEN-CONX59

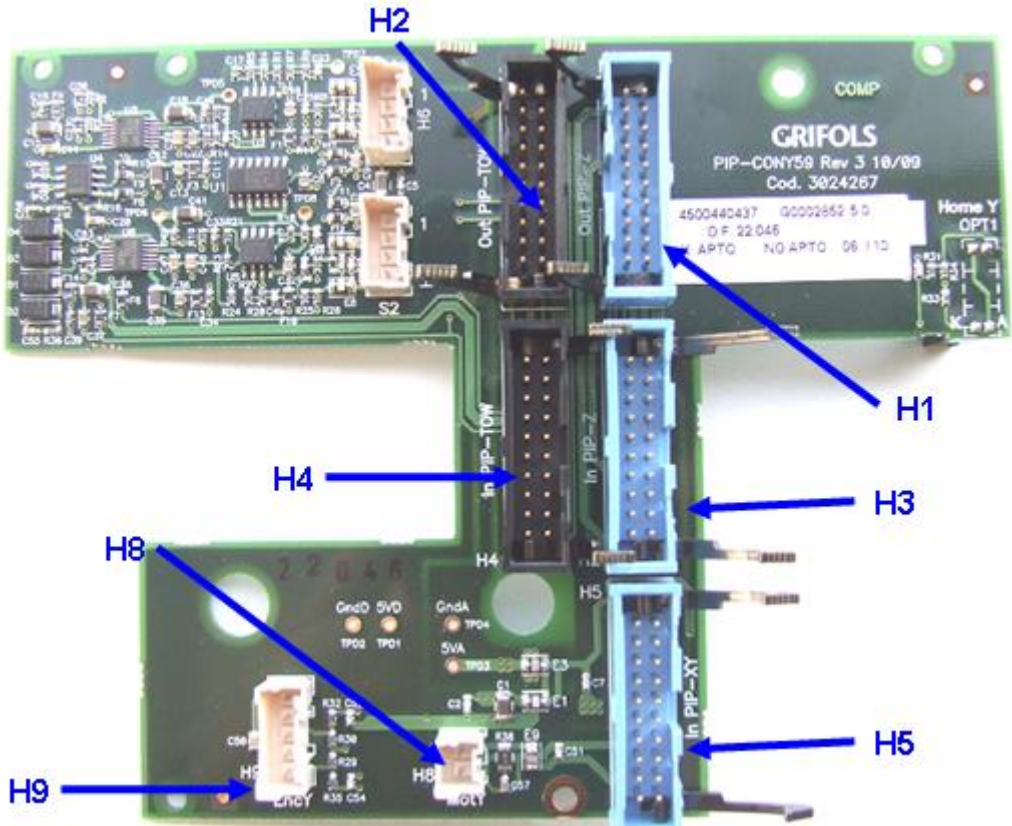


Figure 323. Location of the connectors in PCA PIP-CONY59 related to PIP module

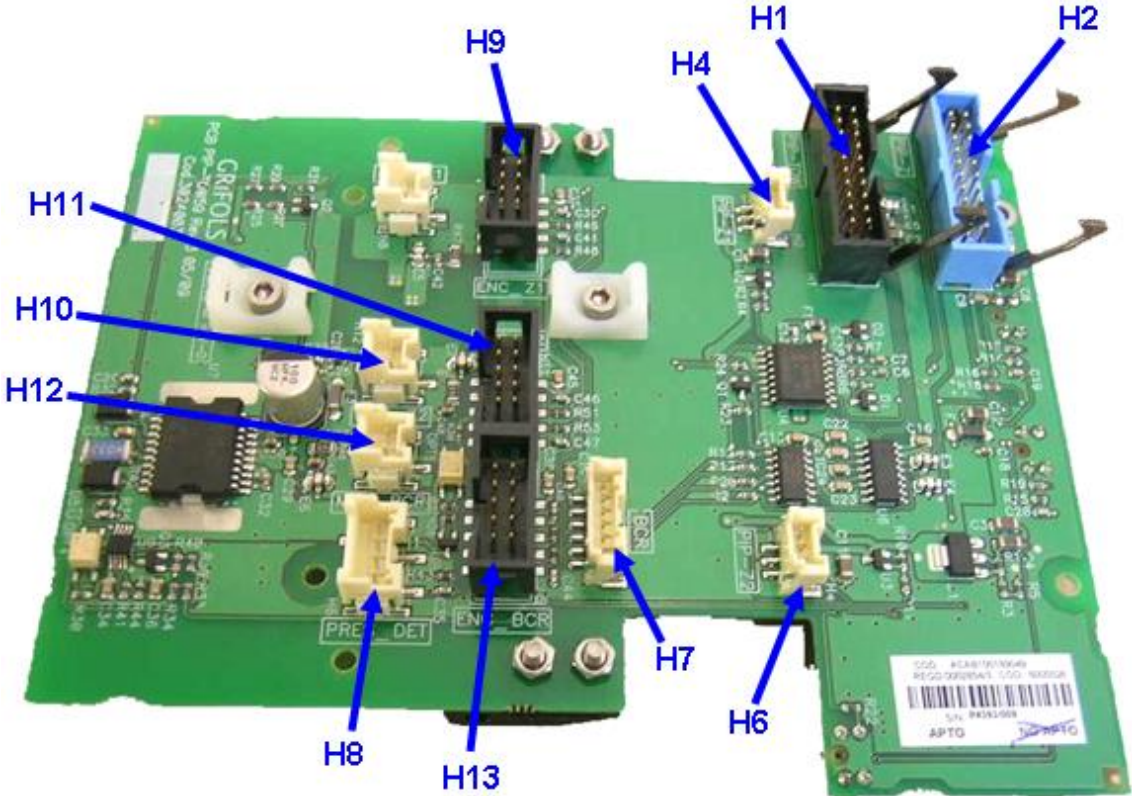


Figure 324. Location of the connectors in PCA PIP-TOW59

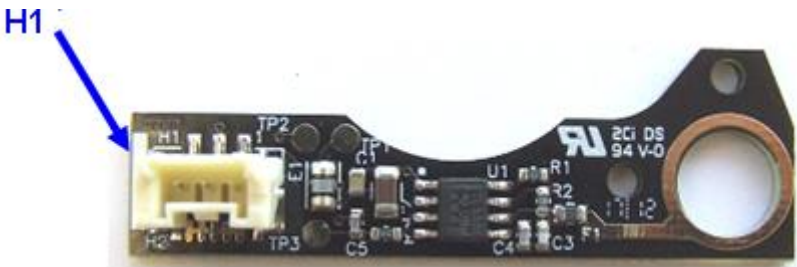


Figure 325. Location of the connector in PCA PIP-P1OSC59 (Impact detector)



Figure 326. Location of the connector in PCA PIP-P2OSC59 (Impact detector)

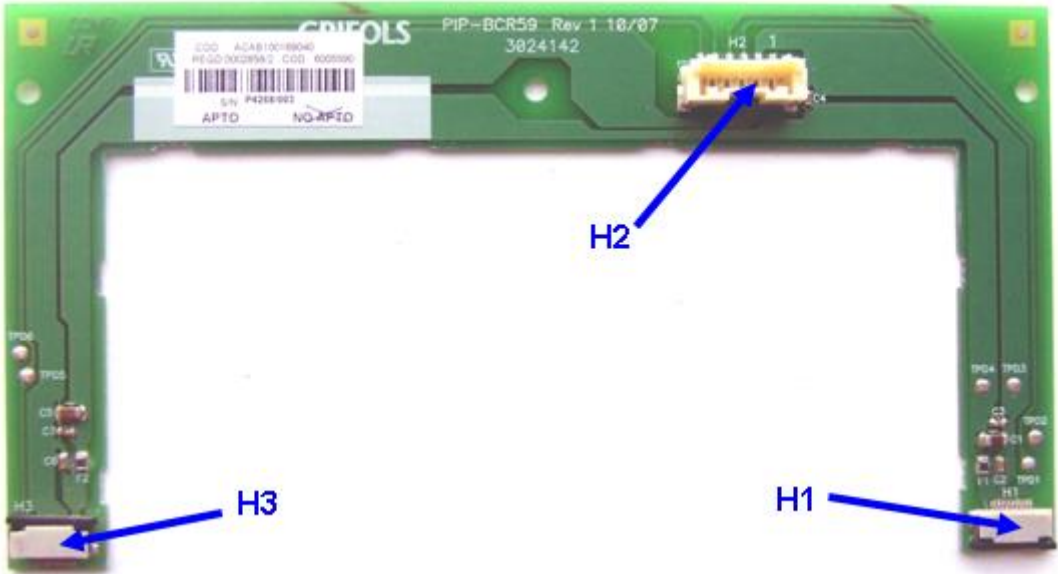


Figure 327. Location of the connectors in PCA PIP-BCR59

7.9 TROUBLESHOOTING

Module	Device	Error code	Error	Cause	Text	Options
PIP (08)	XY Motors (01) X axis (02) Y axis (11) Presence detector during X movements (05) Presence detector during Y movements	003000	ERRORCODE_LEN	FW has received a command with incorrect length.	Error during the upper robotic arm movement. Please, inspect the instrument upper level. Please choose one of the following options:	RETRY CANCEL Works (it does not show any further error. When queueing new samples, <i>The fluidic system is disabled due to previous errors. Please RESTART the instrument before requesting any further test is shown</i>)
		005000	ERRORCODE_ARG	Command has wrong arguments.		
		006000	ERRORCODE_BUSY	Command is addressed to a busy device.		
		007000	ERRORCODE_NOTHOMED	Move command is addressed to a not homed device.		
		008000	ERRORCODE_NOTSAFE	Command is not safe for the instrument.		
		032000	CURRENT_LIMIT	Current limit at servo device.		
		033000	ERRORCODE_SRV_POSITION_ERROR	A position error has been produced at a servo device.		
		034000	MOVEMENT_ERROR	A movement error has been produced at a servo device.		
		035000	HOME_TIMEOUT	A home timeout has been produced at a servo device.		
		036000	ERRORCODE_SRV_SYNC_TIMEOUT	A synchronization error has been produced at a servo device.		
		037000	ERRORCODE_SRV_UNKNOWN_ERROR	An unknown error has been produced at a servo device.		
		041000	ERRORCODE_PD_SYNC_TIMEOUT	Presence detector has been busy for more than its timeout margin.		
		042000	ERRORCODE_PD_UNKNOWN_ERROR	Unknown error produced during a presence detector related operation.		

Module	Device	Error code	Error	Cause	Text	Options
PIP (08)	Probe / BCR Motors (03) Probe 1 (04) Probe 2 (08) BCR	003000	ERRORCODE_LEN	FW has received a command with incorrect length.	Error during the upper robotic arm movement. Please, inspect the instrument upper level. Please choose one of the following options:	RETRY CANCEL Works (it does not show any further error. When queueing new samples, <i>The fluidic system is disabled due to previous errors. Please RESTART the instrument before requesting any further test</i> is shown. In case of BCR motors errors, no error is shown upon cancel error.)
		005000	ERRORCODE_ARG	Command has wrong arguments.		
		006000	ERRORCODE_BUSY	Command is addressed to a busy device.		
		007000	ERRORCODE_NOTHOMED	Move command is addressed to a not homed device.		
		032000	CURRENT_LIMIT	Current limit at servo device.		
		033000	POSITION_ERROR	Position error at a servo device.		
		034000	MOVEMENT_ERROR	Movement error at a servo device.		
		035000	HOME_TIMEOUT	Home timeout at a servo device.		
		036000	ERRORCODE_SRV_SYNC_TIMEOUT	A synchronization error has been produced at a servo device.		
		037000	ERRORCODE_SRV_UNKNOWN_ERROR	An unknown error has been produced at a servo device.		
	BCR (06) BCR 1 (07) BCR 2	003000	ERRORCODE_LEN	FW received incorrect length command.	Error during the upper robotic arm movement. Please, inspect the instrument upper level. Please choose one of the following options:	RETRY CANCEL Works (it does not show any further error. When queueing new samples, <i>The fluidic system is disabled due to previous errors. Please RESTART the instrument before requesting any further test</i> is shown)
		005000	ERRORCODE_ARG	Command has wrong arguments.		
		006000	ERRORCODE_BUSY	Command is addressed to a busy device.		
		011000	ERRORCODE_DATANOTREADY	The data requested is not ready.		
		048000	DRIVER_ERR	Barcode reader driver error.		
		050000	ERRORCODE_BCR_COM_ERR	Unable to communicate with a barcode reader.		
		051000	ERRORCODE_BCR_NO_BARCODE	No barcode is read.		
		052000	ERRORCODE_BCR_SYNC_TIMEOUT	Timeout.		
		053000	ERRORCODE_BCR_UNKNOWN_ERROR	Unknown error.		
		049000	NOT_INIT	Unable to initialize barcode reader.	An error in a barcode reader has been detected. Please, RESTART the system.	OK

Module	Device	Error code	Error	Cause	Text	Options
PIP (08)	Level detector (09) Probe 1 (10) Probe 2	003000	ERRORCODE_LEN	FW has received a command with incorrect length.	Error during the upper robotic arm movement. Please, inspect the instrument upper level. Please choose one of the following options:	RETRY RE-RUN Open (option not available if error occurs in Dilution Cup)
		005000	ERRORCODE_ARG	Command has wrong arguments.		
		006000	ERRORCODE_BUSY	Command is addressed to a busy device.		
		054000	ERRORCODE_LD_IMPRECISE_LEVEL	Level detector is unable to determine a tube or vial level.	Level detection error during pipetting. Please check there is enough volume and the absence of bubbles.	RETRY RE-RUN Open
		055000	ERRORCODE_LD_INSUF_LEVEL	Not enough level detected to perform an aspire command.		
		056000	ERRORCODE_LD_DRIVER_ERR	Level detector driver error.	Error during the upper robotic arm movement. Please, inspect the instrument upper level. Please choose one of the following options:	RETRY RE-RUN Open (option not available if error occurs in Dilution Cup)
		057000	ERRORCODE_LD_NOT_LEVEL	Not level detected in a tube or vial.	Level detection error during pipetting. Please check there is enough volume and the absence of bubbles.	RETRY RE-RUN Open

Module	Device	Error code	Error	Cause	Text	Options
PIP (08)	Level detector (09) Probe 1 (10) Probe 2	058000	ERRORCODE_LD_SYNC_TIMEOUT	Timeout.	Error during the upper robotic arm movement. Please, inspect the instrument upper level. Please choose one of the following options:	RETRY RE-RUN Open (option not available if error occurs in Dilution Cup)
		059000	ERRORCODE_LD_IMPACT	Impact detected.	Probe of upper robotic arm has found an obstacle. Please check: - Samples and reagents suitability. - Samples and reagents racks positioning.	RETRY CANCEL Works Open
		060000	ERRORCODE_LD_UNKNO WN_ERROR	Unknown error.	Error during the upper robotic arm movement. Please, inspect the instrument upper level. Please choose one of the following options:	RETRY RE-RUN Open (option not available if error occurs in Dilution Cup)
Any (xx)	Any (xx)	002000	ERRORCODE_CMD	Unknown command code received.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		003000	ERRORCODE_LEN	Incorrect length command received.		
		004000	ERRORCODE_DEV	Command is addressed to an incorrect device.		
		005000	ERRORCODE_ARG	Command has wrong arguments.		
		014000	ERRORCODE_HW_DEV	Addressed device has a problem with its related HW.		
		015000	ERRORCODE_HW_HWFAL	There is a problem with PCA HW.		
		016000	ERRORCODE_HW_HWNO TSET	PCA setup error.		

Module	Device	Error code	Error	Cause	Text	Options
0	240	000000	ERRORCODE_UNKNOW N	Unknown error	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		008000	ERRORCODE_FW_NOTS AFE	Movement not safe		
		017000	ERRORCODE_FW_UNKN OWN_ERROR	Unknown error		
		061000	ERRORCODE_INVALID_ ARGUMENTS	Invalid arguments		
		062000	ERRORCODE_FWCOMM_ TIMEOUT	Communication timeout		
		063000	ERRORCODE_FWCOMM_ CONNECTION_LOST	Communication lost		
		064000	ERRORCODE_FWCOMM_ COMM_ERROR	Communication error		
		088000	ERRORCODE_PRECOND ITION_FAILED	Previous condition for the action is not met		
		089000	ERRORCODE_STARTUP_ ERROR	Driver startup error		
		091000	ERRORCODE_ALREADY_ STARTED	Another instance of Driver is already opened		
		090000	ERRORCODE_UNEXPEC TED_ERROR	A Driver unexpected error has been produced		

Table 60. List of errors related to PIP module.

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Movement error / current limit error (X axis) 0801032000 0801034000	Fuse blown	Check / replace FUS31 and FUS32
	Large belt too tight	Check tightness of belt / Replace belt
	Incorrect X motor assembly	Check/replace X motor silentblocks
	Lack of lubrication in X linear guide	Lubricate linear guide
	X linear guide damaged	Check / replace linear guide
	Motor damaged	Check / replace motor
	PIP X-Y flat cable damaged	Check / replace PIP X-Y cable
Movement error / current limit error (Y axis) 0802032000 0802034000	Fuse blown	Check / replace FUS31 and FUS32
	Small belt too tight	Check tightness of belt / Replace belt
	Large belt too tight	Check tightness of belt / Replace belt
	Lack of lubrication in Y linear guide	Lubricate linear guide
	Y linear guide damaged	Check / replace linear guide
	Motor damaged	Check / replace motor
	PIP X-Y flat cable damaged	Check / replace PIP X-Y cable / X PIP chain
Movement error / current limit error (Z1/Z2 axis) Z1: 0803032000 Z1: 0803034000 Z2: 0804032000 Z2: 0804034000	Fuse blown	Check / replace FUS31 and FUS32
	Belt too tight	Check tightness of belt / Replace belt
	Lack of lubrication in leadscrew	Lubricate leadscrew
	Leadscrew damaged	Check / replace leadscrew
	Motor damaged	Check / replace motor
	PIP-Z flat cable damaged	Check / replace X - Y PIP chain
Movement error / current limit error (Horns axis) 0808032000 0808034000	Fuse blown	Check / replace FUS31, FUS32, FUSGEN4
	Belt too tight	Check tightness of belt / Replace belt
	Lack of lubrication in leadscrew	Lubricate leadscrew
	Leadscrew damaged	Check / replace leadscrew
	Motor damaged	Check / replace motor
	PIP-TOW flat cable damaged	Check / replace X / Y PIP chain
X axis home is not detected 0801035000	Home flag is not aligned with optical barrier	Check position of the home flag
	Motor does not turn	Check / replace motor
	Failure in optical barrier	Check / replace PCA GEN-CONX59
	PIP X-Y flat cable damaged	Check / replace PIP X-Y cable
Y axis home is not detected 0802035000	Home flag is not aligned with optical barrier	Check position of the home flag
	Motor does not turn	Check / replace motor
	Failure in optical barrier	Check / replace PCA PIP-CONY59
	PIP X-Y flat cable damaged	Check / replace X chain
Z1 / Z2 axis home is not detected Z1:0803035000 Z2: 0804035000	Home flag is not aligned with optical barrier	Check position of the home flag
	Motor does not turn	Check / replace motor
	Failure in optical barrier	Check / replace PCA PIP-TOW59
	PIP-Z flat cable damaged	Check / replace X/ Y PIP chain
Horns axis home is not detected 0808035000	Home flag is not aligned with optical barrier	Check position of the home flag
	Motor does not turn	Check / replace motor
	Failure in optical barrier	Check / replace PCA PIP-TOW59
	PIP-TOW flat cable damaged	Check / replace X / Y PIP chain

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Samples tubes/reagents vials are not detected / incorrectly detected	Fuse blown	Check / replace FUSGEN4
	Tube / vial is not placed correctly	Place correctly tube / vial in the rack
	Sample / reagent rack is not placed correctly over the drawer	Place correctly sample / reagent rack in the drawer
	Drawer is not flat	Check sample/reagent drawer flatness
	Adjustment of the PIP in the drawer is not correct	Check / perform PIP adjustment in the drawer
	Presence detector is not correctly assembled	Check presence detector assembly / PIP laser alignment
	Presence detector adjustment is not correct	Perform presence detector adjustment
	Presence detector damaged	Replace presence detector
	PCA PIP-TOW59 damaged	Replace PCA PIP-TOW59
Diameter of the samples tubes is not detected correctly	Sample tube is not placed correctly	Place correctly sample tube in the rack
	Sample rack is not placed correctly over the drawer	Place correctly sample rack in the drawer
	Sample tube shape provokes this effect	Check / set <i>erytra.properties</i> configuration
	Drawer is not flat	Check sample/reagent drawer flatness
	Adjustment of the PIP in the drawer is not correct	Check / perform PIP adjustment in the drawer
	Presence detector is not correctly assembled	Check presence detector assembly
	Presence detector adjustment is not correct	Perform presence detector adjustment
	Presence detector damaged	Replace presence detector
	PCA PIP-TOW59 is damaged	Replace PCA PIP-TOW59
Presence detector is always ON	PCA PIP-TOW59 is damaged	Replace PCA PIP-TOW59
	Presence detector damaged	Replace presence detector
BCRs are not switched on BC1: 0806050000 BC1: 0806052000 BC1: 0806053000 BC1: 0806049000 BC2: 0807050000 BC2: 0807052000 BC2: 0807053000 BC2: 0807049000	Fuse blown	Check / replace FUS31
	BCRs cables damaged / disconnected	Check / replace BCRs cables
	BCRs are not working	Check / replace BCRs
	PCA PIP-BCR59 damaged	Check / replace PCA PIP-BCR59
	Horns chain damaged	Check / replace horns chain
	PCA PIP-TOW59 damaged	Replace PCA PIP-TOW59
	PIP-TOW / PIP-Z cable damaged	Check / replace X / Y PIP chains
Barcodes are not read BC1: 0806051000 BC2: 0807051000	Sample tube/vial is not placed correctly	Place correctly sample tube/vial in the rack
	Barcode label in bad conditions	Check barcode label
	Mirrors are dirty	Clean mirrors
	Drawer is not flat	Check drawer flatness
	Adjustment of the horns is not correct	Check / perform horns adjustment
	BCR is not properly assembled	Check the assembly of the BCR
	BCR is not working	Check / replace BCR
	BCR cable damaged / disconnected	Replace the BCR cable
	PCA PIP-TOW59 damaged	Replace PCA PIP-TOW59

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Level is not detected properly with probe 1 or probe 2 Z1: 0809054000 Z1: 0809057000 Z2: 0810054000 Z2: 0810057000	Level Detection PCA damaged	Check / replace PCA PIP-P1OSC59 (probe 1) or PCA PIP-P2OSC59 (probe 2)
	Dirt in connection pad of level detection PCA	Clean connection pad of PCA PIP-P1OSC59 (probe 1) or PCA PIP-P2OSC59 (probe 2)
	Incorrect assembly between level detection PCA and probes holder	Check assembly between PCA PIP-P1OSC59 (probe 1) or PCA PIP-P2OSC59 (probe 2) and its probes holder
	Probes washer leakage	Check/clean/replace probes washer
		Check probes washer aspiration circuit
	Probes holder damaged	Check/replace probes holder
	Probe chain damaged	Check / replace probe chain
	Bad ground connection of level detection cables	Check ground connection of level detection cables
	Bubbles on the liquid	Ensure there are no bubbles on the liquid.
	Drops in the tube walls	Ensure there are no drops at the tube walls
Level detected too early	Bent probe	Replace probe
	Probably due to probe/s being wet.	Clean probe. Check for leakages
Probe is not correctly placed over tube / vial / incubator / dilution cup	Possible leak in probes washer	Check tubing and fittings
	Reagent / Sample rack is not placed correctly over the drawer	Place correctly reagent / sample rack in the drawer
	Reagent vial / Sample tube is not placed correctly	Place correctly reagent vial / sample tube in the rack
Probe crashes with the bottom of sample tubes Z1: 0809059000 Z2: 0810059000	Probes are not correctly adjusted	Perform probes adjustment
	Rack not correctly placed	Place correctly reagent rack in the drawer
	Drawer is not flat	Check sample drawer flatness
	Probe adjustment is not performed correctly	Perform probe adjustment in sample drawers position
Probe crashes with the bottom of reagent/diluent vials Z1: 0809059000 Z2:0810059000	Incorrect configuration	Check configuration defined in <i>erytra.properties</i> file
	Rack not correctly placed	Place correctly reagent rack in the drawer
	Rack platform is not correctly assembled	Check rack platform assembly
	Drawer is not flat	Check reagent drawer flatness
Horns crash with drawer during barcode reading	Probe adjustment is not performed correctly	Perform probe adjustment in reagent drawers position
	Horns are not correctly adjusted	Perform horns adjustment
Liquid leaking on top of card foil	Some bubbles are considered normal due to pressurised cards.	Replace probe/s
	Incubator is not flat	Check incubator flatness
	Incubator height adjustment is not performed correctly	Perform incubator height adjustment
	Probe adjustment is not performed correctly	Perform probe adjustment in incubator position

Table 61. PIP module troubleshooting

8 CARD TRANSPORT (CAT) MODULE

8.1 DESCRIPTION

The functions of this module are:

- Card inventory:
 - Card presence detection/location in drawers.
 - Card identification (type, expiration date,...)
- Card transportation (card rack, incubator, centrifuge,...)

To do so, this module consists of a three-axis robotic head. Card transportation is achieved by means of a gripper. This gripper is found in the robotic head and moves along Z axis. Gripper includes a centering system to grab cards from card holder that consists in an H-shaped retractile part. The cards are grabbed by two fingers that are opened/closed by means of an electromagnet. 4 pins with springs contribute to card release/horizontal position. Card grabbing detection is also implemented by means of an optical barrier that detects one of the pins (up) position.

Card presence detection/location is achieved by means of a laser reflection sensor assembled in the robotic head. Robotic head scans the rack from left to right.

Card identification is achieved by means of a barcode reader assembled in the robotic head and it is performed when gripper raises the card to upper position. Barcode reader reference is Symbol SE-955 (s/n < 059-0001335) or Zebra SE-965 (s/n ≥ 059-0001335) and the communication is through RS232.

X-axis is moved by a servo-controlled motor assembled in the left area. This motor turns and provokes that a small belt moves. This belt transmits the movement by means of a gear to a large belt attached to X wagon (assembled on a linear guide). Small belt tension is automatically set by assembly and silentblocks. Large belt tension is adjusted by modifying X end pulley position. Home position is defined at the left and using an L-shaped part assembled in X wagon that will get in an optical barrier. CAT module and PIP module built in same X-axis.

Y-axis is moved by a servo-controlled motor assembled in back axis area. This motor turns and provokes that small belt moves. This belt transmits the movement by means of a gear to a large belt attached to Y wagon (assembled on a linear guide). Small belt tension can be adjusted modifying motor position (slotted hole). Large belt tension is adjusted by modifying Y end pulley position. Home position is defined at the back and using an L-shaped part assembled in Y wagon that will get in an optical barrier.

Z-axis is moved by a servo-controlled motor assembled in robotic head bottom. This motor turns and provokes that a small belt moves. This belt provokes that a pulley fixed to a leadscrew with a set screw turns and this one that the gripper moves. A linear guide is also assembled to assure proper Z movement. Small belt tension can be adjusted modifying motor position (slotted hole). Home position is defined at the Z-axis top and using an L-shaped part assembled in leadscrew nut that will get in an optical barrier.

Encoders resolution for CAT axes is as follows:

CAT Axis	Encoder counts / mm
X	285.714 counts/mm
Y	285.714 counts/mm
Z	1257.1 counts/mm

Table 62. CAT axes encoder resolution

General block diagram of the module is shown in the following figure:

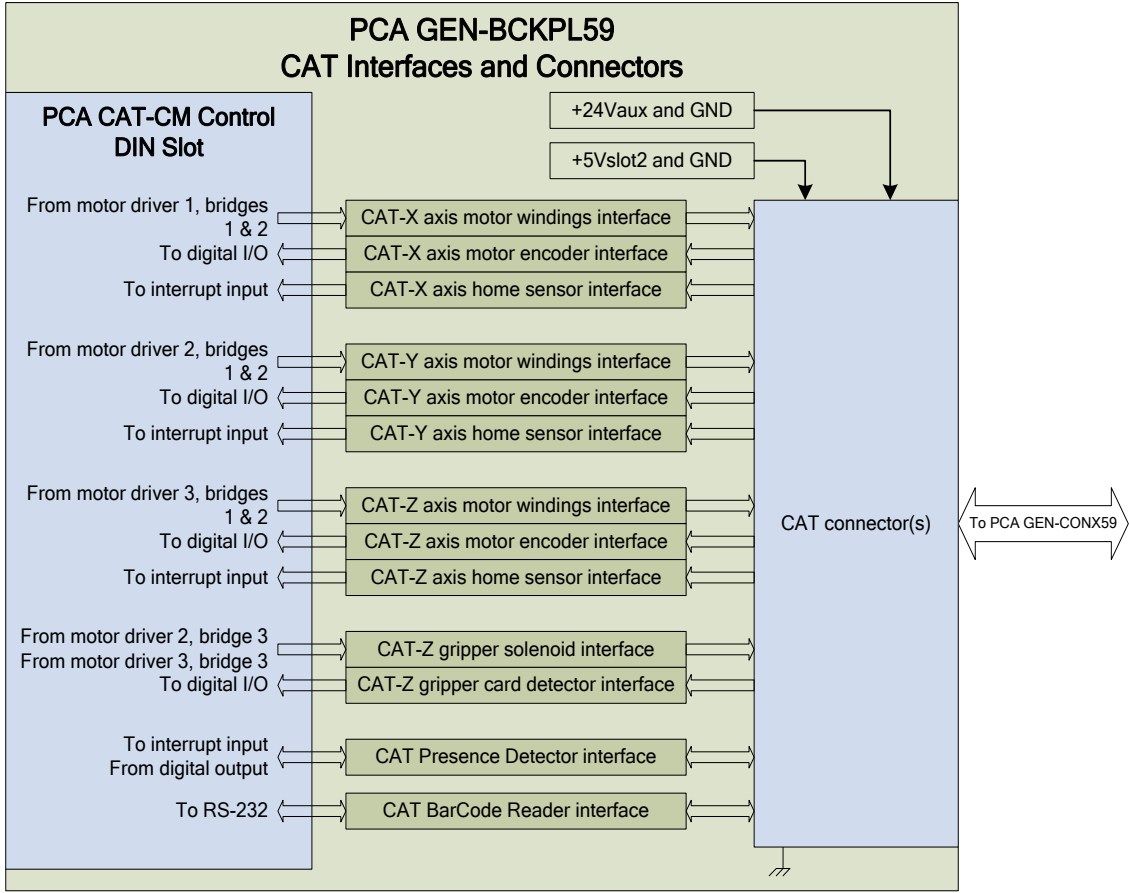


Figure 328. CAT module block diagram

PCAs related to the CAT module are:

- **PCA GEN-BCKPL59:** main PCA of the instrument, located in the backpanel.
- **PCA CAT-CM Control:** located in the backpanel (slot 2), attached to the PCA GEN-BCKPL59.
- **PCA GEN-CONX59:** Located at X axis left side (looking from the front). It contains X axis home optical barrier (power and signal), X axis motor powering and encoder and the three ribbon cables (intermediate) connectors: CAT XY (X and Y motor power and encoder and homes), CATZ (Z motor powering and encoder, electromagnet activation and card presence optical barrier powering and signal) and CAT-TOW (card presence detector and barcode reader signals).
- **PCA CAT-CONY59:** Located at Y axis back (looking from the front). It contains Y axis home optical barrier (power and signal), Y axis motor powering and encoder and two ribbon cables (intermediate) connection: CATZ (Z motor powering and encoder, electromagnet activation and card presence optical barrier powering and signal) and CAT-TOW (Presence detector and barcode reader signals). CATXY ribbon cable transmits Y axis home sensor powering and signal and Y motor powering and encoder signal.
- **PCA CAT-TOW59:** Located inside the robotic head. It contains Z/gripper axis home sensor, Z motor powering and encoder and card presence detector and barcode reader signals.
- **PCA PINZA57:** Located at the top of the gripper. It contains card presence detector and electromagnet powering (open/close fingers)

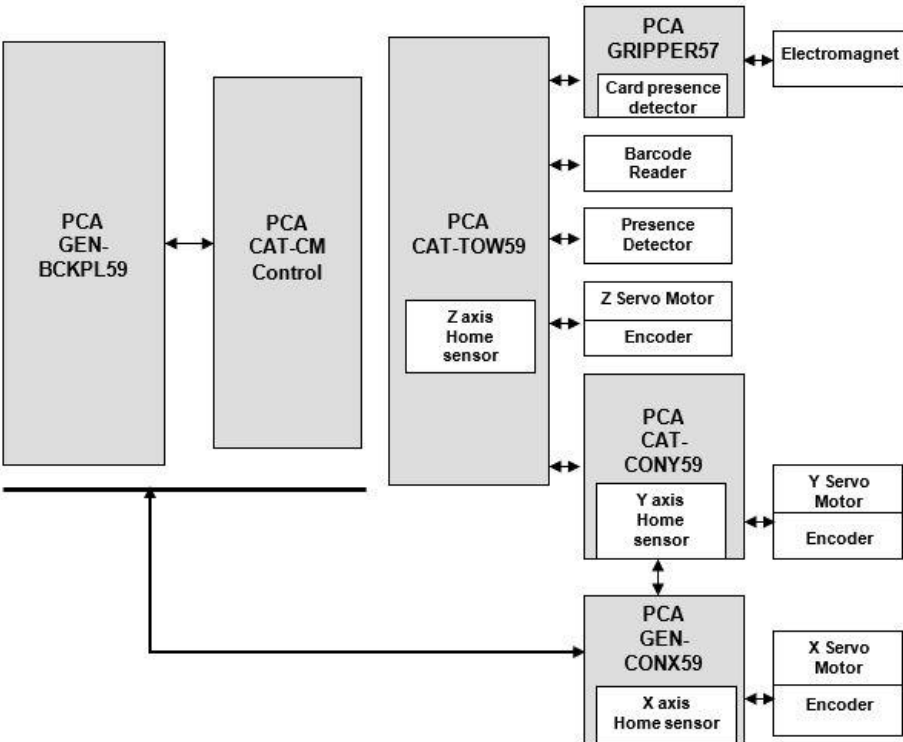


Figure 329. Main blocks of the CAT module

Barcode reader communication is performed through RS232.

8.2 ADJUSTMENT PROCEDURES

In this section, the adjustments related to the CAT module are explained. There are 2 types of adjustments: mechanicals and functionals.

8.2.1 Mechanical adjustments

Mechanical adjustments related to CAT module are detailed.

8.2.1.1 X-Axis small belt tension adjustment

No manual X small belt tension adjustment is required. Assembly and silentblocks assure proper belt tension.

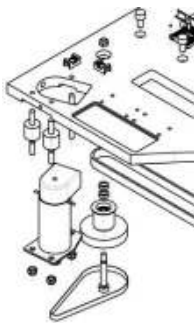


Figure 330. X-axis motor small belt

8.2.1.2 X-Axis large belt tension adjustment

This adjustment is performed to adjust the tension of the belt of the X-axis of the CAT arm.

Prerequisite: NA
Adjustment tool: NA
Procedure:

1. Screw pulley tightening screw to allow pulley movement.
2. Loose both pulley fixing screws.
3. Tighten the belt unscrewing the pulley tightening screw and fix its position tightening both fixing screws.

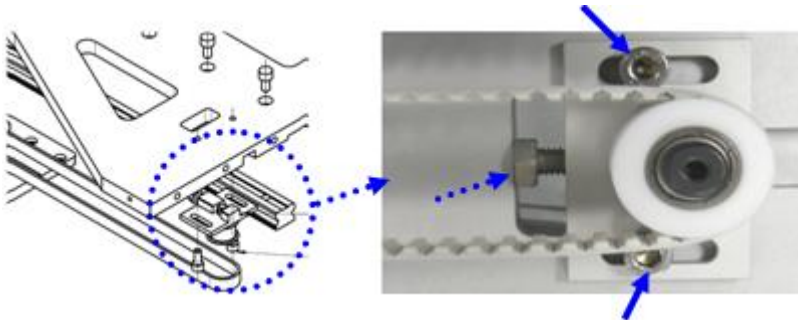


Figure 331. X-axis motor large belt

8.2.1.3 Y-Axis small belt tension adjustment

This adjustment is performed to adjust the tension of the small belt of the Y-axis of CAT arm.

Prerequisite: NA
Adjustment tool: NA
Procedure:

1. Remove Y arm motor cover (remove 2 fixing screws).
2. Unplug Y arm ribbon cables from PCA CAT-CONY59.
3. Remove the screws fixing Y motor-belt assembly to Y arm and remove the assembly.
4. Loose Y motor slotted hole screw.



Figure 332. Y-axis motor small belt

5. Move Y motor to adjust belt tension.
6. Fix motor position tightening slotted hole screw
7. Assemble Y motor-belt assembly to Y arm tightening the tree screws).
8. Connect Y-arm ribbon cables to PCA CAT-CONY59.

9. Assemble Y motor cover.

8.2.1.4 Y-Axis large belt tension adjustment

This adjustment is performed to adjust the tension of the large belt of the Y-axis of CAT arm.

Prerequisite: NA
Adjustment tool: NA
Procedure:

- 1. Screw pulley tightening screw to allow pulley movement.
- 2. Loose both pulley fixing screws.
- 3. Tighten the belt unscrewing the pulley tightening screw and fix its position tightening both fixing screws.

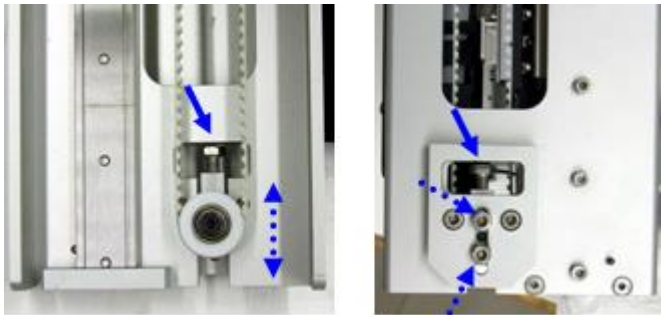


Figure 333. Y-axis motor large belt



NOTE:
Excessively low belt tension can lead to an offset from the intended position in the CAT z-axis.

8.2.1.5 Z-Axis/gripper belt tension adjustment

This adjustment is performed to adjust the tension of gripper belt.

Prerequisite: NA
Adjustment tool: NA
Procedure:

- 1. Loose gripper motor screws.
- 2. Move gripper motor to adjust gripper belt tension.
- 3. Set gripper motor position/belt tension tightening both gripper motor screws.

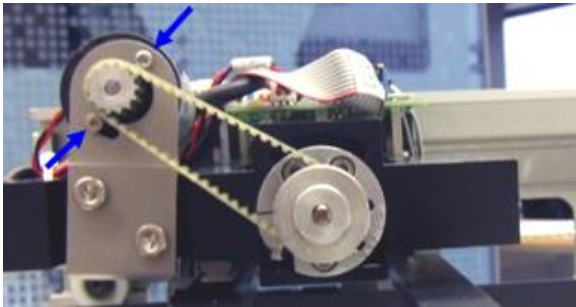


Figure 334. Z-axis motor belt

8.2.1.6 Gripper parallelism

This adjustment must be performed when:

- Gripper (whole) is disassembled.
- Robotic head is disassembled.

Prerequisite: NA

Adjustment tool: NA

Procedure:

1. Unlock CAT robotic head using *unlock/stop* button in *Gripper Arm* tab → *Movement* subtab → *Position Actions* area and selecting all axes.
2. Open card drawer 2.
3. Lower gripper until gripper edges touch gently centrifuge cover surface.

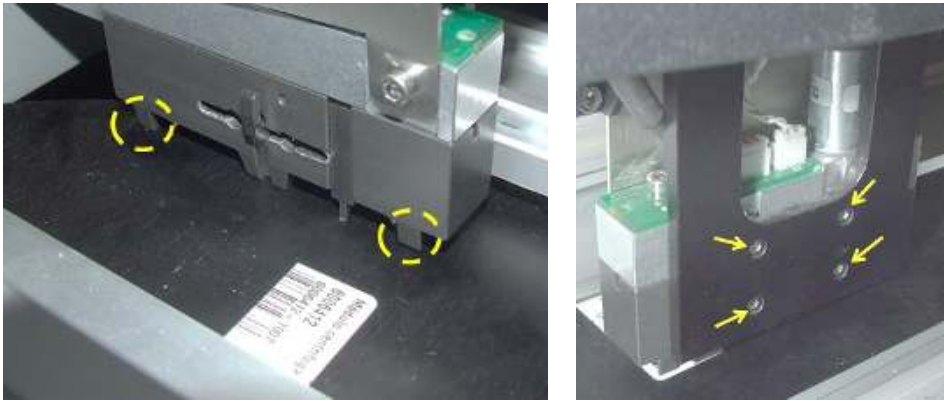


Figure 335. Gripper parallelism adjustment

4. Check that gripper edges (left or right) touch centrifuge cover surface at the same time. Should this not be the case, loose the four screws that fix gripper, align gripper and tighten the screws again.
5. Apply a threadlocking adhesive (such as Loctite 222) to the 4 screws that fix the gripper.



NOTE:

No more than one screw should be removed at a time when applying the sealant. Remove the first screw, apply the adhesive, screw it back in and continue with the next screw.

8.2.1.7 Card presence detector inclination

This adjustment must be performed when card presence detector is disassembled / replaced.

Presence detector inclination is set using two points: screw and presence detector pin. Inclination is assured by assembly if presence detector pin is correctly inserted while screwing. Presence detector bottom is aligned with holder bottom.

Prerequisite: NA

Adjustment tool: NA

Procedure:

1. Loose the screw attaching presence detector to presence detector holder.
2. Move presence detector until pin is properly inserted detector holder and detector bottom is aligned with detector holder.
3. Tighten the screw.

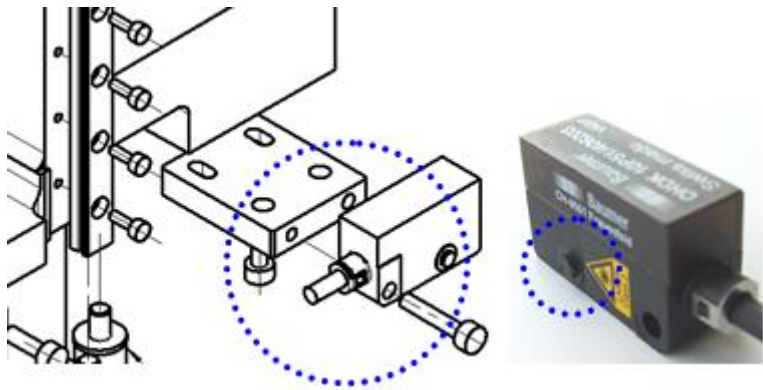


Figure 336. Card presence detector inclination

8.2.2 Functional adjustments

In this section, the functional adjustments related to the card transport module are explained.

All the adjustments determining given set of coordinates are stored in *CATCalibrations.txt* file found in *E:\data\Erytra\conf*.

It is assumed in all the adjustment procedures that the proper conditions for movement are met, that is gripper arm is homed and their movements can be completed without risk of collision.

To home gripper arm, click on *Home* button in *Gripper Arm* tab → *Movement* subtab and perform a home of all gripper arm axes.

CAUTION!

Check incubators position to avoid them to interfere with CAT movement and lead to a crash.

8.2.2.1 Gripper Adjustment in Card drawers area

This adjustment will allow the software to determine coordinates of two card drawer known positions by attaching card drawer adjustment tool to these two drawer positions and inserting gripper in the tool.

This adjustment must be performed when:

- It is disassembled the gripper.
- It is disassembled the CAT robotic head.
- It is modified the position of the latch of the card drawer.

Prerequisite:

- CAT module homed.
- Gripper parallelism verification.
- Card drawer flatness verification.

Adjustment tool: Cards drawer adjustment tool (Ver.2) (PN: 233878)
Procedure:

1. Click on *drawers* button found in Technical Service → *Procedures* tab → *CAT floor Modules* section. Following window is displayed:

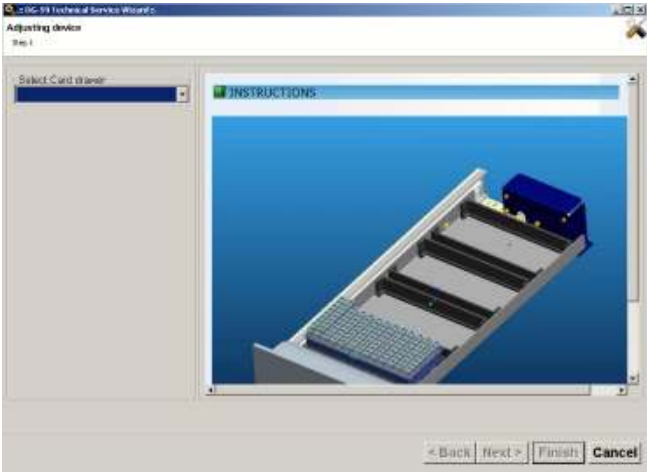


Figure 337. Card drawer selection in TSS

2. Select card drawer to adjust and click on *Next>*.
3. Insert Cards drawer adjustment tool in rear rack position.
4. Move gripper manually or using the cursor buttons to fit it with adjustment tool (pins inserted, gripper bottom lays on adjustment tool). When placing the gripper over the tool, gripper fingers are not involved so they do not grip the tool. When placing the gripper over the tool, be careful to not raise robot (the gripper just rests over the tool).

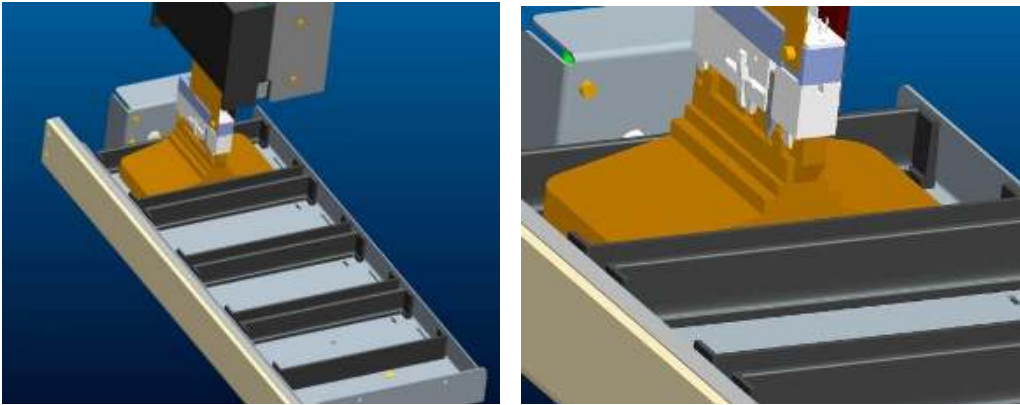


Figure 338. Rear position of card drawer adjustment tool in the drawer

5. Start the automatic adjustment pressing *Next>* button on the screen. A warning message informing that the instrument will perform movements/settings appears on the screen. Press *OK* button. Automatic adjustment is as follows:
 - a. Gripper moves in Z axis until it is detected the position where the gripper stops and no movement is detected (in encoder counts) if the gripper is asked to move further down. The system stores this first Z adjustment in order to allow the system to correctly adjust X and Y axes.
 - b. Then the gripper is moved in X and Y axes until it is detected the position where the affected axis stops when the gripper is discentered and then moved down to fit the adjustment tool.
 - c. Finally, Z axis is adjusted automatically again using the X and Y values of the previous step.

- The system repeats, at least, the previous procedure twice (maximum 6 executions). Obtained values are compared and, if they are smaller than an established threshold, it is considered that the adjustment is completed. Otherwise the procedure is repeated and results compared to previous execution.
- Once the process finishes, the gripper performs a home in Z axis. Remove cards drawer adjustment tool from the rear position and place it in front rack position.
- Move gripper manually or using the cursor buttons to fit it with adjustment tool (pins inserted, gripper bottom lays on adjustment tool). When placing the gripper over the tool, gripper fingers are not involved so they do not grip the tool. When placing the gripper over the tool, be careful to not raise robot (the gripper just rests over the tool).

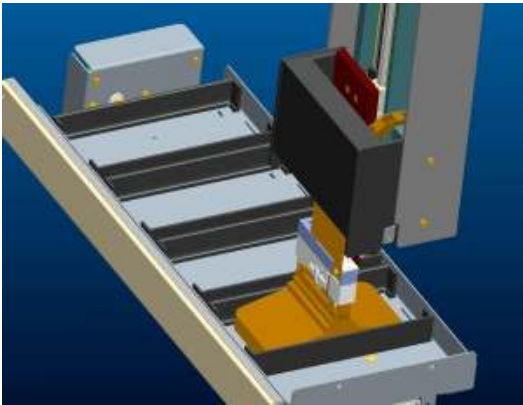


Figure 339. Front position of the cards drawer adjustment tool in the drawer

- Start the automatic adjustment pressing *Next>* button on the screen. Automatic adjustment is performed again.
- Once the process is finished, a screen appears showing the parameters to be stored. Press *Finish* button and press *OK* on the confirmation window. When the adjustments are stored, press *OK* on the information window.
- Press *OK* in the confirmation window that indicates that home movements will be performed. Gripper will perform a home.
- Press *OK* to finish the adjustment procedure.



NOTE:

Adjustment stored in *E:\data\Erytra\conf\CATCalibrations.txt*



CAUTION!

Remove the adjustment tool from the instrument when the adjustment procedure finishes.

8.2.2.2 Gripper Adjustment in incubator area

This adjustment allows the software to determine coordinates of a known position by inserting tool to incubator slot 6 and matching gripper with adjustment tool.

This adjustment must be performed when:

- It is disassembled the gripper.
- It is disassembled the CAT robotic head.
- It is disassembled the incubator module/block.

Prerequisites:

- CAT module homed.
- Gripper parallelism verification.
- Incubator flatness verification.
- Incubator homed.

Adjustment tool: Lower centrifuge / Incubator adjustment tool (Ver.3) (PN: 733068)

Procedure:

1. Click on *incubators* button found in Technical Service *Procedures* tab → *CAT floor Modules* section. Following window is displayed:



Figure 340. Incubator adjustment selection in TSS

2. Select incubator to adjust and click on *Next>*. A warning message informing that the instrument will perform a home of the selected incubator appears on the screen. Press *OK* button and then *Next>* button.



CAUTION!

Check CAT arm position to avoid it to interfere with incubator movement and lead to a crash.

3. Insert incubator section of centrifuge/incubator adjustment tool in incubator position 6.
4. Move gripper manually or using the cursor buttons to fit it with adjustment tool (pins inserted, gripper bottom lays on adjustment tool. Tool has been designed so gripper fingers are not used). Be careful to not raise robot.

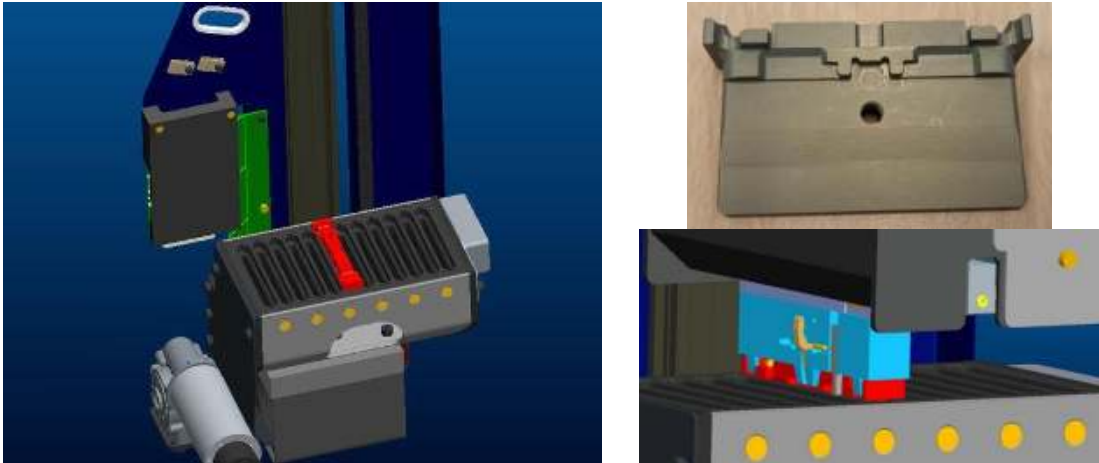


Figure 341. Incubator adjustment in TSS

5. Start the automatic adjustment pressing *Next>* button on the screen. Automatic adjustment is as follows:
 - a. Gripper moves in Z axis until it is detected the position where the gripper stops and no movement is detected (in encoder counts) if the gripper is asked to move further down. The system stores this first Z adjustment in order to allow the system to correctly adjust X and Y axis.
 - b. Then the gripper is moved in X and Y axis until it is detected the position where the affected axis stops when the gripper is discentered and then moved down to fit the adjustment tool.
 - c. Finally, Z axis is adjusted automatically again using the X and Y values of the previous step.
6. The system repeats, at least, the previous procedure twice (maximum 6 executions). Obtained values are compared and, if they are smaller than an established threshold, it is considered that the adjustment is completed. Otherwise the procedure is repeated and results compared to previous execution.
7. Once the process is finished, a screen appears showing the parameters to be stored. Press *Finish* button and press *OK* on the confirmation window. When the adjustments are stored, press *OK* on the information window.
8. Press *OK* in the confirmation window that indicates that home movements will be performed. Gripper will perform a home.
9. Press *OK* to finish the adjustment procedure.



NOTE:

Adjustment stored in *E:\data\Erytra\conf\CATCalibrations.txt*



CAUTION!

Remove the adjustment tool from the instrument when the adjustment procedure finishes.

8.2.2.3 Gripper Adjustment in Card waste container area

This adjustment allows the software to determine coordinates of the card waste container once the gripper has been placed in a known position.

This adjustment must be performed when:

- The gripper is disassembled.
- The CAT robotic head is disassembled.
- The card disposal guide/nozzle is disassembled.

Prerequisites:

- CAT module homed.
- Gripper parallelism verification.

Adjustment tool: NA

Procedure:

1. Click on *waste tray* button found in Technical Service *Procedures* tab → *CAT floor Modules* section. It will open a wizard adjustment. Press *Next>* button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press *OK* button.
2. Move gripper manually or using keypad following the steps listed below:
 - a. Mover the gripper arm so that the rear part of the gripper leans against the top surface of the rear wall of the waste tray.

- b. Move the gripper along X axis as far to the right as possible, until it touches the card waste right wall.

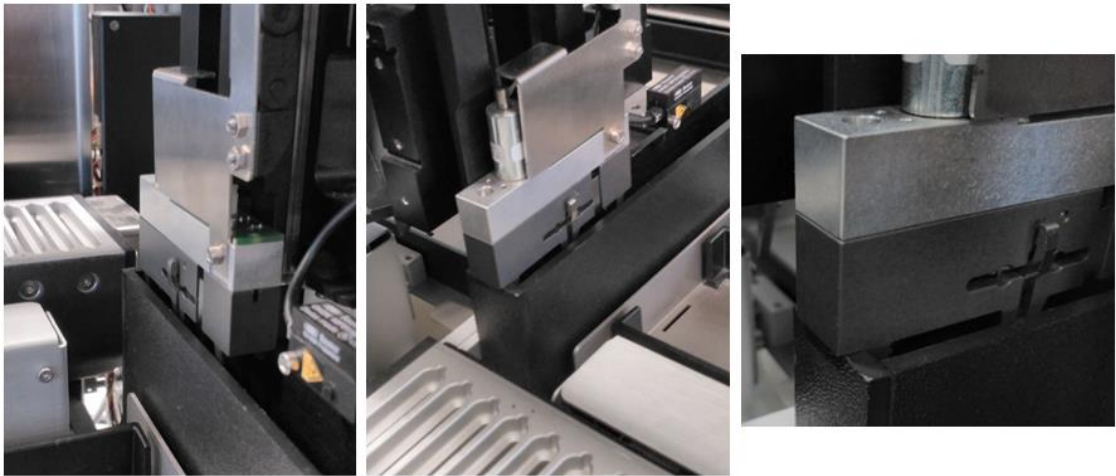


Figure 342. Gripper Adjustment in Card waste container (removed cover)

3. Press *Next>* button. A screen appears showing the parameters to be stored. Press *Finish* button and press *OK* on the confirmation window. When the adjustments are stored, press *OK* on the information window.
4. Press *OK* in the confirmation window that indicates that home movements will be performed. Gripper will perform a home.
5. Press *OK* to finish the adjustment procedure.



NOTE:

Adjustment stored in *E:\data\Erytra\conf\CATCalibrations.txt*

8.2.2.4 Gripper adjustment in centrifuge area

This adjustment allows software to determine coordinates of centrifuge position 1 (gripper and centrifuge).

This adjustment must be performed when:

- It is disassembled the gripper.
- It is disassembled the CAT robotic head.
- It is disassembled the centrifuge.

Prerequisites:

- CAT module homed.
- Centrifuges homed.
- Gripper parallelism verification.
- Centrifuge rotor height adjustment.
- Centrifuge with all the holders.

Adjustment tool: Centrifuge / Incubator adjustment tool (Ver.3) (PN: 733056)

Procedure:

1. Click on *centrifuges* button found in Technical Service *Procedures* tab → *CAT floor Modules* section.



Figure 343. Centrifuge/incubator adjustment tool (left picture) and centrifuge adjustment tab in TS (right picture)

2. Select centrifuge to adjust and click on *Next>*. A warning message informing that the instrument will perform movements/settings appears on the screen. Press *OK* button. Centrifuge will perform a home.
3. Centrifuge will move slot 1 to load/unload position. Insert manually whole/assembled centrifuge/incubator adjustment tool set in the more centred slot.
4. Move gripper manually or using the cursor buttons to fit it with adjustment tool. When placing the gripper over the tool, gripper fingers are not involved so they do not grip the tool. Click on *Next>* button.

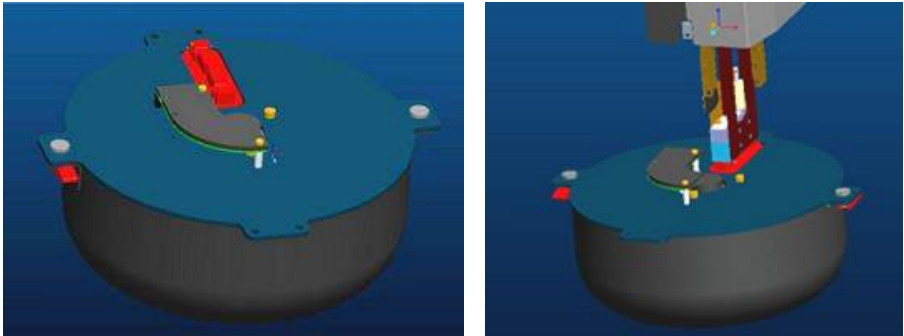


Figure 344. Gripper wizard adjustment in centrifuge in TSS

5. A screen appears showing the parameters to be stored. Press *Finish* button and press *OK* on the confirmation window. When the adjustments are stored, press *OK* on the information window.
6. Press *OK* in the confirmation window that indicates that home movements will be performed. Gripper will perform a home.
7. Press *OK* to finish the adjustment procedure.



NOTE:

Adjustment stored in *E:\data\Erytra\conf\CATCalibrations.txt*



CAUTION!

Remove the adjustment tool from the instrument when the adjustment procedure finishes.

8.2.2.5 Card presence detector – gripper parallelism

Prerequisite: Gripper parallelism verification.

Adjustment tool: Cards drawer adjustment tool (Ver.2) (PN: 233878)

Procedure:

1. Insert cards drawer adjustment tool in the front position of any card drawer.

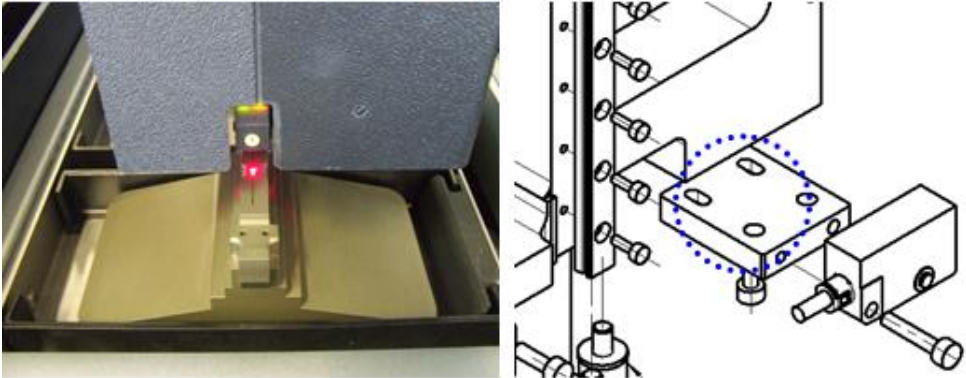


Figure 345. Card presence detector parallelism adjustment

2. Move gripper manually to fit it with adjustment tool (pins inserted, gripper bottom lays on adjustment tool).
3. Perform a home of Z axis in TSS, energize all the CAT axes and then deenergize just Y axis.
4. Move manually robotic head in Y axis so presence detector light spot illuminates (click on *set on* button in *Gripper Arm* tab → *Presence detector* subtab → *presence detector* area) the section of cards drawer adjustment tool that contains central slot.
5. Loose the two screw fixing presence detector holder to Gripper robotic head and move holder backwards/forward until presence detector beam spot illuminates card drawer gripper adjustment tool central slot.
6. Tighten the two screw fixing presence detector holder.



CAUTION!

Remove the adjustment tool from the instrument when the adjustment procedure finishes.

8.2.2.6 Card presence detector height detection

This adjustment must be performed when:

- Card presence detector is disassembled.
- CAT robotic head is disassembled.

Prerequisite: Card presence detection inclination adjustment.

Tools: Cards drawer adjustment tool (Ver.2) (PN: 233878)

Procedure:



NOTE:

The card presence detector device comprises two LEDs:

- Green LED should always be ON as long as the drivers are started, indicating that presence detector device is activated.
- Orange LED will only light (ON mode) when presence is detected.

1. Insert cards drawer gripper adjustment tool in the front position of any card drawer.
2. Activate the presence detector clicking on *set on* button in *Presence detector* area of *Presence detector* subtab of *Gripper Arm* tab.
3. Access the presence detector screw and turn it slowly clockwise (remove the sealant if any) until a 'click' is heard.
4. De-energize gripper arm (*Gripper Arm* tab → *Movement* subtab → *Position Actions* area → *unlock/stop* button and select all axes) and move it so that presence detector laser beam spot illuminates adjustment area 1 (see below picture).

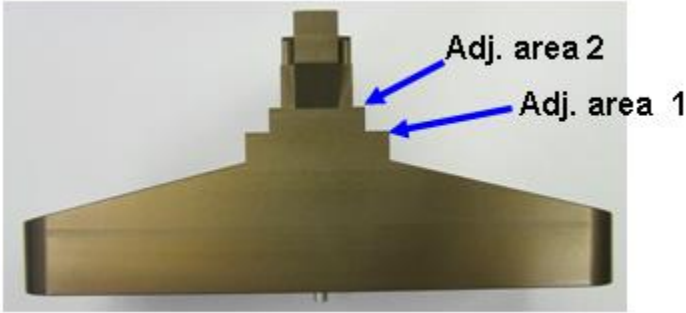


Figure 346. Card presence detector height adjustment areas

5. Turn slowly the presence detector screw counter-clockwise until orange LED is permanently OFF (presence is not detected).
6. Move the gripper arm so that presence detector laser beam illuminates adjustment area 2, checking that presence is detected (orange LED lights uniformly and no blinking is observed). Otherwise, go back to step 3 to start over with the adjustment procedure.

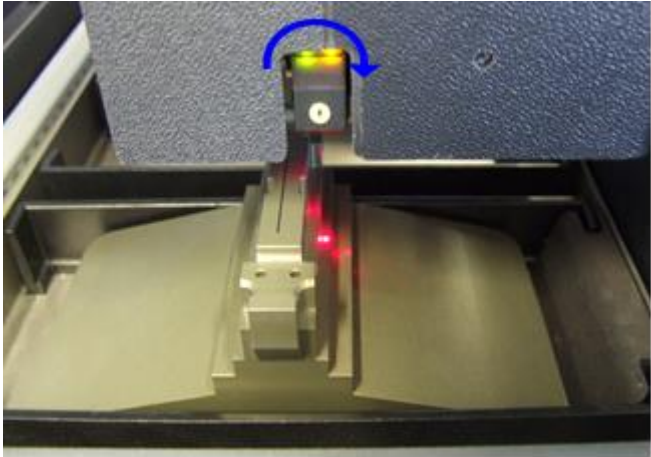


Figure 347. Presence detector over adjustment area 2

7. Move alternatively the CAT's laser beam between adjustment areas 1 and 2, checking that orange LED is OFF over adjustment area 1 and ON over adjustment area 2.
8. Deactivate the presence detector clicking on *set off* button in *Presence detector* area of *Presence detector* subtab of *Gripper Arm* tab.
9. Consider applying a sealant to the potentiometer.



CAUTION!

Remove the adjustment tool from the instrument once the adjustment procedure is finished.

8.2.2.7 Card barcode reader inclination

This adjustment must be performed when:

- It is disassembled the card barcode reader.

Prerequisite: NA

Tools: NA

Procedure:

1. Remove right CAT robotic head cover (fixed by 6 screws, 3 per side).
2. De-attach barcode reader assembly (removing carefully left CAT robotic head cover) to gain access to barcode reader. Keep barcode reader connected to PCA CAT-TOW59.
3. Activate barcode reader: Click on Technical Service *set on* button in *Gripper Arm* tab → *BCR* subtab.
4. Loose the two screws fixing barcode reader position and move barcode reader so laser beam is focused on the center of the reflection mirror.

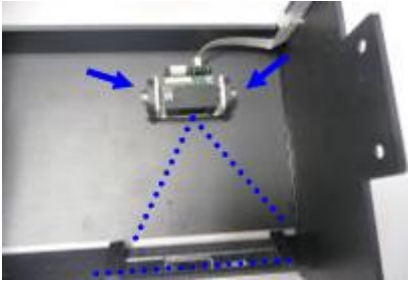


Figure 348. Barcode reader inclination adjustment

5. Tighten the two screws and deactivate barcode reader.
6. Assemble carefully left CAT robotic head cover and then right CAT robotic head cover.

8.2.2.8 Gripper adjustment in Card Reader area

This adjustment must be performed when:

- It is disassembled the gripper.
- It is disassembled the reader module.

Prerequisites:

- CAT homed.
- Incubators homed.
- Gripper parallelism verification.
- Reader adjusted.

Tools: Reader adjustment tool-pattern v2 (PN: 731398).

Procedure:

1. Select *Procedures* tab in Technical Service and in section *CAT floor Modules*, click on *acquisition (central) position* button. Following window will be displayed.



Figure 349. Acquisition (central) position wizard – screen 1

- De-energize Gripper Arm motors, clicking on *unlock/stop* button in *Gripper Arm* tab → *Movement* subtab → *Position Actions* and selecting all axes.
- Open CAT floor door (opening first the user tray and then removing the screw fixing intermediate door).
- Attach manually reader adjustment tool-pattern to gripper as indicated in the picture (pattern side looking at the left, screws heads looking at right).

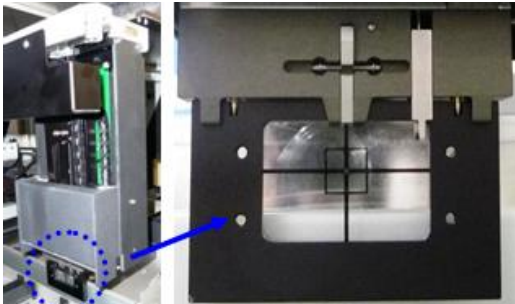


Figure 350. Gripper adjustment in card reader area

CAUTION!

In next step robot head will move without performing automatically a gripper home.

- Perform a home of the gripper (*Gripper Arm* tab → *home* button → *z* axis). Click on *Next*>. A warning message informing that the instrument will perform movements/settings appears on the screen. Press *OK* button. The adjustment is automatic and consists of determining gripper coordinates that sets the pattern in the desired position (current pixels/mm conversion factor matches adjustment one). Wait for adjustment completion.
- At adjustment completion, following window appears:



Figure 351. Acquisition (central) position wizard – screen 2

- Remove reader adjustment tool-pattern from the gripper activating the electromagnet (manually or using Technical Service software). Click on *Next*>.

- 8. A screen appears showing the parameters to be stored. Press *Finish* button and press *OK* on the confirmation window. When the adjustments are stored, press *OK* on the information window.
- 9. Press *OK* in the confirmation window that indicates that home movements will be performed.
- 10. Press *OK* to finish the adjustment procedure.

**NOTE:**
Adjustment stored in *E:\data\Erytra\conf\CATCalibrations.txt*

8.3 VERIFICATIONS

It is assumed in all the verification procedures that the proper conditions for movement are met, that is gripper arm is homed and their movements can be completed without risk of collision.

To home gripper arm, click on *home* button in *Gripper Arm* tab → *Movement* subtab and perform a home of all gripper arm axes.

**CAUTION!**
Check incubators position to avoid them to interfere with CAT movement and lead to a crash.

8.3.1 Gripper parallelism

- 1. De-energize CAT robotic head using *unlock/stop* button *Gripper Arm* tab → *Movement* subtab → *Move Actions* and selecting all axes.
- 2. Open card drawer 2.
- 3. Lower gripper until gripper edges (left or right) touch gently centrifuge cover surface. Check that gripper edges (left or right) touch centrifuge cover surface at the same time.

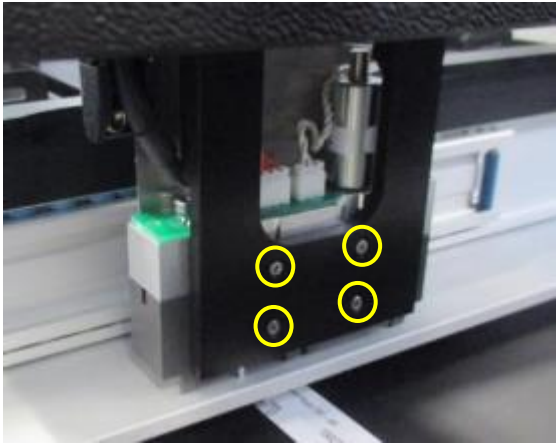


Figure 352. Gripper parallelism verification

Should this not be the case, loose the four screws that fix the gripper (see picture below), align the gripper and tighten the screws again.
Once the alignment is correct a threadlocking adhesive (such as Loctite 222) must be applied to the for screws.



NOTE:
No more than one screw should be removed at a time when applying the sealant. Remove the first screw, apply the adhesive, screw it back in and continue with the next screw.



8.3.2 Card drawers area

1. Insert a card rack with all the cards in the lower floor in the selected card drawer rack 1 position.
2. Scan the selected card drawer – rack 1 (*Gripper Arm* tab → *Presence Detector* subtab).
3. Select desired card drawer, rack 1 and card 1 in *Gripper Arm* tab → *Movement* subtab → *Move Actions* fields and click on *move to* button. Press *Yes* on confirmation window.
4. Click on *get card* button to take the card.
5. Select *CAMERA*, *READ* and *WELLS_1_2* in *Gripper Arm* tab → *Movement* subtab → *Move Actions* section and click on *Move to* button. Press *Yes* on confirmation window.
6. Lower gripper selecting *down* button in *Gripper Arm* tab → *Movement* subtab → *Position Actions* area.
7. Click on *real time image* button in *Procedures* tab → *Checks* section → *Reader Module* area to obtain camera image.

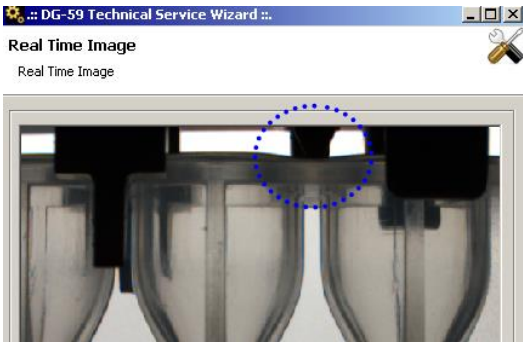


Figure 353. Gripper adjustment in card drawers verification

8. Check that gripper pin is centered in the midpoint between well 1 and 2. Should this not be the case, readjust gripper in that card drawer. Press *Cancel* in the real time image window to close it and then *OK* in the confirmation window.
9. Remove card from reader selecting *up* button in *Gripper Arm* tab → *Movement* subtab → *Position Actions* area.
10. Bring card back to the rack by selecting desired card drawer, rack 1 and card 1 in *Gripper Arm* tab → *Movement* subtab → *move Actions* fields, clicking on *move to* button, *Yes* on confirmation window and then *leave card* button.

11. Move the card rack to rack 4 position of selected drawer and repeat steps 2 to 10 selecting rack 4 instead of rack 1 and card 23 instead of card 1.

8.3.3 Incubators area

1. Select *Incubators* tab and the subtab related to the incubator area to check.
2. Select *Home* button in *Movement Area*.



CAUTION!

Check CAT arm position to avoid it to interfere with incubator movement and lead to a crash.

3. Insert a card rack with all the cards in the lower floor in the selected card drawer rack 1 position.
4. Scan the card drawer 1 – rack 1 (*Gripper Arm* tab → *Presence Detector* subtab).
5. Take a card from the rack selecting card drawer 1, rack 1 and desired card position in *Gripper Arm* tab → *Movement* subtab → *Move Actions* fields, clicking on *move to* button, press *Yes* on confirmation window and then clicking on *get card* button.
6. Insert the card in incubator slot 1 selecting the incubator to check and slot 1 in *Gripper Arm* tab → *Movement* subtab → *Move Actions* fields, clicking on *move to* button, press *Yes* on confirmation window and then clicking on *leave card* button. Check that card is properly leaved.
7. Take another card from the rack selecting card drawer 1, rack 1 and desired card position in *Gripper Arm* tab → *Movement* subtab → *Move Actions* fields, clicking on *move to* button, press *Yes* on confirmation window and then clicking on *get card* button.
8. Insert the card in incubator slot 12 selecting the incubator to check and slot 12 in *Gripper Arm* tab → *Movement* subtab → *Move Actions* fields, clicking on *move to* button, press *Yes* on confirmation window and then clicking on *leave card* button. Check that card is properly leaved.
9. Select incubator to check and slot 1 in *Gripper Arm* tab → *Movement* subtab → *Move Actions* fields and click on *move to* button. Press *Yes* on confirmation window.
10. Click on *get card* button to grip the card.
11. Select *CAMERA*, *READ* and *WELLS_1_2* in *Gripper Arm* tab → *Movement* subtab → *Move Actions* fields and click on *move to* button. Press *Yes* on confirmation window.
12. Lower gripper selecting *down* button in *Gripper Arm* tab → *Movement* subtab → *Position Actions* area.
13. Click on *real time image* button in *Procedures* tab → *Checks* section → *Reader Module* area to obtain camera image.

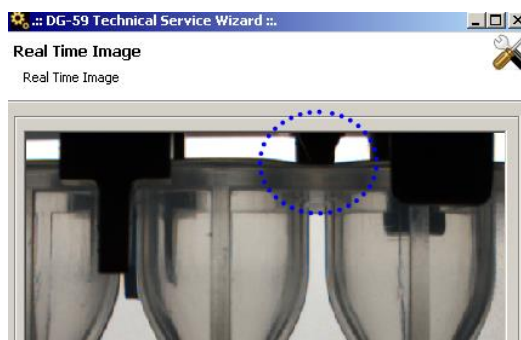


Figure 354. Gripper adjustment in incubator verification

14. Check that gripper pin is centered in the midpoint between well 1 and 2. Press *Cancel* in the real time image window to close it and then *OK* in the confirmation window.
15. Remove card from reader selecting *up* button in *Gripper Arm* tab → *Movement* subtab → *Position Actions* area.

16. Bring card back to the rack by selecting card drawer 1, rack 1 and desired card position in *Gripper Arm* tab → *Movement* subtab → *Move Actions* fields, clicking on *move to* button, press *Yes* on confirmation window and then *leave card* button.
17. Repeat steps 9 to 16 taking card from incubator slot 12 instead of 1 and leaving card in a free card rack position.

8.3.4 Card waste container area

1. Insert a card rack with all the cards in the lower floor in the selected card drawer rack 1 position.
2. Scan the desired rack of the card drawer (*Gripper Arm* tab → *Presence Detector* subtab).
3. Use the robot to get and dispose the cards in the waste container position and check proper performance:
 - a. Select desired card drawer, rack position and card position and click on *move to*. Press *Yes* on confirmation window.
 - b. Click on *get card* button to take the card.
 - c. Select *WASTE_TRAY* position in *Movement* area and click on *move to* button. Press *Yes* on confirmation window.
 - d. Dispose card by clicking on *leave card* button.
 - e. Repeat previous steps until all cards have been disposed.

8.3.5 Centrifuges area

1. Click on *home* button in *Centrifuges* tab → Centrifuge 1/2 subtab and perform a home of the centrifuge.
2. Insert a card rack with all the cards in the lower floor in the selected card drawer rack 1 position.
3. Scan the card drawer 1 – rack 4 (*Gripper Arm* tab → *Presence Detector* subtab).
4. Take a card from the rack selecting card drawer 1, rack 4 and desired card position in *Gripper Arm* tab → *Movement* subtab → *Move Actions* fields, clicking on *move to* button, press *Yes* on confirmation window and then clicking on *get card* button.
5. Insert the card in centrifuge slot 1 selecting the centrifuge to check and slot 1 in *Gripper Arm* tab → *Movement* subtab → *Move Actions* fields, clicking on *move to* button, press *Yes* on confirmation window and then clicking on *leave card* button. Check that card is properly leaved.
6. Move centrifuge slot 6 to loading/unloading position by selecting the centrifuge to check and slot 6 in *Gripper Arm* tab → *Movement* subtab → *Move Actions* fields and clicking on *move to* button. Press *Yes* on confirmation window.
7. Remove the card from the centrifuge slot 1 selecting the centrifuge to check and slot 1 in *Gripper Arm* tab → *Movement* subtab → *Move Actions* fields, clicking on *move to* button, press *Yes* on confirmation window and then clicking on *get card* button.
8. Move card to reader position selecting 'Camera', 'Read' and 'Wells 1_2' in *Gripper Arm* tab → *Movement* subtab → *Move Actions* fields, clicking on *move to* button, press *Yes* on confirmation window and lower gripper selecting *down* button in *Gripper Arm* tab → *Movement* subtab → *Position Actions* area.

- Click on *real time image* button in *Procedures* tab → *Checks* section → *Reader Module* area to obtain camera image.
- Check that gripper pin is centered in the midpoint between well 1 and 2. Press *Cancel* in the real time image window to close it and then *OK* in the confirmation window.

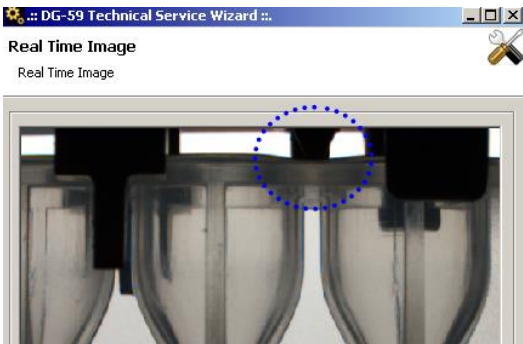


Figure 355. Gripper adjustment in centrifuge verification

- Remove card from reader selecting *up* button in *Gripper Arm* tab → *Movement* subtab → *Position Actions* area.
- Bring card back to the rack by selecting card drawer 1, rack 4 and desired card position in *Gripper Arm* tab → *Movement* subtab → *Move Actions* fields, clicking on *move to* button, press *Yes* on confirmation window and then *leave card* button.

8.3.6 Card presence detector inclination

- Check that presence detector bottom and presence detector holder bottom are aligned.

8.3.7 Card presence detector alignment – gripper parallelism

- Insert cards drawer gripper adjustment tool in the front position of any card drawer.



Figure 356. Card presence detector parallelism checking.

- Move gripper manually to fit it with adjustment tool (pins inserted, gripper bottom lays on adjustment tool).
- Perform a home of Z axis in TSS, energize all the CAT axes and then deenergize just Y axis.
- Move manually robotic head in Y axis so presence detector light spot illuminates (click on *set on* button in *Gripper Arm* tab → *Presence detector* subtab → *presence detector* area) the section of cards drawer adjustment tool that contains central slot.
- Check that presence detector beam spot illuminates card drawer gripper adjustment tool central slot.



CAUTION!

Remove the adjustment tool from the instrument when the verification procedure finishes.

8.3.8 Card presence detector height detection



NOTE:

When presence detector is activated, green LED lights. Orange LED lights for a moment and:

- if no detection occurs, it stops lightning.
- If detection occurs, it keeps lightning.

1. Insert cards drawers gripper adjustment tool in the front position of any card drawer.
2. Activate presence detector clicking on *set on* button in *Gripper Arm* tab → *Presence detector* subtab → *Presence detector* area.
3. De-energize CAT arm (in Technical Service, *unlock/stop* button in *Gripper Arm* tab → *Movement* subtab → *Position Actions* area and selectin all axes) and move it so presence detector laser beam spot illuminates adjustment area 2 (see below picture).

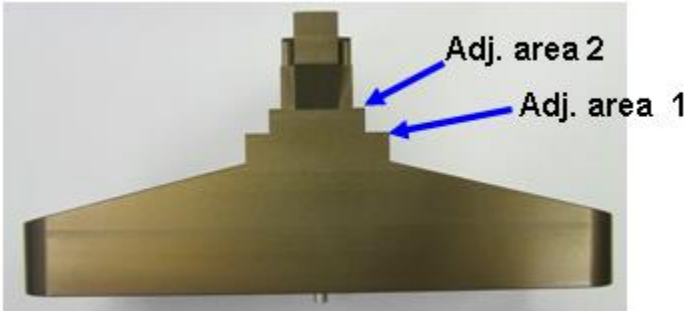


Figure 357. Card presence detector height adjustment

4. Check that presence is detected (orange LED lights).

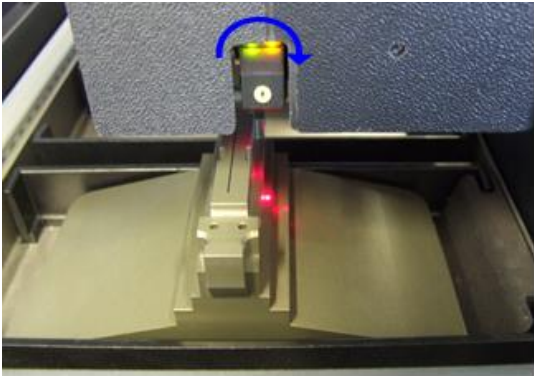


Figure 358. Presence detected over adjustment area 2

5. Move gripper arm so presence detector laser beam spot illuminates adjustment area 1.
6. Check that no presence is detected (orange LED does not light).



CAUTION!

Remove the adjustment tool from the instrument when the verification procedure finishes.

8.3.9 Card barcode reader inclination

1. Activate barcode reader: Click on Technical Service *Set on* button in *Gripper Arm* tab → *BCR* subtab.
2. Check that barcode reader laser beam is focused on the center of the reflection mirror (see section 8.2.2.7).

8.3.10 Reader area

1. Perform a home of the CAT module.
2. Move the incubators to home position, clicking on *Home* button in *Incubators* tab → *Incubator X* subtab.



CAUTION!

Check CAT arm position to avoid it to interfere with incubator movement and lead to a crash.

3. De-energize Gripper Arm motors, clicking on *unlock/stop* button in *Gripper Arm* tab → *Movement* subtab → *Move Actions* and selecting all axes.
4. Open CAT floor door (opening first the user tray and then removing the screw fixing intermediate door).
5. Attach reader adjustment tool-pattern to gripper as indicated in the picture (pattern side looking at the left, screws heads looking at right).



Figure 359. Gripper adjustment in card reader area



CAUTION!

In next step robot head will move without performing a gripper home. Check gripper Z position to avoid a potential crash with any intermediate floor element.

6. Click on *reference check* button in *Procedures* tab → *Checks* area → *Reader Module* section. Instrument will show on screen information of the 3 axes (X, Y and Z) of the adjustment. In case any of the axes return a result of *false*, perform gripper adjustment in reader area.
7. Remove reader adjustment tool-pattern from the gripper activating the electromagnet (manually or using Technical Service software). Click on *Next*.

- Click on *read/check cards test* button in *Procedures* tab → *Checks* area → *Reader Module* section. Following window will be displayed.

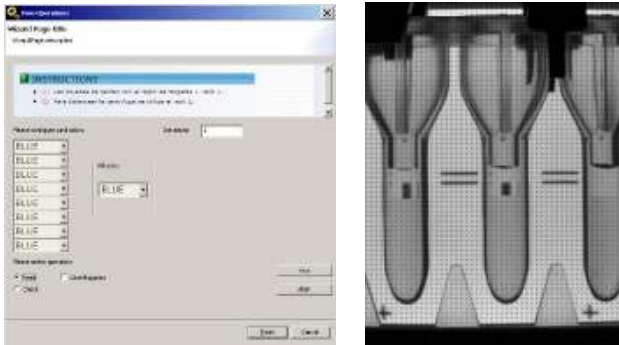


Figure 360. Read/Check card test configuration window and wells picture

- Insert a rack with several cards in card drawer 1, rack 1 position.
- Configure microtubes color and the number of iterations.
- Select *Check* option and click on *start* button. The cards will be moved to reader for initial check positions. The obtained pictures will be displayed/saved. Check proper pattern location inside the wells.

8.4 REPLACEMENT PROCEDURES

8.4.1 Gripper leadscrew replacement



CAUTION!

Do not handle (or gently handle) Y arm by card reader barcode assembly induced stresses may break card barcode mirror. Barcode reader mirror is fragile.

In order to replace gripper leadscrew it is necessary to remove the robotic head from the instrument. To do so, follow the next steps:

- Remove CAT robotic head right cover (fixed by 6 screws, 3 per side).
- De-attach barcode reader assembly from the robotic head (four screws) to gain access to barcode reader. Unplug barcode reader from PCA CAT-TOW59. Be careful with connector flap/flange.
- Remove screws that attach Y chain to Z tower/robotic head.

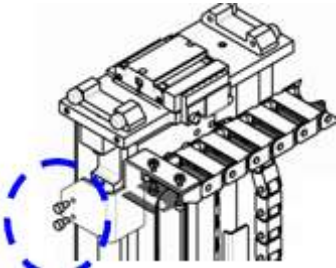


Figure 361. Location of the screws that attach Y chain to Z tower/robotic head

4. Disconnect Y chain ribbon cables from PCA CAT-TOW59.



Figure 362. Location of Y chain ribbon cables at PCA CAT-TOW59

5. Remove Y arm guide front end stopper (remove two screws, see dotted arrows).

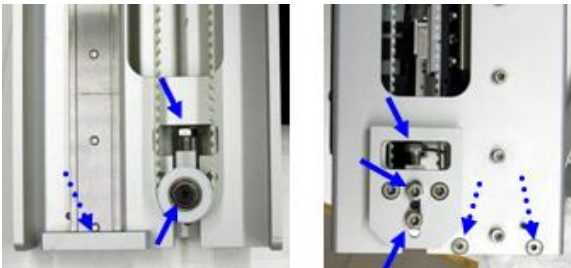


Figure 363. Y arm guide

6. Remove Y belt pulley removing its fixing screw. To ease removal loose belt tension screw and both tensor assembly fixing screws (solid arrows).
7. Remove robot head.

Replace leadscrew

1. Remove two screws fixing gripper guide assembly to leadscrew nut flange.



Figure 364. Gripper guide assembly and leadscrew nut flange

2. Remove leadscrew pinion set screw and remove pinion.
3. Remove three screws fixing leadscrew holder to robotic head bulk. Leadscrew is now free.

4. Extract leadscrew and leadscrew holder from their location, loose holder screw and remove holder. A rubber hammer may be used to ease set release.

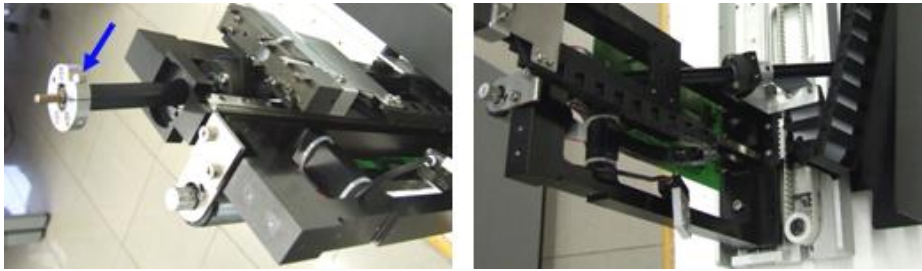


Figure 365. Leadscrew removing process

5. Remove leadscrew. Gripper guide assembly (lineal guide wagon) can be removed from the guide. Wagon has wires to prevent wagon bearings falling.



Figure 366. Leadscrew

6. Remove 3 screws fixing leadscrew nut flange to leadscrew nut and remove flange.
7. Dispose leadscrew.
8. Insert leadscrew nut flange to new leadscrew checking proper orientation. Flange built-in gripper axis home flag.
9. Assemble flange by means of the three screws (check proper screws insertion direction). Do not tighten the screws completely.
10. Insert leadscrew in robotic head.
11. Insert leadscrew holder in leadscrew aligning holder notches and leadscrew and levelling both part external edges (see following picture).



Figure 367. Adjustment of the leadscrew in the leadscrew holder

12. Tighten holder screw.
13. Insert leadscrew and holder in their location. Holder tightening screw head must be located in the side closer to gripper. A rubber hammer may be used to help to insert set.
14. Fix holder to the robotic head.
15. Insert pinion in the leadscrew and level it with leadscrew end. Fix it tightening set screw to the flat side.
16. Attach gripper assembly to the leadscrew: insert and tighten two screws.
17. Tighten screws attaching leadscrew nut flange to nut.
18. Assemble Y arm/Robot head back following removal procedure steps in reverse order.

8.4.2 Card barcode reader mirror replacement

1. Remove Gripper Arm robotic head right lid (fixed by 6 screw, 3 per side)
2. De-attach barcode reader assembly from the robotic head (four screws) to gain access to barcode reader. Unplug barcode reader from PCA CAT-TOW. Be careful with connector flap/flange.

- 3. Remove CAT mirror wedge with pliers (if installed).
- 4. Remove the mirror through the slot pulling it up and then to the right.



Figure 368. Barcode reader mirror

- 5. Insert the mirror with reflective coating side/first side facing barcode reader. Reflective coating side/first side can be easily identified: touch mirror with an object. If the tip of the object seems to touch reflected image, that is mirror coating side/first side.



Figure 369. First face of the mirror (left) and second face of the mirror (right)

- 6. Place CAT mirror wedge (if available) with pliers, checking that it exits from the surface of the cover about 3-4 mm.
- 7. Place left robotic head lid with its screws and connect carefully barcode reader cable to PCA CAT-TOW59.
- 8. Finally, place right robotic head lid with its screws.

8.4.3 CAT X chain replacement

- 1. Switch off the instrument and disconnect it from the mains.
- 2. Remove middle rear instrument cover.
- 3. Open intermediate door.
- 4. Remove interconnection cover (X-Y chains).

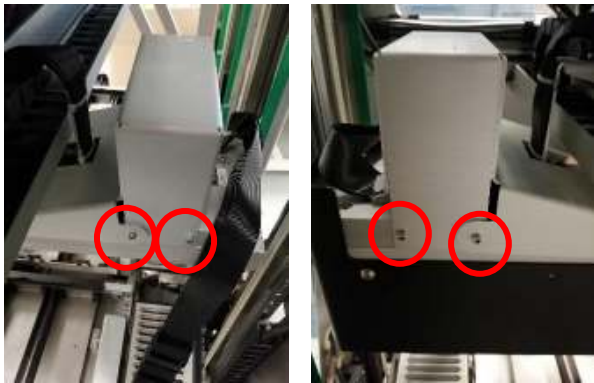


Figure 370. Interconnection cover

5. Disconnect the 3 flat cables from PCA CAT-CONY59 and remove cable ties.



Figure 371. Flat cables and ties

6. Disengage X chain from the last chain slot.



Figure 372. X chain slot

7. Open cable clamp and disconnect the 3 flat cables from PCA GEN-CONX59. If needed remove the left lateral methacrylate cover and the User tray to have a better access to this area (refer to section 4.4.11 User tray replacement).

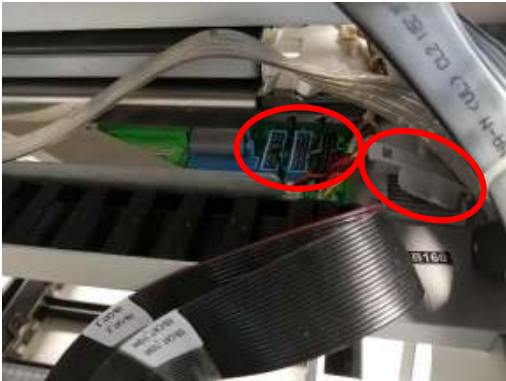


Figure 373. Flat cables and cable clamp

8. Remove 2 screws that attach X chain.



Figure 374. X chain screws

9. Remove X chain.
10. Place new X chain and assemble back all the parts.

CAUTION!

Do not swap ribbon cables in PCA GEN-CONX59 or PCA CAT-CONY59 when connecting them (it could damage different parts of the instrument), specially cables labelled as XB-CAT_XY and XB-CAT_Z that have both a blue connector. Cable labelled as XB-CAT_TOW has a black connector. Cables should match the connector colour when connected to the PCA.

8.4.4 CAT Y chain replacement

1. Switch off the instrument and disconnect it from the mains.
2. Remove middle rear instrument cover.
3. Open intermediate door.
4. Remove right cover of the CAT module (8 screws).

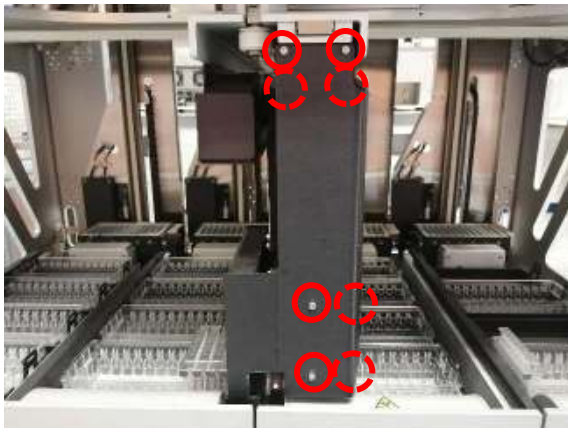


Figure 375. Right cover of the CAT module

5. Disconnect the 2 flat cables from PCA CAT-TOW59 of the CAT module.



Figure 376. Flat cables

6. Disengage Y chain from the last chain slot.



Figure 377. Y chain slot

7. Remove 2 screws from Y chain and open cable clamp.

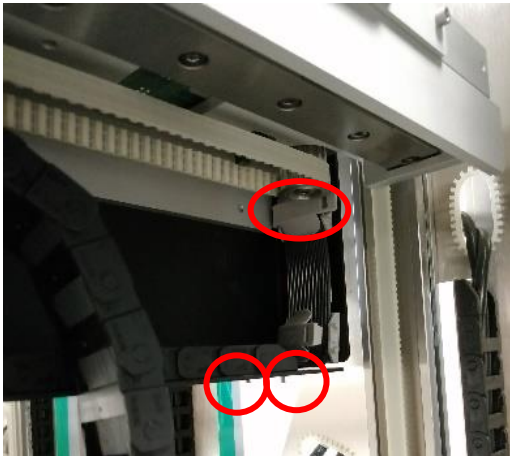


Figure 378. Y chain screws and cable clamp

8. Remove interconnection cover (X-Y chains).

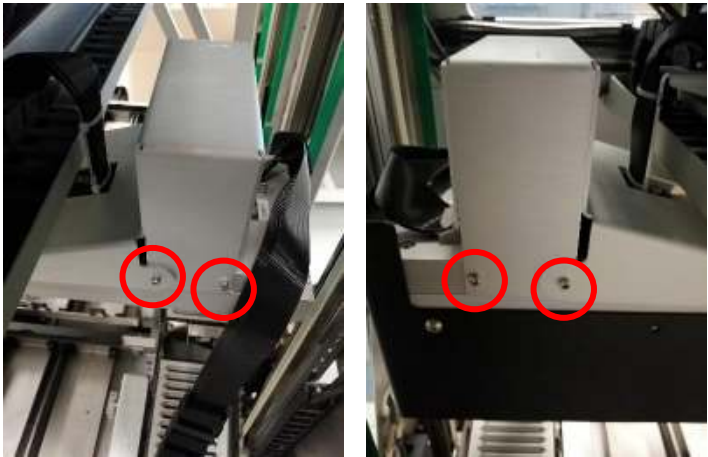


Figure 379. Interconnection cover screws

9. Disconnect 2 flat cables from PCA CAT-CONY59 and open cable clamp.

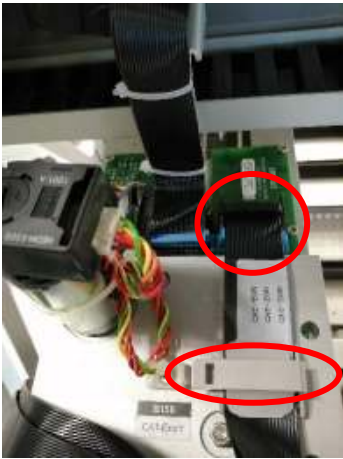


Figure 380. Flat cables and cable clamp

- 10. Remove Y chain.
- 11. Place the new Y chain and assemble back all the parts.

CAUTION!

Do not swap ribbon cables in PCA CAT-CONY59 or PCA CAT-TOW59 when connecting them (it could damage different parts of the instrument).

Cables should match the connector colour when connected to the PCA: cable labelled as CAT-TOW black and CAT-Z blue.

8.4.5 Gripper (Z) chain replacement

- 1. Switch off the instrument and disconnect it from the mains.
- 2. Remove middle rear instrument cover.
- 3. Open intermediate door.
- 4. Remove right cover of the CAT module (8 screws).

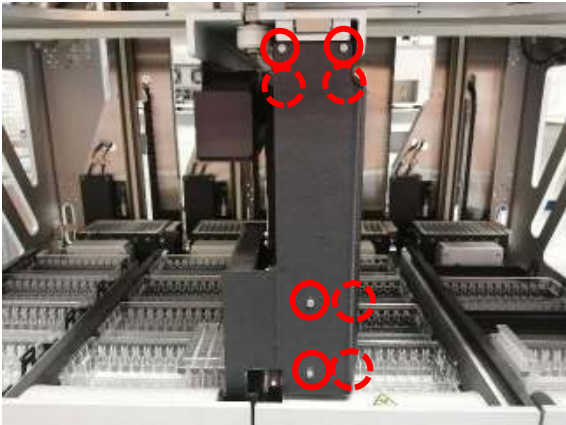


Figure 381. Right cover of the CAT module

- 5. Remove left cover of the CAT module (4 screws)



Figure 382. Left cover of the CAT module

- 6. Pull it apart carefully not damaging the barcode reader cable that is attached to this cover.



Figure 383. Barcode reader cable

7. Remove screw of the metallic plate



Figure 384. Metallic plate screw

8. Disconnect cables



Figure 385. Gripper cabling (1)

9. Disconnect cables.

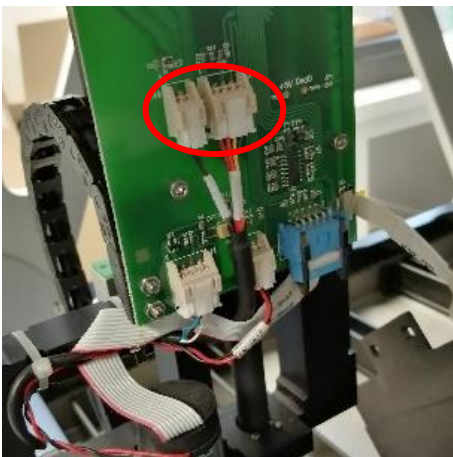


Figure 386. Gripper cabling (2)

- 10. Remove gripper Z chain.
- 11. Place the new Z chain and assemble back all the parts.
- 12. Verify gripper performance.

8.4.6 Gripper replacement

- 1. Switch off the instrument and disconnect it from the mains.
- 2. Open intermediate door.
- 3. Open card drawer 1.
- 4. Remove right cover of the CAT module (8 screws)

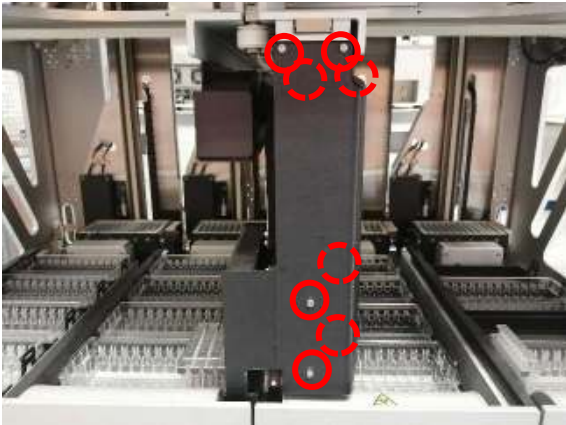


Figure 387. Right cover of the CAT module

- 5. Pull it apart carefully not damaging the barcode reader cable that is attached to this cover.



Figure 388. Barcode reader cable

- 6. Move down the gripper manually by means of the gripper motor pinion.
- 7. Remove screw of the metallic plate



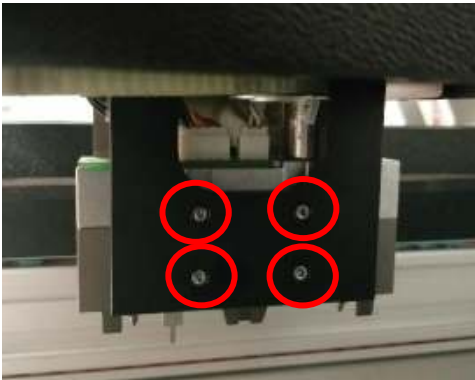
Figure 389. Metallic plate screw

8. Disconnect cables



Figure 390. Gripper cabling

- 9. Remove the 4 screws that attach the gripper to the CAT Module.
- 10. Remove the gripper together with its PCA.
- 11. Remove the PCA from the gripper.
- 12. Attach the PCA to the new gripper.
- 13. Place the new gripper and its PCA to the CAT module and tighten the 4 screws of the gripper. Consider applying a sealant to the 4 screws (once gripper parallelism verification are correct).



- 14. Assemble back all the parts and reconnect cabling.
- 15. Perform Gripper parallelism verification.
- 16. Perform Gripper adjustments in all positions.
- 17. Verify gripper performance.

8.4.7 Toothed belt T2.5/245 (CAT X-axis)

- 1. Switch off the instrument and disconnect it from the mains.
- 2. Open intermediate door.
- 3. Remove small belt.



Figure 391. Small belt

- 4. Place the new small belt.

8.4.8 Toothed belt T2.5/245 (CAT Y-axis)

- 1. Switch off the instrument and disconnect it from the mains.
- 2. Open intermediate door.
- 3. Remove the middle rear cover.
- 4. From the front of the instrument, take out the Y-belt from its position by removing the fixing screw of the pulley.



Figure 392. CAT large belt

- 5. Remove the large belt from the pinion. The belt will be hang on the CAT robotic head.



Figure 393. Large belt (pinion)

6. Remove interconnection cover (X-Y chains).



Figure 394. Interconnection cover screws

7. Remove motor support screws.



Figure 395. Support screws

8. Disconnect 2 flat cables, open cable clamp and pull apart the cables.

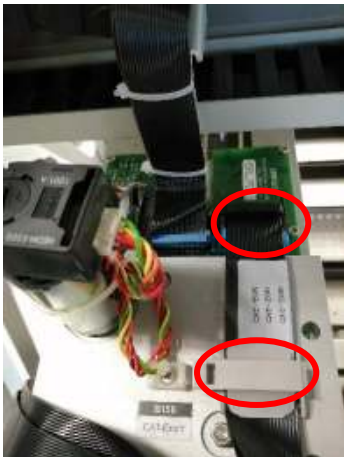


Figure 396. Cabling and cable clamp

9. Remove the motor screw (slotted hole).



Figure 397. Motor screw

10. Move the motor to be able to remove the small belt.
11. Place the new small and large belts and assemble back all the parts.



NOTE:
If the pulley tightening screw is in the same position before disassembly, the belt tension will remain unchanged, else belt tension must be adjusted by modifying the end pulley position.



Figure 398. Pulley tightening screw



NOTE:
Any variation of the belt tension may modify gripper adjustments in all positions.

12. Adjust tension of the small belt.
13. Perform card transport check.

8.5 FUSES

Fuses related to CAT module are detailed in the following table,

PCA	FUSE NAME	DESCRIPTION
PCA GEN-BCKPL59	FUS21	MST250 0034.6615 1A/250V
	FUS22	MST250 0034.6620 3.15A/250V
PCA CAT-CM CONTROL	FUS1	MSF250 0034.6013 0.8A/250V
PCA GEN-CONX59	NA	NA
PCA CAT-CONY59	NA	NA
PCA CAT-TOW59	NA	NA
PCA PINZA57	NA	NA

Table 63. List of fuses of CAT module

8.6 LEDS

LEDs related to CAT module are detailed in the following table,

PCA	LED	DESCRIPTION
PCA GEN-BCKPL59	LED13	Green 5V (FUS21)
	LED14	Green 24V (FUS22)
PCA CAT-CM CONTROL	LED1	Yellow, When the instrument is switched on, LED is ON and a few seconds later is OFF. This means that the FPGA has been loaded correctly with the appropriate data.
	LED2	Green, VMotor Ok. It indicates that the voltage supplied to the motor is $\geq 12V$.
PCA GEN-CONX59	NA	NA
PCA CAT-CONY59	NA	NA
PCA CAT-TOW59	NA	NA
PCA PINZA57	NA	NA

Table 64. List of LEDs of CAT module

8.7 JUMPERS AND TEST POINTS

Neither jumpers nor test points are related to CAT module.

8.8 WIRING TABLE

In the following table, there is detailed all the cables related to CAT module. A wiring diagram can be found attached to Service Manual.

ID	FROM CONNECTOR	FROM PCA/ COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
B-CAT_TOW	CAT_TOW	GEN-BCKPL59	H5 In CAT/PIP-TOW	PCA GEN-CONX59
B-CAT_Z	CAT_Z	GEN-BCKPL59	H4 In CAT/PIP-Z	PCA GEN-CONX59
B-CAT_XY	CAT_XY	GEN-BCKPL59	H3 In CAT/PIP-XY	PCA GEN-CONX59
XB-CAT_TOW	H8 Out CAT/PIP-TOW	PCA GEN-CONX59	H5	PCA CAT-CONY59
XB-CAT_Z	H7 Out CAT/PIP-Z	PCA GEN-CONX59	H4	PCA CAT-CONY59
XB-CAT_XY	H6 Out CAT/PIP-XY	PCA GEN-CONX59	H3	PCA CAT-CONY59
BRX-EncX	H2 EncX	PCA GEN-CONX59	Encoder	X Axis motor
BRX-EXT + BRX-MotX	H1 MotX	PCA GEN-CONX59	Motor powering	X Axis motor
CAT-MotY	H1 MotY	PCA CAT-CONY59	Motor powering	Y Axis Motor
CAT-EncY	H2 EncY	PCA CAT-CONY59	Encoder	Y Axis Motor
CAT-Z	H6 Out CAT-Z	PCA CAT-CONY59	H7 CAT-Z	PCA CAT-TOW59
CAT-TOW	H7 Out CAT-TOW	PCA CAT-CONY59	H6 CAT-TOW	PCA CAT-TOW59
CAT-PRES	H1 PresDet	PCA CAT-TOW59	Presence detector	Presence detector.
CAT-BCR	H8 BarCode	PCA CAT-TOW59	Barcode reader	Barcode reader
CAT-MotZ	H5 MotZ	PCA CAT-TOW59	Motor powering	Gripper motor
CAT-EncZ	H2 EncZ	PCA CAT-TOW59	Encoder	Gripper motor
CAT-CARD	H3 CardDet	PCA CAT-TOW59	J2	PCA PINZA57
CAT-SOL	H4 Sol	PCA CAT-TOW59	J1	PCA PINZA57
NA	Solenoid	Solenoid	J3	PCA PINZA57

Table 65. List of wiring of CAT module

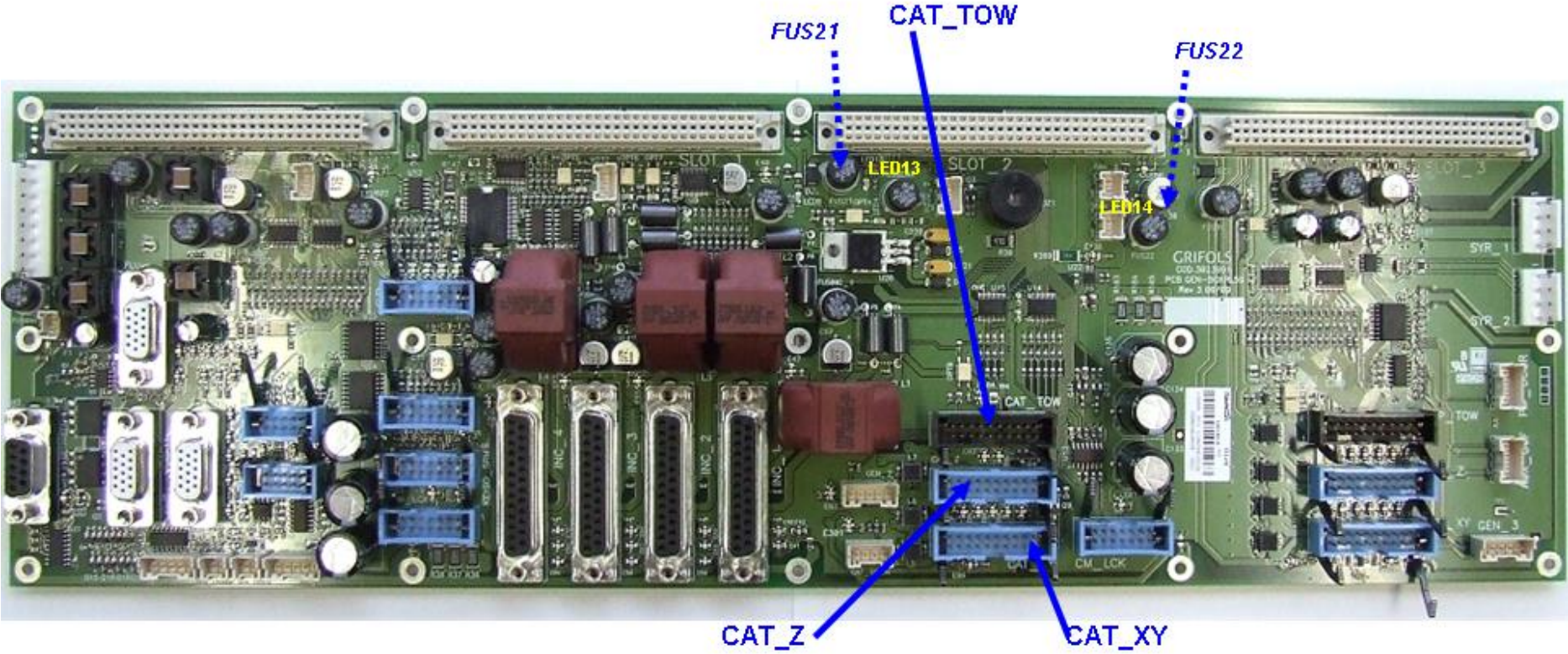


Figure 399. Connectors, fuses and LEDs of PCA GEN-BCKPL59 related to CAT module

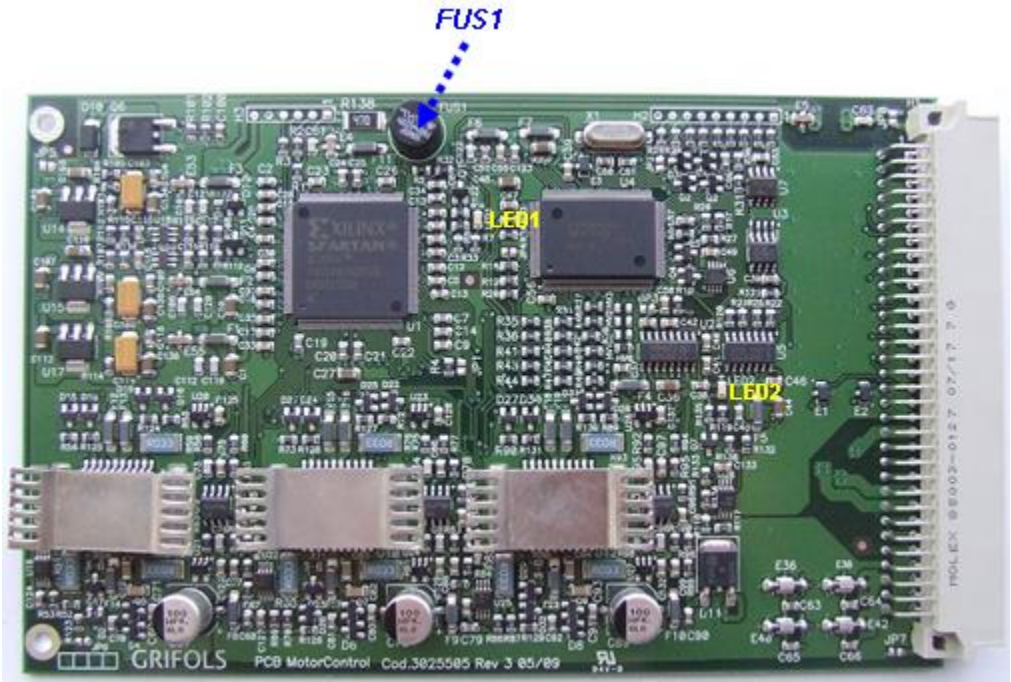


Figure 400. Location of the fuse and LEDs in PCA CAT-CM Control

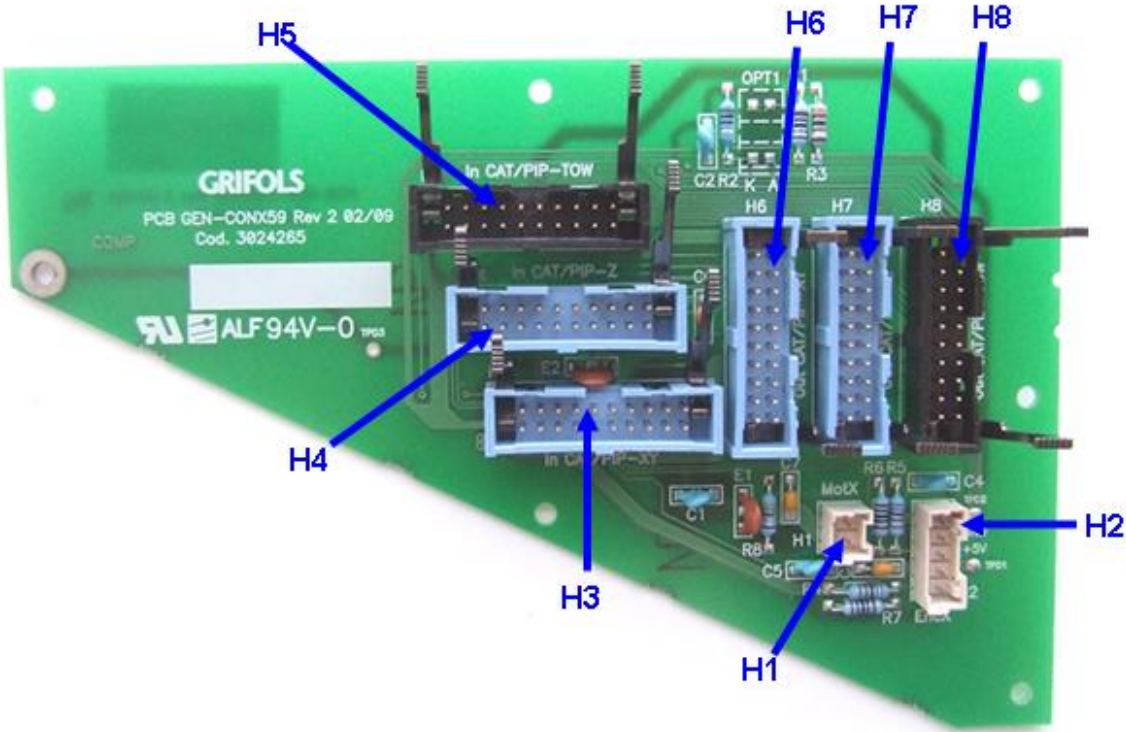


Figure 401. Location of the connectors in PCA GEN-CONX59

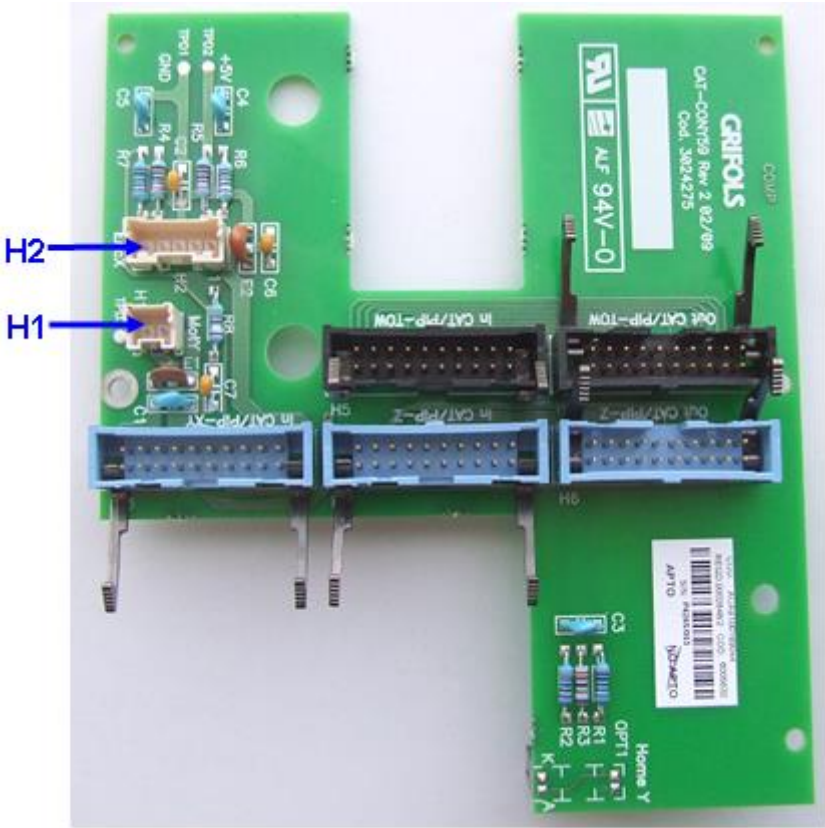


Figure 402. Location of the connectors in PCA CAT-CONY59

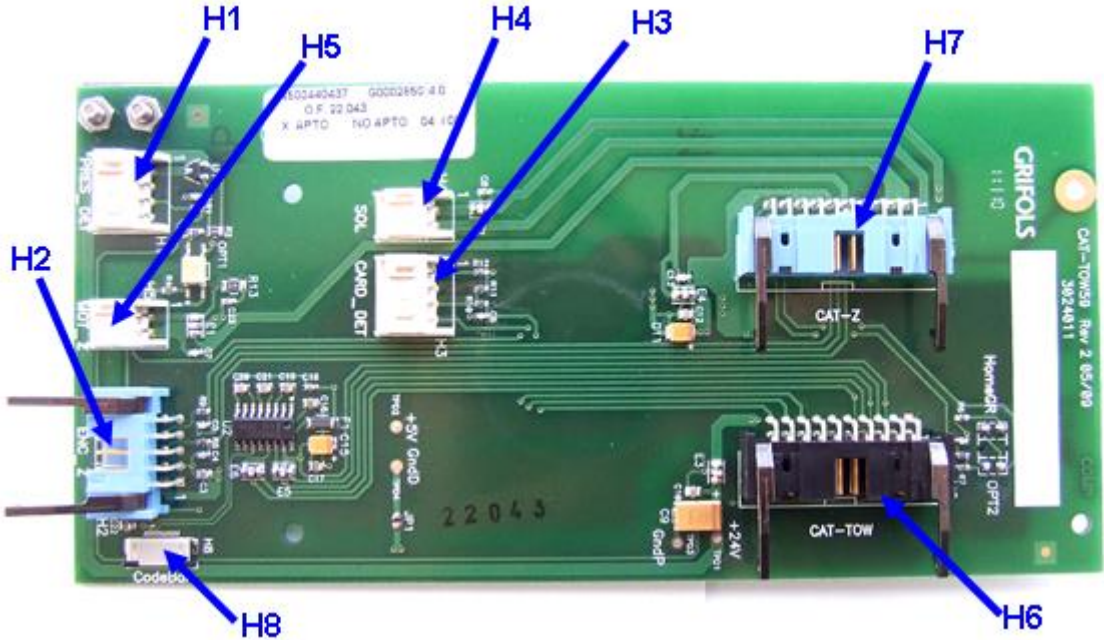


Figure 403. Location of the connectors in PCA CAT-TOW59

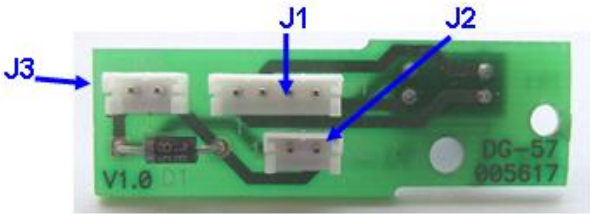


Figure 404. Location of the connectors in PCA PINZA57

8.9 TROUBLESHOOTING

Module	Device	Error code	Error	Cause	Text	Options
CAT (02)	XYZ Motors (01) X axis (02) Y axis (04) Z axis (03) Gripper (05) Presence detector during X movements (07) Presence detector during Y movements	003000	ERRORCODE_LEN	FW has received a command with incorrect length.	Error during the lower robotic arm movement. Please, inspect the instrument middle level. Please choose one of the following options:	RETRY CANCEL Works (it leads to: <i>Gripper disabled</i>)
		005000	ERRORCODE_ARG	Command has wrong arguments.		
		006000	ERRORCODE_BUSY	Command is addressed to a busy device.		
		007000	ERRORCODE_NOTHOMED	Move command is addressed to a not homed device.		
		008000	ERRORCODE_NOTSAFE	Command is not safe for the instrument.		
		011000	ERRORCODE_DATANOTREADY	The data requested is not ready.		
		032000	CURRENT_LIMIT	Current limit at servo device.		
		033000	POSITION_ERROR	A position error has been produced at a servo device.		
		034000	MOVEMENT_ERROR	Movement error produced at servo device.		
		035000	HOME_TIMEOUT	Home timeout has been produced at a servo device.		
		036000	ERRORCODE_SRV_SYNC_TIMEOUT	Synchronization error has been produced at a servo device.		
		037000	ERRORCODE_SRV_UNKNOWN_ERROR	Unknown error has been produced at a servo device.		
		041000	ERRORCODE_PD_SYNC_TIMEOUT	Presence detector has been busy for more than its timeout margin.		
		042000	ERRORCODE_PD_UNKNOWN_ERROR	Unknown error produced during a presence detector related operation.		
		047000	ERRORCODE_GRP_UNKNOWN_ERROR	Unknown error.		

Module	Device	Error code	Error	Cause	Text	Options
CAT (02)	XYZ Motors (01) X axis (02) Y axis (04) Z axis (03) Gripper (05) Presence detector during X movements (07) Presence detector during Y movements	009000	ERRORCODE_CARDPRESENT	Card present in the gripper.	No error reported when reading BCs from cards racks.	NA
					If error is detected in incubator or centrifuge, reported error is Error in the lower robotic arm when taking a card. A card is already present.	RETRY CANCEL Works
		010000	ERRORCODE_NOCARDPRESENT	There is no card present in the gripper.	No error reported when reading BCs from cards racks.	NA
					If error is detected in incubator or centrifuge, following error is reported. Error in the lower robotic arm when releasing a card. Please inspect middle level for lost card.	RETRY CANCEL Works
		044000	CARD_LOST	The gripper has lost a card.	No error reported when reading BCs from cards racks.	NA
					If error is detected in incubator or centrifuge, reported error is Error in the lower robotic arm when taking a card. Please inspect middle level for lost card.	RETRY CANCEL Works
					If error is detected in reader, reported error is Card lost while reading. Please select one of the following options:	

Module	Device	Error code	Error	Cause	Text	Options
CAT (02)	XYZ Motors (01) X axis (02) Y axis (04) Z axis (03) Gripper (05) Presence detector during X movements (07) Presence detector during Y movements	045000	CARD_NOTLEFT	Card not left during a left card operation.	Error in the lower robotic arm when releasing a card. Please inspect middle level for lost card.	RETRY CANCEL Works Release card
		046000	CARD_NOTFOUND	Card not found during a take card operation.	<i>No error is reported when reading BCs from cards racks.</i> <i>If error is detected in incubator or centrifuge, reported error is</i> Error in the lower robotic arm when taking a card.	NA RETRY CANCEL Works
		003000	ERRORCODE_LEN	FW has received a command with incorrect length.	<i>No error is reported when reading BCs from card racks. If the error is detected in incubator or centrifuge, following error is reported if card has not been previously cancelled:</i> Card barcode not expected. Card will be discarded and samples cancelled.	RETRY CANCEL Works (it leads to: Gripper disabled)
		005000	ERRORCODE_ARG	Command has wrong arguments.		
		006000	ERRORCODE_BUSY	Command is addressed to a busy device.		
		011000	ERRORCODE_DATANOT READY	Data requested is not ready.		
	BCR (06)	048000	ERRORCODE_BCR_DRIVER_ERROR	Driver error.		
		049000	ERRORCODE_BCR_NOT_INIT	Barcode reader is not initialized		
		050000	ERRORCODE_BCR_COMM_ERR	Barcode reader communication error.		
		051000	ERRORCODE_BCR_NO_BARCODE	No barcode is returned by the reader.		
		052000	ERRORCODE_BCR_SYNC_TIMEOUT	Barcode reader has been busy more than its timeout margin.		
		053000	ERRORCODE_BCR_UNKNOWN_ERROR	Barcode reader unknown error.		

Module	Device	Error code	Error	Cause	Text	Options
Any (xx)	Any (xx)	002000	ERRORCODE_CMD	Unknown command code received.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		003000	ERRORCODE_LEN	FW has received a command with incorrect length.		
		004000	ERRORCODE_DEV	Command is addressed to an incorrect device.		
		005000	ERRORCODE_ARG	Command has wrong arguments.		
		014000	ERRORCODE_HW_DEV	The addressed device has a problem with its related HW.		
		015000	ERRORCODE_HW_HWF AIL	There is a problem with PCA HW.		
		016000	ERRORCODE_HW_HWN OTSET	PCA setup error.		
Non firmware errors (xx)	Non firmware errors (xx)	000000	ERRORCODE_UNKNOW N	Unknown error.		
		008000	ERRORCODE_FW_NOTS AFE	Movement not safe.		
		017000	ERRORCODE_FW_UNKN OWN_ERROR	Unknown error.		
		061000	ERRORCODE_INVALID_ ARGUMENTS	Invalid arguments.		
		062000	ERRORCODE_FWCOMM TIMEOUT	Communication timeout.		
		063000	ERRORCODE_FWCOMM CONNECTION LOST	Communication lost.		
		064000	ERRORCODE_FWCOMM COMM_ERROR	Communication error.		
		088000	ERRORCODE_PRECOND ITION_FAILED	Previous condition for the action is not met.		
		089000	ERRORCODE_STARTUP ERROR	Driver startup error.		
		091000	ERRORCODE_ALREADY STARTED	Another instance of Driver is already opened.		
		090000	ERRORCODE_UNEXPEC TED_ERROR	A Driver unexpected error has been produced.		
0	240					

Table 66. List of errors related to CAT module

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Movement error / current limit error (X axis) 0201032000 0201034000	Fuse blown	Check / replace FUS21 or FUS22
	Large belt too tight	Check tightness of belt / Replace belt
	Incorrect X motor assembly	Check/replace X motor silentblocks
	Lack of lubrication in X linear guide	Lubricate linear guide
	Motor damaged	Check / replace motor
	CAT X-Y flat cable damaged	Check / replace CAT X-Y cable
	X linear guide damaged	Check / replace linear guide
Movement error / current limit error (Y axis) 0202032000 0202034000	Fuse blown	Check / replace FUS21 or FUS22
	Small belt too tight	Check tightness of belt / Replace belt
	Large belt too tight	Check tightness of belt / Replace belt
	Lack of lubrication in Y linear guide	Lubricate linear guide
	Motor damaged	Check / replace motor
	CAT X-Y flat cable damaged	Check / replace CAT X-Y cable / X CAT chain
	Y linear guide damaged	Check / replace linear guide
Movement error / current limit error (Z axis) 0204032000 0204034000	Fuse blown	Check / replace FUS21 or FUS22
	Belt too tight	Check tightness of belt / Replace belt
	Lack of lubrication in Z linear guide	Lubricate linear guide
	Lack of lubrication in leadscrew	Lubricate leadscrew
	Motor damaged	Check / replace motor
	CAT-TOW flat cable damaged	Check / replace CAT-TOW flat cable
	CAT-Z flat cable damaged	Check / replace X- Y CAT chain
	Z linear guide damaged	Check / replace linear guide
	Leadscrew damaged	Check / replace leadscrew
X axis home is not detected 0201035000	Home flag is not aligned with optical barrier	Check position of the home flag
	Motor does not turn	Check / replace motor
	Faulty belt	Check / replace small / large belt
	Failure in optical barrier	Check / replace PCA GEN-CONX59
	Flat cable damaged	Check / replace CAT X-Y cable
Y axis home is not detected 0202035000	Home flag is not aligned with optical barrier	Check position of the home flag
	Motor does not turn	Check / replace motor
	Faulty belt	Check / replace belt
	Failure in optical barrier	Check / replace PCA CAT-CONY59
	Flat cable damaged	Check / replace X CAT chain (CAT X-Y cable)
Z axis home is not detected 0204035000	Home flag is not aligned with optical barrier	Check position of the home flag
	Motor does not turn	Check / replace motor
	Faulty belt	Check / replace belt
	Failure in optical barrier	Check / replace PCA CAT-TOW59
	Flat cable damaged	Check / replace Z CAT chain (CAT-Z cable)

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Cards are not taken from card drawer/rack	Card rack is not placed correctly	Place correctly card rack in the drawer
		Check / readjust / fix CAT presence detector
	Incorrect card rack detection	Check card drawer flatness
		Check / perform CAT adjustment in card drawer
	Gripper is not parallel	Check gripper parallelism
	Card drawer is not flat	Check card drawer flatness
	Lack of lubrication in gripper	Lubricate gripper
Cards are not taken from incubator	Wrong adjustment	Check / perform CAT adjustment in card drawer
	Gripper is not parallel	Check gripper parallelism
	Incubator is not flat	Check incubator flatness
	Lack of lubrication in gripper	Lubricate gripper
Cards are not taken from centrifuge	Wrong adjustment	Check / perform CAT adjustment in incubator
	Gripper is not parallel	Check gripper parallelism
	Card holder defective	Check / replace card holders
	Rotor is not correctly assembled	Check rotor assembly
	Lack of lubrication in gripper	Lubricate gripper
When a card is taken, it is not detected	Wrong adjustment	Check / perform CAT adjustment in centrifuge
	Home detector or connections broken	Replace PCA PINZA57
Card is lost during transport	Pin bent	Replace gripper pin
	Wrong CAT arm adjustment	Check/perform CAT arm adjustment
Cards tilted in the gripper	Gripper pins bent or springs not equal	Replace pins / springs
	Wrong CAT arm adjustment	Check/perform CAT arm adjustment
Cards are not detected / incorrectly detected	Gripper is not parallel	Check gripper parallelism
	Fuse blown	Check / replace FUSGEN4
	Card rack is not placed correctly	Place correctly card rack in the drawer
	Presence detector is not correctly assembled	Check presence detector assembly
	Adjustment of the presence detector is not correct	Perform presence detector adjustment
	Wrong CAT arm adjustment	Check/perform CAT arm adjustment
	Presence detector is damaged	Replace presence detector
Barcodes are not read	PCA CAT-TOW59 is damaged	Replace PCA CAT-TOW59
	Card rack / card is not placed correctly	Place correctly card rack / rack
	Barcode label in bad conditions	Check barcode label
	Mirror is dirty	Clean mirror
	BCR is not properly assembled	Check BCR assembly
	BCR is not working	Check / replace BCR
	BCR cable damaged / disconnected	Check / replace BCR cable
0206051000	PCA CAT-TOW59 damaged	Replace PCA CAT-TOW59

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
BCR is not switched on	Fuse blown	Check / replace FUS21 or FUS22
	BCR cables damaged / disconnected	Check / replace BCR cables
	BCR is not working	Check / replace BCR
	PCA CAT-TOW59 is damaged	Replace PCA CAT-TOW59
	CAT-TOW / CAT-Z cable damaged	Check / replace X / Y CAT chains

Table 67. CAT module troubleshooting

9 FLUIDICS (FLU) MODULE

9.1 DESCRIPTION

In this section a description of all the parts related to the fluidics module and a description of the fluidics procedures performed by the *Erytra* are detailed.

9.1.1 Parts description

The fluidics module is decomposed by the following submodules:

- **Manifold:** Acrylic block (PMMA) where fluidic components are integrated. These components are electrovalves, expansion vessels, dispensation pumps, sensors and bottles configuration selector.
- **Electrovalves:** There are 18 electrovalves that allow to pass or to block the liquid by the fluidic system. Block mounting body. Nominal voltage: 24 VDC. Electrovalves involved in the dispensing part are closed by firmware 30 seconds after its activation in order to avoid undesired overflows.
EV1, EV2, EV3, EV4, EV5, EV6, EV7, EV8, EV9, EV10, EV13, EV14, EV15, EV16, EV17, EV18, EV19 and EV20.
- **Dispensation pumps:** there are 2 dispensation pumps to drive solution A and B to the fluidic system. The type of the pump is diaphragm for liquids. The motor of the pump is a DC motor with a nominal voltage of 24V. Pump is screwed to a support that holds it in the correct position. The pump is screwed to the manifold by means of clamp. Clamp and seals used are capable to guarantee watertight union in working conditions. PM1 and PM3.



Figure 405. Dispensation pumps

- **Expansion vessels:** Pressure is regulated by means of the dispensation pumps. As the pump head is membrane type, the pressure given to the system is not constant but pulsed. In order to absorb these pulses, an expansion vessel is placed at each pump outlet.



Figure 406. Expansion vessels

- **Pressure / vacuum sensors:** They incorporate an overpressure characteristic that, in case of an electronic failure, it controls pressure levels over nominal range without damage. Working pressure range: ± 15 psi (~ 1034 mbar). S3, S4, S5 and S6.
- **Bottles configuration selector:** change manually the use of 2 bottles of the *Erytra* between waste solution (bottles 1 and 2) and wash solution (solution A and solution B).



Figure 407. Bottles configuration selector

In order to configure the bottles configuration selector if the external drain option is activated (check *DUAL_MODE* parameter in chapter 2), follow next steps:

1. Perform instrument decontamination.
2. Empty fluidic system and then disconnect the bottles.
3. Switch off the instrument and disconnect it from the mains.
4. Remove bottom rear cover of the instrument in order to get access to the manifold.
5. Remove the screw of the bottles configuration selector.
6. Remove the bottles configuration selector with its silicone seal.
7. Dry, if necessary, the location of the bottles configuration selector.
8. Rotate it 90° in counter clockwise direction and place it on its position in the manifold again.



Figure 408. Bottles configuration selector positions – external drain not activated (left picture), external drain activated (right picture)

9. Place the screw of the bottles configuration selector and tighten it to the maximum and then a $\frac{1}{4}$ turn more.
10. Place the bottom rear cover of the instrument.
11. Connect properly the bottles, taking into account that previous waste solution 1 and 2 drawers are considered now as additional solution A and solution B drawers.

12. Connect to the external drain connector of the instrument a tube with a fitting to connect the instrument to an external drain. The specifications for this tube connection are listed below:

- Recommended PVC 6x4 mm.
- Tube connection with no loops.
- Maximum tube length: 5m.
- Maximum drain height: 1m above the external drain connector.
- Minimum drain height: 0.55m below the external drain connector.

13. Perform instrument decontamination after configuring *DUAL_MODE* parameter.

- **Aspiration pump:** to remove waste liquids from probes washer and dilution cup. The type of the pump is diaphragm for liquids. It has two independent heads in order to be associated to each probe circuit. Its flow rate avoids liquid overflow in dilution cup and drops in probes washer. The motor of the pump is a DC motor with a nominal voltage of 24V; an EMI filter is incorporated. Pump is attached to the support by means of 4 silentblocks. PM2.



Figure 409. Aspiration pump

- **Bottles:** There are 4 bottles in the instrument with a capacity of 8 litres, located at the bottom of the instrument. Are allowed two possible configurations (from left to right):
 - **First configuration:** Solution A bottle - Solution B bottle – Waste 1 – Waste 2.
 - **Second configuration:** Solution A bottle - Solution B bottle - Solution A bottle - Solution B bottle, when the system is directly disposing waste liquids to the laboratory drain.

The bottles contain one tube connector to connect it to the fluidic system and it can be manually disconnected by the operator for filling/emptying/removing/replacing purposes. Caps of the bottles are also the exhaust for the fluidic system (bottles are not pressurized).

- **Cards disposal tray:** There is a container located at the bottom of the instrument where the instrument throws the cards that are not necessary for the system. It has a capacity for 425 cards.
- **Bottles and cards disposal container drawers:** there is one drawer per each bottle and card disposal container. The structure of each drawer is formed by the following elements:
 - Moving part: it supports the bottle or cards disposal container and allows the operator to load/unload them. It moves by sliding movement and it includes a cosmetic front panel.

The edges of moving part of each of the fluidic drawers have been hemmed in order to avoid injuries to the field engineer when manipulating it. However there are available edge protectors to be installed in instruments without hemmed edges to mitigate this risk.



Figure 410. Fluidic drawer edge protectors

- Fixed part: it is responsible for allowing the movement of the moving part outside the instrument and to attach the drawer to the base. It also contains the load cell.
- Safety latch: it monitors open/close status, allows opening the drawer electronically and it assures closed position repeatability. The latch is attached to the support, while the bolt is attached to the moving part. The bolt position is adjustable in order to adjust the cosmetic front panel position.

Bottles and card disposal container can be removed when the drawers are fully opened. The amount of liquid in bottles or cards in cards disposal container is monitored by a load cell, located at the rear part of the drawer.

- **Digital pumps:** they are responsible for aspirating and dispensing samples, reagents, diluents and dilutions at the required accuracy. There are 2 digital pumps, one for each probe, located in the upper floor on the left and with a syringe of 1ml. Communication with each dilutor is through 2-wire RS485.
- **Correct Pipetting Detector (CPD):** there are 2 pressure sensors located in the interconnection manifold of the Y axis of the PIP arm that connect each dilutor with its probe. They monitor pressure changes during aspiration / dispensation to detect obstruction in the fluidic system.



Figure 411. CPDs in the interconnection manifold

- **Electronics:** Provide interface between fluidic components (electrovalves, pumps, sensors) and the instrument control. Additionally, they provide interface with lockers and with bottle weighting system. It is located rear the instrument, on the lower floor.

- **Interconnection manifolds:** they interconnect tubing in order to allow mechanical module union. There are 3 units in the *Erytra* :
 - To join tubing that comes from fluidics module to the tubing of X axis module. Located at the rear part of the digital pumps. BK2.
 - To join tubing that comes from X axis module to the tubing of Y axis module. Located at the rear part of the Y axis of the PIP arm. BK3.



Figure 412. Location of the interconnection manifolds, BK2 (left) and BK3 (right)

- To join tubing that comes from Y axis module to the probes. Located at the rear part of the head of the PIP arm. BK6.



Figure 413. Location of the interconnection manifold (c)

- **Tubing:** Provide connection between fluidic components. It is made with the following types (description, dimensions IDxOD):
 - PVC, 4x6 mm
 - Multilumen, 2.3x3.6 mm
 - Silicone, 3x5 mm (not present in fluidic system including expansion vessels)
 - PTFE, 1.1x1.7 mm
 - Shielded PTFE tubing

Multilumen tubing is a set of 8 tubes extruded together in ribbon shape. It is used in chains tubing and when many lines go across the instrument together. The number of tubes of the set varies along the instrument: multilumen inside the chains has 8 tubes, the multilumen of the dilution cup has 4 tubes and the multilumen that goes to the interconnection manifold located in the X-axis has 3 tubes related to probes washer solution B.
- **Module support:** This support integrates module components and gives integration with the rest of the instrument. It is screwed to the base module by means of four silentblocks.



Figure 414. Fluidic module support with all the components

General block diagram of the module is shown in the following figure:

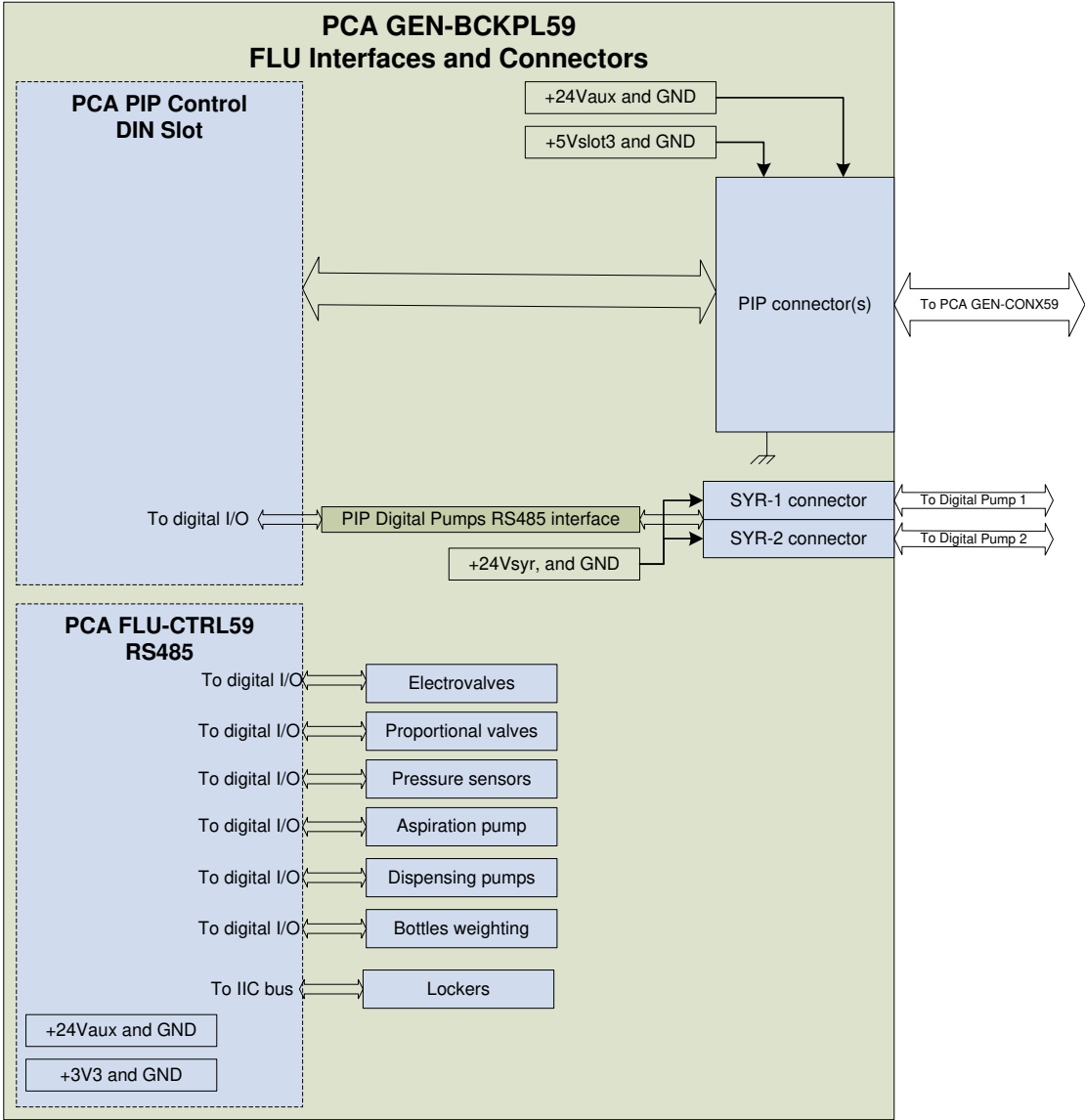


Figure 415. Fluidic block diagram

The PCAs related to the fluidics module are:

- PCA GEN-BCKPL59:** main PCA of the instrument, located in the backpanel. It includes an interface to convert 3 Digital I/O signals of PCA PIP Control (transmission, reception and transmission/reception enable) into a RS485 2-wire bus. This interface supports 38400 bauds communications speed minimum. This bus is used to communicate with both digital pumps. There are 2 connectors, SYR-1 and SYR-2, to connect both dilutors and it interfaces +24V and GND supply rails to SYR-1 and SYR-2 connectors.
- PCA FLU-CTRL59:** located at module support part, close to the manifold. It controls the fluidic system and load cells. It includes a HD15 connector type to connect the module with PCA GEN-BCKPL59.
- PCA GEN-LCK59:** it supports latch electronic control. It contains the safety latch interface (activate / deactivate) and it monitors its status (opened/closed). This PCA is located inside the drawer at the frontal part and it is protected from liquid spills by means of a cover.
- PCA PIP-CONY59:** It contains the sensor and interface to control the CPD of the probes. It is located at Y axis back (looking from the front of the instrument). Other parts are detailed on the PIP arm module.

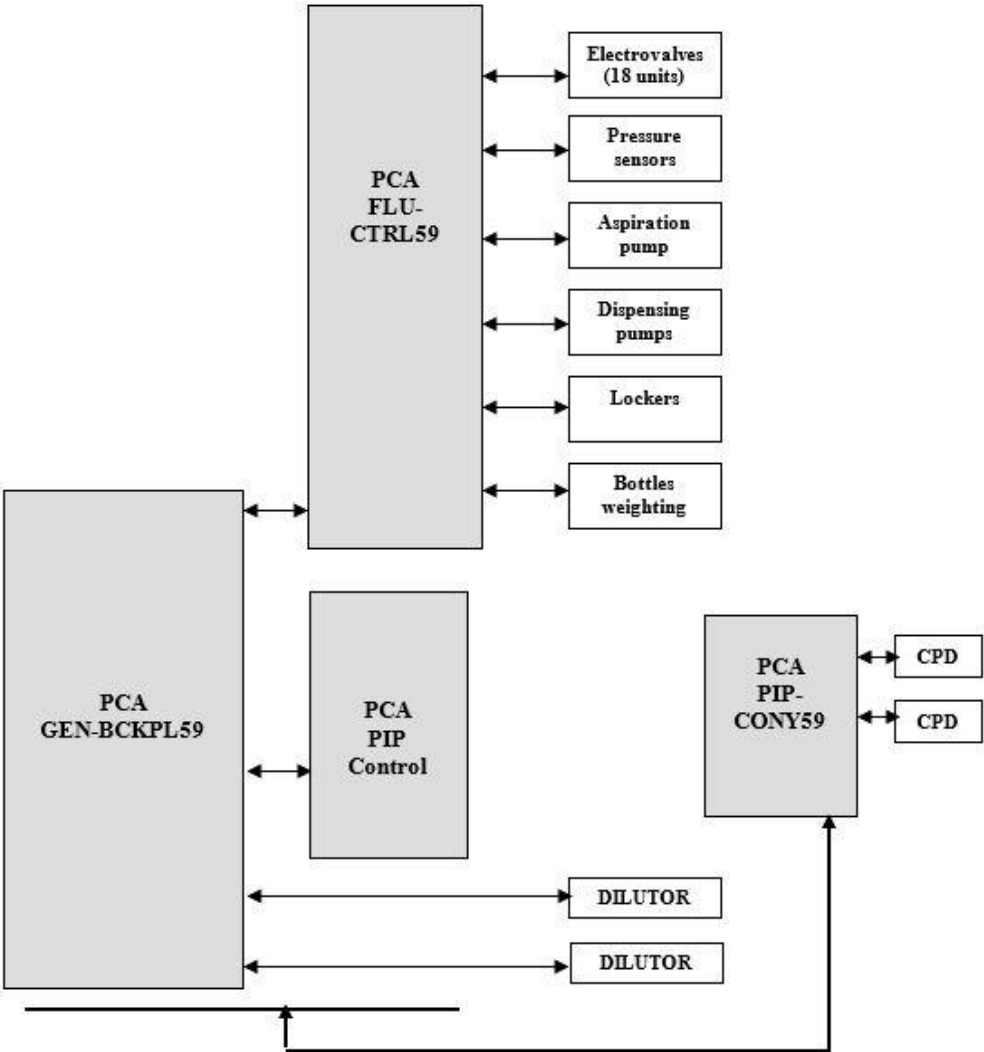


Figure 416. Main blocks of Fluidics module

9.1.2 Fluidic procedures

In this chapter, the different fluidics procedures are explained.

9.1.2.1 Priming

Priming processes can be divided in two main processes: low priming (manifold) and high priming (from manifold to probes, dilution cups, etc.). High priming can also be divided in some smaller processes: high priming with solution B, high priming with solution A, dilution cup priming and syringes priming.

In each fluidic step where S3 and S4 sensors are monitored, a previous checking of all the sensors is performed to monitor their status.

Low priming

The general aim of the low priming is to prime the circuit from bottles to manifold and it can be done for solution A and solution B.

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	383 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

1. *(no external drain)* Check waste levels.
2. *(no external drain)* Select waste bottle:
 - i. In case of using waste bottle 1: Clear EV17 and EV18.
 - ii. In case of using waste bottle 2: Set EV17 and EV18.
3. *(no external drain/external drain)* Check waste circuit.
4. *(external drain)* Clear EV16.
5. *(no external drain/external drain)* Fill with solution B bottle drawer 2, path between EV15 and EV13 and waste bottle/external drain:
 - i. Switch on PM1, EV13, EV19 and EV20. Wait 40s.
 - ii. Check pressure S3 = 200 mbar $\pm$ 150 mbar.
 - iii. Clear EV13, EV19 and EV20.
 - iv. Clear PM1.
6. *(no external drain/external drain)* Fill with solution B bottle drawer 2 path between solution B bottle and waste bottle/external drain (fast prime solution B)
 - i. Set PM3 and EV19. Wait 40s.
 - ii. Check pressure S4 = 180 mbar $\pm$ 100 mbar.
 - iii. Clear EV19.
 - iv. *(external drain)* Clear EV14 and EV16.
 - v. Clear PM3.
7. *(external drain)* Fill with solution B path between solution B bottle drawer 5 and external drain (fast prime solution B):
 - i. Set EV16.
 - ii. Set PM3 and EV19. Wait 40s.
 - iii. Check pressure S4 = 180 mbar $\pm$ 100 mbar.
 - iv. Clear EV19, EV14 and EV16.
 - v. Clear PM3.
8. *(no external drain/external drain)* Fill with solution A path between solution A bottle drawer 1 and waste bottle/external drain (fast prime solution A):
 - i. Set PM1 and EV20. Wait 40s.
 - ii. Check pressure S3 = 180 mbar $\pm$ 100 mbar.
 - iii. Clear EV20.
 - iv. *(external drain)* Clear EV14 and EV16.
9. *(external drain)* Fill with solution A path between solution A bottle drawer 3 and external drain (fast prime solution A):
 - i. Set EV14.
 - ii. Set PM1 and EV20. Wait 40s.
 - iii. Check pressure S3 = 180 mbar $\pm$ 100 mbar.
 - iv. Clear EV20, EV14 and EV16.

High priming solution B (probes washer)

The general aim of the high priming solution B is to prime the circuit from manifold to probes washer.

Print date:	16-May-2023	Purpose:	Uncontrolled Copy
Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization			

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	384 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

1. *(no external drain)* Check waste levels.
2. *(no external drain)* Select waste bottle:
 - i. In case of using waste bottle 1: Clear EV17 and EV18.
 - ii. In case of using waste bottle 2: Set EV17 and EV18.
3. *(external drain)* Check bottle levels.
4. *(external drain)* Select solution B drawer .By default, solution B drawer 2 is used. In case of using solution B drawer 5, set EV16.
5. *(no external drain/external drain)* Set PM2 and PM3.
6. *(no external drain/external drain)* Check pressure/vacuum sensors:
 - o S4 = 180 mbar $\pm$ 30 mbar.
 - o S5 < -500 mbar, S6 < -500 mbar.
7. *(no external drain/external drain)* Store S5 and S6 values.
8. *(no external drain/external drain)* Set EV5 and EV10.
9. *(no external drain/external drain)* Check that vacuum is stable (S5 and S6 values variation is less than 10 mbar).
10. *(no external drain/external drain)* Check that there is a difference in vacuum before and after opening aspiration electrovalves (difference at least of 40% between current S5 and S6 values and the stored ones).
11. *(no external drain/external drain)* Store S5 and S6 values.
12. *(no external drain/external drain)* Set EV1 and EV6. Wait 50s.
13. *(no external drain/external drain)* Check that vacuum increases (S5 and S6 values are lower than the stored ones).
14. *(no external drain/external drain)* Check that liquid arrives to the probes washer (there is a difference at least of 10% between current S5 and S6 values and the stored ones).
15. *(no external drain/external drain)* Clear EV1, EV6, EV5 and EV10.

High priming solution A (dilution cup)

The general aim of the high priming solution A is to prime the circuit with solution A from manifold to dilution cup.

1. *(external drain)* Check bottle levels.
2. *(external drain)* Select solution A drawer .By default, solution A drawer 1 is used. In case of using solution A drawer 4, set EV14.
3. *(no external drain/external drain)* Set PM1 and PM2.
4. *(no external drain/external drain)* Check pressure/vacuum sensors:
 - o S3 = 180 mbar $\pm$ 10 mbar.
 - o S5 < -400 mbar, S6 < -400 mbar.

Print date:	16-May-2023	Purpose:	Uncontrolled Copy
-------------	-------------	----------	-------------------

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	385 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

5. *(no external drain/external drain)* Store S5 and S6 values.
6. *(no external drain/external drain)* Set EV4 and EV9.
7. *(no external drain/external drain)* Check that vacuum is stable (S5 and S6 values variation is less than 10 mbar).
8. *(no external drain/external drain)* Check that there is a difference in vacuum before and after opening aspiration electrovalves (there is a difference at least of 40% between current S5 and S6 values and the stored ones).
9. *(no external drain/external drain)* Store S5 and S6 values.
10. *(no external drain/external drain)* Set EV3 and EV8. Wait 20s.
11. *(no external drain/external drain)* Check that vacuum increases (S5 and S6 values are lower than the stored ones).
12. *(no external drain/external drain)* Check that liquid arrives to the dilution cup (there is a difference at least of 10% between current S5 and S6 values and the stored ones).
13. *(no external drain/external drain)* Clear EV3, EV8, EV4 and EV9.

Syringe priming

The general aim of the syringe priming is to prime the probes and their related fluidic circuit.

1. Perform one obstruction test:
 - Aspire 1ml of air from the probe.
 - Wait 2.5 seconds until the pressure is stable.
 - Check dcp value is ≥ 13.000 counts
 - Syringe returns to its initial position
2. 12 cycles of aspiration and dispensation.
3. 5 cycles of aspiration and dispensation. Check that the mean of S5 and S6 sensors when the cycles have finished is $> 10\%$ than the mean of S5 and S6 sensors before the 5 cycles started.
4. Clear EV5 and EV10. Clear EV14 if activated.

9.1.2.2 Rinse

There are 4 different rinse procedures in the system: low rinse, high rinse solution B, dilution cup rinse and syringe rinse.

In each fluidic step where S3 and S4 sensors are monitored, a previous checking of all the sensors is performed to monitor their status.

Low rinse

The general aim of this procedure is to rinse the circuit from bottles to manifold.

1. *(no external drain)* Check waste levels.

Print date:	16-May-2023	Purpose:	Uncontrolled Copy
-------------	-------------	----------	-------------------

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	386 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

2. *(no external drain)* Select waste bottle:
 - i. In case of using waste bottle 1: Clear EV17 and EV18.
 - ii. In case of using waste bottle 2: Set EV17 and EV18.
3. *(no external drain)* Check waste circuit.
4. *(no external drain/external drain)* Fill with solution B bottle drawer 2, path between EV15 and EV13 and waste bottle/external drain:
 - i. Switch on PM1, EV13, EV19 and EV20. Wait 40s.
 - ii. Check pressure S3 = 200 mbar $\pm$ 150 mbar.
 - iii. Clear EV13, EV19 and EV20.
 - iv. Clear PM1.
5. *(no external drain/external drain)* Fill with solution B bottle drawer 2 path between solution B bottle and waste bottle/external drain (fast prime solution B)
 - i. Set PM3 and EV19. Wait 40s.
 - ii. Check pressure S4 = 180 mbar $\pm$ 100 mbar.
 - iii. Clear EV19.
 - iv. Clear PM3.
6. *(no external drain/external drain)* Fill with solution B bottle drawer 2, path between EV15 and EV13 and waste bottle/external drain:
 - i. Switch on PM1, EV13 and EV20. Wait 40s.
 - ii. Check pressure S3 = 200 mbar $\pm$ 150 mbar.
 - iii. Clear EV13 and EV20.

High rinse probes washer

The general aim of this procedure is to rinse the circuit from manifold to probes washer with solution B.

1. *(external drain)* Check bottle levels.
2. *(external drain)* Select solution B drawer. By default, solution B drawer 2 is used. In case of using solution B drawer 5, set EV16.
3. *(no external drain/external drain)* Set PM2 and PM3.
4. *(no external drain/external drain)* Check pressure/vacuum sensors:
 - o S4 = 180 mbar $\pm$ 10 mbar.
 - o S5 < -500 mbar, S6 < -500 mbar.
5. *(no external drain/external drain)* Store S5 and S6 values.
6. *(no external drain/external drain)* Set EV5 and EV10.
7. *(no external drain/external drain)* Check that vacuum is stable (S5 and S6 values variation is less than 10 mbar).
8. *(no external drain/external drain)* Check that there is a difference in vacuum before and after opening aspiration electrovalves (there is a difference at least of 40% between current S5 and S6 values and the stored ones).
9. *(no external drain/external drain)* Store S5 and S6 values.
10. *(no external drain/external drain)* Set EV1 and EV6. Wait 50s.

Print date:	16-May-2023	Purpose:	Uncontrolled Copy
Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization			

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	387 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

11. *(no external drain/external drain)* Check that vacuum increases (S5 and S6 values are lower than the stored ones).
12. *(no external drain/external drain)* Check that liquid arrives to the probes washer (there is a difference at least of 10% between current S5 and S6 values and the stored ones).
13. *(no external drain/external drain)* Clear EV1, EV6, EV5 and EV10.

High rinse dilution cup

The general aim of this procedure is to rinse the circuit from manifold to dilution cup with solution B.

1. *(external drain)* Check bottle levels.
2. *(external drain)* Select solution B drawer .By default, solution B drawer 2 is used. In case of using solution B drawer 5, set EV16.
3. *(no external drain/external drain)* Set PM1, PM2 and EV13.
4. *(no external drain/external drain)* Check pressure/vacuum sensors:
 - S3 = 180 mbar $\pm$ 10 mbar.
 - S5 < -400 mbar, S6 < -400 mbar.
5. *(no external drain/external drain)* Store S5 and S6 values.
6. *(no external drain/external drain)* Set EV4 and EV9.
7. *(no external drain/external drain)* Check that vacuum is stable (S5 and S6 values variation is less than 10 mbar).
8. *(no external drain/external drain)* Check that there is a difference in vacuum before and after opening aspiration electrovalves (there is a difference at least of 40% between current S5 and S6 values and the stored ones).
9. *(no external drain/external drain)* Store S5 and S6 values.
10. *(no external drain/external drain)* Set EV3 and EV8. Wait 20s.
11. *(no external drain/external drain)* Check that vacuum increases (S5 and S6 values are lower than the stored ones).
12. *(no external drain/external drain)* Check that liquid arrives to the dilution cup (there is a difference at least of 10% between current S5 and S6 values and the stored ones).
13. *(no external drain/external drain)* Clear EV3, EV8, EV4, EV9 and EV13.

Syringe rinse

The general aim of this procedure is to rinse the syringes and their related circuit.

1. Perform one obstruction test:
 - Aspire 1ml of air from the probe.
 - Wait 2.5 seconds until the pressure is stable.

Print date:	16-May-2023	Purpose:	Uncontrolled Copy
-------------	-------------	----------	-------------------

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization

- Check dcp value is ≥ 13.000 counts
 - Syringe returns to its initial position
2. 12 cycles of aspiration and dispensation.
 3. 5 cycles of aspiration and dispensation. Check that the mean of S5 and S6 sensors when the cycles have finished is $> 10\%$ than the mean of S5 and S6 sensors before the 5 cycles started.
 4. Clear EV5 and EV10.
 5. Clear EV13.

9.1.2.3 Emptying circuits

The general aim of this procedure is to empty the fluidic circuit when a shutting down order is given, after the final wash.

EXPANSION VESSELS:

1. *(no external drain)* Check waste level.
2. *(no external drain)* Select waste bottle:
 - a. In case of using waste bottle 1: Clear EV17 and EV18.
 - b. In case of using waste bottle 2: Set EV17 and EV18.
3. *(no external drain/external drain)* Check waste circuit.
4. *(no external drain/external drain)* Set PM2, EV15, EV2 and EV13.
5. *(no external drain/external drain)* Set EV3 and wait 15s.
6. *(no external drain/external drain)* Check pressure level S3 > -250 mbar.
7. *(no external drain/external drain)* Clear EV3.
8. *(no external drain/external drain)* Set EV8 and wait 15s.
9. *(no external drain/external drain)* Check pressure level S3 > -250 mbar.
10. *(no external drain/external drain)* Clear EV8.
11. *(no external drain/external drain)* Set PM2, EV15, EV2, EV13 and EV7.
12. *(no external drain/external drain)* Set EV1 and wait 15s.
13. *(no external drain/external drain)* Check pressure level S4 > -250 mbar.
14. *(no external drain/external drain)* Clear EV1.
15. *(no external drain/external drain)* Set EV6 and wait 15s.
16. *(no external drain/external drain)* Check pressure level S4 > -250 mbar.
17. *(no external drain/external drain)* Clear EV6, EV15, EV2, EV13 and EV7.
18. *(no external drain/external drain)* Set PM2 and EV5 and wait 1s.

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	389 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

19. *(no external drain/external drain)* Clear EV5.
20. *(no external drain/external drain)* Set EV10 and wait 1s.
21. *(no external drain/external drain)* Clear EV10.
22. *(no external drain/external drain)* Set PM2 and EV4 and wait 1s.
23. *(no external drain/external drain)* Clear EV4.
24. *(no external drain/external drain)* Set EV9 and wait 1s.
25. *(no external drain/external drain)* Clear EV9.
26. *(no external drain/external drain)* Set EV13, EV15, EV1 and EV3.
27. *(no external drain/external drain)* Check pressure sensors:
 - S3 > -50 mbar.
 - S4 > -50 mbar.
28. *(no external drain/external drain)* Clear EV1 and EV3.
29. *(no external drain/external drain)* Set PM1, PM3, EV13, EV15, EV4 and EV5.
30. *(no external drain/external drain)* Check pressure sensors:
 - S3 = 180 ± 30 mbar.
 - S4 = 180 ± 30 mbar.
31. *(no external drain/external drain)* Set EV19 and EV20.
32. *(no external drain/external drain)* Check pressure/vaccum sensors:
 - S3 < 150 mbar.
 - S4 < 150 mbar.
 - S5 < 150 mbar.
33. *(no external drain/external drain)* Clear EV19, EV20, PM1, PM3, EV4, EV5, EV13 and EV15.
34. *(no external drain/external drain)* Set dilutor 1 valve to bypass position.
35. *(no external drain/external drain)* Set PM2, EV15 and EV13 and wait 10s.
36. *(no external drain/external drain)* Clear PM2, EV15 and EV13.
37. *(no external drain/external drain)* Set dilutor 1 valve to probe position.
38. *(no external drain/external drain)* Set dilutor 2 valve to bypass position.
39. *(no external drain/external drain)* Set PM2, EV15 and EV13 and wait 10s.
40. *(no external drain/external drain)* Clear PM2, EV15 and EV13.
41. *(no external drain/external drain)* Set dilutor 2 valve to probe position.

Print date:	16-May-2023	Purpose:	Uncontrolled Copy
-------------	-------------	----------	-------------------

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization

9.1.2.4 Washing

There are 5 different washing procedures in the system: internal washing, fast/slow external washing, probe end washing and dilution cup washing.

Internal washing

The general aim of the internal washing is to wash internally the probes with a specific volume and cycles determined by the software.

1. *(no external drain/external drain)* Check bottle levels.
2. *(no external drain/external drain)* Switch on PM2.
3. *(no external drain/external drain)* Set EV5 and EV10.
4. *(external drain)* If using solution A drawer 4, set EV14.
5. *(no external drain/external drain)* Check vacuum sensors: S5 or S6 between -450 mbar and -50 mbar.
6. *(no external drain/external drain)* Move probes up to high position.
7. *(no external drain/external drain)* Aspirate and dispense with both dilutors the number or cycles and volume determined by the scheduler.
8. *(no external drain/external drain)* Clear EV5 and EV10.

Fast/slow external washing

The general aim of this washing is to wash externally the probes

1. *(no external drain/external drain)* Check bottle levels.
2. *(no external drain)* Check waste level.
3. *(no external drain/external drain)* Switch on PM2 and PM3. Set EV5 and EV10.
4. *(no external drain/external drain)* Check vacuum sensors S5 or S6 between -450 mbar and -100 mbar. Timeout = 5s.
5. *(no external drain/external drain)* Check average pressure S4 = 160 mbar $\pm$ 50 mbar.
6. *(no external drain/external drain)* Set EV1 and/or EV6.
7. *(external drain)* If using solution B drawer 5, set EV16.
8. *(no external drain/external drain)* Set washing probe speed.
9. *(no external drain/external drain)* Move probes up.
10. *(no external drain/external drain)* Clear EV1 or EV6.

Probe end washing

The general aim of this washing is to wash internally the end of the probes with a specific number of cycles determined by the software.

1. *(no external drain/external drain)* Check bottle levels.
2. *(no external drain/external drain)* Switch on PM2. Set EV5 and EV10. Set EV1 or EV6. Switch on PM3.
3. *(external drain)* In case of external drain connected and using solution B drawer 5, set EV16.
4. *(no external drain/external drain)* Check average pressure S4 (160 mbar $\pm$ 50 mbar).
5. *(no external drain/external drain)* Check vacuum sensors S5 or S6 between -450 mbar and -100 mbar. Timeout = 5s.
6. *(no external drain/external drain)* Set washing probe speed.
7. *(no external drain/external drain)* Move probes up and down the number of cycles determined by the scheduler.
8. *(no external drain/external drain)* Clear EV1 or EV6.
9. *(no external drain/external drain)* Clear EV5 and EV10.

Dilution cup washing

The general aim of this washing is to wash during routine the dilution cup with a specific number of cycles determined by the scheduler.

1. *(no external drain/external drain)* Check bottle levels.
2. *(external drain)* In case of external drain connected and using solution A drawer 4, set EV14.
3. *(no external drain)* Check waste level.
4. *(no external drain/external drain)* Switch on PM2. Set EV4 and/or EV9.
5. *(no external drain/external drain)* Check vacuum sensors S5 or S6 between -450 mbar and -50 mbar.
6. *(no external drain/external drain)* Clear EV4 and/or EV9.
7. *(no external drain/external drain)* Switch on PM1.
8. *(no external drain/external drain)* Check pressure S3 = 160 mbar $\pm$ 50 mbar.
9. *(no external drain/external drain)* Set EV3 and/or EV8.
10. *(no external drain/external drain)* Clear EV3 and/or EV8.
11. *(no external drain/external drain)* Switch on PM2. Set EV4 and/or EV9.
12. *(no external drain/external drain)* Check vacuum sensors S5 or S6 between -450 mbar and -50 mbar.
13. *(no external drain/external drain)* Clear EV4 or EV9.

14. *(no external drain/external drain)* If there are more cycles programmed by the scheduler, repeat steps 4 to 11 this number of cycles.

9.1.2.5 Overpressure management

Before starting any fluid process, a system status check is carried out.

In the event of a pump being activated in a situation where pressure is already high (should have descended in the previous step but did not achieve it) fluidic procedure may fail and software will keep re-trying, but as long as pressure is not released the error will keep appearing. So, in cases where all EVs related to an active dispensation pump are closed and pressure is > 25 mbar above threshold, it:

- Opens EV20 during 50 ms (related to a high pressure in S3 or Dispensation Pump PM1), or:
- Opens EV19 during 50 ms (related to a high pressure in S4 or Dispensation Pump PM3)

This algorithm is repeated up to 3 times per second in order to allow pressure to descend. It remains in this state until dispensation pump is closed, or any other EV associated to the sensor (thus dispensation pump) is open.



NOTE:

This process implies that sounds of constant electrovalve activation / deactivation might be heard during this operation.

9.2 ADJUSTMENT PROCEDURES

In this section, the adjustments related to the fluidic module are explained. There are 2 types of adjustments: mechanical and functional.

9.2.1 Mechanical adjustments

Mechanical adjustments of the fluidic module are detailed.

9.2.1.1 Safety latch of the drawers of the bottles / cards disposal container

This adjustment is required to set the depth of the drawer front into the instrument in order to align it with the rest of the drawers and the instrument frontal.

Prerequisite: NA

Adjustment tool: NA

Procedure:

1. Open the drawer to be adjusted and remove the bottle/cards disposal container. The safety latch is located underneath the drawer.



Figure 417. Location of the latch in the bottles / cards disposal container drawers

2. Loose the nut attached to the safety latch, and loose or screw the safety latch in order to align the drawer. Once the drawer is aligned, tighten the nut.

9.2.1.2 Cosmetic front panel of the drawers of the bottles / cards disposal container

This adjustment is required to set the cosmetic frontal panel of the drawers in order to align it with the rest of the drawers and the instrument frontal.

Prerequisite: NA
Adjustment tool: NA
Procedure:

1. Open the drawer to be adjusted and remove the bottle/cards disposal container.
2. Loose the 3 screws that are attaching the front panel to the drawer. Move in Z and X axis the front panel in order to align it. Once the frontal panel is aligned, tighten the 3 screws and check its alignment.



Figure 418. Location of the screws that attach the front panel to the drawer

9.2.2 Functional adjustments

Functional adjustments of the fluidic module are detailed in this section.

9.2.2.1 Load cells of the bottle drawers

This adjustment is required to set the full scale and the offset of the load cell.

Prerequisite: Drawers closed.

Adjustment tool: NA

Procedure:

1. Start the adjustment wizard pressing *gauge* button in Technical Service software *Procedures* tab → *Adjustments* section → *FLU floor Modules* area.
2. Select the drawer to be adjusted (solution A, solution B, waste 1 or waste 2). Press *Next>* button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press *OK* button. The selected drawer will be automatically opened.
3. Place an empty bottle in the drawer.
4. Close the drawer and wait until a stable measure is obtained in the software main interface, *Fluidics* tab. Then press *Next>* button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press *OK* button.
5. After drawer opening, place a bottle filled with 8 litres in the drawer.
6. Close the drawer and wait until a stable measure is obtained in TS software main interface, *Fluidics* tab. Then press *Next>* button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press *OK* button.



NOTE:

When adjusting the load cells with the filled bottle (8 litres), wait to press on *Next>* button until liquid does not move to assure a stable measurement.

7. After drawer opening, remove the full bottle and close the drawer.
8. A screen appears showing the parameters to be stored. Press *Finish* button and *OK* on the information window.
9. Press *OK* in the confirmation window that indicates that home movements will be performed. The drawer is automatically opened to place its original bottle.
10. Press *OK* to finish the adjustment procedure.



NOTE:

Load cells adjustments are stored in PCA FLU-CTRL59. In case of replacing this PCA, new adjustments must be performed. In case of restoring the adjustments from a previous backup, check whether the obtained readings are correct.

9.2.2.2 Load cell of the card waste container

This adjustment is required to set the full scale and the offset of the load cell.

Prerequisite: Drawer closed.

Adjustment tool: NA

Procedure:

1. Start the adjustment wizard pressing *gauge* button in Technical Service software *Procedures* tab → *Adjustments* section → *FLU floor Modules* area.
2. Select the card waste tray drawer from the pull-down menu (*WASTE_TRAY*). Press *Next>* button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press *OK* button. The card waste tray drawer is automatically opened.
3. Place an empty card waste container in the drawer.

4. Close the drawer and wait until a stable measure is obtained in the software main interface, *Fluidics* tab. Then press *Next>* button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press *OK* button.
5. Once the drawer is open, place a 2 kg weight into the container by either pouring 2 litres of water (recommended) or placing a known 2 kg weight in the center of the container.
6. Close the drawer and wait until a stable measure is obtained in TS software main interface, *Fluidics* tab. Then press *Next>* button. A warning message informing that the instrument will perform movements/settings appears on the screen. Press *OK* button.
7. Once the drawer is open, empty the container and close the drawer.
8. A screen appears showing the parameters to be stored. Press *Finish* button and *OK* on the information window.
9. Press *OK* in the confirmation window that indicates that home movements will be performed. The drawer is automatically opened. Close the drawer.
10. Press *OK* to finish the adjustment procedure.



NOTE:

Load cells adjustments are stored in PCA FLU-CTRL59. In case of replacing this PCA, new adjustments must be performed. In case of restoring the adjustments from a previous backup, check whether the obtained readings are correct.

9.2.2.3 Pressure/Vacuum sensors

This adjustment is required to set the full scale and the offset of the pressure/vacuum sensors located in the manifold.

Prerequisite:

- Fluidic system is empty.
- Bottles disconnected.

Adjustment tool: Pressure sensor adjustment tool (PN: 233817)

Procedure:



CAUTION!

Be careful when manipulating a sensor already connected to PCA FLU-CTRL59. Avoid that the welded connections of the PCA assembled in the sensor touch any metallic part of the instrument (for example the metallic support plate where the manifold is mounted) as it may damage PCA FLU-CTRL59.

1. Start the adjustment wizard pressing *sensors* button in Technical Service software *Procedure* tab → *Adjustments* section → *FLU floor Modules* area.
2. Select the sensor to be adjusted (S3, S4, S5 or S6) and press *Next>* button.
3. Disconnect sensor from PCA FLU-CTRL59 (to do not damage cables when unscrewing it) and then unscrew the selected sensor (with the O-ring) from the manifold and press *Next>* button on the software.
4. Connect the sensor (with the O-ring) and calibrated manometer to pressure sensor adjustment tool set

5. Connect sensor to the PCA FLU-CTRL59. Press *Next>* button on the software.



Figure 419. Setup of the adjustment of the pressure sensors

6. Apply pressure to the syringe until the manometer measures a pressure of 500 mbar $\pm$ 50 mbar. Press *Next>* button on the software.
7. Write in the textbox the measured pressure in the manometer (without decimals). Press *Next>* button on the software.
8. Disconnect the syringe from the 'T' connector and check that the manometer measures 0 mbar $\pm$ 10 mbar. Press *Next>* button on the software.
9. Write in the textbox the measured pressure in the manometer (without decimals). Press *Next>* button on the software.
10. A screen appears showing the parameters to be stored. Press *Finish* button and *OK* on the information window.
11. Press *OK* in the confirmation window that indicates that home movements will be performed.
12. Press *OK* to finish the adjustment procedure.
13. Disconnect the sensor from PCA FLU-CTRL59.
14. Assemble carefully the sensor (with the O-ring) to the manifold. Screw it as aligned as possible with the thread and manually to the maximum, and then an extra $\frac{1}{4}$ turn.

**CAUTION!**
Applying too much force can break the sensor.

15. Connect the sensor again to the PCA FLU-CTRL59.

**NOTE:**
Pressure/vacuum sensors is stored in PCA FLU-CTRL59. In case of replacing this PCA, the adjustment must be performed. In case restoring the adjustment from a previous backup, then check if the readings are correct.

9.2.2.4 Dilutors configuration

This adjustment is required to set the correct address and configuration of the dilutors. Address selector and configuration microswitches are located at the rear part of the dilutor.

Prerequisite: NA
Adjustment tool: NA

Procedure:

Configure dilutor as detailed in the following table:

Dilutor	Label	Address selector	Microswitch	Jumpers	CONFIGURATION
#1	DIL1	1	3, 7, 8: ON 1, 2, 4, 5, 6: OFF	5-6, 7-8 (factory implemented)	
#2	DIL2	2			

Table 68. Configuration of the dilutors

Dilutor #1 corresponds to the dilutor associated to the probe #1. It is the dilutor on the left, looking the instrument from the front. Dilutor #2 corresponds to the dilutor associated to the probe #2. It is the dilutor on the right looking the instrument from the front.

9.3 VERIFICATIONS

9.3.1 Pressure sensors

CAUTION!

Be careful when manipulating a sensor already connected to PCA FLU-CTRL59. Avoid that the welded connections of the PCA assembled in the sensor touch any metallic part of the instrument (for example the metallic support plate where the manifold is mounted) as it may damage PCA FLU-CTRL59.

1. Disconnect sensor from PCA FLU-CTRL59 (to not damage cables when unscrewing it).
2. Unscrew the selected sensor (with the O-ring) from the manifold
3. Connect sensor to the PCA FLU-CTRL59.
4. Connect the pressure sensor (with the O-ring) and calibrated manometer to pressure sensor adjustment tool set.
5. Apply pressure to the syringe and check that manometer reading and sensor reading (TSS *Fluidics* tab) does not differ more than the following values:
 - Pressure: 0 mbar ± 10 mbar → Reading: 0 mbar ± 10 mbar
 - Pressure: 500 mbar ± 10 mbar → Reading: 500 mbar ± 10 mbar
6. Disconnect sensor from PCA FLU-CTRL59 and from pressure sensor adjustment tool.
7. Assemble carefully the sensor (with the O-ring) to the manifold. Screw it as aligned as possible with the thread and manually to the maximum, and then an extra ¼ turn.

CAUTION!

Applying too much force can break the sensor.

8. Connect sensor to the PCA FLU-CTRL59.

9.3.2 CPD sensors

There are 2 tests to check the circuit between the dilutors and the probes by means of the CPD sensors.

9.3.2.1 Leak-tightness test

- 1. Empty the fluidic circuit (in TSS → *Fluidics* tab → *initializing fluidic* option and then *fluidic empty* option).
- 2. Remove racks of sample drawer 1.
- 3. Run the verification in TSS, *Procedures* tab → *Checks* section → *Others subsection* → *leak-tightness* button. A wizard window is opened:

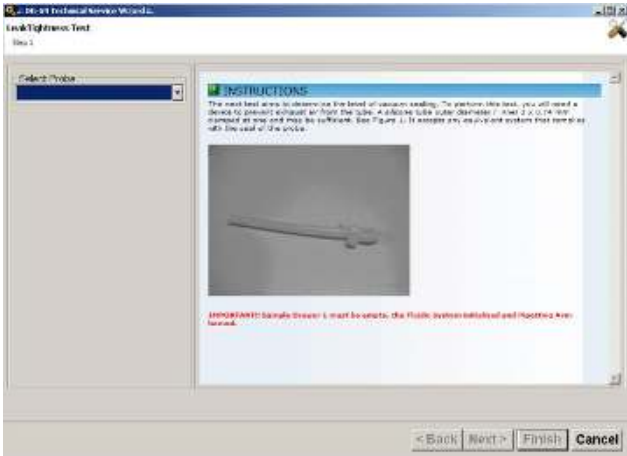


Figure 420. Leak-tightness wizard

- 4. Select the probe of the circuit to be checked and press *Next>* button.
- 5. Press *Next>* button and then *OK* in the warning message informing that the instrument will perform movements/settings to empty the fluidic circuit between the dilutor and the probe selected. PIP robot will move to rack 2 position of sample drawer 1, it will move down the probe and it will perform aspiration/dispensation movements with the dilutor. Then it will check if CPD value is 19000 ± 700 counts.
- 6. If the above condition is met, block the end of the probe so that air cannot enter into the probe. It is recommended to use a piece of silicone tube of 3x0.74 mm (external / internal diameter) tied by one of its sides for this purpose. Place the tube in the probe and make sure that there is no play between the probe and the tube.



Figure 421. Blocking the end of a tube

- 7. Press *Next>* button to start the verification. The instrument will perform an aspiration from the probe, so that vacuum is created inside the circuit. The system will check whether the difference between CPD values before and just after the aspiration is within an expected range. If so, it will compare the latter CPD value with a new one measured 30 seconds later, and then it will calculate the difference between these 2 values, which should be lower than 220 counts.
 - If this test fails, fluidic circuit between dilutor and the probe should be checked.

- If the test is passed, a new step in the adjustment wizard appears on the screen requesting to remove the blockage from the probe. Remove it and press *Next>* button.
- 8. The instrument will then perform movements/settings corresponding to an empty procedure of the fluidic circuit between the dilutor and the probe. At this point, CPD value should be 19000 ± 700 counts.
- 9. If the above condition is met, an aspiration (valve input position) will be made and the software will ask for the silicon tube to be placed again in the probe end so that air cannot enter into the probe. Place the tube in the probe and and make sure that there is no play between the probe and the tube.
- 10. Press *Next>* button in order to start the verification. The instrument will perform a dispensation (valve output position), so a pressure is created inside the circuit. The system will check whether the difference between CPD values before and just after the dispensation is within an expected range. If so, it will compare the latter CPD value with a new one measured 30 seconds later, and then it will calculate the difference between these 2 values, which should be lower than 220 counts.
 - If this test fails, fluidic circuit between dilutor and the probe should be checked.
 - If the test is passed, a new step in the adjustment wizard appears on the screen requesting to remove the blockage from the probe. Remove it and press *Next>* button.
- 11. A screen appears showing the parameters obtained during the test. Press *Finish* button and *OK* on the information window (probe will be homed).
- 12. Press *OK* in the confirmation window to finish the verification procedure.

9.3.2.2 Prime test

Perform the following steps in order to make sure that the circuit from dilutor to probe can be properly primed:

1. Prime the fluidic circuit (in TSS → *Fluidics* tab → *initializing fluidic* option and then *fluidic prime* option)
2. Prime the pipetting circuit from the dilutor to the probe (in TSS → *Fluidics* tab → *prime probe 1* or *prime probe 2* option)
3. Place the probe in home position.
4. With the pressure/vacuum pumps switched off, check that the CPD value is 17700 ± 700 counts.
5. If the CPD value is out of the previous value:
 - a. If CPD value > 18400 , the circuit contains air.
 - b. If CPD value < 17000 , the circuit contains a possible obstruction.

9.3.3 Dispensation pumps and expansion vessels

In order to verify the performance of the dispensation pump-expansion vessel set, follow the next steps:

1. Check that waste bottles or external drain is connected and there is solution A and B liquid.
2. Empty the fluidic circuit (in TSS → *Fluidics* tab → *initializing fluidic* option and then *fluidic empty* option).
3. Check that aspiration electrovalves are closed (activate and deactivate EV5 and EV10 in TSS).
4. Activate PM1 in TSS and check that S3 sensor = $160 \text{ mbar} \pm 10 \text{ mbar}$. Deactivate PM1.
5. Activate PM3 in TSS and check that S4 sensor = $160 \text{ mbar} \pm 10 \text{ mbar}$. Deactivate PM3.
6. Perform a prime of the fluidic circuit (in TSS → *Fluidics* tab → *fluidic prime* option).
7. Perform a rinse of the fluidic circuit (in TSS → *Fluidics* tab → *fluidic rinse* option).

9.3.4 Aspiration pump

To verify the performance of the aspiration pump, follow the next steps:

1. Check that aspiration electrovalves are closed (activate and deactivate EV4, EV5, EV9 and EV10 in TSS).
2. To check the maximum value of vacuum, check that S5 and S6 vacuum values are higher than -600 mbar when it is activated only PM2.
3. To check the minimum value of vacuum, check that S5 and S6 vacuum values are higher than -100 mbar when it is activated PM2 and EV5 and EV10.

9.3.5 Electrovalves

To check the performance of the electrovalves follow the next steps:

1. Activate the EV to be checked in Technical Service software, *Fluidics* tab. Check that a sound must be heard when the EV is activated / deactivated.
2. Check that the LED of the PCA FLU-CTRL59 is activated when the EV is activated in the software.

9.3.6 Manifold

1. Check that no leakages or scratches are observed.

9.3.7 Load cell

In order to verify the performance of a load cell, follow the next steps:

Bottle drawers:

1. Place a bottle with a known volume (4 litres recommended) in the different positions and check that the value shown by the software is the expected one.
2. Repeat the process with an empty bottle and with a one filled with 8 litres.
3. Remove the bottle and check that the software detects that it is not present.

Card waste drawer:

1. Place a 2 kg weight into the container by either pouring 2 litres of water (recommended) or placing a known 2 kg weight in the center of the container, checking that the value shown by the software is close to 410 cards.
2. Repeat the process with an empty card container, checking that the value shown by the software is close to 0 cards.

9.4 REPLACEMENT PROCEDURES

This section describes the replacement procedures for several parts of the fluidic system: pressure and CPD sensors, dispensation and aspiration pumps, electrovalves, manifold, load cell, fitting of the tubes, syringe and valve.

9.4.1 Pressure sensors (S3, S4, S5, S6) replacement

To install a new pressure sensor and the corresponding cable, follow next steps:

**NOTE:**
There are 4 pressure sensors in Erytra instrument.

**CAUTION!**
Be careful when manipulating a sensor already connected to PCA FLU-CTRL59. Avoid that the welded connections of the PCA assembled in the sensor touch any metallic part of the instrument (for example the metallic support plate where the manifold is mounted) as it may damage PCA FLU-CTRL59.

1. Switch on the instrument and enter the Boot Application menu -> *Technical Service* option.
2. Perform an empty of the fluidic system and disconnect all the bottles.
3. Switch off the instrument and disconnect it from the mains.
4. Remove the rear bottom cover of the instrument in order to get access to the manifold.
5. Disconnect the pressure sensor from PCA FLU-CTRL59 and remove the sensor (with the O-ring) from the manifold.



6. Connect the new cable to the new sensor.
7. Connect the new cable and sensor to PCA FLU-CTRL59. Do not place the sensor in the manifold yet.



8. Connect the instrument to the mains, switch it on and enter the Boot Application menu -> *Technical Service* option.

- 9. Perform the pressure sensor adjustment of the new sensor as detailed in the Service Manual.
- 10. Disconnect the new sensor from the new cable.
- 11. Assemble carefully the new sensor (with the O-ring) to the manifold. Screw it as aligned as possible with the thread and manually to the maximum, and then an extra ¼ turn.

CAUTION!

Applying too much force can break the sensor.

- 12. Connect back the new sensor to the new cable.



- 13. Place the lower cover of the rear part.
- 14. Connect the bottles to the fluidic circuit.
- 15. Connect the instrument to the mains.
- 16. Check Pressure sensors (see section 9.3.1 Pressure sensors).

9.4.2 CPD sensors (S1, S2) replacement

- 1. Switch off the instrument and disconnect it from the mains.
- 2. Remove the middle rear cover of the instrument in order to get access to the rear part of the PIP arm.
- 3. Remove the screws of the rear cover of the PIP arm to get a better access to the parts.
- 4. Disconnect the cables of the sensors from the PCA PIP-CONY59.
- 5. Disconnect tube fittings from the interconnection manifold.
- 6. Remove carefully the interconnection manifold in order to do not damage its cables.
- 7. Remove the 2 screws that attach both parts of the interconnection manifold and separate them smoothly.

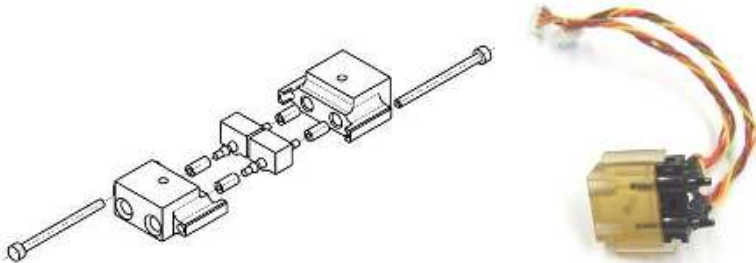


Figure 422. CPD sensors

- 8. Insert silicone tubes (length 8 mm) on both new CPD sensor sides and replace it. Check that the cables of the sensor are correctly placed.
- 9. Join both parts of the interconnection manifold and tighten its screws. Tighten the screws to the maximum, and then a ¼ turn more

10. Place it carefully on its position over the distributor, inserting the cables of the sensors through the distributor and making sure of the CPD orientation is as shown in below picture.

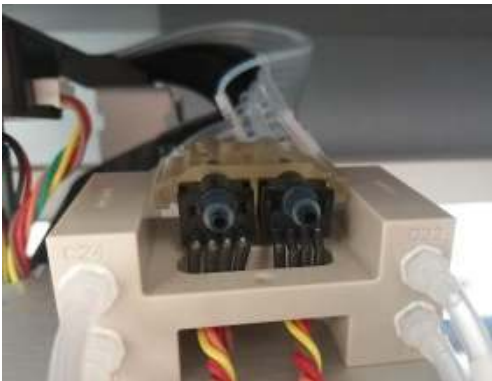


Figure 423. CPD sensors position inside a block



NOTE:

Route carefully CPD sensors cable through the distributor.

11. Connect tube fittings to the interconnection manifold, screwing them until a ‘click’ is heard.
12. Connect the cables of the sensors to the PCA PIP-CONY59.
13. Place the rear cover of the PIP arm.
14. Place the central rear cover of the instrument.
15. Perform Prime test (CPD performance).
16. Perform Leak-tightness test.

9.4.3 Dispensation pumps (PM1, PM3) replacement

1. By means of the Technical Service Software perform an empty of the fluidic system.
2. Disconnect the bottles.
3. Switch off the instrument and disconnect it from the mains.
4. Remove the lower cover of the rear part of the Erytra in order to get access to the manifold.
5. Disconnect the dispensation pump from the PCA FLU-CTRL59.
6. Loose the two screws of the dispensation pumps support and remove it. Take out the dispensation pump.



Figure 424. Dispensation pump screws

- 7. Replace the two o-rings that are placed in the manifold.



Figure 425. O-rings in dispensation pump support

- 8. Place the new dispensation pump, checking that the input and output ports of the dispensation pumps support and tighten the screws to the maximum, and then a ¼ turn more.
- 9. Connect the dispensation pump to the PCA FLU-CTRL59.
- 10. Connect the instrument to the mains and switch it on.
- 11. Check the connection of the bottles.
- 12. Verify the performance of the dispensation pump.
- 13. Place the lower cover of the rear part.

9.4.4 Expansion vessels replacement

- 1. Check that the fluidic system is empty and disconnect the bottles.
- 2. Switch off the instrument and disconnect it from the mains.
- 3. Remove the lower cover of the rear part of the instrument in order to get access to the manifold.
- 4. Remove the 2 screws of the expansion vessel and remove it.



Figure 426. Expansion vessels screws

- 5. Assemble new expansion vessel. Tighten the screws to the maximum, and then a ¼ turn more.
- 6. Place the lower cover of the rear part.
- 7. Check status of the bottles.
- 8. Connect the instrument to the mains and switch it on.

9.4.5 Aspiration pump (PM2) replacement

- 1. By means of the Technical Service Software perform an empty of the fluidic system.
- 2. Disconnect the bottles.
- 3. Switch off the instrument and disconnect it from the mains.
- 4. Remove the lower cover of the rear part of the instrument in order to get access to the manifold.
- 5. Disconnect the aspiration pump from the PCA FLU-CTRL59.



Figure 427. Aspiration pump cable connection

- 6. Disconnect the ground cable of the aspiration pump.
- 7. Disconnect the fluidic tubes from the aspiration pump.



Figure 428. Tubes and ground cable

- 8. Unscrew aspiration pump and plate assembly from the basis of the fluidics and remove the assembly.



Figure 429. Plate assembly

- 9. Unscrew the silentblocks of the aspiration pump and place the new pump.
- 10. Assemble the pump and plate to the basis of the fluidics.
- 11. Connect the ground cable to the aspiration pump.
- 12. Connect the fluidics tubes to the pump.
- 13. Connect the aspiration pump to the PCA FLU-CTRL59.
- 14. Connect the bottles.
- 15. Place the lower cover of the rear part.
- 16. Connect the instrument to the mains and switch it on.
- 17. Verify the performance of the pump.

9.4.6 Electrovalves replacement

1. By means of the Technical Service Software perform an empty of the fluidic system.
2. Disconnect the bottles.
3. Switch off the instrument and disconnect it from the mains.
4. Remove the rear lower cover of the instrument in order to get access to the manifold.
5. Disconnect the electrovalve from the PCA FLU-CTRL59.
6. Remove the electrovalve from the manifold by means of the 2 screws.



Figure 430. Electrovalve screws

7. Place the new electrovalve in its position. Tighten the screws to the maximum, and then a ¼ turn more.
8. Connect the electrovalve to the PCA FLU-CTRL59.
9. Place the lower cover of the rear part.
10. Connect the bottles.
11. Verify the performance of the electrovalve.

9.4.7 Manifold replacement

1. Check that fluidic system is empty and disconnect the bottles.
2. Remove the rear lower cover of the instrument in order to get access to the manifold.
3. Switch off the instrument and disconnect it from the mains.
4. Disconnect all the parts of the manifold from the PCA FLU-CTRL59.
5. Remove the 2 screws that attach the manifold to the fluidic basis.
6. Move up carefully the manifold and disconnect all the tubes that are located at the bottom of the manifold.
7. Place the new manifold, connecting all the tubes and all the parts to the PCA FLU-CTRL59.



NOTE:

The sealant product already present in the fittings might break due to the fittings manipulation when connecting back the tubes.

8. Put a second layer of sealant product around the fittings thread once the tubes have been inserted.
9. Tighten the screws that attach the manifold to the fluidic basis to the maximum, and then a ¼ turn more.
10. Connect the instrument to the mains and switch it on.
11. Verify the performance of the manifold.
12. Place the lower cover of the rear part.

9.4.8 Manifold / distribution block (BK2, BK3) fitting replacement

Manifold /distribution block (BK2, BK3) fittings are sealed with acetoxysilicone, recommended for use on plastic pipe threaded fittings (such as Loctite 5331). To replace a fitting, follow the next steps:

- 13. Check that fluidic system is empty.
- 14. Switch off the instrument and disconnect it from the mains.
- 15. Disconnect the tube from the fitting to be replaced.
- 16. Unscrew the fitting from the manifold/distribution block (BK2, BK3) and clean its allocation.



CAUTION!

Check that no seal product gets into the manifold/distribution block (BK2, BK3) during cleaning process.

- 17. Place the new fitting in its allocation only tightening it a little bit, so most of the thread is out of its allocation.
- 18. Put sealant product around the fitting thread. Sealant product should be spread only on the fitting thread as uniformly as possible from the middle of fitting thread toward the union point with nut. To ensure a better and ease spreading use a pointed end object.



Figure 431. Sealant product application area

- 19. Tighten the fitting manually to the maximum and then ¼ turn more in its allocation.



CAUTION!

Check that the seal product does not block the connection between the fitting and the manifold/distribution block (BK2, BK3).

- 20. Connect the tube to the fitting.



NOTE:

The sealant product already present in the fitting might break due to the fitting manipulation when connecting back the tube.

- 21. Put a second layer of sealant product around the fitting thread once the tube has been inserted.

9.4.9 Load cell replacement

To replace the load cell, follow nex steps:

1. Remove the lower cover of the rear part of the instrument in order to get access to the fluidics area, by removing 11 screws.

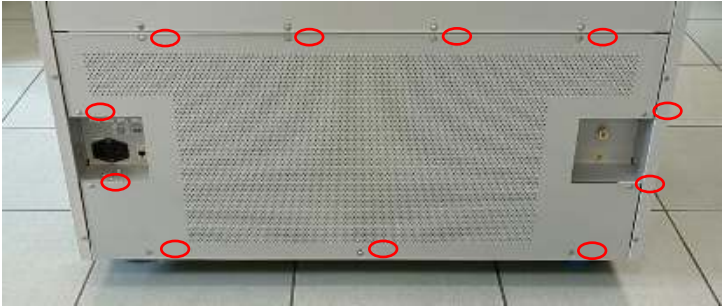


Figure 432. Screws holding the lower rear cover

2. Disconnect from the PCA FLU-CTRL59 the cable of the load cell to replace.

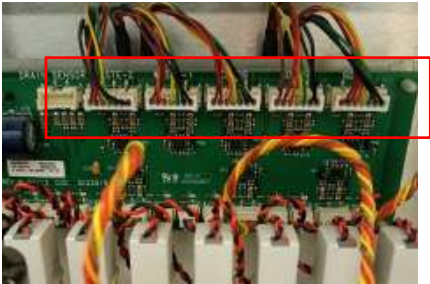


Figure 433. Load cells connections to PCA FLU-CTRL59

3. Using a number 8 open end spanner, remove 4 nuts that attach the manifold basis to the instrument.

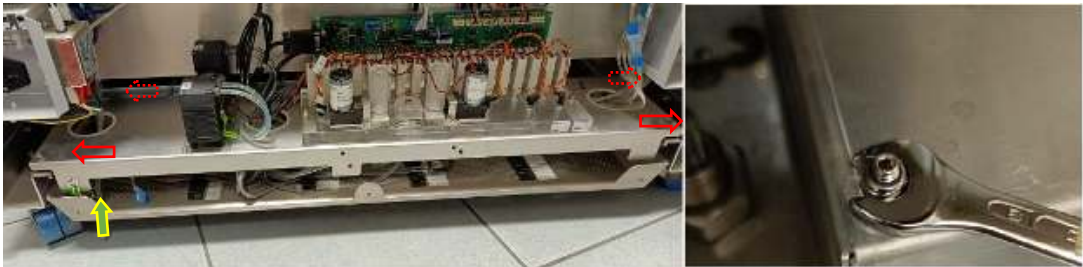


Figure 434. Nuts attaching the manifold basis to the instrument.



CAUTION!

Be careful with the grounding connection when handling the manifold basis.

4. Rotate the manifold basis to access the rear part of the PCA FLU-CTRL59 and unplug the cable of the load cell of its clamps.



Figure 435. Clamps holding load cell cable.

5. Unplug the cable from all the clamps/connectors below the manifold basis.



Figure 436. Connector and clamp below the manifold basis

6. Remove the 3 screws that attach the drawer frame to the basis of the instrument.



Figure 437. Location of screws attaching the drawer frame to the basis of the instrument.

7. Carefully rotate the basis of the drawer to get access to the screws holding the load cell underneath the basis and remove these 2 screws.



Figure 438. Screws holding the load cell

8. Extract the load cell from the front of the instrument.



Figure 439. Load cell removal

9. Position the new load cell on the drawer pushing it to the right to the 2 reference points marked on the image.

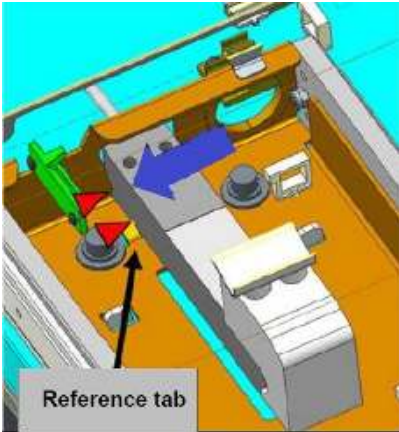


Figure 440. Positioning of the new load cell

10. Attach the load cell to the drawer frame using the 2 previously removed screws.

- 11. Execute steps 1 to 6 in reverse order to assemble the drawer.
- 12. Perform the adjustment of the load cell.

9.4.10 Fitting of the tubes replacement

- 1. Switch off the instrument and disconnect it from the mains.
- 2. Cut the end of the tube square.
- 3. Fit an Omnifit fitting nut onto the tube. Slide an Omni-lok lock ring and ferrule onto the cut tube end. Ensure the lip of the lock ring and ferrule is facing towards the tube end.



Figure 441. Lock ring (left) and ferrule (right) tube assembly

- 4. Place lock ring and ferrule between two pliers and close them until the cone and case are located together but a gap between the lip of the lock ring and the lip of the ferrule remains (see below picture).
- 5. Close slowly both pliers until half of the gap between the lip of the lock ring and the lip of the ferrule is closed.
- 6. Rotate 180° the tube and close slowly the pliers until the lip of the lock ring and the lip of the ferrule are between 0 and 0.2mm apart. Do not deform the lip of the ferrule. The assembly is now ready for use.



Figure 442. Lock ring and ferrule assembly

- 7. Perform a Prime to check the fluidic system.
- 8. Perform Leak-tightness test.

9.4.11 Syringe and valve replacement



NOTE:

In case of replacing only one of the two parts it is necessary to follow the same steps, but replacing only the required part.



NOTE:

If the instrument has the syringe locking system mounted on a MK syringe, it should be removed and discarded.

The MK valve (P/N 734834) and MK syringe (P/N 734835) assemblies do not require the syringe locking mechanism.

1.

Switch on Erytra and click on bottom-left corner grid button . Boot application login window will be displayed.
2.

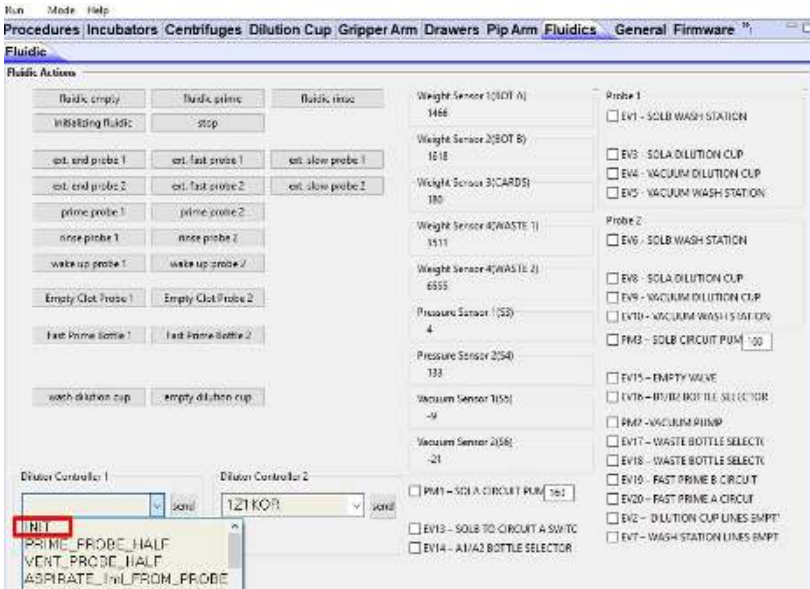
Log in as adminUser (it is case sensitive). Boot Application options will be displayed.
3.

Select Technical Service option .
4.

Once Technical Service software is opened, start the Driver (select Run → Start).
5.

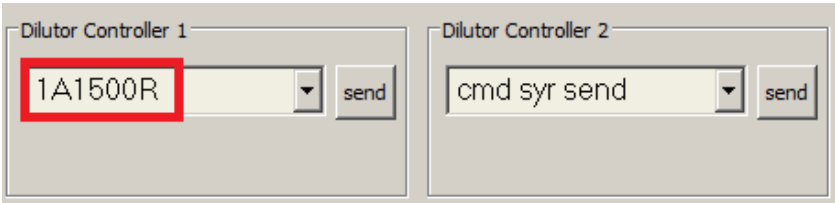
Move the PIP arm to its top-right position in order to obtain a better access to the dilutor module.
6.

Initialize the dilutor by means of Technical Service software (*Fluidics* tab, select *INIT* from the dilutor section for the syringe that is going to be replaced. Then press *Send* button).

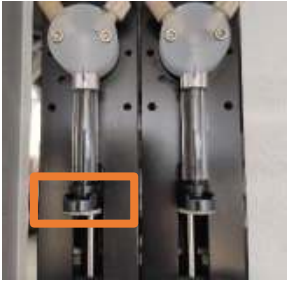


7.

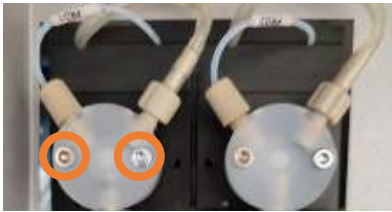
Move the syringe plunger drive to the center of the stroke by means of TSS. Go to *Fluidics* tab and type the command *1A1500R* in the dilutor section for the syringe that is going to be replaced. Press *send* button.



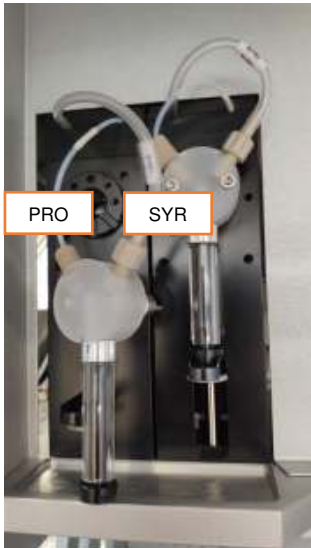
8. Loose the thumbscrew of the plunger.



9. Remove the syringe and valve assembly from the dilutor by loosening the two screws attaching it to the dilutor.



10. Remove the output tube (PRO) first and then the input tube (SYR) while rotating the valve.



11. Insert the upper end of the new syringe into the valve and rotate until tight.



NOTE:
Screw it to the maximum as aligned with the thread as possible



12. Place the new o-ring into the new valve input tubing area.



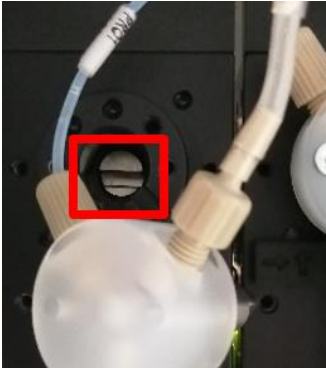
13. Screw the input tube (SYR) rotating the valve and then the output tube (PRO) to the new valve (for output tubing is necessary to screw until a 'click' is heard).



14. Install the new assembly by aligning the coupler on the valve stem with the hole in the encoder and fix the valve using the new screws. Do not tighten the screws completely (a small play should be present when executing the next steps).



NOTE:
A manual valve axis rotation might be needed in order to align the coupling system between the valve stem and the hole in the encoder.



15. Pull the plunger down until it reaches the plunger drive of the dilutor.

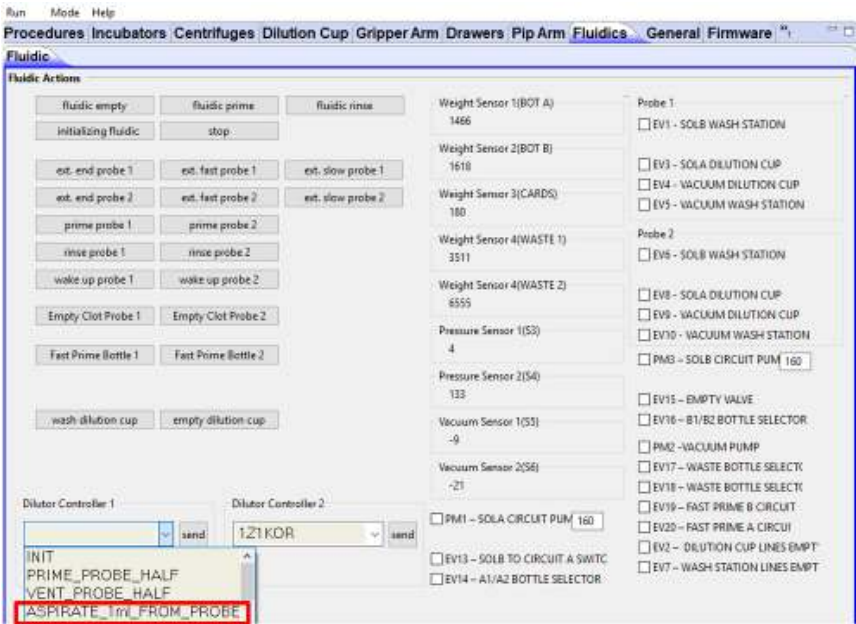
16. Position the plunger so that the plunger button sets into the channel of the drive system.



17. Hold the plunger and tighten the thumbscrew into the plunger. Do not tighten it completely yet.



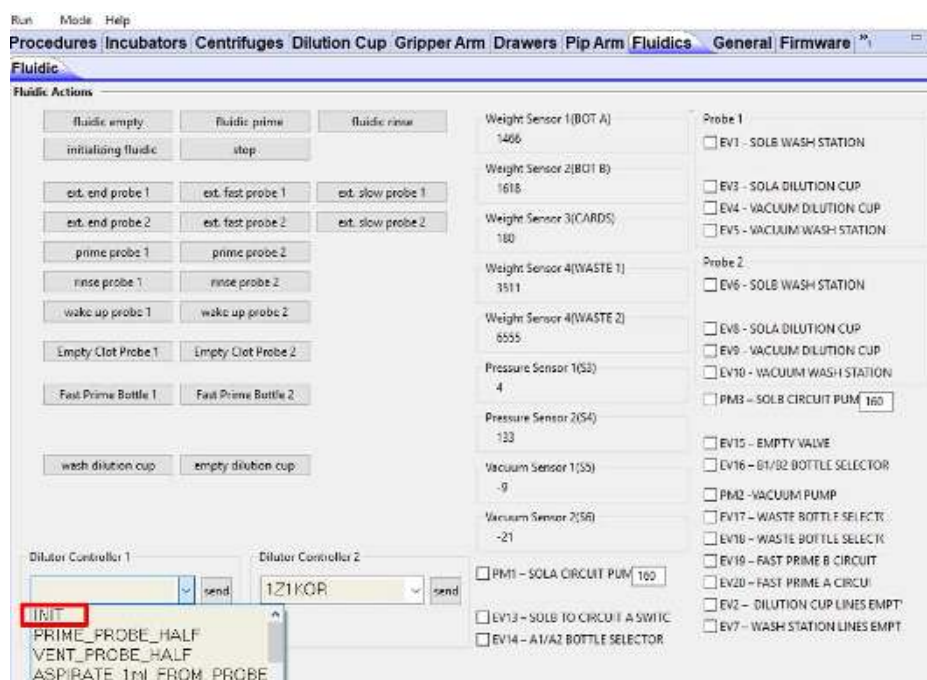
18. Using Technical Service software, bring back the plunger to initial position with an *INIT* command (as in step 6 above) and then position the plunger to the end of the stroke: go to *Fluidics* tab and select *ASPIRATE_1mL_FROM_PROBE* from the dilutor section which syringe is going to be replaced. Press *Send* button.



19. Dispense the aspirated air using the DISPENSE_TO_PROBE command from the dilutor section which syringe is going to be replaced. Press **Send** button

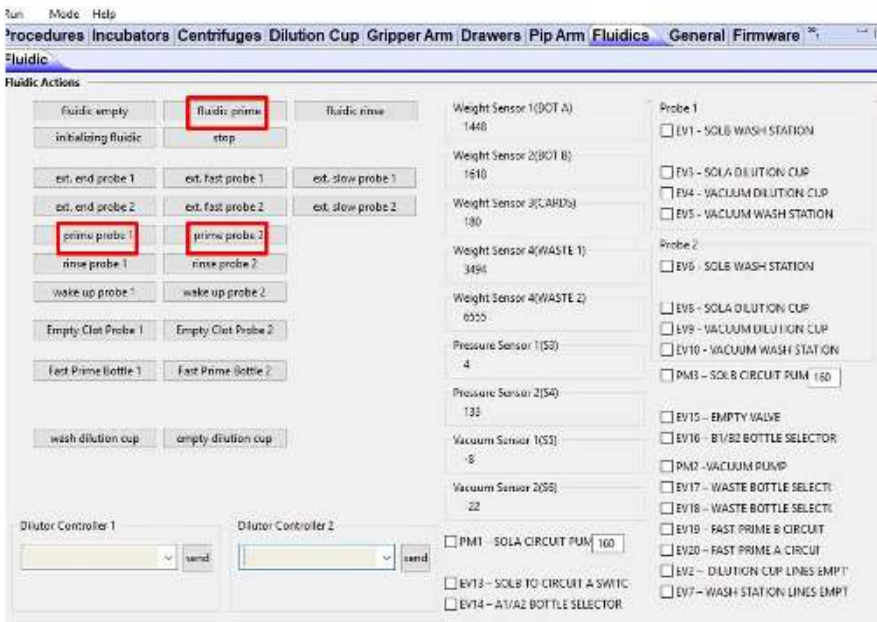


20. Tighten the thumbscrew of the plunger that was not completely tightened previously.
21. Tighten the two screws attaching the valve that were not completely tightened previously.
22. Re-tighten syringe into valve.
23. Initialize again the dilutor by means of Technical Service software (*Fluidics* tab, select *INIT* from the dilutor section for the syringe that is going to be replaced. Then press **Send** button).



24. Repeat steps 6 to 21 above for the syringe related to probe/dilutor 2 circuit.

25. Perform a Prime of the fluidic circuit.



26. Go to *Procedures* tab → *Checks* section → *Others* subsection and run *leak-tightness* test on both circuits to verify that there are no leakages and that the new valve and syringe are correctly tightened.

CAUTION!

If an MK valve already installed in an instrument needs to be replaced, the valve axis may be disengaged while removing the valve from the dilutor.

9.4.12 Fluidic drawer linear guide replacement

1. Open fluidic drawers and remove the 4 fluidic bottles and card waste container.
2. Switch off the instrument and disconnect it from the mains.
3. Remove the 3 Allen screws that fix the fluidic drawer where is needed to replace the linear guide to lower basis.

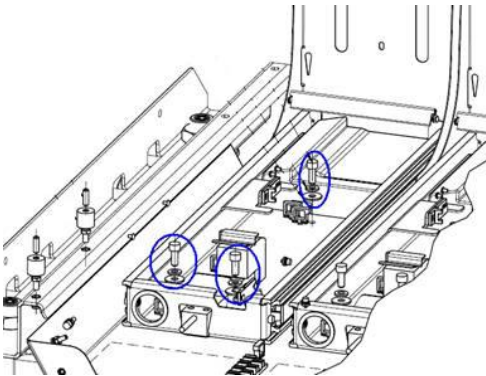


Figure 443. Screws fixing fluidic drawer to lower basis

- 4. Rotate carefully the drawer depending on the linear guide (left or right) you want to replace.
- 5. Remove the 3 Allen screw of the linear guide attached to fluidic drawer lateral.

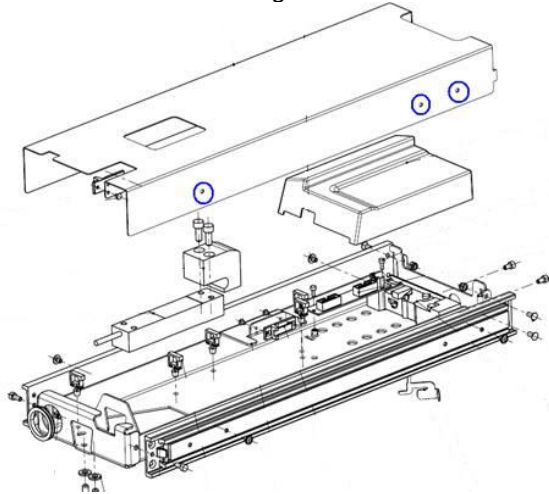


Figure 444. Linear guide screws on fluidic drawer lateral

- 6. Remove the 2 frontal Allen screws (including nuts) and the rear one (including nut) holding the linear guide to main lateral part of the drawer.

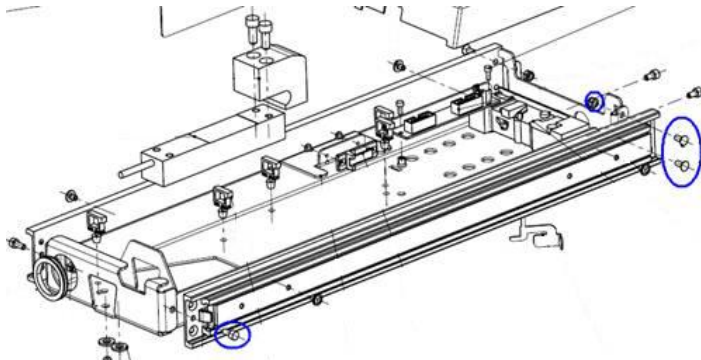


Figure 445. Linear guide screws on drawer main lateral part

- 7. Remove the linear guide and install the new one, lubricating it properly.
- 8. After assembling back everything, check load cell adjustment.

9.4.13 Coupling CPC bottles connector O-ring replacement (F75, F78, F81 AND F84)

- 1. Open fluidic drawers
- 2. Disconnect the 4 fluidic bottles by pressing the lever and pulling the connector.



Figure 446. CPC bottles connector lever

- 3. Remove the coupling CPC bottles connector o-ring.



Figure 447. CPC bottles connector o-ring

- 4. Place the new o-ring to the coupling CPC bottles connector.
- 5. Connect again the 4 fluidic bottles.
- 6. Close the fluidic drawers.

9.4.14 Connection block (BK4, BK5) replacement

To replace the connection block (BK4, BK5) where the CPD sensors are placed follow the next steps:

- 1. Open upper door.
- 2. Check that fluidic system is empty.
- 3. Switch off the instrument and disconnect it from the mains.
- 4. Disconnect the teflon tubes on both sides of the connection blocks.

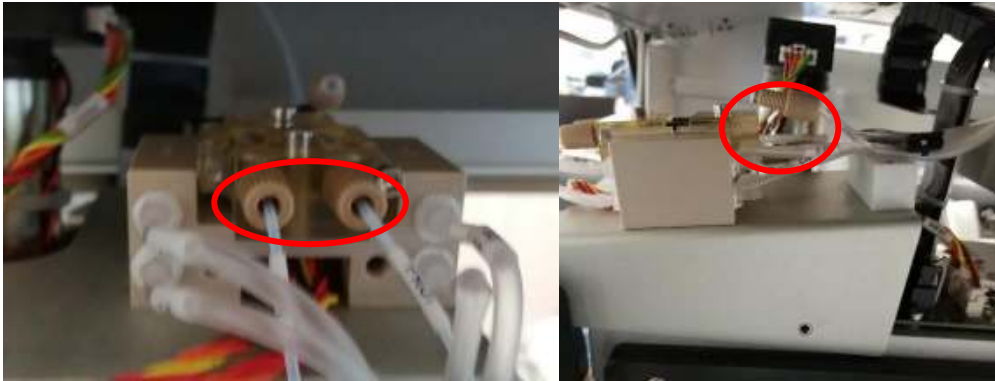


Figure 448. Tubing

- 5. Remove 2 screws that attach both blocks to each other (BK4, BK5).

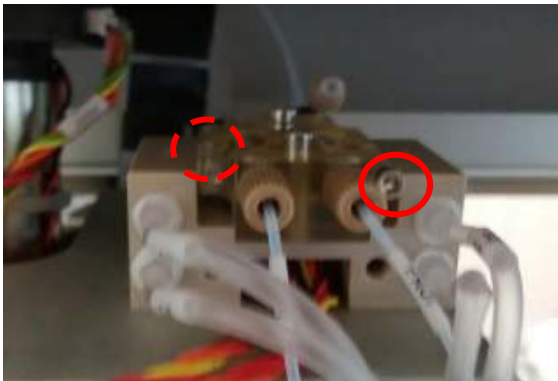


Figure 449. Screws attaching the 2 blocks together

6. Remove screw that attach BK4 to the distributor and also the remove screw that attach BK5 to the distributor.

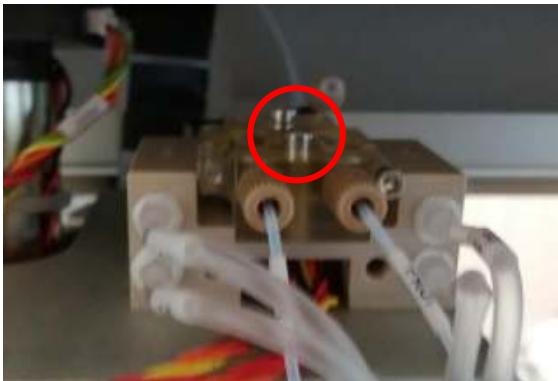


Figure 450. Screws attaching the 2 blocks to the distributor

7. Carefully pull apart both CDP sensors from inside BK4 and BK5 blocks and remove BK4 and BK5 blocks.

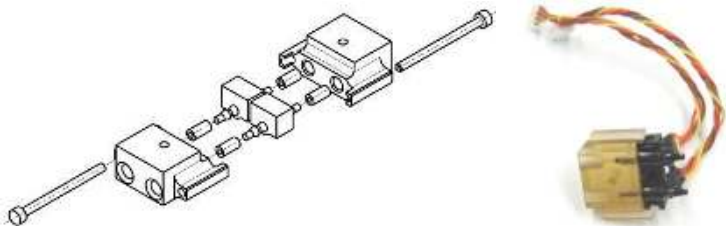


Figure 451. CPD sensors

8. Place the new connection blocks (BK4, BK5).
9. Place the CPD sensors inside the blocks.

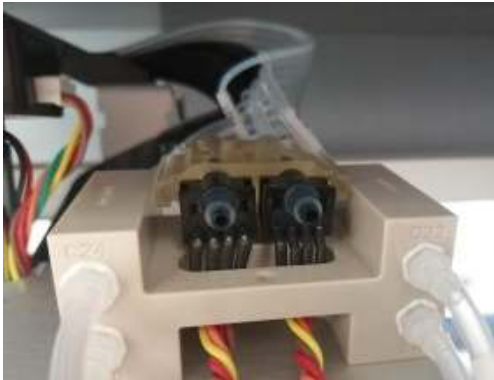


Figure 452. CPD sensors position inside a block



NOTE:

Route carefully CPD sensors cable through the distributor.

10. Join both parts of the interconnection manifold and tighten its screws. Tighten the screws to the maximum, and then a 1/4 turn more
11. Assemble back all the parts.
12. Verify the performance of the CPD sensor.

9.4.15 Vacuum filter replacement

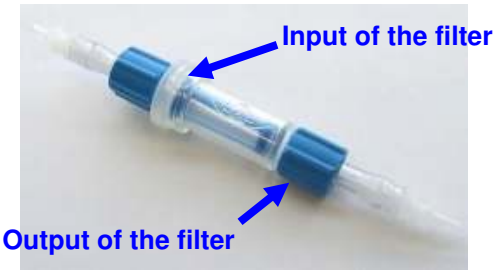
To replace the vacuum filter follow the next steps:

1. By means of the Technical Service Software perform an empty of the fluidic system.
2. Disconnect the bottles.
3. Switch off the instrument and disconnect it from the mains.
4. Remove the lower rear cover.
5. Disconnect the vacuum filters and remove them. There are 4 vacuum filters (tubes C20 / C22 / C24 / C26).



Vacuum filters

6. Place the new vacuum filter and assemble back all the parts. Make sure the filter is correctly oriented (filtering liquid towards aspiration pump).



Vacuum filters orientation

7. Perform a Prime to check the fluidic system.

9.4.16 Aspiration pump tubing replacement

To replace the aspiration pump tubing follow the next steps:

1. By means of the Technical Service Software perform an empty of the fluidic system.
2. Disconnect the bottles.
3. Switch off the instrument and disconnect it from the mains.
4. Remove the lower rear cover.
5. Disconnect the tubing from the aspiration pump (C29,C31).



Figure 453. Aspiration pump tubing

6. Remove the 2 screws that attach the manifold to the fluidic basis. If needed disconnect cabling from PCA FLU-CTRL59.



Figure 454. Manifold screws

7. Disconnect the tubing from the manifold (C29,C31).



Figure 455. Aspiration pump tubing in the manifold

8. Place the new tubing and assemble back all the parts.



NOTE:

The sealant product already present in the fitting might break due to the fitting manipulation when connecting back the tube.

9. Put a second layer of sealant product around the fitting thread once the tube has been inserted.
10. Perform a Prime to check the fluidic system.
11. Check aspiration pump performance.

9.4.17 PCA FLU-CTRL59 board replacement

To replace PCA FLU-CTRL59 follow the next steps. First a backup of the adjustments stored in PCA FLU-CTRL59 eeprom is performed:

1. Switch on the instrument and enter Bootapplication menu
2. Enter Technical Service software
3. Go to *Firmware* tab.

4. Go to *EPROM control* tab and select *FLU*.
5. Click on *Read* button.
6. Click on *Backup* button to save the eprom values with the adjustments of the fluidic module in a file.
7. Close TSS software.

Then PCA FLU-CTRL59 is replaced:

8. Switch off the instrument and disconnect it from the mains.
9. Remove the lower rear cover.
10. Remove all cabling from PCA FLU-CTRL59.
11. Remove PCA FLU-CTRL59 by removing its 6 screws.



12. Place the new PCA FLU-CTRL59 board.
13. Reconnect all the cabling.

Finally the adjustments backup previously performed is restored to the new PCA FLU-CTRL59 eprom:

14. Switch on the instrument and enter Bootapplication menu.
15. Enter Technical Service software
16. Go to *Firmware* tab.
17. Go to *EPROM control* and select *FLU*.
18. Click on *Read* button.
19. Click on *Restore* button and select the previous backup file to restore the eprom values with the adjustments of the fluidic module.
20. Click *Write* button to save the adjustments to the eprom.
21. Go to *PCA control* tab.
22. Select *FLU* and click on *Reset* button.
23. Check all fluidic and load cells adjustments.

9.5 FUSES

Fuses related to Fluidics module are detailed in the following table,

PCA	FUSE NAME	DESCRIPTION
PCA GEN-BCKPL59	FUSGEN3 FUSGEN5	MST250 0034.6621 T4A/250V MST250 0034.6620 3.15A/250V
PCA FLU-CTRL59	FUS1	MST250 0034.6621 T4A/250V
PCA PIP-CONY 59	NA	NA

Table 69. List of fuses of Fluidics module

9.6 LEDS

LEDs related to Fluidics module are detailed in the following table,

PCA	LED	DESCRIPTION
PCA GEN-BCKPL59	LED3 LED17	Green 24V (FUSGEN3) Green 24V (FUSGEN5)
PCA FLU-CTRL59	LED1 LED2 LED3 LED4 LED5 LED6 LED7 LED8 LED9 LED10 LED11 LED12 LED13 LED14 LED15 LED16 LED17 LED18 LED19 LED20 LED21 LED22 LED23 LED24 LED25 LED26 LED27	EV9 activated (Yellow) EV10 activated (Yellow) EV4 activated (Yellow) EV5 activated (Yellow) EV7 activated (Yellow) EV8 activated (Yellow) EV2 activated (Yellow) EV3 activated (Yellow) EV13 activated (Yellow) EV14 activated (Yellow) EV20 activated (Yellow) EV17 activated (Yellow) EV18 activated (Yellow) EV19 activated (Yellow) NA EV16 activated (Yellow) PM1activated (Yellow) PM3 activated (Yellow) EV15 activated (Yellow) EV11 activated (Yellow) EV12 activated (Yellow) EV1 activated (Yellow) EV6 activated (Yellow) PM2 activated (Yellow) Blinks (Yellow) during communication. Problem in the voltage or there is not communication with the lockers (Red). Blinks (Green) : - during normal operation : 1 Hz freq - error in the sensors : 5 Hz freq
PCA PIP-CONY59	NA	NA

Table 70. List of LEDs of Fluidics module

9.7 JUMPERS AND TEST POINTS

Neither jumpers nor test points are related to Fluidics module.

9.8 WIRING TABLE

In the following table, there is detailed all the cables related to Fluidics module. A wiring diagram can be found attached to Service Manual.

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
INF-SOL.A + INF-SOL.A_EXT	H7	PCA FLU-CTRL59	NA	LOAD CELL A SOLUTION DRAWER
INF-SOL.B + INF-SOL.B_EXT	H8	PCA FLU-CTRL59	NA	LOAD CELL B SOLUTION DRAWER
INF-CARD + INF-CARD_EXT	H9	PCA FLU-CTRL59	NA	LOAD CELL CARD DRAWER
INF-WAS1 + INF-WAS1_EXT	H10	PCA FLU-CTRL59	NA	LOAD CELL WASTE 1 DRAWER
INF-WAS2 + INF-WAS2_EXT	H11	PCA FLU-CTRL59	NA	LOAD CELL WASTE 2 DRAWER
B-FLU	H13	PCA FLU-CTRL59	FLU	PCA GEN-BCKPL59
INF-LOCK + BOT-IN	H14	PCA FLU-CTRL59	H1 From Master	PCA GEN-LCK59 A SOLUTION DRAWER
NA	H3	PCA GEN-LCK59 A SOLUTION DRAWER	H	SAFETY LOCK
BOT-OUT + BOT-IN	H2 To Slave	PCA GEN-LCK59 A SOLUTION DRAWER	H1 From Master	PCA GEN-LCK59 B SOLUTION DRAWER
NA	H3	PCA GEN-LCK59 B SOLUTION DRAWER	H	SAFETY LOCK
BOT-OUT + BOT-IN	H2 To Slave	PCA GEN-LCK59 A SOLUTION DRAWER	H1 From Master	PCA GEN-LCK59 CARD DRAWER
NA	H3	PCA GEN-LCK59 CARD DRAWER	H	SAFETY LOCK
BOT-OUT + BOT-IN	H2 To Slave	PCA GEN-LCK59 CARD SOLUTION DRAWER	H1 From Master	PCA GEN-LCK59 WASTE 1 DRAWER
NA	H3	PCA GEN-LCK59 WASTE 1 DRAWER	H	SAFETY LOCK
BOT-OUT + BOT-IN	H2 To Slave	PCA GEN-LCK59 WASTE 1 DRAWER	H1 From Master	PCA GEN-LCK59 WASTE 2 DRAWER
NA	H3	PCA GEN-LCK59 WASTE 2 DRAWER	H	SAFETY LOCK
B-SYR1	H	DILUTOR 1	SYR_1	PCA GEN-BCKPL59
B-SYR2	H	DILUTOR 2	SYR_2	PCA GEN-BCKPL59

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
PIP-S1	H6 S1	PCA PIP-CONY59	S1	PRESSURE SENSOR
PIP-S2	H7 S2	PCA PIP-CONY59	S2	PRESSURE SENSOR
GROUND CABLE	ASPIRATION PUMP	ASPIRATION PUMP	FLUIDIC BASIS	FLUIDIC BASIS
GROUND CABLE	INSTRUMENT BASIS	INSTRUMENT BASIS	FLUIDIC BASIS	FLUIDIC BASIS

Table 71. List of wiring of Fluidics module



Figure 456. Location of the connectors, fuses and LEDs in PCA GEN-BCKPL59 related to Fluidics module

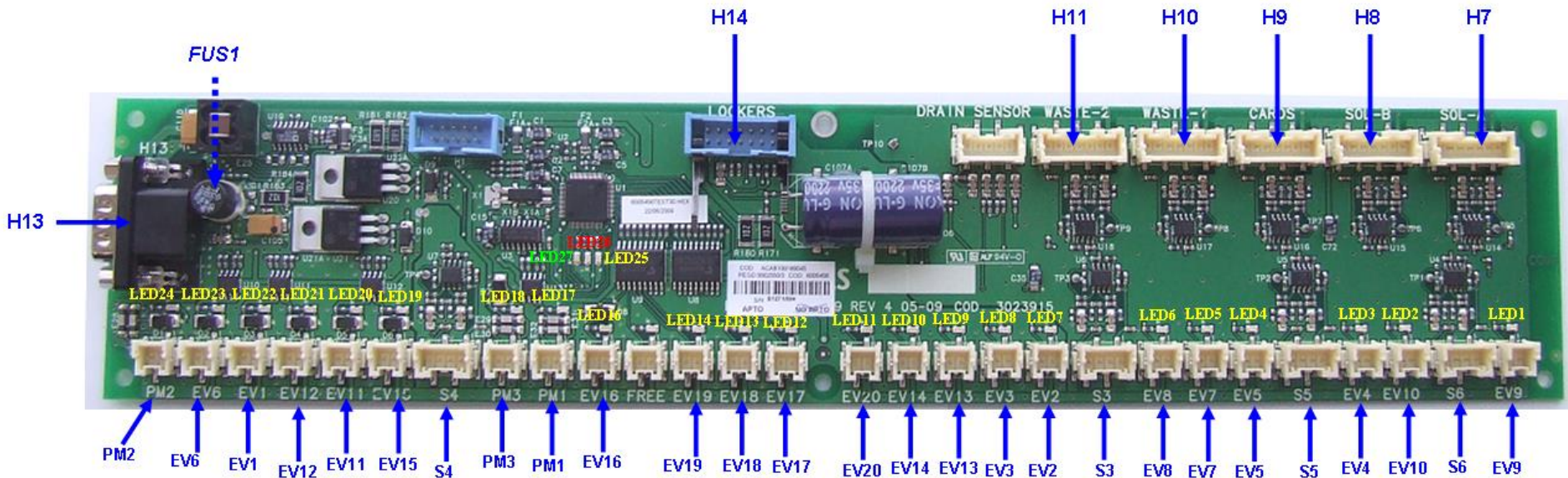
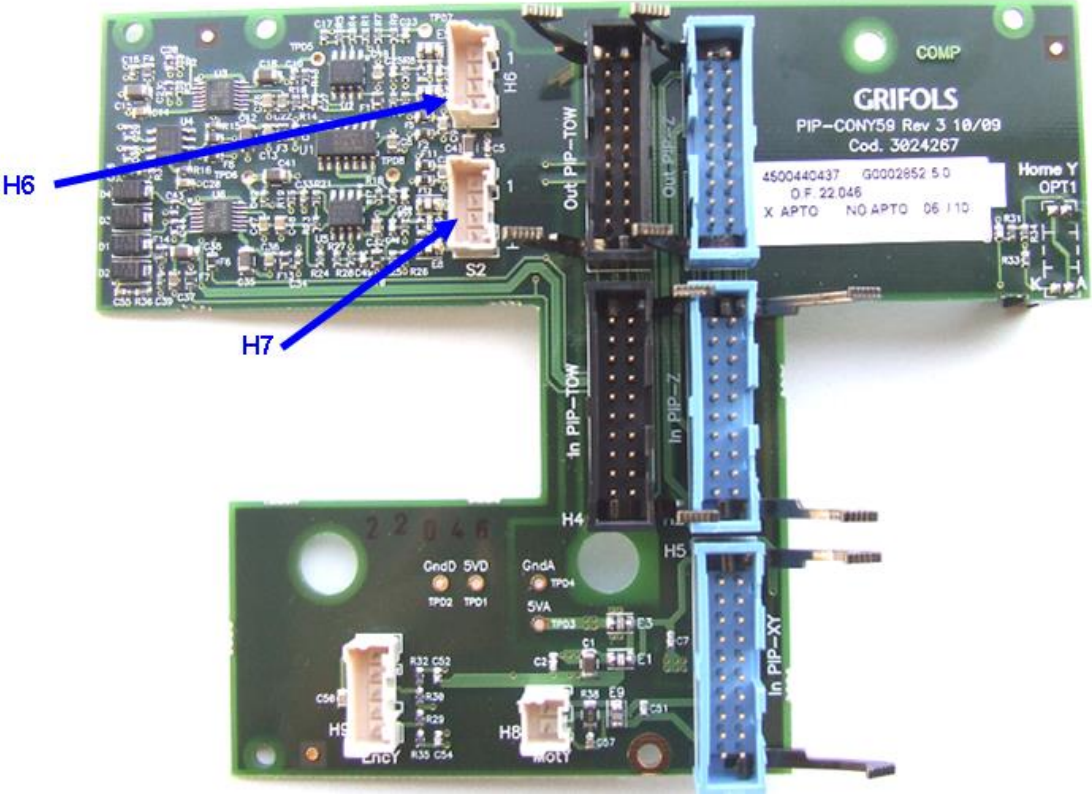


Figure 457. Location of the connectors, fuse and LEDs in PCA FLU-CTRL59



9.9 TROUBLESHOOTING

In fluidics module, last 3 digits errors can be 000 or a different number than 000 (it will indicate the fluidic step where the error has been found, refer to Table 88).

Module	Device	Error code	Error	Cause	Text	Options
FLU (05)	PCA FLU (89)	info	FLG_POWER_VOLTAGE_KO	No power supply at Fluidic PCA.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		info	FLG_SENSOR_VOLTAGE_KO	Fluidic sensors voltage error.		
		info	FLG_I2C_BUS_KO	Error I2C bus.		
	Fluidic drawers lockers (01) Drawer 1 (02) Drawer 2 (03) Drawer 3 (04) Drawer 4 (05) Drawer 5	info	FLG_LOCKER_VOLTAGE_KO	Fluidic lockers voltage error.		
		info	FLG_ACCELEROMETER_KO	NA		
		039xxx	FLG_ERR_LOCK1	Locker error detected.		
	Sol A Lock (01)	077xxx	ERRORCODE_FLU_BOTTLE_A_DRAWER_IS_OPEN	Drawer A is open.	A lower drawer is opened. Please close it and select RETRY. Select CANCEL if the problem persists only.	RETRY CANCEL
		075xxx	FLG_ERR_GAL1	Weight sensor malfunction detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK

Module	Device	Error code	Error	Cause	Text	Options
FLU (05)	Sol B Lock (02)	039xxx	FLG_ERR_LOCK2	Locker error detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		078xxx	ERRORCODE_FLU_BOTTLE_B_DRAWER_IS_OPEN	Drawer B is open.	A lower drawer is opened. Please close it and select RETRY. Select CANCEL if the problem persists only.	RETRY CANCEL
	Sol B Load cell (69)	075xxx	FLG_ERR_GAL2	Weight sensor malfunction detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
	Waste tray Lock (03)	039xxx	FLG_ERR_LOCK3	Locker error detected.		
		081xxx	ERRORCODE_FLU_WASTE_TRAY_DRAWER_IS_OPEN	Cards waste tray is open.	A lower drawer is opened. Please close it and select RETRY. Select CANCEL if the problem persists only.	RETRY CANCEL
	Waste tray Load cell (70)	075xxx	FLG_ERR_GAL3	Weight sensor malfunction detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
	Waste 1 Lock (04)	039xxx	FLG_ERR_LOCK4	Locker error detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		079xxx	ERRORCODE_FLU_WASTE_1_DRAWER_IS_OPEN	Waste 1 drawer is open.	A lower drawer is opened. Please close it and select RETRY. Select CANCEL if the problem persists only.	RETRY CANCEL

Module	Device	Error code	Error	Cause	Text	Options
FLU (05)	Waste 1 Load cell (71)	075xxx	FLG_ERR_GAL4	Weight sensor malfunction detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		083xxx	ERRORCODE_FLU_WASTE_1_CAPACITY_OUT_OF_MARGINS	Waste 1 bottle full.	<i>If there are running works:</i> System has been stopped caused by a container problem and there are pending works. Please, select one of the following options: <i>If there are not running works:</i> System has been stopped caused by a container problem. Please, select one of the following options:	RETRY RE-RUN OPEN
	Waste 2 Lock (05)	039xxx	FLG_ERR_LOCK5	Locker error detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		080xxx	ERRORCODE_FLU_WASTE_2_DRAWER_IS_OPEN	Waste 2 drawer is open.	A lower drawer is opened. Please close it and select RETRY. Select CANCEL if the problem persists only.	RETRY CANCEL

Module	Device	Error code	Error	Cause	Text	Options
FLU (05)	Waste 2 Load cell (72)	075xxx	FLG_ERR_GAL5	Weight sensor malfunction detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		084xxx	ERRORCODE_FLU_WASTE_2_CAPACITY_OUT_OF_MARGINS	Waste 2 bottle full.	<i>If there are running works:</i> System has been stopped caused by a container problem and there are pending works. Please, select one of the following options: <i>If there are not running works:</i> System has been stopped caused by a container problem. Please, select one of the following options:	RETRY RE-RUN OPEN
	Pressure sensor 3 (06)	069xxx	ERRORCODE_FLU_PRESSURE_FAIL_A	Pressure error detected on solution A fluidic circuit.	<i>If external drain is not activated:</i>	RETRY CANCEL Works OPEN solution A drawer
					<i>If external drain is activated:</i>	RETRY CANCEL Works OPEN solution A drawer 1 OPEN solution A drawer 2

Module	Device	Error code	Error	Cause	Text	Options
FLU (05)	Pressure sensor 3 (06)	074xxx	FLG_ERR_PRESSURE_SEN SOR_1	Pressure sensor malfunction detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
	Pressure sensor 4 (07)	070xxx	ERRORCODE_FLU_PRESSURE_FAIL_B	Pressure error detected on solution B fluidic circuit.	<i>If external drain is not activated:</i> Pressure error on solution B circuit. Please check: - Containers are properly connected. - There is no spilling in the upper level.	RETRY CANCEL Works OPEN solution B drawer
					<i>If external drain is activated:</i> Pressure error on solution B circuit. Please check: - Containers are properly connected. - There is no spilling in the upper level.	RETRY CANCEL Works OPEN solution B drawer 1 OPEN solution B drawer 2
		074xxx	FLG_ERR_PRESSURE_SEN SOR_2	Pressure sensor malfunction detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK

Module	Device	Error code	Error	Cause	Text	Options
FLU (05)	Pressure sensor 5 (08)	071xxx	ERRORCODE_FLU_VACUUM_FAIL_1	Vacuum error detected on vacuum 1 fluidic circuit.	<i>If external drain is not activated:</i> Vacuum error. Please check: - Containers are properly connected. - There is no spilling in the upper level.	RETRY CANCEL Works OPEN Waste 1 drawer OPEN Waste 2 drawer
					<i>If external drain is activated:</i> Vacuum error. Please check: - Lab drain tube and connection. - There is no spilling in the upper level.	RETRY CANCEL Works
		Info	FLG_ERR_VACUUM_SENSOR_1	Pressure sensor malfunction detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
	Pressure sensor 6 (09)	072xxx	ERRORCODE_FLU_VACUUM_FAIL_2	Vacuum error detected on vacuum 2 fluidic circuit.	<i>If external drain is not activated:</i> Vacuum error. Please check: - Containers are properly connected. - There is no spilling in the upper level.	RETRY CANCEL Works OPEN Waste 1 drawer OPEN Waste 2 drawer
					<i>If external drain is activated:</i> Vacuum error. Please check: - Lab drain tube and connection. - There is no spilling in the upper level.	RETRY CANCEL Works

Module	Device	Error code	Error	Cause	Text	Options
FLU (05)	Pressure sensor 6 (09)	Info	FLG_ERR_VACUUM_SENSOR_2	Pressure sensor malfunction detected.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
	CPD (17) CPD 1 (18) CPD 2	085xxx	ERRORCODE_FLU_UNABLE_TO_READ_DILUTOR_1_PLUNGER_POSITION	Not possible to read dilutor 1 plunger position. FW aborts the syringe process, due to an overpressure error for example, and this causes this dilutor reading error in the driver.	NA (error is TSS).	NA
		086xxx	ERRORCODE_FLU_UNABLE_TO_READ_DILUTOR_2_PLUNGER_POSITION	Not possible to read dilutor 2 plunger position. FW aborts the syringe process, due to an overpressure error for example, and this causes this dilutor reading error in the driver.		
PIP (08)	CPD (17) CPD 1 (18) CPD 2	003xxx	ERRORCODE_FW_LEN	FW has received a command with incorrect length.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		005xxx	ERRORCODE_FW_ARG	Command has wrong arguments.		
		027xxx	ERRORCODE_DIL_DCP_DRIVER_ERROR	A DCP FW driver error has been produced.		
		028xxx	ERRORCODE_DIL_DCP_OVER_PRESSURE	Pressure value beyond limits.	Obstruction detected in a probe during pipetting.	RETRY RE-RUN OPEN (option not available if error occurs in Dilution Cup)
		029xxx	ERRORCODE_DIL_DCP_TIMEOUT_ERROR	Timeout (waiting for DCP pressure sensor to return to a non-clot value) has expired.	Obstruction detected during pipetting sample(s) x located in position(s) x. Please Open the drawer and remove the obstruction.	RETRY RE-RUN OPEN
		030xxx	ERRORCODE_DIL_DCP_CLOT	DCP value is above threshold.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK

Module	Device	Error code	Error	Cause	Text	Options
PIP (08)	Dilutors (17) Dilutor 1 (18) Dilutor 2	Info	ERRORCODE_DATANOTRE ADY	NA	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		003xxx	ERRORCODE_FW_LEN	FW has received a command with incorrect length.		
		005xxx	ERRORCODE_FW_ARG	Command has wrong arguments.		
		006xxx	ERRORCODE_FW_BUSY	Command is addressed to a busy device.		
		018xxx	ERRORCODE_DIL_CORRUP TED CONFIGURATION FILE	The dilutor configuration file is unreadable.		
		019xxx	ERRORCODE_DIL_CONFIG URATION_FILE_NOT_FOUN D	The dilutor configuration file is unable to be opened.		
		020xxx	ERRORCODE_DIL_CANNOT _LOAD_CONFIGURATION_F ILE	There is a problem during the dilutors configuration file reading.		
		021xxx	ERRORCODE_DIL_UNKNO WN_COMMAND	Dilutor has received an unknown command.		
		022xxx	ERRORCODE_DIL_COMMA ND_TOO_LONG	A command received by the dilutor is too long.		
		023xxx	ERRORCODE_DIL_COM_ER R	Dilutor communication error.		
		024xxx	ERRORCODE_DIL_COMMA ND_NOT_SENT	FW has received a command addressed to a dilutor, but the command has not been send to the dilutor.		
		025xxx	ERRORCODE_DIL_SYNC_TI MEOUT	Dilutor has been busy for more than its timeout margin.		
		026xxx	ERRORCODE_DIL_OEM_ER ROR	A dilutor internal error has been produced.		
		031xxx	ERRORCODE_DIL_UNKNO WN_ERROR	Unknown dilutor error has been produced.		

Module	Device	Error code	Error	Cause	Text	Options
Any (xx)	Any (xx)	002xxx	ERRORCODE_CMD	Unknown command code received.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		003xxx	ERRORCODE_LEN	FW has received a command with incorrect length.		
		004xxx	ERRORCODE_DEV	Command is addressed to an incorrect device.		
		005xxx	ERRORCODE_ARG	Command has wrong arguments.		
		014xxx	ERRORCODE_HW_DEV	The addressed device has a problem with its related HW.		
		015xxx	ERRORCODE_HW_HWFAIL	There is a problem with PCA HW.		
		016xxx	ERRORCODE_HW_HWNOT SET	PCA setup error.		
Non firmware errors (xx)	Non firmware errors (xx)	000xxx	ERRORCODE_UNKNOWN	Unknown error.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		008xxx	ERRORCODE_FW_NOTSAFE	Movement not safe.		
		017xxx	ERRORCODE_FW_UNKNOWN_ERROR	Unknown error.		
		061xxx	ERRORCODE_INVALID_ARGUMENTS	Invalid arguments.		
		062xxx	ERRORCODE_FWCOMM_TIMEOUT	Communication timeout.		
		063xxx	ERRORCODE_FWCOMM_CONNECTION_LOST	Communication lost.		
		064xxx	ERRORCODE_FWCOMM_COMM_ERROR	Communication error.		

Module	Device	Error code	Error	Cause	Text	Options
Non firmware errors (xx)	Non firmware errors (xx)	065xxx	ERRORCODE_FLU_NOT_ENOUGH_SOL_A	Not enough solution A to execute a fluidic operation.	If there are running works: System has been stopped caused by a container problem and there are pending works. Please, select one of the following options:	RETRY RE-RUN Open
					If there are not running works: System has been stopped caused by a container problem. Please, select one of the following options:	
		066xxx	ERRORCODE_FLU_NOT_ENOUGH_SOL_B	Not enough solution B to execute a fluidic operation.	If there are running works: System has been stopped caused by a container problem and there are pending works. Please, select one of the following options:	RETRY RE-RUN Open
					If there are not running works: System has been stopped caused by a container problem. Please, select one of the following options:	
		067xxx	ERRORCODE_FLU_NOT_ENOUGH_WASTE_CAPACITY	Not enough waste capacity to execute fluidic operations.	If there are running works: System has been stopped caused by a container problem and there are pending works. Please, select one of the following options:	
					If there are not running works: System has been stopped caused by a container problem. Please, select one of the following options:	

Module	Device	Error code	Error	Cause	Text	Options
Non firmware errors (xx)	Non firmware errors (xx)	Non firmware errors (xx)	ERRORCODE_FLU_WASTE_TRAY_FULL	The card waste disposal tray is full.	Waste container is full. Please open the drawer and empty it.	OK
		073xxx	ERRORCODE_FLU_ADJUST_ERROR	Impossible to set adjustment.	NA (error is TSS)	NA
		082xxx	ERRORCODE_WASTE_CIRCUIT_ERROR	Error in waste circuit.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		087xxx	ERRORCODE_FLU_UNKNOWN_ERROR	Unknown error.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		088xxx	ERRORCODE_PRECONDITION_FAILED	Previous condition for the action is not met.		
		089xxx	ERRORCODE_STARTUP_ERROR	Driver startup error.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		091xxx	ERRORCODE_ALREADY_STARTED	Another instance of Driver is already opened.		
		093	ERRORCODE_FLU_LOW_PRIMED_SOL_A	Solution A in drawer 1 not primed.	There is a problem in the fluidic system. Please, reintroduce the containers and ensure that they are properly connected and select RETRY. Select CANCEL if the problem persists only.	CANCEL Works
				Solution A in drawer 4 not primed.		CONTINUE
		094	ERRORCODE_FLU_LOW_PRIMED_SOL_B	Solution B in drawer 2 not primed.		CANCEL Works
				Solution B in drawer 5 not primed.		CONTINUE
0	240	090xxx	ERRORCODE_UNEXPECTED_ERROR	A Driver unexpected error has been produced.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK

Module	Device	Error code	Error	Cause	Text	Options
None	None	None	None	Possible dripping detected at PRESSURE_SENSOR_x (where x is the affected sensor).	If error occurs over a sample / reagent drawer: Dripping on drawer error If error occurs over dilution cups or incubator: Dripping on instrument	CANCEL Works CONTINUE

Table 72. List of errors related to Fluidics module

Due to the fluidic procedures consist on different steps, in order to focus properly the troubleshooting it is necessary to know in which step the error appears. This can be done in 2 different ways:

- By means of the last 3 digits of the fluidic error (if they are different than 000).
- By means of *DG59Driver.log*, the error message is stored with the following structure:

[ERROR] ERROR Error Code: (xxyyzzsss)

Fluidic system error

Description: _____

where xxyyzzsss is the error code that can be found in the error lists of this Service Manual. Example (please note that, between these lines of the log file, it is possible that some others lines are registered):

```
[WARN] FLU Starting Initial Low Prime
[INFO] FLU Resetting fluidic device: PUMP_ASP (PM2)
[INFO] FLU Resetting fluidic device: PUMP_DISP_AB (PM1)
[INFO] FLU Resetting fluidic device: PUMP_DISP_B (PM3)
[INFO] FLU Setting fluidic device: PUMP_DISP_AB (PM1)
[INFO] FLU Setting fluidic device: EV_FAST_PRIME_B (EV19)
[INFO] FLU Setting fluidic device: EV_FAST_PRIME_AB (EV20)
[INFO] FLU Setting fluidic device: EV_CH_SOL (EV13)
[INFO] FLU PRIME_LOW_INITIAL_2 - Delay (sec.): 40
[INFO] FLU PRIME_LOW_INITIAL_3
[INFO] FLU Verify Margins failed. Sensor: 3 - Instant value: 48 - Min: 50 Max: 350
[ERROR] FLU Initial low Prime Aborted!!
[INFO] FLU Sensor PRESSURE_SENSOR_3 set to 160 milibars
[INFO] FLU Sensor PRESSURE_SENSOR_4 set to 160 milibars
[INFO] FLU Resetting all Pumps and Valves and terminating Dilutors
[WARN] PIP PROBE_1 dilutor is in mode IDLE and will not be terminated because there is no
instrument flow risk
[WARN] PIP PROBE_2 dilutor is in mode IDLE and will not be terminated because there is no
instrument flow risk
[ERROR] FLUSR Service FLUIDIC_SYSTEM_SERVICE with mode FAST_PRIME failed
[ERROR] ERROR Error Code: (0506069053)
Fluidic system error
Description: Pressure fail on circuit A,
```

Once the error is located in the log file, it is necessary to check the specific step of the fluidic procedure where this error has appeared. Just checking a few lines above the error, it can be found the step and the parts activated in that moment. Further information in all the fluidic procedures can be found in this Service Manual, Fluidics (FLU) Module chapter.

In the log file, the components that are activated are registered with the **Setting fluidic device** command. Components that are deactivated / returned to its original position are registered with **Resetting fluidic device** command.

Once the fluidic step is known (and the parts that are activated), in TSS is possible to simulate this step activating the components in Fluidics tab and checking the value of the fluidic sensors (S3, S4, S5 or S6).



NOTE:

Empty the fluidic circuit every time a simulation of a specific step of the fluidic procedures is performed activating in TSS the different related components.

Last 3 digits error code	Procedure	Fluidic step (refer to section 9.1.2)
001	Check sol. A level – internal drain	Error during check bottle levels step in fluidic procedures
002	Check sol. B level – internal drain	Error during check bottle levels step in fluidic procedures
003	Check waste level – internal drain	Error during check bottle levels step in fluidic procedures
004	Check sol. A level – external drain	Error during check bottle levels step in fluidic procedures
005	Check sol. B level – external drain	Error during check bottle levels step in fluidic procedures
006	Dilution cup fill	Error during step 8 of dilution cup wash procedure
007	Dilution cup drain	Error during step 5 / 12 dilution cup wash procedure
008	Probe Prime	Error during step 3 of syringe prime procedure
009	Probe Rinse	Error during step 3 of syringe rinse procedure
010	Probe Empty	Error during steps 18/ 21 / 22 / 25 of empty procedure
026	Internal wash	Error during step 5 of internal wash procedure
027	Internal wash with sol. B	Error during step 5 of internal wash procedure
028	Slow external wash	Error during step 4 / 5 of slow external wash procedure
029	Fast external wash	Error during step 4 / 5 of fast external wash procedure
052	Low Prime	Error during sensors status monitoring before step 5 of low prime procedure Error during sensors status monitoring before step 4 of low rinse procedure
053	Low Prime	Error during step 5 of low prime procedure Error during step 4 of low rinse procedure
102	Low Prime	Error during sensors status monitoring before step 6/7 of low prime procedure Error during sensors status monitoring before step 5 of low rinse procedure
103	Low Prime	Error during step 6 / 7 of low prime procedure Error during step 5 of low rinse procedure
152	Low Prime	Error during sensors status monitoring before step 8/9 of low prime procedure
153	Low Prime	Error during step 8 / 9 of low prime procedure
252	Probes washer High Prime	Error during step 6 of high prime solution B (probes washer) procedure
254	Probes washer High Prime	Error during step 10 of high prime solution B (probes washer) procedure
256	Probes washer High Prime	Error during sensors status monitoring before step 13 of probes washer prime procedure
257	Probes washer High Prime	Error during step 14 of high prime solution B (probes washer) procedure
260	Dilution cup High Prime	Error during step 4 of high prime solution A (dilution cup) procedure
262	Dilution cup High Prime	Error during step 8 of high prime solution A (dilution cup) procedure
264	Dilution cup High Prime	Error during sensors status monitoring before step 11 of dilution cup prime procedure

Last 3 digits error code	Procedure	Fluidic step (refer to section 9.1.2)
265	Dilution cup High Prime	Error during step 12 of high prime solution A (dilution cup) procedure
302	Low Rinse	Error during sensors status monitoring before step 6 of low rinse procedure
303	Low Rinse	Error during step 6 of low rinse procedure
306	Probes washer High Rinse	Error during step 4 of high rinse (probes washer) procedure
308	Probes washer High Rinse	Error during step 8 of high rinse (probes washer) procedure
310	Probes washer High Rinse	Error during sensors status monitoring in step 10 of high rinse (probes washer) procedure
311	Probes washer High Rinse	Error during step 12 of high rinse (probes washer) procedure
314	Dilution cup High Rinse	Error during step 4 of high rinse (dilution cup) procedure
316	Dilution cup High Rinse	Error during step 8 of high rinse (dilution cup) procedure
318	Dilution cup High Rinse	Error during sensors status monitoring in step 10 of high rinse (dilution cup) procedure
319	Dilution cup High Rinse	Error during step 12 of high rinse (dilution cup) procedure
352	Empty	Error during sensors status monitoring before step 6/9 of empty procedure Error during step 6/9 of empty procedure
355	Empty	Error during sensors status monitoring before step 13/16 of empty procedure Error during step 13/16 of empty procedure
358	Empty	Error during sensors status monitoring before step 19/21 of empty procedure
359	Empty	Error during sensors status monitoring before step 23/25 of empty procedure
360	Empty	Error during sensors status monitoring before step 27 of empty procedure Error during step 27 of empty procedure
361	Empty	Error during step 30 of empty procedure
362	Empty	Error during sensors status monitoring before step 32 of empty procedure Error during step 32 of empty procedure
363	Empty	Error during sensors status monitoring before step 36 of empty procedure
364	Empty	Error during sensors status monitoring before step 40 of empty procedure
402	Check Waste	Error during check waste step in fluidic procedures
404	Check Waste	Error during check waste step in fluidic procedures

Table 73. List of errors (3 last digits) related to Fluidics module

FLUIDIC SYSTEM ISSUES

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
The system is not primed / rinsed	Dilutor/s not initialised	Check that dilutor is initialised
	Dilutor/s valve/s failure	Check / replace dilutor/s valve
	Leak in tube or fitting	Check tubing and fittings
	Dispensation circuit is not working	Check that there is enough liquid in the bottles and they are properly connected Check in DG59Driver log file, the prime / rinse step where the error appears (the fluidic parts activated are detailed in the log file)
	Aspiration circuit is not working	Check that there is enough room in the waste bottles and they are properly connected Check in DG59Driver log file, the prime / rinse step where the error appears (the fluidic parts activated are detailed in the log file)
The system is not emptied	Dilutor/s not initialized	Check that dilutor is initialised
	Dilutor/s valve/s failure	Check / replace dilutor/s valve
	Aspiration circuit is not working	Check that there is enough room in the waste bottles and they are properly connected Check in DG59Driver log file, the empty step where the error appears (the fluidic parts activated are detailed in the log file)
Leak in probes washer when washing probes	Not enough vacuum arrives to the probes washer	Check waste bottles Check aspiration pump (PM2) Check electrovalves EV5 / EV10 Check/adjust/replace vacuum sensors (S5 / S6)
	Tubes/fittings obstructed by dirt / squeezed	Replace tubing, fittings
	Home position of probe/s too low	Check probe home position. Perform probes offset adjustment.
	Probe/s bent	Replace probe/s
	Too much pressure/liquid arrives to the probes washer	Check solution bottles Check/replace dispensation pump (PM3) Check electrovalves EV1 / EV6 Check/adjust/replace pressure sensors (S3 / S4)
	Clot/obstruction in probes washer	Remove clot/obstruction
	Probes washer scratched	Replace probes washer
	Probe/s partially blocked	Replace probe/s
Dirt on probes washer	Probably salt from saline solution	Clean it with distilled water
Electrovalve is not working	Electrovalve broken	Check / replace electrovalve
	PCA FLU-CTRL59 damaged	Check / replace PCA FLU-CTRL59

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Bubbles in sample line	Syringe tube connector defective or wrongly assembled	Replace/fix tube connector
	O-ring (inside syringe valve) lost or defective	Replace O-ring
	Syringe valve defective	Replace dilutor valve
	Syringe plunger seal worn out	Replace plunger
	Syringe glass worn out	Replace glass
	Syringe not properly installed	Install correctly the syringe
	Tubes connected wrongly.	Check tubes connection
	Probe assembly defective	Check probe assembly
	Intermediate manifold BK2, BK4, BK5 and/or BK6 tube connector defective or not properly connected	Replace/fix tube connector/connection
	CPDs assembly defective or incorrectly assembled	Check CPDs assembly
	CDP defective	Replace CDP
	Intermediate manifold BK2, BK4, BK5 and/or BK6 defective	Replace defective interconnection manifold
	Syringe to probe teflon tube defective	Replace defective teflon tube / chain
	Bubbles coming from solution A/B bottles	Check tubing/parts from solution A/B bottle/s to dilutor valves
Overflow in dilution cup	Clot expelled during pipetting in the bottom of the cup/obstruction	Remove clot / obstruction
	Tube blocked or squeezed	Replace / fix tube
	Leak from probes washer	Refer to Leak in probes washer section in this table
	Too much pressure/liquid arrives to dilution cup	Check solution bottles
		Check/replace dispensation pump (PM1)
		Check / replace electrovalves EV3 / EV8
		Check/adjust/replace pressure sensors (S3 / S4)
	Not enough vacuum arrives to dilution cup	Check waste bottles
		Check aspiration pump (PM2)
		Check / replace EV4 / EV9
		Check/adjust/replace vaccumm sensors (S5 / S6)

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Wells not dispensed/incorrect (too low) volume	Incorrect tube diameter detection	Check / readjust / fix PIP presence detector
	Aspiration valves of dilution cup defective	Check / replace EV4 / EV9
	Level detection failure	Check/fix level detection
	Sufficient volume determination failure	Check / readjust probe maximum depth Check / readjust / fix PIP presence detector
	Bubbles in sample line	Refer to bubbles in sample line in this table
	Incorrect probe positioning in dilution cup	Check / adjust probes in dilution cup
	Incorrect probe positioning in samples drawers	Check / adjust probes in samples drawers
	Incorrect probe positioning in reagent drawers	Check / adjust probes in reagent drawers
	Incorrect probe positioning in incubators	Check / adjust incubator flatness Check / adjust incubator height Check / adjust probes in incubators
	Dilutor failure	Replace dilutor
Wells with incorrect (too high) volume	Bubbles in sample line	Refer to bubbles in sample line in this table
	Dilutor failure	Replace dilutor
	Leak from probes washer	Refer to Leak in probes washer section in this table
CPD leakage	CPD sensor is broken	Check / replace CPD sensor (S1, S2)
	Incorrect CPD assembly into interconnection manifold	Check CPD assembly
Plunger overload in dilutor module	Dilutor failure	Replace dilutor
	Syringe failure	Replace syringe
	Input/output valve tubing obstructed	Check/replace related tubing

BOTTLES /CARD WASTE DRAWERS ISSUES

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Drawer is not opened	Lid electric lock damaged	Check / replace lid electric lock
	PCA GEN-LCK59 wiring damaged	Check / replace cables
	PCA GEN-LCK59 damaged	Check / replace PCA GEN-LCK59
	PCA FLU-CTRL59 damaged	Check / replace FLU-CTRL59
Cosmetic front panel is not aligned	Cosmetic front panel screws are loose	Align cosmetic front panel
Drawer is not aligned	Latch depth position is not correct	Adjust latch position
Hard to open / close the drawer	Bad status of linear guides	Replace linear guides
	Incorrect PCA GEN-LCK59 protective cover assembly	Check PCA GEN-LCK59 assembly
	Lack of lubrication	Lubricate
Weight measured is not correct	Drawer not closed properly	Check fluidic drawer
	Incorrect load cell adjustment	Check / perform load cell adjustment

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
No weight is measured	Load cell damaged	Check / replace load cell
	Load cell wiring damaged	Check / replace load cell wiring
	PCA FLU-CTRL59 damaged	Check / replace PCA FLU-CTRL59

Table 74. Fluidics module troubleshooting

10 DILUTION CUP (CUP) MODULE

10.1 DESCRIPTION

Dilution cup has 2 independent teflon wells so 2 independent dilutions can be prepared simultaneously or in any of the 2 wells. In order to homogenize the dilution, one stirring system is provided.

The structure of the module is formed by the following elements:

- Wells: where the liquids are dispensed to make the dilution.
- Stirring system: is the mechanism responsible for homogenize the dilution.
- Supporting structure is responsible for holding the stirring system and the wells, attaching them to the instrument structure. Isolate the dilution cup vibration to the rest of the modules.

Each well has 2 fluidic connectors to connect the tubes required to wash and drain each well. For each well, one connector (located at the upper part of the well) allows filling the well with solution A or B, while the other one (located at the bottom part of the well) allows emptying to the waste bottle. The wells are oriented in a way that allows the 2 probes to access the wells at the same time.

The stirring is accomplished using a motor with a counterweight attached to its axis, and with a silentblock acting as an elastic element to allow the orbital movement. The silentblock is placed between the motor, located at the bottom part of the structure, and the wells located in the upper part of the structure, in order to achieve a proper stirring.

The motor, a brushless type, rotates in only one direction. The control of the velocity is performed by the hall sensors set of the motor and it is overcurrent monitored. Speed and time are determined by the software.

A cover attached to the main structure prevents any spill from reaching the motor or the PCA. The cover has an evacuation channel for the liquid that may come from a well overflow.

General block diagram of the module is shown in the following figure,

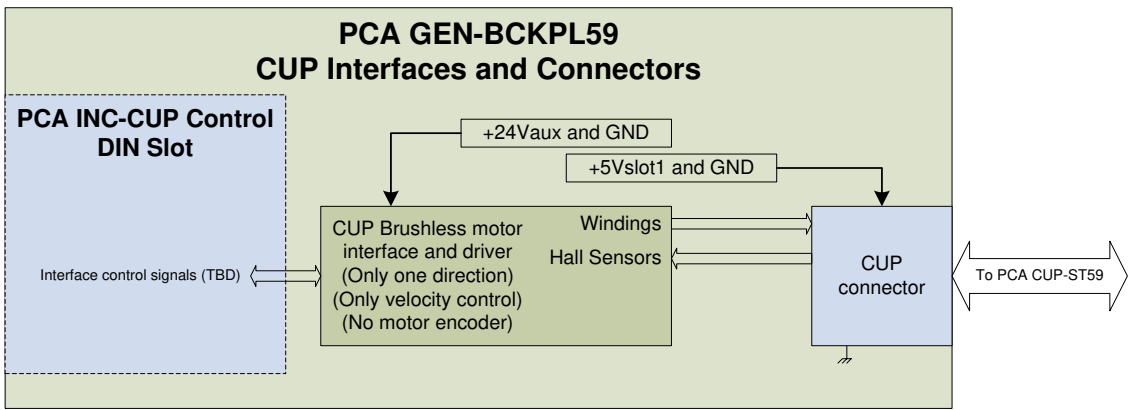


Figure 459. Block diagram of Dilution cup module

PCAs related to the CUP module are:

- **PCA GEN-BCKPL59:** main PCA of the instrument, located at the backpanel.
- **PCA INC-CUP Control:** located at the backpanel (slot 1), attached to PCA GEN-BCKPL59. It includes one brushless motor interface and driver to manage dilution cup motor.
- **PCA CUP-ST59:** located under the dilution cup. It contains the connectors to match with the stirring motor.

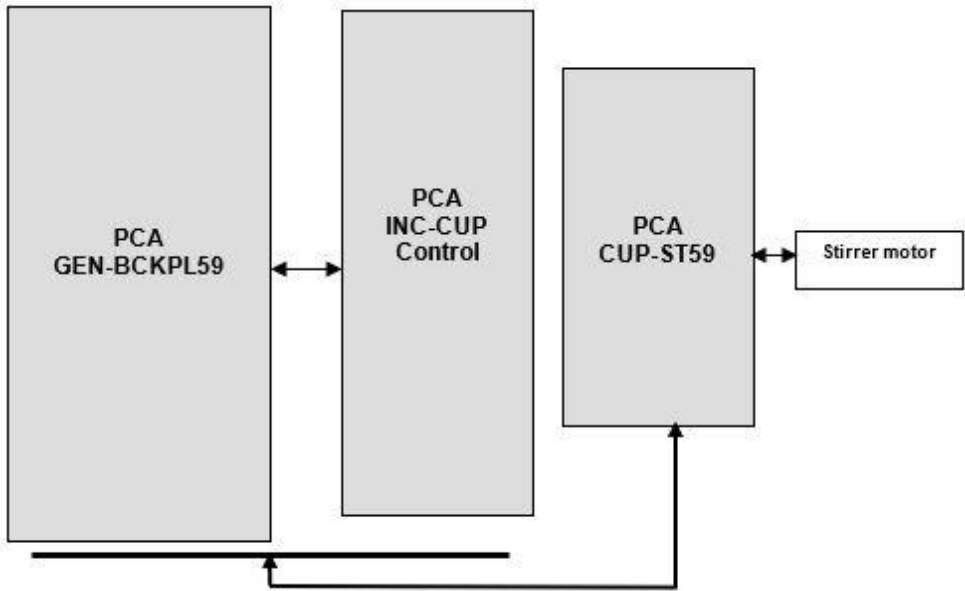


Figure 460. Main blocks of Dilution cup module

10.2 ADJUSTMENT PROCEDURES

In this section, the adjustments related to dilution cup are explained. There are 2 types of adjustments: mechanicals and functionals.

10.2.1 Mechanical adjustments

In this section, dilution cup parallelism is detailed.

10.2.1.1 Dilution cup parallelism

This adjustment is needed to place parallel the dilution cup wells with the probes in order to allow that the probes enter into the dilution cup wells.

Prerequisite: This adjustment can be done outside the instrument. In order to remove the dilution cup from the instrument, the next steps should be followed:

1. Remove the 4 screws of the dilution cup bath
2. Remove the screws of the cover and take out the cover by the front of the instrument.
3. Remove the 2 screws that are attaching the dilution cup module to the *Erytra* chassis.
4. Disconnect the flat cable from the PCA CUP-ST59.
5. Remove the dilution cup wells by means of the 2 screws that attach it to the dilution cup structure. In order to get a better access to the rear screw, it is recommended to rotate a little bit the dilution cup module.
6. Remove the dilution cup bath.
7. Remove the whole module.

Adjustment tool: Dilution cup alignment tool (PN: 233820)

Procedure:

1. Block the setscrew with an Allen key and loose the nut.
2. Check that the silentblock is touching the bottom surface of the dilution cup support and the setscrew is tighten until is going out from the dilution cup support 6mm, as it is shown in the following picture,

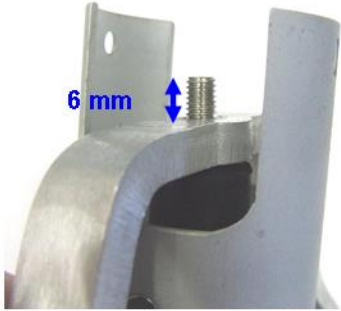


Figure 461. Setscrew inserted in the silentblock

3. Place the spacer in the setscrew and then place the dilution cup parallelism Y axis tool inside the dilution cup structure. Use the screws of the dilution cup wells to fix it.

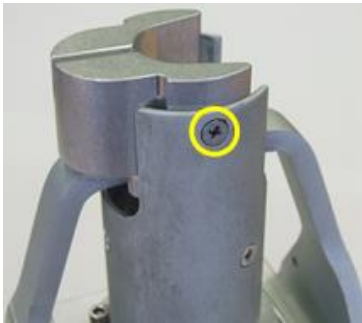


Figure 462. Dilution cup alignment tool fixed into dilution cup structure

4. The setscrew is fixed by means of an Allen key and the tool, as it is shown in the picture.



Figure 463. Parallelism adjustment of the dilution cup module

5. Tighten the nut (check that it is not too tight or or too loose), remove Allen key and the tool and check that the bar of the shaking support is centered in the shaking tube.

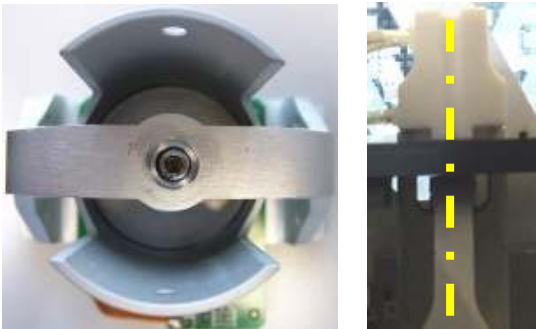


Figure 464. Checking of the correct adjustment of the dilution cup

10.2.2 Functional adjustments

There are not functional adjustments in Dilution cup module.

10.3 VERIFICATIONS

In order to verify that Dilution cup module is working properly, follow the next steps:

- 1. Remove the 4 screws of the dilution cup bath and take out the cover by the front of the instrument.
- 2. Remove the dilution cup wells by means of the 2 screws that attach it to the dilution cup structure and then remove the dilution cup bath.
- 3. Place the dilution cup wells and tighten its 2 screws.
- 4. Stick a reflective adhesive to the basis of the motor and use a tachometer to read the rpm during a stirring. Check that the adhesive does not interfere with anything during the rotation of the motor. In order to start a stirring process, open Technical Service software and go to *Dilution Cup* tab, select *PROFILE_0* in *Stirrer Actions* submenu and press *Spin* button. Expected value of the tachometer measurement is 800 rpm $\pm$ 80 rpm.

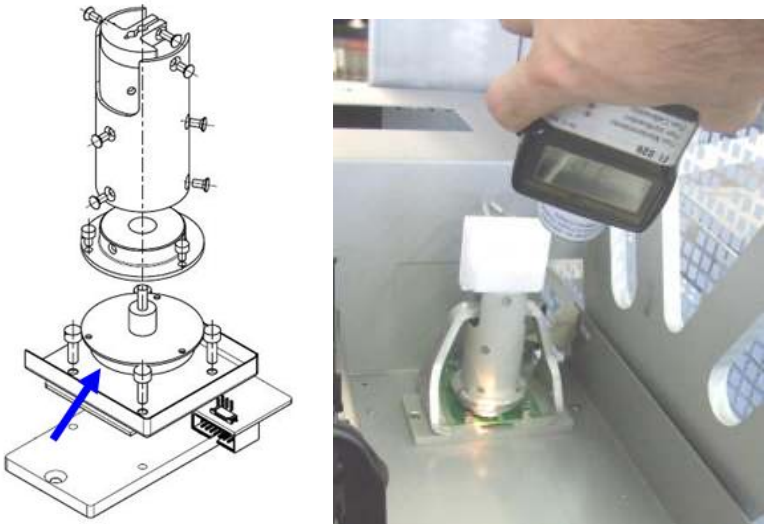


Figure 465. Location of the basis of the motor (left) and rpm measurement with a tachometer during stirring (right)



NOTE:

The use of any tachometer is possible as long as it complies with the following specifications:

Type: Optical
Scale division (rpm): ≤ 1 rpm
Range (rpm): 1 – 99999 rpm
Maximum uncertainty permitted: ± 3 rpm

5. Remove the reflective adhesive.
6. Check the status of the PCA plastic cover.
7. Remove the dilution cup wells and place the cover with its screws.
8. Place the dilution cup bath and tighten its 4 screws.
9. Place the dilution cup wells and tighten its 2 screws.
10. Check that the dilution cup bath is centered in the dilution cup structure to avoid that the dilution cup wells touch the dilution cup bath during the stirring movement.
11. Perform a prime and a wash of the dilution cup in order to check that liquid is dispensed into the wells and it is properly aspirated. To do so, select *fluidic prime* and *wash dilution cup* (selecting desired number of iterations) options in *Fluidics* tab of Technical Service software.

10.4 REPLACEMENT PROCEDURES

In this chapter procedure to replace the motor of the dilution cup is detailed.

10.4.1 Dilution cup wells assembly replacement

1. Switch off the instrument and disconnect from mains.
2. Open the upper door.
3. Remove the dilution cup wells by means of the 2 screws that attach it to the dilution cup structure. In order to get a better access to the rear screw, it is recommended to rotate a little bit the dilution cup module.



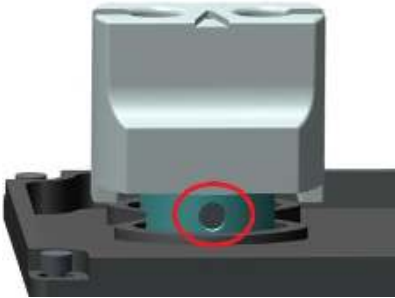
Figure 466. Dilution cup screws

4. Place the new dilution cup wells. After connecting back the tubes, make sure they don't modify/force the dilution cup wells resting position.
5. Perform a Prime to check the fluidic system.
6. Check that the probes don't impact against the dilution cup in both aspiration and dispensing positions when the dilution cup is moving.

10.4.2 Dilution cup motor replacement

In order to replace the motor of the dilution cup follow the next steps:

1. Remove the dilution cup wells unscrewing the 2 screws, one on the front and the other one on the back, attaching it to dilution cup structure.



2. Disconnect the 4 tubes connected to the rear part of the dilution cup.
3. Remove the 4 screws of the dilution cup bath.



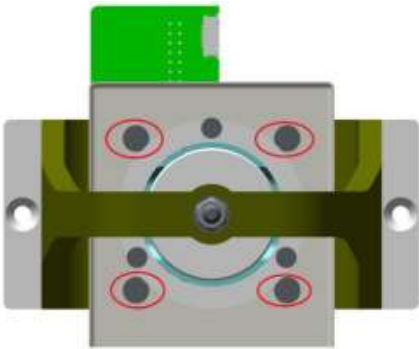
4. Unscrew the 5 screws attaching the cover and remove it by the front of the instrument.



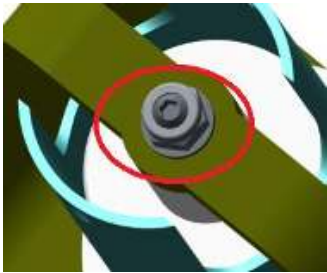
5. Remove the 2 screws that are attaching the dilution cup module to the upper floor, disconnect the flat cable from PCA CUP-ST59 and extract the whole module outside the instrument.



- 6. Carefully disconnect the motor cable from PCA CUP-ST59.
- 7. Remove PCA CUP-ST59 and the plastic protective cover by means of the 4 screws that are attaching them to base of the dilution cup module. Dispose of the plastic cover, PCA and the 4 screws.



- 8. Unscrew the silent block from the dilution cup support.



- 9. Remove the dilution cup structure from the dilution cup support.



- 10. Remove the 3 screws that are located in the lower part of the dilution cup structure fixing the motor assembly and remove it.



11. Remove the 3 screws that are located in the upper part of the dilution cup structure and remove the shock absorber assembly.



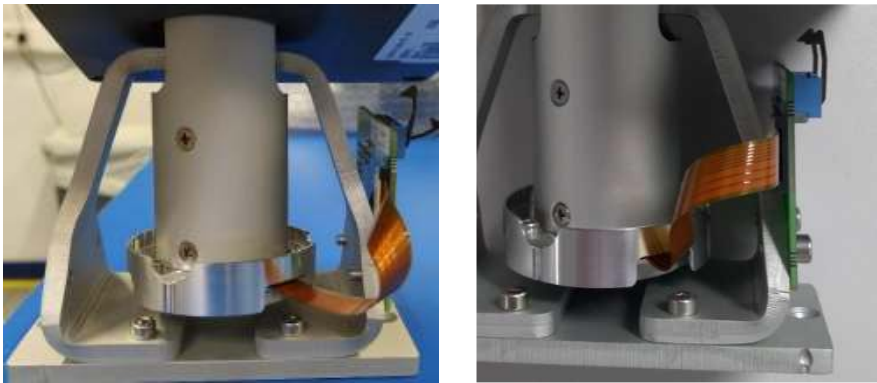
12. Put the new motor with P/N 733699 MOTOR STIRRER DILUTION (Ver. 2) in the same place as the one previously removed, in the lower part of the dilution cup structure. Orientation of the cable must match the one in the following picture:



13. Put the new shock absorber assembly with P/N 733700 SHOCK ABSORBER DILUTION CUP (Ver. 3) in the same place as the one previously removed, in the upper part of the dilution cup structure.
14. Screw the new PCA with P/N 733698 PCA CUP-ST59 v2 to the dilution cup support on any of its sides, as shown on the following image:

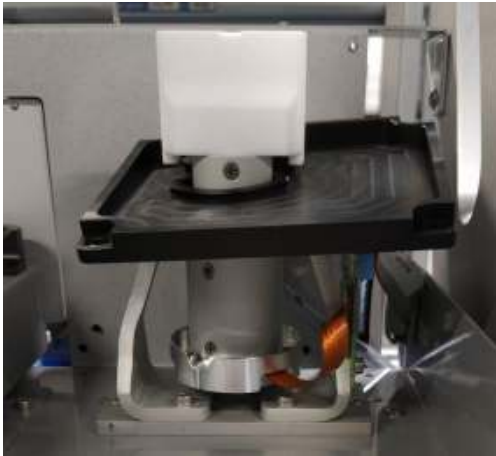


15. Attach the dilution cup structure to the dilution cup support and base using the 4 new screws with P/N 730606 SCREW M4x10 DIN 912, with the motor cable orientated to PCA H1 connector.



NOTE:
At this point, dilution cup parallelism adjustment must be performed.

16. Put back the dilution cup bath in its place, then connect the tubes again on the back on the dilution cup wells and attach this one to the structure, using the previously removed screws.
17. Connect the flat cable to PCA CUP-ST59 v2, then screw back the dilution cup module on its place in the upper floor using the 2 previously removed screws. The PCA must be facing the right lateral cover of the instrument.



18. Place back the cover on its place, using the previously removed screws.



NOTE:
At this point, probes adjustment over dilution cup should be performed.

19. Check that the probes don't impact against the dilution cup in both aspiration and dispensing positions when the dilution cup is moving.
20. Check RPMs of the dilution cup

10.4.3 Fitting replacement

To replace a fitting of the dilution cup wells, follow next steps:

1. Remove dilution cup wells by means of the 2 screws that attach it to dilution cup structure.
2. Disconnect tubing from dilution cup wells.
3. Remove the fitting to be replaced.
4. Place the new fitting. Tighten it manually to the maximum and then ¼ turn more with an open end spanner number 6.
5. Connect tubing, checking the label attached to the tubes and the reference label attached to the rear part of the dilution cup wells.



Figure 467. Rear part of dilution cup wells

6. Place dilution cup wells into the structure and tighten its 2 screws.
7. Perform a Prime to check the fluidic system.

10.5 FUSES

Fuses related to Dilution cup module are detailed in the following table,

PCA	FUSE NAME	DESCRIPTION
PCA GEN-BCKPL59	FUS11	MST250 0034.6615 1A/250V
	FUS12	MST250 0034.6620 3.15A/250V
PCA INC-CUP Control	FUS1	MSF250 0034.6013 0.8A/250V
PCA CUP-ST59	NA	NA

Table 75. List of fuses of Dilution cup module

10.6 LEDS

LEDs related to Dilution cup module are detailed in the following table,

PCA	LED	DESCRIPTION
PCA GEN-BCKPL59	LED7	Green 5V (FUS11)
	LED8	Green 24V (FUS12)
PCA INC-CUP Control	LED1	Yellow When the instrument is switched on, LED is ON and a few seconds later is OFF. This means that the FPGA has been loaded correctly with the appropriate data.
	LED2	Green VMotor Ok. It indicates that the voltage supplied to the motor is $\geq 12V$
PCA CUP-ST59	NA	NA

Table 76. List of LEDs of Dilution cup module

10.7 JUMPERS AND TEST POINTS

Neither jumpers nor test points are related to Dilution cup module.

10.8 WIRING TABLE

In the following table, there are detailed all the cables related to Dilution cup module. A wiring diagram can be found attached to Service Manual.

ID	FROM CONNECTOR	FROM PCA/COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
B-CUP	H2	PCA CUP-ST59	CUP	PCA GEN-BCKPL59
NA	H1	PCA CUP-ST59	MOTOR	CUP MOTOR

Table 77. List of wiring of Dilution cup module

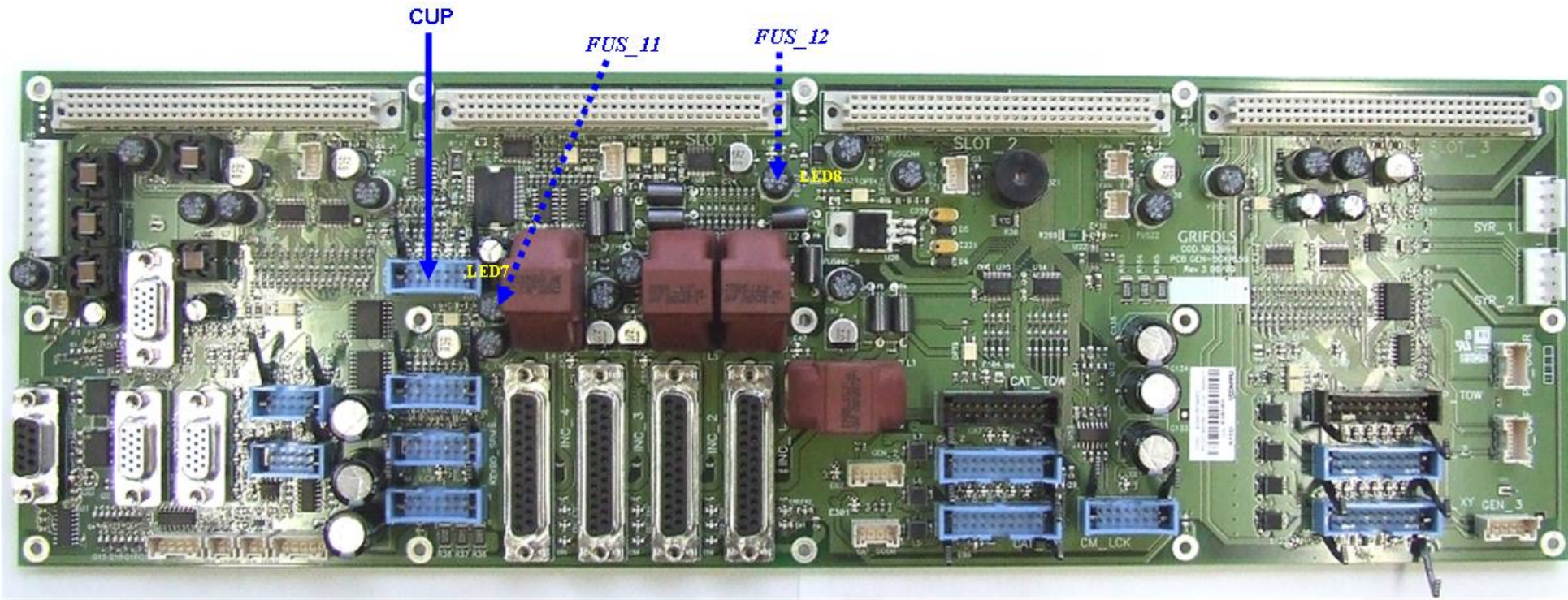


Figure 468. Location of the connector, fuses and LEDs in PCA GEN-BCKPL59 related to Dilution cup module

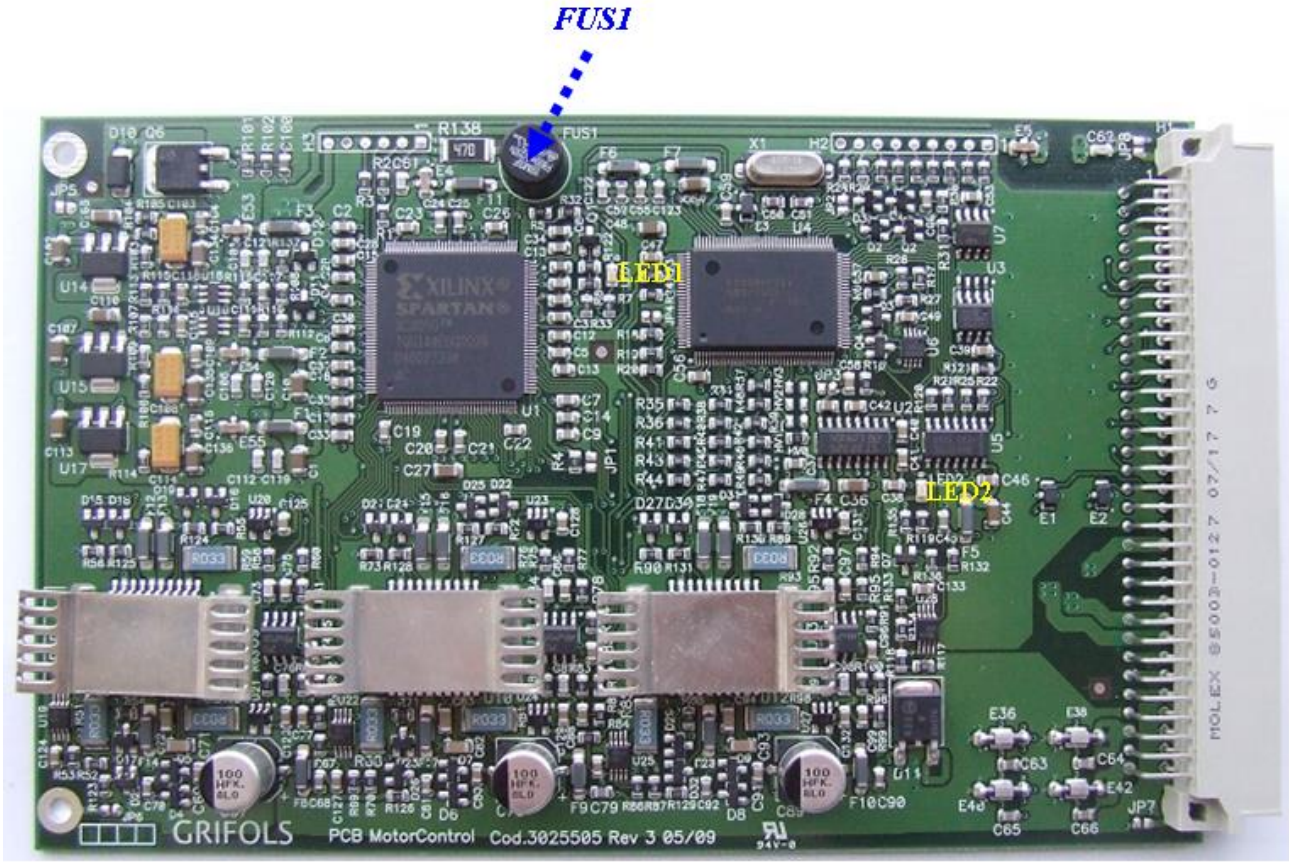


Figure 469. Location of the fuse and LEDs in PCA MotorControl

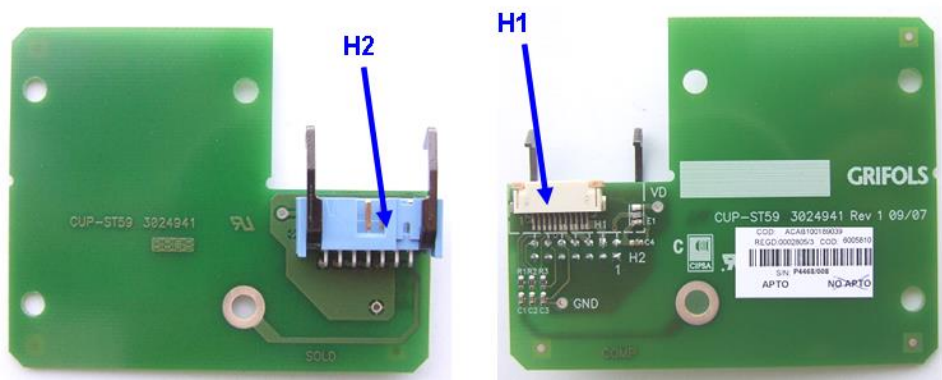


Figure 470. Location of the connectors in PCA CUP-ST59

10.9 TROUBLESHOOTING

Module	Device	Error code	Error	Cause	Text	Options
CUP (04)	Dilution cup (01)	003000	ERRORCODE_LEN	FW has received a command with incorrect length.	Error during reagents stirring process or homogenization in dilution cup. Please choose one of the following options:	RETRY CANCEL Works
		005000	ERRORCODE_ARG	Command has wrong arguments.		
		006000	ERRORCODE_BUSY	Command is addressed to a busy device.		
		012000	ERRORCODE_SPINBADPROFILE	Command includes a spin profile that is invalid for the addressed device.		
		013000	ERRORCODE_NOTSPINNING	The addressed device is not performing the requested spin.		
		032000	CURRENT_LIMIT	Current limit at servo device.		
		034000	MOVEMENT_ERROR	A movement error has been produced at a servo device.		
Any (xx)	Any (xx)	002000	ERRORCODE_CMD	Unknown command code received.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		003000	ERRORCODE_LEN	FW has received a command with incorrect length.		
		004000	ERRORCODE_DEV	Command is addressed to an incorrect device.		
		005000	ERRORCODE_ARG	Command has wrong arguments.		
		014000	ERRORCODE_HW_DEV	The addressed device has a problem with its related HW.		
		015000	ERRORCODE_HW_HWF AIL	There is a problem with PCA HW.		
		016000	ERRORCODE_HW_HWN OTSET	PCA setup error.		

Module	Device	Error code	Error	Cause	Text	Options
0	240	000000	ERRORCODE_UNKNOWN	Unknown error.	The system has a significant error. Please RESTART the instrument before requesting any further test.	OK
		008000	ERRORCODE_FW_NOTSAFE	Movement not safe.		
		017000	ERRORCODE_FW_UNKNOWN_ERROR	Unknown error.		
		061000	ERRORCODE_INVALID_ARGUMENTS	Invalid arguments.		
		062000	ERRORCODE_FWCOMM_TIMEOUT	Communication timeout.		
		063000	ERRORCODE_FWCOMM_CONNECTION_LOST	Communication lost.		
		064000	ERRORCODE_FWCOMM_COMM_ERROR	Communication error.		
		088000	ERRORCODE_PRECONDITION_FAILED	Previous condition for the action is not met.		
		089000	ERRORCODE_STARTUP_ERROR	Driver startup error.		
		091000	ERRORCODE_ALREADY_STARTED	Another instance of Driver is already opened.		
		090000	ERRORCODE_UNEXPECTED_ERROR	A Driver unexpected error has been produced.		

Table 78. List of errors related to Dilution cup module

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Stirring is not performed 0401013000 0401032000 0401034000	Fuse blown	Check / replace FUS11, FUSGEN4
	Motor does not turn	Check / replace motor
	Dilution cup is not stirred but the motor turns	Check assembly of dilution cup
	Cable B-CUP damaged	Check / replace B-CUP cable
	PCA CUP-ST59 failure	Check / replace PCA CUP-ST59
	PCA INC-CUP MotorControl failure	Check / replace PCA INC-CUP MotorControl
	PCA GEN-BCKPL59 failure	Check / replace PCA GEN-BCKPL59
Stirring at different speed that defined one 0401013000 0401032000 0401034000	Improper assembly	Check assembly of dilution cup
	Motor damaged	Check / replace the motor
	PCA CUP-ST59 damaged	Check / replace PCA CUP-ST59
	Wrong firmware	Check / install INC-CUP firmware
	PCA INC-CUP MotorControl failure	Check / replace PCA INC-CUP MotorControl
	PCA GEN-BCKPL59 failure	Check / replace PCA GEN-BCKPL59
Dilution cup is not parallel with probes	Dilution cup incorrectly assembled	Check assembly of dilution cup
Noise when stirring	Dilution cup bath incorrectly assembled	Check assembly of dilution cup bath
	Dilution cup incorrectly assembled	Check assembly of dilution cup

Table 79. Dilution cup module troubleshooting

11 CARD READER (LEC) MODULE

11.1 DESCRIPTION

The functions of this module are card integrity check and results reading:

- Card integrity check/initial identification
 - Initial total volume measurement.
 - Gel volume measurement.
 - Microtube proper condition/artifacts detection.
- Results reading
 - Total volume measurement.
 - Reaction grading.
 - Microtube warnings/incidences: lipemic, haemolysis, artifacts, etc.

To do so, this module consists of a FireWire color camera with optics, a pattern and LED illumination panel controlled by PCA RD-IL421. The camera resolution is 656 (H) x 494 (V).

Reader pattern has four crosses, two vertical lines and two sets of two horizontal lines. The crosses at each corner are used to determine pattern location and orientation (adjustment/checking routines). Vertical lines are used in microtube integrity check/initial check to ease gel volume and total volume calculation (supernatant-air meniscus and gel-supernatant meniscus detection). Horizontal lines sets are used in microtube reading to detect turbidity in non-gel section.

Illumination panel is made of two independent arrays of LED (one array assembled at the top and the other at the bottom). Its light intensity is controlled by software setting current. *Erytra* automatically performs light intensity verification/adjustment at user software start-up. It is also performed in illumination intensity adjustment step of Technical Service software.

Reader takes picture of two microtubes at a time and then four steps/images are required to complete a card processing. Raw images are 640x480 images that contain the pixels information as they were acquired by the sensor (8 bits). Pixels are green, red or blue according to their position on the sensor and the Bayer filter of the camera.

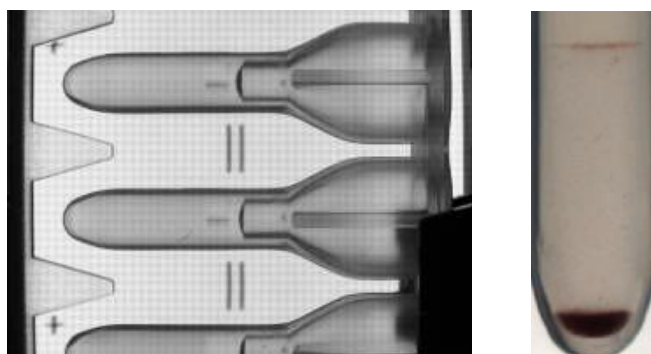


Figure 471. Reader raw image (left) and microtube image (right)

Raw images are saved (images older than 14 days are automatically deleted) in *E:\data\Erytra\Raw* directory with file name:
BBBBBBBBBBBBBBBBBBBB_CCCCCCCC_MXMY_CXCY_PPP_Q.PGM

where:
BBBBBBBBBBBBBBBBBBBB: card barcode
CCCCCCCC: internal unique identification of a test execution performed by the instrument.
MX and *MY*: microtubes numbers associated to that picture (counting from left to right side from non-barcode card side).
CX and *CY*: microtube *MX* and *MY* color.
PPP: Card type (*DGG* for DG Gel cards).
Q: indicates if the picture belong to initial/integrity check (*Q = C*) or to results reading (*Q = R*).

Raw images will be used to obtain each microtube picture when grading reactions (a mask is applied; no reference image exists). 2 files (a *bmp* and a *jpg* file) per microtube will be generated and saved in *E:\data\Erytra\Microtubes* directory. *JPG* reference file will be included in instrument database (upon automatic or manual validation) and reports. These files names will be:
BBBBBBBBBBBBBBBBBBBB_CCCCCCCC_MX_MC_PPP.bmp/jpg

where:
BBBBBBBBBBBBBBBBBBBB: card barcode
CCCCCCCC: internal unique identification of a test execution performed by the instrument.
MX: microtubes numbers associated to that picture (counting from left to right side from non-barcode card side).
MC: microtube *MX* color.
PPP: Card type (*DGG* for DG Gel cards).

General block diagram of the module is shown in the following figure:

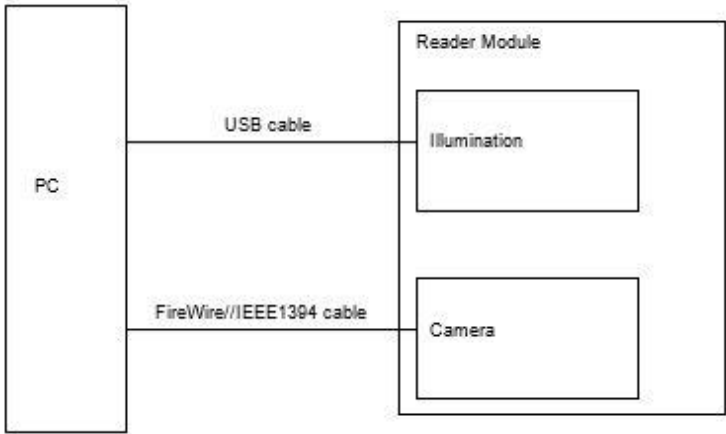


Figure 472. LEC module block diagram

PC - Camera communication is through FireWire/IEEE1394.
PC - PCA RD-IL421 (illumination panel) is through USB.

The PCA related to the LEC module is:

- **PCA RD-IL421**: located in the reader, attached to reader connectors side. It controls reader illumination through PC-RD-IL421 USB cable.

11.2 ADJUSTMENT PROCEDURES

In this section, the adjustments related to the reader module are explained.

11.2.1 Functional adjustments

All adjustments are included in a single wizard in a row. Diaphragm and focus adjustments require access to camera iris and focus and then it is advisable to perform them with reader module disassembled from the instrument; or assembled in the instrument but with the reader lateral cover removed. All other adjustments (illumination and integration adjustments) can be easily completed with the reader assembled inside the instrument.



NOTE:
Reader adjustment tool-support shall be inserted in reader in all the steps of adjustment or verification whenever performed with reader module not assembled inside *Erytra* to avoid external light contribution.



NOTE:
Reader adjustment tool-support shall be inserted in reader together with Reader adjustment tool-pattern in all the steps of adjustment or verification that requires reader adjustment tool-pattern.

Illumination intensity adjustment must be performed whenever reader module is re-assembled in the instrument.

11.2.1.1 Reader Adjustment

Adjustment procedure is described in the following lines assuming that adjustment is performed outside *Erytra* (Reader adjustment tool-support is inserted in the reader in all the steps). Should this not be the case, take into account that Reader adjustment tool-support (Ver. 2) (PN: 234212) should only be inserted when software requires Reader adjustment tool-pattern (PN: 731398).

Whenever an adjustment procedure is performed, the last image acquired during the adjustment routine will be saved in PGM and BMP formats in *E:\Log\DG421\Images* as *LastAdjImage.PGM* and *LastAdjImage.BMP*. Additionally, a log file will be created, containing the values for each parameter, illumination and software version, camera model and serial number. This file is stored in *E:\Logs\DG421* directory with the name: *Adjustment-DD_MM_YYYY-hh_mm_ss.log*, where:

DD: day.
MM: month.
YYYY: year.
hh: hours.
mm: minutes.
ss: seconds.

Summary of adjustments:

Adjustment	Reader adjustment tool-support (module removed)	Reader adjustment tool-pattern	Automatic / Manual	Reader lateral cover	Allowed Range
<i>Optics Adjustments</i>					
Diaphragm Aperture Adjustment	X	-	Manual. Diaphragm adjustment	Removed	160 - 180
Optical Focus Adjustment	X	X	Manual. Focus adjustment	Removed	65 - 100
<i>Illumination Adjustments</i>					
Illumination Position Adjustment	X	-	Automatic	Assembled	Hor: 16 - 79 Ver: 16 - 89
Illumination Dimensions Adjustment	X	-	Automatic	Assembled	Width: 371 - 399 Height: 500 - 523
Illumination Rotation Adjustment	X	-	Automatic	Assembled	Rotation*100: -129 - 129
Illumination Perspective Adjustment	X	-	Automatic	Assembled	HorPersp(%): -83 - 83 VerPersp(%): -80 - 80
Illumination Intensity Adjustment	X	-	Automatic	Assembled	No range; 170
Illumination Color Adjustment	X	-	Automatic	Assembled	Red factor*100: 120 - 229 Blue factor*100: 69 - 160
<i>Integration Adjustments</i>					
Pixels to mm Conversion Factor	X	X	Automatic	Assembled	Pix/mm*100: 2050 - 2150

Figure 473. Summary of adjustments of LEC module

Prerequisite: NA

Adjustment tool: Reader adjustment tool-pattern v2 (PN: 731398) and Reader adjustment tool-support (Ver. 2) (PN: 234212)

Procedure:

Diaphragm Aperture Adjustment

An estimation of the diaphragm aperture is obtained by means of an image process (i.e. mean of the luminance layer) supposing the incident light is constant. To perform this adjustment illumination panel light intensity is set to default (current) value.

1. Click on *Reader* button found in Technical Service *Procedures* tab → *Adjustments* area → *CAT floor Modules* area.
2. Attach Reader adjustment tool-support (Ver. 2) (PN: 234212) to reader without Reader adjustment tool-pattern v2 (PN: 731398).
3. Remove reader lateral cover.
4. Loose screw fixing diaphragm to unlock it (ring further from camera).



Figure 474. Fixing screw of the diaphragm

5. Click on *Start Adjustment* button. Images are acquired and displayed together with maximum, minimum, current light intensity/Aperture values and the area used for measurement.

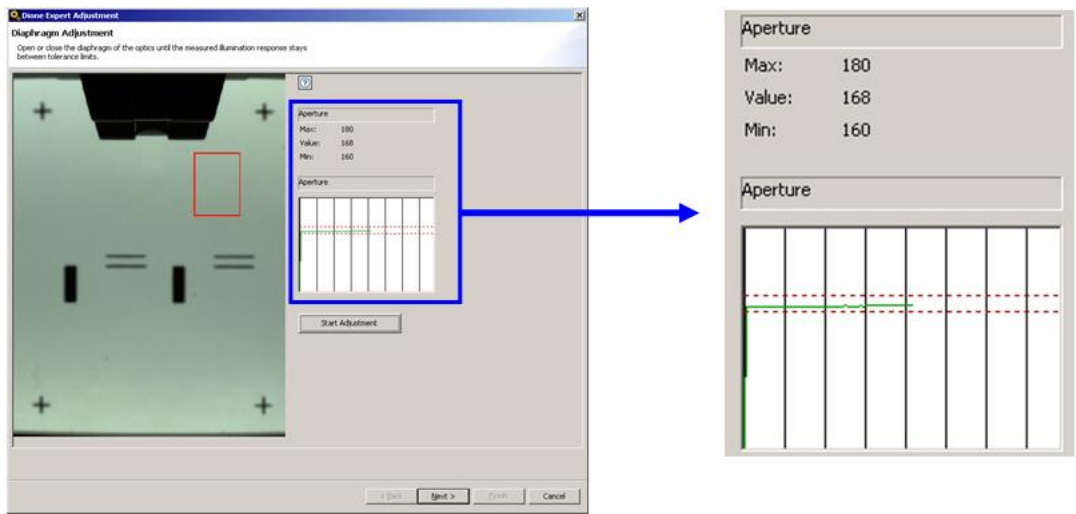


Figure 475. Diaphragm adjustment screen on TSS

6. Adjust diaphragm position so Aperture/current light intensity value is within allowed range (180-160, green line is within dotted red lines. Target value is 170). Then *Next* button is enabled.
7. Tighten diaphragm fixing screw to lock it checking that measured aperture value remains within allowed margin.
8. Click on *Next* button.

Optics Adjustment – Optical Focus Adjustment

Reader adjustment tool-pattern v2 (PN: 731398) provides a reference to measure the level of focus of card/microtubes plane. Contrast of pattern vertical line is measured.

1. A message box indicating that reader adjustment tool-pattern is required for this adjustment is displayed. Click on *Ok* button.
2. Remove reader adjustment tool-support (Ver. 2) (PN: 234212).
3. Attach reader adjustment tool-pattern v2 (PN: 731398) to reader adjustment tool-support (Ver. 2) (PN:234212) as shown in the picture (so screws heads are in illumination panel side).



Figure 476. Reader adjustment tool-pattern v2 (PN: 731398) attached to reader adjustment tool-support v2 (PN: 234212)

4. If reader lateral cover is assembled, remove it.
5. Loose screw fixing focus to unlock it (ring closer to camera).



Figure 477. Fixing screw of the focus

6. Click on *Start Adjustment* button. Images are acquired and displayed together with the area where pattern vertical line is searched and maximum, minimum and current contrast value for pattern vertical line.

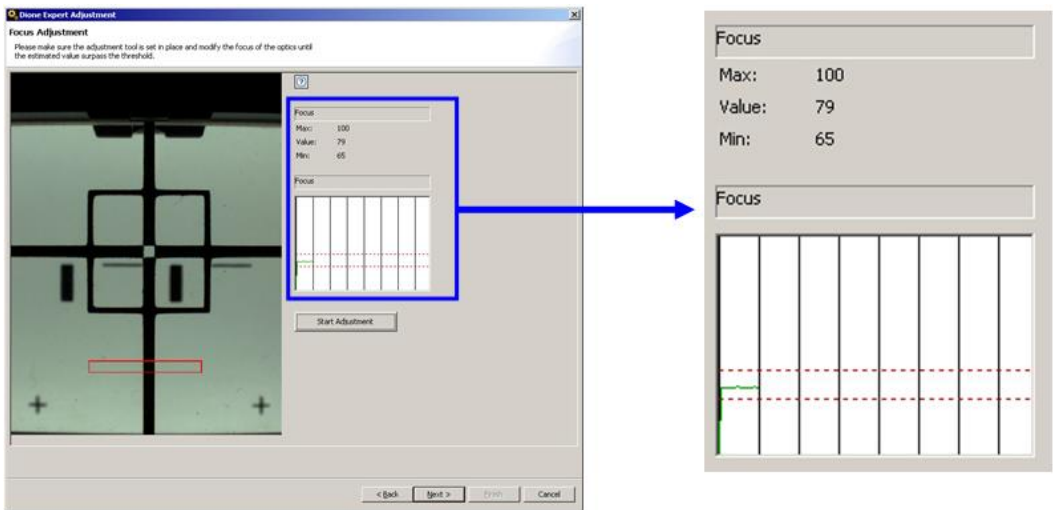


Figure 478. Focus adjustment screen on TSS

7. Adjust focus position so focus current value is the maximum value in the graph and it is within allowed range (green line is within dot red lines). *Next* button is enabled.
8. Tighten focus fixing screw to lock focus checking that measured focus value remains within allowed margin.
9. Assemble reader lateral cover.
10. Click on *Next*.

Illumination Position Adjustment

Software determines illumination position looking for the illumination pattern left-up corner cross within allowed area (red square). The detected cross position are displayed in red color in the picture and 'value' fields are populated with its coordinates.

1. A message box requesting reader adjustment tool-pattern (PN: 731398) removal, so remove it. Click on *Ok* button.
2. Attach reader adjustment tool-support (Ver. 2) (PN: 234212) to reader.
3. Click on *Start Adjustment* button. Images are acquired and left-up corner cross is searched. Acquired image, searching area (red square) and detected cross position (red cross) are displayed together with maximum, minimum and current position values for vertical and horizontal cross position.

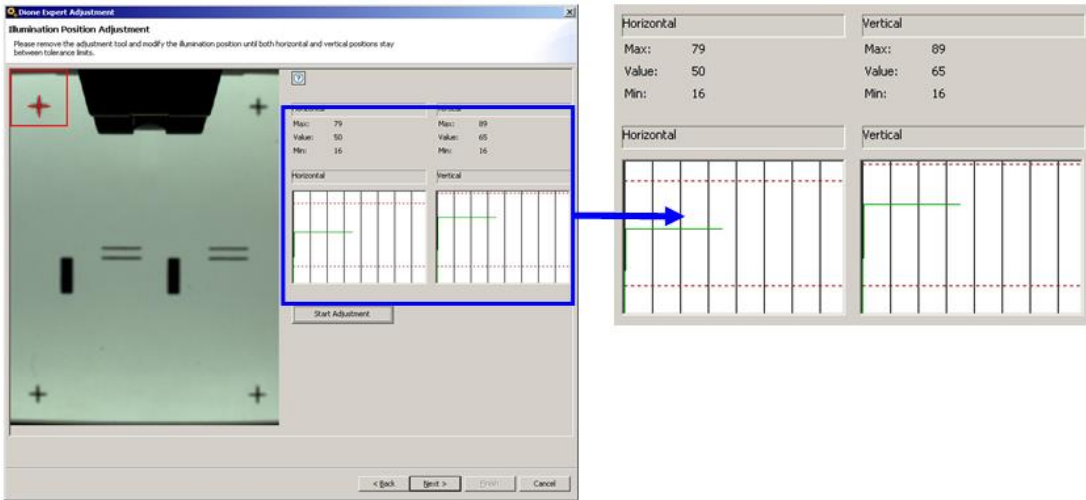


Figure 479. Illumination position adjustment screen on TSS

- Next button is enabled if both position values are within allowed range.
- Click on *Next*.

Illumination Dimensions Adjustment

Software determines illumination panel dimensions/camera-illumination panel distance looking for the four illumination pattern corner crosses within allowed areas (red squares). The detected crosses positions are displayed in red color in the picture and the obtained coordinates are used to calculate illumination pattern horizontal and vertical dimensions. 'Values' fields are populated with obtained dimensions.

- Check reader adjustment tool-support (Ver. 2) (PN: 234212) is attached to reader without reader adjustment tool-pattern v2 (PN: 731398) and that reader lateral cover is assembled.
- Click on *Start Adjustment* button. Images are acquired and displayed together with crosses searching areas (red squares), located crosses positions (red crosses) and maximum, minimum and current height and width values.

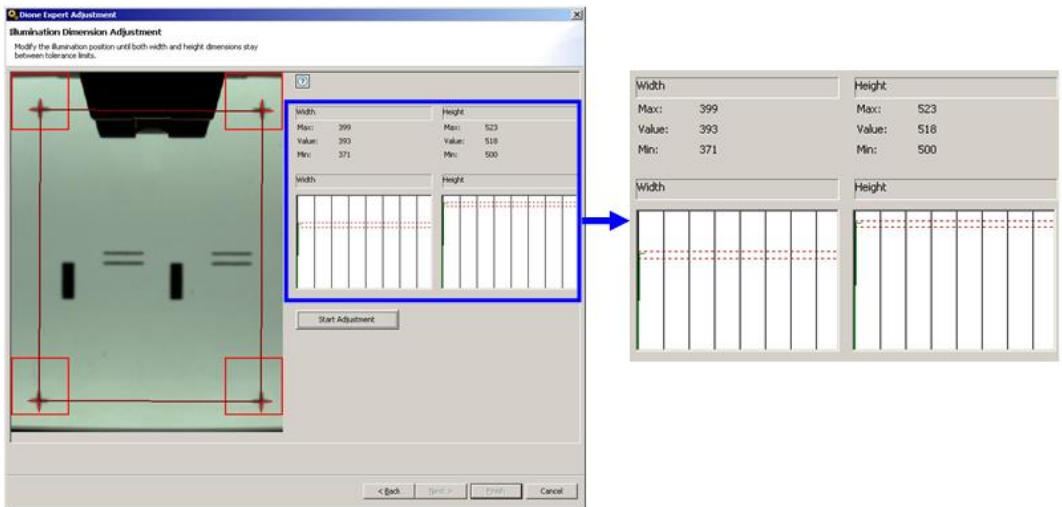


Figure 480. Illumination dimensions adjustment screen on TSS

- Next button is enabled when both dimension values are within allowed range.
- Click on *Next*.

Illumination Rotation Adjustment

Software determines illumination panel rotation comparing vertical and horizontal lines obtained using corner crosses with absolute/complete vertical and horizontal line. Software looks for the four illumination pattern corner crosses within allowed areas (red squares) and displays obtained crosses positions and lines. 'Value' field are populated with obtained rotation.

1. Check reader adjustment tool-support (Ver. 2) (PN: 234212) is attached to reader without reader adjustment tool-pattern v2 (PN: 731398) and that reader lateral cover is assembled.
2. Click on *Start Adjustment* button. Images are acquired and displayed together with crosses searching area, located crosses position and maximum, minimum and current absolute angle value multiplied by 100.

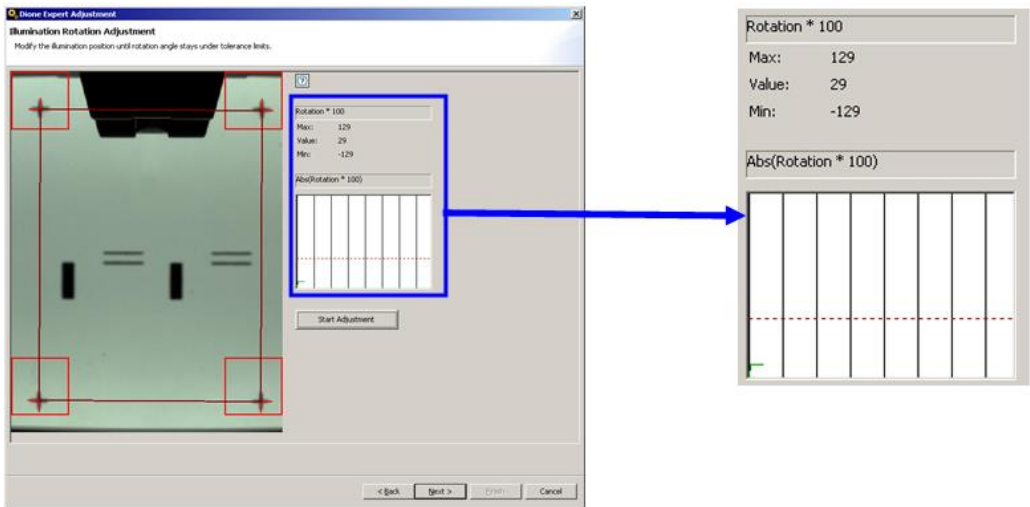


Figure 481. Illumination rotation verification screen on TSS

- Next button is enabled when measured angle value is within allowed range.
3. Click on *Next*.

Illumination Perspective Adjustment

Software determines illumination panel vertical/horizontal perspective comparing both vertical/horizontal lines length obtained using detected corner crosses positions. Software looks for the four illumination pattern corner crosses within allowed areas (red squares) and displays obtained crosses positions and lines. 'Values' fields are populated with obtained dimensions.

1. Check reader adjustment tool-support (Ver. 2) (PN: 234212) is attached to reader without reader adjustment tool-pattern v2 (PN: 731398) and that reader lateral cover is assembled.
2. Click on *Start Adjustment* button. Images are acquired and displayed together with maximum, minimum and current horizontal and vertical perspective values expressed as percentage and multiplied by 100.

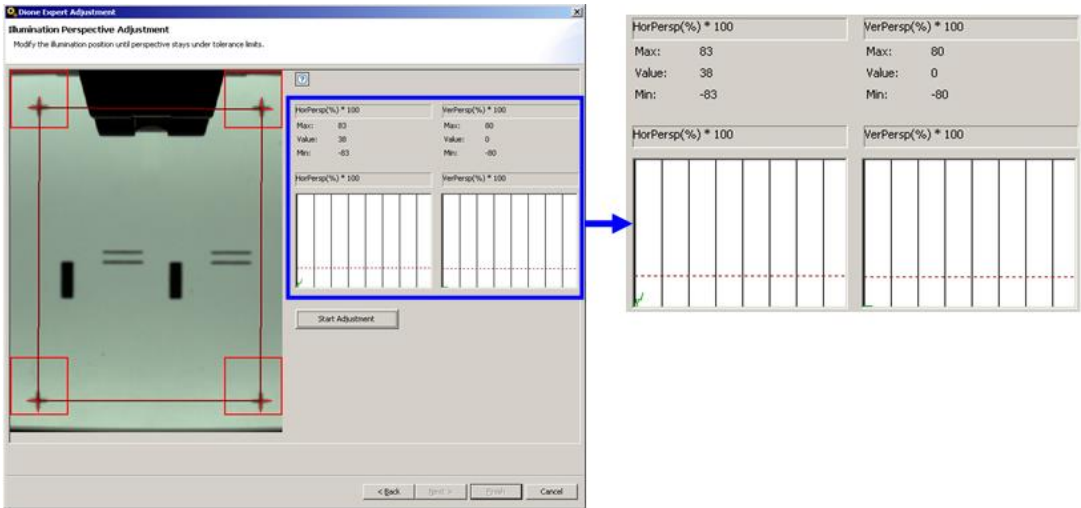


Figure 482. Illumination perspective adjustment screen on TSS

- 0% will indicate no perspective distortion.
Next button is enabled when both perspective values are within allowed range.
- Click on Next button.

Illumination Intensity Adjustment

Software measures the amount of light reaching from certain area (red rectangle) and automatically adjusts illumination panel light intensity (current) to obtain target value (170).

- Check reader adjustment tool-support (Ver. 2) (PN: 234212) is attached to reader without reader adjustment tool-pattern v2 (PN: 731398) and that reader lateral cover is assembled.
- Click on Start Adjustment button. Images are acquired and displayed together with measured area (red rectangle), maximum, minimum and current measured intensity value.

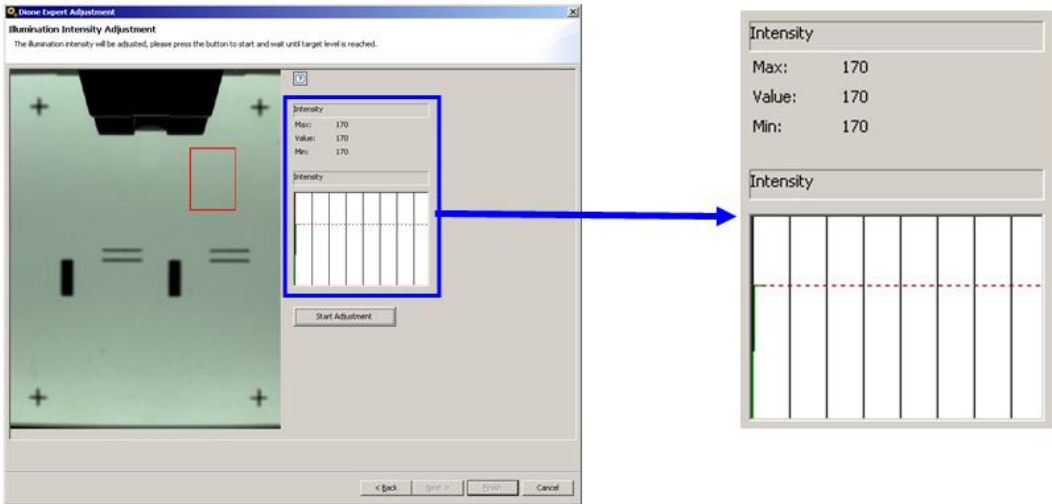


Figure 483. Illumination intensity adjustment screen on TSS

- Software automatically modifies illumination intensity so target value of 170 is obtained.
At that time Next button is enabled.
- Click on Next button.

Illumination Color Adjustment

Software measures the amount of light reaching from certain area (red rectangle are) and calculates red and blue factors (color/white balance).

- 1. Check reader adjustment tool-support (Ver. 2) (PN: 234212) is attached to reader without reader adjustment tool-pattern v2 (PN: 731398) and that reader lateral cover is assembled.
- 2. Click on *Start Adjustment* button. Images are acquired and displayed together with the area used for measurement, maximum, minimum and current measured factors multiplied by 100 for red and blue color.

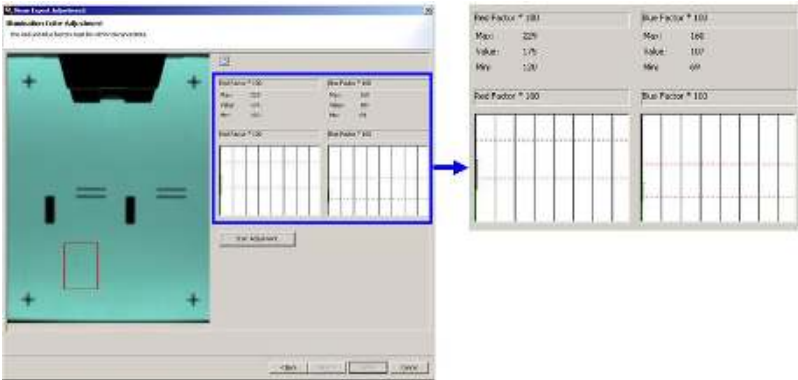


Figure 484. Illumination color adjustment screen on TSS

- Next button is enabled if both factor values are within allowed range.
- 3. Click on *Next* button.

Conversion Factor Adjustment

Software determines pixels/mm conversion factor obtaining the length in pixels of an already known length (in millimetres). The length used is square side (10mm). Software looks for internal edge of square lower horizontal line and external edge of square higher horizontal line within allowed areas (red rectangles). The detected positions are displayed in green color in the picture and used to calculate conversion factor. 'Value' field is populated with obtained conversion factor.

- 1. Attach reader adjustment tool-support (Ver. 2) (PN: 234212) with reader adjustment tool-pattern v2 (PN: 731398) to reader.
- 2. Check that reader lateral cover is assembled.
- 3. Click on *Start Adjustment* button. Images are acquired and displayed together with maximum, minimum and current measured conversion factor multiplied by 100.

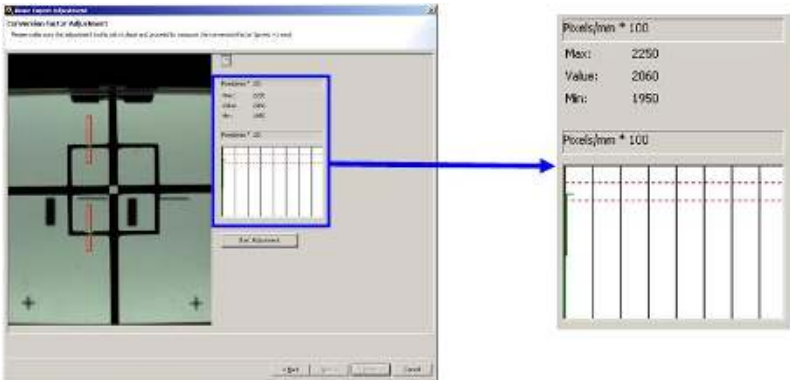


Figure 485. Conversion factor adjustment screen on TSS

- If the obtained conversion factor is within the allowed limit, a message indicating that module has been successfully adjusted will be displayed and the *Finish* button will be enabled.
- Click on *Finish*.
 - Check/perform gripper adjustment in reader area.

11.3 VERIFICATIONS

Following verification can be performed in a row:

Optics Adjustment Checks:

- Diaphragm Aperture Check
- Focus Check (Require reader adjustment tool-pattern)

Illumination Checks:

- Illumination Position Check
- Illumination Dimensions Check
- Illumination Rotation Check
- Illumination Perspective Check
- Illumination Intensity Check
- Illumination color check
- Luminance Uniformity Check
- Color Uniformity Check
- Illumination Cleanliness Check



NOTE:
Reader adjustment tool-support shall be inserted in reader in all the steps of verification whenever performed with reader module not assembled inside *Erytra* to avoid external light contribution.



NOTE:
Reader adjustment tool-support shall be inserted in reader together with Reader adjustment tool-pattern in all the steps of verification that requires Reader adjustment tool-pattern.

Summary of verifications

Check	Reader adjustment tool-support (module removed)	Reader adjustment tool-pattern	Allowed Range
<i>Optics Adjustment Checks</i>			
Diaphragm Aperture Check	X	-	158 – 182
Focus Check	X	X	60 – 100
<i>Illumination Checks</i>			
Illumination Position Check	X	-	Horizontal: 15 – 80 Vertical: 15 – 90
Illumination Dimensions Check	X	-	Width: 370 – 400 Height: 499 – 524
Illumination Rotation Check	X	-	Rotation*100: -135 – 135
Illumination Perspective Check	X	-	Horizontal*100: -110 – 110 Vertical*100: -100 – 100
Illumination Intensity Check	X	-	165 – 175
Illumination color check	X	-	Red factor*100: 119 – 219 Blue factor*100: 59 – 159
Luminance Uniformity Check	X	-	Horizontal: 80 – 100 Vertical: 70 – 100
Color Uniformity Check	X	-	Horizontal: 90 – 100 Vertical: 90 – 100
Illumination Cleanliness Check	X	-	240 – 0

Table 80. Summary of verifications

Verification procedure is described assuming that reader module is assembled inside *Erytra*. Should this not be the case take into account reader adjustment tool-support (Ver. 2) (PN:234212) requirements above stated.

A log file is created whenever a verification procedure is performed. Log file contains the values for each parameter, illumination and software version, camera model and serial number. This file is stored in *E:\Logs\DG421* directory with the name: *Check – DD_MM_YYYY – hh_mm_ss.log*, where

DD:day, *MM*: month, *YYYY*: year, *hh*: hours, *mm*: minutes, *ss*: seconds.

Diaphragm Aperture Check

An estimation of the diaphragm aperture could be obtained by means of an image process (i.e. mean of the luminance layer) supposing the incident light is constant.

1. Click on *Verify* button found in *Technical Service Procedures* tab → *Checks* section → *Reader Module* area. Following window will be displayed:



Figure 486. Diaphragm aperture check initial screen on TSS

2. Click on *Start Check* button. Images will be acquired and displayed together with maximum, minimum, current light intensity/Aperture values, area used for measurement (red rectangle) and check result.

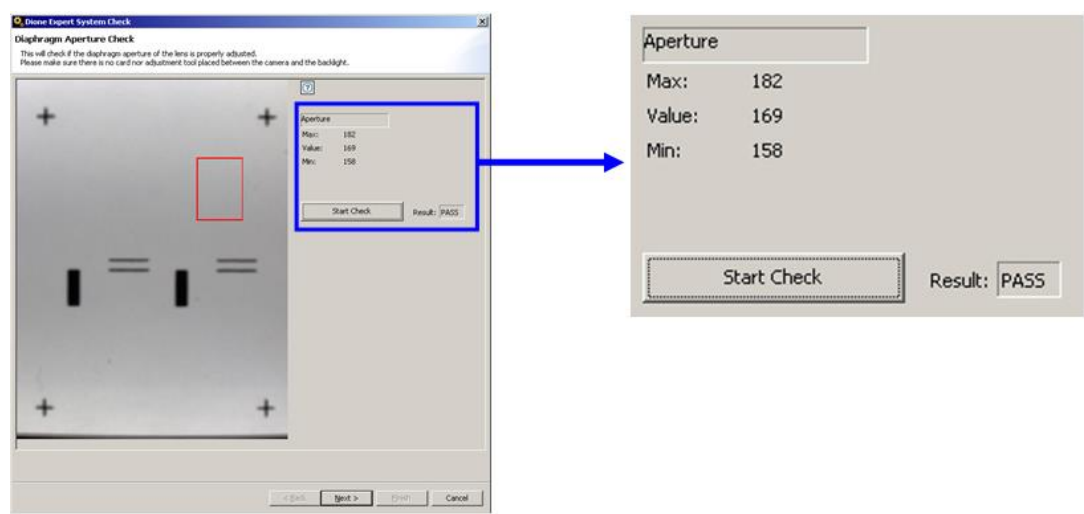


Figure 487. Diaphragm aperture check screen on TSS

- Check result can be *Pass* or *Fail*. If check result is *Fail*, verification process can not go on and *Cancel* button must be selected.
Next button is enabled if test result is *Pass*.
3. Click on *Next/Cancel*.

Optical Focus Check

Reader adjustment tool-pattern will be used to provide a reference to measure the level of focus of card/microtubes plane. Contrast of pattern vertical line will be measured.

1. Attach reader adjustment tool-support (Ver. 2) (PN: 234212) with reader adjustment tool-pattern v2 (PN: 731398) to reader.
2. Click on *Start Check* button. Images will be acquired and displayed together with maximum, minimum, current light intensity/Focus values, area used for measurement and check result.

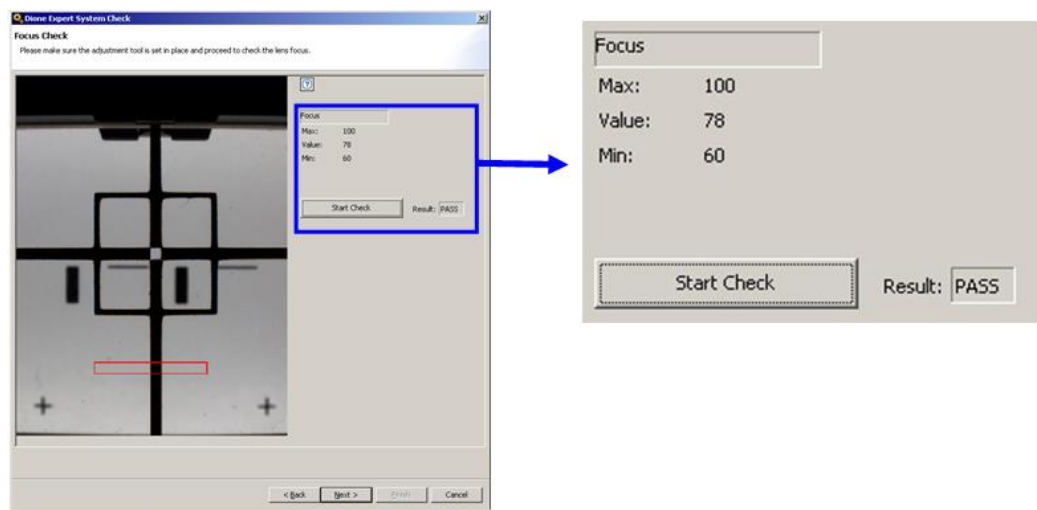


Figure 488. Optical focus check screen on TSS

- Check result can be *Pass* or *Fail*. If check result is *Fail*, verification process can not go on and *Cancel* button must be selected. *Next* button is enabled if test result is *Pass*
3. Click on *Next/Cancel*.

Illumination Position Check

Software will determine illumination position looking for the illumination pattern left-up corner cross within allowed area (red square). The detected cross position will be displayed in red color in the picture and *Value* fields will be populated with its coordinates.

1. Remove reader adjustment tool-support (Ver. 2) (PN: 234212) and reader adjustment tool-pattern v2 (PN: 731398) from reader.
2. Click on *Start Check* button. Images will be acquired and left-up corner cross will be searched. Acquired image, searching area (red square) and detected cross position (red cross) will be displayed together with maximum, minimum, current position values for vertical and horizontal cross position values and check result.

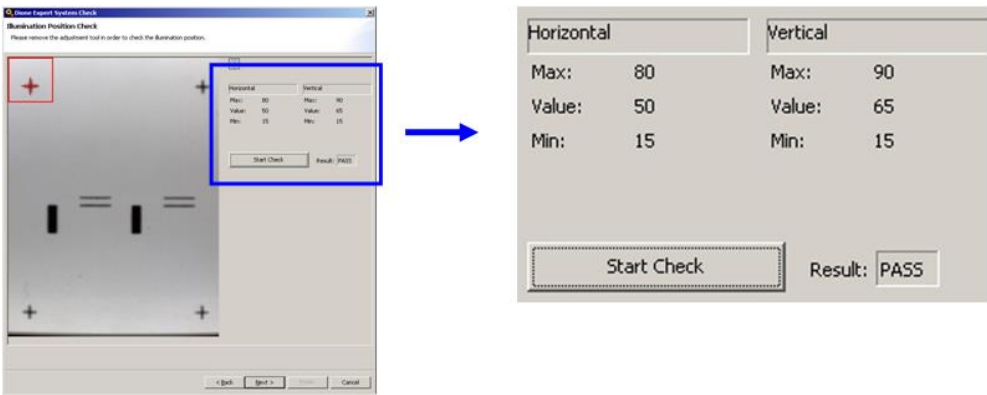


Figure 489. Illumination position check screen on TSS

- Check result can be *Pass* or *Fail*. If check result is *Fail*, verification process can not go on and *Cancel* button must be selected. *Next* button is enabled if test result is *Pass*.
3. Click on *Next/Cancel*.

Illumination Dimensions Check

Software will determine illumination panel dimensions/camera-illumination panel distance looking for the four illumination pattern corner crosses within allowed areas (red squares). The detected crosses positions will be displayed in red color in the picture and the obtained coordinates will be used to calculate illumination pattern horizontal and vertical dimensions. *Values* fields will be populated with obtained dimensions.

1. Click on *Start Check* button. Images will be acquired and displayed together with maximum, minimum, current height and width values and check result.

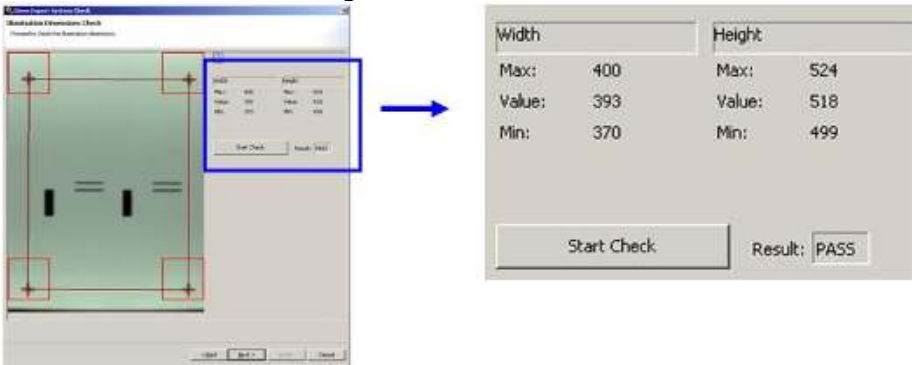


Figure 490. Illumination dimensions check screen on TSS

- Check result can be *Pass* or *Fail*. If check result is *Fail*, verification process can not go on and *Cancel* button must be selected. *Next* button is enabled if test result is *Pass*.
- Click on *Next/Cancel*.

Illumination Rotation Check

Software will determine illumination panel rotation comparing vertical and horizontal lines obtained using corner crosses with absolute/complete vertical and horizontal line. Software will look for the four illumination pattern corner crosses within allowed areas (red squares) and display obtained crosses positions and lines. *Value* field will be populated with obtained rotation.

- Click on *Start Check* button. Images will be acquired and displayed together with crosses searching area, located crosses position and maximum, minimum, current angle value multiplied by 100 and check result.

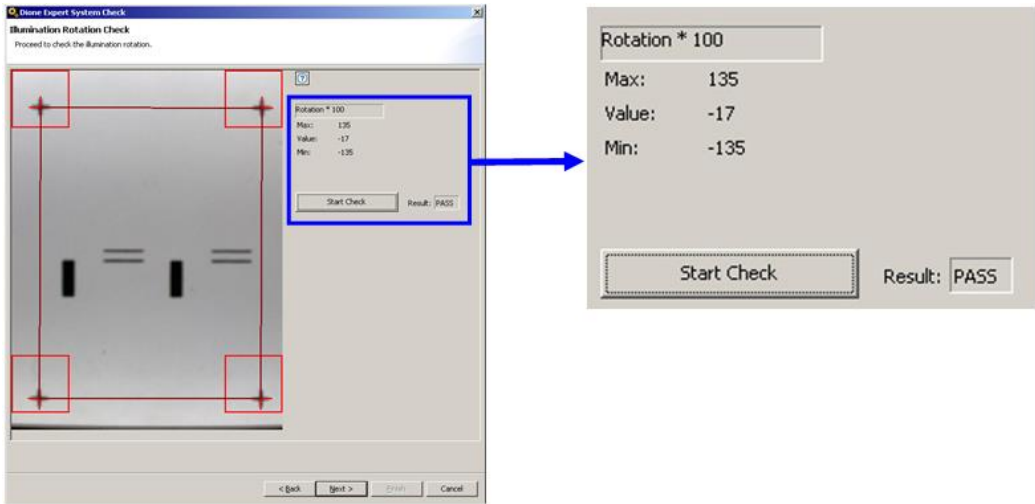


Figure 491. Illumination rotation check screen on TSS

- Check result can be *Pass* or *Fail*. If check result is *Fail*, verification process can not go on and *Cancel* button must be selected. *Next* button is enabled if test result is *Pass*.
- Click on *Next/Cancel*.

Illumination Perspective Check

Software will determine illumination panel vertical/horizontal perspective comparing both vertical/horizontal lines length obtained using detected corner crosses positions. Software will look for the four illumination pattern corner crosses within allowed areas (red squares) and display obtained crosses positions and lines *Values* fields will be populated with obtained dimensions.

1. Click on *Start Check* button. Images will be acquired and displayed together with maximum, minimum, current horizontal and vertical perspective values expressed as percentage and multiplied by 100 and check result.

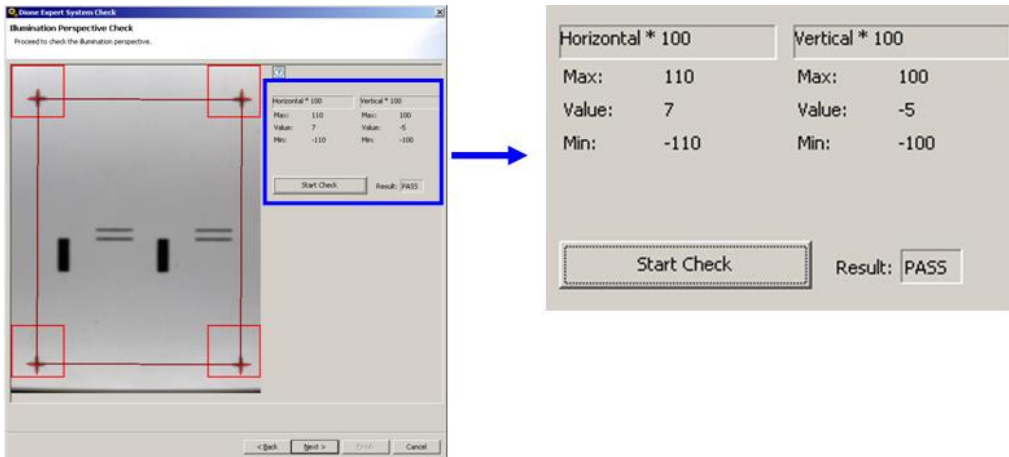


Figure 492. Illumination perspective check screen on TSS

- Check result can be *Pass* or *Fail*. If check result is *Fail*, verification process can not go on and *Cancel* button must be selected. *Next* button is enabled if test result is *Pass*.
2. Click on *Next/Cancel*.

Illumination Intensity Check

Software will measure the amount of light reaching from certain area (red rectangle are) and will automatically adjust light intensity to obtain target value (170).

1. Click on *Start Check* button. Images will be acquired and displayed together with maximum, minimum, current measured intensity value and check result.

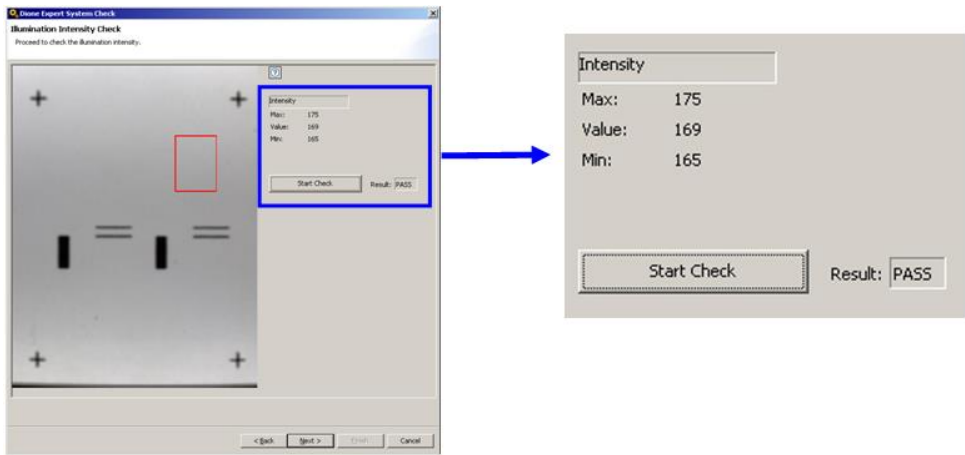


Figure 493. Illumination intensity check screen on TSS

- Check result can be *Pass* or *Fail*. If check result is *Fail*, verification process can not go on and *Cancel* button must be selected. *Next* button is enabled if test result is *Pass*.
2. Click on *Next/Cancel*.

Illumination Color Check

Software will measure the amount of light reaching from certain area (red rectangle), calculate red and blue factors (color/white balance) and compare them with adjustment ones.

1. Click on *Start Check* button. Images will be acquired and displayed together with maximum, minimum, current measured factors variation multiplied by 100 for red and blue color and check result.

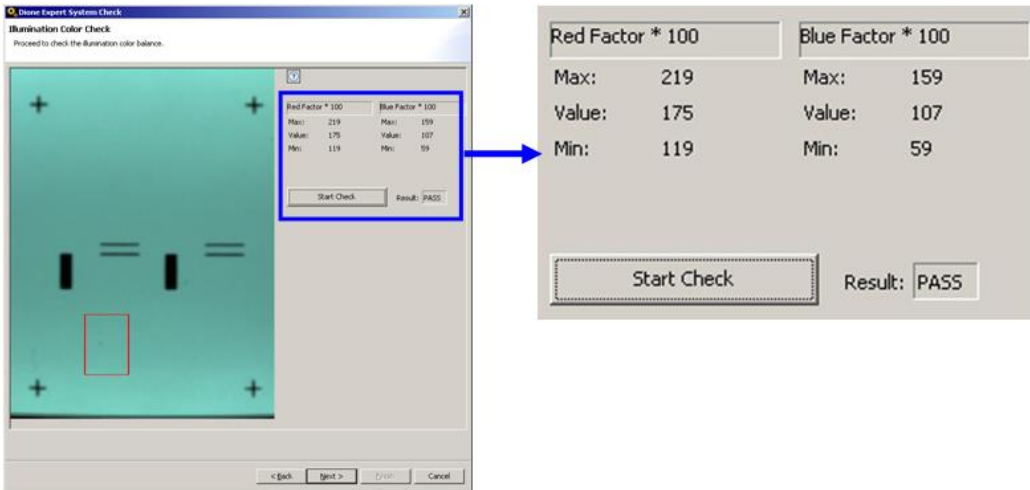


Figure 494. Illumination color check screen on TSS

Check result can be *Pass* or *Fail*. If check result is *Fail*, verification process can not go on and *Cancel* button must be selected. *Next* button is enabled if test result is *Pass*.

2. Click on Next/Cancel.

Luminance Uniformity Check

1. Click on *Start Check* button. Images will be acquired and displayed together with maximum, minimum, current measured factors multiplied by 100 for red and blue color and check result.

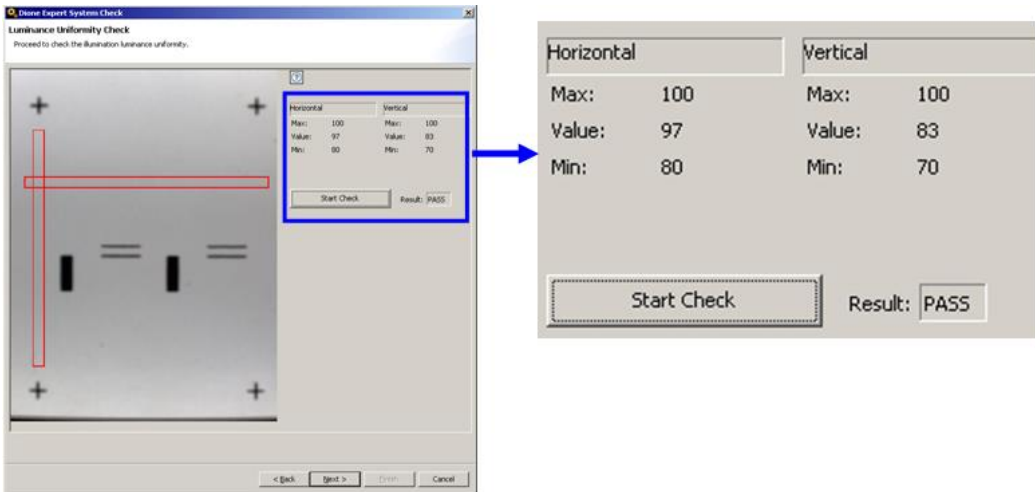


Figure 495. Illumination uniformity check screen on TSS

Check result can be *Pass* or *Fail*. If check result is *Fail*, verification process can not go on and *Cancel* button must be selected. *Next* button is enabled if test result is *Pass*.

2. Click on Next/Cancel.

Color Uniformity Check

1. Click on *Start Check* button. Images will be acquired and displayed together with maximum, minimum, current measured factors multiplied by 100 for red and blue color and check result.

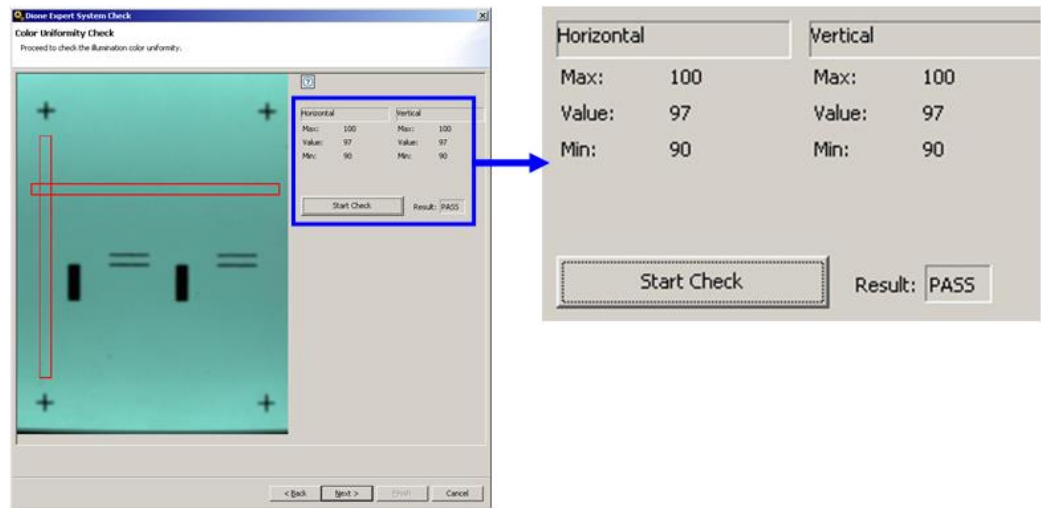


Figure 496. Color uniformity check screen on TSS

Check result can be *Pass* or *Fail*. If check result is *Fail*, verification process can not go on and *Cancel* button must be selected. *Next* button is enabled if test result is *Pass*.

2. Click on *Next/Cancel*.

Illumination Cleanliness Check

Software will look for unexpected elements artifacts, or dirt in illumination/illumination pattern.

1. Click on *Start Check* button. Images will be acquired and displayed together with maximum, minimum, current measured factors multiplied by 100 for red and blue color and check result.

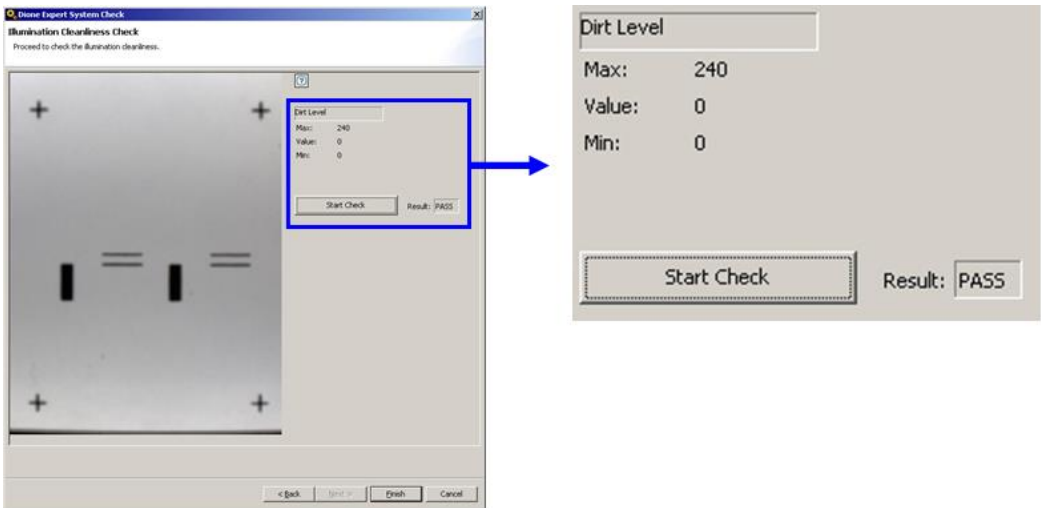


Figure 497. Illumination cleanliness check screen on TSS

Check result can be *Pass* or *Fail*. If check result is *Fail*, verification process can not go on and *Cancel* button must be selected. *Next* button is enabled if test result is *Pass*. A message box with *System Check passed* will be displayed if test result is *Pass*.

11.4 REPLACEMENT PROCEDURES

11.4.1 Reader module removal

1. Open card drawer number 1 and number 2 to have access to reader module fixing screws and wiring.
2. Unplug camera cable (FireWire/IEEE1394) and illumination screen cable (USB) from reader module (illumination panel side).

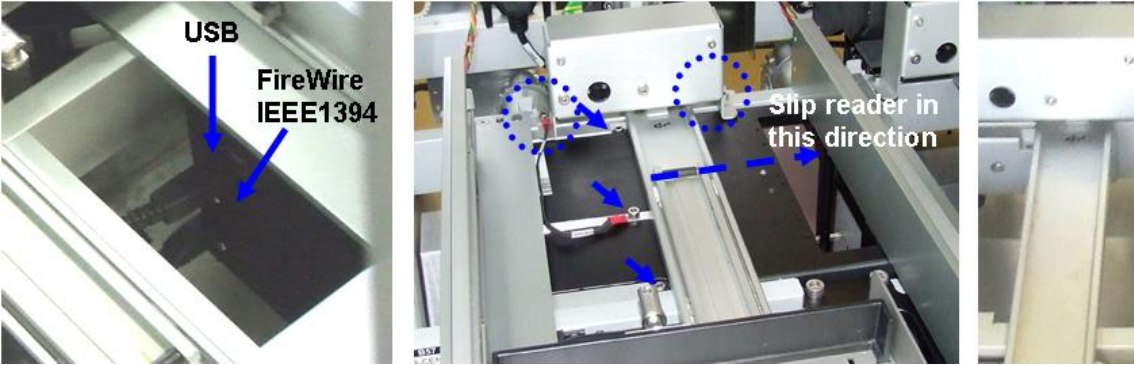


Figure 498. USB and FireWire cables. Reader and grounding fixing screws

3. Remove the screw fixing grounding cable to reader and remove grounding cable.
4. Remove the two screws fixing reader module to the instrument. To ease the removal of the screw closer to the back of the instrument card drawer electric lock might be unscrewed from the instrument.
5. Open solution A and B drawers and remove Solution A and B bottles.
6. Close partially both containers drawers, slip reader to the right so it get out of its location and lower/place it in solution A and B drawers.
7. Open drawers and remove reader.
8. Place Solution A and B bottles back to drawers.



NOTE:

- Once the reader is assembled back:
- Perform reader verification.
 - Check/perform gripper adjustment in reader area.

11.4.2 Camera replacement

1. Remove reader module (refer to section 11.4.1).
2. Remove the eight screws fixing reader cover to reader bulk (two screws per side) and remove the cover.
3. Unplug FireWire/IEEE1394 cable from camera.

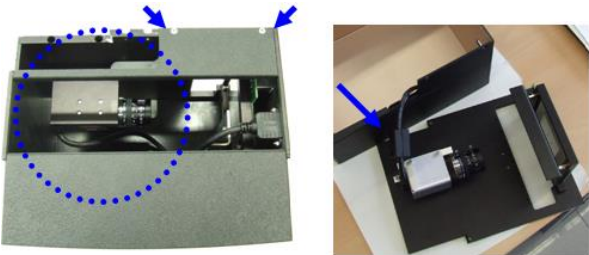


Figure 499. FireWire/IEEE1394 camera cable

4. Remove the two screws fixing the camera case, and then remove the four screws fixing the camera to its support plate so that the camera can be removed.

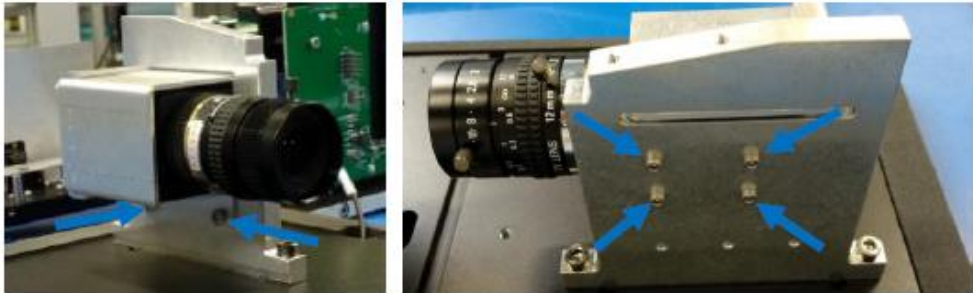


Figure 500. Camera assembly

5. Place new camera so it rests on holder frontal and bottom edges of camera shape area and fix it with the four screws.



Figure 501. Camera and optics set screws assembly

6. Make sure that the optic rings set screws (Iris and focus) are assembled facing the camera holder side and as far as possible from the reader basis (to ease handling during adjustment).
7. Assemble camera cover.
8. Plug FireWire/IEEE1394 cable to camera.
9. Assemble reader cover.
10. Perform reader adjustment. Check that the four crosses lays within expected square area displayed in the adjustment/verification routines. Should this not be the case, loose the four screws fixing the camera and align the camera.
11. Perform reader verification.
12. Check/perform gripper adjustment in reader area.

11.4.3 Reader illumination/PCA RD-IL421 replacement

Reader illumination is part of PCA RD-IL421 and then whole assembly replacement is required. To replace PCA RD-IL421 proceed as follows:

1. Switch off the instrument and disconnect it from the mains.
2. Remove reader module (refer to section 11.4.1).
3. Remove the eight screws fixing reader cover to reader bulk (two screws per side) and remove the cover.
4. Unplug FireWire/IEEE1394 cable from camera.



Figure 502. PCA RD-IL421 v3 assembly

5. Remove the four screws fixing PCA RD-IL421 to reader bulk (two screws per side) and remove PCA RD-IL421.
6. Assemble new PCA RD-IL421 using the four screws and checking proper positioning (see previous figure).
7. Assemble reader cover.
8. Perform reader adjustment.
9. Perform reader verification.
10. Check/perform gripper adjustment in reader area.

11.4.4 Reader illumination pattern replacement

1. Remove reader module (refer to section 11.4.1).
2. Remove the eight screws fixing reader cover to reader bulk (two screws per side) and remove the cover.
3. Unplug FireWire/IEEE1394 cable from camera.
4. Remove the two screws fixing reader splitter to reader bulk remove splitter.

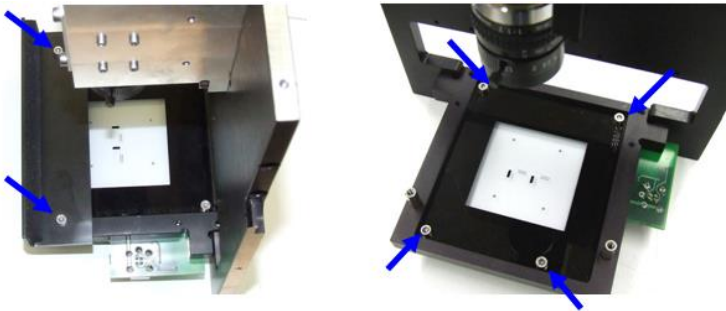


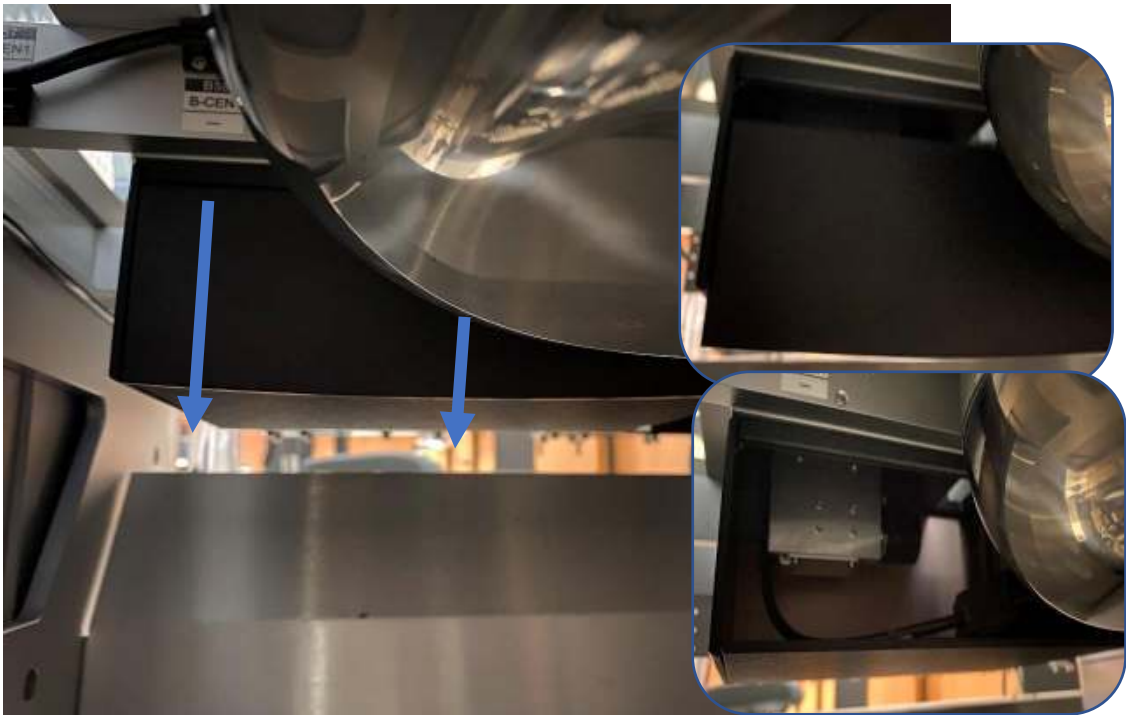
Figure 503. Reader illumination pattern assembly

5. Remove the four screws fixing reader illumination pattern to reader bulk and remove illumination pattern.
6. Remove new illumination pattern protective foil and insert and fix it to reader bulk avoiding fingerprints. Screws positions assure proper insertion/assemble (serigraphy side facing camera).
7. Clean illumination pattern with optics cleaner.
8. Assemble reader splitter.
9. Connect FireWire/IEEE1394 cable from camera.
10. Assemble reader cover.
11. Perform reader adjustment.
12. Perform reader verification.
13. Check/perform gripper adjustment in reader area.

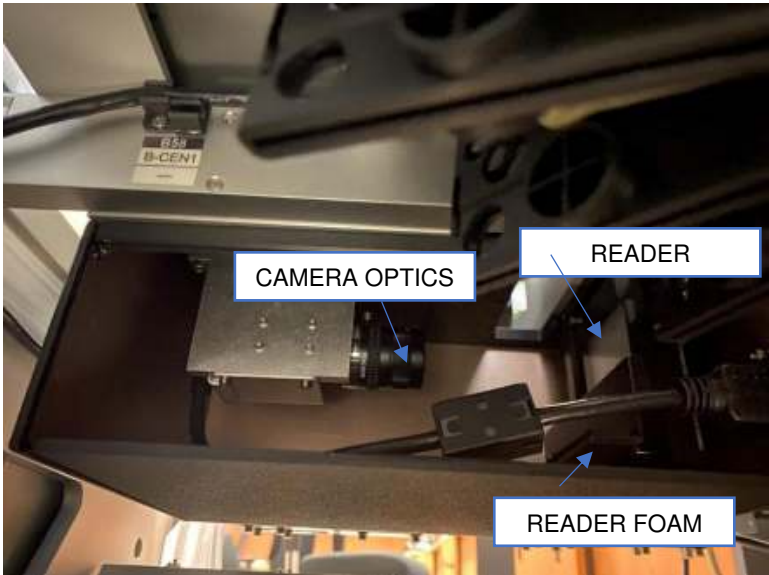
11.4.5 Remove reader cover

In order to access the reader easily, follow the steps below:

- 1. Open fluidic drawers A and B and remove the two solutions.
- 2. Slide down the front cover of the reader, using the side guides until it is completely removed.



It is possible to remove the centrifuge pot for a better access.



NOTE:

If the centrifuge pot has been removed, check that all the card holders are correctly fitted before placing the pot back.

11.5 FUSES

Fuses related to Reader module are detailed in the following table,

PCA	FUSE NAME	DESCRIPTION
PCA RD_IL421	NA	NA

Table 81. List of fuses of Reader module

11.6 LEDS

LEDs related to Reader module are detailed in the following table,

PCA	LED	DESCRIPTION
PCA RD_IL421	NA	NA

Table 82. List of LEDs of Reader module

11.7 JUMPERS AND TEST POINTS

Neither jumpers nor test points are related to Reader module.

11.8 WIRING TABLE

In the following table, there is detailed all the cables related to Reader module. A wiring diagram can be found attached to Service Manual.

ID	FROM CONNECTOR	FROM PCA/ COMPONENT	TO CONNECTOR	TO PCA/COMPONENT
B-READ_FW	PC.FW (FW1)	PC	NA	READER BOX
READ_FW	NA	READER	FIREWIRE	CAMERA
READ_GND	NA	CHASSIS	EARTH	CAMERA
B-READ_USB	PC.USB2	PC	H2	PCA RD-IL421

Table 83. List of wiring of Reader module



Figure 504. Location of the connectors in PCA RD-IL421 v3

11.9 TROUBLESHOOTING

Module	Device	Error code	Error	Cause	Text	Options
Reader	Reader	17409	CAMERA_INITIALIZATION_ERROR	Unable to initialize acquisition system, camera, FireWire bus,...(hardware error).	Error during Reader initialization. Please RESTART the instrument.	Reboot
		17410	ILLUMINATION_INITIALIZATION_ERROR	Unable to initialize the communication with illumination module (hardware error).		
		17413	VERSIONS_DO_NOT_MATCH	Version of the Acquisition Structure stored in EEPROM does not match the software version (versions error).		
		17441	ACQUISITION_HARDWARE_ERROR	Error acquiring image (hardware error).	No error is shown. Card is left in original rack (integrity check) or Service rack (reading).	NA
		17442	ILLUMINATION_COMM_ERROR	Error sending command to illumination module (hardware error).	<i>Instrument initialisation step:</i> Error during Reader initialization. Please RESTART the instrument.	Reboot
					<i>Integrity check/reading step:</i> No error is shown. Card is left in original rack (integrity check) or Service rack (reading).	NA

Module	Device	Error code	Error	Cause	Text	Options
Reader	Reader	17443	ILLUMINATION_HARDWARE_ERROR	Error executing command in the illumination module or an error flag of the status word is on (hardware error).	<i>Instrument initialisation step:</i> Error during Reader initialization. Please RESTART the instrument.	Reboot
					<i>Integrity check/reading step:</i> No error is shown. Card is left in original rack (integrity check) or Service rack (reading).	NA
		17457	UNABLE_TO_ADJUST_LIGHT_INTENSITY	Unable to reach intensity target when adjusting illumination current.	<i>Instrument initialisation step:</i> Error during Reader initialization. Please RESTART the instrument.	Reboot
					<i>Integrity check/reading step:</i> No error is shown. Card is left in original rack (integrity check) or Service rack (reading).	NA
		17458	STARTUP_CHECK_FAILED	Start up check failed. One or more of the checks performed failed.	Error during Reader initialization. Please RESTART the instrument.	Reboot
		17459	STARTUP_CHECK_REQUIRED	A Start up check is required in order to guarantee that the system is within working parameters.		
		17460	NOT_ADJUSTED	Adjustment parameter currently being adjusted is still out of the acceptance range.	NA (only it appears in TSS)	NA

Module	Device	Error code	Error	Cause	Text	Options
Reader	Reader	17473	DISK_ACCESS_ERROR	Error opening, reading or writing a file (xml, image, log, ...) to disk.	Instrument initialisation step: Error during Reader initialization. Please RESTART the instrument.	RESTART
					Integrity check/reading step: No error is shown. Card is left in original rack (integrity check) or Service rack (reading).	NA
		17474	MEMORY_ALLOCATION_ERROR	Error allocating memory (possible memory leak somewhere).	Instrument initialisation step: Error during Reader initialization. Please RESTART the instrument.	RESTART
					Integrity check/reading step: No error is shown. Card is left in original rack (integrity check) or Service rack (reading).	NA
		17489	UNEXPECTED_ERROR	Unexpected error has been found.	Instrument initialisation step: Error during Reader initialization. Please RESTART the instrument.	RESTART
					Integrity check/reading step: No error is shown. Card is left in original rack (integrity check) or Service rack (reading).	NA

Module	Device	Error code	Error	Cause	Text	Options
Reader	Reader	17490	INCORRECT_USE	A method has been called before initializing the module.	No error is shown. Card is left in original rack (integrity check) or Service rack (reading).	NA
		17491	WRONG_ARGUMENT	Wrong argument passed to a method.	Instrument initialisation step: Error during Reader initialization. Please restart the instrument.	RESTART
					Integrity check/reading step: No error is shown. Card is left in original rack (integrity check) or Service rack (reading).	NA
		17505	INVALID_INPUT_IMAGE	Image being processed is not what was expected.	No error is shown. Card is left in original rack (integrity check) or Service rack (reading).	NA
		NA	NA	Reader module does not respond in 15 seconds.	System has an unrecoverable error. Please restart the instrument as soon as possible. If problem persists, please contact your Technical Service.	OK

Table 84. List of errors related to Reader module.

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
System is not able to initialise Reader module / error when software is opened	Software was not closed properly	Switch off the instrument and disconnect from the mains
	Camera cable damaged	Check / replace FireWire cable
	PCA illumination cable damaged	Check / replace USB cable
	Unable to reach illumination intensity	Check / perform reader adjustment
	PCA RD-IL421 failure	Check / replace PCA RD-IL421
	Camera failure	Check / replace camera
Cards are rejected during initial checking 17489	Cards are not in good conditions	Check status of the cards
	Incorrect adjustment of CAT in card drawer	Check adjustment of the gripper in the drawer where the card has been taken
	Camera not correctly assembled	Check camera assembly
	Incorrect adjustment of reader	Check / perform adjustment of the reader
	Incorrect adjustment of CAT in reader	Check / perform adjustment of the gripper in the reader
Cards are not read during final reading 17489	Incorrect adjustment of CAT in centrifuge	Check / perform adjustment of the gripper in the centrifuge where the card has been taken
	Camera not correctly assembled	Check camera assembly
	Incorrect adjustment of reader	Check / perform adjustment of the reader
	Incorrect adjustment of CAT in reader	Check / perform adjustment of the gripper in the reader

Table 85. Reader module troubleshooting

12 MECHANICAL DRAWINGS

This chapter describes the available options when viewing the digital drawings representing all the modules of Erytra instrument, including 3D views and animated assembly procedures that are performed in factory for each of them. As detailed below, a free software is required to view the drawings and assembly procedures, which are contained in '.pvz' files as an attachment to the current Service Manual document.

12.1 Requirements

- 3D drawings ('.pvz' files)
3D drawings are included in the *Drawings* folder as part of the current Service Manual. The file name format of the 3D drawings is *MxxxxxxVn_myy.pvz*, where *xxxxxx* is the module identification number, *n* is a number indicating the drawing version, and *yy* is the modification number for the corresponding module.
- Drawings table
3D drawings table can be found in section 12.2.
This table can be used to find the name of the drawing corresponding to each module.
- Spare parts list
The raw material codes composing a factory module can be found in the 3D drawings. From there, they can be matched to the codes composing an existing spare part in order to know the corresponding spare part reference.
- 3D viewer
Creo View Express is the software used to view the available 3D drawings and the corresponding assembly procedures. It can be downloaded following the instructions provided in section 12.3 and configured as detailed in section 12.4. Version 8.0 or higher is needed to view the drawings.

12.2 Drawings reference table

Module	Assembly	Drawing name
0_ALL	Base (I)	M6007961V0_m06.pvz
	Base (II)	M6011717V0_m03.pvz
	Base (III)	M6006516V0_m03.pvz
2_PACK	Packaging	M210400V0_m00.pvz
4_GEN	User tray	M6018839V0_m00.pvz
	Touchscreen monitor drawer	M6006416V0_m01.pvz
	Chassis	M6006420V0_m03.pvz
	Lower level	M6006421V0_m02.pvz
	Electronics cabinet	M6006424V0_m05.pvz
5_INC	Incubator	M6006413V0_m03.pvz
6_CEN	Centrifuge	M6006412V0_m02.pvz
7_PIP	PIP arm (Y and Z axis)	M6006418V1_m04.pvz
7_PIP 8_CAT	X axis (for PIP and CAT)	M6006419V0_m02.pvz
8_CAT	CAT arm (Y and Z axis)	M6006417V0_m03.pvz
9_FLU	Solution Bottle	M6006428V0_m00.pvz
	Waste Bottle	M6006780V0_m00.pvz
	Bottle drawer	M6006422V0_m07.pvz
10_CUP	Dilution cup	M6006411V0_m04.pvz
11_READ	Reader	M6006414V0_m05.pvz

Table 86. 3D drawings reference table

Additionally, the following mechanical drawings (2D, pdf format) are related to the instrument:

Module	Assembly	Drawing name
2_PACK	External BCR holder	s2018V0_m00.pdf
9_FLU	Fluidic diagram	p2475V0_m09.pdf

Table 87. 2D drawings reference table

12.3 Creo View Express installation

12.3.1 Within Grifols company

Install **Creo View Express** from Grifols Software Center.

12.3.2 External distributors

Look for Creo View Express in <https://www.ptc.com> (free viewer) and follow the downloading and installation instructions.

12.4 Creo View configuration

Once a drawing (‘.pvz’ file) is opened using Creo View Express, the main structure of that module can be displayed by selecting the *Structure* tab on the left panel of the screen, then double-clicking on the main structure code contained in the *Ilustración* field.



NOTE:
Some drawings may contain more than one main structure, so they should be opened individually

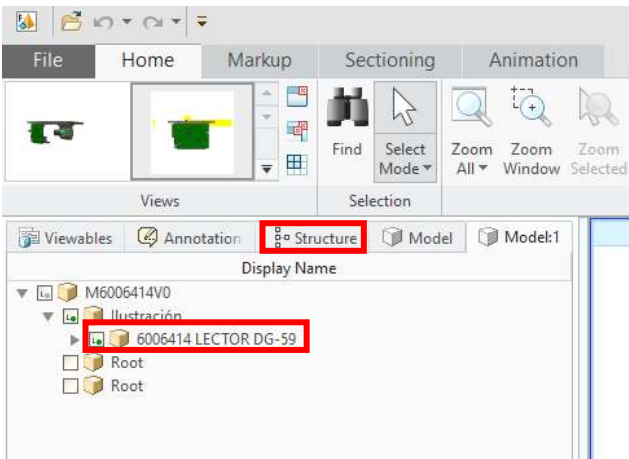


Figure 505. Main structure code

A new tab named *Model* will be created, and the 3D drawing of the selected module will be shown on screen. At this point, several drawing techniques can be used to represent the model by selecting the *Render Mode* button in Creo View Express toolbar (the shaded view is selected by default).

When selecting any of the parts composing the drawing, the corresponding component will be highlighted in the navigation tree on the left side of the screen. In order to see the raw material description and its code (factory reference) instead of the default values, the following configuration should be applied:



NOTE:
This configuration is required if raw material codes need to be searched within the module

1. Go to *File/Options*.
2. Go to *General/Navigation* window.
3. Enable the *Use alternate path names* checkbox and click on *Configure*.
4. Click on *Insert Attribute....*
5. Select 'CODIGO' and click on *Insert value*.
6. Select 'DENOMINACION' and click on *Insert value*.
7. Consider applying a whitespace between both values in the *Configure Alternate Part Names* window.

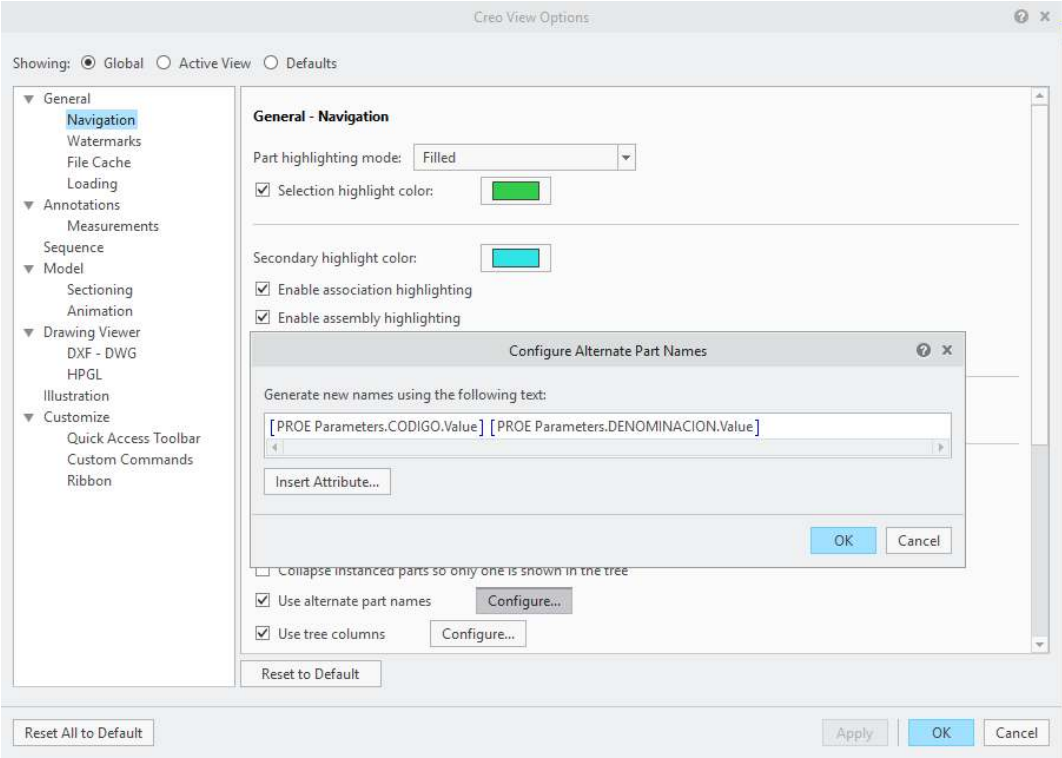


Figure 506. Creo View Options

8. Click on *Close*, then *OK*.
9. Click on *Apply*, then *OK*.
10. Now the raw material description and its factory reference can be seen in the navigation tree for each part composing the drawing. This reference can also be found when searching for *CODIGO* field as described in section 12.7 of present document.

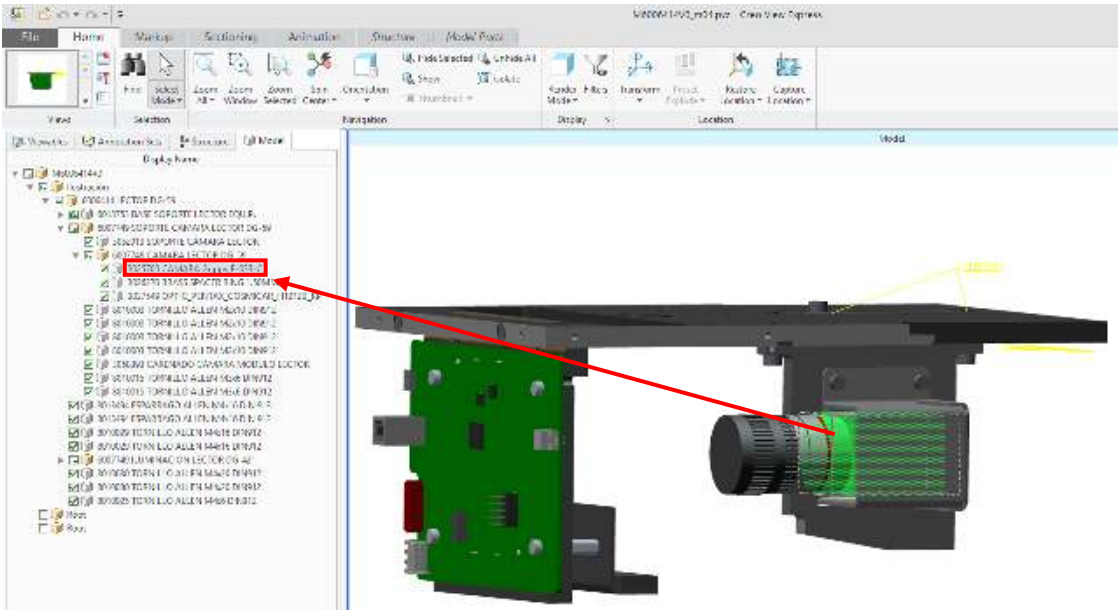


Figure 507. Highlighted part description after selecting a drawing part

12.5 Models structure and navigation

12.5.1 Tree structure

- 1. Click on *Structure* tab and double click on the main structure code contained in the *Ilustración* field.

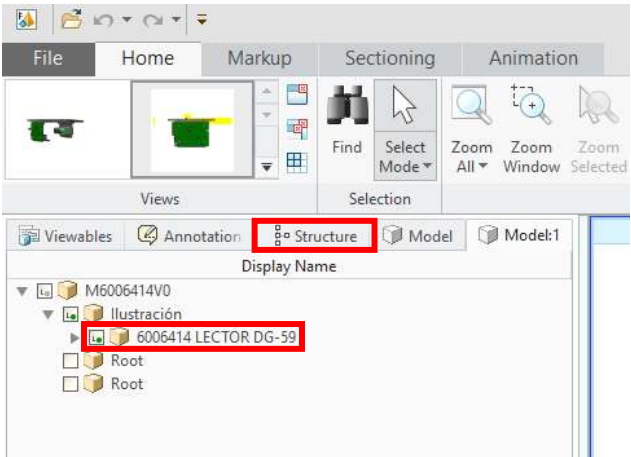


Figure 508. Main structure code

- 2. A new tab named *Model* will be created, and the main model of the selected module will be shown on screen. The tree structure (starting on the main structure code) can be expanded or collapsed by clicking on the triangles next to each field.
- 3. When double clicking in any field of the tree, the corresponding component in the drawing is zoomed in and changes its colour to blue (subassembly containing raw materials) or green (raw materials).

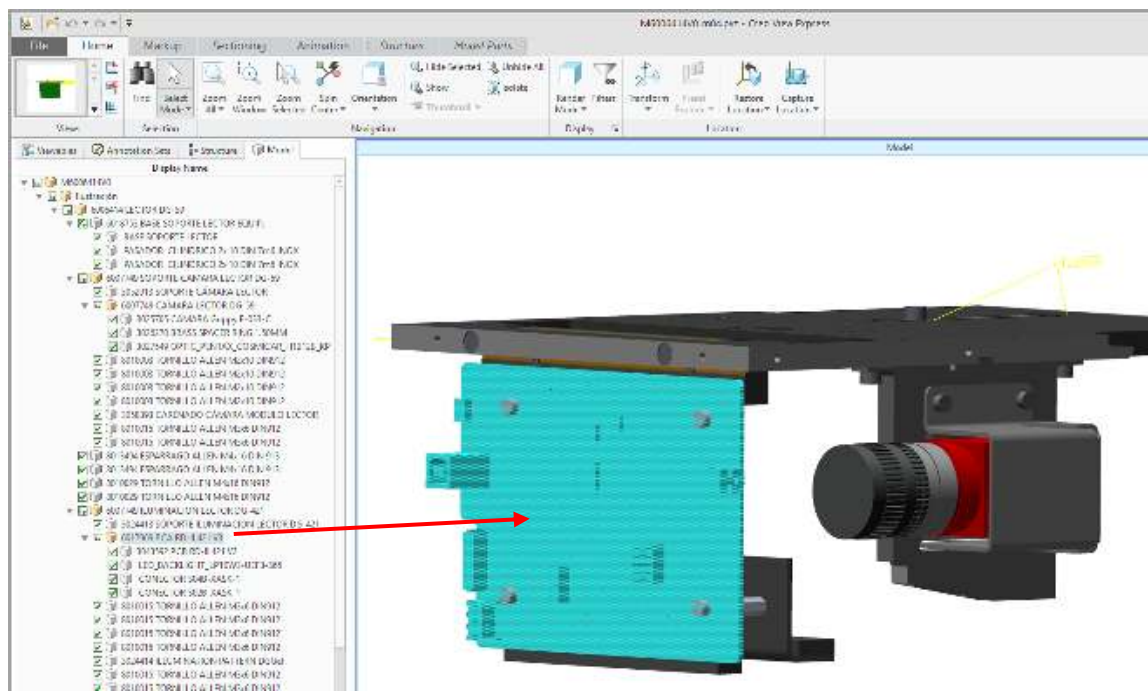


Figure 509. Tree structure component selection

4. Parts can be hidden or isolated by right-clicking on them and selecting the desired action, and then reestablished by right-clicking anywhere within the drawing area and selecting *Unhide All*. They can also be hidden or unhidden by enabling/disabling the check square near each component in the tree structure.

12.5.2 Models navigation

Once the drawing is displayed on screen, it can be rotated, dragged, zoomed in or out, etc. by using the below listed functionalities for Creo View Express:

- Zoom in/out (pointer area taken as reference point):
 - Mouse: rotate the mouse wheel.
 - Pad: put two fingers on pad and move them up or down.
- Rotate model:
 - Mouse: Press and hold right button and drag mouse.
 - Pad:
 - Select *Free Rotation* button in Creo View Express toolbar.
 - Select in the *Ilustración* field in the tree structure, or just select a part of the drawing.
 - Hold left click on the sphere while dragging on the pad.
- Drag model:
 - Mouse: Press the wheel and drag mouse.
 - Pad:
 - Select *Transform* button in Creo View Express toolbar.
 - Select in the *Ilustración* field in the tree structure, or just select a part of the drawing.
 - Hold left click on a sphere's center grey circle while dragging on the pad.



Figure 510. Transform options in Creo View Express toolbar

- Hide/Unhide part:
 - Select the part clicking by right-clicking on it and select *Hide* or *Unhide All*.
 - Select the part clicking on it and enable or disable it in the tree structure.
- Center model on screen: double click on the background
- Restore location: click on *Restore location* button in Creo View Express toolbar. With this, any part that has been moved/rotated from its default position within the drawing area will restore its original position.

12.6 Animated assembly procedures

The *Viewables* tab contains a list of animated assembly procedures detailing the assembly steps as done in factory for each of the modules composing Erytra instrument. Each component being assembled in an animation is identified with its raw material code.

1. When opening a '.pvz' file for a specific module, the *Viewables* tab is selected by default, showing the corresponding animated assembly procedures (*Figures* area).

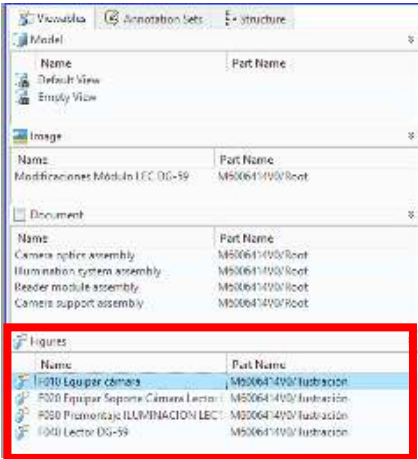


Figure 511. Figures area

2. Double click on the desired assembly and a new tab will be created, showing the tree structure with the parts involved along with the corresponding animated assembly.
3. The animation can be played and paused for each of the steps composing it by using the *Playback* control buttons at the bottom left corner of the animation screen, or those at the *Play* section in Creo View Express toolbar.

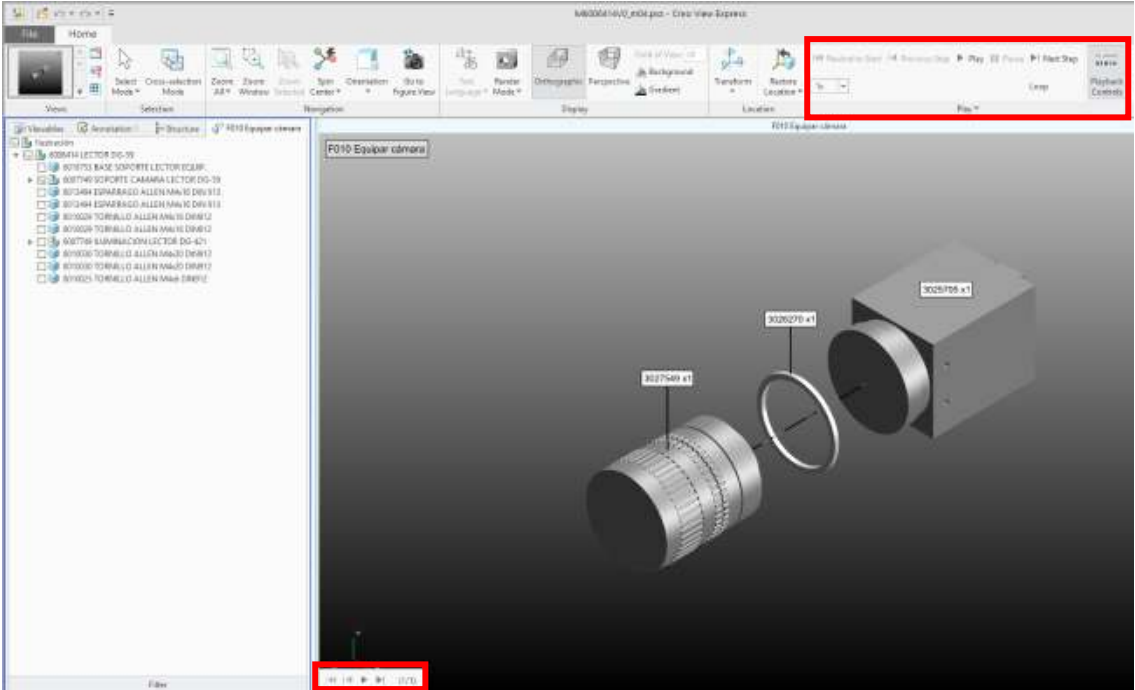


Figure 512. Playback control buttons for an animated assembly procedure

- Parts can be hidden/unhidden during the animation by enabling/disabling the square near each component in the tree structure or selecting the corresponding option after right-clicking on it.
- The background colours can also be modified as desired by editing the *Background* and *Gradient* options of the *Display* section in Creo View Express toolbar.

12.7 Spare parts searching

The raw codes composing a factory module can be found in the 3D drawings. From there, they can be matched to the codes composing an existing spare part in order to know the corresponding spare part reference.



NOTE:
Before searching for a specific raw material code, make sure that the configuration detailed in section 12.4 has been applied

12.7.1 Finding a drawing part for a spare parts list's reference code

- Copy the desired drawing reference code from the Spare Parts List 'Drawing reference' column.
- Search in the drawings table the drawing of the module where the part is assembled.
- Click on *Find* button in Creo View Express toolbar.
- Paste the previously copied code in the new window and click on *Find next*.
- The raw material corresponding to the selected drawing reference will be highlighted both in the tree structure and in the model drawing.

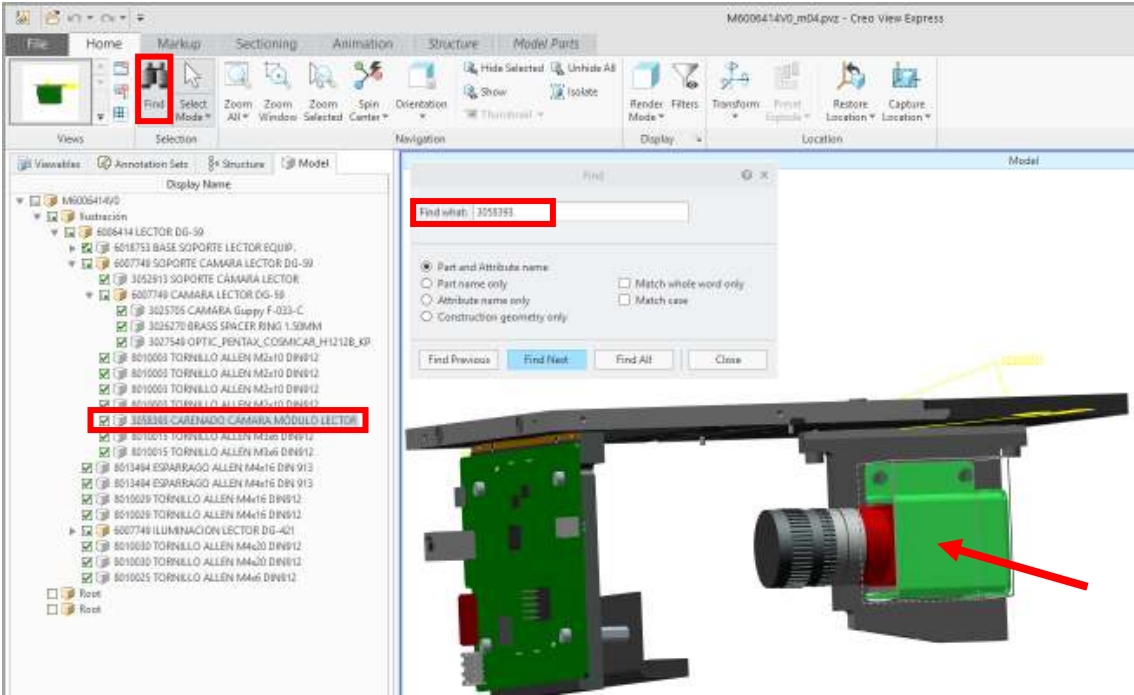


Figure 513. Finding a drawing part from a known code

12.7.2 Finding the spare part list’s reference code for a drawing part

- 1. Select the part of the drawing which raw material code needs to be known.
- 2. The part will be selected in the tree structure. If the view configuration detailed in section 12.4 has been applied, the raw material code (CODIGO field) will be listed as part of the material description. The raw material code (CODIGO field) can also be seen by enabling the lower data panel option on the lower left side of the screen.

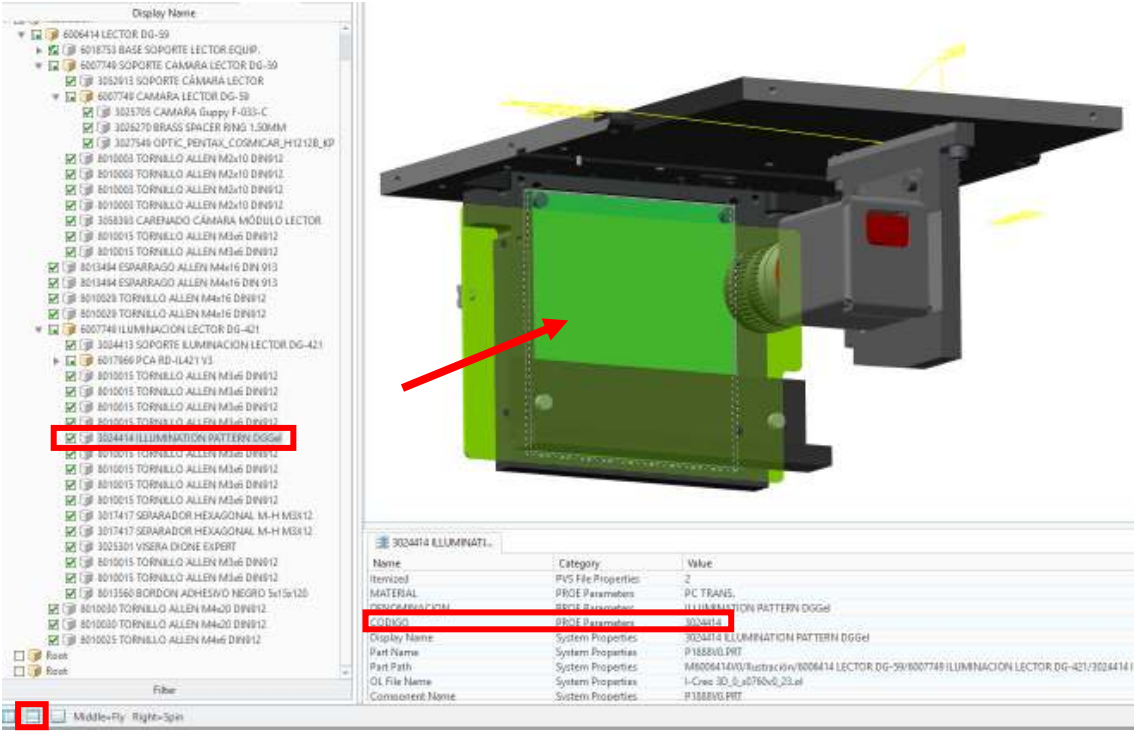


Figure 514. Find a code by selecting a drawing part

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	503 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

3. Search for the obtained raw material code in Erytra Spare Parts List. The code will be listed in the *Drawing Reference* column for one or more spare part codes, so the desired reference can be obtained. If the raw material code cannot be found in the Spare Parts list:
 - a) If the raw material code is part of a set of codes included in a 'parent' code on top of it in the tree structure, try to look for the parent code in the Spare Parts list. The parent code of a specific part can be automatically selected by selecting it and then clicking on the *Select Parent* button than can be found in the *Structure* tab in Creo View Express toolbar.
 - b) If the raw material code is not part of a set of codes included in a 'parent' code, or if the parent code cannot be found in the Spare Parts list, it means that no part number reference has been assigned to the desired raw material code.

Print date:	16-May-2023	Purpose:	Uncontrolled Copy
--------------------	-------------	-----------------	-------------------

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization

13 DIAGNOSTIC APPLICATION

13.1 INTRODUCTION

This chapter contains an overview of the *Diagnostic* software that allows a **QUALIFIED TECHNICIAN / SUPERVISOR / OPERATOR** to run a self-test.

Diagnostic application can help to diagnose a problem in the instrument or just to check the proper functioning as a usual maintenance. Tubes, vials and a diluent bottle required to execute the *Diagnostic* application tests are included in P/N 732944 IH DIAGNOSTIC KIT.

In order for a **QUALIFIED TECHNICIAN** to access *Diagnostic* application, the boot menu must be accessed. For that, switch on the PC, wait for the Erytra logo to appear on screen and then press the grid (■) button. Log in as *adminUser* (case sensitive, *admin* password) and the Boot Application options will be displayed. Once in there, press the arrow in the right side of the screen to switch to page 2/2 of the menu, then select the *Diagnostic* icon among the Boot Application options.



Figure 515. Diagnostic icon

13.2 OVERVIEW

Once *Diagnostic* application is opened, the following window appears on screen:

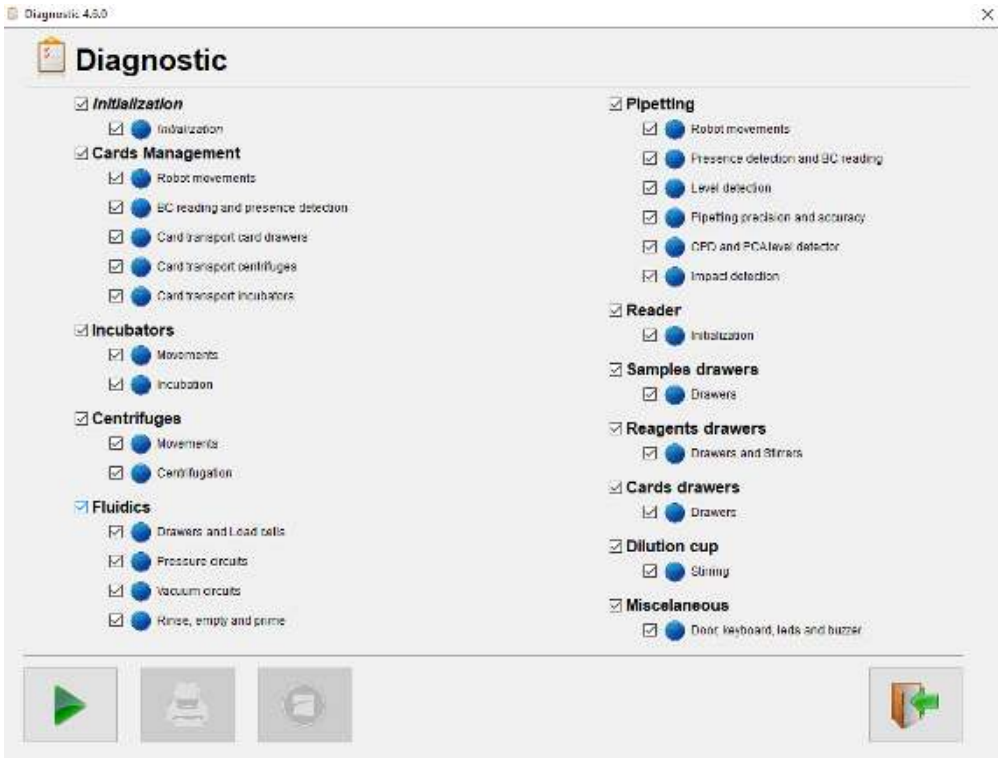


Figure 516. Diagnostic application

Available tests are put into groups depending on the module that are related.

- Initialization
- Cards Management
- Pipetting
- Fluidics
- Reader
- Incubators
- Centrifuges
- Samples drawers
- Reagents drawers
- Cards drawers
- Dilution cup
- Miscellaneous

Diagnostic application can check completely the equipment or only some parts. This can be chosen in the main screen of the software, just selecting or unselecting the modules to be checked.

At the bottom of the main screen there are 4 buttons



Start button

Run the selected tests.



Stop button

Enabled when the test is started. It stops the test.



Printer button

Enabled when the test has finished. It prints the results of the test.



Exit button

To exit from the software

Table 88. Available buttons in Diagnostic application

13.3 TESTS DESCRIPTION

In main screen of the software, it is possible to select the tests that are required to be performed. Once tests are selected, *start* button must be pressed in the main screen of the software in order to run them.

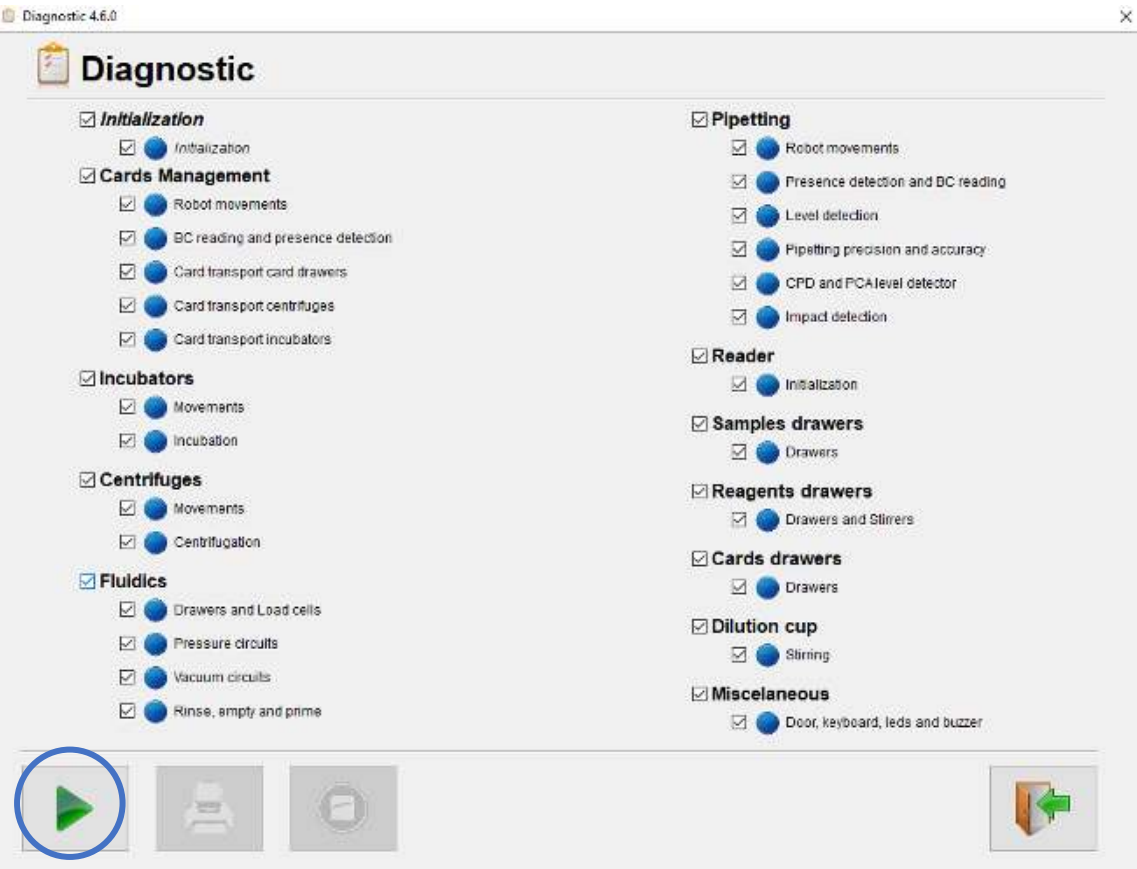


Figure 517. Start button in Diagnostic application

Once is pressed, if any prerequisite is necessary to perform the selected tests, a window message on screen appears with them.

13.3.1 Initialization

This test is executed in each run. If the test fails, no further tests will be executed.

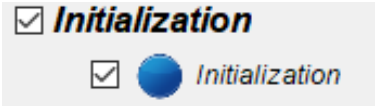


Figure 518. Initialization option in Diagnostic application

Initialization

Prerequisites:

- None.

Action:

- Reset the 4 PCA MotorControl (CEN-SRM, CAT-CM, PIP and INC-CUP) and the PCA FLU-CTRL59.
- Home CAT (Z, Y and X axis).
- Home PIP (Z1, Z2, Horns, X and Y axis).
- Home of the incubators.
- Home of the centrifuges.
- Home of the stirring racks of the reagents drawers.



WARNING!

Performing motor homes implies movement of the robotic arms. Make sure that the door is closed before running the test.

Expected result:

- PCAs can be reset without any error.
- All homes are performed without errors.

13.3.2 Cards Management

It includes the *Robot movements*, *BC reading and presence detection* and *Card Transport* tests.

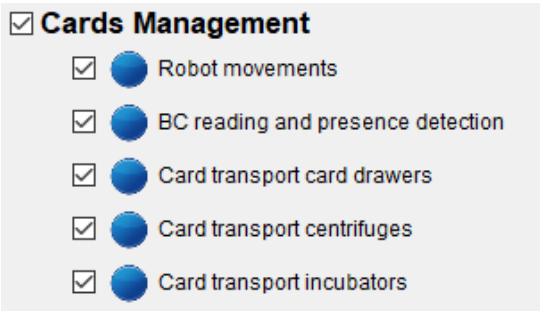


Figure 519. Cards Management options in Diagnostic application

Reference:	REGD-0006730	Title:	Version:	17.0; Most-Recent; Effective; CURRENT	Page	508 of 623
Status:	Effective					
	Service Manual Erytra				Effective Date:	15-May-2023

Robot movements

Prerequisites:

- No card in gripper.
- No cards in drawers 2 to 4.
- No cards in incubators .
- No cards in centrifuges.
- Card drawers and lower door closed.

Action:

- Checking that all card drawers and lower door are closed.
- Checking that there is no card in gripper.
- Performing presence detection in card drawer 2 to 4, incubators and centrifuges.
- Perform 50 random movements.
- After each random movement, execute a Take card command (ignore no card error).

Expected result:

- Card drawers and lower door are closed.
- No card is detected in gripper.
- No cards are detected in card drawer 2 to 4, incubators and centrifuges.
- No error is detected during the execution.

BC reading and presence detection

Prerequisites:

- Card holder with first 12 lower cards in card drawer 1 - rack 4.
- Card drawers and lower door closed.

Action:

- Checking that all card drawers and lower door are closed.
- Detect presence of placed holder.
- Take all cards and read their barcode .

Expected result:

- Card drawers and lower door are closed.
- No cards are detected in holders' upper floor.
- At least 12 cards are detected in the holders' lower floor.
- All cards are read and no error is reported by the BC reader.

Card Transport card drawers

Prerequisites:

- Card drawers and lower door closed.
- Card holder with first 12 lower cards in card drawer 1 - rack 4.
- Empty card holders in card drawer 1 - rack 1 to 3, and drawer 2, 3, and 4, rack 1 to 4.
- No card in gripper.

Action:

- Checking that all card drawers and lower door are closed.
- Checking there is no card in Gripper.
- Detect presence of drawer 1, rack 1 to 3, and drawer 2, 3 and 4, rack 1 to 4.
- Detect presence of placed holder.
- Cards are moved:
 - One by one, move 12 cards to drawer 1, rack 3.
 - One by one, move 12 cards to drawer 1, rack 2.
 - One by one, move 12 cards to drawer 1, rack 1.
 - One by one, move 12 cards to drawer 2, rack 4.
 - One by one, move 12 cards to drawer 2, rack 3.
 - One by one, move 12 cards to drawer 2, rack 2.
 - One by one, move 12 cards to drawer 2, rack 1.
 - One by one, move 12 cards to drawer 3, rack 4.
 - One by one, move 12 cards to drawer 3, rack 3.
 - One by one, move 12 cards to drawer 3, rack 2.
 - One by one, move 12 cards to drawer 3, rack 1.

Print date:	16-May-2023	Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization		Purpose:	Uncontrolled Copy
--------------------	-------------	--	--	-----------------	-------------------

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	509 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

- One by one, move 12 cards to drawer 4, rack 4.
- One by one, move 12 cards to drawer 4, rack 3.
- One by one, move 12 cards to drawer 4, rack 2.
- One by one, move 12 cards to drawer 4, rack 1.
- One by one, move 12 cards to drawer 1, rack 4.

Expected result:

- Card drawers and lower door are closed.
- No card is detected in gripper.
- No cards are detected in holders' upper floor.
- At least 12 cards are detected in the holders' lower floor.
- No cards are detected in drawer 1, rack 1 to 3 and drawer 2, 3 and 4, rack 1 to 4.
- All cards are transported without errors.

Card Transport centrifuges

Prerequisites:

- Card drawers and lower door closed.
- Card holder with first 12 lower cards in card drawer 1 - rack 4.
- Empty card holders in card drawer 1 - rack 1 to 3, and drawer 2, 3, and 4, rack 1 to 4.
- No card in gripper.
- No cards in centrifuges.

Action:

- Checking that all card drawers and lower door are closed.
- Checking there is no card in Gripper.
- Detect presence of drawer 1, rack 1 to 3, and drawer 2, 3 and 4, rack 1 to 4.
- Detect presence of placed holder.
- Detect presence of centrifuges.
- Cards are moved:
 - One by one, move 12 cards to Centrifuge 1.
 - One by one, move 12 cards to Centrifuge 2.
 - One by one, move 12 cards to Cards drawer 1, rack 4..

Expected result:

- Card drawers and lower door are closed.
- No card is detected in gripper.
- No cards are detected in holders' upper floor.
- At least 12 cards are detected in the holders' lower floor.
- No cards are detected in drawer 1, rack 1.
- No cards are detected in centrifuges.
- All cards are transported without errors.

Card Transport incubators

Prerequisites:

- Card drawers and lower door closed.
- Card holder with first 12 lower cards in card drawer 1 - rack 4.
- No card in gripper.
- No cards in incubators.

Action:

- Checking that all card drawers and lower door are closed.
- Checking there is no card in Gripper.
- Detect presence of placed holder.
- Detect presence of incubators.
- Cards are moved:
 - One by one, move 12 cards to Incubator 1.
 - One by one, move 12 cards to Incubator 2.
 - One by one, move 12 cards to Incubator 3.

Print date:	16-May-2023	Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization		Purpose:	Uncontrolled Copy
--------------------	-------------	--	--	-----------------	-------------------

- One by one, move 12 cards to Incubator 4.
- One by one, move 12 cards to Cards drawer 1, rack 4.

Expected result:

- Card drawers and lower door are closed.
- No card is detected in gripper.
- No cards are detected in holders' upper floor.
- At least 12 cards are detected in the holders' lower floor.
- No cards are detected in incubators.
- All cards are transported without errors.

13.3.3 Pipetting

It includes the *Robot movements*, *Presence detection and BC reading*, *Level detection*, *Pipetting precision and accuracy*, *CPD and PCA level detector* and *Impact detection*.

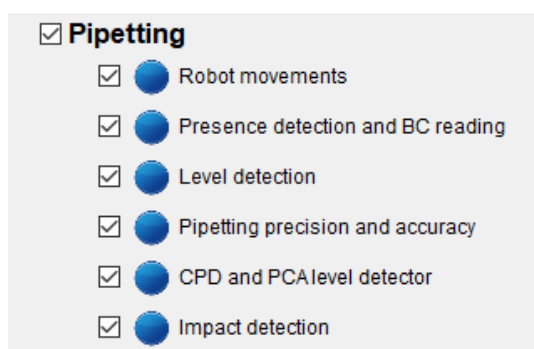


Figure 520. Pipetting options in Diagnostic application

Robot movements

Prerequisites:

- Sample drawers and upper door closed.
- No samples tubes in samples drawers 4, 5 and 6.

Action:

- Checking that sample drawers and upper door are closed.
- Performing presence detection in drawers 4, 5 and 6.
- 50 random movements in those positions where there are not tubes in drawers 4 to 6, odd positions (1, 3, ...) with probe 1.
- After each movement executing a Down command with both probes (ignore level error).
- Performing a barcode reading of tubes in drawer 4 rack 2.
- Moving Horns 50 times up/down.

Expected result:

- Sample drawers and upper door are closed.
- No sample tubes are detected in drawers 4, 5 and 6.
- No movement error is detected.

Presence detection and BC reading

Prerequisites:

- Reagents and samples drawers and upper door closed.
- No samples in drawers 4, 5 and 6.
- Drawer 3, rack 2 full of 10 mm (positions 1 to 4), 12 mm (positions 5 to 8) and 15 mm (positions 9 to 12) tubes with BC label. Tubes filled with saline solution 0.9%.

- Reagent drawer 1, rack 1 with 1 diluent in position 4 and 2 vials in positions 2 and 9 (with BC label).
 - Reagent drawer 1, rack 2 with 4 vials in positions 1, 6, 11 and 18 (with BC label).
- Action:
- Checking that reagents and samples drawers and upper door are closed.
 - Detect presence of sample tubes.
 - Read barcodes of the tubes.
 - Detect presence of vials/bottles.
 - Read barcodes of vials/bottles.
- Expected result:
- Reagents and samples drawers and upper door are closed.
 - Tubes or vials are detected in all required positions of drawer 3, rack 2 and drawer 1.
 - No tubes are detected in drawers 4, 5 and 6.
 - All diameters match expected setting (+0 mm, -2 mm).
 - BCs are read for all tubes.
 - Vials/bottles are detected in all rack positions.
 - BCs are read for all vials/bottles.



NOTE:

If one or many diameters do not match with the expected setting (+0 mm, -2 mm) the test shall be considered as Warning.

Level detection

Prerequisites:

- Samples drawers and upper door closed.
- No samples in drawers 4, 5 and 6.
- Drawer 3, rack 2 full of 10 mm (positions 1 to 4), 12 mm (positions 5 to 8) and 15 mm (positions 9 to 12) tubes with BC label. Tubes filled with saline solution 0.9%.
- Circuit primed (if not primed, prime is performed).

Action:

- Checking that samples drawers and upper door are closed.
- Detect presence of all the sample tubes.
- Check that the fluidic system is primed. If not primed, prime is performed.
- Perform 50 level detections with each probe (probe 1 and 2 in parallel) in all samples tubes (positions 1 and 2, 5 and 6, 9 and 10).



NOTE:

If one or many diameters do not match with the expected setting (+0 mm, -2 mm) the test shall be considered as Warning.



NOTE:

If any of the probes encounters an error that prevents from continuing, the system shall continue with the other probe.

Expected result:

- Samples drawers and upper door are closed.
- Tubes are detected in all required positions of drawer 3, rack 2.
- No tubes are detected in drawers 4, 5 and 6.
- All diameters match expected setting (+0 mm, -2 mm).
- All level detections are performed without error.
- No error is reported during process.

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	512 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

Pipetting precision and accuracy

Prerequisites:

- Reagents and samples drawers, card drawers and doors closed.
- Reagent drawer 1 - rack 1 with 1 diluent bottle in position 4 and 2 vials in positions 2 and 9. All of them must include barcode labels.
- Reagent drawer 1 - rack 2 with 4 vials in positions 1, 6, 11 and 18. All of them must include barcode labels. Tubes filled with saline solution 0.9%.and vial in position 1 must also be colored with tartrazine.
- Cards holder in card drawer 1 – rack 4 with 4 cards in lower floor (minimum).
- Circuit primed (if not primed, prime is performed).
- No cards in Incubators, centrifuges and card drawers 2, 3 and 4.
- No card in gripper.

Action:

- Checking that reagents and samples drawers, card drawers and doors are closed.
- Checking there is no card in Gripper.
- Detect presence of cards.
- Detect presence of vials.
- If fluidic system is not primed, prime is performed.
- Move 4 cards from card drawer 1 – rack 4 to incubator 1, performing an integrity check per each card.
- Move incubator 1 to PIP floor.
- Aspirate 720µl of colouring dilution with each probe.
- Dispense 75µl of colouring dilution in each well. Probe 1 dispenses into card 1 and probe 2 dispenses into card 3.
- Aspirate 920µl of colouring dilution with each probe.
- Dispense 100µl of colouring dilution in each well. Probe 1 dispenses into cards 2 and probe 2 dispenses into card 4.
- Move incubator 1 to the CAT floor.
- Move the 4 cards from the incubator 1 to centrifuge 1.
- Perform a centrifugation of 2min.
- Read all cards and calculate dispensed volume.
- Move all cards to the card drawer 1 – rack 4.

Expected result:

- Reagents and samples drawers, card drawers and doors are closed.
- No card detected in gripper.
- Vials are detected in all required positions.
- No cards are detected in holders' upper floor.
- At least 4 cards are detected in the holders' lower floor.
- No cards are detected in not required positions of card drawers, incubators and centrifuges.
- Cards integrity check gives no error.
- Card read gives no error. Dispensed volume lies within margins (± 12.5 µl).
- No error is reported during process.

CPD and PCA level detector

Prerequisites:

- Circuit primed (if not primed, prime is performed).

Action:

- Set on CPD of probe 1.
- Wait 5 s.
- Check that value of the CPD sensor is between 17000 and 18400 counts.
- Set off CPD of probe 1.
- Set on CPD of probe 2.
- Wait 5 s.
- Check that value of the CPD sensor is between 17000 and 18400 counts.
- Set off CPD of probe 2.

Print date:	16-May-2023	Purpose:	Uncontrolled Copy
Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization			

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	513 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

- Set on level detector of probe 1.
- Wait 5 s.
- Check that value of the level detector is between 7300 and 7900 counts.
- Set off level detector of probe 1.
- Set on level detector of probe 2.
- Wait 5 s.
- Check that value of the level detector is between 9900 and 10900 counts.
- Set off level detector of probe 2.

Expected result:

- Value of the CPD sensors is within margins.
- Value of the level detector sensors is within margins.
- The test is executed without errors.

Impact detection

Prerequisites:

- Samples drawers, reagent drawers and upper door closed.

Action:

- Checking that reagents and samples drawers and upper door are closed.
- Ask operator to insert one diluent bottle with its cap in reagent drawer 2, rack 1, position 4, remove the holder from Samples Drawer 2, Rack 2 and close drawers.
- Performing presence detection in reagent drawer 2, rack 1.
- Move pipetting arm, probe 1, to reagent drawer 2, rack 1, position 4.
- Perform 3 level detections with probe 1, with depth 0 mm, offset max depth 0 mm and vial profile level.
- Move the probe up between level detections.
- Move pipetting arm, probe 1, to Samples Drawer 2, Rack 2, Position 6.
- Move probe 1 down.
- If one or more level detections have been performed with probe 1, ask operator for confirmation of three impacts and if the probe 1 can continue executing tests.
- If operator answers No, and there are pending tests related with PIP arm to execute in current execution, stop tests related with PIP zone.
- If operator answered that the probe is not bent, move probe 1 up.
- Move pipetting arm, probe 2, to reagent drawer 2, rack 1, position 4.
- Perform 3 level detections with probe 2, with depth 0 mm, offset max depth 0 mm and vial profile level.
- Move the probe up between level detections.
- Move pipetting arm, probe 2, to Samples Drawer 2, Rack 2, Position 6.
- Move probe 2 down.
- If one or more level detections have been performed with probe 2, ask operator for confirmation of three impacts and if the probe 2 can continue executing tests.
- If operator answers No, and there are pending tests related with PIP arm to execute in current execution, stop tests related with PIP zone.
- If operator answered that the probe is not bent, move probe 2 up.
- If operator answered that the probe is not bent, ask the operator to remove the cap and place the diluent bottle to reagent drawer 1, rack 1, position 4, insert the holder in Samples Drawer 2, Rack 2 and close drawers.
- Check reagent drawer 1, reagent drawer 2 and Samples Drawer 2 are closed and if at least one of them is open, show an information message indicating to close them.

Expected result:

- Reagents and samples drawers and upper door are closed.
- A vial is detected in reagent drawer 2, rack 1, position 4.
- The impact or movement error has been detected 3 times for each probe.
- The operator has confirmed that both probes are not bent.
- The test is executed without errors.
-

Print date:	16-May-2023	Purpose:	Uncontrolled Copy
-------------	-------------	----------	-------------------

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization

13.3.4 Fluidics

It includes the *Drawers and Load cells*, *Pressure circuits*, *Vacuum circuits* and *Prime, rinse and empty* tests.

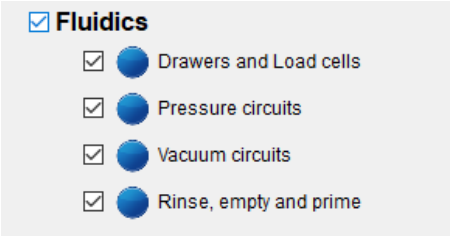


Figure 521. Fluidics options in Diagnostic application

Drawers and Load cells

Prerequisites:

- It is necessary an empty bottle, a full bottle (8l) and a known 2kg weight (2 litres of water recommended).
- Fluidic drawers closed.

Action:

- Checking that all fluidic drawers are closed.
- Solution A drawer is opened. Place an empty bottle, close the drawer, wait 2 seconds and press *OK* on the screen message. The system measures the weight and checks whether values are correct.
- Solution A drawer is opened. Place a full (8l) bottle, close the drawer, wait 2 seconds and press *OK* on the screen message. The system measures the weight and checks whether values are correct.
- Solution B drawer is opened. Place an empty bottle, close the drawer, wait 2 seconds and press *OK* on the screen message. The system measures the weight and checks whether values are correct.
- Solution B drawer is opened. Place a full (8l) bottle, close the drawer, wait 2 seconds and press *OK* on the screen message. The system measures the weight and checks whether values are correct.
- Waste bottle 1 drawer is opened. Place a full bottle (8l), close the drawer, wait 2 seconds and press *OK* on the screen message. The system measures the weight and checks whether values are correct.
- Waste bottle 1 drawer is opened. Place an empty bottle, close the drawer, wait 2 seconds and press *OK* on the screen message. The system measures the weight and checks whether values are correct. Close the drawer
- Waste bottle 2 drawer is opened. Place a full (8l) bottle, close the drawer, wait 2 seconds and press *OK* on the screen message. The system measures the weight and checks whether values are correct.
- Waste bottle 2 drawer is opened. Place an empty bottle, close the drawer, wait 2 seconds and press *OK* on the screen message. The system measures the weight and checks whether values are correct. Close the drawer
- Card waste drawer is opened. Empty the container, close the drawer, wait 2 seconds and press *OK* on the screen message. The system measures the weight and checks whether values are correct.
- Card waste drawer is opened. Place a 2 kg weight into the container by either pouring 2 litres of water (recommended) or placing a known 2 kg weight in the center of the container, wait 2 seconds and press *OK* on the screen message. The system measures the weight and checks whether values are correct.
- Card waste drawer is opened in order to allow the possibility of placing an empty container. Close the drawer.

Expected result:

- All obtained values are correct.
 - Empty bottle 0 ± 0.5 L.
 - Full bottle 8 ± 0.5 L.
 - Empty card waste container: 0 ± 50 cards.
 - Full card waste container: 410 ± 50 cards.
- All drawers are closed and close detection is performed correctly.

Prime, rinse and empty

Prerequisites:

- In mono mode: At least 8l capacity in waste bottles and 4l available in both solution A and solution B bottles.
- In dual mode: At least 2L available in a bottle of Sol A and 2L in a bottle of Sol B.
- All fluidic drawers are closed.

Action:

- In mono mode, checking that there is at least 1,5L +/- 200ml of capacity in a waste bottle and 2L +/-200ml available in both Sol A and Sol B bottles.
- In dual mode, checking that there is at least 2L +/- 200ml available in a bottle of Sol A and 2L +/-200ml in a bottle of Sol B.
- Checking that fluidic drawers are closed.
- Move probes to dilution cups.
- Initialize probes
- Perform an empty of the fluidic circuit.
- Perform a rinse of the fluidic circuit.
- Perform an empty of the fluidic circuit.
- Perform a prime of the fluidic circuit.

Expected result:

- Waste and solution bottles capacity are the expected ones.
- Fluidic drawers are closed.
- All processes performed without errors.

Pressure circuits

Prerequisites:

- In mono mode: At least 8l capacity in waste bottles and 4l available in both solution A and solution B bottles.
- In dual mode: At least 2L available in a bottle of Sol A and 2L in a bottle of Sol B.
- All fluidic drawers are closed.

Action:

- Checking that fluidic drawers are closed.
- In mono mode, checking that there is at least 1,5L +/- 200ml of capacity in a waste bottle and 2L +/-200ml available in both Sol A and Sol B bottles. Related electrovalves shall be opened or closed in order to select the bottle with the needed available capacity or volume.
- In dual mode, checking that there is at least 2L +/- 200ml available in a bottle of Sol A and 2L +/- 200ml in a bottle of Sol B. Related electrovalves shall be opened or closed in order to select the bottle with the needed volume.
- Close all pumps and electrovalves, except electrovalves related with containers switching.
- Open EV20, EV19, EV5 and EV10.
- Wait 10 seconds.
- Close EV10, EV5, EV19 and EV20.
- Switch on Pressure pump A (flowrate to 160).
- Wait 500 ms.
- Check that once the sensor 3 is stabilized (5 seconds with a variability of +/- 20 mbar), pressure value between 170 and 150 mbar is obtained in sensor 3.
- Open EV20.
- Wait 40 seconds.

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	516 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

- Check that once the sensor 3 is stabilized (5 seconds with a variability of +/- 20 mbar), pressure value between 80 and 280 mbar is obtained in sensor 3.
- Switch off Pressure pump A.
- Close EV20.
- Switch on Pressure pump B.
- Configure flowrate of Pressure pump B to 160.
- Wait 500 ms.
- Check that once the sensor 4 is stabilized (5 seconds with a variability of +/- 20 mbar), pressure value between 170 and 150 mbar is obtained in sensor 4.
- Open EV19.
- Wait 40 seconds.
- Check that once the sensor 4 is stabilized (5 seconds with a variability of +/- 20 mbar), pressure value between 80 and 280 mbar is obtained in sensor 4.
- Close EV19.
- Switch off Pressure pump B.

Expected result:

- All fluidic drawers are closed.
- Waste and solution bottles capacity are the expected ones.
- All pressure values obtained are correct.
- Sensors are stabilized in less than 50 seconds.
- The test is executed without errors.

Vacuum circuits

Prerequisites:

- In mono mode: At least 8l capacity in waste bottles and 4l available in both solution A and solution B bottles.
- In dual mode: At least 2L available in a bottle of Sol A and 2L in a bottle of Sol B.
- All fluidic drawers are closed.

Action:

- Checking that fluidic drawers are closed.
- In mono mode, checking that there is at least 1,5L +/- 200ml of capacity in a waste bottle and 2L +/-200ml available in both Sol A and Sol B bottles. Related electrovalves shall be opened or closed in order to select the bottle with the needed available capacity or volume.
- In dual mode, checking that there is at least 2L +/- 200ml available in a bottle of Sol A and 2L +/- 200ml in a bottle of Sol B. Related electrovalves shall be opened or closed in order to select the bottle with the needed volume.
- Close all pumps and electrovalves, except electrovalves related with containers switching.
- Open EV20, EV19, EV5 and EV10.
- Wait 10 seconds.
- Close EV10, EV5, EV19 and EV20.
- Switch on Vacuum pump.
- Wait 500 ms.
- Check that once the sensor 5 is stabilized (5 seconds with a variability of +/- 20 mbar), vacuum absolute value of sensor 5 is greater than or equal to 600 mbar.
- Check that once the sensor 6 is stabilized (5 seconds with a variability of +/- 20 mbar), vacuum absolute value of sensor 6 is greater than or equal to 600 mbar.
- Open EV5 and EV10.
- Wait 500 ms.
- Check that once the sensor 5 is stabilized (5 seconds with a variability of +/- 20 mbar), vacuum absolute value of sensor 5 is greater or equal than 100 mbar.
- Check that once the sensor 6 is stabilized (5 seconds with a variability of +/- 20 mbar), vacuum absolute value of sensor 6 is greater or equal than 100 mbar.
- Close EV5 and EV10.
- Open EV4 and EV9.

Print date:	16-May-2023	Purpose:	Uncontrolled Copy
-------------	-------------	----------	-------------------

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization

- Wait 500 ms.
 - Check that once the sensor 5 is stabilized (5 seconds with a variability of +/- 20 mbar), vacuum absolute value of sensor 5 is greater or equal than 100 mbar.
 - Check that once the sensor 6 is stabilized (5 seconds with a variability of +/- 20 mbar), vacuum absolute value of sensor 6 is greater or equal than 100 mbar.
 - Close EV4 and EV9.
 - Switch off Vacuum pump.
- Expected result:
- All fluidic drawers are closed.
 - Waste and solution bottles capacity are the expected ones.
 - All vacuum values obtained are correct.
 - Sensors are stabilized in less than 50 seconds.
 - The test is executed without errors.



NOTE:
If sensor 5 or sensor 6 are between -450 mbar and -599 mbar in the first check, the test shall be considered as Warning.

13.3.5 Reader

It includes the *Initialization* test.

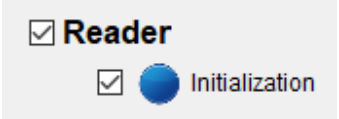


Figure 522. Reader option in Diagnostic application

- Prerequisites:
- None
- Action:
- Perform an initialization of the reader.
- Expected result:
- No errors are detected during the execution.

13.3.6 Incubators

It includes the *Movements* and *Incubation* tests.

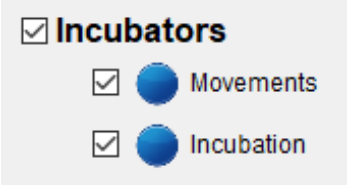


Figure 523. Incubators options in Diagnostic application

Movements

- Prerequisites:
- Doors closed.
- Action:
- Checking that doors are closed.
 - Move CAT robot arm to centrifuge 2.

- Perform 25 movements up and down with incubators 1 and 2.
- Move CAT robot arm to centrifuge 1.
- Perform 25 movements up and down with incubators 3 and 4.

Expected result:

- Doors are closed.
- No error is detected during the execution.

Incubation

Prerequisites:

- None.

Action:

- Start temperature regulation 37°C in all incubators.
- Check that temperature is reached within 20 min.
- Start temperature regulation 24°C in all incubators.
- Check that temperature is reached within 20 min.

Expected result:

- No error is detected during the execution.
- Temperatures are reached within expected time.

13.3.7 Centrifuges

It includes the *Movements* and *Centrifugation* tests.

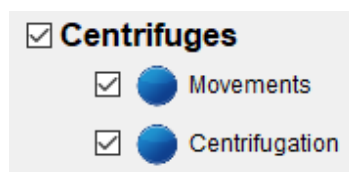


Figure 524. Centrifuges options in Diagnostic application

Movements

Prerequisites:

- Lower door closed.

Action:

- Checking that lower door is closed.
- Perform 50 random movements per each centrifuge.

Expected result:

- Lower door is closed.
- No error is detected during the execution.

Centrifugation

Prerequisites:

- Lower door closed.

Action:

- Checking that lower door is closed.
- Perform a centrifugation (PROFILE_1) in each centrifuge.
- Check that average speed is within limits (1206 rpm $\pm$ 3%).
- When centrifugation finishes, check that centrifugation duration longed as expected (120s - 144s, it includes centrifuge home action after centrifugation).

Expected result:

- Lower door is closed.
- No error is detected during the execution.
- Speed and time lie within expected limits.

13.3.8 Samples drawers

It includes the *Drawers* test.

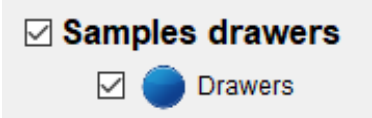


Figure 525. Samples drawers option in Diagnostic application

Prerequisites:

- Samples drawers closed.

Action:

- Check that all samples drawers are closed.
- Software opens automatically, one by one, all samples drawers.
- Close sample drawer 1 (drawer 3) and click Accept.
- Close sample drawer 2 (drawer 4) and click Accept.
- Close sample drawer 3 (drawer 5) and click Accept.
- Close sample drawer 4 (drawer 6) and click Accept.

Expected result:

- All drawers are closed.
- All drawers are opened without error.
- All drawers are closed and close detection is performed correctly and in the correct order.

13.3.9 Reagents drawers

It includes the *Drawers and Stirrers* test.

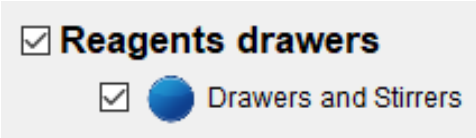


Figure 526. Reagents drawers option in Diagnostic application

Prerequisites:

- Reagents drawers closed.

Action:

- Check that all reagents drawers are closed.
- Software opens automatically, one by one, all reagents drawers.
- Close reagent drawer 1 and click Ok.
- Close reagent drawer 2 and click Ok.
- Perform 10 stirrings with each reagent drawer.

Expected result:

- All drawers are closed.
- All drawers are opened without error.
- All drawers are closed and close detection is performed correctly and in the correct order.
- Stirrings are performed without errors.

13.3.10 Cards drawers

It includes the *Drawers* test.

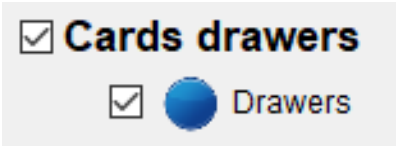


Figure 527. Cards drawers option in Diagnostic application

- Prerequisites:
- Cards drawers closed.
- Action:
- Check that all cards drawers are closed.
 - Software opens automatically, one by one, all cards drawers.
 - Close card drawer 1 and click Accept.
 - Close card drawer 2 and click Accept.
 - Close card drawer 3 and click Accept.
 - Close card drawer 4 and click Accept.
- Expected result:
- All drawers are closed.
 - All drawers are opened without error.
 - All drawers are closed and close detection is performed correctly and in the correct order.

13.3.11 Dilution cup

It includes the *Stirring* test.

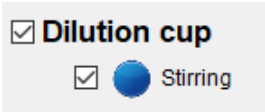


Figure 528. Dilution cup option in Diagnostic application

- Prerequisites:
- None.
- Action:
- Perform 25 dilution cup stirrings.
- Expected result:
- Stirrings are performed without errors.

13.3.12 Miscellaneous

It includes the *Door, keyboard, leds and buzzer* test.

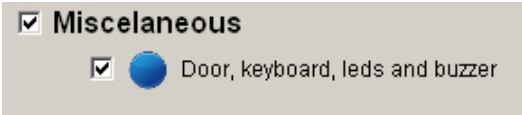


Figure 529. Miscellaneous option in Diagnostic application

- Prerequisites:
- Remove intermediate door screw.
- Action:
- Switch on ambient lights and general LED in red color.
 - Check if ambient lights and general LED in red color are switched on. Press Yes/No in the software.
 - Switch off ambient lights.
 - Switch on samples drawers LEDs and general LED in yellow color.

- Check if samples drawers LEDs and general LED in yellow color are switched on. Press Yes/No in the software.
- Switch off samples drawers lights.
- Switch on general LED in green color and buzzer.
- Check if general LED is in green color and the buzzer is on. Press Yes/No in the software.
- Press all the upper keyboard buttons, one by one, from reagents drawers to stat button.
- Press all the intermediate keyboard buttons, one by one, from card drawer 1 to card drawer 4.
- Open the upper door and press Ok.
- Close upper door and press Ok.
- Open intermediate door and press Ok.
- Close intermediate door and press Ok.

Expected result:

- All LEDs are confirmed to work correctly.
- Buzzer is confirmed to work correctly.
- All buttons are pressed.
- Doors open/close is detected.

13.4 REPORT

Once the selected tests are finished, a report can be obtained pressing the *printer* button in the main screen of the software.



Figure 530. Printer button in Diagnostic application

Report document is opened in a different window software, including information of instrument serial number, final status of tests, date and time, name and signature.

Possible status of tests in the report are the following ones:

- Pass: test has finished without errors.
- Fail: during the running of the test some errors have been detected.
- Warning: during the running of the test some errors have been detected but they are not critical.
- Not executed: test has not been selected.
- Stopped: running of the test has been stopped during its execution.



Figure 531. Report example in Diagnostic application

A list of possible causes that provoke a warning or fail status of the test can be checked in the following table. In order to verify exactly the cause of the problem, *diagnostic.log* must be checked (it can be found in *E:\Program Files\Erytra Diagnostic*).

	TEST	WARNING	FAIL
Initialization	Initialization		-Identify which PCA could not be reset. -Identify which servo failed to home and the error. -Identify if CAT arm or an incubator can not be homed because they are disturbing each other
Cards Management	Robot movements		- Home and no home positions are not correctly detected. -Identify if card is detected in gripper. -Identify which card drawer is open or if lower door is open. -Identify if card drawer 2 to 4, which incubator or which centrifuge has present cards. -Identify which servo failed in any movement and the error, and which movement is performed.
	BC reading and presence detection		-Identify which card drawer is open or if lower door is open. -Cards have been detected in upper floor. -Number of cards detected is lower than 12. -Card BC has not been read. Identify the position of the card in the rack. -Identify which servo failed a movement and the error.
	Card transport drawers		-Identify which card drawer is open or if lower door is open. -Card is detected in gripper. -Cards have been detected in upper floor. -Number of cards detected is lower than 12. -Cards have been detected in drawer 1 rack 1, or any rack of drawer 2, 3 or 4. -Identify which servo failed a movement and the error, and which movement is performed.
	Card Transport centrifuges		-Identify which card drawer is open or if lower door is open. -Card detected in gripper. -Cards have been detected in upper floor. -Number of cards detected is lower than 12. -Cards have been detected in a centrifuge. -Identify which servo failed a movement and the error, and which movement is performed.

	TEST	WARNING	FAIL
Cards Management	Card Transport incubators		-Identify which card drawer is open or if lower door is open. -Card detected in gripper. -Cards have been detected in upper floor. -Number of cards detected is lower than 12. -Cards have been detected in an incubator. -Identify which servo failed a movement and the error, and which movement is performed.
Pipetting	Robot movements		-Home and no home positions are not correctly detected -Identify which sample drawer is open or if upper door is open. -Samples detected in drawers 4, 5 or 6. -Identify which servo failed a movement and the error, and which movement is performed..
	Presence detection and BC reading	Tubes diameters do not match with the expected ones (0/-2 mm).	-Identify which reagent or sample drawer is open or if upper door is open. -Not all BC have been read. -Tube/vial has not been detected in a required position (position). -Tube detected in a not required position (position). -Diameters different from expected (position, obtained diameter and expected diameter). -Identify which servo failed a movement and the error, and which movement is performed.
	Level detection	Tubes diameters do not match with the expected ones (0/-2 mm).	-Identify which sample drawer is open or if upper door is open. -Tube has not been detected in a required position.. -Tube detected in a not required position. -Diameters different from expected +0 mm, -2 mm (position, obtained diameter and expected diameter). -Probe and encoder position if level detection fails. -Identify which servo failed a movement and the error, and which movement is performed. -Identify fluidic error performing prime.

	TEST	WARNING	FAIL
Pipetting	Pipetting precision and accuracy		-Identify which reagent or sample drawer or card drawer is open or which door is open. -Card detected in gripper. -Number of deteted cards smaller than 4 or in holder upper floor in card drawer 1, rack 4. -Identify if card drawer 2, 3 or 4, which incubator or which centrifuge has present cards. -Card integrity fail. -Vials not detected in drawer 1. -Dispensed volume is outside limits (detected volume, expected volume). -Identify which servo failed in any movement and the error, and which movement is performed. -Identify fluidic error performing prime.
	CPD and PCA level detector		-Identify any fluidic error with process, device and error. -Sensor value does not lie within margins (sensor, expected value, obtained value).
	Impact detection		-Identify which reagent or sample drawer is open or if upper door is open. -Vial not detected in reagent drawr 2, rack 1, position 4. -Impact or movement error has not been detected three times for each probe. -Operator notes that a probe is bent. -Identify which servo failed in any movement and the error, and which movement is performed.

	TEST	WARNING	FAIL
Fluidics	Drawers and Load cells		-Measured values are not between margins ($\pm 0.5l$ for containers or ± 25 cards for waste tray). -Drawers are open before starting test . -Drawer is not closed .
	Prime, rinse and empty		-Waste and solution bottles capacity are not the expected ones. -Identify which fluidic drawer is opened. -Identify which servo failed in any movement and the error, and which movement is performed. -Identify any fluidic error with process, device and error.
	Pressure circuits		-A fluidic drawer is open. -Waste or solution bottles capacity are not the expected ones. -P sensor value does not lie within margins. -Sensor not stabilized in less than 50 seconds. -Identify any fluidic error with process, device and error.
	Vacuum circuits		-A fluidic drawer is open -Waste and solution bottles capacity are not the expected ones. -V sensor value does not lie within margins. -Sensor not stabilized in less than 50 seconds. -Identify any fluidic error with process, device and error.
Reader	Initialization		-The reader could not be initialised. -Start-up and check returned an error.
Incubators	Movements		-Identify which door is open. -Identify which servo failed a movement and the error, and which movement is performed.
	Incubation		-Identify what incubator failed to reach temperature in the specified time (less than 20 min).

	TEST	WARNING	FAIL
Centrifuges	Movements		-Lower door is open. -Identify which centrifuge failed in any movement and the error, and which movement is performed.
	Centrifugation		- Lower door is open - Average speed does not lie within limits (1206 rpm $\pm$ 3 %). -Centrifugation time does not correspond to expected value (120s - 144s , it includes centrifuge home action after centrifugation).
Samples drawers	Drawers		-Drawers are not closed at the beginning of the test. -Drawer is not opened. -Drawer is not closed and close detection is not performed correctly or in the correct order.
Reagents drawers	Drawers and Stirrers		-Drawers are not closed at the beginning of the test. -Drawer is not opened. -Drawer is not closed -Close detection is not performed correctly or in the correct order. -Stirring errors, identifying the servo and the error.
Cards drawers	Drawers		-Drawers are not closed at the beginning of the test. -Drawer is not opened (drawer). -Drawer is not closed and close detection is not performed correctly or in the correct order.
Dilution cup	Stirring		-Identify any servo error with the stirrer.
Miscelaneous	Door, keyboard, LEDs and buzzer		-LEDs and Buzzer: user answers NO to a question . -Buttons: *User pressed Cancel. *A button pressed was not detected or is not detected in the specified order.. *Door open/close has not been detected.LED does not work properly.

Table 89. Warning and Fail results in Diagnostic application

14 PREVENTIVE AND CORRECTIVE MAINTENANCE

14.1 ADJUSTMENT TOOLS

All the adjustment tools are detailed in each section where an adjustment is explained. In the following table there are all of them:

ADJUSTMENT TOOL	MODULE	PN
Probes/wagon adjustment tool (Ver. 2)	PIP	233901
Probes adjustment tool (Ver. 4)	PIP	234213
Horns adjustment tool (BCR)	PIP	233819
Omnifit tool (for tube connection Ver 2)	PIP	233898
Cards drawer adjustment tool (Ver.2)	CAT	233878
Centrifuge / Incubator adjustment tool (Ver.3)	CAT	733056
Lower centrifuge / Incubator adjustment tool (Ver.3)	CAT	733068
Dilution cup alignment tool (Ver. 2)	CUP	233938
Centrifuge rotor adjustment tool	CEN	233821
Digital thermometer HIBOK-15 MiniDIN 4v	INC	230903
Temperature probe PT-100 for cards – MiniDIN 4v	INC	230905
Reader adjustment tool – support (Ver. 2)	LEC	234212
Reader adjustment tool – pattern v2	LEC	731398
USB reader cable tool (USB 2.0 cable 1.8m)	LEC	233443
FireWire reader cable tool	LEC	233822
Pressure sensor adjustment tool	FLU	233817
Digital Manometer DIGITRON 2023	FLU	247032
Drawer front panel alignment tool	GEN	233824

Table 90. Erytra adjustment tools

14.2 ITEMS AND CALENDAR

Preventive Maintenance (PM) form defines the procedures that must be carried out periodically on *Erytra* instruments in order to ensure that they are running under proper operating conditions. The period of the preventive maintenance plan (PMP) is defined **annual** under a standard use of the instrument (**65 samples/hour, 8 hours/day, 250 days/year, referred to ABO-Rh Group test**).

The parts to be replaced are decided taking into account the number of tests carried out during each period, under the standard use of the instrument. The table below shows the equivalence between the years and the number of tests. Therefore, Preventive Maintenance parts replacement can be done by periodicity (years) or by use of the instrument (number of tests).

YEAR	NUMBER OF TESTS
1	130.000
2	260.000
3	390.000
5	650.000
6	780.000
7	910.000

Table 91. Equivalence between years in PM and tests under standard instrument use

PM form related to this Service Manual can be found in a separate document under the reference *REGD-0023017*, and must always be executed before the corresponding Operational Qualification (OQ) procedure.

14.3 LUBRICATION POINTS

During some preventive maintenance actions or even after some corrective maintenance actions, lubrication procedures are required to apply to some parts of the equipment. Recommended lubrication products are:

- Grease: P/N 232697 SUPER LUBE GREASE LUBRICANT 85g (5 units)
- Oil: P/N 232696 SUPER LUBE OIL LUBRICANT 7ml (5 units)



NOTE:

Deenergize the motors before performing lubrication procedures in order to distribute the grease or oil. When necessary, perform a home of the module and move it from edge to edge to distribute the grease or oil. Do not apply grease/oil in excess.

14.3.1 GENERAL MODULE

Touchscreen drawer

Lubricate with grease the tilting latch lock.



Figure 532. Lubrication point in touchscreen drawer

Lubricate with grease the guides of the touchscreen monitor.

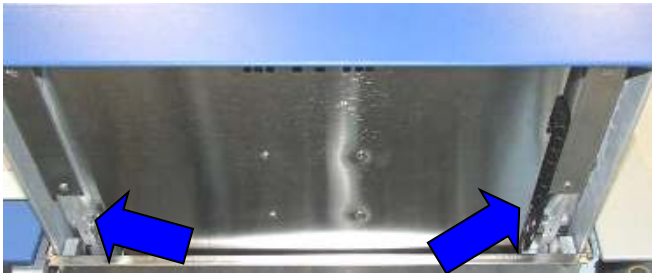


Figure 533. Lubrication points in touchscreen drawer

User tray



NOTE:
Not necessary for instruments with SN $\geq$ 059-0001675 or any Erytra unit where P/N 733671 USER TRAY (Ver. 4) is assembled.

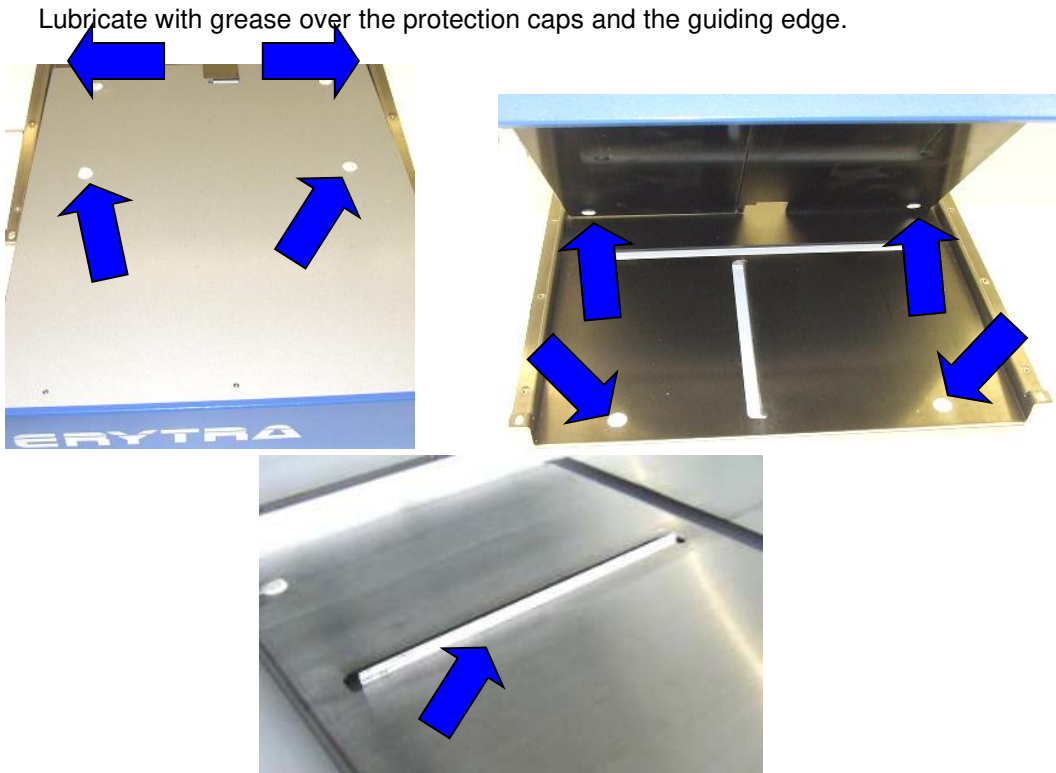


Figure 534. Lubrication points in user tray

Upper door

Lubricate with a drop of oil in joints of the hinges.

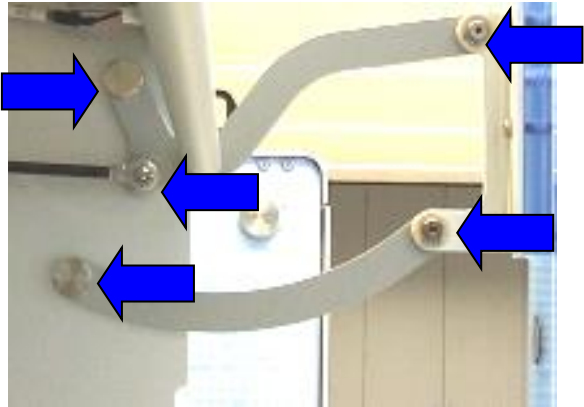


Figure 535. Lubrication points in upper door

Intermediate door

Lubricate with grease the surface of the hinges.

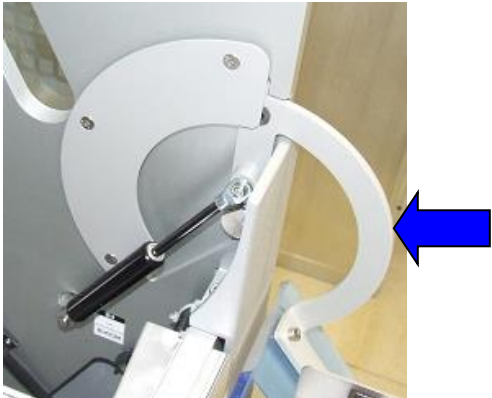


Figure 536. Lubrication points in intermediate door

Reagent drawers

Lubricate the end of the latch with grease.



Figure 537. Lubrication point in reagent drawer (latch)

Lubricate the linear guides with grease (use a brush).

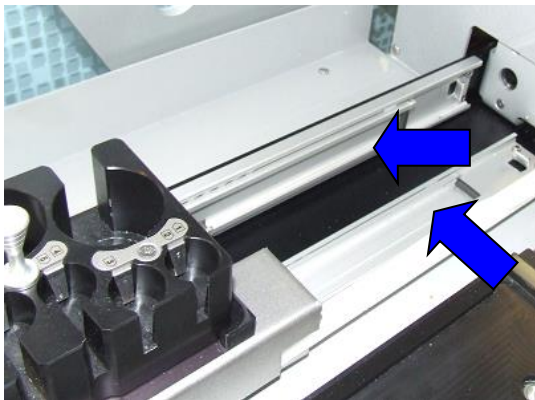


Figure 538. Lubrication point in reagent drawer (linear guides)

Lubricate with grease underneath the foot of the stirred reagents rack platform.

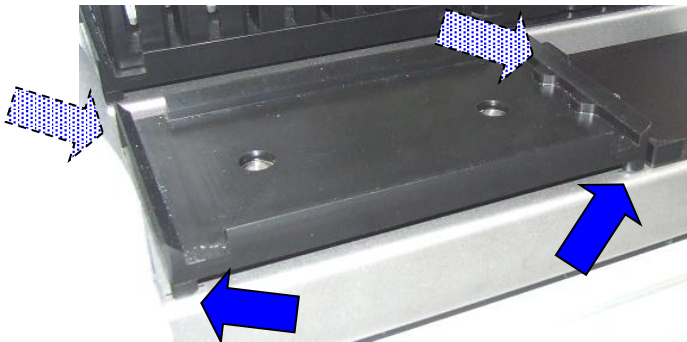


Figure 539. Lubrication point in reagent drawer (stirred reagents platform)

Samples drawers

Lubricate with grease the end of the latch.



Figure 540. Lubrication point in sample drawer (latch)

Lubricate the linear guides with grease (use a brush).

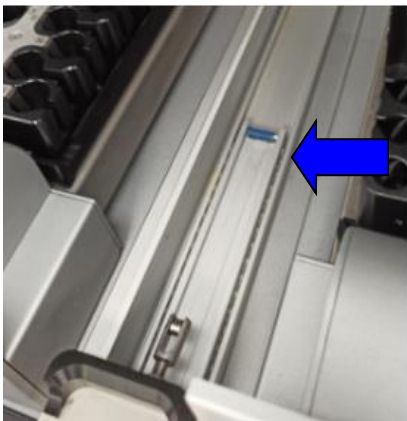


Figure 541. Lubrication point in sample drawer (linear guide)

Cards drawers

Lubricate with grease the end of the latch.



Figure 542. Lubrication point in card drawer (latch)

Lubricate the linear guides with grease (use a brush).

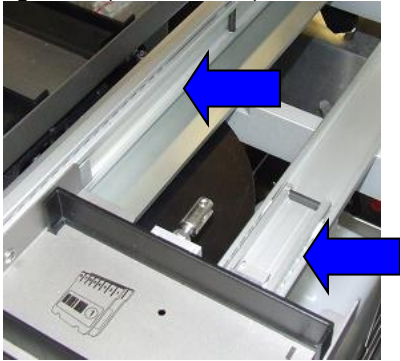


Figure 543. Lubrication point in card drawer (linear guides)

14.3.2 PIPETTING ARM (PIP) MODULE

Horns structure

Lubricate with grease.



Figure 544. Lubrication point in PIP module (horns)

X-axis linear guide

Lubricate with oil.



Figure 545. Lubrication point in PIP module (X-axis linear guide)

Y-axis linear guide

Lubricate with oil.



Figure 546. Lubrication point in PIP module (Y-axis linear guide)

Leadscrews

Lubricate with oil the probes leadscrews and horns leadscrew.

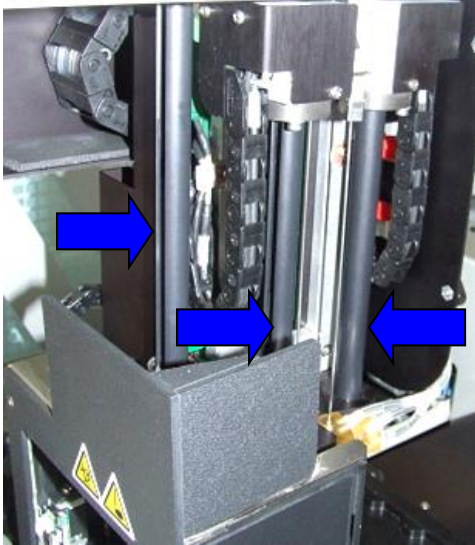


Figure 547. Lubrication points in PIP module (leadscrews)

14.3.3 CARD TRANSPORT ARM (CAT) MODULE

X-axis linear guide

Lubricate with oil.



Figure 548. Lubrication point in CAT module (X-axis linear guide)

Y-axis linear guide

Lubricate with oil.

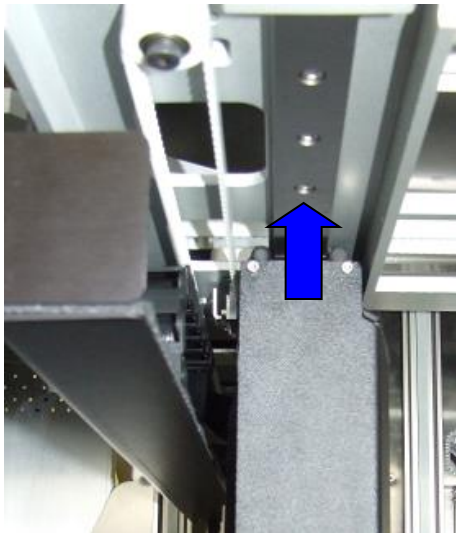


Figure 549. Lubrication point in CAT module (Y-axis linear guide)

Leadscrew

Lubricate with oil.

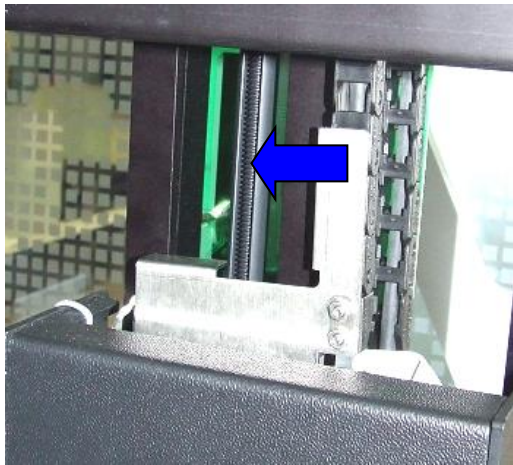


Figure 550. Lubrication point in CAT module (leadscrew)

Gripper linear guide

Lubricate with oil.

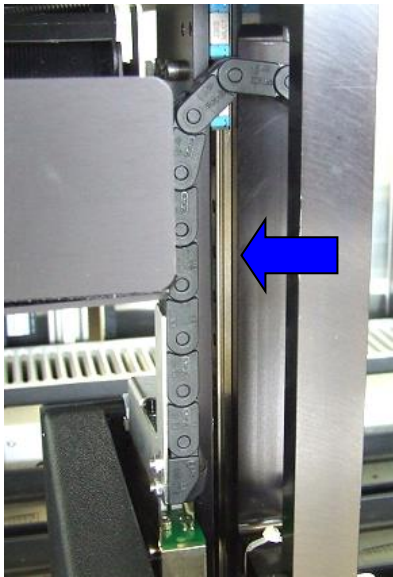


Figure 551. Lubrication point in CAT module (gripper linear guide)

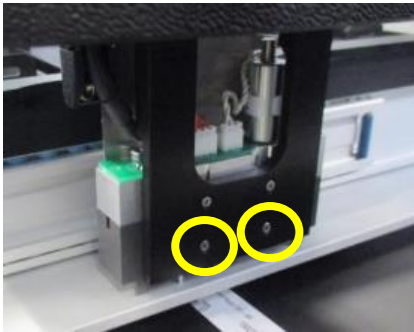
Gripper



NOTE:
Consider removing only the lower part of the gripper for lubrication purposes.
In case of complete removal of the gripper it is necessary to carry out all adjustments of the gripper again. Including the gripper parallelism.

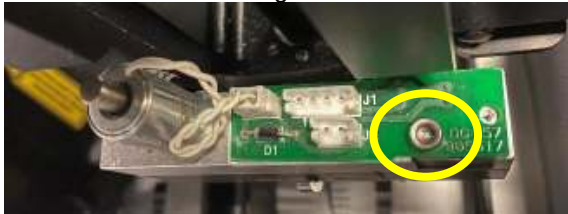
In order to disassemble the lower part of the gripper follow the 8 first steps described in section 8.4.5 Gripper (Z) chain replacement. Then:

-Remove the two lower screw

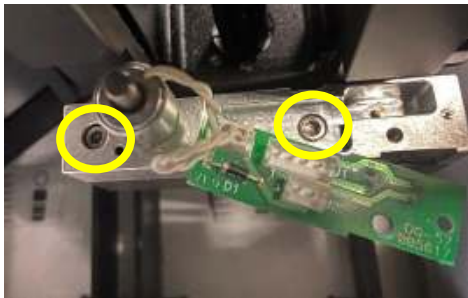


S.

- Remove the screw holding the PCA and disassemble it.



- Completely unscrew the two screws that hold the lower part to the upper part of the gripper. A flat screwdriver can be used to prise the two parts apart.



Once the two parts are separated, lubricate with a small drop of oil the 4 gripper pins.



Figure 552. Lubrication point in CAT module (gripper pins)

Lubricate with grease the surface of the centering system in contact with the lower part of the gripper.



NOTE:
Make sure the spring is placed over the centering part when assembling back all the parts.

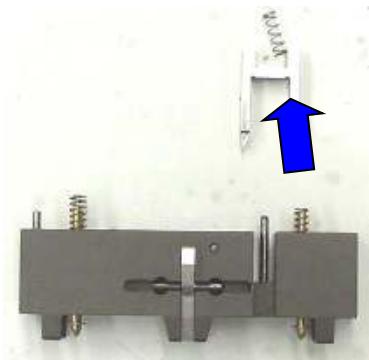


Figure 553. Lubrication point in CAT module (gripper centering system)



NOTE:
Threadlocking adhesive (such as Loctite 222) must be applied to the two removed screws when assembling the gripper in the instrument.

14.3.4 FLUIDICS MODULE

Bottle drawers / Cards disposal tray drawer

Put some grease in load cell ramp.

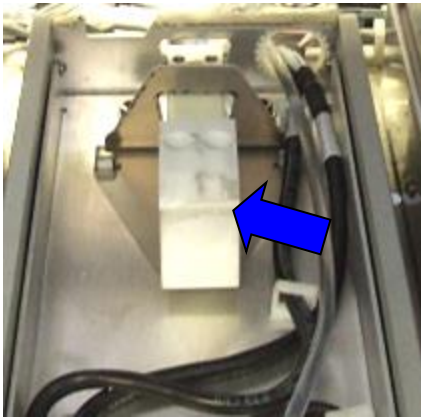


Figure 554. Lubrication point in fluidics drawer (load cell)

Lubricate with grease the tilting base joint.



Figure 555. Lubrication point in fluidics drawer (tilting base joints)

Put grease at the end of the latch.

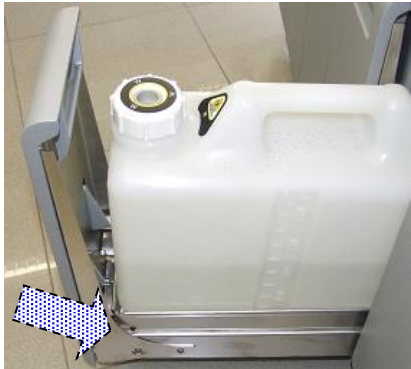
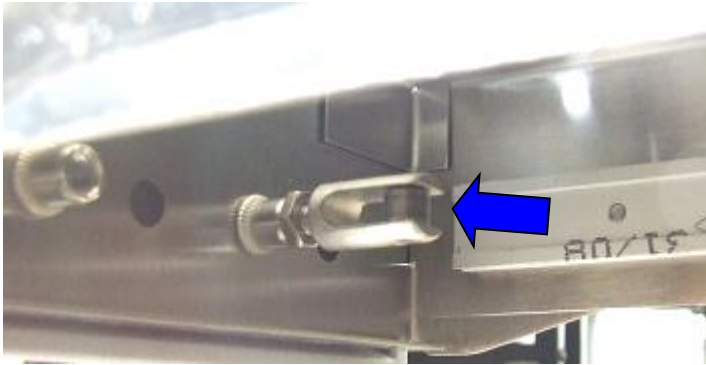


Figure 556. Lubrication point in fluidics drawer (latch)



If the linear guide of the drawer is cleaned, put some grease in the front and rear part.

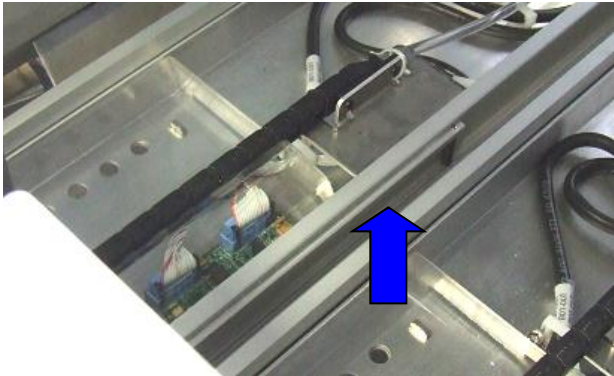


Figure 557. Lubrication point in fluidics drawer (linear guide)

14.4 REPORT OF CORRECTIVE MAINTENANCE SERVICES

Model: _____ Serial Number: _____ Installation date: _____

MAINTENANCE SERVICES CARRIED OUT			
DATE	WORK CARRIED OUT AND RESULTS	TECHNICIAN / OPERATOR	TECHNICIAN / RESPONSIBLE AUTHORITY

Table 92.Maintenance services list

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	538 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

14.5 HOW TO REPORT

The necessary information needed to analyze different incidences occurring during the daily routine can be compiled using the Support Pack feature, which allows the creation of a zipped file with all the data required for support. The generated data will not contain patient data in order to meet the General Data Protection Regulation (GDPR) compliance, and will be required to be stored in a USB drive, using 7-Zip software for its compression.

Further information on how to create a Support Pack and the generated contents is detailed in section 15.17.

Print date:	16-May-2023	Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization	Purpose:	Uncontrolled Copy
-------------	-------------	--	----------	-------------------

15 ADDITIONAL INFORMATION

This chapter provides information on several procedures such as Operating System and software installation, update the firmware of the PCAs, touchscreen drivers installation, LIS configuration, a brief description on how to program macros and a description of the log files.

15.1 Guidelines on error codes location

User error messages include an error code on the top right corner of the error window.



Figure 558. Example of user error message

This identification is generated as *MMDDEEESSS*, where *MM* is the module, *DD* is the device and *EEESSS* is the error. Reader module errors are generated as a 5-digits error code and LIS errors are generated as a 7-digits error code.

In order to match the error code shown to the user with the error codes included in this Service Manual, we recommend performing a search with the last 6 digits in this document. In case of fluidic errors, it is recommended first to check the last 3 digits to check in which step of the fluidic procedures appears the error.

15.2 Restoring procedure



NOTE:

The Operating system used in Erytra software versions $\geq 4.3.0$ is not compatible with PC Embedded versions 1 and 2 (Kontron model), and should be installed in PC Embedded versions ≥ 3 (ASRock model).

Two restoration options are available:

- Full recovery: to restore all SSD partitions.
- OS recovery: to restore the *C:* drive contents and update/replace system files on *E:* drive.

Restauration can be done:

- From USB pen drive
- From Boot application; without the need of connecting a USB pen drive. This feature allows update or recovery of the OS remotely.

In both cases it is recommended to backup the most relevant data:

- with Boot menu procedure
- with
- manual backup



NOTE:

Windows license activation must be performed every time the OS is restored, otherwise, Windows license watermark will be displayed.

The Windows license watermark does not imply that the license distributed by Diagnostic Grifols is not valid. It simply states that the license could not be activated in this instrument.

15.2.1 From USB pen drive

15.2.1.1 Create a bootable device

To create a bootable USB drive, it is necessary:

- External computer with Windows 10 Operating System preferably (Windows 7 32/64-bit or Windows XP Operating Systems can also be used) and USB ports.
- Access to instrument OS and Software programs (*artifactory.grifols.com* or *DxNet*).
- Zip tool.
- A USB drive (16 GB minimum/recommended – 32 GB maximum) with FAT32 format.



CAUTION!

Scan the USB drive for viruses before connecting it to Erytra.

Follow the next steps to create a bootable USB drive:

1. Download Windows 10 v2.2.1 zip file in the external computer.
2. Connect the USB pen drive to the external PC.
3. Open the zip file using 7.zip program.
4. Copy the content of the zip file into the USB pen drive root directory.
5. If is required to install Beyond Trust, copy the latest version of Grifols Secure Remote Connectivity Agent installation file in the following path of the USB: 'USB:\BeyondTrust\'. The version to be installed must be the latest available.



NOTE:

If desired to install automatically Grifols Secure Remote Connectivity Tool during Operating System installation process, copy the agent to USB:\BeyondTrust folder. The version to be installed must be the latest available.

15.2.1.2 **Configure *Erytra* to boot from USB**

Before using the bootable USB drive, BIOS/UEFI of the computer must be configured to allow that the system boots from the USB drive.

**NOTE:**
It is necessary to connect a USB keyboard and a USB mouse to the instrument.

**CAUTION!**
Scan the USB drive for viruses before connecting it to Erytra.

1. Connect the bootable USB drive to *Erytra*.
2. Switch on the instrument and press *F11* key during the initialization of the computer to enter the UEFI boot device selection option.

**NOTE:**
A password to access the BIOS has been set.
When requested enter password: TSDGIH.

3. Select 'UEFI: USB' option among the available boot devices.

System will boot from the USB drive and the Operating System installer application will be opened. Full recovery or OS recovery can be selected:



Figure 559. Erytra Recovery USB software

If the USB drive is not displayed/detected, perform one of the following actions and go back to step 1.

- a. Enter OS Shell mode (or another mode) and check Windows can detect the USB drive.
 - b. Disconnect the instrument from the mains during 5 – 10 seconds.
 - c. Connect the USB drive to another USB port of the instrument.
4. Select an option:
 - Full recovery
 - Operating System recovery

15.2.2 From Boot application

This feature allows to update or recovery the OS remotely. The execution of this process saves the files on the SSD and then reboot automatically without user interaction. In addition, in order to restore the remote connection, this process also preserves the PC name and the active network configuration.



NOTE:
This process can only be executed on instruments with an OS version 2.2.1 installed.

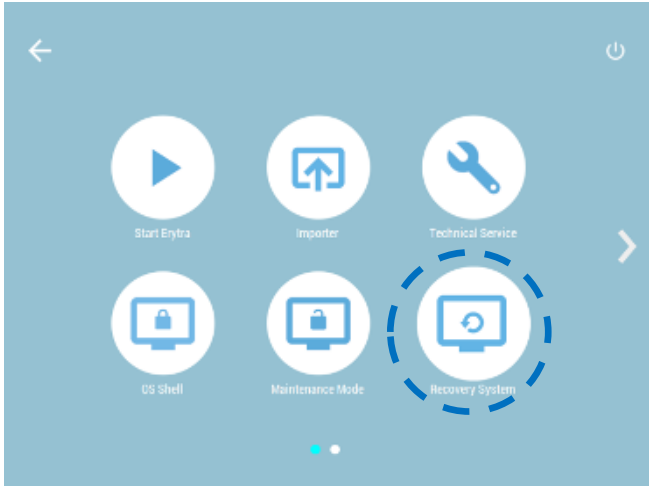


NOTE:
If the Recovery system is run remotely from Beyond Trust it is necessary to check before starting that the installer is present on the R:\ drive.

Grifols Secure Remote Connectivity tool agent will now be automatically installed during OS Restore or OS Full Restore process when the agent is copied in a folder named BeyondTrust in the root of the unit letter containing the OS image (that is USB:\ or R:\).

To restore the system from this option, follow the steps below:

- 1. Select Recovery option from Boot application

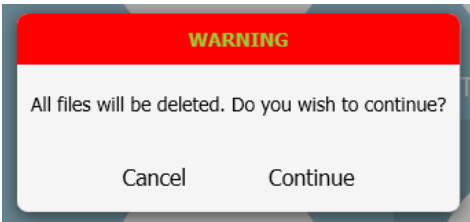


- 2. Once *Recovery System* option is selected, a window asking to choose the desired option will be displayed:

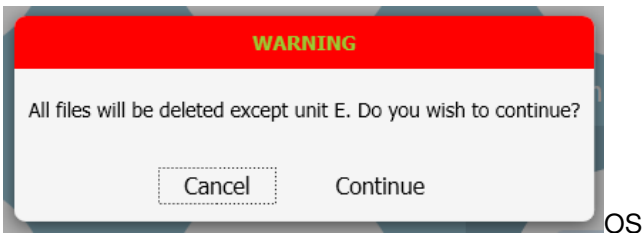


3. A warning and confirmation window will be displayed when selecting:

- Full recovery



- Operating System recovery



Once the action is confirmed, the system will proceed.

**NOTE:**
Once this action is confirmed, it is not possible to stop the process.

15.2.3 Operating System recovery

In case it is only needed to restore the *C:* drive contents and update/replace system files on *E:* drive, it is possible to use the *OS Recovery* option in *Erytra Recovery USB* application.

This procedure will not delete user data, nevertheless, it may be advisable to create a backup.

1. In the menu, select *OS RECOVERY* and press *Enter* in the keyboard.

**CAUTION!**
Do not select *FULL RECOVERY* as with this option the SSD image (C and D drive) is fully restored. Previous information in the SSD is deleted.

2. After some minutes, the process finishes and the computer is automatically switched off.
3. Remove the USB drive and switch on the instrument.
4. Once PC reboots, system configuration continues and AppLocker application in configured. Computer will be rebooted automatically.
5. Once PC reboots, configuration is completed and Boot Application initial window is displayed.



Figure 560. Boot Application initial window

1. Press the grid () button. The Boot Application login window will be displayed.

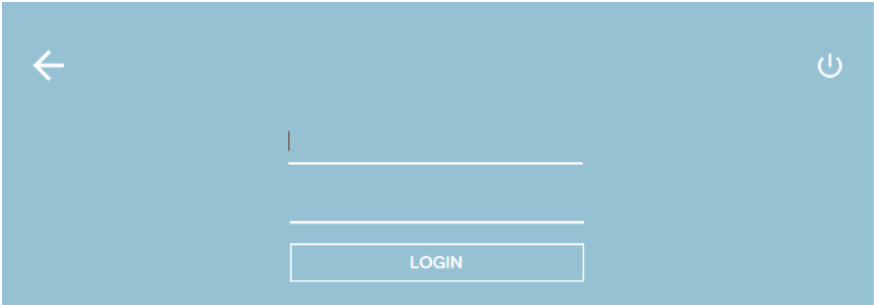


Figure 561. Boot Application login window

2. Log in as *adminUser* (case sensitive, *admin* password) and the Boot Application options will be displayed.
3. Select the *Information* option and check that the installed O.S. version is the expected one.
4. Install *Erytra* software (section 15.3).
5. Restore Diana System configuration in Windows registry, if necessary.
6. For each particular instrument, restore those configurations that have been erased due to the *C:* drive restoring process (i.e. date and time, peripheral installation, touchscreen calibration, etc.)

15.2.4 Full recovery

This option restores the complete hard disk image. All previous information in the hard disk is deleted.

The following steps describe a full restore procedure:



NOTE:
Before restarting the instrument to boot from the USB drive, disconnect any other USB drive/hardisk from the instrument. Also disconnect the instrument from the network (if connected).

1. In the menu, select *FULL RECOVERY* and press Enter in the keyboard.
 - a. The selection menu will be removed and GRIFOLS will remain on screen two minutes approximately.
 - b. The integrity check of files will be performed (it will last 2 minutes approximately).
 - c. The relevant data files backup (instrument adjustment files, configuration files, system tools,..) will be created.
 - d. The contents of each partition will be copied to the instrument SSD.
 - e. The relevant data files backup (instrument adjustment files, configuration files, system tools,..) is restored to instrument SSD.
 - f. The OS image is copied to Recovery partition and the integrity check of the files (that are not inside a compressed file) copied to the SSD recovery partitions is performed.



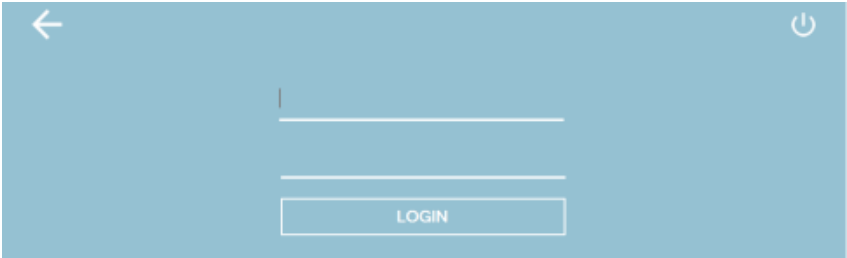
CAUTION!
Full Recovery is mandatory when installed OS v2.2.1 for the first time

- 1 After some minutes, the process finishes and, the computer will reboot automatically.
If computer hasn't switch off automatically after 1 hour, check PC wiring, switch it off manually using the instrument button .
 - 2 Remove USB pen drive, if the recovery action is performed from USB pen drive.
 - 3 Once PC reboots, OS installation continues (installing devices,...). The message *Getting ready* is displayed (screen will also remain black and showing no message for some seconds). Computer will be rebooted automatically.
If computer hasn't reboot automatically after 1 hour, check PC wiring and reboot it manually using the instrument button .
- If installation process doesn't continue and/or it is corrupted, go back and install the Operating System again.

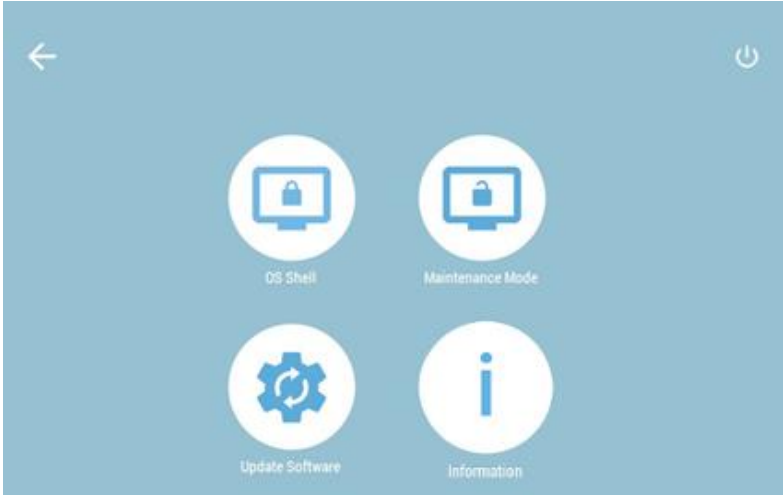
- 4 Once PC boots, OS installation and configuration continues. The message *Getting ready* is displayed (screen will also remain black and showing no message for some seconds).
Computer will be rebooted automatically.
If computer hasn't reboot automatically after 1 hour, check PC wiring and reboot it manually using the instrument button .
If installation process doesn't continue and/or it is corrupted, go back and install the Operating System again.
- 5 Once PC boots, system configuration continues, AppLocker is configured.
Computer will be rebooted automatically.
If computer hasn't reboot automatically after 1 hour, check PC wiring and reboot it manually using the instrument button .
If installation process doesn't continue and/or it is corrupted, and install the Operating System again.
- 6 Once PC reboots, system configuration continues to allow Beyond Trust installation.
Computer will be rebooted automatically.
If computer hasn't reboot automatically after 1 hour, check PC wiring and reboot it manually using the instrument button .
If installation process doesn't continue and/or it is corrupted, go back and install the Operating System again.
- 7 Once PC reboots, system configuration continues and Beyond Trust application is configured.
Computer will be rebooted automatically.
If computer hasn't reboot automatically after 1 hour, check PC wiring and reboot it manually using the instrument button .
If installation process doesn't continue and/or it is corrupted, go back and install the Operating System again.
- 8 Once PC reboots, configuration is completed, and Boot Application initial window is displayed.



- 9 Click on bottom-left corner/grid button . Boot Application login window will be displayed.



- 10 Log in as adminUser (it is case sensitive). Boot Application options will be displayed.



- 11 Select *Information*  option and check that O.S. version installed is 2.2.1.

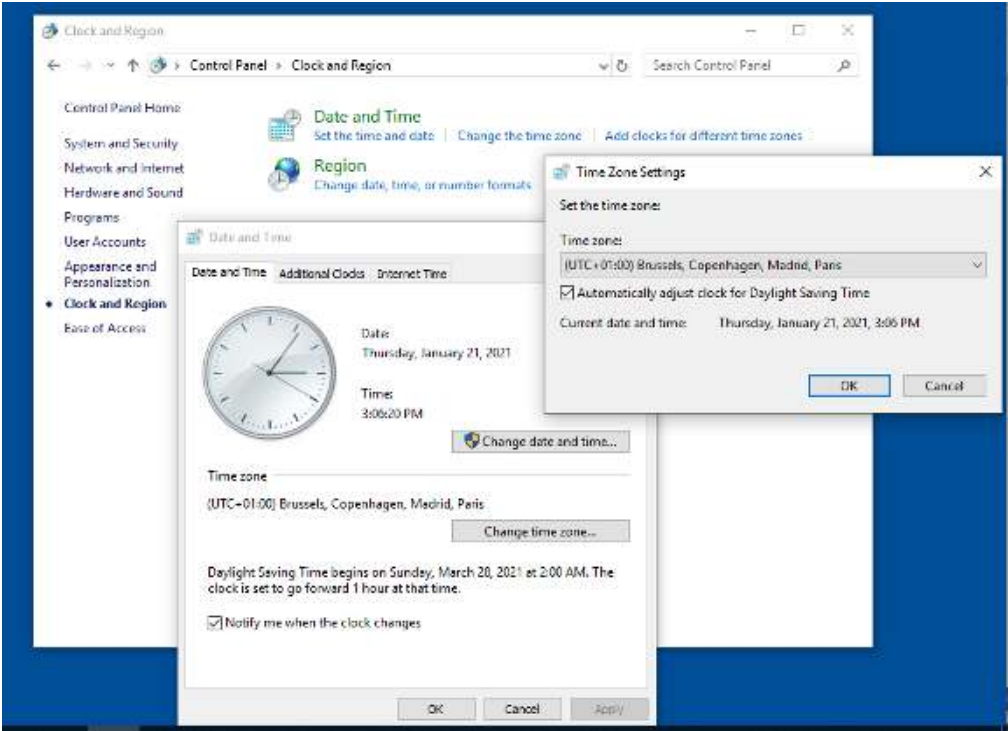


- 12 Close information window.

15.2.5 Restoring configurations after a full recovery

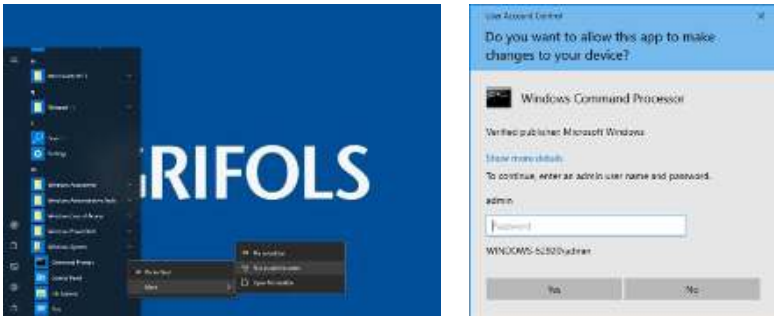


- 1 Select *Maintenance Mode* option.
- 2 Once PC reboots and desktop is available, set:
- The appropriate Time Zone.
How to: *Control Panel – Clock and Region – Date and Time – Change Time zone...*



Keep ticked *Automatically adjust clock for Daylight Saving Time* option.
Untick the option *Notify me when the clock changes*, if displayed.

- Date and Time
How to: *Control Panel – Clock and Region – Date and Time – Change Date and Time....*
- 3 Restore Diana System configuration in Windows registry (Import the backup of Windows 10 *HKEY_LOCAL_MACHINE\SOFTWARE\WOW6432Node\Grifols* or the backup of WES7 *HKEY_LOCAL_MACHINE\SOFTWARE\Grifols*), if necessary. To do so:
- Run *Command Prompt* as Administrator.
How to: *Start → Windows System*
Click right mouse button on *Command Prompt* and select *more > Run as administrator*.
Enter admin user password



- Enter following command (replacing *E:\config.reg* with the location and name of the registry backup file to restore):
C:\Windows\SysWOW64\regedit.exe E:\config.reg
- Confirmation window will be displayed. Press *Yes* button (window will close).
- A message window informing that the values have been successfully added to the registry will be displayed. Press *OK* (window will close).
- Close the *Command Prompt* window.

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	549 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

- 4 If LIS connection is enabled and communication mode is set to serial port, review/check (and configure, if necessary) that the current COM port selected is the one connected to LIS (that COM port name is the same after installing new operating system).
- 5 If LIS connection is enabled and communication mode is set to TCP/IP, create a new outbound firewall rule to open the selected port as indicated in section 15.19.4
- 6 If LIS connection is enabled, review/configure LIS encoding to match current LIS encoding.
Note: HostDemo program uses (only) ANSI encoding.
- 7 If LIS connection is enabled and characters different from Latin alphabet ones are used, enter *Maintenance mode* and set the suitable language in the *Language for non-Unicode programs* option (*Start* → *Control Panel* → *Clock and Region* → *Region* → *Change date, time or number formats* → *Administrative tab*. Administrator password will be requested):
Note: *Language for non-Unicode programs* configuration of DianaServer PC, when not running in instrument PC, shall match Erytra configuration.
- 8 If necessary for backup feature or LIS connection, configure Erytra to share or access a shared folder. as indicated in section 0.
Remark: R and V drives now belong to instrument SSD.
- 9 Restore the backup/set the PC name and network configuration
- 10 Shut down Erytra®.

Print date:	16-May-2023	Purpose:	Uncontrolled Copy
--------------------	-------------	-----------------	-------------------

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization

15.2.6 Activate Windows License

Activate Window license shall be performed every time the Operating System is installed (OS Recovery or Full Recovery), or the computer has some hardware components modifications, such as SSD, CPU or motherboard and when the instrument is connected to a network.



NOTE:

Windows license activation must be performed if the instrument is connected to a network with no access to Microsoft license servers. Otherwise, Windows license watermark will be displayed.

The Windows license watermark does not imply that the license distributed by Diagnostic Grifols is not valid. It simply states that the license could not be activated in this instrument.

This section provides two available methods to activate Windows License. Active Windows License using the method that matches installation characteristics.

Activate Windows License by phone



NOTE:

It is not necessary to be connected to a network.



NOTE:

This activation procedure is detailed for the 'United States' region.

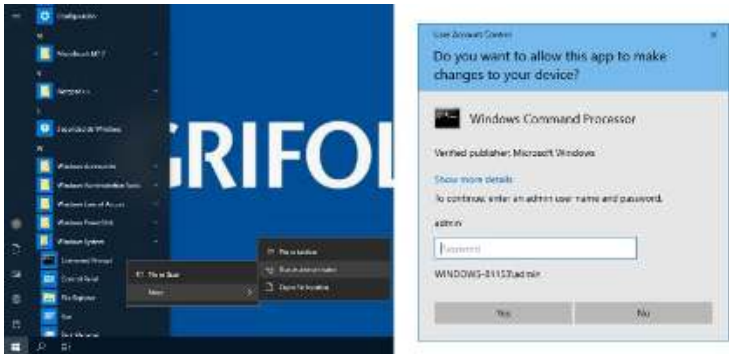
The region selection may be changed based on language preferences. In this case, the detail of the instructions may change.

- 6. Open Maintenance mode  option.
- 7. Open CMD as Administratos.

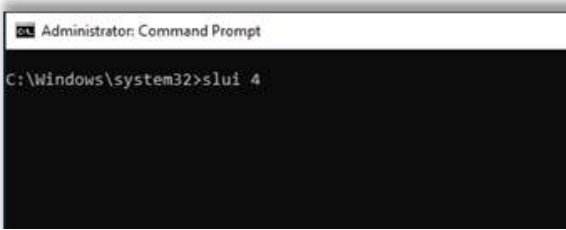
How to: Start → Windows Administrative Tools

Click right mouse button on Command Prompt and select more > Run as administrator.

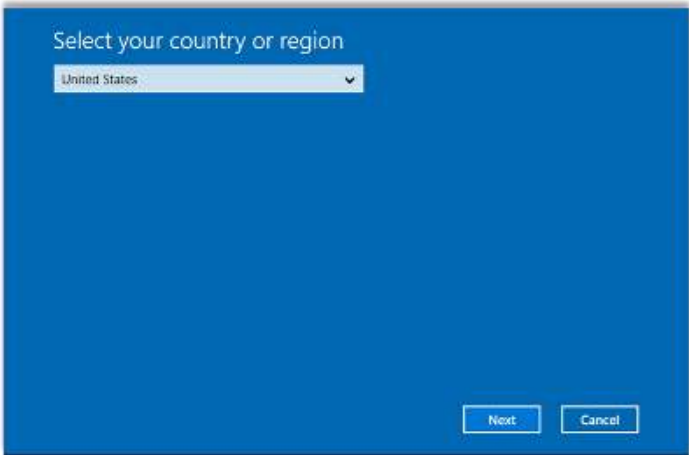
Enter admin user password



8. Type the following command: slui 4



9. Select region: <instrument region>.



5. Call the number and follow instructions to activate Windows 10.

Voice recorded message:	'Product activation center' welcome message. Consent about the call recording. Press '1' to consent. Press '2' to do not consent.
5.1.	Press phone key '2'.
Voice recorded message:	Introduce phone code for security purposes. Random security code of 3 digits is provided.
5.2.	Enter the provided random security code.
Voice recorded message:	Welcome to 'Microsoft Product Activation'. If you are trying to activate Windows 10, press '1'. Otherwise press '2'.
5.3.	Press phone key '1'.
Voice recorded message:	Request about product upgrade. If you previously upgraded to Windows 10, press '1'. Otherwise press '2'.
5.4.	Press phone key '2'.
Voice recorded message:	If you got an error code, press '1'. Otherwise press '2'.
5.5.	Press phone key '2'.
Voice recorded message:	Confirmation about you are with the activation screen open. Say 'Yes' or 'No'.
5.6.	Say 'Yes'.

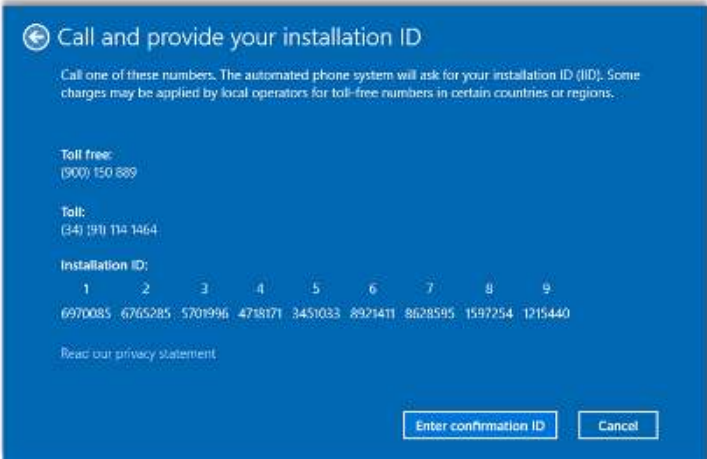
Voice recorded message:	Explanation about the 'Installation ID' format. Long number split in groups. It will request one group at a time.
-------------------------	---

5.7. Enter 'Installation ID' by groups with phone keyboard.

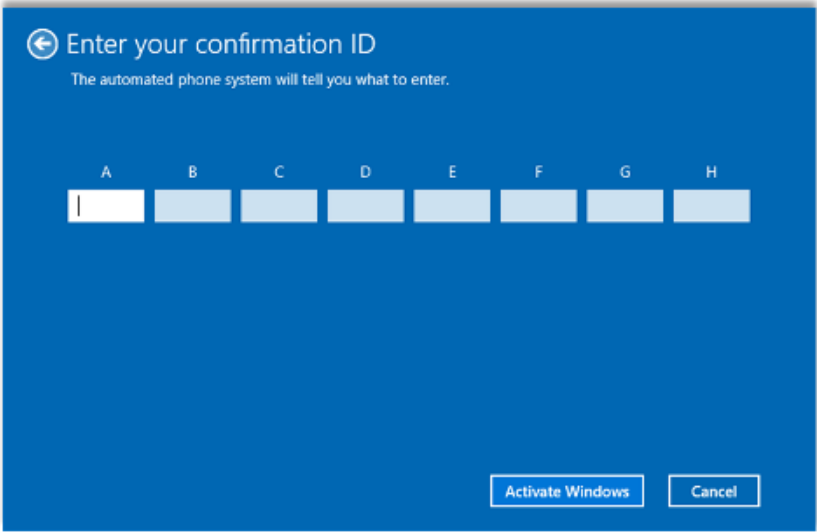
**NOTE:**
Installation ID will change on each license activation.

**NOTE:**
Introduce one group at a time and wait for the license activation assistant before introducing a new one.

Voice recorded message:	Request about waiting to look up the provided 'Installation ID'. Confirmation of verified status.
-------------------------	---



5.8. Select 'Enter confirmation ID' option.
'Enter your confirmation ID' window is displayed.

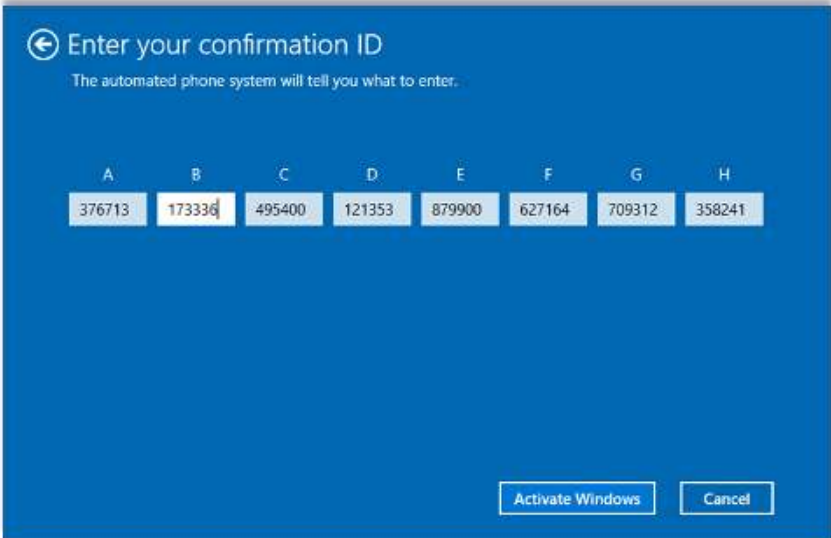


Voice recorded message:	Assistant will read the 'Confirmation ID'. Use your PC keyboard to introduce the provided ID in the empty blocks.
-------------------------	--

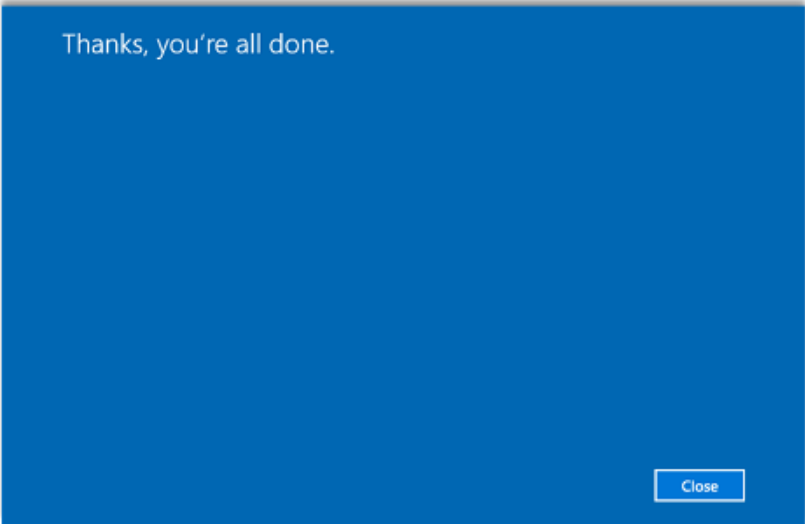
	Assistant will read one block of 6 digits at a time. You will have the opportunity to listen again the block at the end. Say 'Repeat' at the end, if needed. Say 'Yes' if you are ready to listen the first block.		
--	---	--	--

5.9. Enter the confirmation numbers provided by phone.

**NOTE:**
'Confirmation ID' will change on each license activation.
The screenshot is just an example..



6. Select 'Activate Windows' option.
'Thanks you're all done' window is displayed.



7. Select 'Close' option.
Operating System license activation is performed.
8. Shut down Erytra instrument.

Activate Windows License by accessing to Microsoft license servers

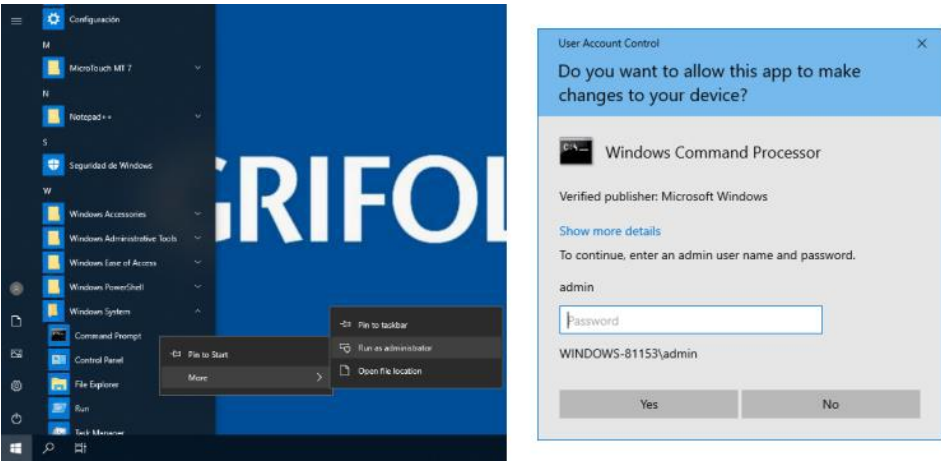


NOTE:
'Confirmation ID' will change on each license activation.
The screenshot is just an example..

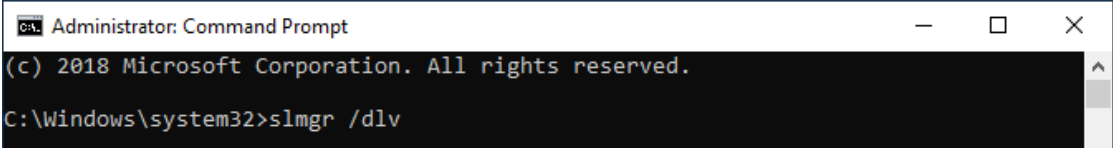


NOTE:
This procedure requires coordination with IT department to enable this connection. These ports that are required for Windows Activation: TCP port 80, TCP port 443.

- 
1. Open Maintenance Mode option.
 2. Connect Erytra PC to network with Internet connection.
 3. Open CMD as Administratos.
How to: Start → Windows Administrative Tools
 Click right mouse button on Command Prompt and select more > Run as administrator.
 Enter admin user password

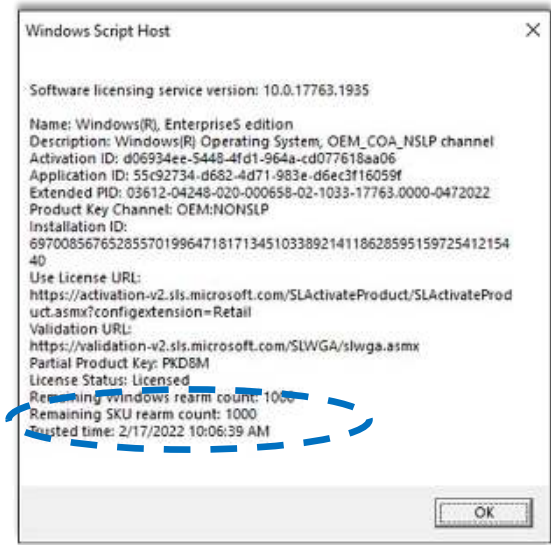


4. Type following command: slmgr /dlv



This command verifies that the activation of the Windows licenses is performed correctly.

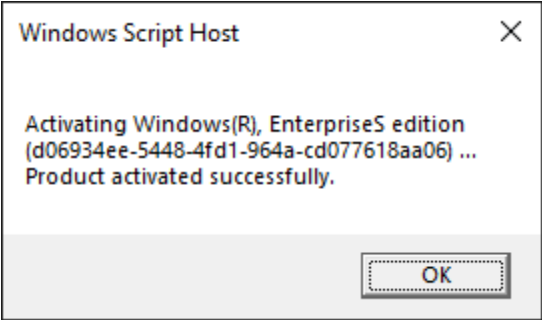
5. The following window is displayed.



6. Check that *License Status* field status is *Licensed*. Should this not be the case, wait 15 minutes and check the status again.

If the status does not change to Licensed execute following command to prompt Windows to try online activation (at that time): `slmgr /ato`.

The following window should be displayed:



Press OK and then check again License Status field.

If an activation error displayed and if the status does not change to Licensed go to section 4.6.1 and activate Windows by phone.

7. Shut down Erytra instrument.

15.2.7 OS and Software downgrade

Operating System version 2.2.1 includes a new system partition that it is not compatible with Operating Systems version v2.1.0 and below.

To be able to install a previous Operating System, version 2.2.1, SSD partitions shall be removed using a third-party tool.

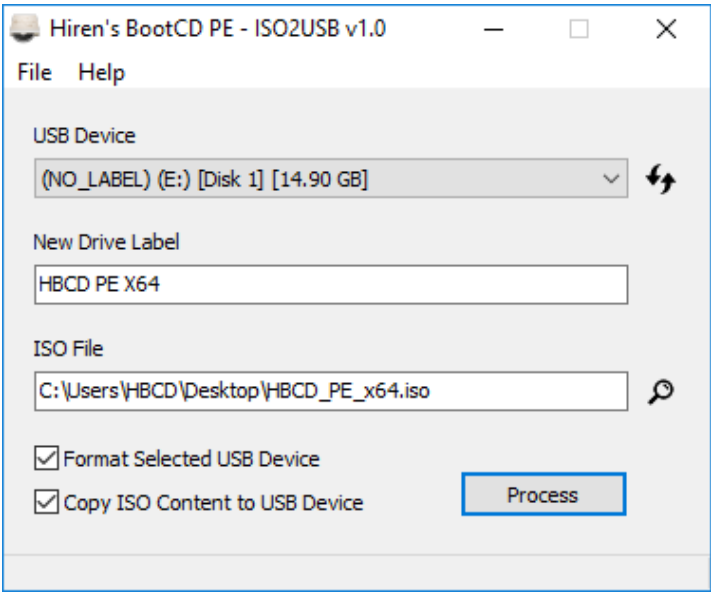
The recommended third-party tool to remove unit partitions is *Hiren’s Boot CD PE Windows 10*.

To install an Operating System version < 2.2.1 in an Erytra with Operating System version 2.2.1, proceed as follows:

Create Hiren’s bootable USB

**NOTE:**
You need Administrator privileges to run this process.

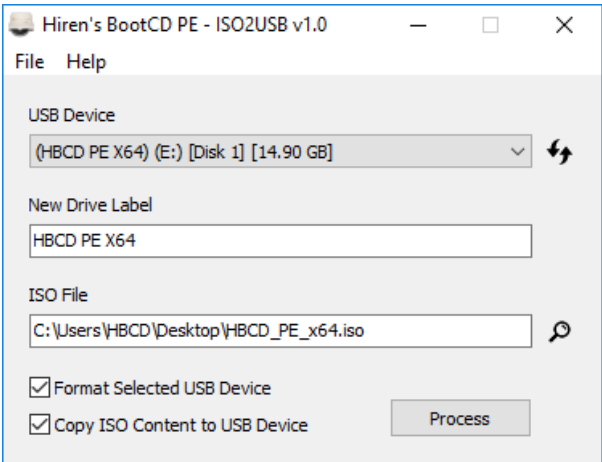
- Download the following files in a local path of an external computer:
 - ISO2USB.exe (Application to create Hiren’s bootable USB)
 - HBCD_PE_x64.iso (ISO file of Hiren’s system)
- Execute the file *ISO2USB.exe*. *Hiren’s BootCD PE - ISO2USB v1.0* window is displayed.



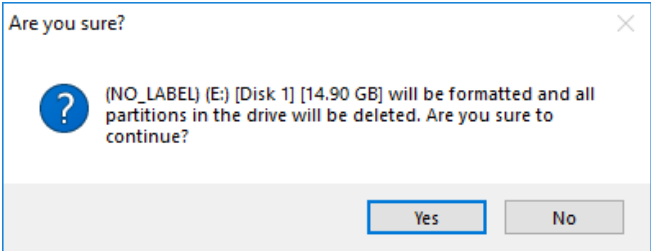
**NOTE:**
Only USB flash drives will be listed in the *USB Device* list.

3. Configure the parameters as follows:
- New Drive Label Set a label for the USB Drive up to 11 characters.

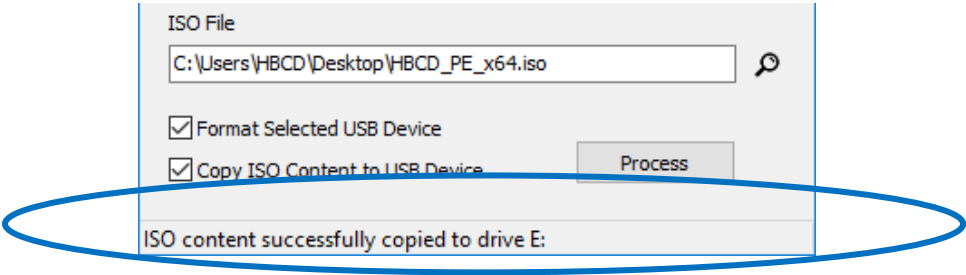
The characters must be alphanumeric and
uppercased.
 - ISO File HBCD_PE_x64.iso
 - Format Selected USB Device Checked
 - Copy ISO Content Checked



4. Select *Process* option. Following confirmation window will be displayed.



5. Select 'Yes' option.
The process will format the USB Flash Drive, remove the partitions and data in the drive.
The Hiren's ISO image will be copied in the USB Flash Drive.
6. Wait until process is finished. The message ISO content successfully copied to drive E: will be displayed.



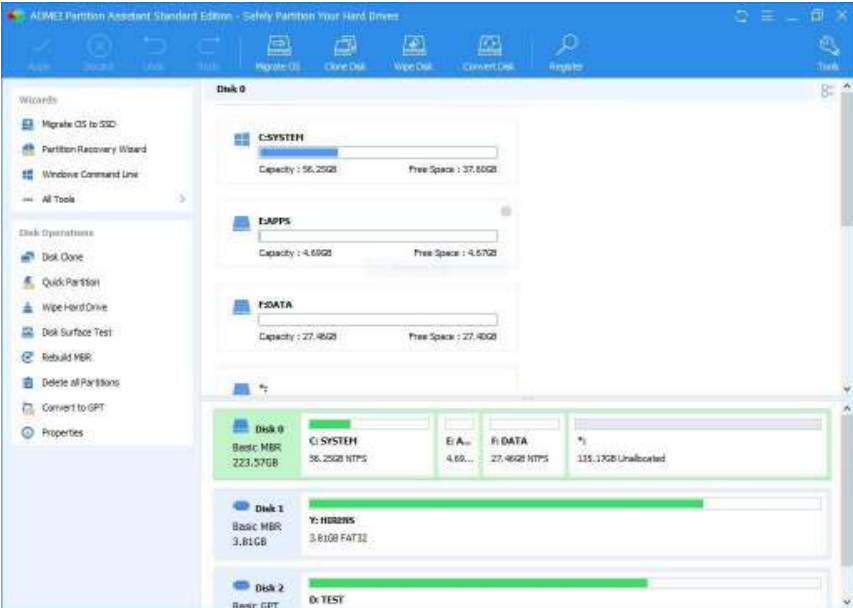
7. Close *Hiren's BootCD PE - ISO2USB v1.0*.

Delete instrument SSD partitions

- 8. Connect bootable USB with *Hiren's BootCD PE Windows 10* in the instrument computer.
- 9. Switch on the instrument and press F11 key during the initialization of the computer to enter the UEFI boot device selection option. Enter password, if required.
BIOS Boot mode window is displayed.

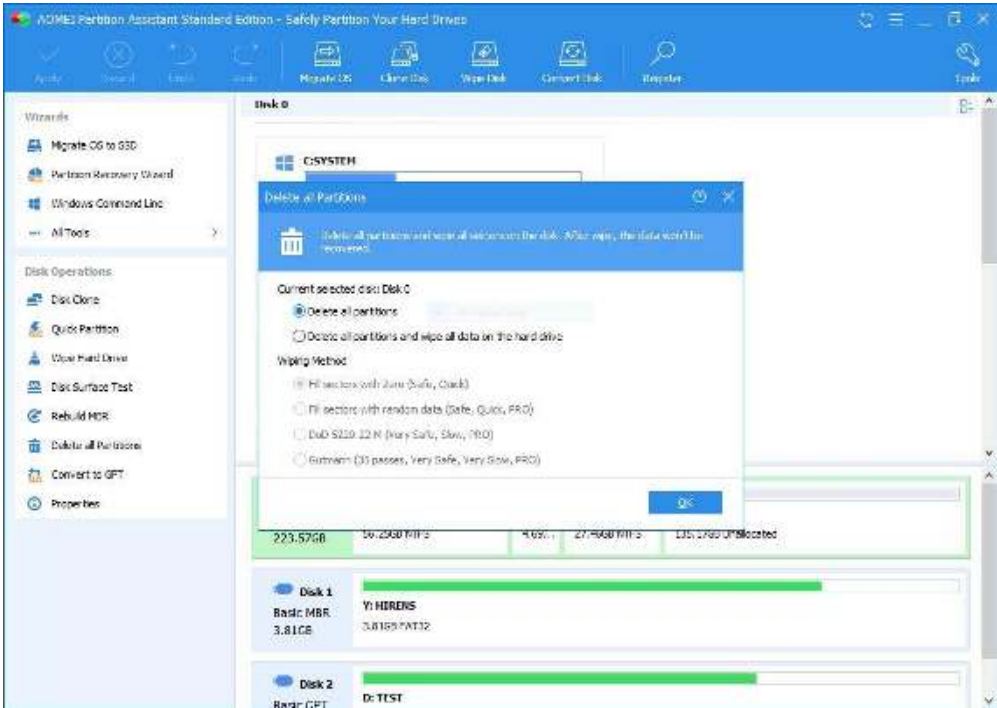


- 10. Select *UEFI: USB* drive name option.
- 11. Wait until Hiren's starts.
- 12. Select Start Windows.
- 13. Select *All Programs – Hard Disk Tools – Partition Tools*.
- 14. Open program: *AOMEI Partition Assistant*.
- 15. Select 'Disk 0'.

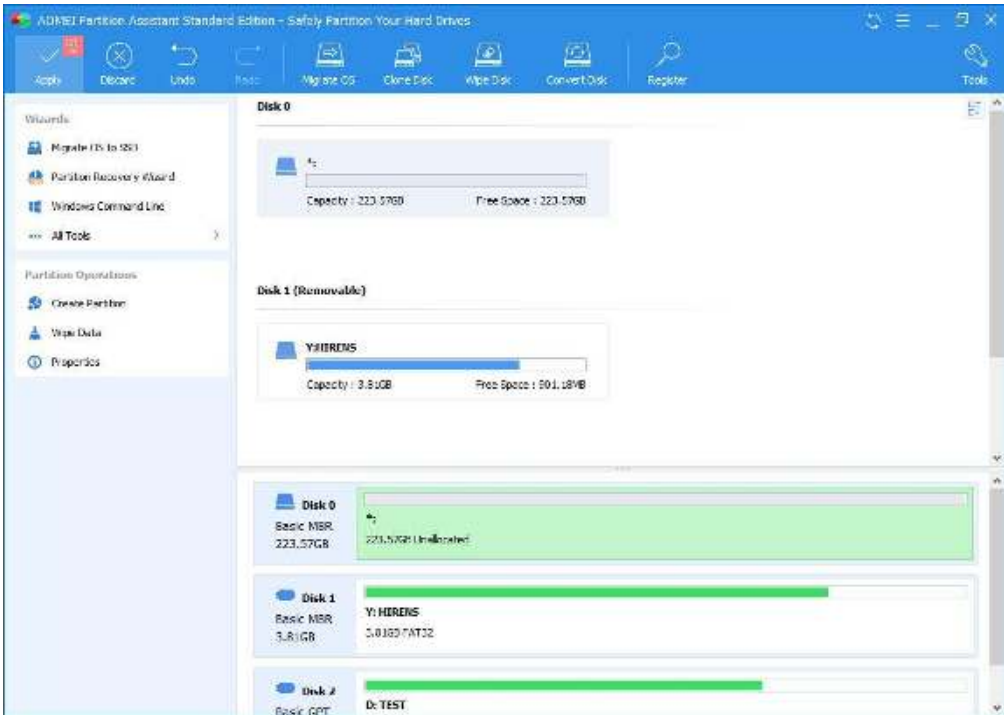


- 16. Click mouse right button and select *Delete all Partitions*.

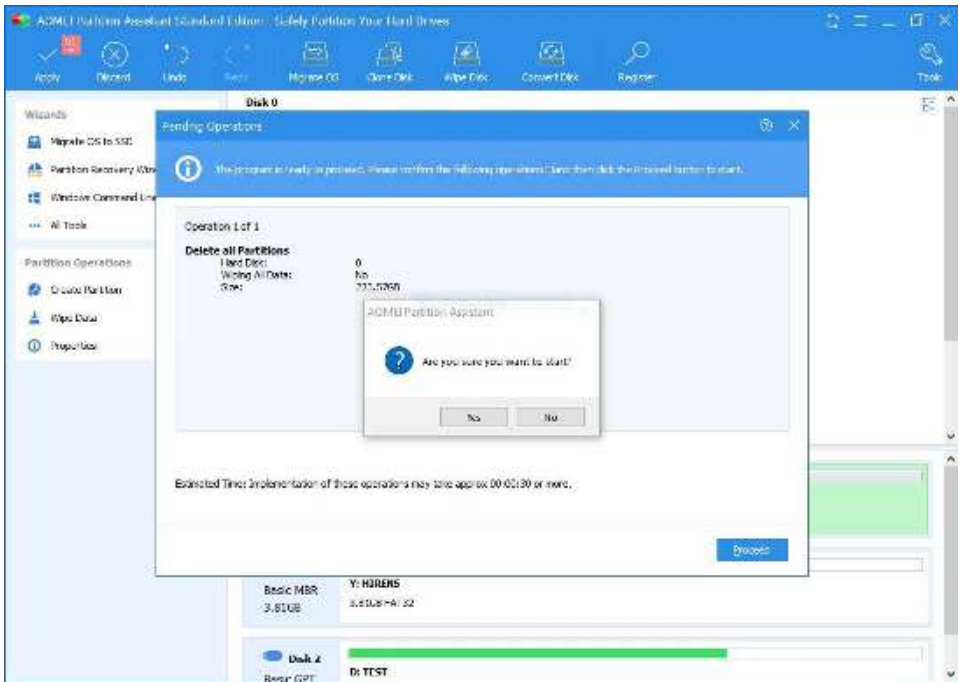
17. Confirm with *OK*.



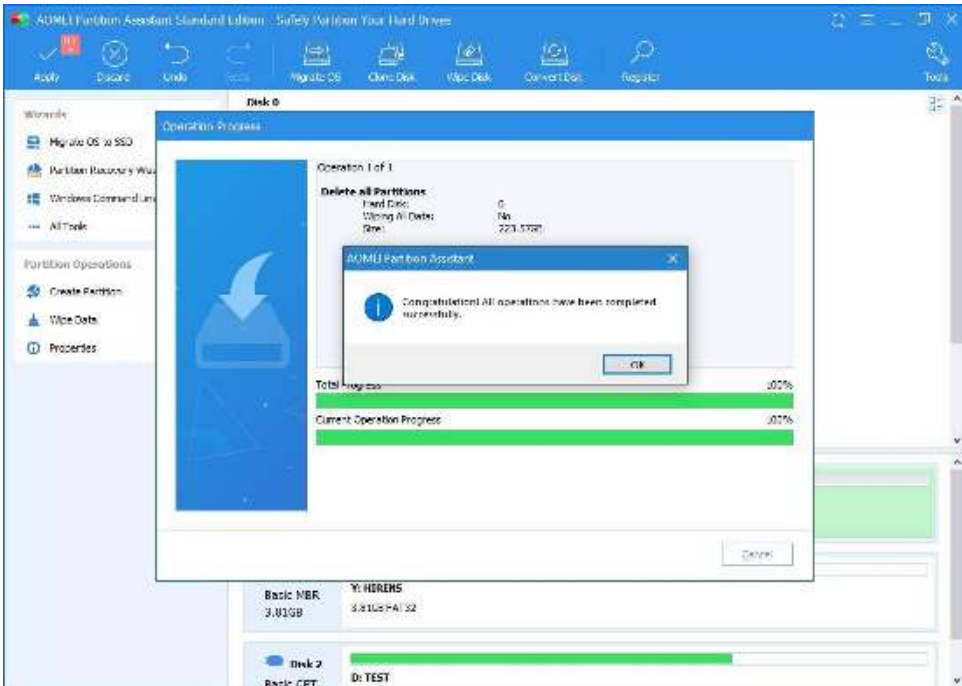
18. Select *Apply* option from changes and restart.



19. Select *Proceed* and *Yes* to start process.



20. Confirmation window is displayed as follows:
Congratulations! All operations have been completed successfully.



Select *Ok* option. Confirmation windows is closed.

- 21. Restart the system.
- 22. At this point, previous Operating System version 2.2.1 can be installed to complete the downgrade.

15.3 Erytra software installation



NOTE:

On this section the general steps for a software installation are described. Depending on the version previously installed and the version that it is going to be installed, some additional steps may be needed (check relevant INR/FSB).

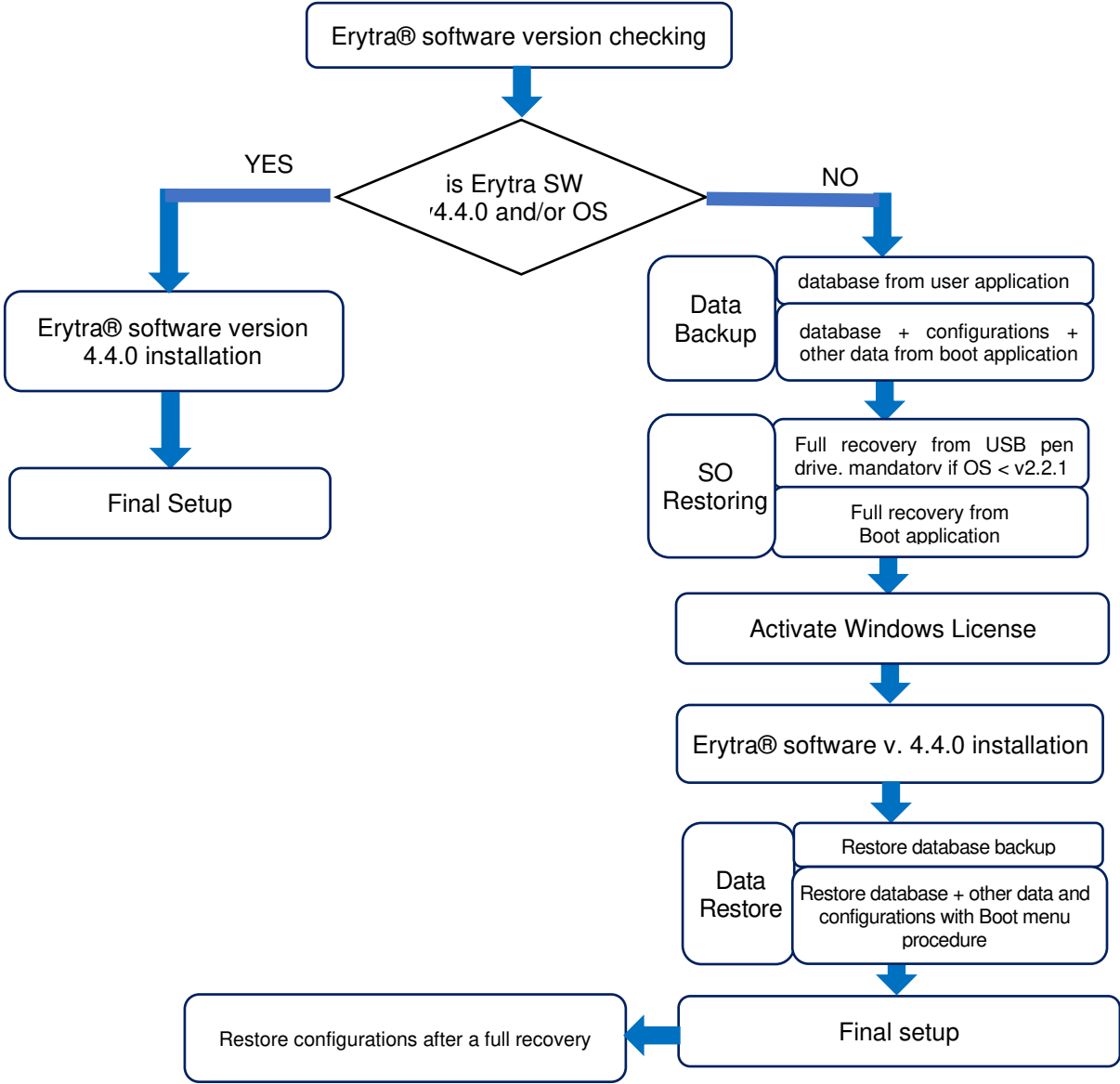


NOTE:

Erytra® software version 4.4.0 requires the following Reagents Pack versions or higher:

- IH Reagents Pack AU v1.5.
- IH Reagents Pack CE v9.0.
- IH Reagents Pack US v2.3..

The following workflow details the procedure to upgrade a properly installed Erytra® software version 4.4.0:



15.3.1 Installation steps

- 1 Switch on the instrument and press the grid () button on the Boot Application initial window. Boot Application login window will be displayed.

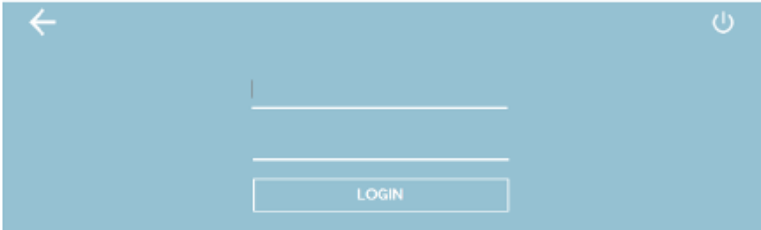


Figure 562 Boot Application login window

- 2 Log in as adminUser (it is case sensitive). Boot Application options will be displayed. Select OS

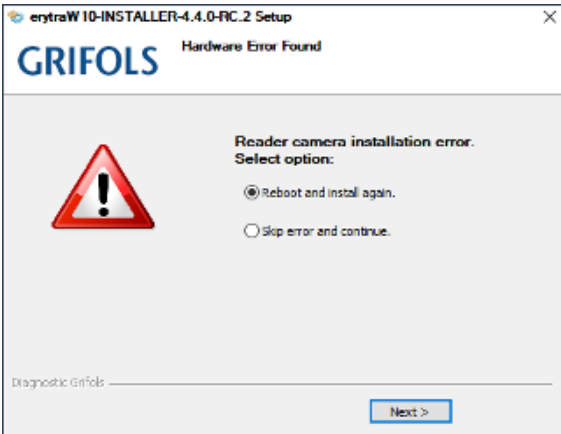


Shell option.Copy *erytraW10-INSTALLER-4.4.0.exe* to an USB drive using an external computer.

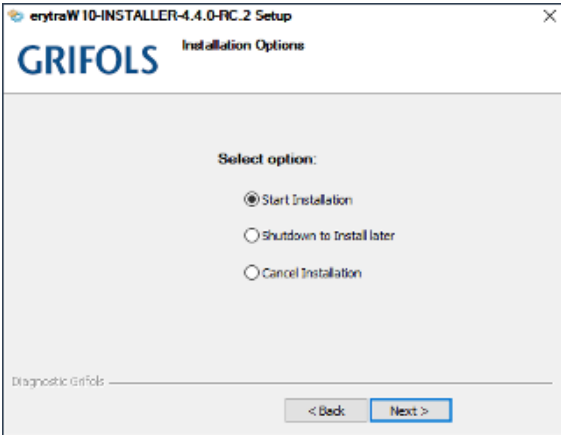
- 3 Connect the USB drive to Erytra PC. It is recommended to use USB 3.0 ports (rear part of the instrument).
- 4 Copy *erytraW10-INSTALLER-4.4.0.exe* file to instrument *E:\update* directory.
- 5 Restart Erytra.
- 6 System will unprotect the volumes, reboot, open AppLocker audit session and run automatically the installer (from E drive). Welcome screen will be displayed:



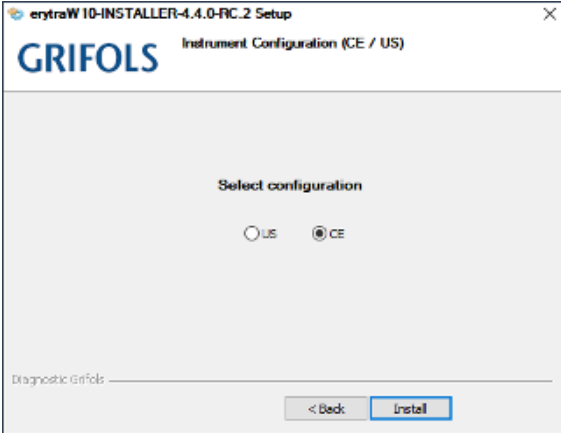
- 7 Click on *Next >*.
- 8 If *Hardware Error Found* window is displayed, check PC BIOS version (refer to FSB-0059#0032), check firewire PCA and camera and retry software installation.



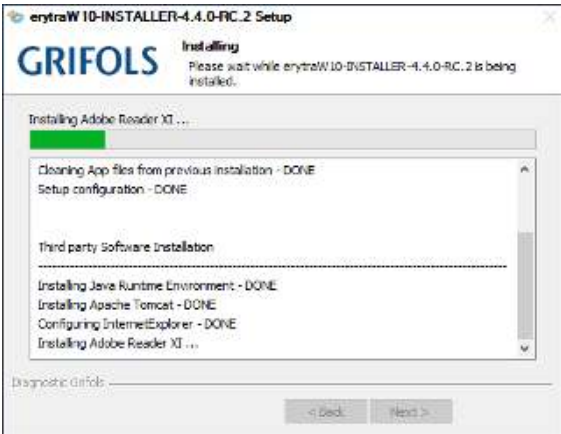
9 Keep *Start Installation* option selected and click on *Next >*.



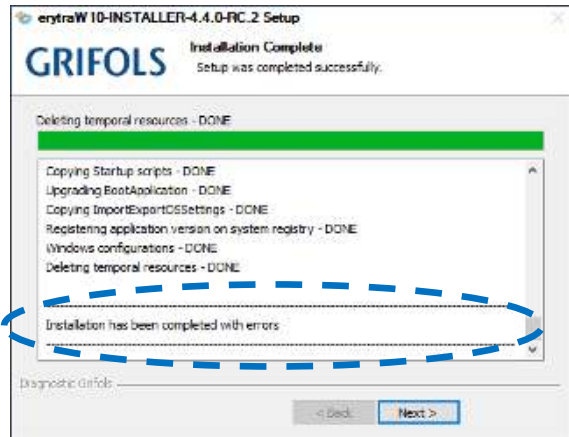
10 Select desired configuration (CE or US) and press *Install*.



11 A window informing about installation progress will be displayed:



- 12 Once completed, review the result and proceed accordingly:
- a. *DONE* is displayed for all the steps. When all steps are *DONE* and the message *Installation has been completed with errors* is not displayed, click on *Next >*.
 - b. *FAILED* is displayed in any step. When a step is *FAILED* and the message *Installation has been completed with errors* is displayed, click on *Next >*, continue until step 19 and then go back to step 3 (retry software installation).



- 13 Following window indicating that installation will be completed will be displayed.



- 14 Click on *Finish*. System will close AppLocker audit session and apply new rules, protect the volumes and restart.

- 15 Once PC reboots, click on bottom-left corner grid  button so Boot Application login window is displayed.



- 16 Log in as adminUser (it is case sensitive). Boot Application options will be displayed. If in step 15 the message *Installation has been completed with errors* was displayed and/or *FAILED* was displayed for a given installation step, go back to step 6 (retry software installation).

15.3.2 Final setup

- During the restart of the instrument, press the grid  button during system startup (login credentials will be required), then select the *OS Shell* icon among the Boot Application options.
- If the LIS nomenclature selected is Erytra, copy DSC.def file from *E:\data\Erytra\Wadiana* to Diana System Defs directory (default directory is *E:\data\SistemaDiana\Defs*).
- Check in the related software version FSB/INR whether any firmware must be installed or additional configuration should be applied.
- If the related software version FSB/INR requires the use of *Importer* application, restart the instrument and press the grid  button during system startup (login credentials will be required), then select the *Importer* icon among the Boot Application options.
 - Import Reagent Language Pack and Erytra Language Pack (although the latter is automatically installed by the Erytra software installer, a more recent Erytra Language Pack version may be preferred instead). Then enable the desired profiles.
 - Install the desired Reagent Pack following the instructions provided in the corresponding FSB/INR.
- Close *Importer* application and launch *Erytra* user software and check software version displayed when clicking on upper-left area (User login information and instrument name area) is the expected one.
- Check the instrument is properly levelled (and perform instrument levelling, if necessary).
- Perform following adjustments:
 - Reader adjustments – Camera.
 - Gripper adjustment in Card Reader area



NOTE:

Camera adjustment shall be performed the first time the SW is installed to guarantee that the instrument conversion factor is within the SW 4.4.0 allowed range.

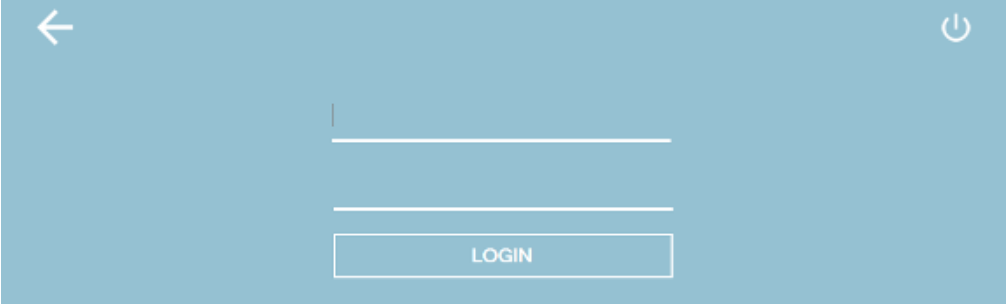


NOTE:

Gripper adjustment in Card Reader area shall be performed as Card Reader adjustment has been performed.

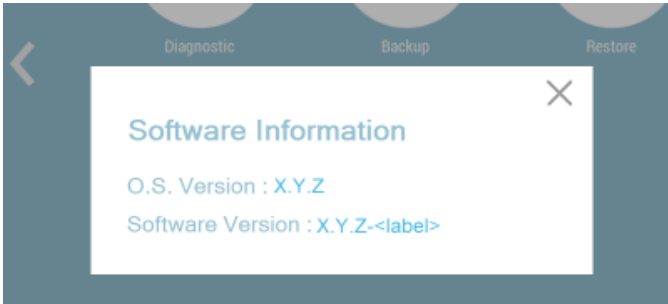
15.3.3 Software version verification

- 1 Switch on the instrument.
- 2 Click on bottom-left corner grid button . Boot Application login window will be displayed.



- 3 Log in as adminUser (it is case sensitive). Boot Application options will be displayed.

- 4 Select Information  option.



- 5 Check that Erytra Eflexis software version is v4.4.0. Should this not be the case, go back to section 15.3.
- 6 Close information window.

15.4 Windows Security Updates

The security packages included in OS image v2.2.1 up to December 2022 (Windows version 10.0.17763.3770)

Name	Description
KB5020866	December 13, 2022 Cumulative Update for .NET Framework 3.5 and 4.7.2
KB4512577	2019-09 Servicing Stack Update for Windows 10 Version 1809 for x64-based Systems.
KB4523204	2019-11 Servicing Stack Update for Windows 10 Version 1809 for x64-based Systems.
KB4562562	2020-06 Servicing Stack Update for Windows 10 Version 1809 for x64-based Systems.
KB5003243	2021-05 Servicing Stack Update for Windows 10 Version 1809 for x64-based Systems.
KB5005112	2021-08 Servicing Stack Update for Windows 10 Version 1809 for x64-based Systems
KB5012170	2022-12 Security Update for Windows 10
KB5021237	2022-12 Cumulative Update for Windows 10 Version 1809 for x64-based Systems. OS Build 17763.3770
KB5009642	2022-01 Security Update for Windows 10
KB5020374	November 8, 2022 Cumulative Update for .NET Framework 4.8 for Windows 10

Table 93 Security Updates packages

Perform the following steps to check which updates are installed:

1. Open a "cmd" with admin rights and execute the following command:

wmic qfe get hotfixid

The installed security packages are displayed.

```

Administrator: Command Prompt

Microsoft Windows [Version 10.0.17763.3770]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\Windows\system32>wmic qfe get hotfixid
HotFixID
KB5020866
KB4512577
KB4523204
KB4562562
KB5003243
KB5005112
KB5012170
KB5021237
KB5009642
KB5020374
  
```

15.5 Access to Windows 10 File Explorer

This option allows qualified Field Service Engineers to access Windows File Explorer and corresponding taskbar, as well as additional functionalities, from Technical Service, Diagnostic application and User application modes.

**NOTE:**
EI SW 4.4.0 includes this functionality.
The executable file launched through INR-0059#0125 is incompatible with this SW version. Should both have been installed the application won't work.

The details on all the functionalities available are the following:

TECHNICAL SERVICE MODE

- Permanent access to Windows 10 File Explorer (taskbar always present at the bottom of the screen).
- Enabled right-click functionality when using the mouse on File Explorer.
- Enabled access to *E:* drive (DATA).
- Enabled shortcut keys, except for *CTRL* and *Fx* keys (where *x* is a number between 1 and 12).
- Enabled Windows () and left *ALT* keys.

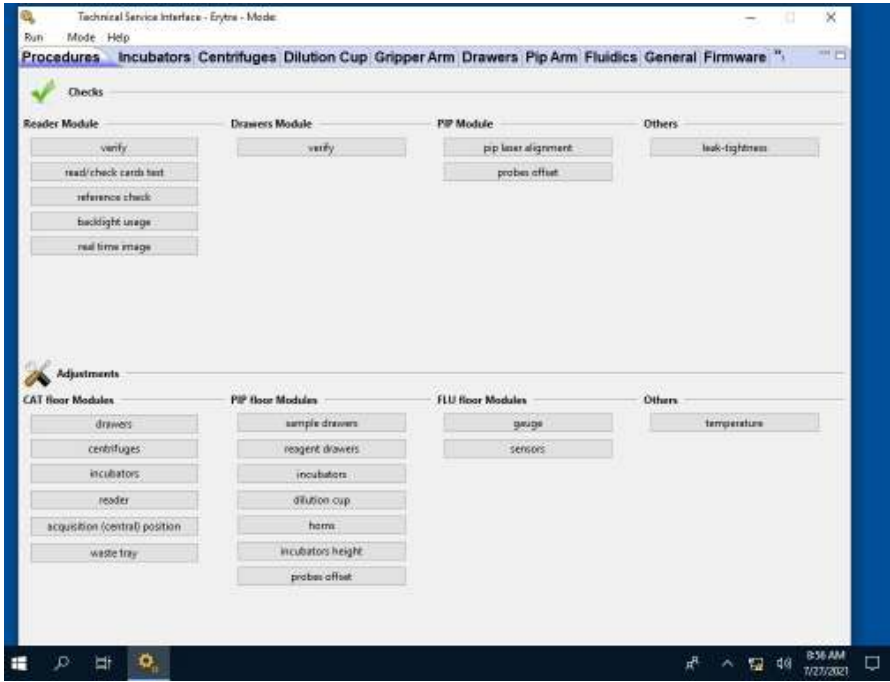


Figure 563 View of Technical Service software with access to Windows 10 File Explorer

When closing Technical Service software, Boot application will be launched again as in a usual start of the instrument (grid button should be pressed if no access to user application is intended for the session).

DIAGNOSTIC APPLICATION MODE



NOTE:

A physical keyboard and admin user password is needed to enable the new functionalities in Diagnostic application mode.

- Permanent access to Windows 10 file explorer (taskbar always present at the bottom of the screen).
- Enabled right-click functionality when using the mouse on File Explorer.
- Enabled access to *E:* drive (DATA).
- Enabled shortcut keys, except for *CTRL* and *Fx* keys (where *x* is a number between 1 and 12).
- Enabled Windows () and left *ALT* keys.

In order to enable the above listed functionalities, *LEFT SHIFT* and *↑* keys should be pressed simultaneously using a physical keyboard. Then, a window will be prompted asking for the Administrator user of the system (same as the one used for OS actions requiring administration privileges).

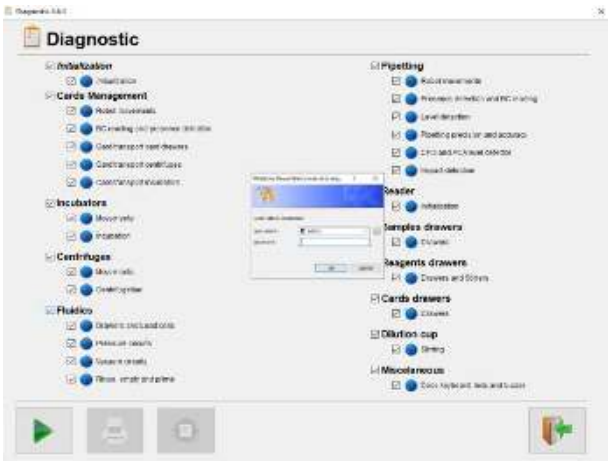


Figure 564 Request for admin user credentials in Diagnostic application mode

Once credentials have been properly introduced, the new functionalities will be enabled during the current session.



Figure 565 View of Diagnostic application with access to Windows 10 File Explorer

Once the troubleshooting actions have been performed by the Field Service Engineer, access to Windows 10 File Explorer in Diagnostic application must be disabled by pressing *LEFT SHIFT* and *↑* keys simultaneously. At that point, Diagnostic application will not be closed and can be executed as usual.

When closing Diagnostic application after having accessed Windows 10 File Explorer, Boot application will be launched again as in a usual start of the instrument (grid button should be pressed if no access to user application is intended for the session).

USER APPLICATION MODE



NOTE:
A physical keyboard and admin user password is needed to enable the new functionalities in Diagnostic application mode.

- Permanent access to Windows 10 file explorer (taskbar appearing when required at the bottom of the screen).
- Enabled right-click functionality when using the mouse.
- Enabled access to *E:* drive (DATA).
- Enabled shortcut keys, except for *CTRL* and *Fx* keys (where *x* is a number between 1 and 12).
- Enabled Windows () and left *ALT* keys.

In order to enable the above listed functionalities, *LEFT SHIFT* and *↑* keys should be pressed simultaneously using a physical keyboard. Then, a window will be prompted asking for the Administrator user of the system (same as the one used for OS actions requiring administration privileges).

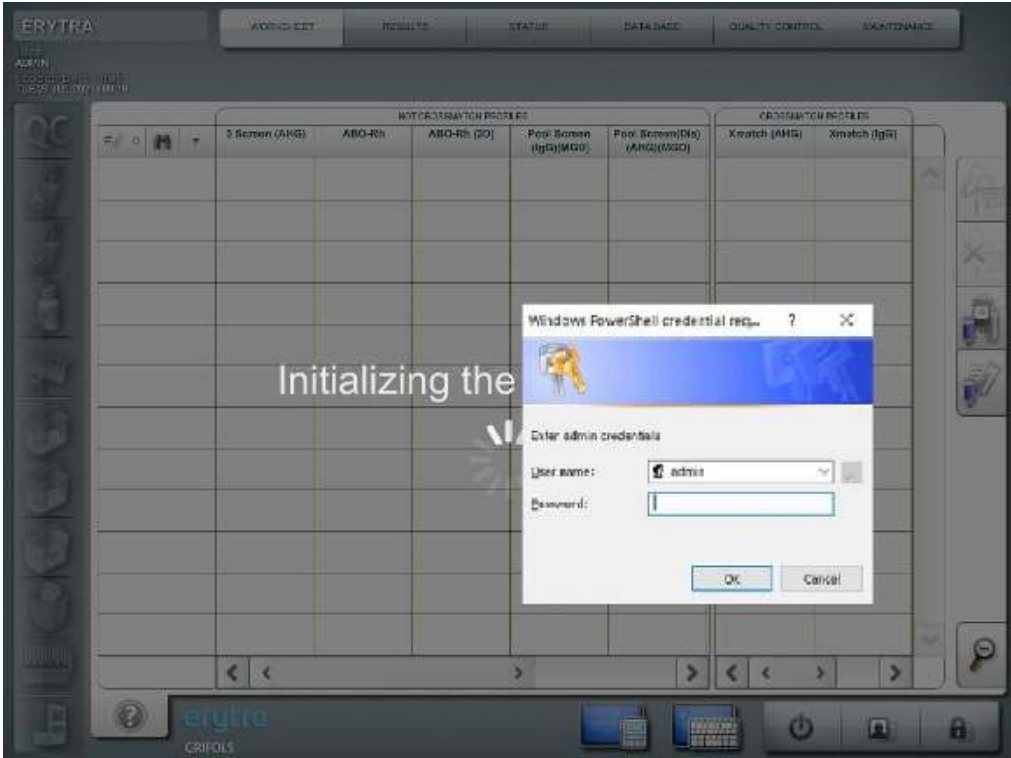


Figure 566 Request for admin user credentials in User application mode

Once credentials have been properly introduced, the new functionalities will be enabled during the current session.

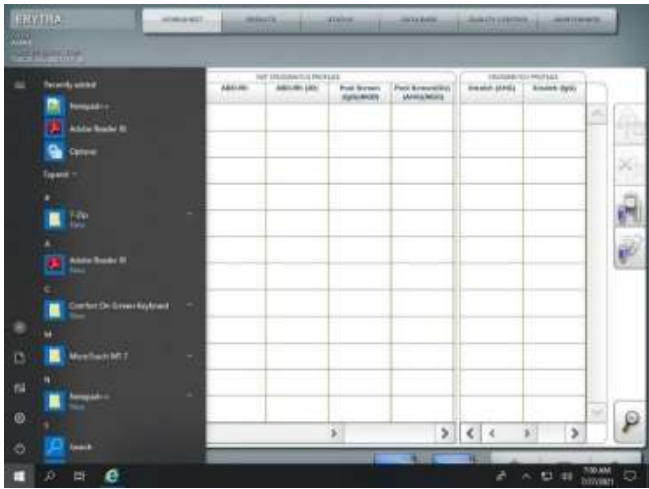


Figure 567 View of User application with access to Windows 10 File Explorer

Once the troubleshooting actions have been performed by the Field Service Engineer, access to Windows 10 File Explorer in User application must be disabled by pressing *LEFT SHIFT* and *↑* keys simultaneously. At that point, User application will not be closed and the user will be able to keep working with the instrument.

OS SHELL MODE

- When modifying LIS connection settings (use of DianaServer program, Listas.exe or Config.exe files) the applied changes will be stored persistently in the corresponding Windows registry keys (no need to do it through Maintenance mode).
- Enabled right-click functionality when using the mouse.
- Enabled shortcut keys, except for *CTRL* and *Fx* keys (where *x* is a number between 1 and 12).
- Enabled Windows () and left *ALT* keys.

15.6 Touchscreen driver installation and configuration

15.6.1 Touchscreen driver installation

During the Operating System installation process, the touchscreen model in use (version 2 or version 3) is automatically detected by the system and the corresponding touchscreen drivers are used.

15.6.2 Touchscreen v2 driver configuration

Settings for touchscreen v2 (SN < 059-0001515) can be modified by means of its *Control Panel* utility:

1. Switch on the instrument and press the grid () button during system startup (login credentials will be required), then select the *Maintenance Mode* icon among the Boot Application options.

2. Once PC reboots and desktop is available, press Windows *Start* button and select *MicroTouch MT 7* program.

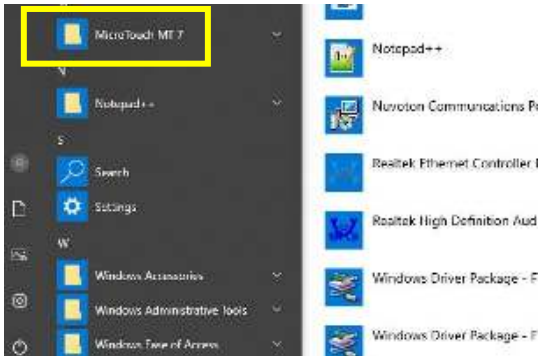


Figure 568. MicroTouch MT 7 program

3. Click on *Control Panel* among the drop down options.

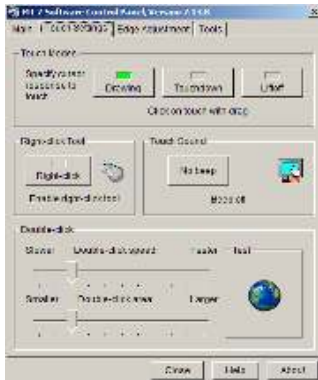


Figure 569. MT7 Control panel

4. Select *Drawing* touch mode in *Touch Settings* tab.
5. Make sure that the *Right-click* option in *Right-click Tool* area is disabled, and *Touch Sound* is set to *Beep off*.
1. Set *Double-click* to:
- Speed: level 3
 - Area: level 3
2. Switch off the instrument.

15.7 Backup

15.7.1 with Boot menu procedure

In order to perform manual databases backup from the Boot Application, switch on the PC, wait for the Erytra logo to appear on screen and then press the grid () button. Log in as *adminUser* (case sensitive, *admin* password) and the Boot Application options will be displayed. Once in there, press the arrow in the right side of the screen to switch to page 2/2 of the menu, then select the *Backup* icon among the Boot Application options and follow the steps listed below.

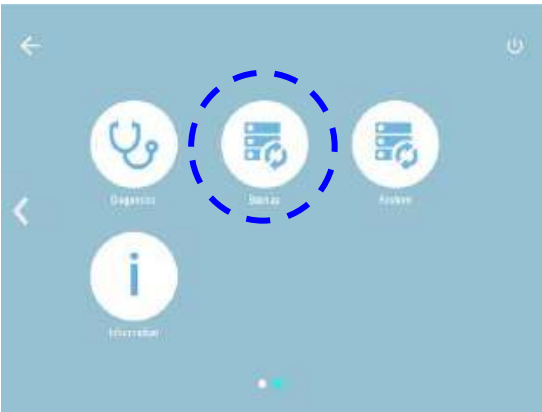


Figure 570. Backup icon

There are two backup options, full backup or custom backup. The two options are listed below:

Full backup

This option creates a backup with following data:

- All the databases (IHManager, IHExecutor and UserAccess)
- Instrument configuration files:
 - PIPCalibrations.xml and CATCalibrations.xml
 - PIP barcode readers (left and right) customized configuration
 - erytra.properties and erytra.properties.backup
- Operating System:
 - Installed Drivers (not including the OS image default ones)
 - Local configured printers (not network printers)
 - Active networks configurations
 - Time zone
 - Language for non-unicode programs configuration.
 - Start menu
 - Touchscreen calibration
 - Language
 - Scheduled Tasks
 - Firewall rules
 - Diana System section of Windows registry
- Diana System data:
 - Log and Backups folder
 - Def folder
 - Plantillas folder
 - Tandas folder
- Logs, if selected, from the selected date to current one.

Customized backup

This option creates a backup with following data:

- Databases sections that contain IH Reagents Pack data and profiles configuration (IHManager and IHExecutor).
- Instrument configuration files:
 - PIPCalibrations.xml and CATCalibrations.xml
 - PIP barcode readers (left and right) customized configuration
 - erytra.properties and erytra.properties.backup
- Operating System:
 - Installed Drivers (not including the OS image default ones)
 - Local configured printers (not network printers)
 - Active networks configuration
 - Time zone
 - Language for non-unicode programs configuration.
 - Start menu
 - Touchscreen calibration
 - Language
 - Scheduled Tasks
 - Firewall rules
 - Diana System section of Windows registry
- If Diana System option is selected (checkbox is ticked), Diana System data:
 - Log and Backups folder
 - Def folder
 - Plantillas folder
 - Tandas folder
- If Users Database option is selected (checkbox is ticked), UserAcces database.



NOTE:

Admin password will be requested to proceed with OS system configuration backup.



NOTE:

DSC.def and *DSC_Custom.def* files are no included in the backup.

The application will copy temporaly all the data in E:\tmp directory, generate a hash file to check backup integrity and then compress all the data in a 7zip file and move it to destination folder.

A log named *ihExport.log* is created in *E:\Program Files\Backup\logs* folder. Once this file reaches 2MB it compressed (gnu zip) and renamed to *ihExport-backup-YYYY-MM-DD.log*, where YYYY is the year, MM the month and DD the day.

Follow the steps listed below:

1. Choose:
 - The backup destination.
 - Whether log files should be included in the backup or not. If log files should be included, indicate the desired starting date.
 - The type of backup (full or customized)
2. Enter a backup description and *press Backup*. Once backup button is pressed, a progress window will be displayed. Describing current step and the total amount of steps.

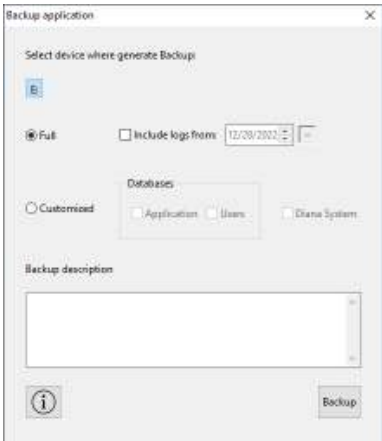


Figure 571. Backup description and log files choice



NOTE:

Information button  displays a new window with the backup instructions. It describes the available options, and which will be the content of the backup.

3. A subfolder named *BAK_yyyy-dd-MM_hh_mm_ss* (where yyyy is year, dd is day, MM is month, hh is hour, mm is minute and ss is second when the backup was performed) will be created in the selected backup destination .

15.7.2 with user interface or OS Shell backup option

It is also possible to perform the Backup from the path "E:\Program Files\Backup" (entering the 'Operating System Shell' option from the Boot application) if the current version is higher than 4.3.2 (Operating System is Windows 10). To perform the Backup from this route, execute the run.bat and press Backup button and the file will be stored in E:\Backup\BAK_YYYY-MM-DD_HH_MM_SS folder.

15.7.3 manual backup

The most relevant data can be stored manually on a USB drive. The following data are considered relevant:



CAUTION!

Scan the USB drive for viruses before connecting it to Erytra.

- 1 Microtubes jpg images (E:\Data\Erytra\Microtubes subfolders).
- 2 Raw (E:\Data\Erytra\Raw) folder.

- Instrument configuration file (*erytra.properties* found in *E:\conf*).
- Backup path configuration file (*E:\Program Files\System Tools\backup.conf*).
- Diana System definition files (*E:\data\SistemaDiana\defs*), if necessary.
- Customized Diana System templates (default directory *E:\Data\SistemaDiana\Plantillas*), if necessary.
- DSC.def* file and, if profiles *gru* names or templates have been customized, *DSC_Custom.def* file (*E:\Data\Erytra\Wadiana*).
- Technical Service software Macros, if necessary.
- Erytra log files (*E:\Log directory and its subdirectories*) and Diana System log files (default location *E:\Data\SistemaDiana\log*), if necessary.
- Diana System *Tandas* and *Traceability* folders, if necessary.
- Number of tests of the instrument: *testcounter.txt* file (*E:\data\Erytra\conf*).
- Create a backup and copy/save it in an external device of Diana System configuration in Windows registry (run *regedit* and export *HKEY_LOCAL_MACHINE\SOFTWARE\WOW6432Node\Grifols*), if necessary.
- Create a backup the current TCP/IP IPv4 MS Windows configuration and computer name if LIS connection is enabled and communication mode is set to TCP/IP.

This procedure will not delete mentioned data below. Nevertheless, it may be advisable to create a backup of this relevant data:

- Instrument adjustment files *PIPCallibrations.txt* and *CATCallibrations.txt* found in *E:\Data\Erytra\conf*.
- Instrument configuration file (*erytra.properties* found in *E:\conf* in Erytra software version 4.3.0 and higher and *D:\conf* in Erytra software versions < 4.3.0).

15.8 Database restore procedure

This procedure is intended to allow loading databases from previous software versions before updating them to the structure of the current one. In order to do this, follow the steps listed below:

- Switch on the instrument and press the grid () button during system startup (login credentials will be required), then select the *OS Shell* icon among the Boot Application options.
- Run *E:\Program Files\ErytraDB Apps\Restore_DataBase.exe*. Restore DataBase setup welcome window is displayed.
- Click on *Next>* button.
- License agreement window is displayed. Click on *I agree*.
- A window allowing to choose the actions to perform is displayed:

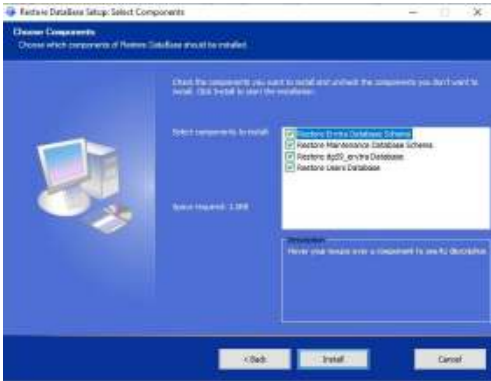


Figure 572. *Restore_DataBase* software window

- 6. Select the databases and/or schemas to restore and click on *Install* button.
- 7. A window allowing selecting the name and location of the *Erytra* database backup file will be displayed.
- 8. Choose the file containing the backup to be restored.

**NOTE:**

In case no backup exists or an empty database is going to be installed select the backup of an empty database file: *DG-59_ErytraBBDD.backup* or *DG-userAccessBBDD.backup* in folder *E:\Program Files\ErytraDB* (**ALL DATA WILL BE LOST!**).

**NOTE:**

The restoring procedure used by this program will not preserve the test count sequence in *testcounter.txt* file (*E:\data\Erytra\conf*). This sequence will only be preserved when restoring databases using the *Restore* option from Boot Application menu. If a different database is restored from this option when opening the user application the last line of *testcounter.txt* file will be modified with the new database data. For example, if the original file is:

```
10.000
restored db
2.500
```

When restoring a database with 222 processed samples, and the user application is launched it will collect the available database data and will update the line with this date.

```
10.000
restored db
222
```

The preventive maintenance replacements are calculated taking into account the processed samples reflected in this file.

- 9. Click on *Open*. Database restore process will start.
- 10. If more than one database should be restored, a window allowing to select the name and location of the other databases will be displayed. Choose the file containing the backup to be restored and click on *Open*.
- 11. Once completed, click on *Next>* and then on *Close*.
- 12. If necessary, update the databases (run *E:\Program Files\liquibase\updatedb.bat*)

15.9 Restore with Boot menu procedure

In order to restore databases backups from the Boot Application, switch on the PC, wait for the Erytra logo to appear on screen and then press the grid () button. Log in as *adminUser* (case sensitive, *admin* password) and the Boot Application options will be displayed. Once in there, press the arrow in the right side of the screen to switch to page 2/2 of the menu, then select the *Restore* icon among the Boot Application options and follow the steps listed below.

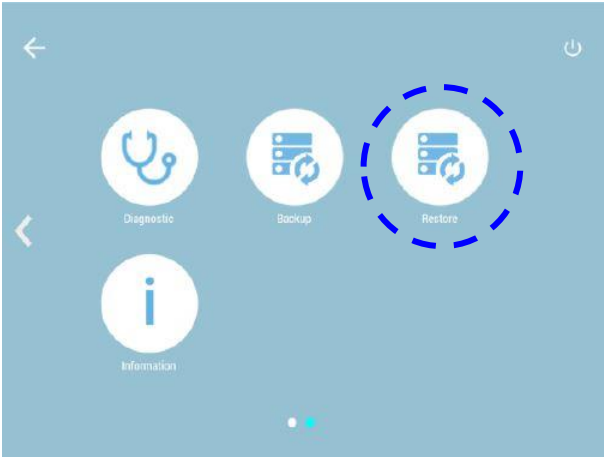


Figure 573. Restore icon

- The optional data to restore, is available, is:
- Application Databases: IHManager and IHExecutor databases
 - User Database: UserAccess database
 - erytra.properties: erytra.properties and erytra.properties.backup
 - Calibrations: PIPCalibrations.xml and CATCalibrations.xml
 - Barcode reader: PIP left and right customized barcode reader configuration
 - Diana System: Diana System folders Log, Backups, Def, Plantillas and Tandas.
 - System: Operating System configuration: Drivers, Local configured printers, networks configuration, Time zone, Language for non-unicode programs configuration, Start menu, Touchscreen calibration, Language, Scheduled Tasks and Firewall rules.

A log named *ihImport.log* is created in *E:\Program Files\Restore\logs* folder. Once this file reaches 2MB it compressed (gnu zip) and renamed to *ihExport-backup-YYYY-MM-DD.log*, where YYYY is the year, MM the month and DD the day.

1. Select the backup to restore.

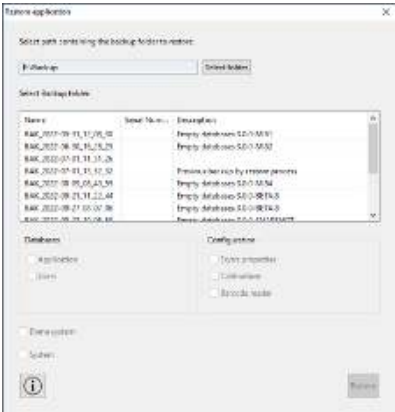


Figure 574. Restore backup window

**NOTE:**

Information button  displays a new window with the backup instructions. It describes the available options, and which will be the content of the backup.

2. Select the components to be restored and press *Restore*. A progress window will be displayed describing current step and the total amount of steps. Once OK button is pressed, the process is resumed.



**NOTE:**

Admin password will be requested to proceed with OS system configuration backup.

When the backup has been created the data will be uncompressed, and the hash will be checked. If it is ok, the process will continue and a Full backup without log files will be created before starting restore process. Its description will be *Full Backup before Restore Backup*.

Once restored, the program will update the file registering the number of executed tests in the current database (*testcounter.txt* file in *E:\data\Erytra\conf*), adding a line indicating that the database has been restored. From there, a new entry will be created and executed tests will be registered starting a new count sequence. For instance, when a database has been restored using the *Restore* option from Boot Application in an instrument where 10.000 tests were previously executed, and 2.500 tests have been executed since the database restoring procedure, the following lines will be registered in *testcounter.txt* file:

```
10.000
restored db
2.500
```




NOTE:

It must be taken into account that in case of an operation that requires restoring the previous database, such as a full recovery and this is restored from this option, a new line will be created in the file testcounter.txt and once the user application is started, the last line will be updated with the existing database data.

If the testcounter.txt file before the restore is:

1.000

In the event the same database is being restored using this backup option:

1.000
restoredb

And once the user applications is started, it will be updated with the database available in the instrument:

1.000
restoredb
1.000

The preventive maintenance replacements are calculated taking into account the processed samples reflected in this file.

15.10 Erytra Database Manager application

Required specifications for any computer where Erytra Database Manager Application will be installed are listed below:

- Operating System: Windows 7 (32 bits), Windows 10 (32 and 64 bits).
- 2 GB RAM minimum.
- erytra-databasemanager-x.x.x.exe installer is available.
- Minimum free space on SSD: 922.6 MB.
- Adobe Reader 9.4 or higher.



NOTE:

Erytra Database Manager software should be installed in a computer where neither Postgres application, Tomcat server or BTM are installed.



NOTE:

User to install/run the software must have an administrator role in the used computer.

INSTALLATION PROCEDURE

In order to install Erytra Database Manager in an external computer, follow next steps:

1. Execute installer “Erytra-DATABASEMANAGER-X.X.X.exe”.
 - o Select “Browse” button to choose an install location. Navigate to an existing folder or create a new one and then select “Install” button.



NOTE:

When the installation begins it opens the Erytra application installer.

- When the installation completes select “Finish” button.

2. Load Erytra v4.4.0 database backups.



NOTE:

This procedure is intended to be performed on a PC where PostgreSQL is not running.

- Open the folder where the application has been installed.
- Execute RESTORE_DB.vbs and in the application:
 - Select “Next” button. Check “I accept the terms of the License Agreement” checkbox. Select “Install” button.
 - Choose dg59_erytra database backup file.
 - Choose userAccess database backup file.
 - Select “Finish” button.



NOTE:

If the application shows the message error: “Error trying to delete database.” then the file “Data\Database\pg_hba.conf” has to be modified: Comment the last line of the file adding the “#” symbol at the beginning of the line:

```
#host all all ::1/128 md5
```



NOTE:

Software version 4.4.0 databases shall be restored in Erytra Database Manager version 4.4.0. The compatibility with previous Erytra Database Manager or software versions is not guaranteed.

15.11 EEPROM backup/restore adjustments

In Erytra most of the adjustments are stored in two files (*PIPCalibrations.txt* and *CATCalibrations.txt*) located in *E:\data\Erytra\conf*, except for the following ones:

- Probes offset adjustment: it is stored in EEPROM of PCA PIP-Control.
- Incubator temperature: it is stored in EEPROM of PCA INC-CUP Control.
- Load cells of the fluidic drawers adjustment: it is stored in EEPROM of PCA FLU-CTRL59.
- Fluidic sensors adjustment: it is stored in EEPROM of PCA FLU-CTRL59.
- Reader/camera adjustment: it is stored in EEPROM of PCA RD-IL421.

When replacing any of these PCAs:

- PCA PIP-Control and PCA INC-CUP Control: the stored adjustment must be performed again or restored from a file (before replacing the PCA, a backup of the adjustment must be performed).
- PCA FLU-CTRL59: adjustment backup can be also restored but is recommended to perform the adjustments.
- PCA RD-IL421 it is recommended to perform the adjustments (not recommended restoring them).

EEPROM BACKUP

- 1. Switch on the instrument and press the grid () button during system startup (login credentials will be required), then select the *Technical Service* icon among the Boot Application options.
- 2. Start the *Driver* and go to *Firmware* tab → *Eprom control* section.

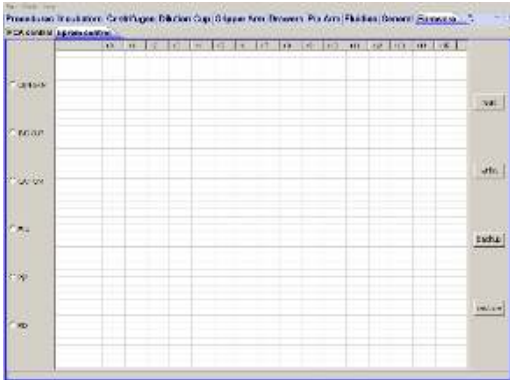


Figure 575. Eprom control window in TSS

- 3. Select PCA to save the EEPROM content.
- 4. Press backup button. A new window appears in order to select the folder and the name of the file to be saved.
- 5. After the generation of the EEPROM file (*filename.txt* file), another file is also created with the electronic signature (*filename_signature.txt* file)



NOTE:
Do not rename the name of the files generated by the backup procedure.

EEPROM RESTORE

- 1. Switch on the instrument and press the grid () button during system startup (login credentials will be required), then select the *Technical Service* icon among the Boot Application options.
- 2. Start the *Driver* and go to *Firmware* tab → *Eprom control* section.
- 3. Select PCA to restore the EEPROM content.
- 4. Press restore button. A new window appears in order to select the folder and the file to be restored.



NOTE:
EEPROM content file and electronic signature file must be in the same folder.

15.12 Firmware update



CAUTION!
Do not operate with the instrument while a firmware loading operation is being performed

There are 3 possible scenarios: update the firmware in a valid PCA, installing the firmware in a spare part PCA and installing the firmware in a PCA with a non valid firmware, in which a previous firmware loading process failed.



Figure 576. Firmware update window in Technical Service software

15.12.1 Update firmware in a valid PCA



CAUTION!

Make sure that only one firmware of the resultant type will be present at the Erytra slots after loading process. Otherwise, first remove the necessary cards in order to achieve this.

1. Switch off the instrument and disconnect it from the mains.
2. Connect the instrument to the mains and switch on the instrument.
3. Press the grid () button during system startup (login credentials will be required), then select the *Technical Service* icon among the Boot Application options.
4. Without starting the *Driver*, go to *Firmware* tab → *PCA control* section. Press load firmware button.
5. Select the PCA to update the firmware in PCA section.
6. Press *Browse* button in Program File section and select the proper program file for the selected PCA (only the ones related to the previous PCA selected are shown). Program files can be found in *E:\Program Files\Firmwares DG-59*.
7. Press *Browse* button in Kernel File section and select (only the one related to the previous PCA selected is shown) the proper kernel file (For CEN-SRM, INC-CUP, CAT-CM and PIP select the H8S2329; for the FLU select the Texas). If the PCA selected is the RD, this option is not activated because is not necessary to indicate any kernel file. Kernels can be found in *E:\Program Files\Firmwares DG-59*.
8. Press *Update Firmware* button and wait until process finishes.

15.12.2 Update firmware in a spare part PCA (CEN-SRM, INC-CUP, CAT-CM or PIP MotorControl)



CAUTION!

Make sure that only one firmware of the resultant type will be present at the Erytra slots after loading process. Otherwise, first remove the necessary cards in order to achieve this.

1. Switch off the instrument and disconnect it from the mains.
2. Connect the instrument to the mains and switch on the instrument.
3. Press the grid (■) button during system startup (login credentials will be required), then select the *Technical Service* icon among the Boot Application options.
4. Without starting the *Driver*, go to *Firmware* tab → *PCA control* section. Press load firmware button.
5. Select the PCA to update the firmware in PCA section and then select Test option in PCA section.
6. Press *Browse* button in Program File section and select the proper program file for the selected PCA (only the ones related to the previous PCA selected are shown). Program files can be found in *E:\Program Files\Firmwares DG-59*.
7. Press *Browse* button in Kernel File section and select (only the one related to the previous PCA selected is shown) the proper kernel file (For CEN-SRM, INC-CUP, CAT-CM and PIP select the H8S2329; for the FLU select the Texas). If the PCA selected is the RD, this option is not activated because is not necessary to indicate any kernel file. Kernels can be found in *E:\Program Files\Firmwares DG-59*.
8. Select the proper slot of the PCA in Selected PCA slot section:
 - a. Slot 0: CEN-SRM
 - b. Slot 1: INC-CUP
 - c. Slot 2: CAT-CM
 - d. Slot 3: PIP
9. Press *Update Firmware* button and wait until process finishes.

15.12.3 Update firmware in a PCA with non valid firmware (CEN-SRM, INC-CUP, CAT-CM or PIP MotorControl)



CAUTION!

Only one PCA with non valid firmware must be present at the *Erytra* slots. Never execute this process if two or more PCAs with no valid firmware are connected at the same time because process will fail.



CAUTION!

Make sure that only one firmware of the resultant type will be present at the *Erytra* slots after loading process. Otherwise, first remove the necessary cards in order to achieve this.

1. Switch off the instrument and disconnect it from the mains.
2. Connect the instrument to the mains and switch on the instrument.
3. Press the grid (■) button during system startup (login credentials will be required), then select the *Technical Service* icon among the Boot Application options.
4. Without starting the *Driver*, go to *Firmware* tab → *PCA control* section. Press load firmware button.
5. Select the PCA to update the firmware in PCA section.
6. Press *Browse* button in Program File section and select the proper program file for the selected PCA (only the ones related to the previous PCA selected are shown). Program files can be found in *E:\Program Files\Firmwares DG-59*.
7. Press *Browse* button in Kernel File section and select (only the one related to the previous PCA selected is shown) the proper kernel file (For CEN-SRM, INC-CUP, CAT-CM and PIP select the H8S2329; for the FLU select the Texas). If the PCA selected is the RD, this option is not activated because is not necessary to indicate any kernel file. Kernels can be found in *E:\Program Files\Firmwares DG-59*.
8. Activate *Enabled* box in *Boot Mode* section.
9. Press *Update Firmware* button and wait until process finishes.

15.13 BIOS settings

This section explains how to check the BIOS configuration.

**NOTE:**
When installing a new BIOS firmware version, previously configured BIOS settings will be erased.

15.13.1 Check BIOS version

1. Connect a physical USB keyboard. Boot up the system and press [Del] when the boot logo page is displayed.

**NOTE:**
A password to access the BIOS has been set.
When requested enter password: TSDGIH.



Figure 577 Boot logo page

2. In the *Main* tab, check the *UEFI Version*.

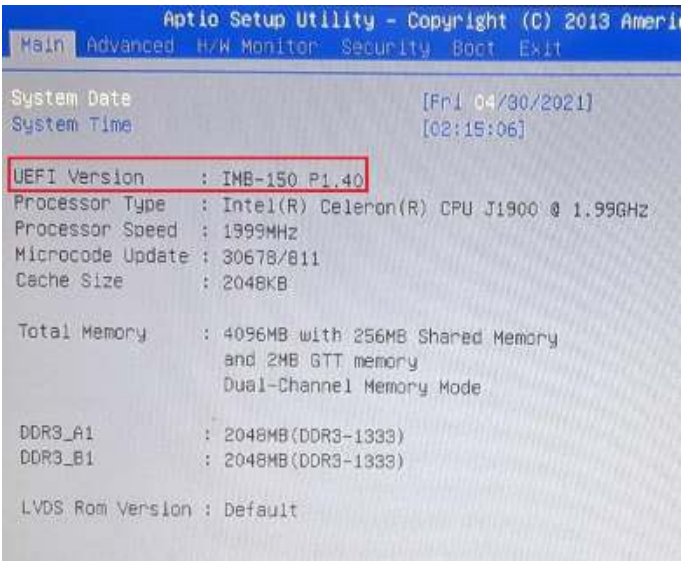


Figure 578 UEFI version

15.13.2 BIOS configuration

1. Enter to BIOS configuration. Boot up the system and press [Del] when the new boot logo page is displayed.



NOTE:
A password to access the BIOS has been set.
When requested enter password: TSDGIH.



Figure 579 New boot logo page

2. Go to *Advanced* tab and select the option *Super IO Configuration*.

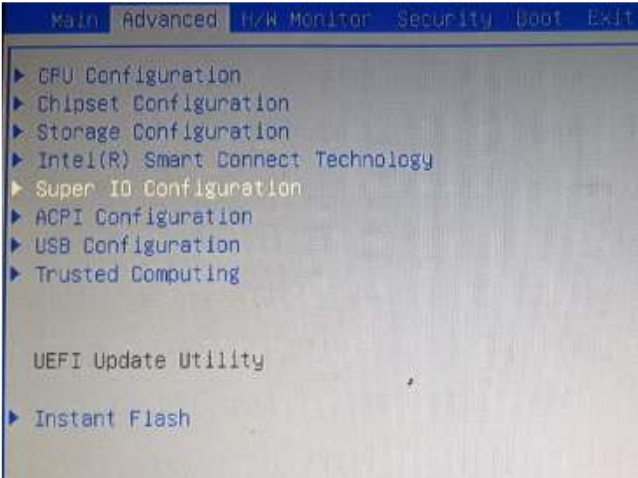


Figure 580 Super IO configuration selection

3. Check *COM4 Configuration* is set to *Disabled*.



Figure 581 COM 4 port disabling

4. Check *COM6 Configuration* is set to *Disabled*.

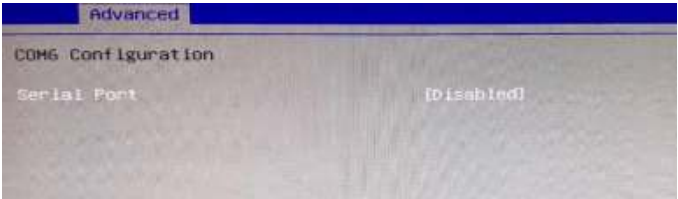


Figure 582 COM 6 port disabling

5. Check *LPT1 Port Configuration* is set to *Disabled*.

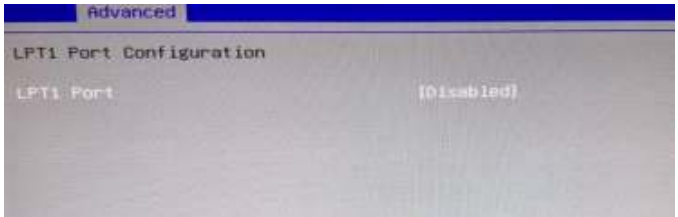


Figure 583 LPT1 port disabling

6. Go to "Exit" tab and select "Save Changes and Exit". Click "Yes".

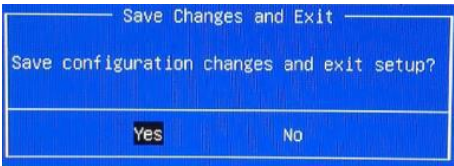


Figure 584 BIOS exit tab

15.13.3 Windows COM ports assignment

COM ports configuration can be checked using Windows Device Manager.

WINDOWS 10

1. Switch on the instrument and press the grid button () during system startup (login credentials will be required). Then select Maintenance Mode option.
2. Once in Maintenance Mode, type "devmgmt.msc" in Windows Search Box. devmgmt.msc appears as a Command option. Select Run as administrator (password for admin user will be required).

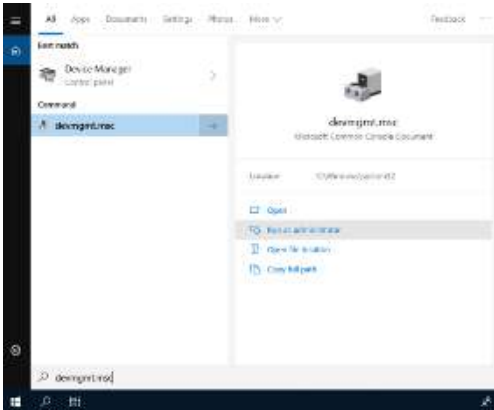


Figure 585 Running "devmgmt.msc" as administrator

3. Check the Ports (COM & LPT) assignment:

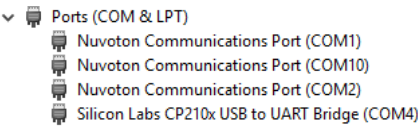


Figure 586 Ports assignment

Silicon Labs CP210x USB to USART Bridge should be assigned to COM4. Should that not be the case, follow the steps listed in the Modify COM ports assignment section below.

Modify COM ports assignment

- 1. Go to View tab in Device Manager and select Show hidden devices.
- 2. If Nuvoton Communications Port (COM4) is present, right click and uninstall it.

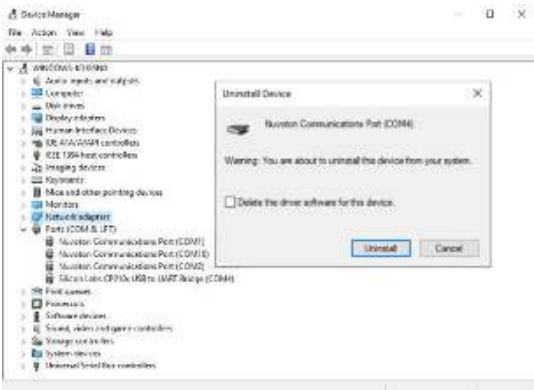


Figure 587 Uninstalling Nuvoton Communications port (COM4)

- 3. If Nuvoton Communications Port (COM6) is present, right click and uninstall it.

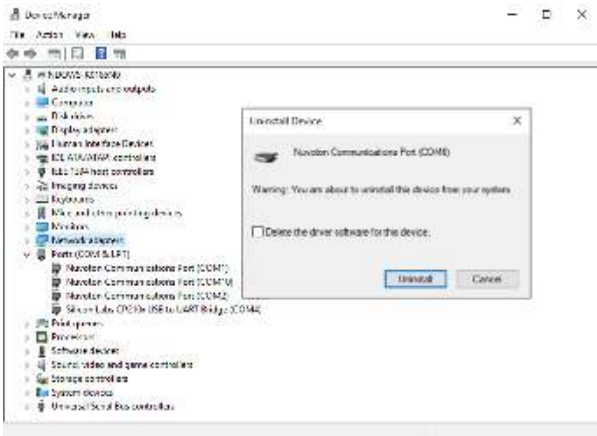


Figure 588 Uninstalling Nuvoton Communications port (COM6)

4. If ECP Printer Port (LPT1) is present, right click and uninstall it.

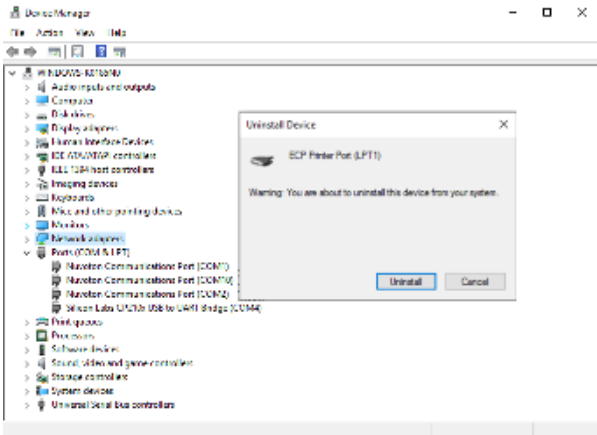


Figure 589 Uninstalling ECP Printer Port (LPT1)

5. Right-click on the COM port assigned to the device Silicon Labs CP210x USB to UART Bridge (COM5 in the example below) and click on Properties.

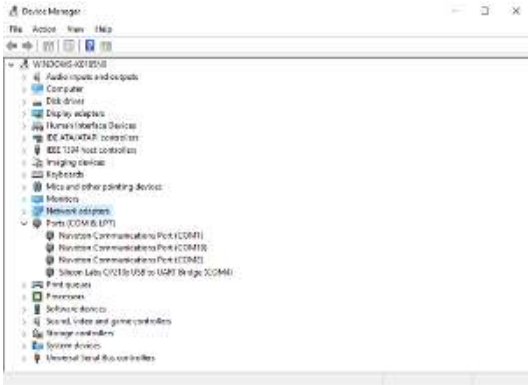


Figure 590 Silicon Labs CP210x USB to UART Bridge selection

6. Go to Port Settings tab and click on Advanced.

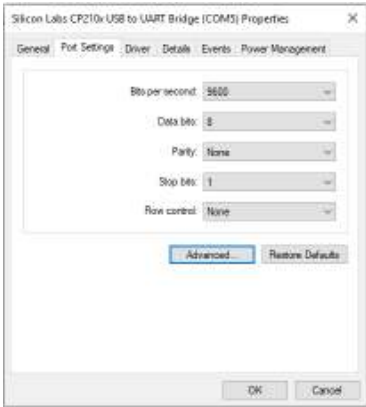


Figure 591 Port settings tab

7. Select COM4 in COM Port Number combo box. Click OK.

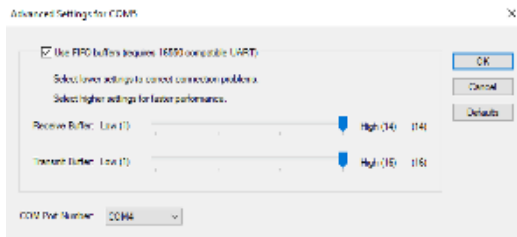


Figure 592 COM 4 selection in COM port number combo

8. If exists, delete the file E:\data\Erytra\conf\serialPort.xml.
9. Make sure that the Ports (COM & LPT) assignment is now the expected one:

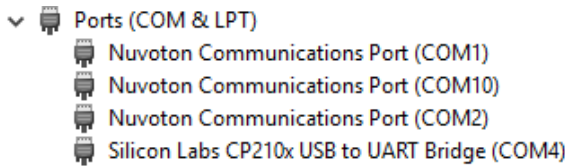


Figure 593 Expected port assignment in Windows 10

15.14 AppLocker application

AppLocker is an application whitelisting technology included in Windows operating systems, used to block unauthorized executables on running on Erytra. It uses a dynamic trust model, preventing attacks from unknown malware by allowing only known-good applications to run, registering them on an approved list of entities known as *whitelist*.

15.14.1 Updating the whitelist

In order to perform new installations or updates of authorized applications or drivers, the whitelist can be dynamically updated by executing the following procedure:

3. Disconnect the instrument from the network (if connected).
4. Switch on the instrument and press the grid () button during system startup (login credentials will be required), then select the *Maintenance Mode* icon among the Boot Application options.
5. PC will be rebooted and Maintenance Mode will be accessed in the subsequent startup.

**CAUTION!**
Once the computer has been restarted, drives are not protected.

6. Press *Windows Start* button and type *Command Prompt*. Rick click at the top result and select *Run as administrator*, then type administrator password *dgroot*.
7. Enter the command: *applocker bu* (Begin Update). The AppLocker Manager graphical interface will be opened in a separate window.

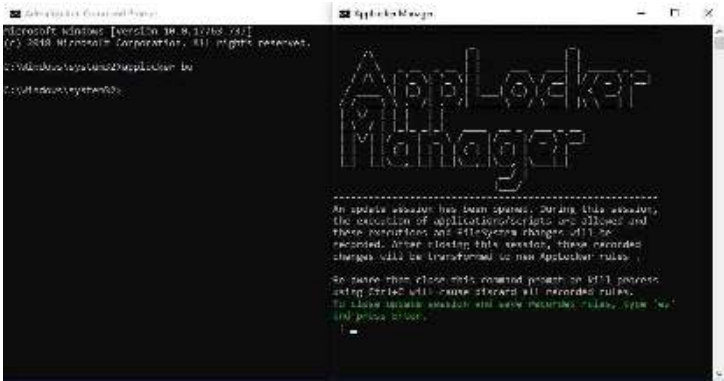


Figure 594. AppLocker application entering update mode

8. Wait 10 seconds before proceeding with the following installation steps.
9. Check that AppLocker has successfully entered update/audit mode: open *E:\AppLockerWorkspace\session.log* file with notepad and check that following lines can be found:

```
... [APPLOCKER] [INFO] Enabling Audit mode on AppLocker ...
... [APPLOCKER] [INFO] Audit mode has been enabled.
... [APPLOCKER] [INFO] Opening File system listeners...
... [APPLOCKER] [INFO] A FileSystem watcher for Created in C:\ has been created.
... [APPLOCKER] [INFO] A FileSystem watcher for Changed in C:\ has been created.
... [APPLOCKER] [INFO] A FileSystem watcher for Created in D:\ has been created.
... [APPLOCKER] [INFO] A FileSystem watcher for Changed in D:\ has been created.
... [APPLOCKER] [INFO] A FileSystem watcher for Created in E:\ has been created.
... [APPLOCKER] [INFO] A FileSystem watcher for Changed in E:\ has been created.
... [APPLOCKER] [INFO] Creating FSRuleCreation Watcher ...
```

Should this not be the case, it AppLocker has not entered update mode successfully, if following lines are found:

```
... [APPLOCKER] [INFO] Enabling Audit mode on AppLocker ...
... [APPLOCKER] [INFO] Audit mode has been enabled.
... [APPLOCKER] [INFO] Opening File system listeners...
... [APPLOCKER] [ERROR] Cannot create type. Only core types are supported in this language mode.
```

then type *eu* in *AppLocker Manager* window, wait 10 seconds and go back to step 7.

10. Install/update the desired application or device driver per usual procedure. Once finished, the installed software will be automatically added to the whitelist.
11. Type *eu* (End Update) in the AppLocker Manager graphical interface window.
12. Wait for 10 seconds so that the change to protected mode is completed.

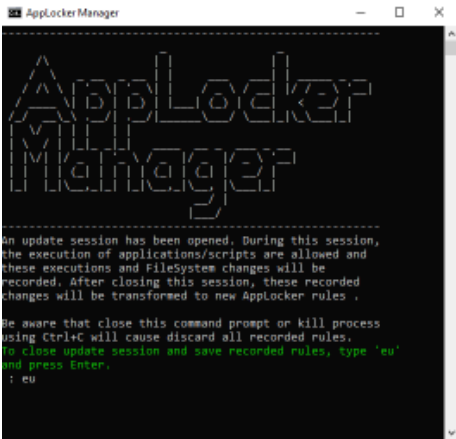


Figure 595. AppLocker application exiting update mode

- 13. Restart the system to exit *Maintenance mode*.
- 14. Connect (if applies) the instrument to the network.

15.14.2 Blocked executables

When the execution of an unauthorized application file has been blocked by AppLocker application, a warning dialog informing of the blocked application is prompted on screen.

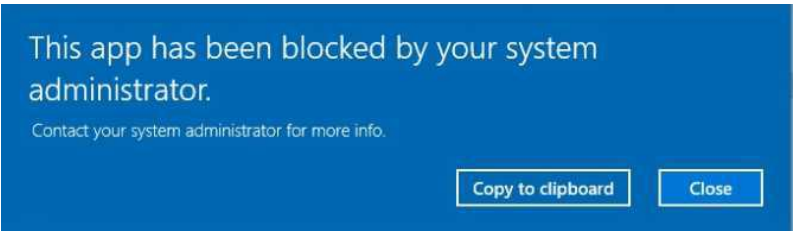


Figure 596. Example of blocked application (I)

If using Windows Command Prompt in order to execute a script or application which is not included in the whitelist, the following messages are shown:

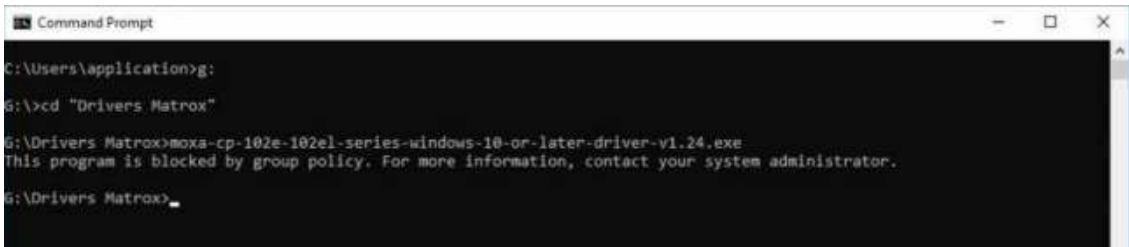


Figure 597. Example of blocked application (II)

When an application or script has been blocked by AppLocker, the corresponding event is captured to Application event log. In order to see them, go to *E:\log\appLocker.log*. It is also registered in Windows events log.

Additionally, the information of each update session is stored in *E:\AppLockerWorkspace*. It contains the log, the original rules and the final status of the rules.

15.15 Erytra Macro Executor

Technical Service software incorporates an option to run test files in order to automate movements and functions of the *Erytra* for testing purposes.



CAUTION!

Be careful with the options programmed in order to do not damage the instrument.

In order to run the Macro Executor, check chapter 3, *Technical Service software overview*.

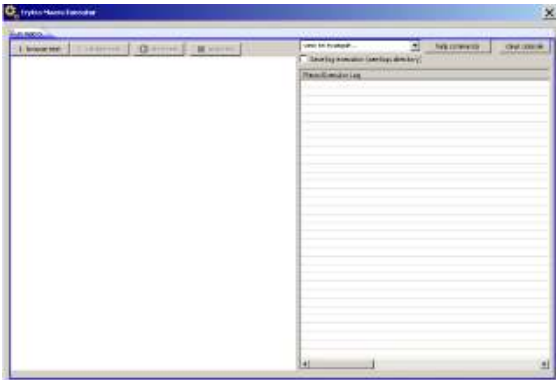


Figure 598. Erytra Macro Executor main screen

Test files are .xml files. In order to run a test program, follow the next steps:

- 1. Load the test file pressing 1.browse test
- 2. Once is loaded, press 2.validate test.
- 3. If no errors appears in the Macro Executor log window, press start test to execute the test file.
- 4. It is possible to stop at any time the test pressing stop test button.

List of options to include in the test file are detailed in the file manual.txt, that it is possible to select in the combo box. Also in the combo box there are detailed examples of all available commands.

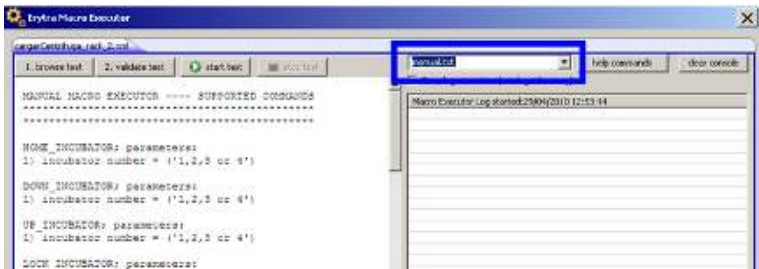


Figure 599. Combo box in Erytra Macro Executor main screen

Template of the test file is as follows,

```
<?xml version="1.0" encoding="UTF-8"?>
<erytraTest xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:noNamespaceSchemaLocation=".\xsd\ErytraTestXMLSchema.xsd">

<!-- // section where the possible initializations for the test are introduced -->
```



```
<init>
  <commands>
    .
    .
    .
  </commands>
</init>

<!-- // // section where the commands are introduced // -->
  <commands executions="xxxx"> where xxxx are the times to be executed

    <command>
      <operations>
        .
        .
        .
      </operations>
    </command>
  </commands>

</erytraTest>
```

15.16 Description of log files

Log files of the instrument are stored in E:\Log folder. Different log files are the following ones:

- E:\Log: contains log files of the instrument:
 - DG59Driver.log: registers Driver events and it is set to INFO level when registering events form the user application, and set to DEBUG level when registering events from Technical Service software. Once it reaches 100MB, it is compressed, moved to E:\Log\Backup folder and renamed to DG59Driver_yyyymmdd_hhmmss.zip, using the date and time of the last entry.
 - erytra.log: registers user interface events and is set to INFO level. Once it reaches 100MB, a compressed backup file is created in E:\Log folder and renamed to erytra_yyyymmdd_hhmmss.zip, using the date and time of the last entry.
 - DSC.log: Diana System Connectivity log file and it is set to DEBUG level. Once it reaches 1MB, it is moved to E:\Log\Backup folder and renamed to DSC_YYYYMMDD_hhmmss.log, using the date and time of the last entry.
 - erytraDebug.log: registers user interface events and it is set to DEBUG level. Once it reaches 100MB, a backup file is created. Only the last 20 backup files are kept. This log does not contain patient demographic data.
 - erytraError.log: registers user interface events and it is set to ERROR level. Once it reaches 100MB, a compressed backup file is created in E:\Log folder and renamed to erytraError_yyyymmdd_hhmmss.zip, using the date and time of the last entry.
 - stdout.log file: registers current day web application (Tomcat) console events. When the day is over it is compressed, moved to E:\Log\YYYY-MM folder and renamed to stdout.log.YYYY-MM-DD-hh-mm-ss.zip, using the date and time of the last entry.
 - dbupdate.log: registers events related to database updates (software installation, patches).
 - appLocker.log: registers events related to AppLocker application.
 - firewire.log: registers events related to reader module initialization.

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	595 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

- *startup.log*: registers the access to Windows file Explorer.
- *erytraAir.log*: Registers application events at the AIR runtime level. Once it reaches 100MB, it is compressed, moved to *E:\Log\Backup* folder and renamed to *erytraAir_yyyymmdd_hhmmss.zip*, using the date and time of the last entry.

- E:\Log\DG421: contains reader active log file (DioneExpert.log). Once it reaches 1,7MB it is compressed and moved to *E:\Log\DG421\Backup* and renamed to *DioneExpert_YYYYMMDD_hhmmss.zip* using the date and time of the last entry.
- E:\Log\Backup: contains backup of log files.
- E:\Program Files\tomcat\logs: contains web server application logs:
catalina.YYYY-MM-DD.log (where YYYY=year, MM=month and DD=day): *Web Server servlet container log*.
localhost.YYYY1-MM-DD.log (where YYYY=year, MM=month and DD=day).

If running Diana System / LIS connection in erytra PC (default configuration):

- E:\data\SistemaDiana\log:
DianaASTM.log: it registers DianaServer (LIS communication) events.

15.17 Support Pack

The Support Pack feature allows the creation of a zipped file with all the data required for support, which will not contain patient data in order to meet the General Data Protection Regulation (GDPR) compliance. This data is required to be stored in a USB drive, using 7-Zip software for its compression.

Print date:	16-May-2023	Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization	Purpose:	Uncontrolled Copy
-------------	-------------	--	----------	-------------------

15.17.1 Generating support pack from user software

The Support Pack can be created from the user software (Maintenance – Analyser – *Technical Service Action* tab) and can be requested by Technical Service, Operator plus, Specialist, Supervisor and Administrator users.



Figure 600.Support Pack button in Technical Service Action tab (user software)

Once the Support Pack option is selected and a USB drive has been connected to the instrument, the following menu will be displayed, allowing to choose the below listed options:

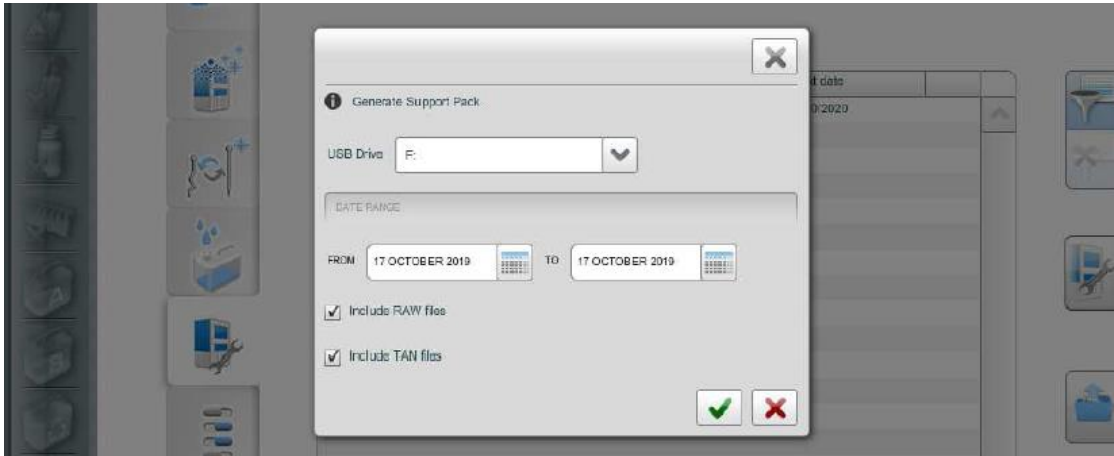


Figure 601.Support pack menu

- Destination USB drive.
- Time frame for the required information (Date FROM/TO).
- Checkbox allowing the inclusion of RAW images (if available) for the selected time frame.
- Checkbox allowing the inclusion of tan files (if available) for the selected time frame.

When pressing on the *Support Pack* button, a progress dialog with the text *Generating Support Pack* will be displayed and the system will collect all the data for the required time frame, erasing patient data in the created database and replacing it by asterisk characters ("****") in the created log files related to communication with LIS.

Data will be temporarily stored in E: drive, naming the temporary folder as *ERYTRA_ssn_YYMMDDhhmm_SupportPack*, where *ssnn* is the instrument serial number extracted from *erytra.properties* and *YYMMDDhhmm* is the date and time when the pack has been created. Then, a folder named *ERYTRA_ssn_SupportPack* will be created in the selected USB drive, and a 7-Zip file will be created inside with the contents of the temporary folder, being named *ERYTRA_ssn_YYMMDDhhmm_SupportPack.7z*. Once created, the temporary folder will be deleted from the E: drive.

The 7-Zip file created by the Support Pack feature will contain:

- *SupportPack.txt* file indicating pack creation date, selected date range, extended date range and the inclusion (YES or NO) of the RAW images and tan files in the pack.
- *Erytra* folder with all the files in *E:\conf* directory, dg59_erytra database backup (without demographic data) and the following subfolders:
 - o Conf: all the files in *E:\data\Erytra\conf*.
 - o CONFIG_DG421: all the files in *E:\data\Config_DG421*.
 - o Log: all the logs in *E:\Log* folder and its subfolders (keeping subfolders structure) belonging to extended date range. It also includes a subfolder named *Tomcat* with all Tomcat logs (*E:\Program Files\Apache Software Foundation\Tomcat 6.0\logs*) belonging to the extended date range.
 - o RAW: all the raw images from *E:\data\Erytra\Raw* belonging to the selected date range, as long as RAW images have been selected.
 - o Wadiana: all the files in *E:\data\Erytra\Wadiana*.
- *SistemaDiana* folder with the following subfolders:
 - o defs: all the files in *E:\data\SistemaDiana\defs*.
 - o Host: all the files in *E:\data\SistemaDiana\Host* except *LisTest.def* and *LisTest.bak* files.
 - o log: all the files in *E:\data\SistemaDiana\log* included in the extended date range.
 - o Plantillas: all the files in *E:\data\SistemaDiana\Plantillas*.
 - o Registry: *config.reg* file with the Windows registry information from *HKEY_LOCAL_MACHINE\SOFTWARE\WOW6432Node\Grifols*.
 - o Tandas: all tan files from *E:\data\SistemaDiana\Tandas* belonging to the extended date range, as long as TAN files have been selected.

Once completed, a message indicating that Support Pack was generated successfully will be displayed, and a Technical Service Action record will be created with the comment *Support Pack*. The Support Pack creation (and errors, if any) are logged in *erytra* and *erytraDebug* log files. The beginning, end and options selected are logged as *WARN* traces and the detail of all the steps of the process are logged as *DEBUG* traces.

15.17.2 Generating support pack from OS shell

The Support pack generator tool, installed with software version, allows to perform a Support Pack, same as User Application but with no need to start the User Application, neither the need for a USB drive or the presence of technical service personnel.

The use of Grifols Secure Remote Connectivity remote desktop software and the possibility of remotely accessing the instruments allows a faster extraction of the Support Pack with no disturbance to the user work. Additionally, this action can be performed with no need to close the User Application by accessing Windows 10 file explorer using the corresponding scripts, both through local access to the instrument or via Grifols Secure Remote Connectivity software.

15.17.2.1 Support pack creation

In order to create a Support Pack using Support Pack generator follow the next steps:

- 1. Run the “E:\Program Files\SupportPack\run.bat” file.

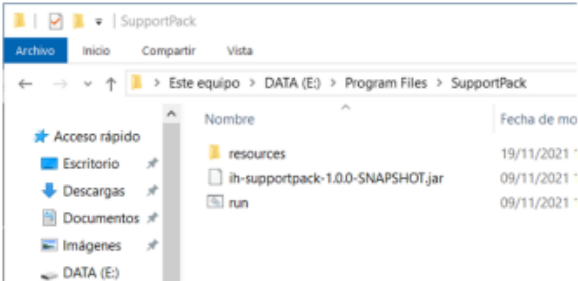


Figure 602 Support pack folder

- 2. A command window will be shown and the “Start date” for the desired time frame is requested. This field must be entered with the format YYYY-MM-DD where:
 - a. YYYY: Year (Example: 2021)
 - b. MM: Month (Example: 10)
 - c. DD: Day (Example: 21)

As long as these requirements are not met, the “Start date” will still be requested.



Figure 603 Support pack start date introduction

- 3. After entering a correct “Start date”, the “End date” for the desired time frame is requested, expecting the same format as when introducing the “Start date”. Additionally, the “End date” must be higher than the “Start date”. As long as these requirements are not met, the “End date” will still be requested.

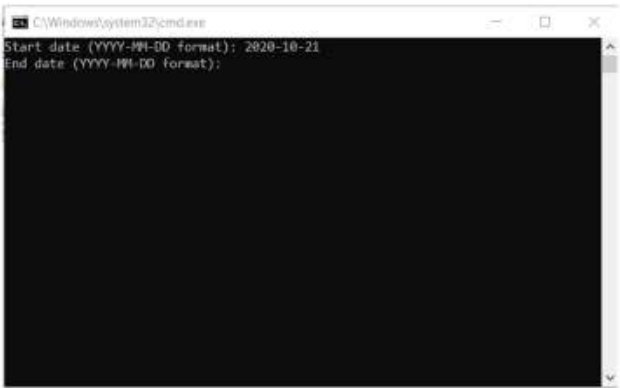


Figure 604 Support pack end date introduction

4. After entering a correct “End date”, the application will ask whether RAW images should be included in the Support Pack or not. In case of an affirmative answer by “Y or y” keys, the available RAW images in the instrument will be packed into the Support Pack. Otherwise, keying “N or n” means that the RAW images will not be included in the Support Pack.

Only the keys “Y, y, N and n” are allowed as an answer. As long as these requirements are not met, the “Include RAW image files” question will still be requested.

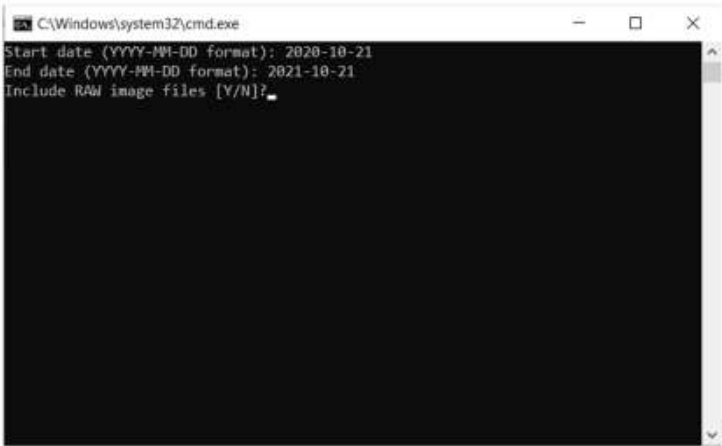


Figure 605 Support pack "Include RAW image files" question

5. After entering a valid parameter for the “Include RAW image files” question, the application will ask whether TAN files should be included in the Support Pack or not. In case of an affirmative answer by “Y or y” keys, the available TAN files in the instrument will be packed into the Support Pack. Otherwise, keying “N or n” means that the RAW images will not be included in the Support Pack.

Only the keys “Y, y, N and n” are allowed as an answer. As long as these requirements are not met, the “Include TAN files” question will still be requested.

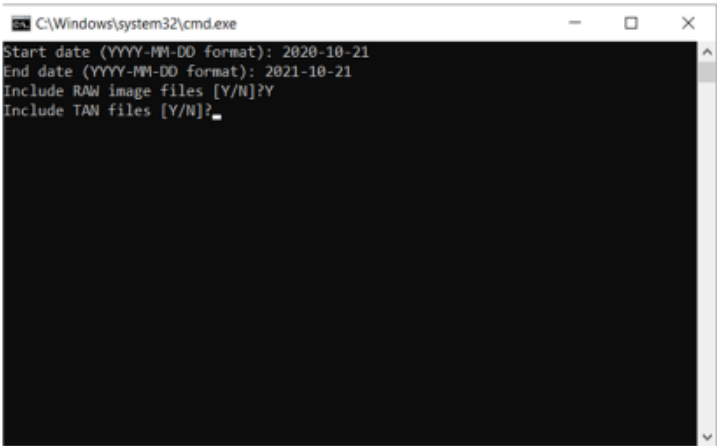


Figure 606 Support pack "Include TAN files" question

6. After entering a valid parameter for the “Include TAN files” question, the Support Pack generator will start to generate the pack.

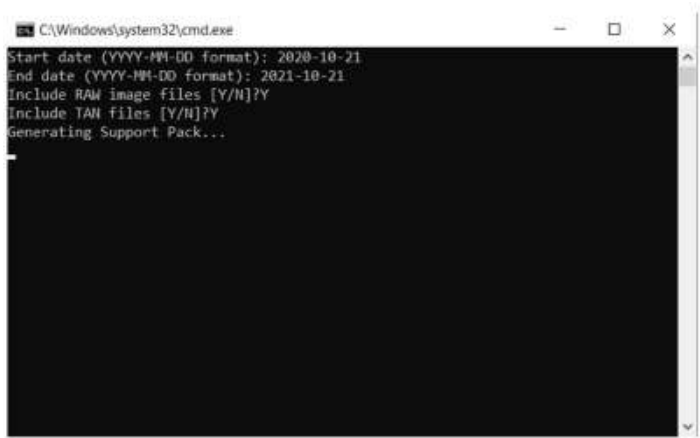


Figure 607 Support pack generation

7. At this moment, the Support Pack generator will perform the following actions:
- a. Determine the extended date range. That is, determine the date range to include the beginning of the session related to the selected “Start date” and the end of the session related to the selected “End date” parameters.
 - b. A temporary folder with name “E:\<INSTRUMENT>_<SERIALNUMBER>_<YYMMDDHHMM>_SupportPack” is then created to generate the pack, where YYMMDDHHMM is the YEAR (YY), month (MM), day (DD), hour (HH) and minutes (MM) at which the pack has been created. Additionally, <INSTRUMENT> is “ERYTRA” for Erytra instruments. For instance:

“E:\ERYTRA_0001123_2110211515_SupportPack”
 - c. If “E:\SupportPack” folder does not exist, it will be created.
 - d. The temporary folder is zipped with the name “<INSTRUMENT>_<SERIALNUMBER>_<YYMMDDHHMM>_SupportPack.7z” And copied into “E:\SupportPack\<INSTRUMENT>_<SERIALNUMBER>_SupportPack” folder. For instance:

“E:\SupportPack\ERYTRA_0001123_SupportPack\ERYTRA_0001123_2110211515_SupportPack.7z”
 - e. The temporary folder is deleted.
8. When these actions are finished, the Support Pack generator will inform to the user that “SupportPack generated successfully”
9. Press any key to close the cmd window.

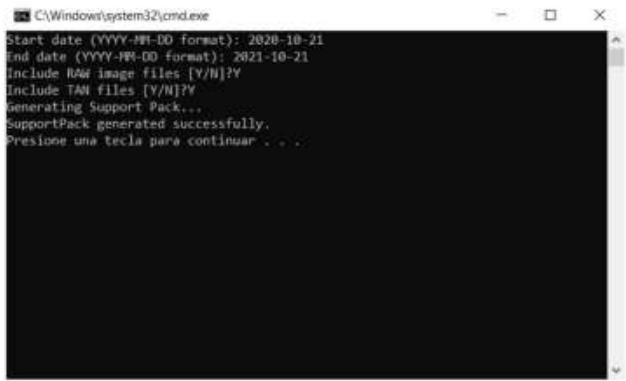


Figure 608 Support pack generated successfully

At this point, the zipped Support Pack is ready to be copied or transferred, and subsequent packs will be created in the “E:\SupportPack” folder following the above- mentioned criteria.

Additionally, the Support Pack generator will create a separate log file, located at “E:\log” folder and named “SupportPack.log” where all the steps related to the Support Pack generation process will be traced.

15.18 Mapping a network drive

If *Erytra* needs to access a shared folder of a remote computer, it can be done in two ways:

PROCEDURE 1:

1. Connect an external mouse to the instrument.
2. Switch on the instrument and press the grid () button during system startup (login credentials will be required), then select the *Maintenance Mode* icon among the Boot Application options.
3. Once PC reboots and desktop is available, press Windows *Start* button and type *Services.msc*. Rick click at the top result and select *Run as administrator*, then type administrator password *dgroot*..
4. Map a unit in *Erytra* computer to the destination shared folder of the remote computer. Open *This PC* in an explorer browser, then select *Map Network Drive* option on the toolbar. In the *Folder* box, type the path to the server.

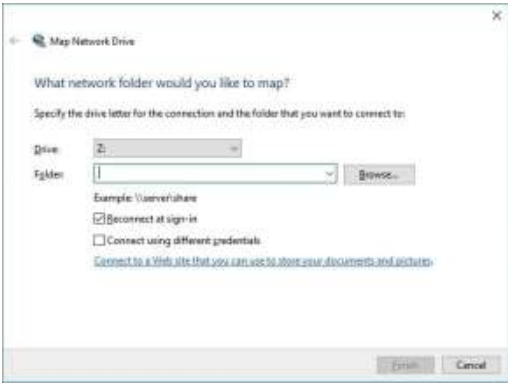


Figure 609. Mapping a network unit

5. Reboot the instrument.

PROCEDURE 2:

A template script named mappedDrive.template can now be found in the folder E:\Program Files\System Tools.

The system will execute all files named MappedDrive (except the template) at startup.

Each unit to be mapped require an individual MappedDrive file properly filled.

Follow the next steps:

1. Enter the Boot Application.
2. Select ‘Maintenance Mode’ option.
The PC loads the ‘Maintenance Mode’ desktop of the operating system.
3. Go to the following path:

- E:\Program Files\System Tools\
4. Open 'mappedDrive.template' file.
 5. Edit the file as follows:

Template: 'mappedDrive.template'

localPath=X:
remotePath=\\server\folder
user=username
password=password

Example: 'mappedDrive.M'

localPath=M:
remotePath=\\server01\folderA
user=dan
password=1234



NOTE:

Character = is not allowed in any of the MappedDrive file variables (Path, username or password).

6. Save the changes.

Mapping attempts and result will be logged in installation log (E:\log).



NOTE:

Duplicate the template as many mapped drive units are required. Replace filename and 'localPath' with a different unit name for each mapped drive.
C: D: E: and R: units are system units and cannot be used.



NOTE:

Ensure there are not USB drives connected with the assigned drive units that are configured.

15.19 Erytra- LIS connection

The connection of *Erytra* to the *LIS* is established through *Diana System* (DianaServer program) and Diana System names (tests, microtubes,...) will be used.

Erytra/Diana System to LIS communication has been implemented in accordance with the *CLSI LIS01-A2* and *CLSI LIS02-A2* standards (formerly *ASTM E1381-02* and *ASTM E1394-97*, respectively). *CLSI LIS02-A2* standard establishes a format for the messages that must be sent from the LIS to the instrument and vice versa, but does not specify the way in which they must be sent. *CLSI LIS01-A2* standard describes the way in which messages should be physically sent to the LIS, but does not specify a format.

Erytra can communicate with the *LIS* in 3 ways:

- Following *CLSI LIS01-A2* standard, where the computers serial channel is used.
- By saving messages in files on local or remote disk units that can be accessed by both, *LIS* and *Erytra*.
- By using TCP/IP protocol in a network domain.

Diana System Connectivity (DSC) is the *Erytra* software section that translates/links *Erytra* and Diana System names.

DianaServer will temporarily store the requests sent by the LIS in *LisTests.def* file. When a result has to be transmitted to the LIS:

- bi-directional transmission: the existing record in the list is updated
- uni-directional/re-send transmission: a new record in is created in the list with the information to be transmitted and, once the transmission has been completed successfully, DianaServer deletes the transmitted record.

15.19.1 DianaServer program

DianaServer is the Diana System program that communicates with the LIS. In order to use it, it should be previously enabled by ticking the corresponding checkbox in Diana O. program (Listas.exe) located in E:\data\SistemaDiana → Program configuration → Others tab. Software installer will set it as enabled when no previous configuration is found. If a previous configuration is found, it will be kept. As the communication between DianaServer and erytra eflexis software uses IP protocol, the IP Address and Port fields of the PC where DianaServer is going to be used should also be configured in this window (default values are localhost and 8070, respectively).

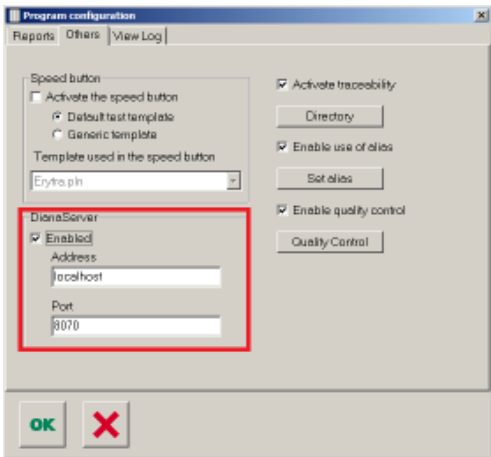


Figure 610.DianaServer configuration options in Diana O.

When executing DianaServer, an icon appears in the bottom-right edge of the Windows taskbar:



Figure 611.DianaServer icon

Right clicking with the mouse brings up a context menu with the options *Configuration* and *Close*. Clicking on the *Configuration* option opens following window:

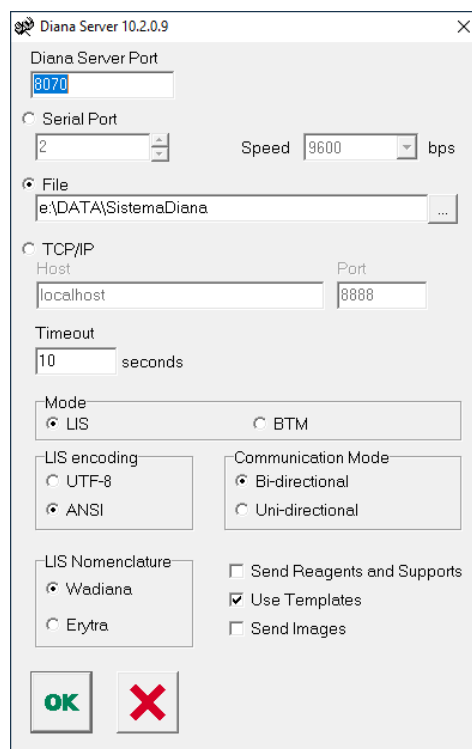


Figure 612.DianaServer configuration window

The diferents options to configure are:

- DianaServer port: Must match with the *Port* number indicated in Diana O. configuration (default value is 8070).
- Communication mode (*Serial Port*, *File* or *TCP/IP*): See below for a detailed explanation of each mode.
- Timeout: See File communication mode for more detail.
- Mode: Connection mode selection, LIS or BTM. See below for detailed explanation.
- LIS encoding: Character encoding system. Default option is ANSI, but UTF-8 can also be selected, making sure that the selected system matches the corresponding language in Windows, *Region and Language* settings.
- LIS Nomenclature: Nomenclature of interpretations, microtubes, results,... transmitted to LIS. Options are *Diana System* (nomenclature used in previous software versions) or *Erytra system* (the names displayed in user interface will be the ones transmitted to LIS).
- Communication mode (Bi-directional / Unidirectional): It does not apply/affect *erytra efflexis* - LIS communication.
- Send reagents and supports: If enabled the barcodes of the cards and reagents used to process each sample are transmitted to LIS. This new option will modify the already existing Windows registry parameter SendBCReacSupp.
- Use templates: When enabled, LIS must indicate/send a template and this template will be used to transmit the results to LIS. When disabled, LIS must not indicate/send a template and all the interpretations and microtubes results displayed in user interface results detailed view will be transmitted to LIS.
The default status is enabled.
- Send images: Option to include microtubes images in the ASTM message to LIS. The default status is disabled.

Serial communication mode

The following settings may be selected:

- Serial port of *Erytra* PC that is connected to the LIS (**COM2** expected for PC EMBEDDED v3, v4 and v5, **COM3** expected for PC EMBEDDED v2 / PC EMBEDDED - USA).
- Transmission speed in bauds.

The rest of settings are fixed (and set when capturing the port):

- Parity not used.
- 1 start bit.
- 1 stop bit.
- 8 data bits.

File communication mode

It allows setting the files exchange directory.



NOTE:

The exchange directory shall be unique. That is, if several systems should be connected to host and more than one DianaServer is used, each DianaServer must be configured with a different exchange directory.

TCP/IP communication mode

It allows setting the host IP address and the related port.

Timeout

This setting is the time in seconds that the DianaServer should wait before receiving an answer once a message has been sent to the LIS.

LIS or BTM connection mode

LIS: when this option is selected, LIS encoding, LIS Nomenclature, Send Reagents and Supports, Use Templates and Send images can be configured.

When LIS and Erytra options are selected, the names of the segment, etc will be transmitted in the Eflexis configured application language.

When LIS and Wadiana options are selected, the names of the segment, etc will be transmitted in the Diana System configured applications language.

Default LIS mode is: LIS.

BTM: When this option is selected, the following parameters are set to required values:

- Encoding: UTF-8
- Nomenclature: Erytra
- Send Reagents and Supports: Yes
- Use Templates: No
- Send Images: Yes

And they are greyed out (non-modifiable).

The names of the segment, etc will be transmitted in English and Comments field will transmitted as 'Remarks'.

15.19.2 Serial communication mode

Using the serial channel system, messages are sent to the Host using the CLSI LIS01-A2 standard.

Serial communication is made through serial port/cable directly to the LIS. The message is chopped into frames.

After each frame is sent, the receiver has to reply with an 'OK' message in order to continue the transmission.

Each frame is numbered so a flow control can be established.

15.19.3 Files communication mode

File exchange mode is as follows: When *Erytra* has to do a request/send a message to the LIS, DianaServer writes the request/message (following the CLSI LIS02-A2 standard) in a file named *TX_TMP.DAT* in the configured exchange directory. Once the complete message is written, this file is renamed to *OUT.DAT*.

LIS shall read this *OUT.DAT* file and then delete it (to indicate that it has been read) within the allowed time set in time out parameter (default time 10 seconds). If the file is not deleted within the allowed timeout, an error is generated.

The LIS shall write the answer message in a file named *IN.DAT* located in the configured exchange directory. DianaServer will read this message and delete it to inform the LIS that the message has been read.

The LIS can not initiate the communication with the *Erytra*.

15.19.4 TCP/IP communication mode

TCP/IP communication mode allows using TCP/IP protocol when exchanging information with the host. In order to use it, the host IP address and port must be indicated in the corresponding textboxes.

Data is encapsulated according to the TCP/IP protocol and transmitted through the network once the latter has been configured accordingly.

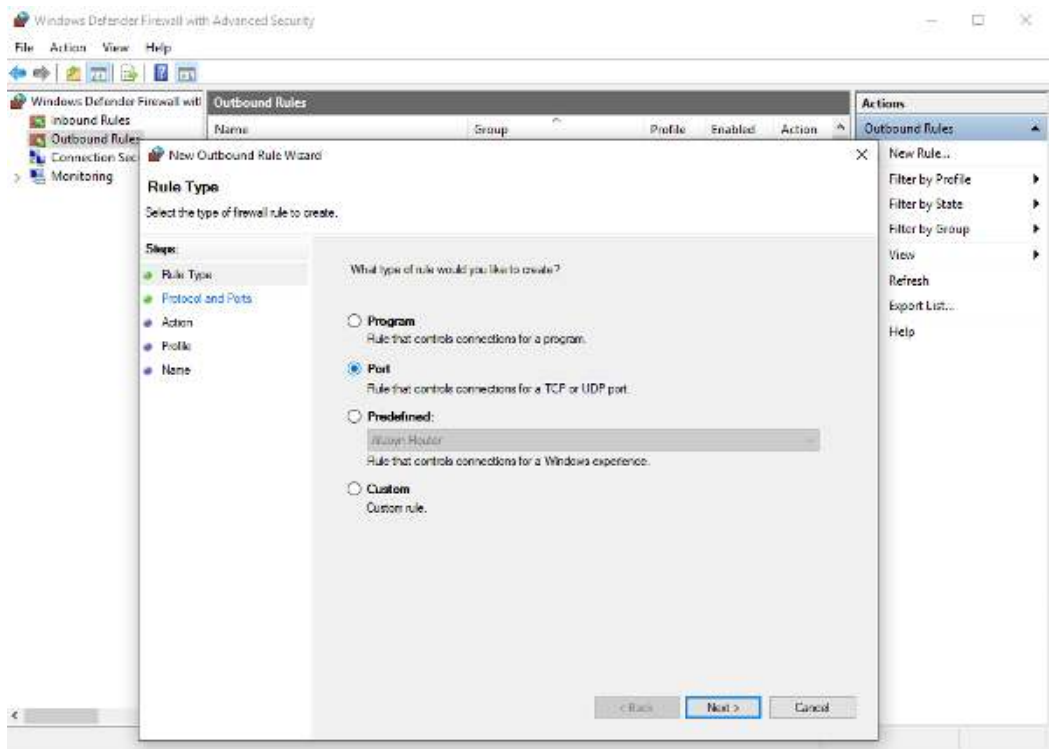
In order to allow TCP/IP communication it is necessary follow the next steps to create a firewall rule:

1. Connect an external mouse to the instrument.
2. Switch on the instrument.
3. Press the grid  button to enter the boot application and log in as adminUser.

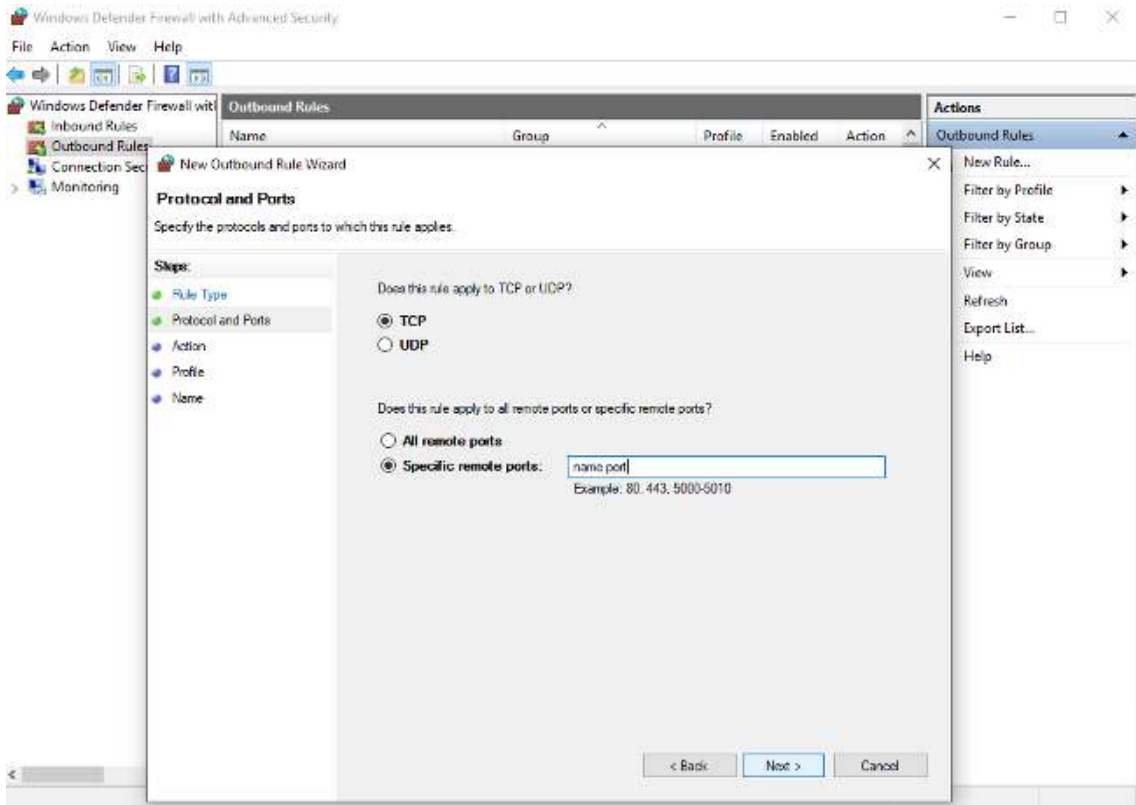


4. Select *Maintenance Mode* option.
5. Open Windows Defender Firewall advanced settings.
How to: Select Start → *Windows System* → *Control Panel*.
Once Control panel is opened, select *System and Security* → *Windows Defender Firewall* → *Advanced settings*
6. Select *Outbound Rules* and *New Rule...* option in *Actions* section.

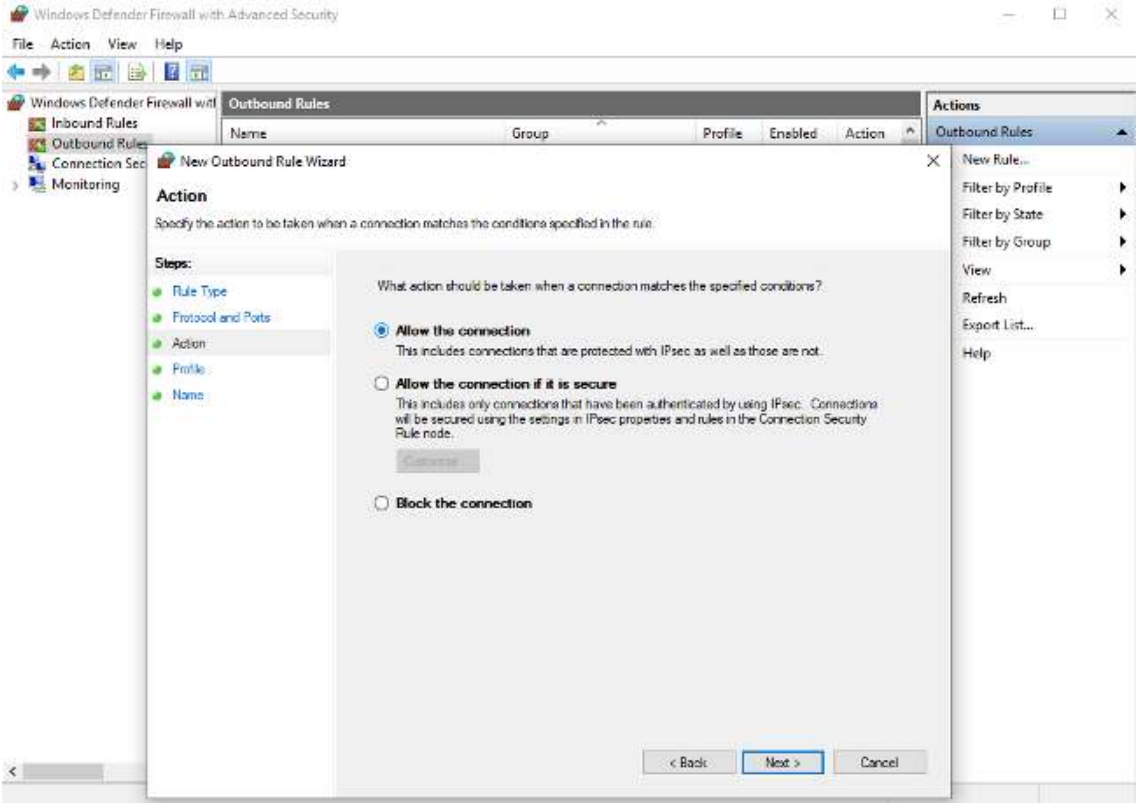
7. Select *Port* in the drop-down menu. Press Next >.



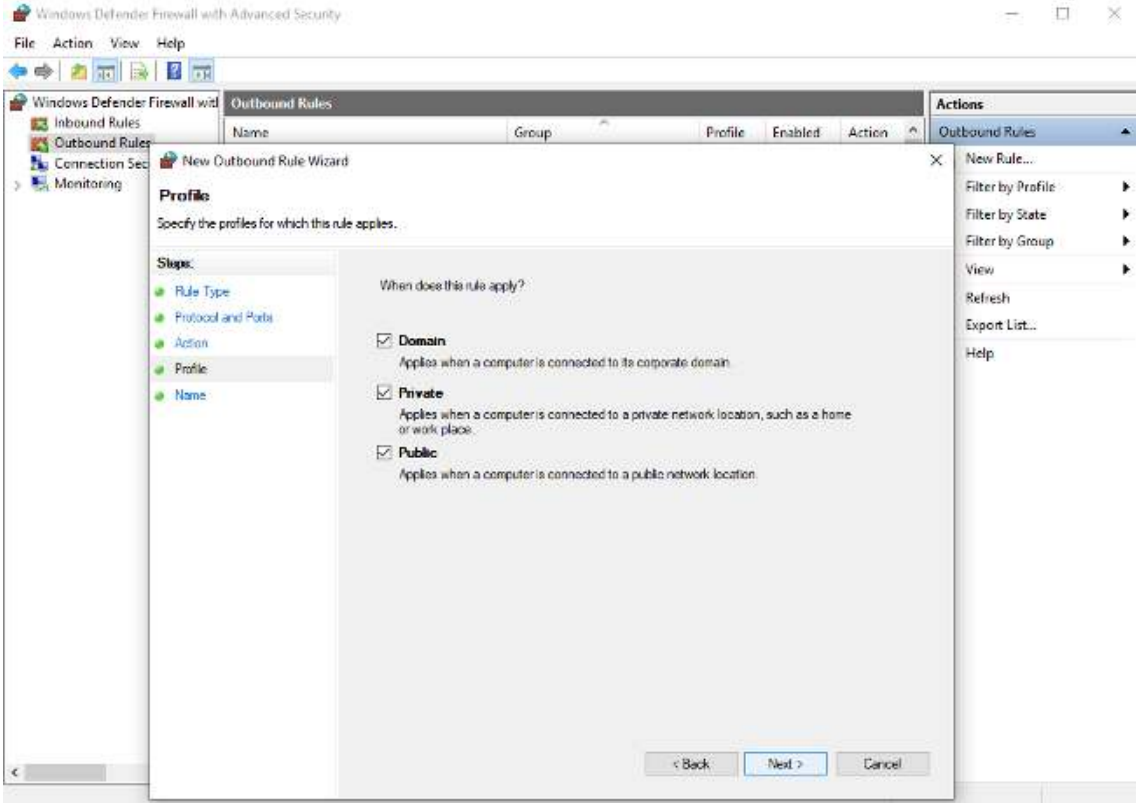
8. Select TCP option and fill the field with the specific remote port. Press Next



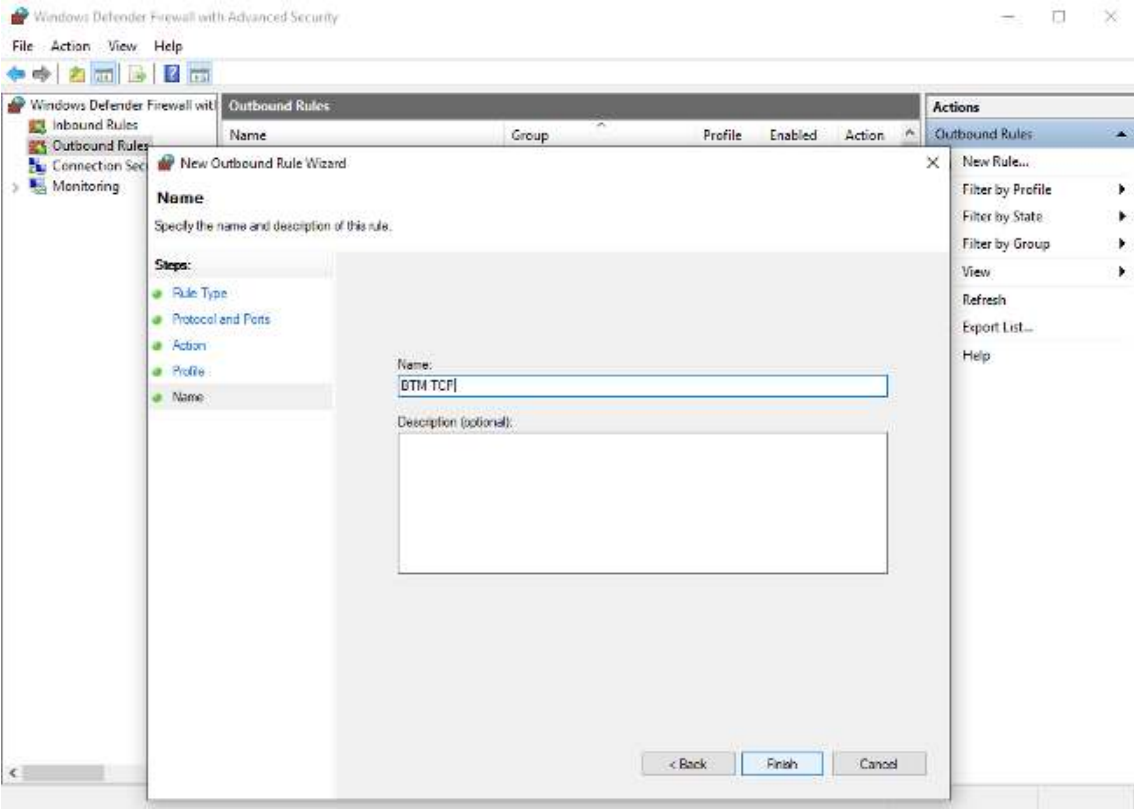
9. Select *Allow the connection* or *Allow the connection if it is secure* and select Next.



10. Do not change the default settings and click on Next.



11. Enter the name of the exception and press *Finish*. The selected rules will be created and enabled



12. Close Windows Defender Firewall with Advanced Security window.

15.19.5 HostDemo program

HostDemo.exe is a technical support tool, the aim of which is to check that the LIS connection is properly configured and information can be exchanged between Diana System and LIS.



NOTE:

While the character encoding system can be configured in DianaServer program (ANSI/UTF-8), HostDemo can only use ANSI system.

HostDemo.exe file is automatically installed when installing/upgrading *Erytra* software in *E:\data\SistemaDiana* directory. Upon opening the program, the software expects the following credentials :

- **User:** ST
- **Password:** d+g_diana

Once user name and password are introduced, the following window will appear:

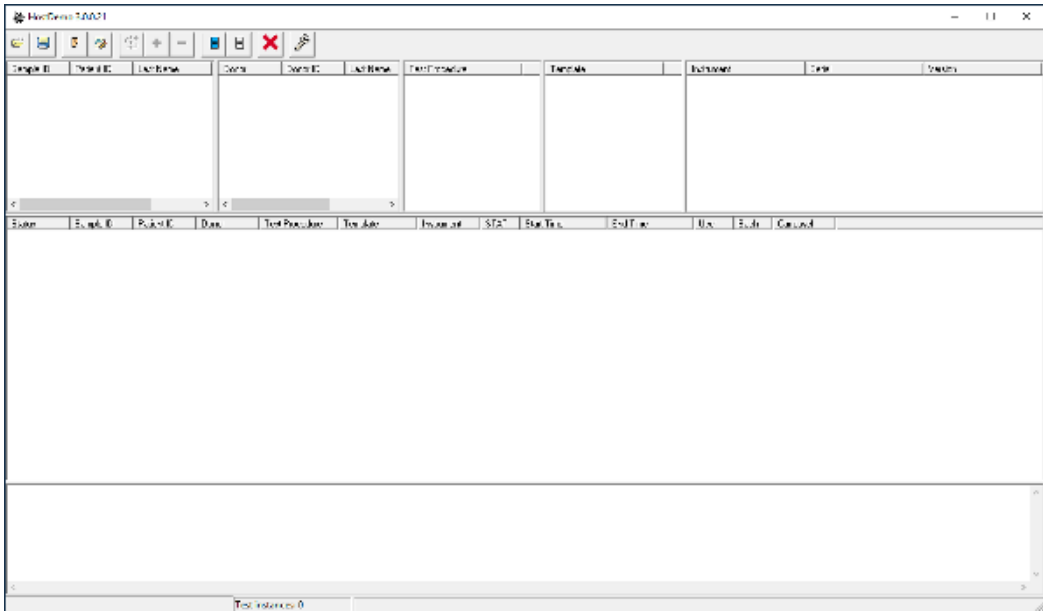


Figure 613.HostDemo program

The HostDemo user interface includes 6 different informative sections:

- List of available samples.
- List of available donors.
- List of available test procedures.
- List of available templates.
- Historical list of connected instruments.
- List of test request.

.hst or .dat file extension are allowed.

Option to create samples and donors.

Option to create automatically a random list of samples and donors.

Option to change the STAT flag of a test requests.

Option to import test procedures and templates from a Diana System installation.

Option to delete test procedures.

Option to delete templates.

Option to ignore requests from REPORT (Diana O.) instruments.

Option to automatically change the status of Cancelled tests to New.

Option to view the detailed results of finished test:

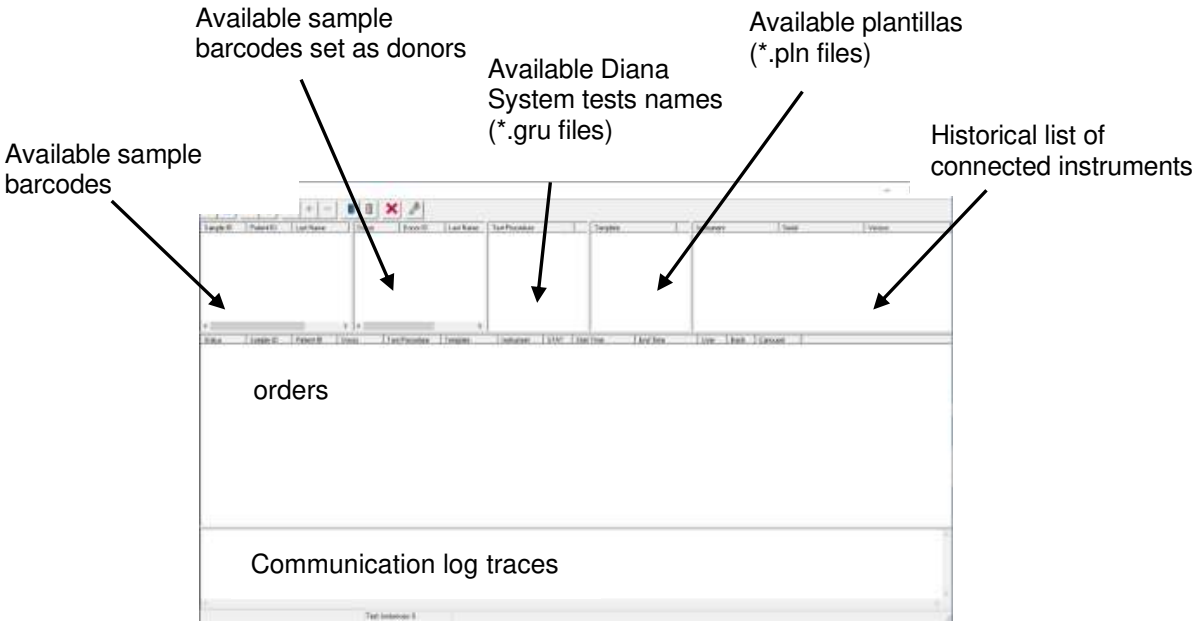
- Wells name.
- Segments name.
- Well images.
- Comments.
- Wells results.
- Segments result.
- Reagent and card barcodes.

New HostDemo.log in the (installation folder) subdirectory named *log*.

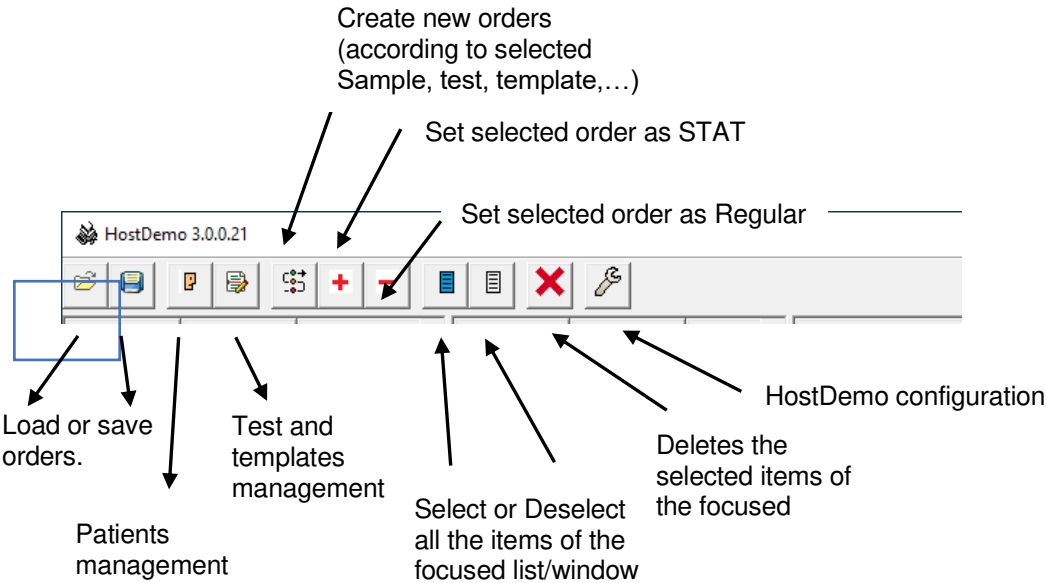
The HostDemo does not allow to assign two different sample barcodes to a given patient.

An *Access violation* error may be displayed when the loading test procedures and template from Diana System directory has been completed successfully.

New HostDemo program main window has now following sections:



And following main menu buttons:



Patients Management:

Patients management window is opened when Patients Management button is selected:

New single patient creation Window

New multiple patient creation Window

Deletes selected patients

Sets or Removes donor status selected patients

Selects or Deselects all patients

People Database Viewer

Patient ID	Sample ID	Title	First Name	Middle Name	Last Name	Suffix	Gender	Donor	Birth date
P00000001	00000001		Cheryl	Madison	Cisneros		Female	No	27-05-1988
P00000002	00000002		Brooke	Cassidy	Rhodes	PhD	Female	No	14-03-1979
P00000004	00000004	Miss	Carol	Kelsey	Lutz		Female	No	05-04-1983
P00000005	00000005	Mr.	Bryce	Lawrence	Martin		Male	No	16-12-1914
P00000006	00000006		Martin	Frederick	Stout		No	No	17-06-2003
P00000007	00000007	Miss	Peggy	Alexa	Sullivan		No	No	30-04-1982
P00000008	00000008	Miss	Ebony	Joyce	Wang		No	No	09-07-1984
P00000009	00000009		Lindsey	Bonnie	Mannin		No	No	15-03-2007
P00000010	00000010	Dr.	Paige	Tracey	Bradsh		No	No	16-07-2018
P00000011	00000011	Mr.	Bernard	Joshua	Allison		No	No	03-07-2011
P00000012	00000012	Dr.	Vicki	Jenna	Torres		No	No	15-07-2003
P00000013	00000013		Chase	Norman	Salas		No	No	17-04-1956
P00000014	00000014		Phillip	Eugene	Ramirez		No	No	17-07-1986
P00000015	00000015	Miss	Joanna	Ana	Marsh		No	No	22-10-2015
P00000016	00000016		Kristopher	Noah	Perry		Male	No	02-05-1975

Mouse right button will show a contextual menu with several options when a patient is selected.

Double clicking left mouse button opens in this window or in the main one Patient data can be edited.

Edit Person

Patient ID

Sample ID

P00000013

00000013

Title

First Name

Middle Name

Last Name

Suffix

Chase

Norman

Salas

Gender

Birth date

Donor

☐ Female

☒ Male

☐ Undefined

4/17/1956

☐ Donor

OK

X

And Multiple Patients creation window is opened when selecting this option, allowing to define starting Patient, how many new patients will be created and Patient and Sample pattern:

Create Multiple Patients

Start

Count

108

100

Patient Pattern

Sample Pattern

P%08d

%08d

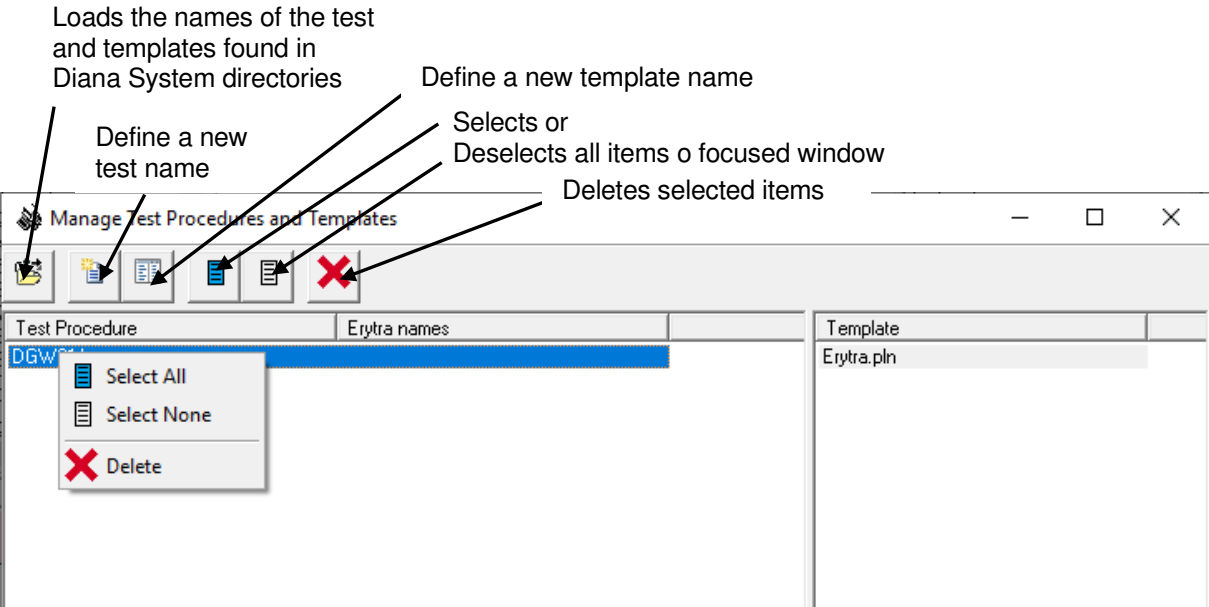
☐ Donor

OK

X

Test and Templates management

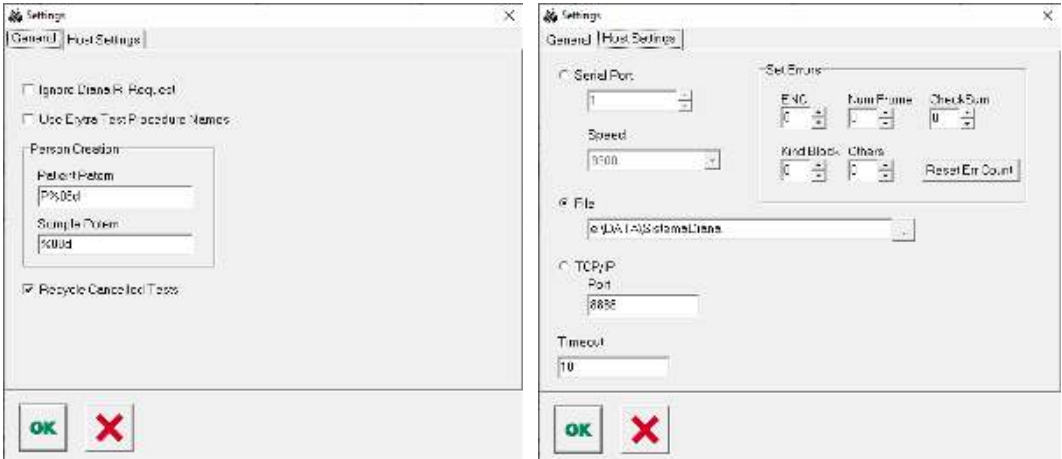
Test and Templates management window will open with following options:



Mouse right button will show a contextual menu with several options when a patient is selected.

Configuration:

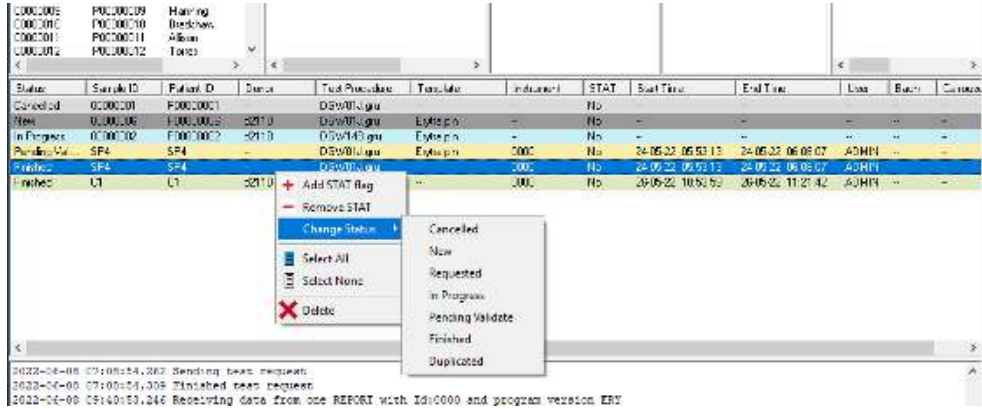
Configuration window includes two tabs, the one related to general options and the on related to Host Settings:



Orders section:

The orders are displayed in this section of main window.

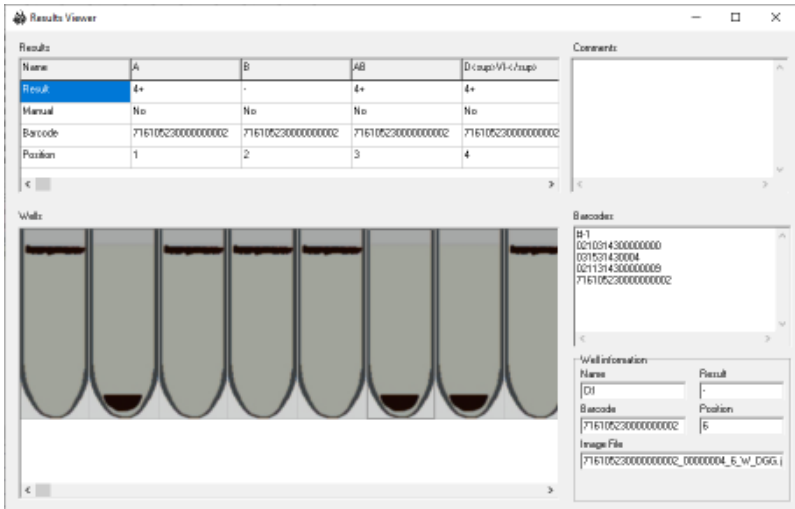
Order color indicates order status.



Mouse right button will show a contextual menu with several options when an order is selected.

One of this option is *Change Status*, that allows to manually modify order status.

Double clicking left mouse button on the selected order will open *Test Results* window, that allows reviewing microtubes results, images, reagents,...



Hostdemo created DianaASTM.log file in E:\data\SistemaDiana\log\log folder

15.19.6 How to enable and configure LIS connection

To enable and configure *Erytra*-LIS conection, proceed as listed below. If connected, it is also recommended to disconnect the instrument from the network during the configuration procedure.

Erytra software *configuration*

1. Switch on the instrument and press the grid () button during system startup (login credentials will be required), then select the *Maintenance Mode* icon among the Boot Application options.
2. PC will be rebooted and Maintenance Mode will be accessed in the subsequent startup.

3. Once desktop is available, press *Windows Start* button and type *Command Prompt*. Right click at the top result and select *Run as administrator*, then type administrator password *dgroot*.
4. Enter the command: *applocker bu* (Begin Update). The AppLocker Manager graphical interface will be opened in a separate window.
5. Wait for 10 seconds so that the change to update mode is completed.
6. Edit *E:\conflerytra.properties* file using *Notepad++*.
7. Set *LIS_ACTIVATED* parameter to **true**.
8. Check/set *LIS_SHARED_MODE* parameter:
 - false:** sample requests where the sample is missing will be held waiting for the missing sample to be identified (default value).
 - true:** sample request where the sample is missing will be sent back to LIS as cancelled.
9. Check/set *RECEIVE_AND_SHOW_DEMOGRAPHIC_DATA* parameter (only for *CE* instrument configuration):
 - false:** do not store in Erytra database and do not show demographic data in Erytra application (default value).
 - true:** store in Erytra database and show demographic data in Erytra application.
10. Check/set *LIS_AUTO_QUERY_BY_SAMPLE* parameter:
 - false:** no automatic request to LIS will be performed after a sample drawer is identified or a sample tube is manually identified (default value).
 - true:** once a sample drawer is identified or a sample tube is manually identified, automatically request to LIS all pending work for every identified sample inside the instrument.
11. *AUTO_LIS_QUERY_ENABLED*: It allows enabling and disabling the periodic automatic query to LIS.
 - false:** Auto LIS query feature is disabled (default value).
 - true:** Auto LIS query feature is enabled.
12. *AUTO_LIS_QUERY_MODE*: It allows to set the query mode to Query All or to Query by Sample (when *AUTO_LIS_QUERY_ENABLED*=true).
 - QALL:** Query All will be sent to LIS (when *AUTO_LIS_QUERY_ENABLED*=true)
 - QBSAMPLE:** Query by Sample will be sent to LIS. (when *AUTO_LIS_QUERY_ENABLED*=true) (default value).
13. *AUTO_LIS_PERIOD_UPDATE*: It allows defining the amount of time in minutes that will elapse between each periodic automatic query (when *AUTO_LIS_QUERY_ENABLED*=true).
 - Values allowed: integers from 1 to 60.
 - Default value: 5.
(Default value will be applied if a value out of the allowed range is found)
14. Save the file and close the *Notepad++*.

DianaServer configuration

15. Enable DianaServer and set the desired IP address and port number in the *Program Configuration* options in Diana O. Then start DianaServer and configure all the parameters. See section 15.19.1 for details on each of the expected parameters.



NOTE:

When connecting 2 or more *Erytra* computers to the same network domain and workgroup, it is recommended to rename the name of the computers to avoid conflicts between them.

Configure DianaServer to automatically start at PC startup

Software installer will configure the system to start DianaServer automatically when user interface starts. In case this configuration is disabled, please follow the steps below to enable it:

Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	616 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

16. Go to *E:\Program files\Startup* and create an empty text file called *DIANASERVER.START*. Make sure that the original file extension '.txt' is also removed from the file name (in order to delete it, make the extension visible by clicking on the *Organize* button in the current explorer window, select *Folder and search options* → *View* tab and uncheck the *Hide extensions for known file types* option. Once the '.txt' extension is deleted, enable the option again so that file extensions remain hidden).
17. Type *eu* (End Update) in the AppLocker Manager graphical interface window.
18. Wait for 10 seconds so that the change to protected mode is completed.
19. Reboot the system to exit Maintenance Mode.

Confidential Information. It is not allowed the total or partial reproduction of this document without prior authorization			Purpose:	Uncontrolled Copy
Print date:	16-May-2023			

15.19.7 Erytra – LIS errors

In this section, Erytra and LIS communication errors are listed. Following tables include Diana System errors and Erytra software errors related to LIS communication.

Module	Error code	Cause
Message contents errors (ASTM-1394 standard)	Host format error #1	Heading error: wrong separators
	Host format error #2	Heading error: first line of the message does not begin with 'H'
	Host format error #3	Error Reading Patient No patient ID
	Host format error #4	Test: Not enough fields
	Host format error #5	Test: No specimen field
	Host format error #6	Test: No test name fields
	Host format error #7	Error, an O field before a P field
	Host format error #8	Error, an R field before an O field
	Host format error #9	Results: Not enough fields
	Host format error #10	File received has no lines, it is empty
File communication errors	Host file error #1	Exchange file TX_TMP.DAT / OUT.DAT can not be written in the working directory
	Host file error #2	Exchange file TX_TMP.DAT / IN.DAT has been created but it is not possible to fill it
	Host file error #3	IN.DAT file can not be deleted
	Host file error #4	OUT.DAT file can not be deleted
TCP/IP communication errors	Reomote host not ready	LIS server is not ready when trying to establish communication with DianaServer.
Serial or TCP/IP communication errors	Host serial error #1	Time-out when receiving frame
	Host serial error #2	Incorrect frame received
	Host serial error #3	Incorrect frame received, STX fault
	Host serial error #4	Neither ETB or ETX being received in the frame
	Host serial error #5	Frame received without checksum
	Host serial error #6	Frame received with wrong checksum
	Host serial error #7	Frame received with wrong ID
	Host serial error #8	Communication error in establishing the connection while awaiting reception
	Host serial error #9	Undetermined error
	Host serial error #10	Frame sent 6 times and returned by LIS as faulty (NAK)
	Host serial error #11	ENQ received while establishing communication
	Host serial error #12	Strange character received while establishing communication
	Host serial error #13	EOT missing

Table 94. Diana System errors related to LIS communication

Module	Error code	Error	Cause	Text	Options
LIS communication	5000002	ERROR_POCILLOS_DEF	Unable to access Pocillos.def file.	LIS error: Unable to initialize file Pocillos.def. Please contact Technical Service	OK
	5000003	ERROR_INTERP_DEF	Unable to access INTERP.def file.	LIS error: Unable to initialize file Interp.def. Please contact Technical Service	
	5000004	ERROR_ALIAS_DEF	Unable to access alias.def file.	LIS error: Unable to initialize file Alias.def. Please contact Technical Service	
	5000005	ERROR_IDIOMA_L_DEF	Unable to access Idioma_l.def file.	LIS error: Unable to initialize file Idioma_l.def. Please contact Technical Service	
	5000006	ERROR_DSC_DEF	Unable to access DSC.def file.	LIS error: Unable to initialize file DSC.def. Please contact Technical Service	
	5000007	ERROR_ERYTRA_DATA	Unable to find erytra_data environment O.S. variable.	LIS error: Unable to find erytra_data environment variable. Please contact Technical Service	
	5000008	ERROR_TIMEOUT_REQUEST	Timeout exceeded during a request to DianaServer.	LIS error: Timeout exceeded during a request to DianaServer. Please contact Technical Service	
	5000009	ERROR_REQUEST	DianaServer request error.	LIS error. DianaServer request error. Please contact Technical Service	
	5000010	ERROR_NO_INIT	Unable to initialize dsc.dll file.	LIS error: DLL has not been initialized. Please contact Technical Service	
	5000011	ERROR_GRU_NOT_FOUND	Unable to find gru in DSC.def file.	LIS error: Gru x not found in DSC.def. Please contact Technical Service	
	5000012	ERROR_TEMPLATE	Unable to load template.	LIS error: Error loading template x. Please contact Technical Service	
	5000013	ERROR_XML	Error when dsc.dll and java try to communicate.	LIS error: Error loading XML: x. Please contact Technical Service	

Module	Error code	Error	Cause	Text	Options
LIS communication	5000014	ERROR_RESULT_SEGMENT_NOT_FOUND	Unable to find result or segment in DSC.def file.	LIS error: Result of segment x not found in DSC.def. Please contact Technical Service	OK
	5000015	ERROR_WELL_NOT_FOUND	Unable to find well in DSC.def file.	LIS error: Well x not found in DSC.def. Please contact Technical Service	
	5000016	ERROR_RESULT_WELL_NOT_FOUND	Unable to find result in DSC.def file.	LIS error: Result of well x not found in DSC.def. Please contact Technical Service	
	5000017	ERROR_INCIDENCE_WELL_NOT_FOUND	Unable to find incidence well.	Program error: Well incidence not found in DSC.def. Please contact Technical Service.	
	5000018	ERROR_IMAGE_WELL_NOT_FOUND	Unable to find image well.	LIS error: Well x not found in DSC.def. Please contact Technical Service.	
	5000019	ERROR_DELETING_BATCH	Error when deleting batch.	Program error: Error deleting batch. Please contact Technical Service.	
	5000020	ERROR_OPEN_DATABASE	Error when accessing database.	Program error: Error open database. Please contact Technical Service.	
	5000021	ERROR_ERYTRA_LOG	Error when accessing log file.	Program error: Error in Erytra log. Please contact Technical Service.	
	5000022	ERROR_UNKNOWN	Unknown error.	Program error: Unknown LIS error. Please contact Technical Service.	
	5000023	ERROR_DSC_CUSTOM_DEF	Error when accessing DSC_custom.def file.	Program error: Error in DSC_Custom.def. Please contact Technical Service.	
	5000024	ERROR_COMMUNICATION_DIANASERVER	Error in communication with DianaServer.	Program error: Error in communication with DianaServer. Please contact your local Grifols service representative.	
	5000025	ERROR_INTERPRETATION_NOT_FOUND	Interpretation not found in DSC.def.	Program error: Interpretation xxx not found in DSC.def. Please contact your local Grifols service representative.	
			Interpretation not found in INTERP.DEF.	Program error: Interpretation xxx not found in INTERP.DEF. Please contact your local Grifols service representative.	

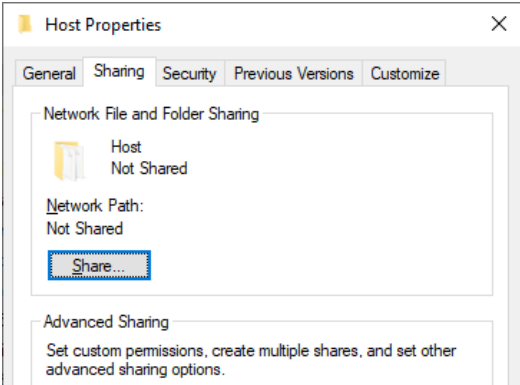
Table 95. List of errors related to Erytra – LIS communication module

SYMPTOM	POSSIBLE CAUSES	RECOMMENDED ACTIONS
Error LIS 5000008 (ERROR_TIMEOUT_REQUEST)	Erytra does not communicate with <i>DianaServer</i>	Check if <i>DianaServer</i> is running
		Check that Erytra has access to the computer where the <i>DianaServer</i> is running
Error LIS 5000009 (ERROR_REQUEST)	<i>DianaServer</i> does not communicate with LIS	Check that LIS is running and it has access to the working directory
		Communication by files
		Check that <i>DianaServer</i> is running and it has access to the working directory
		Increase <i>DianaServer</i> timeout
		Check that serial cable is connected
		Configure serial ports buffer to minimum
		Communication by serial port
		Check/replace serial cable
Error LIS 5000012 (ERROR_TEMPLATE)	Error when trying to use a template to send the results	Check that LIS is running
		Increase <i>DianaServer</i> timeout
		Check that template sent by LIS exists in <i>Plantillas</i> folder of the working directory defined in Erytra (default in <i>E:\data\SistemaDiana\Plantillas</i>)
		Check that defined template by means of <i>Importer</i> application for the related tests exists in <i>Plantillas</i> folder of the working directory defined in Erytra (default in <i>E:\data\SistemaDiana\Plantillas</i>)
No results are sent to LIS	Incorrect definition of the template	Check that default template defined in O.S. registry (<i>HKEY_LOCAL_MACHINE\SOFTWARE\WOW6432Node\Grifols\SistemaDiana\Diana O.\PlantillaDefecte</i>) exists in <i>Plantillas</i> folder of the working directory defined in Erytra (default in <i>E:\data\SistemaDiana\Plantillas</i>)
		Define correctly the wells/segments to send in the template being used
Results sent are truncated	Incorrect template definition	Template is empty
		No wells/segments are defined in the template being used
Well/segments are sent with another name that defined one	Incorrect template definition	Check template definition
		Results field in ASTM message allocates 200 characters maximum including separators
Segments names are sent in another language that desired one	Diana System language is not configured properly	Template field in ASTM message allocates 255 characters maximum including separators
		Check/configure alias.def located in defs folder of the working directory (default location in <i>E:\data\SistemaDiana\defs</i>)
Segments names are sent in another language that desired one	Diana System language is not configured properly	Check/configure Diana System language by means of <i>Config.exe</i> located in the working directory (default location in <i>E:\data\SistemaDiana</i>)

Table 96. LIS troubleshooting

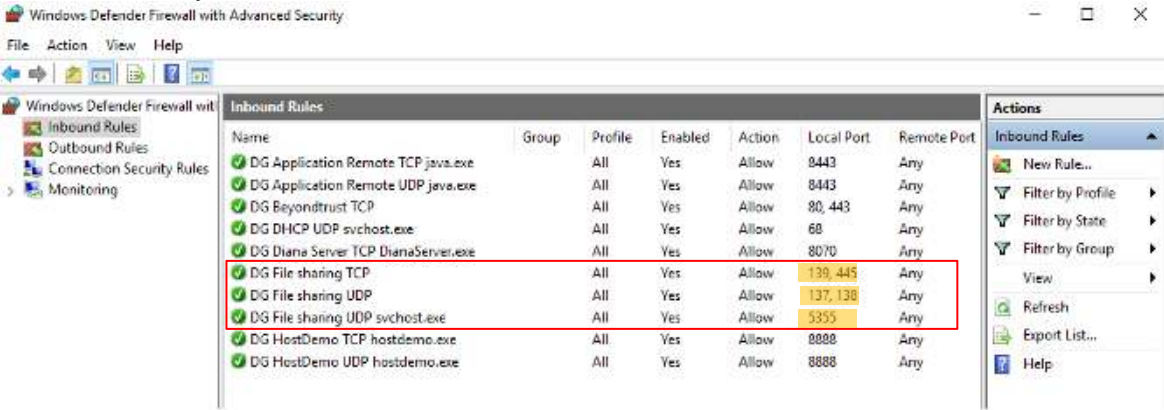
15.20 Share or access a shared folder

The Operating System image version 2.2.1 has been configured to be able to share folders without the need to modify any configuration.

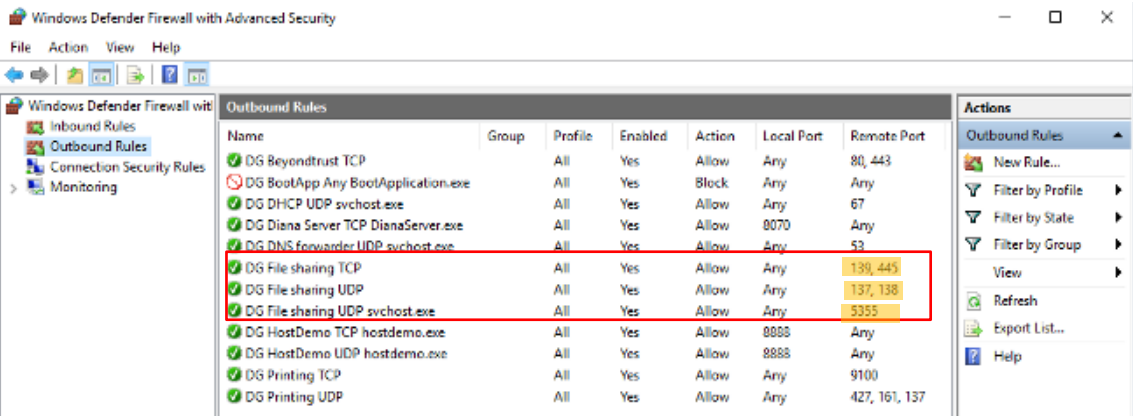


Required Server service and TCP/IP NetBIOS Helper service Startup type are already running (they are set to automatic start up. In previous Operating System versions the Startup type was set to disabled).
The minimum required firewall rules to share or access a share folder have been created and are enabled for all network profiles (Private, Public and Domain).

The automatically enabled inbound firewall rules to share files are:



The automatically enabled outbound firewall rules to share files are:



Then, no additional action is required to be able to access or share a folder in Operating System version 2.2.1 and Erytra software version 4.4.0.

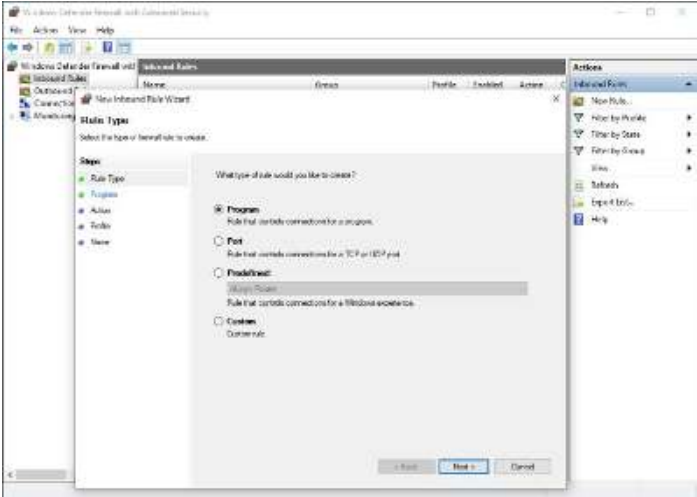
In the event of not being able to share the desired folder with the default configuration, it is possible to create new rules manually as follows:

1. Connect an external mouse to the instrument.
2. Switch on the instrument.
3. Press the grid  button to enter the boot application and log in as adminUser.

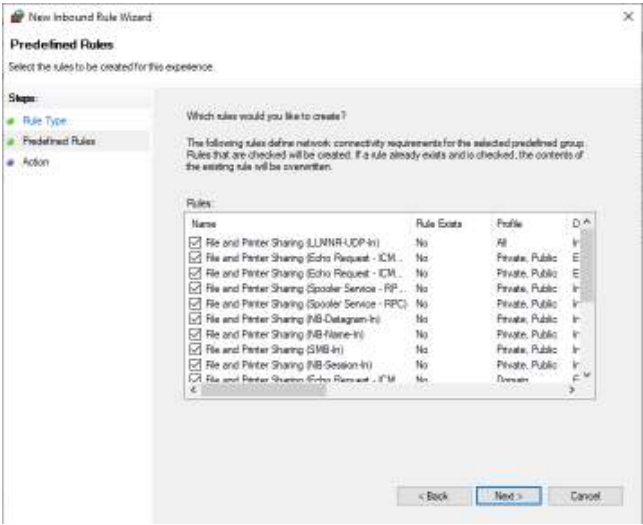
4. Select *Maintenance Mode*  option.

5. Open Windows Defender Firewall advanced settings.
How to: Select Start → *Windows System* → *Control Panel*.
Once Control panel is opened, select *System and Security* → *Windows Defender Firewall* → *Advanced settings*

6. Select *Inbound Rules* and *New Rule...* option in *Actions* section.



7. Select *Predefined* option and *File Printer Sharing* option in the drop-down menu. Press Next >.
8. Review the proposed rules, select the desired ones and then select Next >



Reference:	REGD-0006730	Version:	17.0; Most-Recent; Effective; CURRENT	Page	623 of 623
Status:	Effective	Title:	Service Manual Erytra	Effective Date:	15-May-2023

9. Select *Allow the connection* or *Allow the connection if it is secure* and select *Finish*. The selected rules will be created and enabled.
10. Select *Outbound Rules* and *New Rule...* option in *Actions* section
11. Select *Predefined* option and *File Printer Sharing* option in the drop-down menu. Press Next >.
12. Review the proposed rules, select the desired ones and then select Next >
13. Select *Allow the connection* or *Allow the connection if it is secure* and select *Finish*. The selected rules will be created and enabled.
14. Close Windows Defender Firewall with Advanced Security window.

To keep access to a shared folder, see section 15.18 Mapping a network drive, procedure 2.

Print date:	16-May-2023	Purpose:	Uncontrolled Copy
--------------------	-------------	-----------------	-------------------